The Pearson Guide to Quantitative Aptitude and Data Interpretation for the CAT



CAT

Nishit K. Sinha

ALWAYS LEARNING

PEARSON

THE PEARSON GUIDE TO

Quantitative Aptitude and Data Interpretation for the CAT

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THE PEARSON GUIDE TO

Quantitative Aptitude and Data Interpretation for the CAT

NISHIT K. SINHA
CAT CONSULTANT

PEARSON
Chandigarh • Delhi • Chennai

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ISBN 9788131774120
eISBN 9788131799550

Head Office: A-8(A), Sector 62, Knowledge Boulevard, 7th Floor, NOIDA 201 309, India
Registered Office: 11 Local Shopping Centre, Panchsheel Park, New Delhi 110 017, India

To

Papa and Ma

Mr Kumar Kalyan Pd Sinha

and

Mrs Sanjila Sinha

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Preface

The fact that there was no surprise element in CAT 2010 with respect to CAT 2009 was itself a surprise. Probably IIMs/CAT committee wanted to regain the ground that they might have lost during CAT 2009 online test glitches.

So everybody was expecting some changes in CAT pattern in 2011, and CAT committee obliged. The CAT changed again in 2011—with the introduction of (a) 2-Section format instead of 3-section format and (b) sectional time limit. This is a welcome change as now the CAT is akin to other global exams like the GMAT and GRE. They also comprise only two sections and have sectional time limit.

This book covers everything that you need to prepare for the Quantitative Aptitude and Data Interpretation Section as per CAT 2011 pattern. Before we move ahead, let us see the cut-off percentile of different IIMs for CAT 2011 (for general category):

S. No	*IIM's*	*Overall score*	*QA/DI*	*VA/LR*
1	IIM-A	99	94	94
2	IIM-B	90	80	90
3	IIM-C	99.55	94.24	93.73
4	IIM-L	90	85	85
5	IIM-I	90	85	85
6	IIM Trichy	80	70	70
7	IIM Kashipur	75	70	70
8	IIM Udaipur	80	70	70
9	IIM Ranchi	90	70	70
10	IIM Shillong	***	65	65
11	IIM Kozhikode	85	55	55

*** Not given.

This book is divided into four parts: Part 1—Quantitative Aptitude; Part 2—Data Interpretation; Part 3—CAT Papers; and Part 4—Section Tests. Each of the first two parts is further distributed among modules to facilitate the learning of students. Once a student is through with chapters, there are benchmarking tests with calibrated percentile. Moving on further, there are CAT papers to know the level of your preparedness. Finally there are four section tests to help you assess your preparation level.

Further, to facilitate the online testing, a CD containing Section Tests and Full Length Tests is appended.

Though I have taken utmost care in preparing the manuscript and going through the subsequent proofs, some errors may have crept in the book. Kindly email me your valuable constructive advice on my email id: nsinha.alexander@gmail.com

NISHIT K. SINHA

Acknowledgements

This book bears the imprint of many people— my colleagues, my students and my teachers who have had a significant impact on my thought process and have generously extended help whenever I needed.

I am thankful to my present organization for giving me enough time to work on this project.

I would like to thank my teachers Mr Anoop Singhania, Mr Vinay Singh, Mr M. K. Alam Bhutto, Mr Jairam Singh and Mr Arun Sharma.

I would also like to thank Mr Anil Kumar, Mr Saral Choudhary, Mr Sachin Goyal, Mr Shayank Bariar, Mr Manoj Kumar, Mr Parichay Shrestha, Mr Mohit Berry, Mr Sandeep Singh, Mr Siddharth Tripathi, Mr Tanzar Aftab, Mr Anup Singh, Mr Chandrabhal Rathod, Abhishek Anand and of course Ma'am Parul Jaiswal for their valuable inputs.

My special thanks to my brothers Ravi Shankar Prasad, Sharat Chandra Mayank, Amit Kumar and Vinit Kumar.

I extend heartfelt thanks to all my students, past and present, who have helped me improve the content and the presentation of the book.

I would like to thank Mrs Preeta Priyamvada for her painstaking efforts in going through the manuscript. A special mention of thanks to Mr Raza Khan, Sr Managing Editor, Pearson, who played a pivotal role in getting this book printed in such a short span of time.

I may have forgotten some names here. I wish to express my gratitude towards all who have contributed in the making of this book.

Nishit K. Sinha

CAT Demystified

CAT stands for the Common Admission Test. It is a test conducted by IIMs for admission into several programs offered by them. Besides IIMs, there are a good number of colleges which accept CAT score in their first round of selection process. As of now, there are 13 IIMs offering PGP at following places: Ahmedabad, Bangalore, Calcutta, Lucknow, Indore, Kozhikode, Shillong, Ranchi, Rohtak, Raipur, Udaipur, Trichy, Kashipur.

History of the CAT

For last almost three decades, since the CAT has been started, it has changed its colours many a time in terms of number of questions, sections asked and orientation of those questions. Here we will discuss the pattern of CAT 2000 onwards.

Chart 1

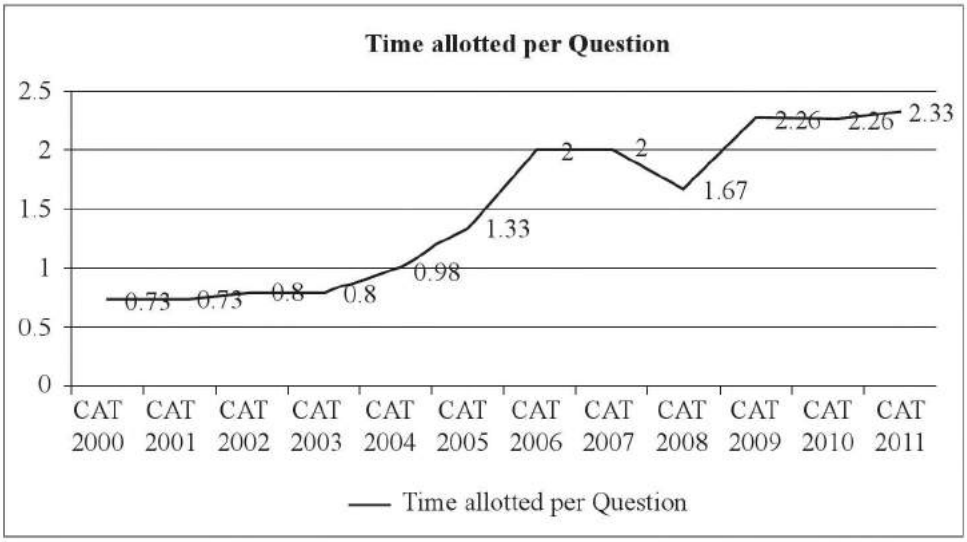
	Number of sections	*Total number of questions*	*Total Marks*	*Time allowed*
CAT 2000	3	165	N.A.	120 minutes
CAT 2001	3	165	N.A.	120 minutes
CAT 2002	3	150	N.A.	120 minutes
CAT 2003	3	150	N.A.	120 minutes
CAT 2004	3	123	150	120 minutes
CAT 2005	3	90	150	120 minutes
CAT 2006	3	75	300	150 minutes
CAT 2007	3	75	300	150 minutes
CAT 2008	3	90	360	150 minutes
CAT 2009	3	60	450 (scaled score)	135 minutes
CAT 2010	3	60	450 (scaled score)	135 minutes
CAT 2011	2	60	450 (scaled score)	140 minutes

CAT 2011 has two sections: (a) Quantitative Ability & Data Interpretation (b) Verbal Ability & Logical Reasoning with 30 questions in each section. CAT 2011 also has sectional time limit of 70 minutes for each section.

Before CAT 2004, the CAT did not mention how many marks one question carried? Marks carried per question were announced for the first time in CAT 2004.

Quite obvious from the above table that time allotted per question has risen sharply from CAT 2000 to CAT 2011. One possible conclusion drawn from here is that the CAT is focussing more on accuracy than speed, and secondly, it expects students to gain a certain level of competence across all the areas in a particular section. With the number of questions going down and time going up, students do not have much choice of questions to choose from.

Chart 2



Sectional Breakups and Getting an IIM Call

One thing that has remained constant during this period of CAT 2000 - CAT 2010 is the number of sections and the way these sections have been joined—Quantitative Aptitude (QA), Logical Reasoning and Data Interpretation (LR/DI), and English Usage/Reading Comprehension (EU/RC).

Though CAT 2011 changed it all:

Chart 3

Year	*QA*	*LR DI*	*EU RC*	*Total no. of questions*
CAT 2000	55	55	55	165
CAT 2001	50	50	50	150
CAT 2002	50	50	50	150
CAT 2003	50	50	50	150
CAT 2004	35	38	50	123
CAT 2005	30	30	30	90
CAT 2006	25	25	25	75
CAT 2007	25	25	25	75
CAT 2008	25	25	40	90
CAT 2009	20	20	20	60
CAT 2010	20	20	20	60
CAT 2011	30 (QA+DI)		30 (Verbal + LR)	60

One possible reason predicted for the 2-section format is the fact that the CAT is trying to emulate the standard global papers like the GRE/GMAT.

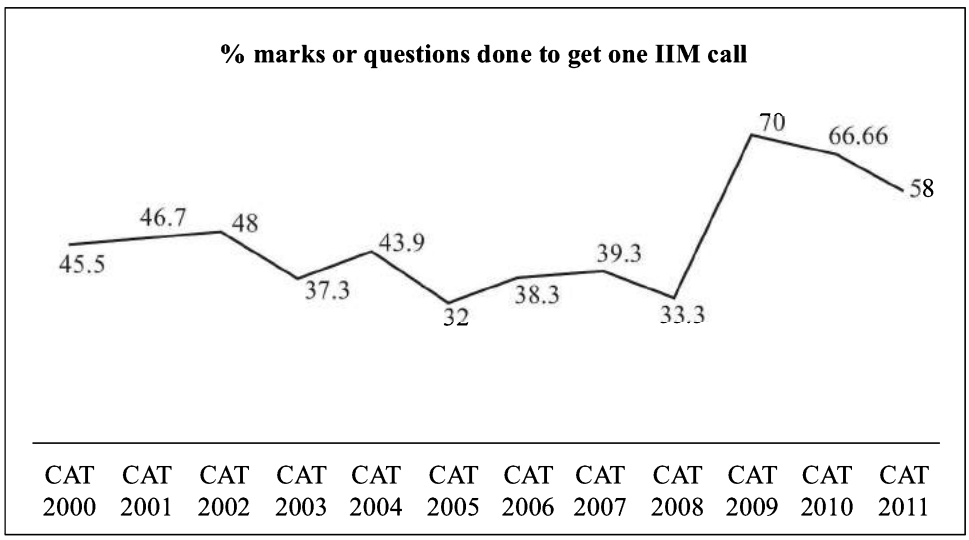
However, past CAT trends show that a student is required to get around 70% marks to get at least one IIM call (with clearing the sectional cut-off). The following table and bar chart give us some clarity regarding the same:

Chart 4

Year	*Total marks or questions*	*Marks/Qs required to get at least one IIM call*
CAT 2000	165	75
CAT 2001	150	70
CAT 2002	150	72
CAT 2003	150	56
CAT 2004	123	54
CAT 2005	150	48
CAT 2006	300	115
CAT 2007	300	118
CAT 2008	360	120
CAT 2009	60	42
CAT 2010	60	40
CAT 2011	60	35

Following line chart gives questions solved or marks required as a percentage of total marks or total questions (as applicable):

Chart 5



Note: Above calculation is based upon the data collected from the students who got IIM calls in that particular year.

So, to get at least one IIM call in CAT 2008, a student was required to get 33.33% marks out of the total with clearing the cut-off across the sections. Though in the online format of the CAT (since 2009), percentage questions to be done to get at least one IIM call has gone up, it is primarily because the exam is perceived to be easier in its totality than pen and paper based exams. A student might have got 10 easy questions out of 20 questions in a section.

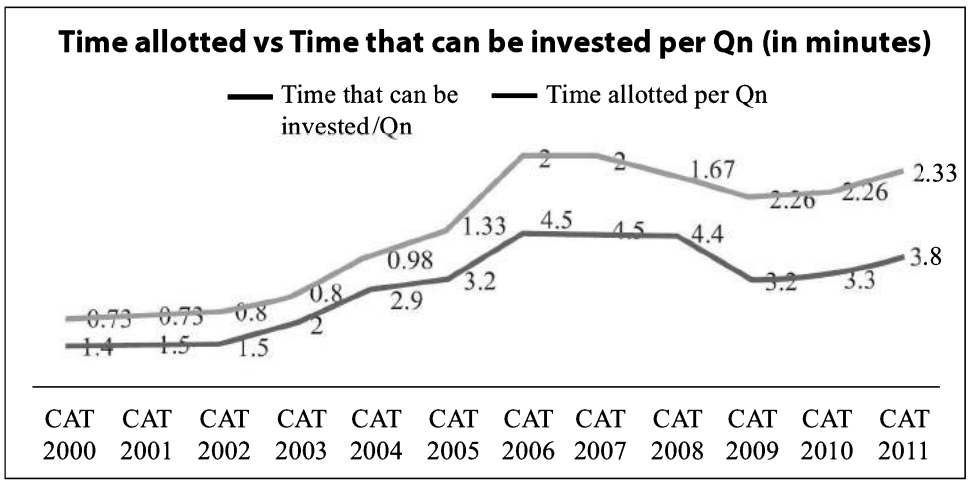
If we convert the requirement of marks to be obtained from the above line chart into questions to be done, we get the following table:

Year	*No. of questions to be solved*	*Time allotted*	*Time invested per qn*
CAT 2000	85	120 minutes	1.4
CAT 2001	80	120 minutes	1.5
CAT 2002	81	120 minutes	1.5
CAT 2003	60	120 minutes	2.0

Year	*No. of questions to be solved*	*Time allotted*	*Time invested per qn*
CAT 2004	42	120 minutes	2.9
CAT 2005	38	120 minutes	3.2
CAT 2006	33	150 minutes	4.5
CAT 2007	33	150 minutes	4.5
CAT 2008	34	150 minutes	4.4
CAT 2009	42	135 minutes	3.2
CAT 2010	40	135 minutes	3.3
CAT 2011	35	135 minutes	3.8

Chart 6—For CAT 2000 to CAT 2008, at 90% accuracy with 1/4 negative marking, these are the approximate number of questions to be done. For CAT 2009 to CAT 2011, net of these many questions is to be done.

To summarize this whole discussion till now, we will compare chart 2 with chart 6 and present them in a unified line chart given below (chart 7).



This line brings to us an important information—For CAT 2006, CAT 2007 or CAT 2008, even if a student has taken approximately 4.5 minutes to solve a question with 90% accuracy, s/he has got enough marks to get at least one IIM call (provided s/he clears the sectional cut-off too). For online formats for CAT, it is around 3.3 minutes per question.

So, it's more about accuracy now than speed.

How to Prepare for QA

Purpose of QA Being Included in the CAT

Have you ever thought why quantitative-aptitude questions are asked in the CAT? Is it to check your math ability with questions from algebra, or permutation and combination? If that were the case, they would have asked questions also from group theory and applied mathematics, which they don't. The questions asked in quantitative aptitude help test the student's reasoning ability based on mathematical skills. They also test the student's comprehension of basic mathematical concepts. And even if somebody has not studied mathematics after class 10, they can solve these problems with a little practice. What is required is an understanding of the basic concepts. It is, therefore, clear that through the test of QA, the examiners intend to check functional ability pertaining to basic mathematical operations.

Five-point Action Plan for QA

1. Address your math-phobia first You will not be in the best state of mind to crack the CAT if past failures or inabilities in mathematics haunt you. As it is said, it's okay to lose a couple of battles. Keep your eyes set on the war!

2. Make a roadmap 'Divide and win' is the key. Your preparation should be divided into topics and every topic should be further divided into sub-topics. Once you have a topic-centric roadmap of the entire QA syllabus, carefully distribute the time you want to devote to each section on a daily basis.

3. Strengthen your fundamentals Till you have understood the concepts and their various applications thoroughly, do not start solving the problems. Let this process take some time. Now, before you start solving questions, go over the fundamental concepts once again and see which concept should be applied to get the right answer.

4. Adopt a systematic practice technique Don't jump to solving problems immediately. Make an effort to understand the basic theories behind the mathematical concepts, however trivial they may seem. And only then solve the problems.

5. Focus on weaknesses If students are weak in a particular area (say, geometry or permutations and combinations), they tend to neglect these questions and leave the solutions to choice. This is not only unproductive but also damaging. Identify the areas with which you are not comfortable. Numbers, geometry and algebra account for a major portion of the questions in the CAT; hence, they deserve due attention.

Consistency with Perseverance is the Key

If you do not find ways to solve a problem on the first day, sleep on it and tackle it the next day. Sometimes even the simplest of answers elude you and lead to frustration. Even the best of mathematicians face this problem; therefore, leave it for another day. And if you are unable to solve this problem even on the second day, try it on the third day or the third time. If you are still not able to solve it, check the solution provided at the end of the book or take help. The bottom line is that only consistent efforts and practice can bring in positive results. If you want to go through this book in 10 days' time, you can do that also. And you will be able to learn the fundamentals, but remember it will be only short-term. You will be required to go through the same chore again after a few days and repeat this exercise.

How to Use This Book

Let me share with you what my feelings were before I started working on this book. I thought of the kind of book I would prefer if I had to learn the theories of public administration when I do not have any background of this subject. I realized I would want a book which is strong in fundamental concepts but lucid in language; I would need a book which is self-explanatory but not verbose; I would need a book which is all-encompassing but not irrelevant; I would need a book with a good number of practice questions but no repetitive questions (if I cannot learn a concept with 200 relevant questions, chances are low that I will be able to learn that concept with 250 questions having the same level of relevance); I would need a book which can help me judge my progress from time to time. And this is what I have sincerely tried to provide in this book.

How to Go Ahead with This Book

Step 1 Go through the learning objectives and remain focused on them. After completing the chapter, check if these objectives have been fulfilled.

Step 2 Start with the concepts. Before moving on to the next concept, go through the worked-out examples related to that concept. Move ahead only when you have internalized them. Sometimes this might appear to be drudgery, but you must do it.

Step 3 Do the warm-up exercise (in QA part only). It is a precursor to the problems. Most of the questions in the warm-up exercise will help you check only your understanding of the concept and not the application of it. Ideally, you should not give more than 2 minutes to any question in the warm-up exercise. If you get less than 75 per cent questions correct, go over the concepts for which you got the answers wrong.

Step 4 Do the foundation exercise. Solve the problems in this set without any time constraint. This level tests your comprehension of the concept, and is a precursor to the application-based questions. Try to solve all the questions in one sitting, whatever time it takes (one or two breaks of 10 minutes can be taken). If you don't get the correct answers, attempt the same question once again in the next sitting (preferably the next day). This will help you develop:

- A thorough understanding of the concepts
- An experience of the problems being asked at the basic level
- Confidence building

Step 5 Next is the moderate exercise. This level tests your ability to apply a particular concept, or a combination of concepts, to a problem. You might find that concepts of geometry are being used in problems on time, speed and distance. This will help you develop:

- The ability to identify easy and difficult questions
- Mental images of the problems
- A logical connection between concepts and their application.

Step 6 Do not go for the advanced level immediately. Relax. Do not solve any QA question for one day. The following day, take the benchmarking test. Your performance in this test can tell you a lot about your progress. Ideally, you should aim for 85 percentile or more.

Step 7 After you have done all the moderate exercises and benchmarking tests, go over the topics again. Only then attempt the advanced questions. Most of the questions here are above the regular CAT level. The idea is to prepare you for a higher level. If you excel at this level, CAT will be a cakewalk for you.

Step 8 Take the Review Tests. Target anything above 98 percentile. If you get it consistently in all the RTs, you are prepared for the CAT. If not, repeat the process from Step 5. If in any one of the RTs, you get less than 85 percentile, repeat the process from Step 4.

All the best.

About the Author

Nishit Sinha is a postgraduate from IIM Lucknow. For the last ten years, he has been training students for the CAT. Many of his students have now graduated from premier B-Schools across the globe. He is also the author of widely praised books *The Pearson Guide to Quantitative Aptitude and Data Interpretation for the CAT, The Pearson Guide to Verbal Ability and Logical Reasoning for the CAT* and *Demystifying Number System*.

He is currently running the test preparation organization dueNorth Academics, based in Dehradun.

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PART 1

Quantitative Aptitude

Module 1

- Number and Its Properties

Module 2

- Arithmetic

Module 3

- X+2 Maths

Module 4

- Measurement

Module 5

- Miscellaneous

Module 1

Numbers and Their Properties

1

CHAPTER

Vedic Mathematics



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- How to do Faster Calculations?
- Multiplications
- Squares
- Cubes
- Properties of Squares and Cubes

Introduction to the topic

Vedic mathematics is the ancient system of mathematics drawn from the Vedas. The Vedas are ancient texts that encompass a broad spectrum of knowledge, covering all aspects of life. These include the sutras (verses) pertaining to mathematics. In the early 20th century, Swami Shri Bharati Krishna Tirthaji Maharaja claimed to have rediscovered a collection of 16 ancient mathematical sutras from the Vedas and published it in a book titled *Vedic Mathematics*. Historians, however, do not agree on whether or not these were truly a part of the Vedic tradition. If these sutras dated back to the Vedic era, they would be a part of an oral rather than a written tradition. Despite controversies, they are a novel and useful approach to computation; they are flexible in application and easy to remember. They can often be applied in the algebraic contexts and in simple arithmetic as well.

TYPES OF CALCULATIONS

The different types of calculations that form the basis of mathematics are:

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Ratio comparison
6. Percentage calculations

When we talk about the techniques of calculations, addition and subtraction can simply not have any short-cuts. Since addition and subtraction are the basic units, we can at best only approximate the values.

In case of multiplication, the techniques of vedic maths can be used.

Ratio comparison techniques are discussed in the chapter on ratio, proportion and variation and the percentage calculations in the chapter on percentage.

VEDIC MATH TECHNIQUES IN MULTIPLICATION

There are several techniques of multiplication. We will discuss them one by one.

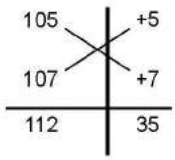
Method 1: Base Method

In this method, one number is used as a base; for example, 10, 50, 100 etc. The number that is closer to both the numbers should be taken as the base.

Example 1 105×107

Solution In this case, both the numbers are close to 100, so 100 is taken as the base. We will now find the deficit/surplus from the base.

Base = 100, Surplus = 5 and 7



The right part (after slash) ⇒ this is the product of the surplus. Since the base = 100 and the surpluses are 5 and 7, the product would be $5 \times 7 = 35$.

The left part (before slash) ⇒ It could be either of the numbers plus the surplus of the other multiplicand. Hence, the left part would be either (105 + 7) or (107 + 5) = 112 (both will always be the same) i.e., 112.

The left part would be equivalent to the number × v100. In this case, $112 \times 100 = 11200$

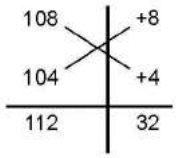
Now, we add both the right part and the left part = 11200 + 35 = 11235

Hence, the result of the multiplication would be 11235.

To know more about Vedic maths, go to *www.pims.math.ca/pi/issue6/page15-16.pdf*

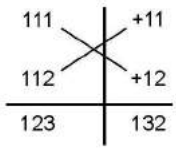
Example 2 108×104

Solution



Example 3 111×112

Solution

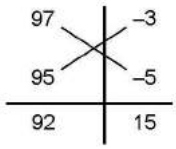


Here, it is $11 \times 12 = 132$. But it can have only two digits. Thus, 1 will be carried over to the left part and the right part will be only 32. Left part will be either 111 + 12 + 1 (1 for the carry over) or (112 + 11 + 1), i.e., 124. So, the result will be 12432.

For 102×104, the answer will be 10608. Please note that the right part will be 08 and not simply 8.

Example 4 97×95

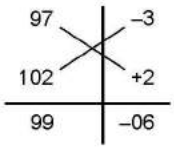
Solution



Base = 100, Deficit = 97 − 100 = −3 and 95 − 100 = −5

Example 5 97×102

Solution



96×108

Base = 100, Deficit = 97 − 100 = −3,

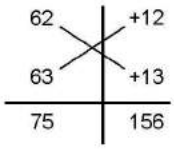
Surplus = 102 − 100 = 2

The right part will now be $(-3) \times 2$, i.e., −06. To take care of the negative, we will borrow 1 from the left part, which is equivalent to borrowing 100 (because we are borrowing from the hundred digits of the left part). Thus, this part will be 100 − 06 = 94.

So, the answer = 98 94

Example 6 62×63

Solution



We will assume here the base as 50 owing to the fact that the numbers are close to 50.

Base = 50, Surplus = 62 – 50 = 12,

Surplus = 63 – 50 = 13

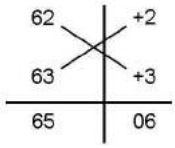
The left hand side = 156 and the right hand side = 75. Since the base is assumed to be equal to 50, so the value of the right hand side = 75 × 50 = 3750. Besides, only two digits can be there on the right hand side, so 1(100) is transfered to the left hand side leaving 56 only on the left hand side.

So, the value on the right hand side = 3750 + 100 = 3850

Value on the left hand side = 56

Net value = 3850 + 56 = 3906

Let us do the same multiplication by assuming 60 as the base.



Base = 60, Surplus = 62 – 60 = 2, Surplus = 63 – 60 = 3

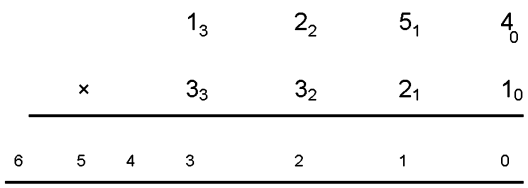
Since the base is assumed to be equal to 60, the value of the right hand side = 65 × 60 = 130 × 30 = 3900

So, net value = 3906

Method 2: Place Value Method

In this method of multiplication, every digit is assigned a place value and the multiplication is done by equating the place values of multiplicands with the place value of the product.

Let us see this with some examples:

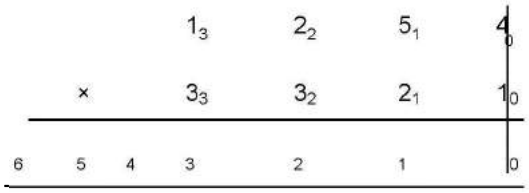


Conventionally, the unit digit is assigned a place value 0, the tens place digit is assigned a place value 1, the hundreds place digit is assigned a place value 2, the thousands place digits is assigned a place value 4 and so on.

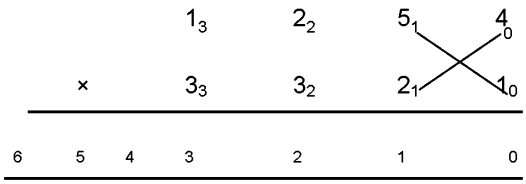
This multiplication is a two-step process.

Step 1 Add the place values of the digits of the numbers given (1254 × 3321) to obtain the place value of the digits of the product.

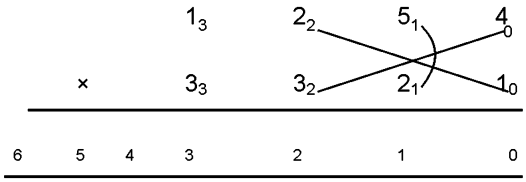
For example, using the place values of the multiplicands, i.e., using 0, 1, 2 and 3 of the number 1254 and the same place values 0, 1, 2 and 3 of the another multiplicand 3321, we can get 0 place value in the product in just one way, i.e., adding 0 and 0.



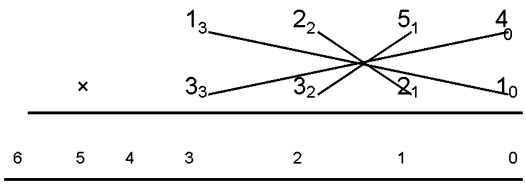
Place value 1 in the product can be obtained in two ways.



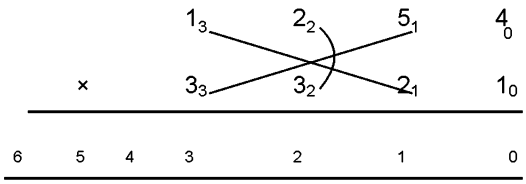
Place value 2 can be obtained in three ways.



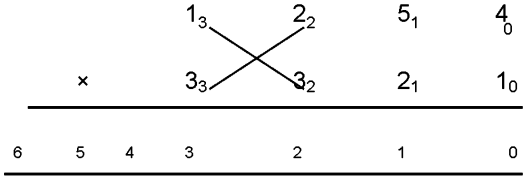
Place value 3 can be obtained in four ways.



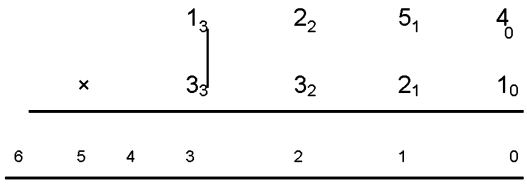
Place value 4 can be obtained in three ways.



Place value 5 can be obtained in two ways.

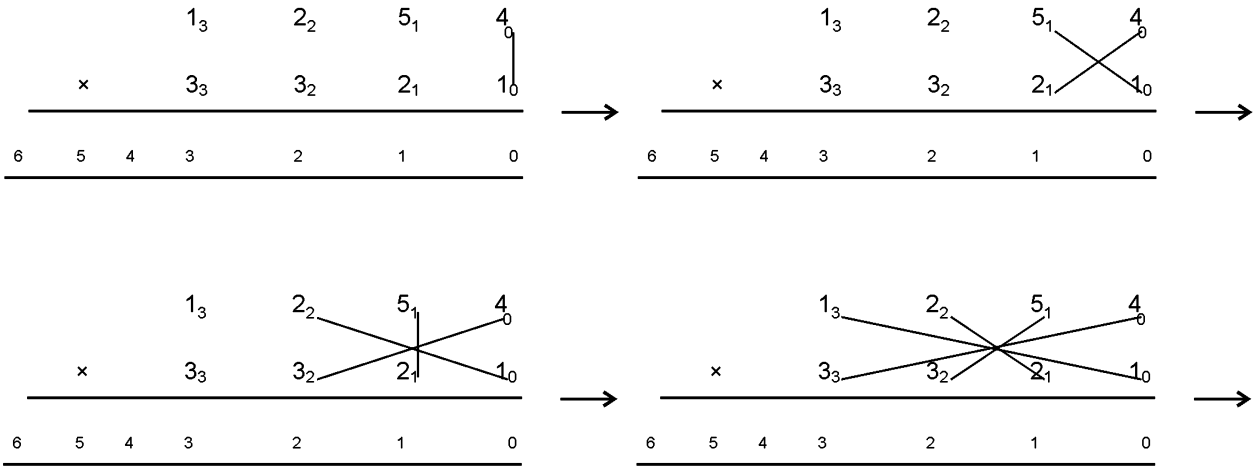


Place value 6 can be obtained in one way.



And this is the maximum place value that can be obtained.

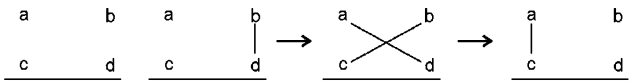
Step 2 Multiply the corresponding numbers one by one.



In this manner, we can find the product = 4164534

This method is most useful in case of the multiplications of 2 digits × 2 digits or 2 digits × 3 digits or 3 digits × 3 digits multiplication.

E.g., *ab* × *cd*



Similarly, we can have a proper mechanism of multiplication of 2 digits × 3 digits or 3 digits × 3 digits also using the place value method.

Method 3: Units Digit Method

This method of multiplication uses the sum of the units digit, provided all the other digits on the left hand side of the unit digit are the same.

Example 7 75 × 75

Solution

1.0+7	5
7	5
56	25

The sum of the units digit = 10, so we add 1.0 in one of the digits on the left hand side.

Example 8 62 × 63

Solution

0.5+6	3
6	2
39	06

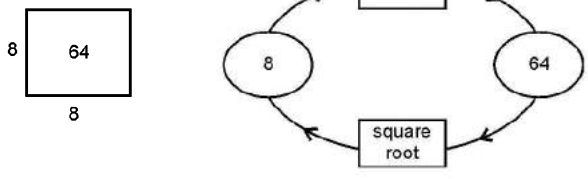
The sum of the units digit = 5, so we add 0.5 in one of the digits on the left hand side.

SQUARING

A **square number**, also called a **perfect square**, is an integer that can be written as the square of some other integer. In other words, a number whose square root is an integer is known as the square number of a perfect square.

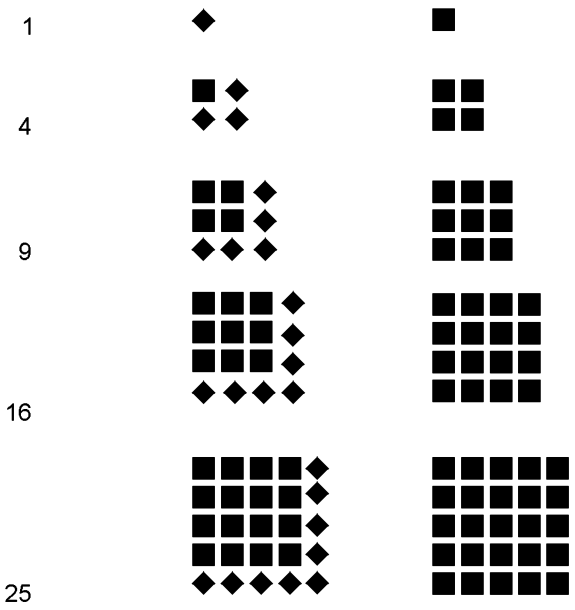
For example, 9 is a square number since it can be written as 3 × 3.

This can be seen through the following flow-chart also.



Properties of a Square Number

1. The number N is a square number if it can be arranged as N points in a square:



Thus, it can be deduced that the formula for the nth square number is n^2. This is also equal to the sum of the first n odd numbers $n^2 = \sum_{k=1}^{n}(2k-1)$, as can be seen in the above figure, where a square results from the previous one by adding an odd number of points (marked as '◆'). For example, $5^2 = 25 = 1+3+5+7+9$.

It should be noted that the square of any number can be represented as the sum $1+1+2+2+\ldots+n-1+n-1+n$. For instance, the square of 4 or 4^2 is equal to 1 + 1 + 2 + 2 + 3 + 3 + 4 = 16. This is the result of adding a column and row of thickness 1 to the square graph of three. This can also be useful for finding the square of a big number quickly. For instance, the square of $52 = 50^2 + 50 + 51 + 51 + 52 = 2500 + 204 = 2704$.

2. A square number can only end with digits 00, 1, 4, 6, 9, or 25 in base 10, as follows:
3. If the last digit of a number is 0, its square ends in 00 and the preceding digits must also form a square.
4. If the last digit of a number is 1 or 9, its square ends in 1 and the number formed by its preceding digits must be divisible by four.
5. If the last digit of a number is 2 or 8, its square ends in 4 and the preceding digit must be even.
6. If the last digit of a number is 3 or 7, its square ends in 9 and the number formed by its preceding digits must be divisible by four.
7. If the last digit of a number is 4 or 6, its square ends in 6 and the preceding digit must be odd.
8. If the last digit of a number is 5, its square ends in 25 and the preceding digits (other than 25) must be 0, 2, 06, or 56.
9. A square number cannot be a perfect number. (If the sum of all the factors of a number excluidng the number itself is equal to the number, then the number is known to be a perfect number.)
10. The digital sum of any perfect square can be only 0, 1, 4, 9, 7. (Digital sum of any number is obtained by adding the digits of the number until we get a single digit. Digital sum of 385 = 3 + 8 + 5 = 1 + 6 = 7)

An easy way to find the squares is to find two numbers which have a mean of it. This can be seen through the following example:

To find the square of 21, take 20 and 22, then multiply the two numbers together and add the square of the distance from the mean: $22 \times 20 = 440 + 1^2 = 441$. Here, we have used the following formula

$(x-y)(x+y) = x^2 - y^2$

known as the difference of two squares. Thus,
$(21-1)(21+1) = 21^2 - 1^2 = 440$.

Odd and Even Square Numbers

Squares of even numbers are even, since $(2n)^2 = 4n^2$.

Squares of odd numbers are odd, since $(2n+1)^2 = 4(n^2+n)+1$.

Hence, we can infer that the square roots of even square numbers are even, and square roots of odd square numbers are odd.

Methods of Squaring

As we have seen in the case of multiplication, there are several methods for squaring also. Let us see the methods one by one.

Method 1: Base 10 Method

Understand it by taking few examples:

- Let us find out the square of 9. Since 9 is 1 less than 10, decrease it still further to 8. This is the left side of our answer.
- On the right hand side put the square of the deficiency that is 1^2. Hence, the answer is 81.
- Similarly, $8^2 = 64$, $7^2 = 49$.
- For numbers above 10, instead of looking at the deficit we look at the surplus. For example,
  $11^2 = (11+1).\ 10 + 1^2 = 121$
  $12^2 = (12+2).\ 10 + 2^2 = 144$
  $14^2 = (14+4).\ 10 + 4^2 = 18\ .\ 10 + 16 = 196$
  and so on.

This is based on the identities $(a+b)(a-b) = a^2 - b^2$ and $(a+b)^2 = a^2 + 2ab + b^2$.

We can be use this method to find the squares of any number, but after a certain stage, this method loses its efficiency.

Method 2: Base 50*n* Method here, (*n* is any natural number)

This method is nothing but the application of $(a + b)^2 = a^2 + 2ab + b^2$.

This can be seen in the following example:

Example 9 Find the square of 62.

Solution Because this number is close to 50, we will assume 50 as the base.

$(62)^2 = (50 + 12)^2 = (50)^2 + 2 \times 50 \times 12 + (12)^2 = 2500 + 1200 + 144$

To make it self explanatory a special method of writring is used.

$(62)^2$ = [100's in (Base)]^2 + Surplus | Surplus^2

= 25 + 12 | 144 = 38 | 44 [Number before the bar on its left hand side is number of hundreds and on its right hand side are last two digits of the number.]

$(68)^2 = 25 + 18 \mid 324 = 46 \mid 24$

$(76)^2 = 25 + 26 \mid 676 = 57 \mid 76$

$(42)^2 = 25 - 8 \mid 64 = 17 \mid 24$ [$(a - b)^2 = a^2 - 2ab + b^2$]

Example 10 Find the square of 113.

Solution Since this number is closer to 100, we will take 100 as the base.

$(112)^2 = (100 + 12)^2 = (100)^2 + 2 \times 100 \times 12 + (12)^2 = 10000 + 2 \times 1200 + 144$

$(112)^2$ = [100's in (Base)]^2 + 2 × Surplus | Surplus^2

$= 100 + 2 \times 12 \mid 12^2 = 125 \mid 44$

Alternatively, we can multiply it directly using base value method.

Had this been 162, we would have multiplied 3 in surplus before adding it into [100's in (Base)]^2 because assumed base here is 150.

$(162)^2$ = [100's in (Base)]^2 + 3 × Surplus | Surplus^2

$= 225 + 3 \times 12 \mid 12^2 = 262 \mid 44$

Method 3 : 10^n Method

This method is applied when the number is close to 10^n.

With base as 10^n, find the surplus or deficit (×)

Again answer can be arrived at in two parts

$(B + 2x) \mid x^2$

The right-hand part will consist of *n* digits. Add leading zeros or carry forward the extra to satisfy this condition.

$108^2 = (100 + 2 \times 8) \mid 8^2 = 116 \mid 64 = 11664$

$102^2 = (100 + 12 \times 2) \mid 2^2 = 104 \mid 04 = 10404$

$93^2 = (100 - 2 \times 7) \mid (-7)^2 = 86 \mid 49 \Rightarrow 8649$

$1006^2 = (1000 + 2 \times 6) \mid 6^2 = 10 \mid 12 \mid 036 = 1012036$

The right-hand part will consist of 2 digits. Add leading zeros or carry forward the extra to satisfy this condition.

$63^2 = (25 + 13) \mid 13^2 = 38 \mid 169 = 3969$

$38^2 = (25 - 12) + (-12)^2 = 13 \mid 144 = 1444$

Square Mirrors

$14^2 + 87^2 = 41^2 + 78^2$
$15^2 + 75^2 = 51^2 + 57^2$
$17^2 + 84^2 = 71^2 + 48^2$
$26^2 + 97^2 = 62^2 + 79^2$
$27^2 + 96^2 = 72^2 + 69^2$

Some Special Cases

1. Numbers ending with 5

If a number is in the form of n5, the square of it is n (n+1) | 25

E.g., $45^2 = 4 \times 5 \mid 25 = 2025$

$135^2 = 13 \times 14 \mid 25 = 18225$

This is nothing but the application of the multiplication method using the sum of units digits.

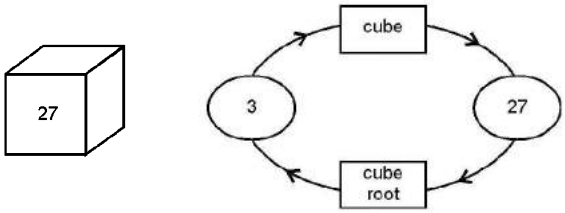
We can use this method to find out the squares fractions like $1\frac{1}{2}, 2\frac{1}{2}, 3\frac{1}{2} \ldots$ also.

Process: Multiply the integral portion by the next higher integer and add $\frac{1}{4}$.

For example, $\left(6\frac{1}{2}\right) = 6 \times 7 + \frac{1}{4} = 42\frac{1}{4}$

CUBING

A number whose cube root is an integer is called a perfect cube.



Properties of a Cube

1. The sum of the cubes of any number of consecutive integers starting with 1 is the square of some integer. (For example, $1^3 + 2^3 = 9 = 3^2$, $1^3 + 2^3 + 3^3 = 36 = 6^2$, etc.)
2. Unit digit of any cube can be any digit from 0-9.

Methods of Cubing

We can find the cube of any number close to a power of 10 say 10^n with base = 10^n by finding the surplus or the defict (*x*). The answer will be obtained in three parts. $B + 3x \mid 3 \cdot x^2 \mid x^3$

The left two parts will have n digits.

104^3

Base B = 100 and surplus = $x = 4$

$(100 + 3 \times 4)|3 \times 4^2|4^3 = 112|48|64 = 1124864$

109^3

Base B = 100 and $x = 9$

$(100 + 3 \times 9)|3 \times 9^2|9^3 = 127|243|729 = 1295029$

98^3

Base B = 100 and $x = -2$

$(100 - 3 \times 2) \mid 3 \times (-2)^2 \mid (-2)^3 = 94 \mid 12 \mid -8 = 94 \mid 11 \mid 100 - 8 = 941192$

VEDIC MATHS TECHNIQUES IN ALGEBRA

1. If one is in ratio, the other one is zero

This formula is often used to solve simple simultaneous equations which may involve big numbers. But these equations in special cases can be visually solved because of a certain ratio between the coefficients. Consider the following example:

$6x + 7y = 8$

$19x + 14y = 16$

Here, the ratio of coefficients of y is the same as that of the constant terms. Therefore, the "other" is zero, i.e., $x = 0$. Hence, the solution of the equations is $x = 0$ and $y = 8/7$.

Alternatively,

$19x + 14y = 16$ is equivalent to. $(19/2)x + 7y = 8$.

Thus, x has to be zero and no ratio is needed, just divide by 2!

Note that it would not work if both had been "in ratio":

$6x + 7y = 8$

$12x + 14y = 16$

This formula is easily applicable to more general cases with any number of variables. For instance,

$ax + by + cz = a$

$bx + cy + az = b$

$cx + ay + bz = c$

which yields $x = 1, y = 0, z = 0$.

2. When samuccaya is the same, that samuccaya is zero

Consider the following symbols: N_1 – Numerator 1, N_2 – Numerator 2, D_1 – Denominator 1, D_2 – Denominator 2 and so on.

This formula is useful for solving equations that can be solved visually. The word "samuccaya" has various meanings in different applications. For instance, it may mean a term, which occurs as a common factor in all the terms concerned. For example, an equation "$12x + 3x = 4x + 5x$". Since "x" occurs as a common factor in all the terms, therefore, $x = 0$ is the solution. Alternatively, samuccaya is the product of independent terms. For instance, in $(x + 7)(x + 9) = (x + 3)(x + 21)$, the samuccaya is $7 \times 9 = 3 \times 21$, therefore, $x = 0$ is the solution. It is also the sum of the denominators of two fractions having the same numerical numerator, for example:

$1/(2x - 1) + 1/(3x - 1) = 0$ means $5x - 2 = 0$.

The more commonly used meaning is "combination" or total. For instance, if the sum of the numerators and the sum of denominators are the same then that sum is zero. Therefore,

$$\frac{2x+9}{2x+7} = \frac{2x+7}{2x+9}$$

Therefore, $4x + 16 = 0$ or $x = -4$.

This meaning ("total") can also be applied in solving the quadratic equations. The total meaning not only imply sum but also subtraction. For instance, when given $N_1D_1 = N_2/D_2$, if $N_1 + N_2 = D_1 + D_2$ (as shown earlier) then this sum is zero. Mental cross multiplication reveals that the resulting equation is quadratic (the coefficients of x^2 are different on the two sides). So, if $N_1 - D_1 = N_2 - D_2$ then that samuccaya is also zero. This yields the other root of a quadratic equation.

The interpretation of "total" is also applied in multi-term RHS and LHS. For instance, consider

$$\frac{1}{x-7} + \frac{1}{x-9} = \frac{1}{x-6} + \frac{1}{x-10}$$

Here, $D_1 + D_2 = D_3 + D_4 = 2x - 16$. Thus $x = 8$.

There are several other cases where samuccaya can be applied with great versatility. For instance, "apparently cubic" or "biquadratic" equations can be easily solved as shown below:

$$(x - 3)^2 + (x - 9)^3 = 2(x - 6)^3.$$

Note that $x - 3 + x - 9 = 2(x - 6)$. Therefore, $(x - 6) = 0$ or $x = 6$.

Consider

$$\frac{(x+3)^3}{(x+5)^3} = \frac{x+1}{x+7}$$

Observe: $N_1 + D_1 = N_2 + D_2 = 2x + 8$.

Therefore, $x = -4$.

2

CHAPTER

Number System



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Numbers and their different types
- Definitions and properties of those numbers
- Concepts attached to these numbers
- Kinds of questions which are asked in the CAT
- Methods of solving

Introduction to the topic

With respect to preparation for the CAT, number system has been seen to be one of the most important topics in the quantitative appitude (QA) paper. During the last 15 years, it has been observed that the QA paper has had 20%–30% questions from the chapter on number system. This chapter will give you a clear understanding of the definitions and the concepts, and help you develop a keen insight about numbers and their properties.

TYPES OF QUESTIONS

Questions asked from this topic are of the following two kinds:

1. Based on definitions or properties of properties of numbers
2. Based on concepts.

1. Based on definitions Questions of this kind are based on definitions and properties of different kinds of numbers, involving formulae and some very basic calculations.

2. Based on concepts The other kind of questions are those that are based on the following concepts:

i. LCM and HCF
ii. Divisibility
iii. Divisibility rules (for base other than 10)
iv. Number of divisors
v. Number of exponents
vi. Remainders
vii. Base system
viii. Units digit
ix. Tens digit
x. Pigeon-hole principle

CLASSIFICATION OF NUMBERS/ INTEGERS

Natural Numbers

Natural numbers are counting numbers, i.e., the numbers which we use to count. For example, 1, 2, 3, ... The lowest natural number is 1.

Whole Numbers

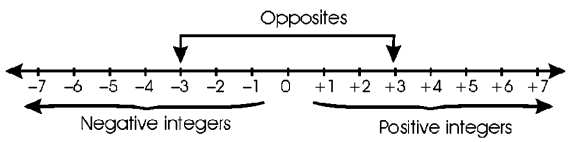
When zero is included in the list of natural numbers, the numbers are known as whole numbers. For example, 0, 1, 2, ... The lowest whole number is 0.

Integers

Integers are whole numbers, the negative of whole numbers and zero. For example, 43434235, 28, 2, 0, −28, and −3030 are integers, but numbers like 1/2, 4.00032, 2.5, Pi, and −9.90 are not whole numbers.

Number Line

The number line is used to represent the set of real numbers. A representation of the number line is given below:



Properties of Number Line

- The number line goes on till infinity in both directions. This is indicated by the arrows.
- Integers greater than zero are called positive integers. These numbers are to the right of zero on the number line.
- Integers less than zero are called negative integers. These numbers are to the left of zero on the number line.
- The integer zero is neutral. It is neither positive nor negative.
- The sign of an integer is either positive (+) or negative (−), except zero, which has no sign.
- Two integers are the opposites if each of the two are at the same distance from zero, but on the opposite side of the number line. One will have a positive sign while the other a negative sign. In the number line above, +3 and −3 are labelled as opposites. In other words, the whole negative number scale looks like a mirror image of the positive number scale, with a number like −15 being at the same distance away from 0 as the number 15 is.
- The number half-way between −1 and −2 is −1.5; just as the number halfway between 1 and 2 is 1.5.
- We represent positive numbers without using a +ve sign. For example, we would write 29.1 instead of +29.1. But when we talk of negative numbers, the sign must be there.

Prime Numbers and Composite Numbers

There are two types of natural numbers, prime and composite, except 1.

Prime numbers

The numbers which are divisible by 1 and itself only are known as prime numbers.

In other words, we can say that all the numbers which have only two factors are known as prime numbers.

Prime numbers can also be seen as the building blocks. We combine two or more than two prime numbers, same or distinct, to create numbers bigger than these prime numbers. For example, $3 \times 2 = 6$.



To know more about primes, go to *www.primes.utm.edu/largest.html*

List of all the prime numbers less than 1000

2	3	5	7	11	13	17	19
23	29	31	37	43	47	53	59
61	67	73	79	83	89	97	101
103	107	109	113	127	131	137	139
149	151	157	163	167	173	179	181
193	197	199	211	223	227	229	233
239	241	251	257	263	269	271	277
281	283	293	307	311	313	317	331
337	347	349	353	359	367	373	379
383	389	397	401	409	419	421	431
433	439	443	449	457	461	463	467
479	487	491	499	503	509	521	523
541	547	557	563	569	571	577	587
593	599	601	607	613	617	619	631
641	643	647	653	659	661	673	677
683	691	701	709	719	727	733	739
743	751	757	761	769	773	787	797
809	811	821	823	827	829	839	853
857	859	863	877	881	883	887	907
911	919	929	937	941	947	953	967
971	977	983	991	997			

Number of prime numbers for every 100 numbers

Nos. from-to	1-100	101-200	201-300	301-400	401-500	501-600	601-700	701-800	801-900	901-1000
No. of primes	25	21	16	16	17	14	16	14	15	14

Largest prime number till date and the history of prime number The largest known prime number, today, is 7,816,230-digit prime number, $2^{25964951}-1$, found in early 2005. But how big have the "largest known primes" been historically and when can we see the first billion-digit prime number?

Table 1.1 Records before the advent of electronic computers

Number	*Digits*	*Year*	*Prover*	*Method*
$2^{17}-1$	6	1588	Cataldi	Trial division
$2^{19}-1$	6	1588	Cataldi	Trial division
$2^{31}-1$	10	1772	Euler	Trial division
$(2^{59}-1)/179951$	13	1867	Landry	Trial division
$2^{127}-1$	39	1876	Lucas	Lucas sequences
$(2^{148}+1)/17$	44	1951	Ferrier	Proth's theorem

The prime number found by Lucas in 1876 was accepted as the largest prime number till 1951. In 1951, Ferrier used a mechanical desk calculator and techniques based on partial inverses of Fermat's little theorem (see the pages on remainder theorem) to slightly better this record by finding a 44-digit prime.

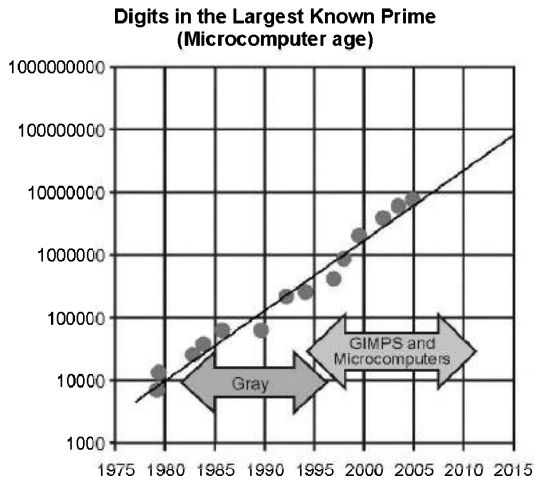
In 1951, Ferrier found the prime $(2^{148}+1)/17 = 20988\,936657440586486151264256610222593863921$. However, this record was very short-lived. In the same year, the advent of electronic computers helped in finding a bigger prime number. Miller and Wheeler began the electronic computing age by finding several primes as well as the new 79-digit record: $2^{127}-1$.

During this period, everybody was working hard to find out the primes with the help of computers. Records were broken at a never-before pace. Sometimes, within a day.

When will we have a one-billion digit prime?

Using the regression line given above, we can expect

- 10,000,000-digit prime by 2007,
- a 100,000,000-digit prime by 2016, and
- a 1,000,000,000-digit prime by early 2027.



Formula for prime numbers

Till now, all attempts made to arrive at a single formula to represent all prime numbers have proved to be fruitless. It is all because there is no symmetricity between the differences among the prime numbers. Sometimes, two consecutive prime numbers differ by 2, sometimes by 4, and sometimes even by 10,000 or more.

However, there are some standard notations, which give us a limited number of prime numbers.

$N^2 + N + 41 \rightarrow$ For all the values of N from –39 to +39, this expression gives us a prime number.

$N^2 + N + 17$ is again a similar example.

Properties of Prime Numbers

1. All prime numbers end in 1, 3, 7, or 9, except for 2 and 5. (Numbers ending in 0, 2, 4, 6 or 8 can be divided by 2 and numbers ending in 5 can be divided by 5)
2. If p is a prime number and p divides a product ab of integers, then p divides a or p divides b.
3. If p is prime and a is any integer, then $a^p - a$ is divisible by p. This is knows as **Fermat's Little theorem.**
4. If p is a prime number other than 2 and 5, 1/p is always a recurring decimal, with a period of p−1 or a divisor of p−1.
5. An integer $p > 1$ is prime if and only if the factorial $(p-1)! + 1$ is divisible by p. This is known as **Wilson's theorem.**
   Conversely, an integer $n > 4$ is composite if and only if $(n-1)!$ is divisible by n.
6. If n is a positive integer greater than 1, then there is always a prime number p with $n < p < 2n$.
7. All the prime numbers P > 3 give a remainder 1 or 5 when divided by 6; that is, all the prime numbers > 3 will be of the format 6N ± 1. But vice versa is not always true.
8. There is only one set of three prime numbers with a gap of 2 between two prime numbers and that set is 3, 5 and 7.

Composite numbers

A number is composite if it is the product of two or more than two distinct or same prime numbers. For example, → 4, 6, 8,…

$4 = 2^2$

$6 = 2^1 \times 3^1$

Lowest composite number is 4.

Even and Odd Numbers

Suppose N is an integer. If there exists an integer P such that N = 2P + 1, then N is an *odd number*. If there exists an integer P such that N = 2P, then N is an *even number*.

Putting in a simple language, even numbers are those integers which are divisible by 2 and odd numbers are those integers which are not divisible by 2.

Even and odd numbers can be positive as well as negative also.

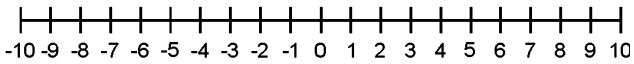
In other words, if x is any integer (even or not), then $2x$ will be an even integer, because it is a multiple of 2. Also x raised to any positive integer power will be an even number, so x^2, x^3, x^4 etc. will be even numbers.

Any integer that is not a multiple of 2 is called an *odd* number. For instance, −1, 3, 6883 and −8147 are all odd numbers. Any odd number raised to a positive integer power will also be an odd number, so if x is an odd number, then so will x^2, x^3, x^4 etc. be odd numbers.

The concept of even and odd numbers are most easily understood in the binary base. The Above definition simply states that even numbers end with a 0, and odd numbers end with a 1.

Comparing Integers

We can compare two different integers by looking at their positions on the number line. For any two different places on the number line, the integer on the right is greater than the integer on the left. Note that every positive integer is greater than any negative integer.



E.g., $9 > 4$, $6 > -9$, $-2 > -8$, and $0 > -5$
$-2 < 1$, $8 < 11$, $-7 < -5$, and $-10 < 0$

Points to Remember

- 1 is neither prime nor composite.
- 0 is neither positive nor negative.

Example 1 Two of a, b, c and d are even and two are odd, not necessarily in order. Which of the following is definitely even? **(CAT 1997)**

(a) $a + b + c - 2d$
(b) $a + 2b - c$
(c) $a + b - c + d$
(d) $2a + b + c - d$

Solution Since we do not know which two are even and which two are odd, we will have to do a bit of hit-and- trial to solve this problem, with the help of options.

In option (a), if a and b are even, and c and d are odd, then this will lead us to odd number.

In option (b), if a and b are even, and c is odd, then this will lead us to odd number.

In option (c), whatever is the value of a, b, c and d, it is always going to be an even number.

In option (d), if a and b are odd, and c and d are even, then this will lead us to odd number.

Explanation Whatever kind of calculation we do with two even and two odd numbers, we will always get an even result.

So, the answer is option (c).

Example 2 If N, N + 2 and N + 4 are prime numbers, then the number of possible solutions for N is are: **(CAT 2003)**

(a) 1 (b) 2 (c) 3 (d) None of these

Solution There is only one triplet of prime numbers where difference between any two prime number is 2, that is 3, 5 and 7. So, N = 3 is the only solution.

Hence, the answer is (a).

Proof of above example We know that prime numbers are of the form 6M ± 1 (except 2 and 3). Now if N is of the format 6M + 1, then N + 2 will of 6M + 3 format and N + 4 will be of 6M + 5 format. Out of these three numbers, since N + 2 is of 6M + 3 format, it will be divisible by 3.

Similarly, if N is of the format 6M – 1, then N + 2 will of 6M + 1 format and N + 4 will be of 6m + 3 format. Out of these three numbers, since N + 4 is of 6M + 3 format, it will be divisible by 3.

In both the cases, we find that one number out of given three number is divisible by 3. In the example given above (3, 5 and 7), one of the given three numbers is divisible by 3.

Example 3 Let x and y be positive integers such that x is prime and y is composite. Then which of the following is true? **(CAT 2003)**

(a) $y - x$ cannot be an even integer
(b) xy cannot be an even integer
(c) $\frac{x+y}{x}$ cannot be an even integer
(d) None of these

Solution Eliminating the options,

To eliminate option (a), If $y = 4$ and $x = 2$, then $y - x$ can be even.

To eliminate option (b), If $y = 4$ and $x = 2$, then yx can be even.

To eliminate option (c), If $y = 6$ and $x = 2$, then it can be even.

So, the answer is option (d).

QUESTIONS BASED ON CONCEPTS

In this section, we will discuss the concepts and questions based upon them.

LCM and HCF

Meaning of LCM

A common multiple is a number that is a multiple of two or more than two numbers. The common multiples of 3 and 4 are 12, 24,…

The least common multiple (LCM) of two numbers is the smallest positive number that is a multiple of both.

Multiples of 3 → 3, 6, 9, 12, 15, 18, 21, 24,…

Multiples of 4 → 4, 8, 12, 16, 20, 24, 28, …

So, the LCM of 3 and 4 will be 12, which is the lowest common multiple of 3 and 4.

First of all, the basic question is what kind of numbers we can use for LCM.

Let us see this through an example— LCM of 10, 20 and 25 is 100. It means that 100 is the lowest number which is divisible by all these three numbers.

But cannot the LCM be (–100)? Since (–100) is lower than 100 and divisible by each of 10, 20 and 25. Or, it can be zero also.

Or, What will be the LCM of (–10) and 20?

Will it be (–20) or (–200) or (–2000) or the smallest of all the numbers, i.e., $-\infty$

Answer to all these questions is very simple. LCM is a concept defined only for positive numbers, be it an integer or a fraction i.e., LCM is not defined for negative numbers or zero.

Now, we will define a different method of finding out the LCM of two or more than two positive integers.

Process to find out LCM

Step 1 Factorize all the numbers into their prime factors.

Step 2 Collect all the distinct factors.

Step 3 Raise each factor to its maximum available power and multiply.

Example 4 Find LCM of 10, 20, 25.

Solution *Step 1* $10 = 2^1 \times 5^1$
$20 = 2^2 \times 5^1$
$25 = 5^2$

Step 2 2, 5

Step 3 $2^2 \times 5^2 = 100$

The biggest advantage of using this method lies in the fact that we can find out the LCM of any number of numbers in a straight line without using the conventional method. It can be understood in the following way with the previous example.

First of all, find out the LCM of 10, 20 = 20, and now the LCM of 20 and 25 = 100 (For this you will have to check which factor of 25 is not present in 20 and then multiply by this factor. Since 25 has 5^2 and 20 has 5^1 only, so we will multiply 20 by 5.)

Example 5 Find LCM of 35, 45, 55.

Solution First of all, find out the LCM of 35 and 45.

Now $35 = 5^1 \times 7^1$ and $45 = 3^2 \times 5^1$.

So, it can be observed here that 35 does not have 3^2 in it, so we will multiply 35 by 3^2.

So, LCM of 35 and 45 = 35×3^2. (You can start with 45 also to find out about the missing factors of 35 in 45.)

Now, we will find out the LCM of 35×3^2 and 55

$55 = 5^1 \times 11^1$

Now, 11^1 is not there with 35×3^2. So, we will multiply 35×3^2 with 11^1.

So, finally the LCM = $35 \times 3^2 \times 11^1 = 3465$.

Relationship between factors of a number and its LCM We can use the factors of a number in several ways to produce that under as the LCM.

Example 6 N = 72. How many sets of two values (a, b) are there for which the LCM (a, b) is 72?

Solution $N = 72 = 2^3 \times 3^2$

Values of a and b have to be chosen in such a way that both 2^3 and 3^2 are contained in either a or b.

Possible sets are (72 ,1) (72, 2) (72, 3) (72, 4) (72, 6) … (72, 36), (72, 72). These are a total of 12 sets.

The other possible sets are (9, 8), (9, 24), (8, 18), (8, 36),…

These are a total of six sets.

So, the total number of possible sets are 12 + 6 = 18 sets

In general, if $N = a^p \times b^q \times c^r \times \ldots$, then the set of values (x, y) such that LCM (x, y) is N can be summed up as:

Number of sets $(x, y) = \frac{(2p+1)(2q+1)(2r+1)\ldots-1}{2}+1$

Meaning of HCF

Factors are those positive integral values of a number, which can divide that number. HCF, which is known as GCD (Greatest Common Divisor) also, is the highest value which can divide the given numbers.

Factors of 20: 1, 2, 4, 5, 10, 20.

Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30.

So, 10 will be the HCF of 20 and 30.

Process to find out HCF

Step 1 Factorize all the numbers into their prime factors.

Step 2 Collect all the common factors.

Step 3 Raise each factor to its minimum available power and multiply.

Example 7 Find HCF of 100, 200 and 250.

Solution

Step 1 $100 = 2^2 \times 5^2$
$200 = 2^3 \times 5^2$
$250 = 5^3 \times 2^1$

Step 2 2, 5

Step 3 $2^1 \times 5^2 = 50$

Alternatively, to find out HCF of numbers like 100, 200 and 250, it is required to observe the quantity which can be taken out common from these numbers. By writing these numbers as $(100x + 200y + 250z)$, it can be very easily observed that 50 is common out of these numbers.

Summarizing LCM and HCF

It is very essential to understand the mechanism of finding out LCM and HCF. We can simply understand the mechanism to find out lowest common multiple and highest common factor through this example.

Example 8 Find out LCM and HCF of 16, 12, 24.

Solution

No.	Multiples	Factors
16	16, 32, 48, 64, 80, 96, 112, 128,…	1, 2, 4, 8, 16
12	12, 24, 36, 48, 60, 72, 84, 96, 108,…	1, 2, 3, 4, 6, 12
24	24, 48, 72, 96, 120, 144, 168, 192,…	1, 2, 3, 4, 6, 8, 12, 24
	Common Multiple	Common Factor
	48, 96, 144	1, 2, 4
	Lowest Common Multiple	Highest Common Factor
	48	4

Standard Formulae

1. LCM × HCF = Product of two numbers.
   This formula can be applied only in case of two numbers. However, if the numbers are relatively prime to each other (i.e., HCF of numbers = 1), then this formula can be applied for any number of numbers.
2. LCM of fractions = LCM of numerator of all the fractions/HCF of denominator of fractions.
3. HCF of fractions = HCF of numerator of all the fractions/LCM of denominator of fractions.
4. HCF of (sum of two numbers and their LCM) = HCF of numbers.

Example 9 HCF of two natural numbers A and B is 120 and their product = 10,00 0. How many set of values of A and B is/are possible?

Solution HCF (A, B) = 120 ⇒ 120 is a common factor of both the numbers (120 being the HCF). Hence 120 is present in both the numbers. So the minimum product of A and B = 120 × 120 = 14400. Hence no set of A and B are possible satisfying the conditions.

QUESTIONS BASED ON STANDARD APPLICATION OF LCM AND HCF

Case 1 Time and work

Example 10 Tatto can do a work in 10 days and Tappo can do the same work in 12 days. How many days will it take if both of them start working together?

Solution Let us assume total work = LCM of (10, 12) units = 60 units. Now, since 60 units of work is being done by Tatto in 10 days, so Tatto is doing 6 units of work per day and similarly, Tappo is doing 5 units of work per day. Hence, together they are doing 11 units of work in one day. So, finally they will take $\frac{60}{11} = 5\frac{5}{11}$ days to complete the work.

Case 2 Time, speed and distance: Circular motion

Example 11 The speed of A is 15 m/s and the speed of B is 20 m/s. They are running around a circular track of length

1000 m in the same direction. After how much time will they meet at the starting point if they start running at same time?

Solution Time taken by A in taking one round = 66.66
Time taken by B in taking one round = 50
LCM (66.66, 50) = 200

Case 3 Number system: Tolling the bell

Example 12 There are two bells in a temple. Both the bells toll at a regular interval of 66.66 sec and 50 sec respectively. After how much time will they toll together for the first time?

Solution Time taken by the 1st bell to toll = 66.66
Time taken by the 2nd bell to toll = 50
LCM (66.66, 50) = 200

It can be observed here that mathematical interpretation of both the questions are same, just the language has been changed.

Case 4 Number system: Number of rows

Example 13 There are 24 peaches, 36 apricots and 60 bananas and they have to be arranged in several rows in such a way that every row contains the same number of fruits of one type. What is the minimum number of rows required for this to happen?

Solution We can put one fruit in one row, and in (24 + 36 + 60) 120 rows, we can arrange all the fruits. Or, we can put two fruits in one row and can arrange all the fruits in 60 rows. But for the rows to be minimum, the number of fruits should be maximum in one row.

HCF of 24, 36, 60 = 12, so 12 fruits should be there in one row.

Hence, the number of rows = 10

Case 5 Number system: Finding the remainder

Example 14 There is a number which when divided by 4 and 5 gives 3 as the remainder. What is the lowest three-digit number which satisfy this condition?

Solution Let us assume that there is no remainder. So, the number has to be a multiple of LCM of 4 and 5. Now, LCM (4, 5) = 20

But there is a remainder of 3 when divided by 4 and 5. So, the number will be of the form (20N + 3).

Hence, the numbers are 23, 43, 63, 83, 103 and so on …

So, 103 is the answer.

DIVISIBILITY RULES (FOR DECIMAL SYSTEM)

Divisibility rules are quite imperative, because with the help of these rules we can infer if a particular number is divisible by another number or not, without actually dividing it. Divisibility rules of numbers are specific to that particular number only. It simply means that divisibility rules of different numbers will be different. We shall now see a list of divisibility rules for some of the natural numbers.

Divisibility Rules

For 2 If unit digit of any number is 0, 2, 4, 6 or 8, then that number will be divisible by 2.

For 3 If sum total of all the digits of any number is divisible by 3, then the number will be divisible by 3. (e.g., 123, 456 etc.)

Example 15 How many values of A are possible if 3245684A is divisible by 3?

Solution Sum total of the number = 32 + A

For this number to be divisible by 3, A can take three values, namely, 1 or 4 or 7. (No other values are possible since A is the unit digit of the number.)

For 4 If last two digits of a number are divisible by 4, then that number will be divisible by 4 (e.g., 3796, 248, 1256).

For 5 If last digit of the number is 5 or 0, then that number will be divisible by 5.

For 6 If last digit of the number is divisible by two and sum total of all the digits of the number is divisible by 3, then that number will be divisible by 6.

For 7 The integers are divisible by 7 if and only if, the difference of the number of its thousands and the remainder of its divisible by 1000, is divisible by 7.

For 7 If the difference between the numbers of tens in the number and twice the unit digits is divisible by 7, then the given number is divisible by 7.

E.g., Let us take the number 795. The units digit is 5 and when it is doubled it, we get 10. The remaining part of the number (i.e., the tens) is 79. If 10 is subtracted from 79 we get 69. Since this result is not divisible by 7, the original number 795 is not divisible by 7.

For 8 If last 3 digits of the number are divisible by 8, then the number itself will be divisible by 8 e.g., 128, 34568, 76232 etc.

For 9 If sum of the digits of the number is divisible by 9, then the number will be divisible by 9 e.g., 129835782.

1 + 2 + 9 + 8 + 3 + 5 + 7 + 8 + 2 = 45. Since 45 is divisible by 9, then the number will be divisible by 9.

Example 16 How many pairs of A and B are possible in number 89765A4B if the number is divisible by 9, given that last digit of the number is even?

Solution Sum of digits of number is 8 + 9 + 7 + 6 + 5 + A + 4 + B = 39 + A + B.

So, A + B should be 6 or 15. Next value should be 24 but since A and B are digits so it can't be more than 18. Possible pairs of A and B are

A	B
0	6

1	5
2	4
3	3
4	2
5	1
6	0
7	8
8	7
9	6
6	9

Since B is even, six possible set of values of A and B are there.

For 11 A number is divisible by 11, if the difference between the sum of the digits at the even places and the sum of the digits at the odd places is divisible by 11 (zero is divisible by 11) e.g., 6595149 is divisible by 11 as the difference of 6 + 9 + 1 + 9 = 25 and 5 + 5 + 4 = 14 is 11.

For 12 If the number is divisible by 3 and 4, then the number will be divisible by 12, e.g., 144, 348.

For 13 **(A + 4B)**, where B is the units place digit and A is all the remaining digits.
E.g., Checking the divisibility of 1404 by 13: Here, A=140 and B = 4, then A+4B=140 +4×4=156. 156 is divisible by 13, so 1404 will be divisible by 13.

For 14 If the number is divisible by both 2 and 7, then the number will be divisible by 14.

For 15 A number is divisible by 15, if the sum of the digits is divisible by 3 and unit digit of the number is 0 or 5, e.g., 225, 450, 375 etc.

For 16 A number is divisible by 16, if the number formed by the last 4 digits of the given number is divisible by 16 e.g., 12578320 is divisible by 16, since last 4 digits of the number, 8320 is divisible by 16.

For 17 **(A – 5B),** where B is the units place digit and A is all the remaining digits.

For 18 Number should be divisible by both 9 and 2.

For 19 **(A + 2B),** where B is the units place digit and A is all the remaining digits.

If the sum of the number of tens in the number and twice the unit digit is divisible by 19, then the number is divisible by 19.

For example, let us take the number 665. The units digit is 5 and when it is doubled, we get 10. The remaining part of the number is 66. If 10 (which is the unit digit doubled) is added to 66 we get 76. Since, 76 is divisible by 19, it means the original number 665 is also divisible by 19.

For 20 The number should be divisible by 4 and 5.

Process to find out the divisibility rule for prime numbers

Process for the divisibility rule for P, a prime number is as follows:

Step 1 Find the multiple of P, closest to any multiple of 10. (This essentially will be of the form 10K + 1 or 10K –1.)

Step 2 If it is 10 K – 1, then the divisibility rule will be A + KB, and if it is 10K + 1, then the divisibility rule will be A – KB, where B is the units place digit and A is all the remaining digits.

E.g., Finding out the divisibility rule of 23: Lowest Multiple of 23, which is closest to any multiple of 10 $= 69 = 7 \times 10 - 1$

So, rule is A + 7 B.

Divisibility rule for any number of the format $10^n \pm 1$, where *n* is any natural number

In this method, we will be considering two factors, the first one being the value of n and the second one being the sign '+' or '–'. The value of '*n*' signifies how many digits will be taken one at a time and the sign signifies in what manner these digits will be taken. Let us see this with the help of some examples.

When *n* = 1

Case 1 $10^n + 1 = 11$

This method tells us that now we will be considering one digit at a time of the number which is to be divided by 11 and since the sign is '+' between 10^n and 1, we will alternately subtract and add starting from the right hand side.

E.g., To check if 123345 is divisible by 11, we will start from the right hand side, by subtracting the first number and then adding the third and then subtracting the fourth and so on…

Case 2 $10^n - 1 = 9$

Since the value of *n* is one, we will take one digit at a time and due to the '–' sign between 10^n and 1, we will keep on adding the digits from the right hand side (or, can be done from the left hand side also in this case).

When *n* = 2

Case 3 $10^n + 1 = 101$

Since the value of *n* here is two, we will make the pairs of digits from right hand side, and then we will alternately subtract and add each of the pairs.

E.g., Checking if 12345678 is divisible by 101: Make the pairs first – 12 34 56 78. Now we will do this calculation – (78) – (56) + (34) – (12) = 44, which is not a multiple of 101, so this number 12345678 is not divisible by 101.

Case 4 $10^n - 1 = 99$

Since the value of *n* is two, we will take two digit at a time after making the pairs and due to the '–' sign between 10^n and

1, we will keep on adding the pairs from right hand side (or, can be done from left hand side also in this case).

E.g., Checking if 122166 is divisible by 99: Let us make the pairs first: 12 21 66. Now add the pairs 12 + 21 + 66 = 99, so this number 122166 is divisible by 99.

And so on, we extend it for any natural number value of n.

NUMBER OF DIVISORS/FACTORS

If one integer can be divided by another integer an exact number of times, then the first number is said to be a *multiple* of the second, and the second number is said to be a *factor* of the first.

For example, 48 is a *multiple* of 6 because 6 goes into 48 an exact number of times (8 times in this case). Similarly, 6 is a *factor* of 48. On the other hand, 48 is not a multiple of 5, because 5 does not go into 48 an exact number of times. So, 5 is not a factor of 48.

When we talk about the number of divisors of any number, we are basically talking about positive integral divisor of that number. Likewise, it can be observed that 20 is having six divisors namely 1, 2, 4, 5, 10 and 20.

Its very essential here to understand the mechanism of formation of divisors.

Since $20 = 2^2 \times 5^1$

We will take three power of 2 viz., 2^0, 2^1 and 2^2 and two powers of 5 viz., 5^0 and 5^1.

Divisors will come from all the possible arrangements of powers of 2 and powers of 5.

$2^0 \times 5^0 = 1$
$2^0 \times 5^1 = 5$
$2^1 \times 5^0 = 2$
$2^1 \times 5^1 = 10$
$2^2 \times 5^0 = 4$
$2^2 \times 5^1 = 20$

Now, to find out the number of divisors of any number, we can use the following formula

If N is any number which can be factorised like $N = a^p \times b^q \times c^r \times \ldots$, where a, b and c are prime numbers.

Number of divisors = $(p + 1)(q + 1)(r + 1)\ldots$

Example 17 Find the number of divisors of $N = 420$.

Solution $N = 420 = 2^2 \times 3^1 \times 7^1 \times 5^1$

So, the number of divisors =
$(2 + 1)(1 + 1)(1 + 1)(1 + 1) = 24$

Example 18 Find the total number of even and prime divisors of $N = 420$.

Solution $N = 420 = 2^2 \times 3^1 \times 7^1 \times 5^1$

Odd divisors will come only if we take zero power of 2 (since, any number multiplied by any power (≥ 1) of 2 will give us an even number)

So, odd divisors will come if we take $N_1 = 2^0 \times 3^1 \times 7^1 \times 5^1$

So, number of odd divisors = $(0 + 1)(1 + 1)(1 + 1)(1 + 1) = 8$

So, total number of even divisors = total number of divisors – number of odd divisors = 24 – 8 = 16

Alternatively, we can also find out the number of even divisors of $N = 420$ directly (or, in general for any number).

$420 = 2^2 \times 3^1 \times 7^1 \times 5^1$

To obtain the factors of 420 which are even, we will not consider 2^0, since $2^0 = 1$

So, the number of even divisors of 420 = (2) (1 + 1) (1 + 1) (1 + 1) = 16

(We are not adding 1 in the power of 2, since we are not taking 2^0 here, i.e., we are not taking one power of 2.)

Prime divisor = 4 (namely 2, 3, 5 and 7 only)

Example 19 $N = 2^7 \times 3^5 \times 5^6 \times 7^8$. How many factors of N are divisible by 50 but not by 100?

Solution All the factors which are divisible by 50 but not divisible by 100 will have atleast two powers of 5, and one power of 2.

And its format will be $2^1 \times 5^{2+y}$.

So, number of divisors = $1 \times 6 \times 5 \times 9 = 270$

Alternatively, this is equal to (Number of factors divisible by 50) – (Number of factors divisible by 100).

Condition for two divisors of any number *n* to be co-prime to each other

Let us see it for $N = 12$

Total number of factors of 12 = 6 (namely 1, 2, 3, 4, 6, 12)

Now, if we have to find out the set of factors of this number which are co-prime to each other, we can start with 1.

Number of factors which are co-prime to 1 = 5 (namely 2, 3, 4, 6, 12)

Next in line is the number of factors which are co-prime to 2 = 1 (namely 3)

Number of factors co-prime to 3 = 1 (namely 4)

So, the total number of set of factors for 12 which are co-prime to each other = 6

So, we can induce that if we have to find out the set of factors which are co-prime to each other for $N = a^p \times b^q$, will be equal to $[(p + 1)(q + 1) - 1 + pq]$.

If there are three prime factors of the number, i.e., $N = a^p \times b^q \times c^r$, then set of co-prime factors can be given by $[(p+1)(q + 1)(r + 1) - 1 + pq + qr + pr + 3\,pqr]$

Alternatively, we can find out the set of co-prime factors of this number by pairing it up first and then finding it out with the third factor.

Example 20 Find the set of co-prime factors of the number N = 720.

Solution $720 = 2^4 \times 3^2 \times 5^1$

Using the formula for three prime factors $[(p + 1)(q + 1)(r + 1) - 1 + pq + qr + pr + 3\,pqr]$

We get, $[(4 + 1)(2 + 1)(1 + 1) - 1 + 4.2 + 2.1 + 4.1 + 3.4.2.1] = 67$

Alternatively, let us find out first for $2^4 \times 3^2 = [(4 + 1)(2 + 1) - 1 + 4.2] = 22$

Now $p^{22} \times 5^1$ will give us $[(22 + 1)(1 + 1) - 1 + 22.1] = 67$

Sum of Divisors

Like the number of divisors of any number, we can find out the sum of divisors also.

If N is any number which can be factorised like $N = a^p \times b^q \times c^r \times \ldots$, where a, b and c are prime numbers. Then sum of the divisors

$$= \frac{(a^{p+1}-1)(b^{q+1}-1)\ (c^{r+1}-1)}{(a-1)(b-1)(c-1)}$$

REMAINDERS

Dividend = Quotient × Divisor + Remainder

Basics of Remainder

1. When any positive number A is divided by any other positive number B, and if B > A, then the remainder will be A itself.
   For example remainder of 5/12 = 5
   Remainder of 212/678 = 212
2. Remainder should always be calculated in its actual form i.e., you can not reduce the fraction to its lower ratio.
   For example, remainder of 1/2 = 1
   Remainder of 2/4 = 2
   Remainder of 3/6 = 3
   It can be observed that despite all the fractions being equal, remainders are different in each case.

Example 21 What is the remainder when 5×10^5 is divided by 6×10^6?

Solution As we know that we cannot reduce the fractions to its lower terms, the remainder obtained will be equal to 5×10^5.

3. Concept of negative remainder: As obvious from the name, remainder implies that something has been left out. So, remainder simply can never be negative. Its minimum value can be zero only and not negative.

Example 22 What is the remainder when 50 is divided by 7?

Solution $-50/7 = (-56 + 6)/7$; which gives a remainder of 6.

Or, when we divide −50 by 7, we get −1 as the remainder. Now, since remainder has to be non-negative, so we add 7 (quotient) to it which makes final remainder as→ −1 + 7 = 6.

There are two methods to find out the remainder of any expression:

- Cyclicity Method
- Theorem Method

Cyclicity method

For every expression of a remainder, there comes attached a specific cyclicity of remainders.

Example 23 What is the remainder when 4^{1000} is divided by 7?

Solution To find the cyclicity, we keep finding the remainders until some remainder repeats itself. It can be understood with the following example:

Number/7 →	4^1	4^2	4^3	4^4	4^5	4^6	4^7	4^8
Remainder →	4	2	1	4	2	1	4	2

Now, 4^4 gives us the same remainder as 4^1, so the cyclicity is of 3 (Because remainders start repeating themselves after 4^3.)

So, any power of 3 or a multiple of 3 will give a remainder of 1. So, 4^{999} will give 1 as the remainder.

Final remainder = 4

Example 24 What is the remainder when 4^{96} is divided by 6? **(CAT 2003)**

Solution Finding out the cyclicity,

Number/6 →	4^1	4^2	4^3	4^4	4^5	4^6	4^7	4^8
Remainder →	4	4	4	4	4	4	4	4

The remainder in all the cases is 4, so the final remainder will be 4. Actually, there is no need to find remainders upto 4^8 or even 4^3. 4^2 itself gives us a remainder of 4 when divided by 6, which is the same as the remainder obtained when 4^1 is divided by 6. So, we have got the cyclicity here, which is of 1.

Hence, final remainder = 4

It also can be observed here that if we write $4^{100}/6 = 2^{200}/6 = 2^{199}/3$, then remainder obtained will be 2, which is not the right answer (as given in CAT brochure of next year i.e., CAT 2004)

Remainder Theorem

The product of any two or more than two natural numbers has the same remainder when divided by any natural numbers, as the product of their remainders.

Let us understand this through an example.

Example 25 Remainder $\frac{12\times13}{7}$ = Remainder $\frac{156}{7} = 2$

Solution Normal way of doing this is – Product →→→ Remainder

Theorem method – Remainder →→→ Product →→→ Remainder

So, first of all we will find the remainders of each number individually and then multiply these individual remainders to find out the final remainder.

Remainder 12/7 = 5

Remainder 13/7 = 6

Remainder $\frac{12\times13}{7}$ = Remainder $(5 \times 6)/7$

= Remainder $30/7 = 2$

Example 26 What is the remainder obtained when $(1421 \times 1423 \times 1425)$ is divided by 12? **(CAT 2000)**

Solution Remainder of $1421/12 = 5$

Remainder of $1423/12 = 7$

Remainder of $1425/12 = 9$

Remainder $(1421 \times 1423 \times 1425)/12$

= Remainder $(5 \times 7 \times 9)/12$

= Remainder $(5 \times 63)/12$

= Remainder $(5 \times 3)/12 = 3$

Concept of Successive Division

Suppose N is any number which is divided successively by 3 and 5, which means – first, we divide N by 3 and then the quotient obtained is divided by 5.

E.g., Let us see the case when 50 is divided by 5 and 3 successively.

50 divided by 5 gives 10 as the quotient. Now, we will divide 10 by 3. It gives finally a quotient of 3 and remainder of 1.

Example 27 When a number N is divided successively by 3 and 5, remainder obtained are 1 and 2 respectively. What is the remainder when N is divided by 15?

Solution It can be seen that we are required to calculate it from back-end.

Family of numbers which when divided by 5 gives remainder $2 = 5\ S + 2$

So, $N = 3(5\ S + 2) + 1 = 15\ S + 7$

Now, when N is divided by 15, remainder = 7

Fermat's remainder therorem

Let P be a prime number and N be a number not divisible by P. Then the remainder obtained when A^{P-1} is divided by P is 1.

Example 28 What is the remainder when 2^{100} is divided by 101?

Solution Since it satisfies the Fermat's theorem format, remainder = 1

Euler's theorem of remainder

Let $f(n)$ be the number of integers less than n and co-prime with n, then the remainder obtained when $m^{f(n)}$ is divided by n is 1, where m and n are co-prime to each other.

As we can see:

$f(2) = 1, f(3) = 2, f(4) = 2, f(5) = 4$

So, the remainder obtained when $8^{f(5)=4}$ is divided by 5 is 1. It can also be seen that the remainder obtained when any power of 8 divisible by 4 such as 8^4 or 8^8 or 8^{12} and so on will give the same remainder when divided by 5.

Similarly, the remainder obtained when $12^{f(5)=4}$ is divided by 5 is 1;

or, the remainder obtained when $8^{f(7)=6}$ is divided by 7 is 1.

Derivations

1. $\frac{(A+1)^N}{A}$ will always give 1 as the remainder.

   (For all natural values of A and N)

Example 29 What is the remainder when 9^{100} is divided by 8?

Solution For A = 8, it satisfies the above condition. So, remainder = 1

Alternatively, we can apply either cyclicity or theorem method to find the remainder. (Do this yourself).

2. $\frac{A^N}{A+1}$ When N is even, remainder is 1 and when N is odd, remainder is A itself.

Example 30 What is the remainder when 2^{10} is divided by 3?

Solution Since here N is even, so remainder = 1

3. i. $(a^n + b^n)$ is divisible by $(a + b)$, if n is odd.
   Extension of the above formula – $(a^n + b^n + c^n)$ is divisible by $(a + b + c)$, if n is odd and a, b, c, are in arithmetic progression.
   Similarly, the above situation can be extended for any number of terms.
   ii. $(a^n - b^n)$ is divisible by $(a + b)$, if n is even.
   iii. $(a^n - b^n)$ is divisible by $(a - b)$, if n is either odd or even.

Example 31 What is the remainder when $(15^{23} + 23^{23})$ is divided by 19? **(CAT 2004, 2 marks)**

Solution It can be observed that $(15^{23} + 23^{23})$ is divisible by 38, so it will be divisible by 19 also. Hence, remainder = 0.

Alternatively, this problem can be done either by cyclicity method or theorem method.

Example 32 What is the remainder when $(16^3 + 17^3 + 18^3 + 19^3)$ is divided by 70? **(CAT 2005, 1 mark)**

Solution We know, this is a basic multiplication and division question. But using the above approach makes it a lot simple.

We know that $(a^n + b^n)$ is divisible by $(a + b)$, if n is odd. Taking cue from this, we can say that $(a^n + b^n + c^n)$ is divisible by $(a + b + c)$, if n is odd and similarly $(a^n + b^n + c^n + d^n)$ is divisible by $(a + b + c + d)$. Now $16 + 17 + 18 + 19 = 70$, so remainder is zero. (As a, b, c, d are in AP)

UNIT DIGIT

As we have seen the cyclicity of remainders above, cyclicity exists for unit digit of the numbers also. (But there is no relation between the cyclicity of remainders and unit digit.)

Taking a very simple example– $2^5 = 32$, and so we know that unit digit of 2^5 is 2. But problem occurs when we start getting big numbers like 25678^{2345} etc. To find out the unit

digit of these kinds of numbers, we have some standard results, which we use as formulae.

(Any even number)^{4n} = … 6

It means that any even number raised to any power, which is a multiple of 4, will give us 6 as the unit digit.

(Any odd number)^{4n} = … 1

It means that any odd number raised to any power, which is a multiple of 4, will give us 1 as the unit digit.

Exception 0, 1, 5, 6 [these are independent of power, and unit digit will be the same respectively]

Example 33 Find the unit digit of $25678^{2345} \times 3485^{4857}$.

Solution Unit digit of 25678^{2345} = Unit digit of 8^{45}

(To find out unit digit, we need to have unit digits only. And similarly, to find out tens digit we need to have the tens and units digit only. In the present case, we are considering only last two digits of the power because divisibility rule of 4 needs only the last two digits of the number.)

$8^{45} = 8^{44+1} = 8^{44} \times 8^1 = (\ldots 6) \times 8 = \ldots 8$

Example 34 What is the unit digit of $_{32}32^{32}$?

Solution 32 is an even number which is having a power of the form $4n$. So, it will give 6 as the unit digit.

Example 35 When 3^{32} is divided by 50, it gives a number of the format (asdf… ·xy) (xy being the last two digits after decimal). Find y.

Solution It can be observed that unit digit of $3^{32} = 1$. Now any number having 1 as the unit digit will always give 2 at the unit place when divided by 50.

So, answer is 2.

Example 36 What is the last non-zero digit of the number 30^{2720}? **(CAT 2005, 2 marks)**

Solution $30^{2720} = [30^4]^{680} = \ldots 10000 \ldots . 00$

Unit digit can also be found out by cyclicity method as well.

It can be seen that

Unit digit of $2^1 = 2$

Unit digit of $2^2 = 4$

Unit digit of $2^3 = 8$

Unit digit of $2^4 = 6$

Unit digit of $2^5 = 2$

So, it can be inferred that unit digit of 2^1 = unit digit of 2^5 = unit digit of 2^9

Hence, cyclicity of 2 = 4, i.e., every fourth power of 2 will give the same unit digit.

Similarly,

Cyclicity of 3 = 4

Cyclicity of 4 = 2

Cyclicity of 7 = 4

Cyclicity of 8 = 4

Cyclicity of 9 = 2

Cyclicity of 0 or cyclicity of 1 or cyclicity of 5 or cyclicity of 6 = 1. To know more about Unit Digid, refer to the book. Demystifying Number System.

TENS' DIGIT

There lies the cyclicity of tens' place digit of all the digits. This is given below:

Digits	*Cyclicity*
2, 3, 8–	20
4, 9 –	10
5–	1
6–	5
7–	4

Example 37 What is the tens' place digit of 12^{42}?

Solution For this, we need to break 12^{42} first by using binomial theorem as $(10+2)^{42}$. Obviously this expression will have 43 terms, and out of these 43 terms first 41 terms will have both of their tens and units place digit as 0.

Last two terms will be $\rightarrow {}^{42}C_{41} \times 10^1 \times 2^{41} + {}^{42}C_{42} \times 10^0 \times 2^{42}$

Now, we will find the tens' place digit of all these terms individually.

Tens digit of ${}^{42}C_{41} \times 10^1 \times 2^{41} = 42 \times 10 \times (02)$ [Cyclicity of 2 is 20, so 2^{41} will have same tens digits as 2^1] = 840, so 40 are the last two digits.

Similarly, ${}^{42}C_{42} \times 10^0 \times 2^{42} = 1 \times 1 \times 04 = 04$

So, finally last two digits are $\rightarrow 40 + 04 = 44$, so 4 is the tens' place digit.

Note: $(25)^n$ and $(76)^n$ will always give 25 and 76 as the units and tens digit for any natural number value of n.

NUMBER OF EXPONENTS

Let us take a simple number – 10^5

This is read as – 10 to the power 5, or we say that exponent of 10 is 5.

In simple terms, exponents are also known as Power.

Example 38 What is the maximum value of s if $N = (35 \times 45 \times 55 \times 60 \times 124 \times 75)$ is divisible by 5^s?

Solution If we factorize $N = (35 \times 45 \times 55 \times 60 \times 124 \times 75)$, then we can see that 5 appears 6 times, it means N is divisible by 5^6.

So, maximum value of $s = 6$

Exponent of any prime number P in $n!$

$$= \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \ldots + \left[\frac{n}{p^x}\right], \text{ where } n \geq p^x \text{ and } [.]$$

denotes the greatest integer value i.e., we have to consider only the integral value.

Let us find the exponent of 5 in 1000! = $1000/5 + 1000/5^2 + 1000/5^3 + 1000/5^4 = 200 + 40 + 8 + 1 = 249$

Example 39 What is the highest power of 5 which can divide $N = (22! + 17894!)$?

Solution Number of times this number is divisible by 5 is same as the number of zeroes at the end of this number. Since 22! have 4 zeroes at its end, so N will also be having only four zeroes at its end. Hence, highest power of 5, which can divide N is 4.

Process to find out the exponent of any composite number in *n*!

There are three different kinds of composite numbers.

1. Product of two or more than two prime numbers with unit power of all the prime numbers
   For example, 15 (5×3), 30 ($2 \times 3 \times 5$) etc.
2. (Any prime number)^n, where $n > 1$
   For example, 4 (2^2), 27 (3^3)
3. Product of two or more than two prime numbers with power of any one prime number more than 1.
   For example, 12 ($2^2 \times 3$), 72 ($2^3 \times 3^2$) etc.

Let us find out the exponents of the above-written composite numbers one by one:

- Find out the exponent of 15 in 100!

15 is the product of two distinct prime numbers 5 and 3. So, to find out the exponents of 15, we need to find out the exponents of 5 and 3 individually.

So, by applying the same formula of finding out the exponents for any prime number in both of these cases individually, minimum of those two will be the answer.

$100/5^x = [100/5] + [100/5^2] = 20 + 4 = 24$

$100/3^x = [100/3] + [100/3^2] + [100/3^3] + [100/3^4]$
$= 33 + 11 + 3 + 1 = 48$

Obviously, 24 is going to be the answer.

- Find out the exponent of 25 in 100
  $25 = 5^2$

In this case, we first find out the exponents of 5 and then divide it by 2 (actually the power) to find the exponents of 25.

$100/5^x = [100/5] + [100/5^2] = 20 + 4 = 24$
So, $100/25^x = 24/2 = 12$

- Similarly, we can find out for the third category of numbers also.

BASE SYSTEM

In our decimal system of writing the numbers, we use 10 digits (0 – 9). In this system, largest number of single digit = 9, and the moment we have to form a number bigger than this number we need two-digit numbers starting from 10. Similarly, largest number of two digits = 99 and after this we have 100, a number of three digits.

Lets assume a system of writing where we use only 6 digits (0 – 5). Largest single digit number in this system will be 5 and next to this will be 10. Similarly, largest two digit number will be 55 and next to this is 100.

This whole procedure can be summed up in the following table:

$(0\text{-}9)_{10}$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$(0\text{-}8)_{9}$	0	1	2	3	4	5	6	7	8	10	11	12	13	14	15	16	17	18
$(0\text{-}7)_{8}$	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17	20	21
$(0\text{-}6)_{7}$	0	1	2	3	4	5	6	10	11	12	13	14	15	16	20	21	22	23
$(0\text{-}5)_{6}$	0	1	2	3	4	5	10	11	12	13	14	15	20	21	22	23	24	25
$(0\text{-}3)_{4}$	0	1	2	3	10	11	12	13	20	21	22	23	30	31	32	33	100	101
$(0\text{-}2)_{3}$	0	1	2	10	11	12	20	21	22	100	101	102	110	111	112	120	121	122

Types of Questions

Questions from this concept are asked in three different ways:

1. $(\text{Base})_{10}$ to any other base and vice-versa
2. $(\text{Base})_x$ to $(\text{Base})_y$ and vice-versa; none of x and y being equal to 10 but x and y will be given.
3. $(\text{Base})_x$ to $(\text{Base})_y$, value of x and y will not be given.

1. $(\text{Base})_{10}$ to any other base and vice-versa

Method 1

Let us see in case of $(74)_{10}$.

$(74)_{10} = 7 \times 10^1 + 4 \times 10^0$, since the base is 10.

Now, if we have to convert this number in 9 base, then we will try to write it in terms of powers of 9.

$(74)_{10} = 8 \times 9^1 + 2 \times 9^0 = (82)_9$

$(74)_{10} = 1 \times 8^2 + 1 \times 8^1 + 2 \times 8^0 = (112)_8$

$(74)_{10} = 1 \times 7^2 + 3 \times 7^1 + 4 \times 7^0 = (134)_7$

$(74)_{10} = 2 \times 6^2 + 0 \times 6^1 + 2 \times 6^0 = (202)_6$

While converting the numbers from decimal system to any other system of writing the numbers, we should be concerned with the following two rules

i. Take maximum possible power of the base and then keep writing rest of the number with the help of lesser power of base (as illustrated in the above example).
ii. Once we have used (base)^n; where n is the maximum power, we will be required to

E.g., Write the co-efficients of all the powers of base from 0 to $(n-1)$ as in the case of $(74)_{10} = (202)_6$.

Now, suppose we have to convert $(356)_7$ in the base of 10.

$(356)_7 = 3 \times 7^2 + 5 \times 7^1 + 6 \times 7^0 = (188)_{10}$

Method 2

Converting $(74)_{10}$ to the base of $(\ )_9$

Base	74	Remainder
9	8	2 ↑
	→	

So, $(74)_{10} = (82)_9$

Converting $(74)_{10}$ to the base of $(\ )_8$

Base	74	Remainder
8	9	2 ↑
8	1	1 ↑
	→	

So, $(74)_{10} = (112)_8$

Converting $(74)_{10}$ to the base of $(\ )_7$

Base	74	Remainder
7	10	4 ↑
7	2	3 ↑
	1 →	

So, $(74)_{10} = (134)_7$

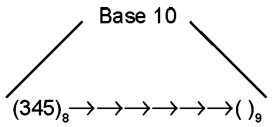
Converting $(74)_{10}$ to the base of $(\ )_6$

Base	74	Remainder
6	12	2 ↑
6	2	0 ↑
	↑ →	
	Quotient	

So, $(74)_{10} = (202)_6$

2. (Base)$_x$ to (Base)$_y$ and vice-versa; none of *x* and *y* being equal to 10 but *x* and *y* will be given

Converting $(345)_8$ to the base of $(\ )_9$



We will do this problem with the help of creating a bridge of base 10 between base 8 and base 9.

Step 1 Convert $(345)_8$ into base 10.

$345 = 3 \times 8^2 + 4 \times 8^1 + 5 \times 8^0 = (229)_{10}$

Step 2 Now convert this number in base 10 into base 9.

$(229)_{10} = 2 \times 9^2 + 7 \times 9^1 + 4 \times 9^0 = (274)_9$

However, if new base is a power of old base and vice-versa, then it can be converted directly also in the new base, i.e., we are not needed to go to base 10 for these kind of conversions. E.g., $(\text{Base})_2$ to $(\text{Base})_4$ or $(\text{Base})_2$ to $(\text{Base})_8$ conversion does not require a bridge of base 10.

Converting $(101110010)_2$ to Octal $(\ )_8$ system

At first, we will club three digits of binary number into a single block and then write the decimal equivalent of each group (left to right).

So, $(101110010)_2$ is now $(101)_2(110)_2(010)_2$

Now, $(101)_2 = 1 \times 2^2 + 0 + 1 \times 2^0 = 5$

$(110)_2 = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 6$

$(010)_2 = 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 2$

So, $(101110010)_2 = (562)_8$

Converting $(101110010)_2$ to Hexa-decimal $(\ )_{16}$ system

At first, we will club four digits of binary number into a single block and then write the decimal equivalent of each group (left to right).

So, $(101110010)_2$ is now $(0001)_2(0111)_2(0010)_2$

Now, decimal equivalent of $(0001)_2 = 1$

Decimal equivalent of $(0111)_2 = 7$

Decimal equivalent of $(0010)_2 = 2$

$(101110010)_2 = (172)_{16}$

3. (Base)$_x$ to (Base)$_y$, value of x and y will not be given

In these type of questions, normally some calculation is given in some unknown system of writing numbers and on the basis of that we will be required to solve the questions.

Example 40 In a system of writing of *N* digits, $4 \times 6 = 30$ and $5 \times 6 = 36$. What will be the value of $N = 3 \times 4 \times 5$ in the same system of writing?

Solution Let us assume that there are *N* digits in this system of writing.

So, $(30)_N = 3 \times N^1 + 0 \times N^0 = 24$

$\Rightarrow 3N = 24$

$\Rightarrow N = 8$

So, this system of writing has 8 digits.

In this system $3 \times 4 \times 5 = 60$ will be written as 74. $(60 = 7 \times 8^1 + 4 \times 8^0)$

Alternatively, since this system is having 6 as one of its digits, minimum value of *N* will be 7. Again, 24 is written as 30 in this system, so *N* is less than 10. Now use hit and trial for $N = 7$ or 8 or 9 to find out *N* in $24 = (30)_N$.

DECIMAL CALCULATION

So far we have seen the calculations involving natural numbers only. Let's work now with decimals.

Converting decimal system numbers to any other system

Suppose (12.725) is a number in decimal system which is required to be converted into octal system (8 digits)

We will first convert 12 into octal system.

$(12)_{10} = (14)_8$

Now to convert $(0.725)_{10}$ into $(\ )_8$, we will apply the following method:

$0.725 \times 8 = 5.8$ Take out integral part from here.
$0.8 \times 8 = 6.4$ Take out integral part from here.
$0.4 \times 8 = 3.2$ Take out integral part from here.
$0.2 \times 8 = 1.6$ Take out integral part from here.

And keep doing this till we get decimal part as zero, i.e., the product should be an integer.

$(0.725)_{10} = (0.5632\ldots)_8$

So, $(12.725) = (14.5632\ldots)_8$

Converting any other system numbers to decimal system

Now suppose if $(15.453)_7$ is to be converted into decimal system, then the process is as follows:

We will first convert $(15)_7$ into decimal system.

$(15)_7 = 1 \times 7^1 + 5 \times 7^0 = (12)_{10}$

Now $(0.453)_7 = 4 \times 7^{-1} + 5 \times 7^{-2} + 3 \times 7^{-3} = 0.6821$

So, $(15.453)_7 = (12.6821)_{10}$

Basic Algebraic Calculations Involving Base Systems

Addition

$$\begin{array}{r} 325_7 \\ +\ 456_7 \\ \hline \end{array}$$

Start with the units place digit, $5 + 6 = 11$ which is 14_7. So, unit digit is 4 and carry over is 1.

Next is tens place digit, 2 + 5 + 1 (carry over) = 8 which is 11_7. So, tens digit is 1 and carry over is again 1.

Next is 3 + 4 + 1 (carry over) = 8 which is 11_7.

$$\begin{array}{r} 325_7 \\ +\ 456_7 \\ \hline 1114_7 \end{array}$$

Subtraction

$456_8 - 367_8$

Starting with the units digit, since 6 is smaller than 7, we will borrow 1 from the tens place digit. So, now it is 14 (when the base is 10, we get 10 but here base is 8, so we get 8) and 7 subtracted from it = 14 – 7 = 7, which is the units digit.

Next, tens digit is now 4 and we have to subtract 5 from it. We again borrow 1 from hundred's place digit. So, now it is 12 and 12 – 6 = 6, which is the tens place digit.

Now, hundred's place digit is 3 (4 – 1), so 3 – 3 = 0

$$\begin{array}{r} 456_8 \\ -\ 367_8 \\ \hline 67_8 \end{array}$$

Note: Another method of doing these kinds of calculations is to convert these values (in whatever base) into decimal system, then do the actual calculation in decimal system itself and finally converting the numbers into the required or given system.

Some standard systems of writing

1. Decimal system
   Digits used: 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.
   Total digits used = 10 digits
2. Hexa-decimal system
   Digits used: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E and F.
   Total digits used = 16
3. Octal system
   Digits used: 0, 1, 2, 3, 4, 5, 6 and 7.
   Total digits used = 8
4. Binary system
   Digits used: 0, 1
   Total Digits used = 2

Comparison of Different Base Systems

It should be understood here that using different bases creates a difference only in the numbers (more than single digit) and not in the digit. This means in that $(5)_{10}$ is same as $(5)_9$ or any other system of writing the numbers.

Example 41 How many different values of N are possible in the following calculation

$(2)_N \times (4)_N = (8)_N$?

Solution Any value of N which has 8 as a digit in that system will satisfy this calculation. So, any value of $N \geq 9$ is possible. Hence infinite values of N are possible.

This concept can be further understood through the mechanism of multiplication. Multiplication is basically the condensed form of addition. So, 2 added up four times will give 8, and hence all the base systems which have the following can be one of the values of N:

0 1 2 3 4 5 6 7 8.

Example 42 How many different values of N are possible in the following calculation

$(4)_N + (5)_N = (9)_N$?

Solution In this case also, the digits used can be of any system that has the digit 9. So, all the inegral values $N \geq 10$ are possible.

Example 43 How many different values of N are possible in the following calculation.

$(4)_N \times (5)_N = (24)_N$?

Solution Obviously, there can be either just one possible value of N or no possible value of N.

$(4)_{10} \times (5)_{10} = (20)_{10}$ [We know $(4)_N = (4)_{10}$ and $(5)_N = (5)_{10}$]

Now, $(24)_N = 2 \times N^1 + 4 \times N^0 = 20$

Hence, $N = 8$. And this will be the only value of N.

Divisibility Rules For Systems Other Than Decimal System

I would like to emphasize that different number systems are just different ways to write numbers. Thus the divisibility of one number by another does not depend on the particular system in which they are written.

At the same time, in each system there are some tricks to determine divisibility by certain specific numbers. These are the divisibility tests.

Let us investigate the tests for divisibility by 3 and 9. We will try to generalize these tests for any number base system.

Is 1234565642317 divisible by 6?

We know the divisibility rule for 9 – Sum of digits of the number should be divisible by 9.

Sum of the digits of this number is 42.

Now, we can answer this question easily, since the sum of the digits (which is 4210) is divisible by 6, the number itself is also divisible by 6.

In general, the sum of the digits of a number written in the base n system is divisible by $(n-1)$ if and only if the number itself is divisible by $(n-1)$.

So, divisibility rule for 4 in a base system of 5 in that the sum of the digits of the number should be divisible by 4. For example, 31_5 is divisible by 4.

Similarly, if we have to find out the divisibility rule of 12 in the base of 11, it will be nothing but same as the divisibility rule of 11 in the base of 10. Generalizing this whole concept, divisibility rule of any natural number N in the base of $(N-1)$ will be same as divisibility rule of 11 on base 10.

Before you proceed, you should have a clear understanding of how to find the divisibility rule for any prime number in base 10.

The only difference is that in case of base 10, we find the multiples of prime numbers in the format $10n \pm 1$, and in case of base B, we find the multiples of prime numbers in the format $Bn \pm 1$.

Let us find out the divisibility rule of $(5)_8$.

Multiples of 5 in base 10

Multiples of 5 in base 10	5	10	15	20	25	30
Multiples of 5 in base 8	5	12	17	24	31	26

Obviously, 17 is the first number in the format of $8n \pm 1$.

$17 = 8 \times 2 - 1$ [Yes, it is correct. 17_8 and RHS is in base 10]

So, divisibility rule of 5 on base 8 = A + 2B, where A and B have the same meaning as that in divisibility rules of 10.

We can check this by the following method:

First few numbers in base 8							
0	1	2	3	4	5	6	7
10	11	12	13	14	15	16	17
20	21	22	23	24	25	26	27
30	31	32	33	34	35	36	37

Every 5th number in this series will be divisible by 5. So, 5 is the first number divisible by 8. The next few numbers are 12, 17, 24, 31, 36.

We can use base to generate good concepts in remainders also.

Example 44 What is the remainder when $(1234)_6$ is divided by 5?

Solution Divisibility rule of $(5)_6$ will be similar to that of the divisibility rule of $(9)_{10}$. So, if the sum of digits of the number in base 6 is divisible by 5, then it will be divisible by 5.

Adding the digits of $(1234)_6 = 10$, which is divisible by 5, the remainder obtained when $(1234)_6$ is divided by 5 is 0.

Example 45 What is the remainder when $(12341)_6$ is divided by 5?

Solution The sum of the digits of $(12341)_6 = 11$, so the remainder obtained will be 1.

Pigeon-hole Principle

Although only a few questions have been asked from this concept so far in CAT, which is also true for some of the arithmetic topics like Profit and Loss; Calendar etc., the importance of this topic lies in the fact that this concept is purely logical and having a good practice over this concept gives a logical edge over others.

The concept is that if we put $(N+1)$ or more pigeons in N holes (nests), then at least one hole will be there which will have 2 or more pigeons.

Example 46 What is the minimum number of people in any group of five people who have an identical number of friends within the group, provided if A is a friend of B, then B is also a friend of A?

Solution Since there are five persons in the group, so possible number of friends is 0, 1, 2, 3, 4. It seems here that everybody is having a different number of friends, so the answer is zero. But anybody having four friends ensures that nobody is having 0 friends. So, at least two persons must have the same number of friends.

PRACTICE EXERCISES

WARM UP

Q.1. Which of the following is the smallest?
(a) $5^{1/2}$ (b) $6^{1/3}$ (c) $8^{1/4}$ (d) $12^{1/5}$

Q.2. A number N is divisible by 6 but not divisible by 4. Which of the following will not be an integer?
(a) N/3 (b) N/2 (c) N/6 (d) N/12

Q.3. If a, b and c are consecutive positive integers, then the largest number which always divides $(a^2 + b^2 + c^2)$
(a) 14 (b) 55
(c) 3 (d) None of these

Q.4. $\dfrac{(3.134)^3 + (1.866)^3}{(3.134)^2 - 3.134 \times 1.866 + (1.866)^2} = ?$
(a) 25 (b) 2.68 (c) 1.038 (d) 5

Q.5. If n^2 is a perfect cube, then which of the following statements is always true?
(a) n is odd
(b) n is even
(c) n^3 is a perfect square
(d) n is a perfect cube

Q.6. If $(5x + 11y)$ is a prime number for natural number values of x and y, then what is the minimum value of $(x + y)$?
(a) 2 (b) 3 (c) 4 (d) 5

Q.7. For what values of x is $25^x + 1$ divisible by 13?
(a) All real values of x
(b) Odd natural values of x
(c) Even values of x
(d) All the integral values of x

Q.8. Which of the following numbers lies between 5/6 and 6/7?
(a) 71/84 (b) 31/42 (c) 129/168 (d) 157/339

Q.9. By multiplying with which of the following numbers does the product of $8 \times 9 \times 10 \times 11 \times 12$ become a perfect square?
(a) 55 (b) 11 (c) 165 (d) 310

Q.10. What is the difference between the sum of the cubes and that of squares of the first ten natural numbers?
(a) 5,280 (b) 2,640 (c) 3,820 (d) 4,130

Q.11. If $3 - 9 + 15 - 21 + \ldots$ up to 19 terms $= x$, then x is a/an
(a) odd number (b) even number
(c) prime number (d) irrational number

Q.12. What is the unit's digit of $21^3 \times 21^2 \times 34^7 \times 46^8 \times 77^8$?
(a) 4 (b) 8 (c) 6 (d) 2

Q.13. If the unit's digit in the product $(47n \times 729 \times 345 \times 343)$ is 5, what is the maximum number of values that n may take?
(a) 9 (b) 3 (c) 7 (d) 5

Q.14. In how many ways can 846 be resolved into two factors?
(a) 9 (b) 11
(c) 6 (d) None of these

Q.15. If a number is divided by 15, it leaves a remainder of 7. If thrice the number is divided by 5, then what is the remainder?
(a) 5 (b) 6 (c) 7 (d) 1

Q.16. A number when divided by 391 gives a remainder of 49. Find the remainder when it is divided by 39.
(a) 10 (b) 9
(c) 11 (d) Cannot be determined

Q.17. p and q are two prime numbers such that $p < q < 50$. In how many cases would $(q + p)$ be also a prime number?
(a) 5 (b) 6
(c) 7 (d) None of these

Q.18. How many distinct factors of 1,600 are perfect cubes?
(a) 3 (b) 4 (c) 6 (d) 2

Q.19. The LCM of 96,144 and N is 576. If their HCF is 48, then which of the following can be one of the values of N?
(a) 168 (b) 192 (c) 144 (d) 244

Q.20. If p and q are consecutive natural numbers (in increasing order), then which of the following is true?
(a) $q^2 < p$ (b) $2p > p$
(c) $(q + 1)^2 > p^2$ (d) $(p + 2)^3 < q^3$

Q.21. $(17^{21} + 19^{21})$ is not divisible by
(a) 36 (b) 8 (c) 9 (d) 18

Q.22. Which of the following will divide $11^{12296} - 1$?
(a) 11 and 12 (b) 11 and 10
(c) 10 and 12 (d) 11 only

Q.23. If a, b, c and d are consecutive odd numbers, then ($a^2 + b^2 + c^2 + d^2$) is always divisible by
(a) 5 (b) 7 (c) 3 (d) 4

Q.24. Four bells toll at intervals of 14, 21 and 42 min respectively. If they toll together at 11.22 a.m., when will they toll together for the first time after that?
(a) 11.56 a.m. (b) 12.04 p.m.
(c) 12.06 p.m. (d) 11.48 a.m.

Q.25. When x is divided by 6, remainder obtained is 3. Find the remainder when $x^4 + x^3 + x^2 + x + 1$ is divided by 6.
(a) 3 (b) 4 (c) 1 (d) 5

Q.26. I have 7^7 sweets and I want to distribute them equally among 2^4 students. After each of the student got maximum integral sweets, how many sweets are left with me?
(a) 8 (b) 5
(c) 1 (d) None of these

Q.27. When I distribute some chocolates to my 40 students, three chocolates will be left. If I distribute the same number of chocolates to my students and my colleague Manoj Dawrani, seven chocolates are left. Find the minimum number of chocolates I have.
(a) 1,443 (b) 1,476
(c) 1,480, (d) None of these

Q.28. The LCM of two numbers is 40 times their HCF. The sum of the LCM and HCF is 1,476. If one of the numbers is 288, find the other numbers?
(a) 169 (b) 180 (c) 240 (d) 260

Q.29. 1010101 ... 94 digits is a 94 digits number. What will be the remainder obtained when this number is divided by 375?
(a) 10 (b) 320
(c) 40 (d) None of these

Q.30. Chandrabhal adds first N natural numbers and finds the sum to be 1,850. But actually one numbers was added twice by mistake. Find the difference between N and that number.
(a) 40 (b) 33 (c) 60 (d) 17

Q.31. When I distribute a packet of chocolate to 7 students, I am left with 4 chocolates. When I distribute the same packet of chocolate to 11 students, I am left with 6 chocolates. How many chocolates will be left with me if I distribute the same packet of chocolate among 13 students (a packet of chocolate contains total number of chocolates N, $1000 < N < 1050$)?
(a) 2 (b) 0 (c) 6 (d) 7

Q.32. How many prime numbers are there between 80 and 105?
(a) 3 (b) 4 (c) 5 (d) 8

Q.33. If x and y are consecutive natural numbers in an increasing order, then which of the following is always true?
(a) $x^y > y^x$ (b) $y^x > x^y$
(c) $x^x > y^y$ (d) $y^y > x^x$

Q.34. What is the remainder when 5^{79} is divided by 7?
(a) 1 (b) 0 (c) 5 (d) 4

FOUNDATION

Q.1. LCM of two natural numbers is 590 and their HCF is 59. How many sets of values are possible?
(a) 1 (b) 2 (c) 5 (d) 10

Q.2. MUL has a waiting list of 5005 applicants. The list shows that there are at least 5 males between any two females. The largest number of females in the list could be
(a) 920 (b) 835 (c) 721 (d) 1005

Q.3. HCF of two numbers A and B is 24. HCF of two other numbers C and D is 36. What will be the HCF of A, B, C and D?
(a) 12 (b) 24
(c) 36 (d) 6

Q.4. How many zeroes will be there at the end of $25 \times 35 \times 40 \times 50 \times 60 \times 65$?
(a) 6 (b) 8 (c) 5 (d) 7

Q.5. What is the unit digit of $576847 \times 564068 \times 96467 \times 458576$?
(a) 2 (b) 4 (c) 6 (d) 8

Q.6. What is the unit digit of $1! + 2! + 3! + \ldots + 99! + 100!$?
(a) 3 (b) 1 (c) 5 (d) 6

Q.7. How many divisors will be there of the number 1020?
(a) 12 (b) 20
(c) 24 (d) 36

Q.8. In question 7, what is the difference between the number of even divisors and number of prime divisors?
(a) 13 (b) 12
(c) 11 (d) None of these

Q.9. $N = 7!^3$. How many factors of N are multiples of 10?
(a) 736 (b) 1008 (c) 1352 (d) 894

Q.10. A number N has odd number of divisors. Which of the following is true about N?
(a) All the divisors of this number will be odd.
(b) There will be atleast $(N - 11)$ prime divisors.
(c) N will be a perfect square.
(d) Atleast one divisor of the number should be odd.

Q.11. How many zeroes will be there at the end of the expression $N = 2 \times 4 \times 6 \times 8 \times \ldots \times 100$?
(a) 10 (b) 12
(c) 14 (d) None of these

Q.12. How many zeroes will be there at the end of the expression $N = 10 \times 20 \times 30 \ldots \times 1000$?
(a) 1280 (b) 1300
(c) 1320 (d) None of these

Q.13. How many zeroes will be there at the end of the expression $N = 7 \times 14 \times 21 \times \ldots \times 777$?
(a) 24 (b) 25
(c) 26 (d) None of these

Q.14. The number from 1 to 33 are written side by side as follows: 123456 ... 33. What is the remainder when this number is divided by 9?
(a) 0 (b) 1 (c) 3 (d) 6

Q.15. The number 444444 ... (999 times) is definitely divisible by
(a) 22 (b) 44
(c) 222 (d) All of these

Q.16. Find the unit digit of $N = {}_{17}27!^{37!}$
(a) 1 (b) 3 (c) 7 (d) 9

Q.17. How many divisors of $N = 420$ will be of the form $4n + 1$, where n is a whole number?
(a) 3 (b) 4 (c) 5 (d) 8

Q.18. $N = 2^3 \times 5^3 \times 7^2$. How many sets of two factors of N are co-prime?
(a) 72 (b) 64
(c) 36 (d) None of these

Q.19. What is the unit digit of $2^{3^{45}}$?
(a) 2 (b) 4 (c) 8 (d) 6

Q.20. How many zeroes will be there at the end of $1003 \times 1001 \times 999 \times \ldots \times 123$?
(a) 224 (b) 217
(c) 0 (d) None of these

Q.21. How many zeroes will be there at the end of $36!^{36!}$?
(a) $7 \times 6!$ (b) $8 \times 6!$ (c) $7 \times 36!$ (d) $8 \times 36!$

Q.22. The number formed by writing any digit 6 times (e.g., 111111, 444444, etc) is always divisible by
(i) 7 (ii) 11 (iii)13
(a) i and ii (b) ii and iii
(c) i and iii (d) i, ii and iii

Q.23. What is the maximum value of HCF of $[n^2 + 17]$ and $(n + 1)^2 + 17]$?
(a) 69 (b) 85
(c) 170 (d) None of these

Q.24. What is the number of pairs of values of (x, y), which will satisfy $2x - 5y = 1$, where $x \leq 200$, and x and y are positive integers?
(a) 38 (b) 39 (c) 40 (d) 41

Q.25. $N = 2^3 \times 5^3$. How many sets of two factors of N are co-prime to each other?
(a) 12 (b) 24 (c) 23 (d) 11

Q.26. What is the sum of digits of the least multiple of 13, which when divided by 6, 8 and 12 leave 5, 7 and 11 as the remainder?
(a) 5 (b) 6 (c) 7 (d) 8

Q.27. What is the unit digit of $7^{11^{22^{33}}}$?
(a) 1 (b) 3 (c) 7 (d) 9

Q.28. What is the remainder when $(1! + 2! + 3! + \ldots 1000!)$ is divided by 5?
(a) 1 (b) 2 (c) 3 (d) 4

Q.29. $A = 3^{150} \times 5^{76} \times 7^{140}$, $B = 3^{148} \times 5^{76} \times 7^{141}$, $C = 3^{148} \times 5^{80} \times 7^{139}$, $D = 3^{151} \times 5^{80} \times 7^{142}$, then the order of A, B, C and D from largest to smallest is
(a) DACB (b) CDBA
(c) CDAB (d) DCAB

Q.30. The HCF of 0.3, 0.15, 0.225, 0.0003 is
(a) 0.0003 (b) 0.3 (c) 0.15 (d) 0.0015

Q.31. How many numbers between 1 to 550 are divisible by 5 but not by 9?
(a) 98 (b) 97
(c) 101 (d) None of these

Q.32. A and B are two distinct digits. If the sum of the two-digit numbers formed by using both the digits is a perfect square, what is the value of (A + B)?
(a) 9 (b) 11 (c) 13 (d) 17

Q.33. A number N = 897324P64Q is divisible by both 8 and 9. Which of the following is the value of P + Q?
i. 2 ii. 11 iii. 9
(a) either i or ii (b) either ii or iii
(c) either i or ii or iii (d) None of these

Direction for questions 34 and 35: Read the passage below and solve the questions based on it.

A = Set of first N positive numbers. There are 16 numbers in A which are divisible by both X and Y. There are 50 numbers in A which are divisible by X but not by Y and 34 numbers in A divisible Y but not by X.

Q.34. How many numbers in A are divisible by any of the two numbers?
(a) 100 (b) 50
(c) 200 (d) None of these

Q.35. How many numbers in N are divisible by X?
(a) 42 (b) 56
(c) 66 (d) None of these

Q.36. Nitin had forgotten his 6 digit bank account number but only remembered that it was of the form X515X0 and was divisible by 36. What was the value of X?
(a) 4 (b) 7 (c) 8 (d) 9

Q.37. Students from the Delhi Public School are writing their exams in Kendriya Vidyalaya. There are 60 students writing their Hindi exams, 72 students writing there French exam and 96 students writing their English exam. The authorities of the Kendriya Vidyalaya have to make arrangements such that each classroom contains equal number of students. What is the minimum number of classrooms required to accommodate all students of Delhi Pubic School?
(a) 19 (b) 38 (c) 13 (d) 6

Q.38. In the Jyotirmayi school, all classes started at 9.00 am. The school has three sections: primary, middle and secondary. Each class for the primary section lasts for half an hour, for the middle section for forty five minutes and for the secondary section for half an hour. A lunch break has to be given for the entire school when each of three sections have just finished a respective class and are free. What is the earliest time for the lunch break?
(a) 11.00 am (b) 10.30 am
(c) 12.00 pm (d) 12.30 pm

Q.39. In the firing range, 4 shooters are firing at their respective targets. The first, the second, the third and the fourth schooter hits the target once every 5s, 6s, 7s, 8s respectively. If all of them hit their target at 10.00 am, when will they hit their target together again?
(a) 10.14 am (b) 10.28 am
(c) 10.30 am (d) 10.31 am

Q.40. Two friends Harry and Jayesh were discussing about 2 numbers. They found the two numbers to be such that one was twice the other. However, both had the same number of prime factors while the larger one had 4 more factors than the smaller one. What are the numbers?
(a) 40,80 (b) 20,40
(c) 30,60 (d) 50,100

Q.41. To celebrate their victory in the World Cup, the Sri Lankans distributed sweets. If the sweets were distributed among 11 players, 2 sweets were left. When the sweets were distributed among 11 players, three extra's and 1 coach, even then 2 sweets were left. What is the minimum number of sweets in the box?
(a) 167 (b) 334 (c) 332 (d) 165

Q.42. The first 20 natural numbers from 1 to 20 are written next to each other to form a 31 digit number N = 12345 67891011121314151617181920. What is the remainder when this number is divided by 16?
(a) 0 (b) 4 (c) 7 (d) 9

Q.43. Two friends Kanti and Sridhar were trying to find the HCF of fifty distinct numbers. If they were finding the HCF of two numbers at a time, how many times this operation should be repeated to find the HCF of 50 numbers?
(a) 20 (b) 25 (c) 49 (d) 50

Q.44. How many zeroes will be there at the end of N = 18! + 19!?
(a) 3 (b) 4
(c) 5 (d) Cannot be determined

Q.45. Manish was dividing 2 numbers by a certain divisor and obtained remainders of 437 and 298 respectively. When he divides the sum of the two numbers by the same divisor, the remainder is 236. What is the divisor?
(a) 499 (b) 735
(c) 971 (d) None of these

Q.46. I purchased a ticket for the football match between France and Italy in the World Cup. The number on the ticket was a 5 digit perfect square such that the first and the last digit were the same and the 2nd and 4th digit were the same. If the 3rd digit was 3, then what was the ticket number?
(a) 24,342 (b) 12,321
(c) 21,312 (d) None of these

Q.47. How many integers N in the set of integers {1, 2, 3, …, 100} are there such that $N^2 + N^3$ is a perfect square?
(a) 5 (b) 7 (c) 9 (d) 11

Q.48. In a birthday party, all the children were given candy bars. All the children got three candy bars each except the child sitting at the end who got only 2 candy bars. If each child had been given only 2 candy bars there would have been 8 candy bars remaining. How many children were there and how many candy bars were distributed?
(a) 9, 26 (b) 6, 26 (c) 9, 18 (d) 6, 15

Q.49. A natural number N satisfies following conditions.
(A) Number is having all the 9's
(B) It is divisible by 13.
How many digits are there in N?
(a) 5 (b) 6 (c) 7 (d) 8

Q.50. What is the minimum number of identical square tiles required to cover a floor of dimension 3.78 m × 4.8 m?
(a) 2,520 (b) 3,780
(c) 5,040 (d) 6,480

Q.51. What is the smallest five-digit number which when divided by 7, 11 and 21 leaves a remainder of 3 in each case?
(a) 10,019 (b) 10,001
(c) 10,111 (d) 10,167

Q.52. A milkman has 3 jars containing 57 litres, 129 litres and 177 litres of pure milk respectively. A measuring can leaves the same amount of milk unmeasured in each jar after a different number of exact measurements of milk in each jar. What is the volume of largest such can?
(a) 12 litres (b) 16 litres
(c) 24 litres (d) 48 litres

Q.53. A boy was carrying a basket of eggs. He fell down and some of the eggs were broken. The boy has 10 eggs left with him. When asked by his mother how many eggs were broken, the boy could not recall. However he recalled that when he counted the total number of eggs 3 at a time 1 egg was left. When counted 4 at a time, 1 egg was left and when counted 5 at time a no egg was left. How many eggs were broken?
(a) 15 (b) 25 (c) 30 (d) 35

Q.54. How many prime numbers are there which, when divided by another prime number, gives a quotient which is same as the remainder?
(a) 0 (b) 1
(c) 2 (d) More than 2

Q.55. The number of natural numbers n such that $\frac{(n+1)^2}{n+7}$ is an integer, is
(a) 4 (b) 5
(c) 6 (d) None of these

Q.56. Find the product of all the factors of 3^{16}.
(a) 3^{33} (b) 3^{68} (c) 3^{136} (d) 3^{128}

Q.57. What is the remainder when 90^{91} is divided by 13?
(a) 0 (b) 7 (c) 12 (d) 1

Q.58. Find the remainder when the product of 10 consecutive natural numbers starting from 8641 is divided by 8640.
(a) 1 (b) 55 (c) 10 (d) 0

Q.59. Palindromes are numbers which when read from both the ends will give us the same number, for example, 123321 etc. In $(P)_N$, P is a palindrome of S digits, where S is even, written in base N. Which of the following number will definitely divide P?
(a) N − 1 (b) N + 1
(c) 2 N + 1 (d) None of these

Q.60. How many numbers are there between 400 and 600 in which 8 occurs only once?
(a) 36 (b) 18 (c) 19 (d) 38

Q.61. If $n^2 = 123.45654321$, which of the following is the exact value of n.
(a) 11.1001 (b) 11.1101
(c) 11.1111 (d) 11.1011

Q.62. A mule said to a horse, "If I take one sack off your back, my load will be double of yours and if you take one off my back our loads will be the same." How many sacks in all were they carrying?
(a) 5 (b) 7 (c) 12 (d) 14

Q.63. Divide 45 into 4 parts such that if the first is increased by 2, the second is decreased by 2, the third multiplied by 2 and the fourth divided by 2, the result is the same.
(a) 20, 8, 5, 12 (b) 12, 5, 20, 8
(c) 5, 8, 12, 20 (d) 8, 12, 5, 20

Q.64. Find the remainder when $3x^2-x^6+31x^4+21x+5$ is divided by $x + 2$.
(a) 10 (b) 12
(c) 11 (d) None of these

Q.65. Four prime numbers are in ascending order of their magnitudes. The product of the first three is 385 and that of last three is 1001. The largest given prime number is
(a) 11 (b) 13 (c) 17 (d) 19

Q.66. What is the remainder when 4^{44} is divided by 15?
(a) 1 (b) 2 (c) 3 (d) 4

Q.67. LCM of two integers P and Q is 211. What is the HCF of P and Q?
(a) 37 (b) 1
(c) 3 (d) Cannot be determined

Q.68. How many times does the digit 6 appear when we count from 11 to 400?
(a) 34 (b) 74 (c) 39 (d) 79

Q.69. In Q.68, how many numbers will be having 8 as its digit?
(a) 74 (b) 75 (c) 76 (d) 77

Q.70. S is a number formed by writing 8 for 88 times. What will be the remainder of this number when divided by 7?
(a) 4 (b) 5 (c) 8 (d) 1

Q.71. We are writing all the multiples of 3 from 111 to 324. How many times will we write digit 3?
(a) 18 (b) 19 (c) 21 (d) 22

Q.72. What is the remainder when 7 + 77 + 777 + 7777 + ... (till 100 terms) is divided by 8?
(a) 0 (b) 2 (c) 4 (d) 6

Q.73. A number has exactly 15 composite factors. What can be the maximum number of prime factors of this number?
(a) 2 (b) 3 (c) 4 (d) 5

Q.74. $N = 204 \times 221 \times 238 \times 255 \times ... \times 850$. How many consecutive zeroes will be there at the end of this number N?
(a) 8 (b) 10 (c) 11 (d) 12

Q.75. 1st 126 natural numbers are put side by side in the ascending order to create a large number N = 123456... 125126. What will be the remainder when N is divided by 5625?
(a) 5126 (b) 26 (c) 126 (d) 156

Q.76. When a number S is divided by 3, 4 and 7 successively, remainders obtained are 2, 1 and 4 respectively. What will be the remainder when the same number is divided by 84?
(a) 43 (b) 53 (c) 63 (d) 73

Q.77. What is the remainder when $1714 \times 1715 \times 1717$ is divided by 12?
(a) 3 (b) 8 (c) 2 (d) 9

Q.78. N^2 = 12345678987654321. Find N.
(a) 101010101 (b) 11111
(c) 111111111 (d) 1000000001

Q.79. If a, b, c and d are distinct integers in the range 10 to 15 (both inclusive), the greatest value of $(a + b)(c + d)$ is—
(a) 750 (b) 731 (c) 700 (d) 729

Q.80. The smallest natural number which is a perfect square and is of the form abbb lies in between
(a) 1,000 to 2,000 (b) 2,000 to 3,000
(c) 3,000 to 4,000 (d) 4,000 to 5,000

MODERATE

Q.1. How many number of zeroes will be there at the end of 12! expressed in base 6?
(a) 4 (b) 5 (c) 6 (d) 7

Q.2. Find the remainder when $2222^{5555} + 5555^{2222}$ is divided by 7.
(a) 1 (b) 3 (c) 0 (d) 5

Q.3 LCM of first 100 natural numbers is N. What is the LCM of first 105 natural numbers?
(a) 5! × N (b) 10403 N
(c) 105N/103 (d) 4N

Q.4. How many divisors of 10^5 end with a zero?
(a) 1 (b) 3 (c) 9 (d) 16

Q.5. Following expression holds true if we replace some of '+' signs by '×' signs.
1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 +10 = 100
How many '+' signs are needed to be replaced by '×'?
(a) 2 (b) 3 (c) 4 (d) 1

Q.6. In a particular country, all the numbers are expressed with the help of three alphabets a, b and c.
15 is written as abc.
6 is written as bc.
60 is written as bcbc.
How would one write 17 in that country?
(a) abb (b) bab (c) baa (d) aba

Q.7. When a certain two-digit number is added to another two digit number having the same digits in reverse order, the sum is a perfect square. How many such two-digit numbers are there?
(a) 4 (b) 6 (c) 8 (d) 10

Q.8. What is the remainder when ${}_{32}32^{32}$ is divided by 7?
(a) 2 (b) 3 (c) 4 (d) 6

Q.9. N is a 1001 digit number consisting of 1001 sevens. What is the remainder when N is divided by 1001?
(a) 7 (b) 700
(c) 777 (d) None of these

Q.10. Find four positive numbers such that the sum of the first, third and fourth exceeds the second by 8; the sum of the squares of the first and second exceeds the sum of the squares of the third and fourth by 36; the sum of the products of the first and second, and of the third and fourth is 42; the cube of the first is equal to the sum of the cubes of the second, third, and fourth.
(a) 2, 1, 9,3 (b) 2, 4, 6, 8
(c) 6, 5, 4, 3 (d) None of these

Q.11. Digital sum of a number is obtained by adding all the digits of a number until a single digit is obtained. Find the digital sum of 19^{100}.
(a) 1 (b) 4 (c) 7 (d) 9

Q.12. Find the HCF of $(2^{100} - 1)$ and $(2^{120} - 1)$.
(a) $2^{10} - 1$ (b) $2^{20} - 1$
(c) 1 (d) None of these

Q.13. What is the remainder when $22^{33} + 10^{35}$ is divided by 45?
(a) 2 (b) 11
(c) 8 (d) None of these

Q.14. An intelligence agency forms a code of two distinct digits selected from 0, 1, ... ,9 such that the first digit of the code is non-zero. The code, handwritten on a slip, however, can potentially create confusion when read upside down, e.g., the code 91 may appear as 16. How many codes are there for which no such confusion can arise?
(a) 80 (b) 63
(c) 71 (d) None of these

Q.15. If p, p + 2, p + 4 are prime numbers, then the number of possible solutions for p is
(a) 0 (b) 1
(c) 2 (d) None of these

Q.16. Suppose N is an integer such that the sum of the digits of N is 2, and $10^9 < N < 10^{10}$. How many values of N are possible?
(a) 11 (b) 10 (c) 9 (d) 8

Q.17. Ten students solved a total of 35 questions in a maths Olympiad. Each question was solved by exactly one student. There is at least one student who solved exactly one problem, at least one student who solved exactly two problems and at least one student who solved exactly three problems. What is the minimum number of students who has/have solved at least five problems?
(a) 1 (b) 2
(c) 3 (d) None of these

Q.18. N! is having 37 zeroes at its end. How many values of N is/are possible?
(a) 0 (b) 1 (c) 5 (d) Infinite

Q.19. In the above question, how many values of N will be even?
(a) 0 (b) 2 (c) 3 (d) Infinite

Q.20. N! is having 30 zeroes at its end. How many values of N is/are possible?
(a) 0 (b) 1 (c) 5 (d) Infinite

Q.21. What is the remainder when $(1^1 + 2^2 + 3^3 + \ldots + 100^{100})$ is divided by 4?
(a) 0 (b) 1 (c) 2 (d) 3

Q.22. A 3-digit number in which all the 3-digits are odd is such that if the cubes of the digit are added, the sum would be equal to the number itself. If one of the digit is 7, find the number.
(a) 171 (b) 371 (c) 575 (d) 775

Q.23. A teacher said that there were 100 students in his class, 24 of whom were boys and 32 were girls. Which base system did the teacher use in this statement?
(a) 9 (b) 5 (c) 6 (d) 8

Q.24. What is the remainder when 3^{450} is divided by 108?
(a) 3 (b) 1 (c) 27 (d) 81

Q.25. P is a natural number. 2P has 28 divisors and 3P has 30 divisors. How many divisors of 6P will be there?
(a) 35 (b) 40 (c) 45 (d) 48

Q.26. pqr is a three digit natural number such that $pqr = p^3 + q^3 + r^3$. What is the value of r?
(a) 0 (b) 1
(c) 3 (d) Cannot be determined

Q.27. There are two three-digit numbers. When one number is divided by another number, quotient obtained is 6 and remainder is 0. Sum total of both the numbers is a multiple of 504. What is the difference between the numbers?
(a) 720 (b) 360 (c) 120 (d) 420

Q.28. LCM of 12^{24}, 16^{18} and N is 24^{24}. Number of all the possible values of N = S. What is the value of S?
(a) 25 (b) 1,800
(c) 1,825 (d) None of these

Q.29. Each of P, Q, R and S equals either 0 or 1. It is given that
If Q = 0, then R = 1
If R = 0, then P = S
If S = 0, then P = 1
Assume R = 0, find the value of (P + Q + R + S)?
(a) 0 (b) 1 (c) 2 (d) 3

Q.30. When asked about his date of birth in 1996, Mayank replied that "last two digits of my birth year stands for my age." When Siddharth was asked about his age, he also replied the same. But Siddharth is older to Mayank. What is the difference in their age?
(a) 46 (b) 50
(c) 0 (d) Cannot be determined

Direction for questions 31 to 35: Read the passage below and solve the questions based on it.

There is a prison with 100 cells inside it. Cells are numbered from 1 to 100 and every cell is occupied by one prisoner only. One day jailer decides to release some of the prisoners and for this he defines an algorithm of 100 steps which follows:

Step 1 Reverse the position of all the cells which are divisible by 1.

Step 2 Reverse the position of all the cells which are divisible by 2.

Step 3 Reverse the position of all the cells which are divisible by 3.

……

……

……..

Step 99 Reverse the position of all the cells which are divisible by 99.

Step 100 Reverse the position of all the cells which are divisible by 100.

Initially all the cells are closed. After executing all these steps, prisoners of all the cells which remain open are released.

Q.31. How many prisoners are released?
(a) 25 (b) 10
(c) 5 (d) None of these

Q.32. Which of the following cell number will be open at the end?
(a) Cell number 56 (b) Cell number 64
(c) Cell number 72 (d) Cell number 84

Q.33. Which of the following is true about the family of cell numbers N that will be open at the end?
(a) All the elements of N will be having only two factors including the number itself.
(b) All the elements of N will be having odd number of factors excluding the number itself.
(c) All the elements of N will be having odd number of factors including the number itself.
(d) None of these

Q.34. If number of prisoners in each cell is equal to the cell number, then how many prisoners will be released?
(a) 285 (b) 385
(c) 485 (d) None of these

Q.35. There are many islands on earth for which we do not have much information, or sometime we do not have any information about those islands. On such an island, which uses a system of S digits ($S \neq 10$) to write the numbers, selling price of a goat is Rs 1,143. Parul bought a goat and paid the shopkeeper Rs 1,150. Now shopkeeper returned Parul Rs 5. How many digits are used to write the numbers in that system?
(a) 9 (b) 8 (c) 7 (d) 6

Q.36. What is the value of N in the following expression $345_6 + 632_7 + 487_9 = (N)_5$?
(a) 11,412 (b) 11,214
(c) 10,412 (d) 21,412

Q.37. In a certain system of writing of numbers we write $(12\Phi)_4 = (124)_\Phi$, where Φ is the unit digit in the first place and number of digits in that system in the second place.
How many values of Φ are possible?
(a) 0 (b) 1 (c) 3 (d) Infinite

Q.38. Digital sum of a number is obtained by adding all the digits of a number until a single digit is obtained. What is the digital sum of 4444^{4444}?
(a) 2 (b) 6 (c) 7 (d) 3

Direction for questions 39 to 41: Read the passage below and solve the questions based on it.

$N = A^2 \times B^3 \times C^4$, where A, B and C are prime numbers.

Q.39. How many factors of N are perfect squares?
(a) 6 (b) 12
(c) 8 (d) None of these

Q.40. How many factors of N are cube of any natural number?
(a) 2 (b) 3 (c) 4 (d) 6

Q.41. How many factors of N will be having exactly 3 factors?
(a) 2 (b) 3
(c) 4 (d) None of these

Direction for questions 42 and 43: Read the passage below and solve the questions based on it.

M and N are two natural numbers such that M + N = 949. LCM of M and N is 2628.

Q.42. What is the HCF of M and N?
(a) 23 (b) 73
(c) 69 (d) None of these

Q.43. When M and N are divided by P, remainder obtained is same. How many values of P are possible?
(a) 0 (b) 1 (c) 2 (d) 3

Q.44. What is the difference between the largest four digit number and the smallest four digit number, both written in hexadecimal system?
(a) FFFF (b) EFFF
(c) EEEE (d) EFEF

Q.45. A ten digit number is said to be interesting if its digits are all distinct and it is a multiple of 11111. How many interesting integers are there?
(a) 1,001 (b) 101
(c) 10,001 (d) None of these

Q.46. Sum of a three-digit number and its mirror image is a multiple of 111. What is this sum (provided 6 is one of the digits of this three-digit number)?
(a) 999 (b) 666 (c) 888 (d) 777

Q.47. Consider the following statements:
(i) If $(2)_N \times (4)_N = (8)_N$, where $(X)_N$ is a number written in a system of writing having N digits, then N can have infinite values.
(ii) If $(4)_N \times (5)_N = (24)_N$, where $(X)_N$ is a number written in a system of writing having N digits, then N will have more than 1, but finite values.
(iii) If $(5)_N \times (6)_N = (3A)_N$, where $(X)_N$ is a number written in a system of writing having N digits and A is the unit digit of this number (3A) written in a system of writing the numbers having N digits, then A can have four values.
How many of the above written statements are true?
(a) 1 (b) 2
(c) 3 (d) None of these

Q.48. How many numbers between 1 to 250 can be represented in the form x^y, where $y>x>1$?
(a) 6 (b) 7 (c) 8 (d) 9

Q.49. What will be the ten's place digit in the following expression:
$1! + 2! + 3! + \ldots + 70!$?
(a) 1 (b) 3 (c) 5 (d) 6

Q.50. A two-digit number is K times the sum of its digits. When these numbers are interchanged, it equals N multiplied by the sum of digits. Which of the following is equal to N?
(a) K – 11 (b) 99 – K
(c) 11 – K (d) K – 99

Q.51. What is the remainder when ${}_{32}33^{34}$ is divided by 7?
(a) 4 (b) 2 (c) 3 (d) 1

Q.52. 1's are given 100 times, 2's are given 100 times and 3's are given 100 times. Now numbers are made by arranging these 300 digits in all possible ways. How many of these numbers will be perfect squares?
(a) 10101 (b) 1001001
(c) 100010001 (d) None of these

Q.53. Which of the following number is the greatest number dividing a family of numbers $(N^5 - N)$, where N is any natural number?
(a) 0 (b) 15
(c) 30 (d) 120

Q.54. When Rahul Ghosh distributes Rs 5 to each of his employees, he is left with Re 1. And when he distributes Rs 6 to each of his employees, he is left with Rs 5. What is the sum of digits of the minimum possible sum with Rahul Ghosh if he is having more than Rs 100?
(a) 1 (b) 2 (c) 3 (d) 4

Q.55. The integers 34041 and 32506,when divided by a three digit integer N, leave the same remainder. What can be the value of N?
(a) 289 (b) 307 (c) 317 (d) 319

Q.56. In the above question, how many values of N is/are possible?
(a) 2 (b) 1
(c) 4 (d) None of these

Q.57. What is the remainder when 77777... upto 56 digits is divided by 19?
(a) 1 (b) 7 (c) 9 (d) 13

Q.58. There are 3 consecutives odd integers such that the product is a prime number. Find the largest of the prime number.
(a) 1 (b) 3 (c) 5 (d) 7

Q.59. A square floor of the dimensions 72 cm × 72 cm has to be laid with rectangular tiles whose length and breadth are in the ratio 3 : 2. What is the difference between the maximum number of tiles and minimum numbers of tiles, given that the length and the breath are integers?
(a) 858 (b) 864 (c) 868 (d) 872

Q.60. A man took a 5 digit number ending in 9 and raised it to an even power greater than 50. He then multiplied it with 17 raised to a multiple of 4. What is the last digit of the resulting number?
(a) 1 (b) 9 (c) 3 (d) 7

Q.61. Rahul took an odd number of prime numbers and added them. He found the result to be even. Which of the following could be concluded?
(a) The numbers include an even number.
(b) The numbers do not include an even number.
(c) There are even number of even numbers.
(d) None of the above.

Q.62. When asked as to how many guavas Lovely has, she replies, "if you add three quarters of the number of guavas that I have to three quarters of one, guava, you will get the numbers of guavas I have". How many guavas does she have?
(a) 2 (b) 5
(c) 7 (d) None of these

Direction for questions 63 to 65: Read the passage below and solve the questions based on it.

Rohan purchased some pens, pencils and erasers for his young brothers and sisters for the ensuing examinations. He had to buy at least 11 pieces of each items in a manner that the number of pens purchased be more than the number of pencils, which is more than the number of erasers. He purchased a total of 38 pieces.

Q.63. How many erasers did Rohan purchase?
(a) 11 (b) 10
(c) 8 (d) Cannot be determined

Q.64. If each eraser costs Rs 3, each pencil Rs 2 each pen Rs 10, what is the maximum amount that Rohan could have spent?
(a) 207 (b) 255 (c) 288 (d) 300

Q.65. If the number of pencils cannot be equally divided among his 4 brothers and sisters, how many pens did he purchase?
(a) 11 (b) 12 (c) 13 (d) 14

Q.66. What is the remainder when [100! + 1] is divided by 101?
(a) 100 (b) 1 (c) 0 (d) 2

Q.67. A is a set of first ten natural numbers. B is a subset of A such that B consists of exactly 2 numbers, both co-primes to each other. Find the maximum number of such possible subsets.
(a) 29 (b) 28 (c) 33 (d) 31

Q.68. When written in the lowest term, rational numbers A and B, have denominators 60 and 70 respectively. Now A and B have been added. What is the smallest possible denominator of (A + B)?
(a) 72 (b) 84 (c) 105 (d) 148

Direction for questions 69 and 70: Read the passage below and solve the questions based on it.

The multiplication of two numbers is shown below.

$$\begin{array}{r} AD4 \\ \times\ \ xE \\ \hline A206 \\ \hline \end{array}$$

where A, D, and E are all distinct digits.

Q.69. The value of D is
(a) 3 (b) 9 (c) 2 (d) 7

Q.70. The value of A + E is
(a) 6 (b) 9 (c) 11 (d) 10

Q.71. P, Q, R, S and T are five prime numbers, where P < Q < R < S < T. It is also given that P + Q + R + S + T = 482. What is the value of P^5?
(a) 243 (b) 32
(c) 16807 (d) More than one value

Q.72. If n is a natural number greater than 1, then $1075^n - 1075^{n-1}$ is not divisible by
(a) 15 (b) 23 (c) 43 (d) 179

Direction for questions 73 and 74: Read the passage below and solve the questions based on it.

(a, b) is an ordered pair such that a and b are two-digit numbers a is the number obtained by interchanging the digits of b.

Q.73. How many ordered pairs are there such that $(a+b)$ is a perfect square?
(a) 4 (b) 8
(c) 3 (d) None of these

Q.74. How many ordered pairs are there such that $(a-b)$ is a perfect square greater than zero?
(a) 3 (b) 5
(c) 13 (d) None of these

Q.75. N is an even number 8 when written in the decimal system. If S is the sum of the digits of N, when it is written in base 7, we can surely say that S would always be
(a) Even (b) Odd
(c) Composite (d) Both (a) and (c)

ADVANCED

Q.1. Find the last non zero digit of 96!.
(a) 2 (b) 4 (c) 6 (d) 8

Q.2. What will be remainder when 1212121212...300 times, is being divided by 99?
(a) 18 (b) 81 (c) 54 (d) 36

Q.3. Two players A and B are playing a game of putting '+' and '−' signs in between any two integers written from 1 to 100. A starts the game by putting a plus sign anywhere between any two integers. Once all the signs have been put, the result is calculated. If it is even then A wins and if it is odd then B wins, provided they are putting signs by taking turns one by one and either of them can put any sign anywhere between any two integers. Who will win at the end?
(a) A (b) B
(c) Either A or B (d) Cannot be determined

Q.4. What is the remainder when 2^{100} is divided by 101?
(a) 1 (b) 100
(c) 99 (d) None of these

Q.5. If $S = \{X_1, X_2, X_3 \ldots X_n\}$, where $n > 0$, $X_1 = 1$ is a set of positive numbers where $X_n = X_{n-1} + 2n - 1$. Which of the following sets of numbers does S resemble?
(a) Fibonacci (b) Cube
(c) Square (d) Prime

Q.6. A three-digit number ABC is a perfect square and the number of factors of this number is also a perfect square. If (A + B + C) is also a perfect square, then what is the number of factors of the 6-digit number ABCABC?
(a) 32 (b) 52
(c) 72 (d) Cannot be determined

Q.7. How many divisors of 10^5 will have atleast one zero at its end?
(a) 9 (b) 12 (c) 15 (d) 25

Q.8. Let $V_1, V_2, V_3, \ldots V_{100}$ be hundred positive integers such that $V_i + V_{i+1} + V_{i+2} + V_{i+3} = K$, where K is a constant and i = 1, 2, 3, ... 97. If $V_3 = 9$, then what is the value of V_{99}?
(a) 9 (b) K – 9
(c) (K/2 – 9) (d) Cannot be determined

Q.9. In the above question, if $V_5 = 7$, then what is the value of V_{90}?
(a) 7 (b) K – 7
(c) (K – 7)/2 (d) Cannot be determined

Q.10. *abcd* is a four-digit number in base 7 such that 2 (*abcd*) = *bcda*. (*a*, $b \neq 0$) Find the value of *a*.
(a) 1 (b) 2
(c) 3 (d) Cannot be determined

Q.11. If K is any natural number, such that $100 \leq K \leq 200$, how many values of K exist such that K! has 'z' zeroes at its end and (K + 2)! has 'z + 2' zeroes at its end?
(a) 2 (b) 4
(c) 6 (d) None of these

Q.12. Tatto bought a notebook containing 96 pages leaves and numbered them which came to 192 pages. Tappo tore out the latter 25 leaves of the notebook and added the 50 numbers she found on those pages. Which of the following is not true?
(a) She could have found the sum of pages as 1990
(b) She could have found sum of pages as 1275
(c) She could have got sum of pages as 1375
(d) None of these

Direction for questions 13 to 15: Read the passage below and solve the questions based on it.

There are 50 integers a1, a2, a3, ... ,a50; not all of them necessarily different. Let the greatest integer of these 50 integers be referred to as G and smallest integer be referred to as L. The integers a1 to a24 form a sequence S1 and the rest form a sequence S2. Each member of S1 is less than or equal to each member of S2.

Q.13. All values in S1 are changed in sign, while those in S2 remain unchanged. Which of the following statements is true?
(a) Every member of S1 is greater than or equal to every member of S2
(b) G is in S1
(c) If all the numbers originally in S1 and S2 had the same sign, then after the change of sign, the largest number of S1 and S2 is in S1
(d) None of these

Q.14. Elements of S1 are in ascending order and those of S2 are in descending order. a24 and a25 are interchanged then which of the following is true?
(a) S1 continues to be in ascending order
(b) S2 continues to be in descending order
(c) Both (a) and (b)
(d) Cannot be determined

Q.15. Every element of S1 is made greater than or equal to every element of S2 by adding to each element of S1 an integer *x*. Then, *x* cannot be less than
(a) 2^{10}
(b) The smallest value of S2
(c) The largest value of S2
(d) (G-L)

Q.16. Twenty-five boxes of sweets are delivered to Mr Roy's home. Mr Roy had ordered sweets of three different types. What is the minimum number of boxes of sweets which are having sweets of same type?
(a) 1 (b) 8
(c) 9 (d) Cannot be determined

Q.17. A warehouse contains 200 shoes of size 8, 200 shoes of size 9 and 200 shoes of size 10. Of these 600 shoes, there are 300 left shoes and 300 right shoes. What is the minimum number of usable shoes?
(a) 50 (b) 100
(c) 200 (d) None of these

Q.18. A teacher was doing some calculation exercise on the blackboard. When the teacher went out, a naughty student Chunmun erased some of the numbers written on the blackboard. Now it appeared like this

$$\begin{array}{r} 23_5_ \\ +1_642 \\ \hline 42423 \end{array}$$

When teacher entered the room, he realized that still this calculation was right, but in some other system of writing (i.e., not 10). How many digits are there in that system?
(a) 11 (b) 9 (c) 7 (d) 8

Q.19. Tatto, Tappo and Bubbly were solving problems from a problem book. Each solved exactly 60 problems, but they solved only 100 problems altogether. Any problem is known as "easy" if it was solved by all of them, and "difficult" if it was solved by only one of them. What is the difference between the number of "difficult" problems and number of "easy" problems?
(a) 10 (b) 20
(c) 30 (d) 40

Q.20. LCM of two numbers A and B = $P^x \times Q^y$, where P and Q are prime numbers and x and y are positive whole numbers. How many set of values are possible for A and B?
(a) $xy(x+y)$ (b) $xy(x-y)$
(c) $x^2y^2(x+y)$ (d) None of these

Q.21. When 7179 and 9699 are divided by another natural number N, remainder obtained is same. How many values of N will be ending with one or more than one zeroes?
(a) 24 (b) 124
(c) 46 (d) None of these

Q.22. There exists a 5 digit number N with distinct and non-zero digits such that it equals the sum of all distinct three digit numbers whose digits are all different and are all digits of N. Then the sum of the digits of N is a necessarily.
(a) Perfect square (b) Cube
(c) Even (d) None of these

Q.23. Starting with 1, positive integers are written one after the other. What is the 40,0000th digit that will be written?
(a) 3 (b) 6
(c) 8 (d) None of these

Q.24. Which of the following would always divide a six-digit number of the form ababab?
(a) 10,101 (b) 11,111
(c) 10,001 (d) None of these

Q.25. If in the number system of a particular country, 25 means 5 tens and 2 units, 467 means 7 hundreds, 6 tens and 4 units. Then find the value of 173 × 425?
(a) 4,04,491 (b) 7,35,255
(c) 6,22,744 (d) 5,25,376

Q.26. Let A be the set of integers N such that
i. $100 \le N \le 500$
ii. N is even
iii. N is divisible by either 2 or 3 or 4 but not by 7.
How many elements are there in set A?
(a) 171 (b) 172 (c) 170 (d) 173

Q.27. Three distinct prime numbers, less than 10 are taken and all the numbers that can be formed by arranging all the digits are taken. Now, difference between the largest and the smallest number formed is equal to 495. It is also given that sum of the digits is more than 13. What is the product of the numbers?
(a) 30 (b) 70 (c) 105 (d) 315

Q.28. What is the remainder when ${}_{55}56^{57}$ is divided by 17?
(a) 1 (b) 4 (c) 13 (d) 17

Q.29. What is the remainder when ${}_{20}51^{97}$ is divided by 17?
(a) 10 (b) 9 (c) 15 (d) 7

Q.30. $[111\ldots111(200\text{ digits}) - 222\ldots22(100\text{ digits})]^{1/2}$ is equal to
(a) 1313 ... 1313(100 digits)
(b) 2121...2121(100 digits)
(c) 1111...1111(100 digits)
(d) 3333...333(100 digits)

Q.31. N is a number which when divided by 10 gives 9 as the remainder,
when divided by 9 gives 8 as the remainder,
when divided by 8 gives 7 as the remainder,
when divided by 7 gives 6 as the remainder,
when divided by 6 gives 5 as the remainder,
when divided by 5 gives 4 as the remainder,
when divided by 4 gives 3 as the remainder,
when divided by 3 gives 2 as the remainder,
when divided by 2 gives 1 as the remainder.
What is N?
(a) 2519 (b) 841 (c) 839 (d) 2521

Q.32. What is the remainder when $(10^3 + 9^3)^{752}$ is divided by 12^3?
(a) 1 (b) 729 (c) 752 (d) 1000

Q.33. $f(x, x) = x^x$ and $f(f(x, x) = {}_x x^x$ and so on its keeps on going. What is the value of $f(f(f(f(7,7))))$ is divided by 5?
(a) 1 (b) 2 (c) 3 (d) 4

Q.34. The students of class 10th of Morgan High School took a test which had a maximum of 50 marks. The teacher misplaced the text notebooks of two of the students— Robin and Garry, but remembered that Garry had scored something between 10 and 15 and Robin something between 32 to 40. She also remembered that the product of the marks obtained by the two students was also equal to 10 times the marks obtained by two of them. How many marks did Garry scored?
(a) 11 (b) 12 (c) 13 (d) 14

Q.35. The History teacher was referring to a year in the 19th century. Rohan found an easy way to remember the year. He found that the number, when viewed in a mirror, increased 4.5 times. Which year was the teacher referring to?
(a) 1,801 (b) 1,810
(c) 1,818 (d) More than one value

Q.36. Srini wrote his class 10th board examination this year. When the result came out he searched for his hall ticket to see his roll number but could not trace it. He could remember only the first three digits of the 6 digit number as 267. His father, however, remembered that the number was divisible by 11. His mother gave the information that the number was also divisible by 13. They tried to recollect the number when all of a sudden Srini told that the number was a multiple of 7. What was the unit digits of the number?
(a) 5 (b) 7
(c) 2 (d) Cannot be determined

Q.37. Prof. Mathur and Prof. Singh attended the All India Historian's meet last week. Prof. Mathur told Prof. Singh, "I found out that your teaching experience is twice that of mine". Prof. Singh replied in the affirmative. Prof. Mathur continued, "But last time when both of us came for the same meet, I remember that your teaching experiencr was thrice that of mine". "That was 2 years ago." Prof. Singh said. How many years has Prof. Singh been working?
(a) 8 yrs (b) 10 yrs (c) 12 yrs (d) 16 yrs

Direction for questions 38 and 39: Read the passage below and solve the questions based on it.

ABCDEF is a 6-digit number with distinct digits. Further, the number is divisible by 11 and the sum of its digits is 24. Further A > C > E and B > D > F.

Q.38. The sum A + C + E is equal to
(a) 12 (b) 6
(c) 8 (d) Cannot be determined

Q.39. A + B is always
(a) 10 (b) 9
(c) 6 (d) Cannot be determined

Q.40. Raju had to divide 1080 by N, a two-digit number. Instead, he performed the division using M which is obtained by reversing the digits of N and ended up with a quotient which was 25 less than what he should have obtained otherwise. If 1080 is exactly divisible both by N and M, find the sum of the digits of N.
(a) 6 (b) 8
(c) 9 (d) None of these

Q.41. Let S = {1, 2, 3, ... n} be a set of N natural numbers. Let T be a subset of S such that the sum of any three elements of T is not less than N. Find the maximum number of elements in any such subset T for N = 40?
(a) 26 (b) 27
(c) 28 (d) None of these

Q.42. The last digit of the LCM of $(3^{2003} - 1)$ and $(3^{2003} + 1)$ is
(a) 8 (b) 2 (c) 4 (d) 6

Q.43. a, b and c are positive integers such that, $a + b + c = 2003$. Let $E = (-1)^a + (-1)^b + (-1)^c$. Find the number of possible values of E.
(a) 2004 (b) 3 (c) 1003 (d) 2

Q.44. Ajay took a 4-digit number in base 5 notation. He subtracted the sum of the digits of the numbers from the number. From the result, he struck off one of the digits. The remaining 3 digits were 1, 0 and 2. Then the digit struck off by Ajay was:
(a) 2 (b) 1
(c) 4 (d) Cannot be determined

Direction for questions 45 and 46: Read the passage below and solve the questions based on it.

N is a single digit integer satisfying the following two conditions.
i. N is non-zero.
ii. N is the right most digit of the number $(n!)^4$, where n is a natural number greater than 1.

Q.45. What is the number of possible values of N?
(a) 1 (b) 2
(c) 0 (d) None of these

Q.46. If condition (a) is relaxed, the number of possible values of N is.
(a) 1 (b) 2
(c) 0 (d) More than 2

Q.47. A teacher wrote a number on the blackboard and the following observations were made by the students
The number is a four-digit number.
The sum of the digits equals the product of the digits.
The number is divisible by the sum of the digits.
The sum of the digits of the number is
(a) 8 (b) 10 (c) 12 (d) 14

Q.48. The N of odd numbers are taken. Product of these odd numbers is of the form $(4n + 1)$, where n is any natural number Which of the following is true regarding the number of numbers?
(a) There must have been an odd number of numbers of the form $4n + 1$
(b) There must have been an even number of numbers of the form $4n + 1$
(c) There must have been an even number of numbers of the form $4n + 3$
(d) None of these

Q.49. 16 students were writing a test in a class. Rahul made 14 mistakes in the paper, which was the highest number of mistakes made by any student. Which of the following statements is definitely true?
(a) At least two students made the same number of mistakes
(b) Exactly two students made the same number of mistakes
(c) At most two students made the same number of mistakes
(d) All students made different number of mistakes.

Q.50. The sum of the factorials of the three-digits of a 3-digit number is equal to the three-digit number formed by these three digits, taken in the same order. Which of the following is true of the number of such three-digit numbers, if no digit occurs more than once?
(a) No such number exists
(b) Exactly one such number exists
(c) There is more than one such number, but they are finite in number
(d) There are infinite such numbers

Direction for questions 51 and 52: Read the passage below and solve the questions based on it.

The inhabitants of planet Romeo, a remote planet in the milky way, use a number system which is similar to the decimal system used here on the earth, except that it has eleven distinct digits instead of the usual ten digits. The extra digit is an alien digit called x, which is inserted between the digits 7 and 8. Now $8 - 7 = 2$ and x is the digit, which is equidistance from 7 and 8, i.e., $x - 7 = 8 - x = 1$. All the algebric signs and operations carry the same meaning as in the usual sense.

Q.51. Romeo, an inhabitant of planet Romeo, evaluated an expression and arrived at an answer of '$6x$'. Then which of the following can be the expression that Romeo evaluated?
(a) $x^2 + x - 1$ (b) $x^2 + x + 1$
(c) $x^2 + x + 2$ (d) None of these

Q.52. What is the decimal equivalent of the two digit number '$9x$'?
(a) 98 (b) 107
(c) 118 (d) None of these

Q.53. If P is the product of four consecutive positive integers, then which of the following statements is not true?
(a) P is perfectly divisible by 24
(b) P is not a perfect square
(c) (P + 1) is a perfect square
(d) (P – 2) is a perfect square

Q.54. Let S be a two-digit number such that both S and S^2 end with the same digit and none of the digits in S equals zero. When the digits of S are written in the reverse order, the square of the new number so obtained has the last digit as 6 and is less than 3000. How many values of S are possible?
(a) 3 (b) 4 (c) 5 (d) 6

Q.55. How many different four digit numbers are there in the octal (Base 8) system, expressed in that system?
(a) 3584 (b) 2058 (c) 6000 (d) 7000

Q.56. How many numbers less than or equal to 500 are there, each of which is the product of more than three distinct prime numbers?
(a) 3 (b) 4 (c) 5 (d) 6

Q.57. How many numbers below 100 can be expressed as a difference of two perfect squares in only one way?
(a) 15 (b) 25 (c) 35 (d) 45

Q.58. The LCM of the two quotients, obtained when two numbers a and b, are divided by their HCF is 77. LCM of another pair of quotients, when other two numbers c and d have the same HCF as that of a and b, and when divided by it, is 221. Which of the following could be, ratio of the sum of a and b to that of the difference between c and d?
(a) 4 : 17 (b) 11:4 (c) 2:9 (d) 39:2

Q.59. Let p, q and s be integers such that $p^2 = sq^2$. Then it follows that
(a) p is an even number
(b) if s divides p, then s is a perfect square
(c) s divides p
(d) q^2 divides p

Q.60. Let N be a positive integer not equal to 1. Then none of the numbers 2, 3,…., N is a divisor of $(N! - 1)$. Thus, we can conclude that
(a) $(N! - 1)$ is a prime number
(b) At least one of the numbers $N + 1, N + 2, \ldots, N! - 2$ is a divisor of $(N! - 1)$
(c) The smallest number between N and N! which is a divisor of $(N! + 1)$, is a prime number
(d) None of the foregoing statement is necessarily correct

TRUE/FALSE

1. A natural number M is divided by another natural number N and the remainder obtained = R ≠ 0 and M > N. It is always possible to find out the exact remainder when M is divided by P (where P is a factor of N).
State whether True or False.

2. A natural number M is divided by another natural number N and the remainder obtained = R "" 0 and M > N. It is always possible to find out the exact remainder when M is divided by P (where P is a multiple of N).
State whether True or False.

3. A, B, C are three natural numbers. If we know the LCM of A, B and LCM of B, C separately, then it is always possible to find out the LCM of A, B, C.
State whether True or False.

4. A, B, C are three natural numbers. If we know the HCF of A, B and HCF of B, C separately, then it is always possible to find out the HCF of A, B, C.
State whether True or False.

5. A, B, C, D are four natural numbers. If we know the LCM of A, B and LCM of C, D separately, then it is always possible to find out the LCM of A, B, C, D.

   State whether True or False.

6. A, B, C, D are four natural numbers. If we know the HCF of A, B and HCF of C, D separately, then it is always possible to find out the HCF of A, B, C, D.

   State whether True or False.

7. If we know the total number of odd factors of a number, then we can always find out the total number of factors of that number.

   State whether True or False.

8. If we know the total number of even factors of a number, then we can always find out the total number of factors of that number.

   State whether True or False.

9. If a number is odd, then it cannot have total number of factors as an even number.

   State whether True or False.

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(d)	(d)	(d)	(d)	(d)	(b)	(a)	(c)	(b)	(a)	(a)	(d)	(c)	(d)	(d)	(b)	(a)	(b)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34						
Answer	(b)	(c)	(d)	(b)	(c)	(d)	(d)	(b)	(c)	(a)	(b)	(c)	(d)	(c)						

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(a)	(a)	(a)	(a)	(c)	(b)	(b)	(c)	(b)	(d)	(c)	(c)	(c)	(a)	(b)	(d)	(a)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(d)	(a)	(c)	(b)	(d)	(c)	(c)	(d)	(a)	(b)	(b)	(a)	(a)	(c)	(c)	(a)	(b)	(a)	(c)
Q. No.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	(a)	(a)	(c)	(b)	(a)	(b)	(c)	(a)	(b)	(c)	(d)	(c)	(a)	(b)	(a)	(c)	(c)	(d)	(b)	(a)
Q. No.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
Answer	(c)	(c)	(d)	(d)	(b)	(a)	(b)	(d)	(b)	(b)	(d)	(d)	(a)	(b)	(c)	(b)	(c)	(c)	(d)	(a)

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(c)	(b)	(c)	(b)	(a)	(c)	(c)	(b)	(c)	(a)	(b)	(c)	(c)	(b)	(b)	(a)	(c)	(c)	(a)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(a)	(b)	(c)	(d)	(a)	(d)	(a)	(c)	(c)	(b)	(b)	(b)	(c)	(b)	(b)	(a)	(a)	(c)	(b)	(c)
Q. No.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	(d)	(b)	(c)	(b)	(d)	(c)	(a)	(b)	(a)	(c)	(d)	(d)	(c)	(b)	(b)	(c)	(a)	(b)	(a)	(a)
Q. No.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75					
Answer	(a)	(d)	(a)	(a)	(c)	(c)	(d)	(b)	(a)	(d)	(b)	(b)	(b)	(b)	(d)					

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(a)	(a)	(a)	(c)	(d)	(d)	(a)	(d)	(d)	(c)	(a)	(d)	(a)	(d)	(c)	(b)	(c)	(b)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(c)	(a)	(d)	(a)	(a)	(b)	(b)	(a)	(a)	(d)	(a)	(a)	(c)	(d)	(c)	(b)	(a)	(a)	(c)	(c)
Q. No.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	(d)	(c)	(d)	(b)	(a)	(b)	(a)	(c)	(a)	(b)	(c)	(c)	(d)	(a)	(d)	(b)	(c)	(d)	(b)	(d)

■■■ True/False

1. True

   To find out the remainder when M divided by P, we simply need to divide R by P.

2. False

   If we divide M by P, we would get a range of remainders in terms of R and not the exact value of remainder in all the cases.

3. True

   LCM, by its meaning, is the lowest number divisible by all the numbers constituting it. Final LCM will be the LCM of the pairs of numbers.

4. True

   HCF, by its meaning, is the highest number that can divide the numbers constituting it. Final HCF will be the HCF of the pairs of numbers.

5. True

   LCM, by its meaning, is the lowest number divisible by all the numbers constituting it. Final LCM will be the LCM of the pairs of numbers.

6. True

   HCF, by its meaning, is the highest number that can divide the numbers constituting it. Final HCF will be the HCF of the pairs of numbers.

7. False

8. False

9. False

   Total number of factors do not have any relationship with the number being odd or even. For example, all the perfect squares (irrespective of being odd or even) have total number of factors = Odd number.

HINTS AND SOLUTIONS

Moderate

1. In case of decimal system, we obtain 10 by multiplying 5 and 2, and then to find the number of zeroes, we search the exponents of 5. In case of base 6, 10 will be obtained by multiplying 3 and 2. So, here we will check for the exponents of 3 to know about the number of zeroes. And obviously it is 5 [12/3 + 12/9].

2. The remainder obtained when $2222^{5555} + 5555^{2222}$ is divided by 7 will be the same as the remainder when $3^{5555} + 4^{2222}$ is divided by 7. Now find the individual remainder and solve it.

3. If we look at the numbers $100 < N \leq 105$, we see only 101 and 103 do not have their factors in N (because these are primes). So, obviously the new LCM will be $101 \times 103 \times N$.

4. $10^5 = 2^5 \times 5^5$

   Now all the factors of 10^5 which will end in one zero will be zero power of 2 and 1-5 powers of 5 and vice-versa. This will be equal to 9.

5. $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$

   So, by replacing the signs we need to make 45 extra. This is possible only if we write in this way:

   $1 \times 2 + 3 \times 4 + 5 + 6 + 7 \times 8 + 9 + 10 = 55 + 45 = 100$

6. The key is the fact that in this country only three symbols are used to write numbers.

   So, $6 = (20)_3 = (bc)_3$

   So, $b = 2$, c = 0 and $a = 1$

   $17 = (122)_3 = abb$

8. Remainder of $({}_{32}32^{32}$ divided by 7) = Remainder of $({}_{4}32^{32}$ divided by 7)

   Now find cyclicity of remainder of $({}_{4}32^{n}$ divided by 7).

   Remainder when ${}_{4}32^{1}$ divided by 7 = 2

   Remainder when ${}_{4}32^{2}$ divided by 7 = 4

   Remainder when ${}_{4}32^{3}$ divided by 7 = 2

   So, the cyclicity is 2, 4, 2, 4 and so on.

   For every even value of n, remainder = 4

   So, answer = (d)

9. $1001 = 7 \times 11 \times 13$

   We know that any digit written 6 times consecutively (like 111111, or 222222 etc.) will be divisible by 3, 7, 11, 13, 37. So, this question is– what is the remainder when 11111 is divided by 1001. Find it out by actual division method.

10. Go through options.

12. Use the formula given in the concepts.

14. Digits which can create confusion = 1, 6, 8, 9 (0 cannot create confusion because passwords has to be two-digit numbers).

    Total two digit numbers with distinct digit = 81

    Two digit numbers created by 1, 6, 8, 9 = 12

    So, total numbers left = 69

    But 69 and 96 wont create confusion (it looks same upside down), so total numbers = 71

16. See the solution of CAT 04 given at the end of this book.

18. N! is having 37 zeroes at its end, so N = 150 (can be arrived at by a guess).

    Obviously, $150 \leq N < 155$ is the answer.

19. Use the data of the Q.18.

23. The question is: In which system of writing, 24 + 32 = 100. Go through options.

25. 2P is having 28 (7×4) divisors but 3P is not having a total divisor which is divisible by 7. So, the first part of the number P will be 2^5.

    Similarly, 3P is having 30 (3×10) divisors but 2P does not have a total divisor which is divisible by 3. So, 2nd part of the number P will be 3^3.

    So, $P = 2^5 \times 3^3$.

26. *pqr* can be 370 or 371. So, it is not possible to arrive at a unique answer.

30. Mayank DOB = 1948 and Siddharth DOB = 1898

Questions 31 to 35

All the numbers which will have odd number of factors will be open at the end. Now all the perfect squares only have the total number of divisors = Odd factors

Now solve the questions.

36. Convert all the numbers to base 10 and then find the value.

37. $(124)_{\Phi}$ shows that value of Φ should be 5 or more than 5 (since 4 is a digit used in this system).

    $(12\Phi)_4$ shows that value of Φ can be at max 3 (since the system = 4).

    Now this is a contradiction. So, no such value exists.

    $N = A^2 \times B^3 \times C^4$, where A, B and C are prime numbers.

39. For the factors to be perfect squares, assume $A^2 = X$, $B^2 = Y$ and $C^2 = Z$

So, for the factors to be perfect square, we can write $N = XYZ^2$
So, total number of factors = (1 + 1) (1 + 1) (2 + 1) = 12

41. All the squares of prime numbers give 3 factors. Use this to solve the question.

44. Largest number = FFFF and smallest number = 1000

52. The digital sum of any perfect square can be 1, 4, 9, 7, 0 only. 100 times 1's are given, 100 times 2's are given and 100 times 3's are given. Digital sum of all these digits taken all at a time will be 6. So, this will not give any perfect square.

55. The difference will be = 34041 – 32506 = 1535

56. Factorize $1535 = 307^1 \times 5^1$

[redacted]

63. Number of pen > number of pencils > number of erasers
Minimum number of each of the items = 11. So, number of erasers = 11 [If we take it 12, then minimum value of pencils = 13, and minimum value of pens = 14, sum of these three exceeds 38].

64. Number of pens = 15, number of pencils = 12, and number of erasers = 11.

65. Different possibilities for the number of pencils = 12 or 13. Since it cannot be divided into his 4 brothers and sisters, it has to be 13. In that case, number of pens = 14

66. Use Wilson's Theorem.

71. P + Q + R + S + T = 482. Sum of five prime numbers is even possible only if four of these are odd and one is even (all these cannot be odd). So, P = 2.

75. Assume some values of N and check it out.

Advanced

1. Factorize 96! into prime factors. Find the unit digit of all the factors individually and multiply to get the unit digit of 96!.
$96! = 2^{92} \times 3^{46} \times 5^{22} \times \ldots$
Now 5^{22} and 2^{22} can be eliminated, since these will result in zeroes. Find the unit digit of all the remaining.

2. This number 1212121212 ... 300 times is divisible by 9. So, we can write 1212121212...300 times = 9 N, where N is the quotient obtained when divided by 9. Now this question is like –

$$\frac{1212121212...300\text{ times}}{99}$$

$$= \text{Remainder}\ \frac{9}{9} \times \frac{134680...\text{written 50 times}}{11}$$

[121212 = 13468 × 9]
Now we will have to find the reminder obtained when 134680134680 ... 50 times is divided by 11.
For this, we are supposed to use the divisibility rule of 11 from right hand side.
[Using the divisibility rule from left hand side might give us the wrong remainder, like if we find out the remainder obtained when 12 is divided by 11, remainder = 1 = (2 –1) ≠ (1 – 2)]

$$\text{Remainder}\ \frac{134680...\text{written 50 times}}{11} = 2$$

So, the total remainder = 18
Alternatively, divisibility rule of $10^n - 1$, $n = 2$ can be used to find the remainder in this case.

3. Whatever is the sign between two consecutive integers starting from 1 to 100, it will be odd. So, we are getting 50 sets of odd numbers. Now, whatever calculation we do among 50 odd numbers, result will always be even. So, A will win always.

4. Using Fermat's Theorem, the remainder will be 1.

6. More than one value is possible, so we cannot determine any unique number.

7. Total number of factors of $10^5 = 36$

Total number of factors of 10^5 ending with no zero = 11

Hence, the total number of factors having at least one zero at its end = 25

8. It can be seen that $V_1 = V_5 = V_9 = \ldots$ and so on
$V_2 = V_6 = V_{10}$ and so on.
So, $V_3 = V_7 = \ldots = V_{99} = 9$

9. $V_5 = V_9 = \ldots = V_{89}$, so we cannot find the value of V_{90}.

10. 2 values of a are possible, so cannot be determined.

12. When we are adding the sum of page numbers of 25 pages, it will always be an odd number. So, she could not have found the sum of pages as 1990.

[redacted]

13. Assume S1 = {1, 2, 3, ..., 24} and S2 = {25, 26, 27, 28, ..., 50} and now check the options. None of the first three options are correct.

14. Assume S1 = {1, 2, 3, ... , 24} and S2 = { 50, 49, 48, 47, ... , 25}. Now even if we have interchanged 24

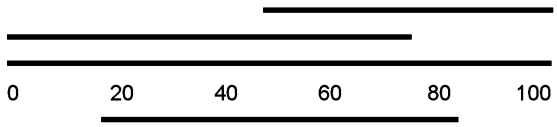
and 25, S1 continues to be in ascending order and S2 continues to be in descending order. However, by choosing negative values of a24 and a25, we can show that S1 continues to be in ascending order, but S2 is no longer in descending order.

15. Do it by the method of elimination.

16. This is one classic example of pigeon-hole principle. Since Mr Roy has ordered for 25 boxes and three different types of sweets, so minimum 9 boxes of sweets will have the same type of sweets.

18.
$$\begin{array}{cccccc} & 2 & 3 & _ & 5 & _ \\ + & 1 & _ & 6 & 4 & 2 \\ \hline & 4 & 2 & 4 & 2 & 3 \end{array}$$
The key here is the numbers written in bold letters. When 5 and 4 are added in decimal system, the value is equal to 9. But its unit digit here is represented by 2. Going through the options, only option (c) is satisfying this equation.

19.



Easy problems = Q.40 to Q.60 [not both including] = 20
Difficult problem = Q.1 to Q.20 + Q.80 to Q.100 [not both including] = 40
So, the answer = 40 – 20 = 20

21. 9699 – 7179 = 2520 and all the factors of this number will be the value of N.

24. $10101 = 3 \times 7 \times 13 \times 37$.
ababab = $10101 \times (10a + b)$

27. Prime numbers less than 10 = 2, 3, 5, 7
So, option (d) cannot be the answer.
Sum of digits is more than 13 – so, set of [2, 3, 5] is not possible.
So, option (a) cannot be the answer.
Now check for options (b) and (c) by taking the values.

30. Go through options.
Checking option (a) 1313 ... 1313 (100 digits). So for 1 digit, it should be equal to 1 only.
Now, taking 1st two digits of [111 ... 111 (200 digits)] and 1 digit from [222 ... 22 (100 digits)] $\rightarrow [11-2]^{1/2} = 3$, so option (a) is not correct.
Similarly, we will check all the options.

31. Check it through the options.
Alternatively, answer will be LCM (2,3,4,..., 9) – 1 = 2520 – 1 = 2519

32. There is no rocket science involved in it. Simply write $(10^3 + 9^3)^{752} = (1729)^{752}$ and now find the remainder from the given method of calculations.

35. Going through the options, 8181/1818 = 4.5

36. His roll number is divisible by 1001 and the first three digits are 267. Hence the last three digits will also be 267.

37. Go through the options.

[redacted]

The number is 125367.

40. $1080 = 2^3 \times 3^2 \times 5^1$.
Now, N × Q = 1080 and M × (Q – 25) = 1080
Do a bit of hit and trial.
N = 27

42. The given numbers are two consecutive even numbers, so their HCF = 2
Now, using LCM × HCF = Product of two numbers
LCM × 2 = (... 6) × (... 8)
IT can be seen now that the unit digit of LCM = 4

43. $a + b + c = 2003$
It is possible only if all the three a, b and c are odd, or two of the three are even and one is odd.
Hence, E can have only two values.

[redacted]

45. In this question, we have to find out the possible unit digits of $(n!)^4$. This can be either 1 or 5 or 6.

47. Using options, the only possible value is 4112. The key here is:
The sum of the digits equals the product of the digits.

49. The number of mistakes made by all the students will be between 0 and 14, i.e., students are having a total of 15 options to make mistakes. Since the number of students = 16, at least two students will have the same number of mistakes (that can be zero also, i.e., two students are making no mistakes). Hence, option 1 is the answer.

50. There is only one number, 145, which exhibits this property.

52. Actually the value of 9 here would become 10 and x represents 8.
So, $(9x)_{11} = 10 \times 11^1 + 8 \times 11^0 = 118$

54. In this question, several restrictions are operating:
If S and S^2 are ending with the same unit digit, then it can be 0, 1, 5, 6, but it is given that none of the digits

is equal to zero, so the unit digit can be only 1, 5, 6.
Next, unit digit of the square of the number written in reverse order is 6, so tens place digit of the actual number should be either 4 or 6.
So, the actual numbers could be 41, 45, 46, 61, 65, 66.
Now, this square is less than 3000, so the only possibilities are 41, 45, 46.

55. The total number of numbers of four digits in octal system = $7 \times 8 \times 8 \times 8 = 3584$
When we convert this number into octal system, this is equal to 7000.

60. Eliminate the options.
For example, option (1) can be eliminated by assuming N = 2

BENCHMARKING TEST 1

This test paper contains 20 questions of 4 marks each. One-third of the mark allotted to a particular question will be deducted in case of wrong answer.

Time Given: 45 Minutes **Total Marks: 80** **Qualifying Marks: 24**

Break-up of the Expected Percentile
(assuming this to be a CAT paper)

Marks	*Expected Percentile*
42+	99+
36	98.5+
30	98
22	92
16	85

Q.1. The remainder when 25! is divided by 10^7 is
(a) 2 (b) 4×10^6
(c) 6×10^6 (d) 2×10^6

Q.2. A box contains 90 balls of different colours: 13 yellow, 19 green, 27 red, 10 black, 7 brown and 14 white. The smallest number N such that any N ball drawn from the box will contain at least 14 balls of the same colour is
(a) 79 (b) 69
(c) 70 (d) 72

Q.3. At a bookstore, 'Modern Book Store', is flashed using neon lights. The words are individually flashed at the intervals of $2\frac{1}{2}$ s, $4\frac{1}{4}$ s and $5\frac{1}{8}$ s, respectively, and each word is put off after a second. The least time after which the full name of the bookstore can be read again is
(a) 49.5 s (b) 73.5 s
(c) 1744.5 s (d) 85.5 s

Q.4. A number has exactly 32 factors out of which 4 are not composite. Product of these 4 factors is 30. How many such numbers are possible?
(a) 2 (b) 6
(c) 3 (d) Not possible

Q.5. 1st 126 natural numbers are put side by side in the ascending order to form a large number: 123456… 125126. What will be the remainder when this large number is divided by 5625?
(a) 5126 (b) 126
(c) 26 (c) None of these

Q.6. P and Q are two distinct whole numbers and P + 1, P + 2, P + 3, … , P + 7 are integral multiples of Q + 1, Q + 2, Q + 3,…, Q + 7 respectively. What is the minimum value of P?
(a) 0 (b) 240
(c) 1080 (d) None of these

Q.7. A child was asked to add first few natural numbers (that is 1 + 2 + 3 +…) so long his patience permitted. As he stopped, he gave the sum as 575. When the teacher declared the result wrong, the child discovered that he had missed one number in the sequence during addition. The number he missed was
(a) Less than 10 (b) 10
(c) 15 (d) More than 15

Q.8. $1000^7 - 17$ is written in decimal notation. What is the sum of its digits?
(a) 181 (b) 182
(c) 183 (d) None of these

Q.9. N is an integer. For how many distinct values of N, $N^4 - 20N^2 + 4$ is a prime number?
(a) 1 (b) 3
(c) 3 (d) None of these

Q.10. Of 128 boxes of oranges, each box contains atleast 120 and at most 144 oranges. The number of boxes containing the same number of oranges is at least:
(a) 5 (b) 6
(c) 103 (d) None of these

Q.11. What is the value of $\sum_{N=1}^{99} \frac{1}{\sqrt{N}+\sqrt{N+1}}$?
(a) 9 (b) $\sqrt{99}$
(c) $\sqrt{100}-\sqrt{99}$ (d) None of these

Q.12. a, b, c, d and e are prime numbers, not necessarily consecutive. If $a + b + c + d + e = 350$, and $a < b < c < d < e$, how many values of a^5 is/are possible?
(a) 1 (b) More than 1, but finite
(c) Infinite (d) Nothing can be said

Q.13. The ten-digit number $9793x\,6160y$ is divisible by 11. If $0 < x < y$, find the sum of remainders when the number is divided by $x + y$ and x successively.
(a) 4 (b) 6 (c) 7 (d) 9

Q.14. What is the remainder when 128^{500} is divided by 153?
(a) 63 (b) 67
(c) 123 (d) None of these

Q.15. What is the minimum number of weights which enable us to weigh any integer number of grams of gold from 1 gm to 100 gms on a standard balance with two pans, weights can be put on either pan of balance?
(a) 4 (b) 5
(c) 6 (d) None of these

Q.16. $N = 2^8 \times 3^{10} \times 5^8 \times 7^2$. What is the number of factors of this number N which are factors of 360 but not the factors of 540?
(a) 120 (b) 210
(c) 114 (d) 144

Q.17. What is the remainder when 2^{104} is divided by 101?
(a) 1 (b) 2 (c) 16 (d) 64

Q.18. Given that:
$X_1 = 5$, $X_2 = 25$ and $X_{n+2} = \text{GCD}\,(X_{n+1}, X_n) + X_n$.
What is the LCM of (X_{19} and X_{20})?
(a) 1840 (b) 2560
(c) 2160 (d) 2440

Q.19. $N_1 + N_2 + N_3 + \ldots + N_N = 100$, where N_1, N_2, N_3 ... and N_N are $n(n>1)$ consecutive natural numbers such that $N_1 < N_2 < N_3 < N_4 \ldots < N_N$. If $N_1 > 8$, then how many values of n is/are possible?
(a) 4 (b) 8 (c) 2 (d) Infinite

Q.20. A teacher writes 101 numbers on the blackboard, of which 50 are zeros, and 51 are ones. A student is asked to perform the following operation 100 times on the board: strike out any two numbers. If they are equal, write another zero. If they are unequal, write a one. What are the numbers left on the board?
(a) Single One (b) Single Zero
(c) Two Ones (d) None of these

ANSWERS

■■■ Benchmarking Test 1

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(c)	(b)	(b)	(b)	(d)	(d)	(b)	(d)	(b)	(a)	(a)	(d)	(b)	(b)	(d)	(c)	(b)	(c)	(a)

Module 2

Arithmetic

- Percentage
- Profit Loss and Discount
- Average and its Applications
- Ratio Proportion and Variation
- Time and Work
- Time Speed and Distance

3

CHAPTER

Percentage



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- What is percentage
- Definitions and properties
- Percentage–ratio equivalence
- Usage of percentage as a calculation device
- Methods of solving questions
- Concept of SI and CI

Introduction to the topic

Percentage is of immense importance when it comes to problem solving in quantitative aptitude (QA) and data interpretation (DI). Despite the fact that the number of questions asked from this concept is not very high in the CAT, a sound understanding and good practice of percentage calculations comes handy in almost all the topics of QA and DI. However, percentage is an important concept for all the other B-school and aptitude examinations.

WHAT IS PERCENTAGE?

Before we talk of the utility of the concept 'percentage', we should be clear regarding when should we read (%) as per cent and when as percentage. Whenever (%) symbol is attached to any value, known or unknown, the word which we use for symbol (%) is per cent, e.g., x% is read as x per cent or 20% is read as 20 per cent, not 20 percentage. The word percentage is used whenever (%) is not attached to any value.

This whole nomenclature can be understood with the help of the following example:

The salary of Abhishek is 20% more than the salary of Abhinav. By how much percentage is the salary of Abhinav less than the salary of Abhishek?

Let us first have some standard meaning of percentage:

- Percentage as a common platform
- Percentage as a rate measurer

Percentage as a Common Platform

The concept of percentage can be further understood with the help of the following table, which gives us the marks obtained by different students in their Class 10 exam:

Student in country	*Marks obtained*
America	100
India	25
China	45
France	50

Right now, by just having this piece of information, we cannot decide that students of which country have performed the best and the data required here is the total marks obtained in each country.

Now, suppose if we add the following data to this question, then the whole situation looks like:

Actually, here we are using percentage as a common platform to compare all the given values.

Student in country	*Marks obtained*	*Total marks*	*Marks obtained/ 100 marks*
America	100	1000	10%
India	25	25	100%
China	45	300	15%
France	50	100	50%



To know more about percentage, go to *www.gomath.com/exercises/percentage.php*

Percentage as a Rate Measurer

The concept of percentage is best-suited to find out the rate of change when two or more than two quantities are attached to each other by any common relationship. In this case, we get the respective change which should be brought in the value of quantities, when the value of other quantities are changing. We will see the application of percentage as a rate measurer while doing product stability ratio.

Before we move ahead, it is important to understand the basic statements used in percentage.

Basic statement 1

What is x% of y?

$$\Rightarrow \frac{x \times y}{100}$$

It can also be seen that x% of y = y% of x

For example 4.5% of 20,000 = 20,000% of 4.5

This one simple fact can be used to divide or multiply any number by 50 or 25 or so. Let us see this with the help of an example: We are trying to find out the value of 25×32, which is nothing but $32 \times 100/4 = 800$. Similarly if we have to divide any number by 50, we should be multiplying the number by 100 and dividing it by 2 finally.

Using this, we can see that if we have to calculate 24% of 25 (or any other calculation of similar nature), it is better to find out 25% of 24 (= $24 \times 100/4$) = 600.

Example 1 What is 20% of 50% of 40% of 20?

Solution Percentage means 'per hundred'.

So, 20% of 50% of 40% of 20 =

$(20/100) \times (50/100) \times (40/100) \times 20 = 0.8$

What we can observe here is that even if we change the order of values here, the final result will be the same.

Example 2 Statistics show that 20% of smokers get lung cancer and 80% of lung cancer patients are smokers. If 30% of the population smokes, then the percentage of population having lung cancer is

(a) 4 (b) 3 (c) 8 (d) 7.5

Solution Assume that the total population is 100.

30 people smoke, and out of them 6 people will be having lung cancer. This 6 represents 80% of lung cancer patients (because they smoke). Hence, total percentage of population having lung cancer = $\frac{6}{80} \times 100 = 7.5\%$

Basic statement 2

What percentage of x is y?

$$\Rightarrow \frac{y \times 100}{x}$$

(This can be easily deduced by assuming Z% of x is y and then proceeding as given in statement 1.)

Some more example of the same kind are given below:

1. What is the percentage growth in the sales value of the year 2002-03 over the sales value of 2003-04?

2. What is the percentage hike in the salary of Manoj this year compared to his salary hike last year?

Basic statement 3

Change in value

(a) Percentage change = $\frac{\text{Change}}{\text{Initial value}} \times 100$

(b) Percentage point change – It is the numerical difference between the values for which we have to calculate change.

Let us assume some values to understand the above written concept

Market share	*2002-03*	*2003-04*
Maruti	40%	48%
Honda	30%	26%

Percentage change in the market share of Maruti over the years $= \frac{48-40}{40} \times 100 = 20\%$

Percentage point change in the market share of Maruti over the years = 48% – 40% = 8%

Similarly, if we have to increase any quantity N by S%, then it is equal to $N\left(1+\frac{S}{100}\right)$ and when the same quantity N is to be decreased by S%, then final quantity $= N\left(1-\frac{S}{100}\right)$

It is worth mentioning here that

100→20%↑→100×1.2=120 100→20%↓→100×0.8=80

150→30%↑→150×1.3=195 150→30%↓→150×0.7=105

210→27%↑→210×1.27=266.7 210→27%↓→210×0.73=153.3

So, if the final value and percentage increase or percentage decrease is given and we have to find out the initial value, then it can be done in similar way.

Using S → 30%↑→ S × 1.3 = 195

So, if the final value 195 and 30%↑ is given, then initial value $S = \frac{195}{1.3} = 150$

Example 3 My Reliance India phone bill for the month of May is Rs B. Moreover, there is a service tax of S% which is to be levied upon this value. But since they are overcharging their customers they offer a discount of D% on it. So, now I have two options to make the payment

Rs B → S%↑→ D% ↓→ Final bill

Rs B → D% ↓→ S%↑→ Final bill

Which option is beneficial for me if S > D?

Solution Prima facie, it might appear that the 1st one is better than the 2nd one or the 2nd one is better than 1st one, but a close and deep inspection will reveal that final bill is same in both the cases.

It can be checked with the help of assuming values also.

B = Rs 100, S% = 20% and D% = 10%

Rs 100 – (20%↑) – Rs 120→ (10%↓) – Rs 108 (Final bill)

Rs 100 – (10%↓) – Rs 90→ (20%↑) – Rs 108 (Final bill)

So, both the values are same at the end.

Importance of base Whenever we are talking about percentage, it is important to specify what it is relative to, i.e., what is that total which corresponds to 100%. The following situation illustrates this point:

In a certain school 60% of all students are male, and 10% of all students are computer science majors. If 5% of males are computer science majors, what percentage of computer science majors are male?

Here, we are asked to calculate the ratio of male computer science majors to all computer science majors. We know that 60% of all students are male, and among hese 5% are computer science majors, so we conclude that .6 × .05 = .03 or 3% of all students are male computer science majors. Dividing this by the 10% of all students that are computer science majors, we arrive at the answer: 3%/10% = .3 or 30% of all computer science majors are male.

While in QA, choosing the right denominator is often not a problem but sometimes it becomes very problematic in Data Inerpretation (DI), because we are unsure as to which value should be taken as denominator or base. However, if we go by some keywords viz., of/than/over/to, the quantity, which is attached to these keywords will be the denominator.

It can be seen through the examples also.

Example 4 A student multiplies a number x by 5 instead of dividing it by 5. What is the percentage change in the result due to this mistake?

Solution Let us assume $x = 5$

So, actual result should have been 100 ÷ 5 = 20

But the result obtained = 100 × 5 = 500

So, percentage change = (500 – 20) × 100/20 = 2400%

(It can be very well observed that if we take base as 500, percentage change = 96%. What we should primarily take care of, with comparison to which number we are calculating percentage change?)

SUCCESSIVE PERCENTAGE CHANGE

Suppose we have to increase a quantity successively by 20% and 30%. Ideally this should be done by taking 100 as the initial value and then changing this initial value first by 20% and then by 30%. It can be seen below:

100 → 20%↑ → 120 → 30%↑→ 156

So, net percentage increase = 56%

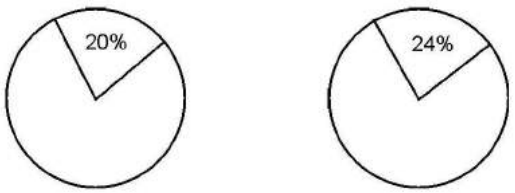
This is known as the **straight line method** of solving the problems.

Alternatively, we can use successive change formula also: $\left(a+b+\frac{ab}{100}\right)\%$

Using this formula, $20+30+\frac{600}{100}=56\%$

This kind of calculation is very much important in DI questions, especially in pie chart which can be seen with the help of following example

The following figure shows the sales figure of Due North Inc. for the year 2002-03 and 03-04. It is also given that the share of north India in total sales figure is 20% and 24% in respective years.



2002-03 2003-04

Total sales for the year 2002-03 is Rs 1202 crores and for the year 2003-04, it is Rs 1381 crores.

Now the question is: What is the percentage growth in the sales value of north India in 2003-04 over the sales value of 2002-03?

Normal way of doing this question is:

Sales value of north India in 2002-03 = Rs 240.4 crores

Sales value of north India in 2003-04 = Rs 331.4 crores

So, growth = Rs 91 crore

So, the percentage growth $=\frac{\text{Growth}}{\text{Initial value}}\times 100=37.93\%$

Alternatively, percentage increases in two factors in the given pie-charts viz., share is increasing from 20% to 24% and then total sales value is increasing from Rs 1202 crores to Rs 1381 crores.

Rather than calculating the sales value of individual years, we can directly calculate the net percentage change by calculating percentage changes independently in the above-mentioned two factors.

Percentage growth in share = 20%

$\left(\frac{\text{Growth}\times 100}{\text{Initial value}};\text{and remember that is not 4\%}\right)$

Percentage growth in total value ≅ Approx. 15%

So, 100 → 20%↑ → 120 → 15%↑→ 138, or

100 → 15%↑→ 115 → 20%↑→ 138

So, net percentage growth = 38%

Example 5 The price of petrol is increased by 20%. But expenses increase only by 10%. What is the percentage increase or decrease in consumption?

Solution If consumption remains constant, then the expenses should have also increased by 20%. But since expenses increase by only 10%, so consumption has been reduced.

These kinds of problems can be done in three ways:

i. Straight line method

100 (Initial expenditure)→ 20%↑ in price →120→ x%↓ in consumption→110 (Final expenditure),

$x\% = \frac{10}{120}\times 100=8.33\%$

ii. 10 × 10 method

We know, Price × Consumption = Expenditure

Assume Initial price = Initial consumption = Rs 10

So, Initial expenditure = Rs 100

	Price		Consumption		Expenditure
Initially ... →	10	×	10	=	100
Finally ... →	12	×	C	=	110

So, C = 110/12 = Rs 9.16

So, percentage reduction = 8.33%

iii. Formula method

Here a = 20%, b is not known and final percentage change = 10%

20 + b + 20b/100 = 10

⇒ 20 + b + b/5 = 10

⇒ b = −8.33%

So, there is a decrease of 8.33% in the consumption.

PERCENTAGE–RATIO EQUIVALENCE

The essence of percentage–ratio equivalence lies in the fact that most of the percentage calculations like 25%, 37.5%, 33.33% etc., hover around some particular ratios only. And having a good command over these ratios is definitely going to give a good percentage calculation speed which is very essential for DI calculations as well as QA calculations. (See more such calculation in ratio, proportion and variation chapter.)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	100%	200	300	400	500	600	700	800	900	1000	1100	1200	1300	1400	1500	1600
2	50	100	150	200	250											800
3	33.33	66.66	100													533.33
4	25	50	75	100												400
5	20	40	60	80	100											
6	16.66	33.33	50	66.66	83.33	100										
7	14.28	28.56	42.85	57.14	71.42	85.71	100									
8	12.5	25	37.5	50	62.5	75	87.5	100								
9	11.11	22.22	33.33	44.44	55.55	66.66	77.77	88.88	100							
10	10	20	30	40	50	60	70	80	90	100						
11	9.09	18.18	27.27	36.36	45.45	54.54	63.63	72.72	81.81	90.90	100					
12	8.33	16.66	25	33.33	41.66	50	58.33	66.66	75	83.33	91.66	100				
13	7.69												100			
14	7.14	14.28	21.42	28.57	35.71	42.85								100		
15	6.66	13.33	20	26.66	33.33	40	46.66	53.33	60	66.66	73.33	80	86.66	93.33	100	
16	6.25	12.5	18.75	25	31.25	37.5	43.75	50	56.25	62.5	68.75	75	81.25	87.5	93.75	100

PRODUCT STABILITY RATIO

In a number of topics/concepts, we encounter the relationship where the product of two quantities equals the third quantity.

For example, Speed × Time = Distance

Price × Consumption = Expenditure

Number of persons × Days = Work done

Length × Breadth = Area of rectangle

Apart form these cases, a number of times we see cases where one quantity is increased to get another quantity, e.g., if we increase cost price to obtain a certain profit, we obtain selling price or if we increase principal, we obtain amount.

If we generalize product stability ratio, it can be written like

$$A \times B = P$$

Now, if A is increased by a certain percentage, then B is required to be decreased by certain percentage so that the product (P) remains stable. For example, if we increase A by 25% and P has to be constant, then B is required to be decreased by 20%.

This procedure can be summed up in the following way:

Change in A	*Change in B*	*Change in P*
100%↑	50%↓	0%
50%↑	33.33%↓	0%
33.33%↑	25%↓	0%
25%↑	20%↓	0%
20%↑	16.66%↓	0%
16.66%↑	14.28%↓	0%
14.28%↑	12.5%↓	0%
12.5%↑	11.11%↓	0%
11.11%↑	10%↓	0%
10%↑	9.09%↓	0%
9.09%↑	8.33%↓	0%
8.33%↑	7.69%↓	0%
And so on.....		

So, if A is increased by 25%, then we need to decrease B by 20% to maintain the product stable.

This one mathematical information can be used in so many forms:

1. **Percentage** If A is 25% more than B, then by how much percentage B is less than A?
2. **Profit and Loss** An article is sold for Rs 125 at a profit of 25%. What is the cost price of the article?
3. **Time, Speed and Distance (TSD)** When speed of a car is increased by 25%, time taken reduces by 20 minutes in covering a certain distance. What is the actual time taken to cover the same distance by actual speed?
4. **TSD** Mayank goes to his office from his home at a speed of 20 kmph and gets late by 10 minutes. However, when he increases his speed to 25 kmph, he is 20 minutes early. What is the distance from his office to his home?
5. **Time and Work** Efficiency of Amit is 25% more than Vinit. Vinit takes 20 days to complete a work. How many days will Amit take to do the same work?
6. **Time and Work** 20 men can do some job in 50 days. In how many days will 25 men do the same job?
7. **Simple Interest (SI)** Rate of interest is 12.5% per annum SI. What is the principal if amount obtained after two years is Rs 1250?
8. **Percentage** Due to a price hike of 25%, 5 kg less sugar is available for Rs 100. What is the original price per kg?
9. **Mensuration** Length of a rectangle is increased by 25%. By what percentage should the breadth be decreased so that area remains constant?

In all the above written situations, just one mathematical information has been used, i.e., if A is increased by 25%, then B decreases by 20%. Let us see the solution of all the questions given above.

Solution 1 Normal Method

Let us assume B = 100, then A = 125

Now, B is 25 less than A.

Percentage of B is less than A = 25/125 × 100 = 20%

Product Stability Ratio Method

Using product stability rule, since A is 25% more than B, so B is 20% less than A.

Solution 2 Normal Method

CP × 1.25 = SP

So, CP = SP/1.25 = 125/1.25 = Rs 100

Product Stability Ratio Method

If we increase CP by 25%, we will get SP.

So, if we decrease SP by 20%, we will get CP.

Hence CP = Rs 100

Solution 3 Normal Method

Since we know S = V × T (Distance = Speed × Time)

New speed = 1.25 V, so new time = T/1.25

So, reduction in time = T − T/1.25 = 0.25 T/1.25 = T/5

T/5 = 20 min ⇒ T = 100 min

Product Stability Ratio Method

Since speed has been increased by 25%, so time will reduce by 20%.

Now, 20% T (Time) = 20 min

So, Total time = 100 min

Solution 4 Normal Method

Let us assume that distance = D

So, D/20 – D/25 = 30/60 hr. = ½

So, D = 50 km

Product Stability Ratio Method

S = V × T

25%↑20%↓

So, 20% T = 30 min

⇒ T = 150 mins = 2½

So, total distance = 20 × 2½ = 50 km

Solution 5 Normal Method

Vinit is taking 20 days to complete the work i.e., Vinit is doing 100% work in 20 days. So, Vinit is doing 5% work in a day. Since efficiency of Amit is 25% more than Vinit, so Amit is doing 6.25% work per day.

So, number of days taken by Amit = 100/6.25 days = 16 days

Product Stability Ratio Method

Efficiency of Amit is 25% more than Vinit. So, Amit will take 20% less days than Vinit.

So, number of days taken by Amit = 16 days

Solution 6 Normal Method

Using Work = Number of persons × Number of days

Work = 20 × 50 = 1000

Now, 1000 = 25 × D

So, D = 40

Product Stability Ratio Method

Number of persons increases by 25%, so number of days will decrease by 20%.

So, number of days = 40 days

Solution 7 Normal Method

Using the formula for SI = PRT/100

P = (SI × 100)/RT

Putting the values gives us P = Rs 1,000

Product Stability Ratio Method

Interest for two years = 25%

So, if we decrease the amount by 20%, then we will get the principal.

Hence, Principal = Rs 1,000

Solution 8 Normal Method

Let us assume that original price per kg = Rs P per kg

So, final price per kg = Rs 1.25 P

Hence, (120/P) – (120/1.25P) = 5
Solving this equation gives P = Rs 4 per kg

Product Stability Ratio Method

Since, the price hike is 25%, 20% less quantity of sugar will be available for Rs 100.
20% = 5 kg ⇒ 100% = 25 kg
So, 25 kgs were available for Rs 100 initially.
So, Price = Rs 4/kg

Solution 9 Normal Method

	Length		*Breadth*		*Area*
Initially… →	10	×	10	=	100
Finally… →	12.5	×	B	=	100

So, B = 8
Percentage decrease = 20%

Product Stability Ratio Method

Till now, it must have become very obvious that the breadth will decrease by 20% to keep the area constant.

Extension of Product Stability Ratio

This table is a two-way table, i.e., if we decrease A by 50%, then B is needed to be decreased by 100%.

If we express the percentage figures given in the product stability table in ratios, then it comes like the following:

Change in A	*Change in B*	*Change in P*
$\frac{1}{1}\uparrow$	$\frac{1}{2}\downarrow$	0 %
$\frac{1}{2}\uparrow$	$\frac{1}{3}\downarrow$	0 %
$\frac{1}{3}\uparrow$	$\frac{1}{4}\downarrow$	0 %

So, $\frac{1}{50}\uparrow$ corresponds to $\frac{1}{51}\downarrow$. It means that if we increase A by 2%, then B is needed to be decreased by 1.96% (approx.) so that P remains constant.

Similarly, it can be done with all the reciprocals.

But the problem which lies with this table is that it has values which are reciprocals only.

So, what are we required to do if we increase A by 15%?

Take this as

Change in A	*Change in B*	*Change in P*
$15\% = \frac{3}{20}\uparrow \Leftrightarrow$	$\frac{3}{23}\downarrow$	0%
$35\% = \frac{7}{20}\uparrow \Leftrightarrow$	$\frac{7}{27}\downarrow$	0%

APPLICATION OF PERCENTAGE

1. Simple Interest and Compound Interest

Interest is defined as the "Time value of money." As the time passes on, value of money keeps on changing. And while factors like inflation or depreciation in money decreases the value of money with the passage of time, interest the counters all these.

The basic difference between simple interest and compound interest lies in the fact that while in case of simple interest, the principal as well as the interest remain same for the entire period given, in case of compound interest, after a certain period, both the principal and the compound interest keep on changing.

Simple interest

In case of simple interest, interest is reckoned at a flat rate for the entire period of time for the amount borrowed.

Simple interest can be calculated by finding the percentage of principal for that period of time.

If the principal is P, rate of interest per annum = R% and time = T years, then Simple Interest will be equal to RT % of P.

So, Simple Interest (SI) = $\frac{P\times(R\times T)}{100}$

Example 6 A sum of money becomes 3 times in 5 years. In how many years will the same sum become 6 times at the same rate of SI?

Solution The sum of money gets 3 times means that 200% is being added to the original sum (principal) in 5 years.

So, 500% will be added up in $12\frac{1}{2}$ years.

Compound interest

In case of compound interest, interest is reckoned on the interest of previous years also apart from reckoning it on the current year.

So, compound interest can be seen as the extension of simple interest in such a way that previous year interest also becomes principal for the next year.

If principal is P, rate of interest per annum = R% and time = n years, then

Compound Interest = $P\left(1+\frac{R}{100}\right)^n - P$

Amount = $P\left(1+\frac{R}{100}\right)^n$

It can be seen that $P\left(1+\frac{R}{100}\right)^n$ is nothing but $P\left(1+\frac{R}{100}\right)\left(1+\frac{R}{100}\right)\ldots n$ times, which is simply the application of successive percentage change.

Example 7 The difference between two years of compound interest and simple interest at 10% over Rs X is Rs 10. What is the value of x?

Solution

	SI	*CI*
At the end of 1st year	10%	10%
At the end of 2nd year	10%	10% + 10% of 10% =11%
	=20%	=21%

So, 1% = Rs 10

$\Rightarrow$ 100% = Rs 1000

It is pertinent to understand here that if the rate of Interest = R% per annum for both CI and SI, then the difference between CI and SI for 2 years will be equal to R% of R% = $R^2/100\%$

In the above case, R = 10%, so the difference between CI and SI for 2 years = 1%

Comparison between SI and CI

Assume that two different sums are getting doubled at their respective rates of SI and CI in 5 years. The following table gives us the mechanism of getting money n times in the above situation.

	After 5 years	*After 10 years*	*After 15 years*	*After 20 years*
At SI	2times	3 times	4 times	5 times
At CI	2 times	4 times	8 times	16 times

2. Simple Annual Growth Rate (SAGR) and Compounded Annual Growth Rate (CAGR)

Consider the following table pertaining to the sales value of Due North Inc. in different years:

Year	*2004-05*	*2005-06*	*2006-07*	*2007-08*
Sales Value (Rs in crores)	100	110	124	131

If we find out the growth over the given period, then it is equal to 31%.

However, there are two more growth models, namely, Simple Annual Growth Rate (SAGR) and Compounded Annual Growth Rate (CAGR).

$$\text{SAGR} = \frac{\text{Overall growth\%}}{\text{No. of years for which growth can be calculated}}$$

$$= \frac{31\%}{3} = 10.33\%$$

To find out CAGR, we are needed to use the approach of CI and we will consider 131 as the amount and 100 as the principal.

So, $131 = 100\left(1+\frac{R}{100}\right)^3$

To find out the value of R, we will apply the method of hit and trial, or otherwise find out the value of $\sqrt[3]{1.31}$.

Since we know, $(1.1)^3 = 1.331$, so value of R will be less than 10% and very close to 10%.

So, it will be around 9.7% – 9.8%.

Compound interest reckoned half-yearly or quarterly If the rate of interest is R% annually and CI is compounded half yearly for n years, then the expression for Compound

$$\text{Interest} = P\left(1+\frac{\frac{R}{2}}{100}\right)^{2n} - P$$

If the rate of interest is R% annually and CI is compounded quarterly, then the expression for Compound

$$\text{Interest} = P\left(1+\frac{\frac{R}{4}}{100}\right)^{4n} - P$$

3. Calculation

Percentage is mostly helpful in multiplication and division. Let us learn this with the help of examples.

Multiplication

To use this percentage method effectively for multiplying two values, we should be thorough with the equivalent ratio of the percentage figures. Besides, having a good addition speed will be an added advantage.

Example 8 63×72.

Solution The moment we see any number, we should start mental scanning of the percentage–ratio equivalence. Here,

$63 \times 72 = (62.5 + 0.5) \times 72$

$= 5/8 \times 100 \times 72 + 0.5 \times 72$ (5% of 72)

$= 4500 + 36$

$= 4536$

Example 9 76×24

Solution $= (75 + 1) \times 24 = ¾ \times 100 \times 24 + 1 \times 24 = 1800 + 24 = 1824$

Division

Example 10 Dividing 243 by 50.

Solution $\frac{243}{50} = \frac{243}{100} \times 2 = 2.43 \times 2 = 4.86$

To divide any number by 50, we will divide it by 100 and multiply by 2.

Example 11 Dividing 312 by 25.

Solution $\frac{312}{25}=\frac{312}{100}\times 4=3.12\times 4=12.48$

To divide any number by 25, we will divide it by 100 and multiply by 4.

Similarly, while dividing any number by any such number for which we can find out a comparable value in terms of 100 $\left(\text{as } 25=\frac{100}{4}\right)$ should be used.

PRACTICE EXERCISES

WARM UP

Q.1. What fraction is equal to 57.12% (approximately)?
(a) 4/8 (b) 5/8 (c) 5/7 (d) 4/7

Q.2. What is 50% of 40% of Rs 1,200?
(a) Rs 450 (b) Rs 100
(c) Rs 300 (d) Rs 240

Q.3. a% of a + b% of b = 2% of ab, then what percentage of a is b?
(a) 50% (b) 75%
(c) 100% (d) Cannot be determined

Q.4. Subtracting 34% of x from x is equivalent to multiplying x by N. Which of the following is the value of N?
(a) 0.66 (b) 66 (c) 0.34 (d) 6.6

Q.5. Subtracting 48% of N from N is equivalent to multiplying N by
(a) 0.48 (b) 1.48 (c) 0.52 (d) 0.42

Q.6. Tipu's salary has increased from Rs 7,200 to Rs 8,100. What is the percentage increase in Tipu's salary?
(a) $33\frac{1}{3}\%$ (b) 25% (c) $16\frac{2}{3}\%$ (d) $12\frac{1}{2}\%$

Q.7. 42% of a number is subtracted from 61% of itself and the resulting value is 38. What is the number?
(a) 100 (b) 250 (c) 200 (d) 300

Q.8. The length of rectangle is increased by 30% and the breadth is decreased by 25%. What is the percentage change in the area of the rectangle due to this?
(a) 2.5% increase (b) 25% decrease
(c) 25% increase (d) 2.5% decrease

Q.9. If a value is increased twice successively by 20%, then what is the net percentage increase in that value?
(a) 20% (b) 40% (c) 21% (d) 44%

Q.10. The numbers are respectively 30% and 40% of a third number. What percentage is the first number of the second number?
(a) 25% (b) 50% (c) 40% (d) 75%

Q.11. Two numbers are 15% and 20% less than a third number. What percentage is the first number as a percentage of the second number?
(a) $116\frac{2}{3}\%$ (b) $83\frac{1}{3}\%$
(c) $106\frac{1}{4}\%$ (d) $123\frac{2}{3}\%$

Q.12. 10 litres of water is added to 50 litres of a solution containing 20% of alcohol in water. What is the concentration of alcohol in the solution now?
(a) 20% (b) $16\frac{2}{3}\%$
(c) $12\frac{1}{2}\%$ (d) $33\frac{1}{3}\%$

Q.13. A man spends 25% of his money on food. After spending 50% of the remaining, he is left with Rs 375. How much money was with that man initially?
(a) Rs 5,000 (b) Rs 4,000
(c) Rs 5,800 (d) Rs 1,000

Q.14. A man earns Rs 4,000 in a month and saves 30% of his income. What is his expenditure in a year?
(a) Rs 28,600 (b) Rs 18,400
(c) Rs 32,000 (d) None of these

Q.15. After spending 20% on clothes, 10% on books, 9% on purchasing gift for girl friend and 7% on others, Chandra has a balance of Rs 2,700. How much money was there with him initially?
(a) Rs 5,000 (b) Rs 6,000
(c) Rs 7,000 (d) None of these

Q.16. A man spends 60% of his income. His income increases by 40% and due to this, he increases his expenditure also by 40%. Find the percentage increase in his savings.
(a) 30% (b) 4%
(c) 4.44% (d) None of these

Q.17. If the numerator of a fraction is increased by 20% and its denominator decreased by 10% the fraction becomes 3/2. What is the value of the original fraction?
(a) $\frac{2}{3}$ (b) $\frac{7}{8}$ (c) $\frac{9}{8}$ (d) $\frac{6}{7}$

Q.18. In an exam of two papers maths and chemistry, 60% of the students pass in maths and 70% pass in chemistry. What is minimum percentage of students who could have failed in both the subjects?
(a) 0% (b) 30%
(c) 40% (d) None of these

Q.19. Everyday a fruit seller sells 50% of his stock. 10% of the stock gets spoiled during the night. If a total of 1983 fruits got spoiled in the three days period, how many fruits did he start with?
(a) 20,000 (b) 25,000
(c) 24,000 (d) None of these

Q.20. A certain sum becomes 7/3 times itself in ten years under simple interest. Find the rate of interest.
(a) $7\frac{1}{2}$% p.a (b) $13\frac{1}{3}$% p.a
(c) 20% p.a. (d) 10% p.a.

Q.21. An amount is lent at x% p.a. simple interest for two years. Instead, had it been lent at $2x$% p.a. simple interest for y more years, then the interest would have been five times the earlier interest. Find the value of y.
(a) 2 (b) 3 (c) 4 (d) 5

Q.22. According to a new plan rolled out by HISP Bank, the rate of simple interest on a sum of money is 8% p.a. for the first two years, 10% p.a. for the next three years and 6% p.a. for the period beyond the first five years. Simple interest accrued on a sum for a period of eight years is Rs 12,800. Find the sum.
(a) Rs 24,000 (b) Rs 16,000
(c) Rs 15,000 (d) None of these

Q.23. Find the least number of integral years in which a sum of money invested at 20% compound interest per annum will become more than double itself.
(a) 1 (b) 4 (c) 2 (d) 3

FOUNDATION

Q.1. In the country of Tattoland, a new term tercentage has been introduced in the place of percentage. While percentage is per 100, tercentage is per 125. What will be the tercentage marks obtained by a student in Tattoland if his percentage of marks is 60?
(a) 48 tercentage (b) 75 tercentage
(c) 60 tercentage (d) Cannot be determined

Q.2. In the country of Tappoland, a new term hercentage has been introduced in the place of percentage. While percentage is per 100, hercentage is per 1000. What will be the hercentage marks obtained by a student in Tappoland if his percentage is 60?
(a) 6 hercentage
(b) 600 hercentage
(c) 166.66 hercentage
(d) Cannot be determined

Q.3. In a ΔABC measurement of $\angle ABC = 15^\circ$. What will be the value of $\angle ABC$ when seen through a magnifying glass which can magnify any object by 20%?
(a) Value reduce by 20%
(b) Same
(c) Value increased by 20%
(d) Value decreased by 16.66%

Q.4. In a maternity centre, 5% of all the childbirth cases result in twins. What is the approximate percentage of twins out of total children born?
(a) 5% (b) 7.6% (c) 9.5% (d) 10.4%

Q.5. Due to a price hike of 20%, 4 kg less sugar are available for Rs 120. What is the initial price per kg of sugar?
(a) Rs 5/kg (b) Rs 4/kg
(c) Rs 6/kg (d) Rs 5.5/kg

Q.6. A person has to pay 20% tax on first Rs 4,500 earned, for amount earned between Rs 4,500 and Rs 7,500 the tax is 25% and for the amount earned more than Rs 7,500, the tax is 30%. If he paid 23.33% of his income as tax, then what is his income?
(a) Rs 9,000 (b) Rs 8,500
(c) Rs 10,000 (d) Rs 10,500

Q.7. A clothes-shop offers a discount to all its customers on the basis of the value of the purchase. If the value of purchase is less than Rs 3,200, the customer gets a discount of 14%. The customer gets 16% discount on

purchase between Rs 3,200 – Rs 6,300. The customer gets a discount of certain % on the purchase exceeding Rs 6,300. A customer gets a total discount of Rs 1,520 which is equal to 16% of the purchase value. How much discount does the customer get for amount exceeding Rs 6,300?
(a) 16% (b) 18% (b) 20% (b) 22%

Q.8. Three friends P, Q and R donate 8%, 7% and 9% of their salary to a charitable hospital in the given order. Salary of P and Q is same and the difference of their donations is Rs 74. The total donation by the P and Q is Rs 525 more than R's donation. What is the percentage of the total salary of the three donations?
(a) 7.95% (b) 8.3%
(c) 6.34% (d) None of these

Q.9. Fresh watermelons contain 90% water by weight whereas dry watermelons contain 20% water by weight. What is the weight of dry watermelon obtained from 20 kgs of fresh watermelon?
(a) 2.4 kg (b) 2 kg
(c) 2.5 kg (d) Cannot be determined

Q.10. In solution of sugar and water the ratio of sugar and water by weight is 1:4. This solution is heated and in the process it loses 50% weight. What is the ratio of weight of sugar and water in the final mixture?
(a) 4:1 (b) 2:1 (c) 2:3 (d) 3.5:2

Q.11. Rate of inflation is 1000% per annum. What is the value of an article two years from now if it costs Rs 6 today?
(a) Rs 66 (b) Rs 126
(c) Rs 660 (d) Rs 726

Q.12. Dataman Infosys Systems is a Lucknow-based software company which is growing tremendously. It doubled its turnover in 1998 from Rs 50 crores in 1997. Then it tripled its turnover in 1999 and grew by 50% in 2000. What is the turnover at the end of 2000?
(a) Rs 250 cr (b) Rs 450 cr
(c) Rs 300 cr (d) Rs 600 cr

Q.13. In the recently concluded CBSE board examination, a total of 6,00,000 students appeared. 40% of them were females while the rest were males. Pass percentage among males is 75% and the overall pass percentage is 70%. What is the pass percentage for females?
(a) 37.5% (b) 50% (c) 62.5% (d) 70%

Q.14. Because of the budget presented by Yashwant Sinha, the price of sugar increased by 40%. The Verma family reduced its consumption so that expenditure on sugar is up by 12%. If the total consumption of sugar before the rise in price was 50 kg. What is the present consumption of sugar (in kg)?
(a) 48 kg (b) 40 kg (c) 36 kg (d) 32 kg

Q.15. Tatto's working hours per day were increased by 25% and his wages per hour were increased by 20%. By how much per cent was his daily earnings increased?
(a) 20% (b) 25% (c) 50% (d) 45%

Q.16. A's income is reduced from Rs 75,000 to Rs 60,000, while B's income for the same period is increased from Rs 60,000 to Rs 75,000. What percentage of decrease in A's income is the increase in B's income?
(a) 125% (b) 75% (c) 133% (d) 100%

Q.17. 18% of A plus 15% of B plus 19% of C is equal to 17% of the sum of A, B, and C. If A – B = 500 and A – C = 3400, what is the value of A + B + C?
(a) 12,400 (b) 11,600
(c) 13,500 (d) None of these

Q.18. Out of 51,600 candidates who appeared in an examination, 35% failed, while 15% passed with honours. What is the number of candidates who passed the examination, but failed to obtain honours, assuming that no student who has failed can obtain honours?
(a) 25,000 (b) 30,000
(c) 27,309 (d) 25,800

Q.19. A and B have some guavas divided among themselves. A says to B "If I give you 25% of the guavas I have, I will still have 2 more guavas than you have." To this, B says "If you give me guavas equal to 70% of what I have now, I will have 4 more guavas than you have." What is the total number of guavas that they have?
(a) 80 (b) 64 (d) 36 (d) 88

Q.20. Raghupati Raghav was able to score a total of 600 in 12 tests. He scored less than or equal to 80% of his average score per test in four of these tests. If he did not score more than 60 in any of the tests, what is the minimum number of tests in which he should have scored more than 50?
(a) 8 (b) 4 (c) 3 (d) 2

Q.21. A locomotive engine runs at a speed of 50 km/h when no compartment is attached to it. For every new compartment that is attached to it, the speed of the train is reduced by 10% of the earlier speed. At most, how many compartments can be attached so that the train can cover a distance of 180 km in a maximum of 8 hours?
(a) 7 (b) 5 (d) 6 (d) 8

Q.22. Babloo organized a party by taking contributions from the participants. 60% of the participants contributed 80% of the funds. The average contribution of all the people who attended is Rs 50. What is the average contribution of the remaining 40% of the people?
(a) Rs 100 (b) Rs 150
(c) Rs 250 (d) Rs 25

Q.23. If the price of an article rose by 25% every odd year and fell by 20% every even year, what would be the percentage change after 180 years?
(a) 10% increase (b) 25% increase
(c) No change (d) 20% decrease

Q.24. Pranav saves 10% of his total salary. Next year he increases his expenses by 20% but his percentage savings remain the same. What is the percentage increase in his salary next year?
(a) 10% (b) 20% (c) 40% (d) 16.66%

Q.25. In an examination, 80% students passed in philosophy and 70% students passed in maths. At the same time 15% failed in both the subjects. If 390 students passed in both the subjects, how many students appeared in the examination?
(a) 500 (b) 400 (c) 800 (d) 600

Q.26. The salary of Anil and Vinit is 20% and 30% less than the salary of Dheeraj respectively. By what percentage is the salary of Anil more than the salary of Vinit?
(a) 33.33% (b) 50%
(c) 10% (d) 14.28%

Q.27. In the recent census report of Patna, it was found that 40% of total male adult population and 55% of total female adult population is married. What percentage of total adult population of Patna is married? (Polygamy is not in practice in Patna.)
(a) 46.32% (b) 43.6%
(c) 49.81% (d) 40%

Q.28. Sona and Ellaraputtu are very close friends. One day they were figuring out about their respective incomes and expenses. It was found that income of Sona and Ellaraputtu is 'S' and 'E' respectively. Sona spends 12% of her income and Ellaraputtu also spends the same amount. What percentage of her income is Ellaraputtu spending?
(a) E/(12S) (b) 12S/E
(c) 12 E/S (d) S/(12E)

Q.29. Sixty students appeared in the physics and math paper of an examination. Of these 60% students passed in physics and 70% passed in math. What is the maximum percentage of students who failed in both the subjects?
(a) 40% (b) 30% (c) 0% (d) 100%

Q.30. In an election, 10% of the voters on the voter's list did not cast their votes and 60 voters cast their ballot papers blank. There were only two candidates. The winner was supported by 47% of all voters in the list and he got 308 votes more than his rival. The number of voters on the list were
(a) 3,600 (b) 6,200
(c) 4,575 (d) 6,028

Q.31. A rainy day occurs once in every 10 days. Half of the rainy days produce rainbows. What in the percentage of all the days when we have no rainbows?
(a) 95% (b) 10% (d) 50% (d) 5%

Q.32. One-third of Mayank's marks in history equals his marks in geography. If he obtained 180 marks in two subjects together, how many marks did he get in geography?
(a) 40 (b) 45 (c) 60 (d) 36

Q.33. If the price of the sugar is increased by 20%, one can buy 2 kg less sugar in Rs 50. What is the quantity of sugar that could be bought before the price hike?
(a) 10 (b) 8 (c) 16 (d) 12

Q.34. The entry fee in an exhibition was Rs 10. Later this was reduced by 25%, which increased the sale of tickets by 20%. Find the percentage increase in the number of visitors.
(a) 54 (b) 57 (c) 60 (d) 66

Q.35. A car bought for Rs 13,350 depreciates each year by 12½%. In how many years will its value be reduced to Rs 6,000?
(a) 4 years (b) 6 years
(c) 8 years (d) 10 years

Q.36. An electrical contractor purchased a certain amount of wire, 10% of which was stolen. After using 85% of the remainder he had 47.25 m of wire left. How much wire did he purchase?
(a) 600 m (b) 103 m
(c) 325 m (d) 350 m

Q.37. Anoop got 273 marks in his graduation exam and got 5% more than the pass percentage. If Siddharth got 312 marks then by what percentage above the pass marks, did he pass the exam?
(a) 9% (b) 12.5%
(c) 20% (d) 25%

Q.38. There were only two contestants A and B in the recent assembly election in Bihar at Raghopur constituency. B got 50% of the votes that A got. Had A got 200 votes less, there would have been a tie. What is the total number of votes polled?
(a) 1,000 (b) 800
(c) 1,200 (d) Cannot be determined

Q.39. A city had a population of 30,00,000 in the beginning of 1999. Its average growth rate is 4% per year, but due to a massive earthquake in 2001, its population is reduced by 8% in that year. But it again maintained the same growth rate of 4% in following years. What will be the approx. population of the city at the end of 2003?
(a) 32,06,460 (b) 34,68,420
(c) 31,52,360 (d) 32,28,810

Q.40. A is an alloy of tin and copper and B is an alloy of copper and zinc. A has 30% copper while B has 50% copper. x gms of A is mixed with 30 gms of B to form another alloy which has 45% copper. What is the value of x?
(a) 5 (b) 10 (c) 15 (d) 20

Q.41. Rajesh Raj has Rs 90,000 with him. He purchases a Bike, a home theatre and a flat TV for Rs 15,000, Rs 13,000 and Rs 35,000 respectively and puts the remaining money in his bank account which pays 15 per cent per annum Compound Interest. After 2 years he sells off the three items at 80% of their original price and also withdraws his entire money from the bank by closing the account. What is the total change in his asset?
(a) −4.5% (b) +3.5%
(c) −4.32% (d) +5.5%

Q.42. After three successive raises, Aftab's salary became equal to $\frac{378}{125}$ of his initial salary. By what percentage was the salary raised the first time if the third rise was twice as high (in percentage) as the second rise was twice as high (in percentage) as the first rise?
(a) 10% (b) 15% (c) 20% (d) 25%

Q.43. One litre of water is evaporated from 6 litres of solution containing 4% of sugar. The percentage of sugar in the remaining solution is
(a) 1.4% (b) 5.2% (c) 4.8% (d) 3.4%

Q.44. A's income is 60% of B's income, and A's expenditure is 70% of B's expenditure. If A's income is 75% of B's expenditure, find the ratio of A's savings to B's savings.
(a) 5:1 (b) 1:5 (c) 3.5:1 (d) 2:7

Q.45. There are 1,10,000 books at Khuda Baksh Oriental Public Library, 40% of which are science books. It was decided to add 20,000 new books to the library. How many of the new books should be science books in order to bring the percentage of science books in the library equal to 45%?
(a) 15,000 (b) 1,500
(c) 1,450 (d) 14,500

Q.46. If 33 untrained labourers can do a piece of work in 15 days of 12 hours each, how many trained labourers can do 50% more work in 11 days of 9 hours each? (Assuming that it takes 2 trained labourers to do the work of 5 untrained labourers.)
(a) 42 (b) 36 (c) 90 (d) 100

Q.47. X's income is 75% of Y's income, and X's expenditure is 80% of Y's expenditure. If X's income is 90% of Y's expenditure, find the ratio of X's savings to Y's savings.
(a) 1:2 (b) 2:1 (c) 1:4 (d) 2:3

Q.48. Kajol went to a fruit market with a certain amount of money. With the money she has, she can buy either 40 apples or 70 mangoes. She retains 15% of her money for auto fare. If she buys 35 mangoes, how many more apples can she buy?
(a) 15 (b) 20
(c) 14 (d) None of these

Q.49. In an election for the post of class representative at Pitampura centre where two candidates were contesting, 10% of the students did not vote (because they were late, as usual) and 60 votes recorded were declared invalid (because stray marks were made on the ballot paper). Winning candidate won the election by 308 votes and it was found that he had been supported by 47% of the whole students of the class. Which of the following can be the number of valid votes recorded for each candidate?
(a) 2308, 2000 (b) 2904, 2596
(c) 2914, 2606 (d) 2464, 2772

Q.50. If the length and breadth of a rectangular plot are increased by 20% and 50% respectively, Then the new area is how many times the original area?
(a) 5/8 (b) 9/5 (c) 3/7 (d) 7/3

MODERATE

Q.1. There are fifty successive percentage discounts given in a series of 2%, 4%, 6%, 8%... and so on. What is the net discount?
(a) 98% (b) 2550%
(c) 100% (d) Infinite

Q.2. In Question 1, after how many such percentage discounts in succession will the effective discount be higher than 50%?
(a) 7th (b) 8th (c) 9th (d) 10th

Q.3. My Alliance India phone bill for the month of May is Rs B. On top of this, there is a service tax of S% which is to be levied upon this value. But since they are having a guilt feeling of overcharging their customers fraudulently, they offer a discount of D% also on top of this. So, my billing goes like this

Rs B→ S%↑→ D% ↓→ Final Bill

But surprisingly, my final bill is equal to Rs B itself.

Now, which of the following is true regarding S and D (assume all the values to be positive)?
(a) S > D (b) S = D
(c) D > S (d) Cannot be determined

Q.4. Assuming percentage to be true for all the real values, which of the following can be true?
40% of A = 60% of B
i. A > B ii. B > A iii. B = A
(a) i and ii only (b) ii and iii only
(c) i and iii only (d) None of these

Direction for questions 5 and 6: Read the passage below and solve the questions based on it.

Alok has certain number of Oranges with him. He sells each orange for Rs 2. Shazmi bought 25% of the oranges and Sandeep bought one-third of what was left. Mohit bought 50% of what was left and Siddharth bought 4 oranges. Total amount paid by Shazmi and Mohit is Rs 24.

Q.5. How many oranges did Sandeep purchase?
(a) 3 (b) 4 (c) 6 (d) 8

Q.6. How many oranges were left with Alok?
(a) 3 (b) 4
(c) 6 (d) None of these

Direction for questions 7 and 8: Read the passage below and solve the questions based on it.

All students of class X took the pre-board test. One third of the class received B, one fourth received B+, one sixth received B–, and one eighth of the class failed (grade C). Everyone else got A.

Q.7. What can be the minimum number of students in class *X*?
(a) 8 (b) 48
(c) 24 (d) Cannot be determined

Q.8. How many students in class X got A?
(a) 3 (b) 6
(c) 4 (d) Cannot be determined

Q.9. A manufacturer of detergent cakes increased the price of his product as a result of which there was a decrease of 20% in its sales. If the sales turnover increases by 20% due to this change what was the percentage increase in the price?
(a) 20% (b) 36% (c) 40% (d) 50%

Q.10. Ravi Shankar wishes to buy an air conditioner (AC) with the money in the bank, which currently is earning interest at the rate of 15 pcpa compounded annually. But Tanzar, his friend forecasts that the inflation rate applicable to AC is going to be 14%; 15% and 16% respectively for the next 3 consecutive years and advises Ravi Shankar to postpone the purchase by 3 years. Does Ravi Shankar gain monetarily, if he takes Tanzar's advice?
(a) Yes
(b) No
(c) He neither gains nor losses
(d) He gains only if the purchase is made in the second year.

Q.11. In the recently concluded referendum in New York, regarding calling back US troops from Iraq, 600 people voted on a referendum, but after some discussion the opponents increased by 150%. The motion was then rejected by a majority two times as great as that by, which it was formerly passed. How many people voted against the initial referendum?
(a) 400 (b) 300 (c) 200 (d) 500

Q.12. The number of students present in the class of Manoj Yadav is M and out of those M students, 22.1875% are stupids. If the number of stupids in the class is N, then what can be the minimum value of M+N?
(a) 1,955 (b) 391
(c) 16,200 (d) 12,21,785

Q.13. If 75% of a class of 40 students can sing and only 20% cannot dance, the maximum number of students who can neither sing nor dance is
(a) 0 (b) 6
(c) 8 (d) Cannot be determined

Q.14. There are five contestants A, B, C and D in the assembly election from the Bihar Sharif constituency. It is given that none of the contestants got less than 1% of the valid vote. Consider the following statements in isolation to each other:
i. A got 49% of the total valid votes
ii. B got 55% of the total votes
iii. C got 46% of the total valid votes
iv. D got 48% of the total valid votes
With how many of the following statements when used independently, it is possible to find out the winner of the election?
(a) 0 (b) 1 (c) 2 (d) 3

Q.15. For admission in a graduation program of Delhi University, 90% of the candidates who appeared for the written test were males and the rest were females, 60% of the males and 80% of the females passed in the written test. What is the total number of students who appeared for the written test, if the total number of passed candidates was 1240?
(a) 2,500 (b) 2,000 (c) 2,750 (d) 3,500

Q.16. In a laboratory experiment, a sample of air, which is a mixture of oxygen and water vapour is taken. Water vapour, contains hydrogen and oxygen gases. If air contains a total of 70% oxygen (including that contained in the water vapour) by weight while water

vapour contains 16·66% of oxygen by weight, how many kilograms of water vapour is present in 1 kilogram of air?
(a) 0.2 (b) 0.36 (c) 0.32 (d) 0.25

Q.17. In Patna University, out of every 7 students learning science, 3 take literature as well; for every student taking at least one of these two courses, there are 3 students who take up neither. If 10% of the students take only science, the percentages who have taken science is:
(a) $17^1/_2$% (b) $23^1/_3$%
(c) $7^1/_2$% (d) Cannot be determined

Q.18. 60% of the students of a class are boys and in an examination, 80% of the girls scored more than 40 marks (out of a maximum possible 150 marks). If 60% of the total students scored more than 40 marks in the same examination, find the fraction of the boys who scored 40 marks or less.
(a) 8/15 (b) 7/15 (c) 4/5 (d) 1/5

Q.19. A thousand people voted on a resolution with 10% of the votes being invalid. After some discussion 1,000 people voted again. This time there were 20% invalid votes. The opponents were increased by 50% while the motion was now rejected by a majority, which is 300% more than it was formerly passed by. How many people voted against the resolution after the discussion?
(a) 400 (b) 200 (c) 300 (d) 500

Q.20. Recently I had gone to a locality called Shadigarh for conducting a survey about the number of married persons in the locality. The population of the locality is 7,200 and $^{11}/_{18}$th of those are males and the rest females. If 40% of the males are married, find percentage of married females in the locality.
(a) $48^1/_7$% (b) $52^4/_7$%
(c) $62^6/_7$% (d) $71^1/_7$%

Q.21. Nitish was to get a 50% hike in his pay but the computer operator wrongly typed the figure as 80% and printed the new pay slip. He received this revised salary for three months before the organization realized the mistake. What percentage of his correct new salary will he get in the fourth month, if the excess paid to him in the previous three months is to be deducted from his fourth month?
(a) 20% (b) 25% (c) 40% (d) 33 1/3%

Q.22. In the US opinion polls held during August, 60% of the respondents favoured Walter Mondale while the rest favoured George W. Bush. It was found in September polls that 10% of Mondale supporters switched their preference to Bush, while the same percentage of Bush's supporters also switched their preference to Mondale. What percentage of the electorate should now switch their preference from Mondale to Bush so that they are at par?
(a) 6% (b) 8% (c) 10% (d) 14%

Q.23. An index of 12 shares contains, among others, the shares of Reliance, HLL and Infosys with weightage of 7%, 13% and 15% respectively. What is the increase in the prices of other shares, if these three rise by 9%, 10% and 4% respectively, while the index rises by 6%?
(a) 5.34% (b) 4.91%
(c) 4.58% (d) Cannot be determined

Q.24. A community service organisation has raised 80% of the donation that it needed for a new building by receiving an average donation of Rs 400 per person from a group of people. To raise the remaining amount, the organisation now intends approaching another group of people which has two-thirds the number of people as the first group. What should be the average donation per person from the new group?
(a) Rs 100 (b) Rs 150
(c) Rs 200 (d) Rs 250

Q.25. 60% of the employees of Due North Inc. are female, and 75% of the men earn more than Rs 25,000 per year. If 45% of the company's employees earn more than Rs 25,000 per year, what fraction of the women employed by the company earn Rs 25,000 per year or less?
(a) 2/11 (b) 1/4 (c) 1/3 (d) 3/4

Q.26. The pollution in normal atmosphere is less than 0.01%. Due to leakage of methyl isocyanide (MIC) from a factory the pollution is increased to 20%. If every day 80% of the pollution in the atmosphere is neutralized, in how many days will the atmosphere become normal?
(a) 5 days (b) 3 days
(c) 4 days (d) None of these

Q.27. The marks obtained by Arpit in English, maths, science and Hindi in Class 10 are as follows (Maximum marks per subject = 100)
(a) The marks obtained in maths is 1.5 times the marks obtained in English
(b) Arpit got 64% in these 4 subjects.
(c) Arpit got the maximum and minimum marks in science and Hindi, respectively, with a difference of 48 marks between them
(d) An addition of 50% of the marks obtained in English to the final score gives an overall percentage of 70%
What would be his percentage of marks if only maths and science marks are counted?
(a) 14 (b) 82 (c) 84 (d) 86

Q.28. Amit puts some amount in bank and obtained an interest of Rs 15 at the end of one year. He added Rs 85 to this amount and put the whole amount again in the bank for another year. At the end of the 2nd year, Amit got Rs 420 as total (interest + principal). What amount did he put in the beginning and what is the rate of interest offered by the bank if the minimum amount that can be deposited in the bank is Rs 75?

(a) 250, 5% (b) 300, 5%
(c) 200, 8% (d) 350, 3.5%

Q.29. Amul, Binit and Chand start a business by investing Rs 70,000 that earns them a profit of Rs 42,000 at the end of the year. Amul invests his share in the profit in a scheme that gives him 10% interest compounded annually and Binit invests his share in a scheme that gives him 20% interest compounded annually. Amul gets Rs 2,520 as interest at the end of 2 years and Binit gets an interest of Rs 4,200 at the end of one year. Find Chand's investment in the business.

(a) Rs 1,60,000 (b) Rs 15,000
(c) Rs 17,520 (d) Cannot be determined

Q.30. A food processing unit received fresh consignment of berries from Himachal Berry Corporation containing 'a' kilogram of berries. After analysis these berries were found to contain 99% water. After 15 days, sample of berries from the same lot was found to contain 98% water. What must be the weight of same lot at that time?

(a) $\frac{a}{2}$ kg (b) $\frac{2a}{3}$ kg
(c) $\frac{4a}{5}$ kg (d) $\frac{a}{3}$ kg

Q.31. Kishore is an employee of a MNC firm. He needs Rs 10 lakhs for buying a new flat in Patel Nagar. His company is offering housing loan at an interest rate of 15% for 5 years. The two banks ICICI and HFDC are offering loan for 5 year period at an interest rate of 18% and 13% respectively. How much should Kishore borrow from the two banks so that he can pay the interest rate same as that of his own MNC firm?

(a) 5 lakhs from HDFC and 5 from ICICI
(b) 3 lakhs from HDFC and 7 from ICICI
(c) 6 lakhs from HDFC and 4 from ICICI
(d) 8 lakhs from HDFC and 2 from ICICI

Q.32. The population of towns A and B is in the ratio of 1:4. For the next 2 years, the population of A would increase and that of B would decrease by the same percentage every year. After 2 years, their population became equal. What is the percentage change in the population?

(a) 33.33% (b) 66.66%
(c) 25% (d) Not possible

Direction for questions 33 to 35: Read the passage below and solve the questions based on it.

Petroleum product sales (in million tones)

Petroleum product	*2002-03*	*2003-04*
Diesel	36.0	36.6
Petrol	7.6	7.9
LNG	8.3	9.3
Crude oil	11.9	11.7
ATF	2.3	2.5
Total	**104.2**	**107.4**

Choose (a) if only A is true.
Choose (b) if only B is true.
Choose (c) if both A and B are true.
Choose (d) if neither A nor B is true.

Q.33. A. The percentage share of ATF in petroleum product sales has increased in 2003-2004 over the previous year.
B. Percentage share of diesel in petroleum product sales has decreased in 2003-2004 over the previous year.

Q.34. A. LNG has experienced the maximum percentage increase in petroleum product sales in 2000-2004 over the previous year.
B. Change in percentage shares of crude oil in petroleum product sales in 2003-2004 over the previous years is approximately 0.2%.

Q.35. A. Percentage decrease in crude oil sales in 2003-2004 over the previous year is more than percentage increase in diesel sales during the same period.
B. LNG sales in 2003-2004 is less than 25% of diesel sales in 2003-2004.

Direction for questions 36 to 38: Read the passage below and solve the questions based on it.

In each question, there are two statements—A and B, either of which can be right or wrong on the basis of the information given below

Direct taxes collected in different years (Rs in crores)

Taxes	*1998-99*	*1999-00*	*2000-01*	*2001-02*	*2002-03*
Corporation tax	22,750	23,460	25,190	27,080	28,880
Income tax	18,210	20,460	23,790	26,110	29,080

Taxes	*1998-99*	*1999-00*	*2000-01*	*2001-02*	*2002-03*
Service tax	220	240	200	230	260
Expenditure tax	300	320	340	350	345
Interest tax	1,280	1,190	1,350	1,410	1,490
Water tax	8	15	20	22	25

Mark your answer as
(a) if only A is right.
(b) if only B is right.
(c) if both A and B are right.
(d) If both A and B are wrong.

Q.36. A. The growth rate for Service Tax from 1998-99 to 1999-2000 was higher than that for income tax for the same period.
B. The growth rate for Service Tax from 1998-99 to 1999-2000 was lower than that for the period 2001-02 to 2002-03.
(a) 4 (b) 2 (c) 3 (d) 1

Q.37. A. The ratio of income tax to corporation tax was higher in 1998-99 than in 1999-2000.
B. Service Tax, as a percentage of expenditure tax, was greater in 2000-01 than in 2001-02.
(a) 1 (b) 2 (c) 3 (d) 4

Q.38. A. The average annual growth rate for income tax during the period 1998-99 to 2002-03 was more than 1.5 times the same for corporation tax during the same period.
B. The average annual growth rate for interest tax was more than 5% during the given period.
(a) 1 (b) 3 (c) 2 (d) 4

Direction for questions 39 and 40: Read the passage below and solve the questions based on it.

Abhay, Bhaskar, Chitra and Danny work in the same office. Abhay's salary is 20% less than Bhaskar's salary. Chitra's salary is 25% more than Abhay's salary. Danny's salary is 20% more than Abhay's salary.

Q.39. If Abhay has a salary of Rs 10,000 per month, who has a salary of Rs 12,500 per month?
(a) Bhaskar (b) Chitra
(c) Danny (d) 1 and 2

Q.40. If Chitra received a raise and her new salary is 10% higher than Danny's, what is her new monthly salary? (use the data from the previous question).
(a) Rs 12,750 (b) Rs 13,200
(c) Rs 13,333 (d) Rs 14,225

Direction for questions 41 to 44: Read the passage below and solve the questions based on it.

There are two statements given in each question: A and B, either of which can be true or false on the basis of the information given below.

Following table gives us the detailed information regarding the admission process of a reputed B-School IIM-Jehanabad for the year 2003 and 2004.

Year	*Gender*	*Number of applicants*	*Number of candidates appeared for written test*	*Number of candidates called for an interview*	*Number of candidates selected for the course*
2003	Male	6,100	5,900	60	12
	Female	1,900	1,500	15	5
2004	Male	6,300	6,000	64	13
	Female	4,000	3,500	30	6

Choose (a) if only A is true
Choose (b) if only B is true
Choose (c) if both A and B are true
Choose (d) if neither A nor B is true

Q.41. A. The success rate of moving from the written test to interview stage for females was worse than for males in 2003.
B. The success rate of moving for the written test to interview stage for males was better in 2003 than in 2004.

Q.42. A. The percentage of absentees in the written test among males decreased from 2003 to 2004.
B. The percentage of absentees in the written test in 2003 among females was larger than that among females in 2004.

Q.43. A. The percentage of the female candidates selected in 2004 from female applicants is more than the percentage of male candidates selected for year 2004 out of total male applicants of that year.
B. In 2003, from among those called for an interview, the males had a greater success rate than females.

Q.44. A. Difference in the total number of male and female applicants is more than 50% of the total candidates who appeared for the written test in 2004.

B. The total number selected for the job for both the years is more than 70% of the total female candidates called for an interview in both the years.

Q.45. In a laboratory, equal number of rats were injected with solution A. In a week, 55% of rats injected with solution A got infected with a disease and of that, equal number of rats were treated with vaccine C and D. Now, percentage of rats treated with vaccine C who recovered was 70% and that with vaccine D was 50%, while others died. Calculate the number of rats on whom the experiment was carried out in the laboratory, if the number of rats who died even after being treated with vaccines due to the disease is 77.
(a) 600 (b) 700
(c) 750 (d) 800

ADVANCED

Direction for questions 1 to 3: Read the passage given below and solve the questions based on it.

The table gives the production of five agricultural crops between 1999-2000 and 2000-2001.

Crops	*Target for 2001-2002 (in MT)*	*Output for 2000-2001 (in MT)*	*% increase in output over 1999-2000*
Rice	162.7	160.4	25
Wheat	18.9	16.2	18
Tea	20.5	22.5	35
Cotton	14.7	18.9	32
Coffee	12.3	14.8	16

Q.1. The production of rice in 1999-2000 (in MT) is approximately
(a) 124 (b) 128
(c) 132 (d) 136

Q.2. Output of which crop in 2000-01 is showing maximum percentage difference from the target set in 2001-02?
(a) Wheat (b) Coffee
(c) Cotton (d) Tea

Q.3. A new parameter is invented to judge the target: output ratio. That parameter is known as efficacy ratio and is defined as the ratio of target for 2001-02 and output for 2000-01. For how many given crops the efficacy ratio is more than 1?
(a) 1 (b) 3 (c) 2 (d) 4

Q.4. Some people form a joint account for one year with a condition that "each person will deposit an amount equal to the number of people in the account that month. Also the person who withdraws from the account in between will get their amount at the end of the year." After 6 months, 25% of the people withdrew and 33.33% of the remaining withdrew after 3 months. At the end of the year, there was an amount of Rs 4,860 in their account before paying off the withdrawn amounts. Find the number of people in the beginning of the year.
(a) 12 (b) 24
(c) 36 (d) None of these

Direction for questions 5 to 9: Read the passage below and solve the questions based on it.

The number of taxi drivers and auto drivers working in three shifts are given below:

Vehicle	*Shift*	*Number of drivers*	*Working hours*	*Total collection (in 1,000 Rs)*
Taxi	M1	425	10	132.1
	M2	672	8	240.2
	M3	135	6	42.3
Auto	M1	350	9	57.3
	M2	425	8	100.8
	M3	95	7	18.9

A taxi driver charges Rs 13 for every 1.6 km, while an auto driver charges Rs 8 for every 1.3 km. For the night shift, charges are 1.5 times the normal rate.

Q.5. By what percentage is the average collection per taxi driver more during the M2 shift than during the M1 shift?
(a) 4% (b) 15% (c) 34% (d) 82%

Q.6. What is the average distance travelled by the auto drivers during the M3 shift?
(a) 1,356 km (b) 2,045 km
(c) 3,054 km (d) 2,672 km

Q.7. Who get the highest returns for their efforts with respect to work hours?
(a) Auto Drivers — M3 Shift
(b) Taxi drivers — M2 shift

(c) Taxi drivers — M3 shift
(d) Auto drivers — M1 shift

Q.8. What is the maximum difference between the average distances covered in an hour by drivers of any two shifts?
(a) 3,100 km (b) 3,300 km
(c) 3,500 km (d) 3,400 km

Q.9. Which of the following is false?
(a) Lowest contribution per driver is by auto drivers in the M1 shift
(b) Lowest returns for the efforts are for auto drivers in the M1 shift
(c) The percentage of drivers during the M3 shift is 21% of the total drivers
(d) None of these

Q.10. Consider a big cube formed of 64 smaller cubes of which 40 are coloured. The smaller cubes are arranged such that the exposure of the coloured cubes to the outside is minimized. What is the percentage of the exposed area that is coloured?
(a) 62.5% (b) 50%
(c) 41.6% (d) None of these

Q.11. On May 1, 2002, a person sold all his investments for Rs 10,8,000, incurring a loss of 10% with respect to the previous month. On April 1, 2002 he had invested all his money in shares, which represented a 20% increase over the sum he had on March 1, 2002 which represented a 20% decrease over the money he had on February 1, 2002 which represented a 25% increase over the money he had on January 1, 2002. The person had maximum money in
(a) February 2002 (b) January 2002
(c) April 2002 (d) None of these

Direction for questions 12 and 13: Read the passage below and solve the questions based on it.

Patna and Lucknow are the only cities of India which are having only adults. Female population in Patna grows by 50% to 60,000, but maintaining the same male to female ratio as previous year. Population in Lucknow increased by 12.5% to 90,000, but the ratio of male to female changed by 1:1 to 4:5. Male population in Patna was 10% more than the male population in Lucknow in the previous year.

Q.12. What is the female population in Lucknow in the second year?
(a) 55,000 (b) 45,000
(c) 4,00,000 (d) 50,000

Q.13. What is the total population in Patna after the increase (i.e., in the second year)?
(a) 80,000 (b) 1,00,000
(c) 20,000 (d) 26,000

Q.14. The owner of a flower shop follows a particular pattern for his business. During a period of inflation, he raises his price by P% and during a slowdown he decreases his existing prices by P%. After a year in which there was inflation first, followed by a slowdown, the cost of a red-rose bouquet decreases by Rs 162. After another year, in which there was inflation once more followed by a slowdown, the cost of this bouquet reduced by a further Rs 147.42. What was the original price of the red-rose bouquet?
(a) Rs 1,500 (b) Rs 1,000
(c) Rs 1,800 (d) Rs 1,111

Direction for questions 15 to 17: Read the passage below and solve the questions based on it.

Three leading drug manufacturers, Arpit Pharma, Bittu Pharma and Chintu Pharma each launch an antiseptic drug. Each company uses all or some of the 5 constituents A, B, C, D and E. The table below gives the percentage composition of these constituents in the respective drugs manufactured by different companies.

	A	*B*	*C*	*D*	*E*
Arpit Pharma	12	15	20	28	25
Bittu Pharma	37	15	13	15	20
Chintu Pharma	26	15	10	34	15

The above values are in percentage of composition. All manufacturers produce their drug as a capsule of 300 mg.

Q.15. The effectiveness of constituents A, B, C and D are in the ratio 5:6:4:3, while E is just a sweetener and does not have any curing effect. Which of the following company's capsule is most effective?
(a) Arpit Pharma (b) Bittu Pharma
(c) Chintu Pharma (d) Data Insufficient

Q.16. If the side effects caused by drugs are proportional to the ratio of C's composition to the sum of B's and E's composition, which drug has least side effect?
(a) Arpit's drug
(b) Bittu's drug
(c) Chintu's drug
(d) All of them have equal side effect

Q.17. Arpit Pharma finds that its sales are dipping because its drug tastes too sweet. So, it decides to decrease the composition of E and increase that of C, keeping other composition unchanged. It also wants to make sure that the side effect is not more than double its previous value. What could be the maximum percentage of C's

composition in Arpit's Drug? (Use data from the previous questions)
(a) 25% (b) 30% (c) 34% (d) 37%

Q.18. If 3000 capsules are produced by these 3 companies per annum, the consumption of component C annually would be
(a) 60 kg (b) 90 kg
(c) 105 kg (d) Data Insufficient

Q.19. In Ranchi, only two newspapers *Dainik Jagran* and *Prabhat Khabar* are published. It is known that 25% of the city population reads *Dainik Jagran* and 20% reads *Prabhat Khabar* while 8% reads both *Dainik Jagran* and *Prabhat Khabar*. It is also known that 30% of those who read *Dainik Jagran* but not *Prabhat Khabar* look into advertisement and 40% of those who read *Prabhat Khabar* but not *Dainik Jagran* look into advertisements while 50% of those who read both *Dainik Jagran* and *Prabhat Khabar* look into advertisements. What is the percentage of the population who read an advertisement?
(a) 10.8% (b) 11.3%
(c) 13.9% (d) 12.7%

Direction for questions 20 to 22: Read the passage below and solve the questions based on it.

The following table gives the distribution of number of B-schools out of a sample of 100 B-Schools based on the scores obtained by them on five different parameters—Infrastructure, Placements, Research, Industry Interface and Intellectual Capitals. The maximum possible score in any of these parameters is 100 points.

Score/ Parameter	*80 points and above*	*60 points and above*	*40 points and above*	*20 points and above*
Infrastructure	16	30	50	65
Placements	12	28	37	50
Research	30	50	70	92
Industry Interface	24	42	57	73
Intellectual Capitals	27	43	47	70

For example, 16 out of 100 schools surveyed have scored 80 points or above in the parameter Infrastructure.

Q.20. What is the maximum number of business schools which could have scored 40 points or more but less than 60 points in at least three of the five parameters?
(a) 20 (b) 18
(c) 12 (d) None of these

Q.21. What is the minimum number of business schools which could have scored 40 points or more but less than 80 points in at least two of the five parameters?
(a) 10 (b) 5
(c) 12 (d) None of these

Q.22. Which of the following is a possible figure for number of schools which scored 20 points or more but less than 80 points in each of the five parameters?
(a) 38 (b) 36
(c) 42 (d) None of these

Q.23. In D-segment car market, only three competitors (Honda, Toyota and Hyundai) exist. Last year the sales of firm Toyota were 10% more than Honda. In a year, both the firms Honda and Toyota increased their respective sales by 20%. This year, the sales of the firm Toyota are five times that of Hyundai. How much were the sales of the firm Hyundai last year, if the total sales remained constant over the two years period?
(a) 46% (b) 25%
(c) 40.33% (d) 38.67%

Q.24. Chintu is given a quadratic equation $ax^2 + bx + c = 0$ and is asked to make another quadratic equation from this with $a = 1$. Also one root of the second quadratic equation is same as one of the roots of the first equation but opposite in sign and the other root of the second equation is two times the second root of the first equation. Find the percentage change in the constant term of the second equation as compared to the first equation?
(a) 200% increase (b) 300% decrease
(c) 400% increase (d) 100% decrease

Q.25. There are two investment schemes offered by two different financial institutions X and Y as follows
Scheme X: Rate of interest = 10% per annum and period of maturity = 2 years.
Amount is returned with compound interest compounded annually.

Scheme Y: Rate of interest = 11% per annum amount is returned with simple interest.
For what duration, should I invest Rs 25,000 in scheme Y, so as to get same amount in return if I had invested in scheme X?
(a) $\frac{21}{11}$ Years (b) $\frac{31}{11}$ Years
(c) $\frac{19}{7}$ Years (d) $\frac{19}{7}$ Years

Direction for questions 26 to 28: Read the passage below and solve the questions based on it.

Given below is the conversion rate of some of the currencies with respect to rupees.

	USD	*Pound*	*Euro*	*Canadian Dollar*	*Australian Dollar*
Rs Price/ unit	Rs 44	Rs 66	Rs 54	Rs 32	Rs 36

Q.26. Mr Kumar Kalyan Prasad Sinha is having two Pounds with him. How many Euros can he purchase for two Pounds?
(a) 2.33 (b) 2.44 (c) 2.55 (d) 2.66

Q.27. Value of USD is expected to rise by 10% against Rupee next year whereas value of Rupee is expected to decline by 10% against Canadian Dollar next year. How many USD can be purchased for 5 Canadian Dollars?
(a) 3.67 (b) 2.97
(c) 4.27 (d) None of these

Q.28. Mrs Sanjila Sinha has Rs 1,000 with her and her husband Mr Kumar Kalyan Prasad Sinha has Rs 2,000 with him. They want to purchase maximum number of currency notes given above but they are not ready to combine their money. What will be the maximum number of currency notes, as given above other than rupees which they can have?
(a) 62 (b) 94 (c) 93 (d) 32

Direction for questions 29 and 30: Read the passage below and solve the questions based on it.

The table given below shows the break-up of the percentage of people of different age groups frequenting bars in 4 different metro cities viz., Delhi, Hyderabad, Bangalore and Patna in the year 2002.

Q.29. Which city has the maximum number of people, upto 25 years of age, frequenting bars?

Cities	*Percentage break-up for age groups (Years) in 2002*						
	Up to 15	*15-20*	*20-25*	*25-30*	*30-35*	*35-40*	*Above 40*
Delhi	8	13	24	21	11	17	6
Hyderabad	3	8	35	23	10	16	5
Bangalore	4	21	27	11	8	14	15
Patna	1	7	43	32	9	5	3

(a) Bangalore (b) Patna
(c) Both 1 and 2 (d) Cannot be determined

Q.30. Which age group contributes to the maximum number of people frequenting bars in given cities?
(a) 20-25 (b) 25-30
(c) 30-35 (d) Cannot be determined

Direction for questions 31 to 34: Read the passage below and solve the questions based on it.

Five friends– Akshay, Binod, Chandu, Dev and Eeshwar– went on a shopping trip. Before shopping, the first person had Rs 400, the second person had 25% less than the first person, the third person and the fourth person had 50% less than the first person and the fifth person had 66.66% less than the second person. While shopping they did not lend or borrow from each other. After the shopping was over, it was observed that they were left with Rs 165, Rs 95, Rs 70, Rs 40 and Rs 10, not necessarily in that order. We don't know who is the first or the second or onwards. Following is known about the money they started with, they spent, or they were left with:

i. Akshay started with more money than Dev.
ii. Binod spent Rs 15 more than Chandu.
iii. Eeshwar started with more money than just one another person of the group.
iv. Akshay spend the most but did not end with the least.
v. Chandu started with 66.66% of the money that Binod started with.
vi. Dev spend the least and ended with more than Akshay and Chandu.
vii. Eeshwar spent Rs 35.

Q.31. Who ended with the maximum amount of money?
(a) Akshay (b) Binod
(c) Chandu (d) Eeshwar

Q.32. How much money did Akshay spend?
(a) Rs 205 (b) Rs 190
(c) Rs 35 (d) Rs 360

Q.33. In ascending order of spending, Eeshwar would rank at which position?
(a) 1 (b) 2 (c) 4 (d) 5

Q.34. Who ended with Rs 40?
(a) Akshay (b) Binod
(c) Chandu (d) Dev

Q.35. In a certain town, at least 50% of the people read a newspaper. Among those who read a newspaper, at most 25% read more than one paper. Only one of the following statements follows from the statements given below. Which one is it?
(a) At the most 25% read exactly one newspaper
(b) At least 25% read all the newspaper
(c) At the most 37½% read exactly one newspaper
(d) At least 37½% read exactly one newspaper

Q.36. Two vessels contain equal quantities of 40% alcohol. Anil changed the concentration of the first vessels to 50% by adding extra quantity of pure alcohol. Balu changed the concentration of the second vessels to 50% replacing a certain quantity of the solution with pure alcohol. By what percentage is the quantity of alcohol added by Anil more than that replaced by Balu?
(a) 20% (b) 25%
(c) 40% (d) Cannot be determined

Direction for questions 37 to 40: Read the passage below and solve the questions based on it.

Six students Abhay, Bhupi, Chandi, Deb, Ellaraputtu and Fastinder appear in CAT 2006. Below is given the percentage marks obtained by them in different headings DS, DI, QA, EU, RC, GD and PI. (GD and PI are the part of CAT 06)

Its also known to us that each of DS, DI, QA, EU and RC carry same weightage. And the weightage given to PI is same as weightage given to GD.

Maximum marks in CAT 06 is 250, including GD and PI.

	Written Exam					*GD*	*PI*	*Total*
	DS	DI	QA	EU	RC			
A	45	50	40	90	80		75	62.4
B	55	60	65	70	75	80		66.4
C	75	72.5		65	82.5	88	84	
D	80	70	85	65	60	84	76	
E	90	60	80	60	50	88	72	70.4
F	85	80	75	45	70	72		

Q.37. What is the score obtained by Abhay in GD?
(a) 15 (b) 16 (c) 17 (d) 18

Q.38. If Deb has scored the maximum among the six students, what can be the maximum score that Chandi can get in QA?
(a) 20 (b) 21 (c) 22 (d) 23

Q.39. What is Bhupi's percentage score in PI?
(a) 42 (b) 56 (c) 64 (d) 72

Q.40. If Fastinder and Chandi end up with the same score, what is the difference between Chandi' QA score and Fastinder's PI score?
(a) 0 (b) 1 (c) 4 (d) 7

TRUE/FALSE

Q.1. Length of a rectangle is increased by 30% and its breadth is increased by 10%. Perimeter of this rectangle can increase by 25%.

State whether True or False.

Q.2. Length of a rectangle is increased by 30% and its breadth is increased by 10%. Perimeter of this rectangle can increase by 32%.

State whether True or False.

Q.3. Length of a rectangle is increased by 40% and its breadth is increased by 20%. Percentage increase in the perimeter of the rectangle over the previous perimeter depends upon the dimension of the rectangle.

State whether True or False.

Q.4. Length of a rectangle is increased by 20% and its breadth is increased by 10%. Percentage increase in the area of the rectangle over the previous area depends upon the dimension of the rectangle.

State whether True or False.

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(d)	(c)	(a)	(c)	(d)	(c)	(d)	(d)	(d)	(c)	(b)	(d)	(d)	(a)	(d)	(c)	(a)	(c)	(b)
Q. No.	21	22	23																	
Answer	(b)	(d)	(b)																	

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(b)	(c)	(a)	(a)	(b)	(a)	(c)	(c)	(d)	(b)	(c)	(b)	(c)	(d)	(c)	(d)	(b)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(a)	(d)	(c)	(b)	(d)	(d)	(a)	(b)	(b)	(b)	(a)	(b)	(d)	(c)	(b)	(d)	(c)	(d)	(d)	(b)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(d)	(c)	(c)	(b)	(d)	(b)	(a)	(c)	(c)	(b)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(b)	(a)	(d)	(c)	(d)	(c)	(d)	(d)	(a)	(c)	(b)	(c)	(b)	(b)	(d)	(a)	(b)	(a)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(c)	(d)	(a)	(b)	(b)	(a)	(b)	(b)	(b)	(a)	(c)	(a)	(c)	(a)	(a)	(c)	(b)	(b)	(d)	(b)
Q. No.	41	42	43	44	45															
Answer	(a)	(b)	(d)	(b)	(b)															

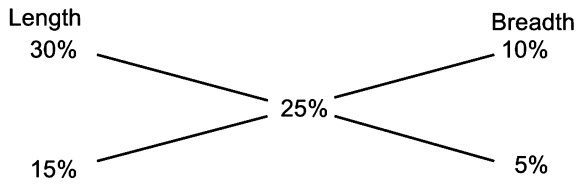
■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(c)	(c)	(b)	(b)	(b)	(b)	(d)	(c)	(c)	(a)	(d)	(d)	(c)	(b)	(c)	(b)	(d)	(c)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(d)	(b)	(b)	(a)	(b)	(a)	(c)	(d)	(a)	(d)	(d)	(b)	(a)	(d)	(a)	(b)	(c)	(c)	(b)

■■■ True/False

1. True

   Increase in perimeter has to be in between 10% and 30%. Hence, 25% increase in perimeter is possible. Actually it's a simple question of alligation. Ratio between length and breadth of rectangle should be 3:1.



   Hence, the ratio between length and breadth = 15 : 5 = 3 : 1

2. False

   Taking a cue from the above problem, 32% increase in perimeter is not possible.

3. True

   Percentage increase in the perimeter of rectangle can be any value between 20% – 40%, and it depends upon the dimension of the rectangle (i.e., the length and breadth).

4. False

   Percentage increase in the area of rectangle does not depend upon the dimension of the rectangle and it remains constant for all the values of length and breadth. In this case, percentage increase in the area = 32%.

HINTS AND SOLUTIONS

Warm up

1. $\frac{4}{7} = 4 \times \frac{1}{7} = 4 \times 14.28 = 57.12\%$

2. $\frac{50}{100} \times \frac{40}{100} \times 1200 = 240$

3. $\frac{a}{100} \times a + \frac{b}{100} \times b = \frac{2}{100} ab$

   $a^2 + b^2 = 2ab$ and $(a-b)^2 = 0$

   a= b

4. $x - 0.34\ x = Nx$

   or, $0.66x = Nx$

   $N = 0.66$

5. $N - 0.48\ N = 0.52\ N$

6. $\frac{900}{7200} = \frac{1}{8} = 12.5\%$

7. $0.61x - 0.42x = 38$

   or, $0.19x = 38$

   $x = 200$

8. Let Length = 100, Breadth = 100

   Hence, Area = 10000

   Now, Length = 130, Breadth = 75

   Hence, Area = 130 × 75 = 9750

   Required Percentage $= \frac{250}{10000} \times 100$

   2.5 % decrease

9. Let number = 100

   1st 20% increment, No becomes = 120

   2nd 20% increment, No becomes = 144

10. Let 3rd No = 100

    1st No. = 30 and 2nd No. = 40

    Hence, 1st is 75% of 2nd No.

11. Let 3rd No. = 100

    1st No. = 85

    2nd No. = 80

    Required Percentage $= \frac{85}{80} \times 100 = \frac{425}{5} = 106\frac{1}{4}\%$

12. Alcohol in solution = 20% of 50 l = 10 l

    Concentration of alcohol in new solution

    $= \frac{16}{40} \times 100 = 16\frac{2}{3}\%$

13. Assume that he had initially Rs 100.

    After spending on food, money left = Rs 75

    After spending 50% of Rs 75, he had Rs 37.5

    Money he had initially = $\frac{100}{37.5} \times 37.5 \times 10 =$ Rs 1000

14. Monthly expenditure = 70% of 4000 = Rs 2800

    Expenditure in a year = Rs 2800×12 = Rs 33600

15. Let Initial Amount = Rs 100

    Money left = 100 – (20+10+9+7)

    = Rs (100–46)

    = Rs 54

    Money he had $= \frac{100}{54} \times 2700 =$ Rs 5000

16. Let Income = Rs 100

    Expenditure = Rs 60

    Savings = Rs 40

    Net Income = Rs 140

    New Expenditure = Rs 84

    New Savings = Rs 56

    Require increment = $\frac{16}{40} \times 100 = 40\%$

17. Require Increment

    $$\frac{3 - \frac{1}{2}}{\frac{18+2}{9}} = \frac{\frac{5}{2}}{\frac{20}{9}} = \frac{5}{2} \cdot \frac{9}{20} = \frac{9}{8}$$

18. Since, we have to find the minimum percentage, it can be possible that no one has failed in both the subjects.

19. The best way of solving this question would be the method of elimination, i.e., going through the options.

    Obviously, option (d) cannot be the answer, because if 500 people are voting in against the resolution, then motion cannot be passed.

    Checking option (a): If 400 people are voting against the resolution for the first time, then 500 are voting in favour. Now, in the second voting,

persons voting against the resolution would be = 600, so persons voting in favour = 200

In the first voting, resolution was passed by 100 votes and in the second voting, resolution was defeated by 400 votes. And it is 300% more than 100.

So, the answer is option (a).

20. $S.I.=\left(\frac{7}{3}-1\right)=\frac{4}{3}$

$S.I.=\left(\frac{7}{3}-1\right)=\frac{4}{3}$

$R=\frac{40}{3}=13\frac{1}{3}\%$

21. $S\left(\frac{P\times x\times 2}{100}\right)=\frac{P\times 2x\times(y+2)}{100}$

$y=3$

22. $\frac{x\times 8\times 2}{100}+\frac{x+10+3}{100}+\frac{x\times 6\times 3}{100}=12800$

$0.64\,x=12800$

$x=20000$

23. $\text{CI}=P\left(1+\frac{20}{100}\right)^n=p\left(\frac{6}{5}\right)^n$

When, n = 4, only then condition given is satisfied.

Foundation

1. The numerator has been increased by 25%. Hence, denominator will also increase by 25%.

   Value in tercentage = 60 × 1.25 = 75 tercentage

2. Numerator has become 10 times, hence denominator will also become 10 times.

3. There will be no change in the measurement of angles. It's the size that will increase, and not the slope or angle.

4. No. of twins cases out of 100 = 5

   No. of twins out of 100 children = 10

   Percentage of twins = $\frac{10}{100}\times 100=\frac{100}{11}=9.5\%$

5. Price Quantity

   $\uparrow\frac{1}{5}$ $\frac{1}{6}\downarrow$

   Or, $\frac{1}{6}x=4$

   $X=24$

   Initially price per kg = $\frac{120}{24}$ Rs 5/kg

6. Going through choices is the best way. (a) is the answer.

8. Salary of P and Q = x and salary of R = y then

   $\frac{8x}{100}-\frac{7x}{100}=74$ and $x=7400$

   then, donation by R = Rs 585

   then $\frac{9}{100}y=585$

   $y=\frac{585\times 100}{9}=$ Rs 6500

   Required percentage = $\frac{1695}{21800}\times 100=7.95\%$

10. Weight of solution = (1+4) = 5 units

    weight of sugar = 1 unit

    weight of water after heating = 1.5 units

    Required ration = 1 : 1.5 = 2 : 3

11. It can be done by either of the methods of compound interest or straight line method of solving questions.

    6 gets increased by 1000% = Final Value = 6 + 60 = 66

    60 gets increased by 1000% = Final Value = 66 + 660 = 726

12. Turnover in 1998 = Rs 100 cr

    Turnover in 1999 = Rs 300 cr

    Turnover in 2000 = Rs 450 cr

13. $\frac{40}{60}=\frac{75-70}{70-x}$ and $\frac{2}{3}=\frac{5}{70-x}$

    and $140-2x=15$

    $2x=125$

    $x=62.5\%$

14. Original price = Rs 100

    Expenditure = Rs 50×100= Rs 5000

    Expenditure after rise = Rs 5600

    Then, $140\times x=5600$

    $X=40$

15. Hours × wage = earnings

    Previously, $x\times y=xy$

    $\frac{5}{4}x\times\frac{6}{5}y=\frac{3}{2}xy$

    Increment = $\frac{\frac{3}{2}xy}{xy}\times 100=\frac{\frac{1}{2}xy}{xy}\times 100=50\%$

16. Decrease in A's income = Rs 15000

    Increase in B's income = Rs 15000

    Required ratio = $\frac{1500}{15000}\times 100=100\%$

17. 0.18A + 0.15B +0.19C = 0.17A+0.17B+0.17C

    0.01A – 0.02B+0.02C=0 (1)

    A – B = 500 (2)

    A – C = 3400 (3)

Solving 1, 2 and 3 we get
A = 5800, B = 5300 and C = 2400
A + B + C = 13500

18. No. of candidates failed = $\frac{35}{100}\times 51600 = 18060$

No. of candidates passed with honour

$= \frac{15}{100}\times 33540 = 5031$

Note: 15% of the remaining

No. of students failed to obtain honour but got passed = 28509

19. Let A has x guavas and B has y guavas, the

$x-\frac{1}{4}x=\left(y+\frac{1}{4}x\right)+2 \qquad (1)$

and $y-\frac{7}{10}y=x+\frac{1}{10}y-4 \quad (2)$

Solving 1 and 2, we get

$x = 44 \quad y = 20$

21. Let n = number of components attached, V_n = speed of train when 'n' compartments are attached

$V_0 = 50$

$V_1 = 45$

$V_2 = 40.5$ and so on

At $n = 7$, we get the answer.

22. x = number of participants

total fund = $50x$

then, $\frac{3}{5}$ x participates contributed $40x$

Required average $= \frac{10x}{\frac{2}{5}x} = \frac{50}{2} =$ Rs 25

23. Let the original price = Rs 100

Price after 1st year = Rs 125

Price after 2nd year = Rs 100

Hence, after 180 year, there will be no change in the price

24. Salary = Rs 100

Savings = Rs 10, Expense = Rs 90

Next year, Expense = Rs 108,

90% of salary = Rs 108

Salary = $\frac{10}{90}\times 10$ Rs 120

Hence, 20% increase in salary

25. 15% failed in both the subjects ⇒ 85% passed in at least one of the subjects.

So, percentage of students passing both the subjects
= 80 + 70 – 85 = 65%

So, 65% = 390 ⇒ 100% = 600

26. Salary of Dheeraj = Rs 100

Salary of Anil = Rs 80

Salary of Vinit = Rs 70

Required percent $= \frac{10}{70}\times 100 = 14.28\%$

27. Let adult males = x and adult females = y

then $0.55y = 0.4x$

$x = \frac{0.55y}{0.40} = \frac{11}{8}y$

Now, required percentage $= \frac{0.4x+0.55y}{x+y} = 46.32\%$

28. Sona spends = $\frac{3}{25}$ S and Ellaraputtu spends

$= \frac{3}{25}$ S

Let Ellaraputtu spends = x% of her income

Then x % of $E = \frac{3}{25}x$

or $x = 12\frac{S}{E}$

30. No. of voters on the voter's list = x

No. of voters participated = $x-\left(\frac{x}{10}+60\right)$

$= \frac{10x-x-600}{10} = \frac{9x-600}{10}$

then $\frac{47}{100}x+\left(\frac{47x}{100}+380\right)+\left(\frac{9x-600}{10}\right)=x$

We get x = 6200

31. Percentage of rainy day = 10%

Percentage of days having rainbows = 5%

Hence, required percentage = 95%

32. $x + y = 180 \qquad (1)$

$x - 3y = 0 (2)$

Solving the equations, we get $x = 135$, $y = 45$

33. Price Quantity

$\uparrow \frac{1}{5}$ $\downarrow \frac{1}{6}$

Hence $x/6 = 2$, or, $x = 12$

34. 10×100=1000, 100 = no. of visitors

Now, 7.5 × No. of visitors = 1200

No. of visitors = 160

35. $6000 = 13350\left(1-\frac{125}{1000}\right)^n$

$\frac{600}{1335} = \left(\frac{7}{8}\right)^n$

36. $\frac{15}{100}\cdot\frac{9x}{10} = 47.25$

$1x = 350$

38. $x - 200 = \frac{x}{2} \rightarrow x = 400$

A got 400 votes and B got 200 votes

But is information 18 given regarding invalid votes.

So (d) is the answer.

39. Population after 2000 = 3244800

Population after 2001 = 2985216

Population at the end of 2003 = 3228810

40. $\frac{\frac{30}{100}x+15}{x+30}\times 100 = 45$

$x = 10$

41. Money spend in buying things = Rs 63000

Money kept in bank = Rs 27000

Money he got after selling the things = Rs 50400

Money he received from bank = Rs 35707.5

Loss = Rs 3892.5

Loss % = 4.32%

42. $\frac{378}{125} = 3 + \frac{3}{125} = 302.4\%$

Let original salary be Rs 100

And now going through option, we get (c) as answer

43. Sugar in original solution = $\frac{4}{100}\times 6 = 0.24$

Required percentage $= \frac{0.24}{5}\times 100 = 4.8\%$

44. Let B's Income = Rs x

A's Income = Rs $\frac{3}{5}x$

And B's expenditure = Rs y

A's expenditure = Rs $\frac{7}{10}y$

Also, $\frac{3}{5}x = \frac{3}{4}\cdot\frac{7}{10}y$

$$\frac{\text{A'savings}}{\text{B'savings}} = \frac{x-y}{\frac{3}{5}x-\frac{7}{10}y} = \frac{\frac{7}{8}y-y}{\frac{3}{5}\cdot\frac{7}{8}y-\frac{7}{10}y} = \frac{-y/8}{\frac{21y}{40}-\frac{7}{10}y}$$

$= \frac{5}{25} \approx 1:5$

45. No. of science book initially = $\frac{40}{100}\times 110000 = 44000$

No. of books later on = 130000

No. of science books = $\frac{45}{100}\times 130000 = 58500$

Required no. of books = (58500 – 44000)

= 14500

46. 2 trained labors = 5 untrained labourers

33 untrained labors= $\frac{2}{5}\cdot 3$ trained labours.

= 13.2 trained labours

No. of labour	Days	Hrs	Works
13.2	15	12	1
x	11	9	1.5

$\frac{13.2\times 15\times 12}{1} = \frac{x\times 11\times 9}{1.5}$

$x = 36$

47. y's income = Rs A

x's income Rs $\frac{3}{4}$A

y' expenditure = Rs B

x' expenditure = Rs $\frac{4}{5}$B

Also, $\frac{3}{4} = \frac{9}{10}\cdot$B

A $= \frac{6}{5}$B.

Required ratio $= \frac{\frac{3}{4}\text{A}-\frac{4}{5}\text{B}}{\text{A}-\text{B}} = 1:2$

48. Let she had Rs 100

70 mangoes – Rs 100 and 40 apples – Rs 100

35 mangoes = Rs 50

Rs 15 retained for auto fare

Remaining money = Rs 35

Number of apples = $\frac{40}{100}\times 35 = 14$

50. Let L = 100, B = 100

A = 10000

New length = 120, new breadth = 150

New Area = 18000

Moderate

1. Let MP = Rs 100

CP after 1st discount = Rs 98

CP after 2nd discount = Rs 94.08

But discount cannot be more than 100%

2. Proceeding the solution of 1, we get (b) as the answer.

3. Let B = Rs 100, service tax = Rs 10

Final Bill = Rs 100, Discount = Rs 10

S% = 10%

D% = 9.09

4. $\frac{2}{5}A = \frac{3}{5}B$

$2A = 3B$

5. & 6. Let no. of oranges with Alok = x

Shazmi bought = $\frac{1}{4}x$

Sandeep = $\frac{1}{3}\cdot\frac{3}{4}x = \frac{1}{4}x$

Siddharth bought = $\frac{1}{4}x$

$\frac{1}{4}x = 4$ and $x = 16$

7. Let the class has x students

No. of students receiving grade B = $\frac{1}{3}x$

No. of students receiving B + = $\frac{1}{4}x$

No. of students receiving B – = $\frac{1}{6}x$

No. of students failed = $\frac{1}{8}x$

No. of students receiving

$$A = x - \left(\frac{1}{3}x + \frac{1}{4}x + \frac{1}{6}x + \frac{1}{8}x\right)$$

$$= x - \left(\frac{21x}{24}\right) = 3x/24$$

For (3x/24) to be an integer, min. value of x = 24.

9. Let original price = Rs 100

No. of Units gold = 100

Turn-over = Rs 10000

Now, No. of Units sold = 80

Turn-over = Rs 12000

Price per unit = $\frac{1200}{80}$ = Rs 150

10. Let Rs x is kept in bank

After 3 years money will become

$$x\left(1+\frac{15}{100}\right)^3 = \text{Rs } 1.521x$$

Price of AC after 3 years = Rs 1.52 x

Hence, Ravi will gain.

11. The best way to solve the question is through options.

12. N = 22.1875% of M

$$N = \frac{221875}{1000000}M$$

Values of M and N should be integer. So, Min. value of M+N = 391

13. No. of students that can sing = 30

No. of students, that cannot dance = 8

Maximum no. of students that can neither sing nor dance = 8

14. Using statement ii only we can find the winner. In case of statement i, there may be two contestants with 49% valid votes each and remaining two getting 1% valid votes each.

15. Let total no. of students appeared = x

No. of males students appeared = $\frac{9}{10}x$

No. of female students appeared = $\frac{x}{10}$

Now, $\frac{3}{5}\cdot\frac{9}{10}x + \frac{4}{5}\cdot\frac{x}{10} = 1240$

$\frac{31x}{50} = 1240$

$x = 2000$

16. Hydrogen = 300 gm

or, $\frac{5}{6}$ of water vapor = 300 gm

water vapour = 360 gm = 0.36 kg

17. No. of students taking only science = 4

or, 10% of total students = 4

or, total students = 40

Required percentage = $\frac{7}{40}\times 100 = 17.5\%$

18. Total no. of students = x

No. of boys = $\frac{3}{5}x$

No. of girls = $\frac{2}{5}x$

No. of girls scoring more than 40 marks

$= \frac{4}{5}\cdot\frac{2}{5}x\cdot\frac{8}{25}x;$

total number of students scoring more than 40 marks

$= \frac{3}{5}x$

Required fraction $= \frac{7}{25}\cdot\frac{5}{3} = \frac{7}{15}$

19. The best way of solving this question would be the method of elimination, i.e., going through the options.

Obviously, option (d) cannot be the answer, because if 500 people are voting in against the resolution, then motion cannot be passed.

Checking option (a); If 400 people are voting against the resolution for the first time, then 500 are voting in favour. Now, in the second voting, persons voting against the resolution would be = 600, so persons voting in favour = 200

In the first voting, resolution was passed by 100 votes and in the second voting, resolution was defeated by 400 votes. And it is 300% more than 100.

20. No. of males $= \frac{11}{18}\times 7200 = 4400$

No. of males married $= \frac{40}{100}\times 4400 = 1760$

No. of females married = 1760

Required percentage $= \frac{1760}{2800}\times = 62\frac{6}{7}\%$

21. Original salary = Rs 100

Actual hiked salary = Rs 150

Salary for three months = Rs 450

Salary to be deducted = Rs 90

Required percentage $= \frac{60}{150}\times 100 = 40\%$

22. Let total no. of respondents = 100

People favoured Mondale = 60

People favoured blesh = 40

New No. of people favouring mandale = 58

New No. of people favouring bush = 42

Required per cent = 14%

23. Let the value of Index = 100

Value of Reliance share = 7

Value of HLL = 13

Value of Infosys = 15

Value of remaining = 65

New value of Reliance = 7.63

New value of HLL = 14.3

New value of Infosys = 15.60

New value of remaining = 68.47

Hence, Required percentage $= \frac{3.47}{65}\times 100 = 5.34\%$

24. Let total donation raised = Rs x

No. of people in group = y

then $\frac{4}{5}x = 400y$

$x = 500\,y$

Now, let z = Average donation per person from the new group

$(2y/3)z = x/5$

$z = 150$

25. Total employees = x

Female employees = $\frac{3}{5}x$, male employees $= \frac{2}{5}x$

No. of male employees who earn more than

Rs $25000 = \frac{3}{4}\times\frac{2}{5}x = \frac{3}{10}x$

No. of employees who earn more than Rs 25000 = $\frac{9}{20}x$

No. of female employees who earn more than Rs $25000 = \frac{3}{20}x$

Required fraction $= \frac{\frac{3x}{5}-\frac{3x}{20}}{\frac{3}{5}x} = \frac{3}{4}$

26. The question is based upon the concept of successive discount.

Percentage pollution at the end of one day = 20% (1 – 0.8) = 4%

Percent6age pollution at the end of two days = 20% (1 – 0.8) (1 – 0.8) = 4%

Similarly, go ahead now.

27. M = 1.5E …… (1)

E + M + S + H = 64…… (2)

S = H + 48 …… (3)

$\frac{3}{2}$ E + M + S + H = 70……(4)

Using all the equations, we get

$M + \frac{2M}{3} + S + S - 48 = 64$

$5M + 6S = 336$ ……..(A)

Similarly, using (1), (2), and (4) we get another equation (B).

Solving both (A) and (B), we get (b) as the answer.

28. Amount = x,

rate = r

$$15 = \frac{x \times r \times 1}{100}, \text{or, } rx = 1500$$

$$\frac{(x+100) \times r \times 1}{100} = \left[420 - (x+100)\right]$$

Solving,

we get $x = 300$, $r = 5$

29. $(x + 2520) = x\left(1 + \frac{10}{100}\right)^2$, x = Amul's Profit

$x = 12000$

y = Binit's Profit

$$4200 = \frac{y \times 20 \times 1}{100}$$

$y = 21000$

Chand's profit = Rs 9000

Ratio of their profit

= 12000 : 21000 : 9000

= 12 : 21 : 9 = 4 : 7 : 3

$$\text{Chand's Share} = \frac{3}{14} \times 70000 = \text{Rs } 15000$$

31. Go through the optoins.

32. $$x\left(1 + \frac{r}{100}\right)^2 = 4x\left(1 - \frac{r}{100}\right)^2$$

$$\left(1 + \frac{r}{100}\right) = 2\left(1 - \frac{r}{100}\right)$$

$$\frac{3r}{100} = 1$$

$$r = \frac{100}{3} = 33\frac{1}{3}\%$$

33. For statement A

$$\text{Share of ATF in } 2002 - 03 = \frac{2.3}{104} \times 100 = 2.32\%$$

Hence, A is true

For statement B

$$\text{Share of diesel in } 2002 - 03 = \frac{36}{104.2} \times 100$$

$= 34.55\%$

$$\text{Share of diesel; in } 2003\text{–}04 = \frac{36.6}{107.4} \times 100 = 34.0\%$$

39. Salary of Abhay = Rs 10000

Salary of Bhaskar = Rs 12500

Chitra's salary = Rs 12500

Dany's Salary = Rs 12000

40. Chitra's New Salary = Rs 13200

Advanced

1. $$\frac{1}{5} \times 160.4 = 32.8$$

Production in 99–00 = (160.4-32.8) MT = 128

2. $$\text{Wheat} = \frac{2.7}{18.9} \times 100 = 14.28\%$$

$$\text{Cotton} = \frac{4.2}{14.7} \times 100 = 28\%$$

$$\text{Coffee} = \frac{2.5}{12.3} \times 100 = 20\%$$

$$\text{Tea} = \frac{2}{20.5} \times 100 = 10\%$$

3. $$\text{Rice} = \frac{162.7}{160.4} > 1, \text{ Coffee } \frac{12.3}{14.3} < 1$$

$$\text{Wheat} = \frac{18.9}{16.2} > 1$$

$$\text{Tea} = \frac{20.5}{22.5} < 1$$

$$\text{Cotton} = \frac{14.7}{18.9} < 1$$

4. $6x^2 + \frac{27x^2}{16} + \frac{3}{4}x^2 = 4860$, x = no. of people at the beginning

$x = 24$

5. $$\text{Required percentage} = \frac{357.5 - 310.8}{310} \times 100$$

$$= \frac{46.7}{310.8} \times 100 = 15\%$$

6. $$\text{Required distance} = \frac{18.9 \times 1000}{1.3 \times 95}$$

7. $$\text{Auto drivers M3 shift} = \frac{18.9 \times 1000}{7} = 2700$$

$$\text{Taxi drivers – M2 shift} = \frac{240200}{8} = 30025$$

$$\text{Taxi drivers M3 shift} = \frac{42300}{6} = 7050$$

$$\text{Auto drivers M1 shift} = \frac{57300}{9} = 6366.66$$

11. Jan 2002 = Rs x

Feb 2002 = Rs $\frac{5}{4}x$

March 2002 = Rs x

April 2002 = Rs 1.2 x

May = Rs $\frac{54x}{50}$

12. Female population in Patna in the previous year = 40,000

For Lucknow

$x+\frac{1}{8}x=90000 \quad \frac{9x}{8}=90000$

$x = 80000$

Male = 40000, female = 40000

Next year

Male = 40000, female = 50000

13. Male population of Patna in the previous year = 44000

Male : Female = 11:10

In the next year, male : female = 11:10

$\frac{\text{Male}}{60000}=\frac{11}{10}$, Hence males = 66,000

Total population = 66000 + 60000 = 126000

15. For Arpit Pharma

Constituent A = $\frac{12}{100}\times 300 = 36$ gm

Constituent B = $\frac{15}{100}\times 300 = 45$ gm

Constitutent C = $\frac{20}{100}\times 300 = 60$ gm

Constituent D = $\frac{28}{100}\times 300 = 84$ gm

For Bittu Pharma:

Constituent A = $\frac{37}{100}\times 300 = 111$ gm

Constituent B = 45 gm

Constituent C = 39 gm

Constituent D = 45 gm

For Chintu Pharma

Constituent A = 78 gm

Constitutent B = 45 gm

Constituent C = 30 gm

Constituent D = 102 gm

The most affective constituent is B, B is present equally in all three capsules.

Now, it is turn of A.

16. For Arpit drug, side effect = $\frac{60}{100}=\frac{1}{2}=0.5$

For Bittu's drug, side effect = $\frac{.39}{105}=\frac{13}{35}=0.37$

For Chintu's drug, side effect = $\frac{30}{90}\times\frac{1}{3}=0.33$

17. Go through options

18. $\frac{3000}{1000}\times[60+39+30]=3\times 129=387\,\text{kg}$

19. Let population = 100

People read Dainik Jagran = 25

People read Prabhat Khabar = 20

People reading both = 8

People reading only Dainik Jagran = 17

People reading only Prabhat Khabar = 12

Required percentage = (5.1 + 4.8 + 4) = 13.9%

23. For last year

Honda = 100 cars

Toyota = 110 cars

This year

Honda = 120 cars

Toyota = 132 cars

Hyundai = 26.4 cars

Last year, Hyundai = 68.4 cars

Required percentage $=\frac{68.4}{278.4}\times 100 = 25\%$

24. Let the equations be

$x^2 - 2x + 1 = 0$ (1)

and $x^2 - x - 2 = 0$ (2)

Required percentage $=\frac{1-(-2)}{1}\times 100 = 300\%$

25. $25000\left(1+\frac{10}{100}\right)^2-25000=\frac{25000\times 11\times x}{100}$

$25000\left[\left(\frac{11}{10}\right)^2-1\right]-\frac{25000\times 11\times x}{100}$

$\left(\frac{11}{10}\right)^2-1=\frac{11x}{100}\Rightarrow\left(\frac{121}{100}-1\right)=\frac{11x}{100}$

$\frac{21}{100}=\frac{11x}{100}$ and $x=\frac{21}{11}$

26. 2 pounds = Rs 132

Rs 54 = 1 euro

Rs $132=\frac{1}{54}\times 132$ Euro$==\left(\frac{66}{27}\right)$Euro$=\frac{22}{9}$Euro

= 2.44 Euro

27. 1 Canadian Dollar = Rs 35.2
1 USD = Rs 48.4

Required No. $= \frac{35.2 \times 5}{48.4} = 3.67$

28. Required no. of notes $= \frac{1000}{32} + \frac{2000}{32} = 93.47 = 93$ notes

29. Data is in %age and question is about numbers, so it cannot be determined.

30. Through observation.

Questions 31 to 34

Total money with them can be either Rs 400, or Rs 300, or Rs 200, or Rs 200, or Rs 100.

From statement (3) Eeshwar = 200 – 35 = 165

From statement (5) Chandu = 200

From statement (4) Akshay = 400 – 360 = 40 (not least)

Now reading the question meticulously, we get to know that we don't need to calculate the sum of money with Bindu and Dev.

31. Five friends have Rs 400, Rs 300, Rs 200, Rs 200 and Rs 100

Eeshwar had Rs 200	Eeshwar left with Rs 165
Binod had Rs 300	Binod left with Rs 95
Chandu had Rs 200	Akshay left with Rs 40
Akshay had Rs 400	Chandu left with Rs 10
Dev had Rs 100	Dev left with Rs 70

35. Let population = 100

At least 50 people read a newspaper

At most 12.5 people read more than a newspaper. Hence, at least 37.5 people read only one newspaper.

36. Solution = 100 ml and Alcohol = 40 ml

For first vessel

$$\frac{40+x}{100+x} = \frac{1}{2},$$

so, x = 20ml

For second vessel

$$\frac{40+\frac{3}{5}y}{100+\frac{2}{5}y} = \frac{1}{2},$$

so, y = 25ml

Required percentage $= \frac{5}{25} \times 100 = 20\%$

Questions 37 to 40

This is basically a question that can be solved by forming equations.

Let DS = QA = EU = RC = x and GD = PI = y

So $5x + 2y = 250$, and similarly generate another equations to solve.

4

CHAPTER

Profit, Loss and Discount



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Basic terms involved
- Mechanism of profit generation
- Honest and dishonest ways
- Kinds of questions which are asked in the CAT
- Methods of solving questions

Introduction to the topic

As far as the CAT is concerned, this chapter is important because the application of the concept of profit, loss and discount ranges from simple mathematical calculations in data interpretation (DI) to complex logical evaluations in QA. This chapter can be considered as an extension of percentage, the only difference being the terminology used in both the concepts. The terms we use in percentage are initial value, final value, percentage increase etc. But when we use the same terms in profit, loss and discount (PLD), the names get converted to cost price, selling price, profit percentage etc. Let us understand the concepts and learn problem solving methods for this concept.

TERMINOLOGY

Cost Price

The price (amount) paid to purchase a product or the cost incurred in manufacturing a product is known as the cost price (CP) of that product.

Types of cost

1. **Fixed cost** As obvious from the name, it is that kind of cost which is fixed in all the cases.
2. **Variable cost** Variable costs are those costs which vary according to the number of units produced.
3. **Semi-variable cost** Semi variable costs are those costs which are fixed in one particular strata, but the costs varies among the different stratas.

One good example of fixed cost, variable cost and semi-variable cost is the bill we receive for the telephone connections at our home. A part of that bill, the rental, is fixed cost; and the rest of the part of the bill is calculated on the basis of the numbers of calls made.

Selling Price

The price at which a product is sold is called the selling price (SP) of the product.

Marked Price

The marked price or the mark-up price (MP) is the price which the shopkeeper/retailer fixes in anticipation of some discount they may be asked by a customer.

List Price

List price or the tag price, as the name suggests, is the price that is printed on the tag of the article.

For our calculations related to the concept of PLD, till the moment nothing is stated in the questions we would not see much difference between Marked Price and List price. This can be understood in the following example.

Suppose, the CP of an article is Rs 100 and the shopkeeper wants to earn a profit of 20%, i.e., he wants to have a profit of Rs 20. Now, he anticipates that a customer will ask for some discount and if he fixes the SP at Rs 120 he would not be able to give any discount to the customer. So, he now marks up his price over Rs 120 so that he can give some discount to his customer.

Remember that if nothing else is stated, then mark-up percentage is always calculated over CP.

$$MP \xrightarrow{\%\text{ decrease}} SP \xrightarrow{\%\text{ profit/\% loss}} CP$$

Example 1 After selling an article at a discount of 50%, profit percentage obtained is 20%. What is the mark-up over CP?

Solution Let us assume that CP = Rs 100.

So, SP = Rs 120.

Now, after giving a discount of 50% over MP, Rs 120 is the SP.

Using product stability ratio, 50%↓ = 100%↑

So, MP = 100%↑ over SP = Rs 240

Hence, percentage mark-up = 140%

Alternatively, 0.5 MP = 1.2 CP

MP/CP = 1.2/0.5 = 2.4

So, percentage mark-up $= \left(\frac{MP}{CP} - 1\right) \times 100$

$= (2.4 - 1) \times 100 = 140\%$

Further, as we know

If SP > CP, then there will be a profit and if SP < CP, then there will be a loss.

SP – CP = Profit

CP – SP = Loss

Profit % = Profit/CP × 100

Loss % = Loss/CP × 100

Profit% or Loss % is always calculated upon CP, unless something else is mentioned.

It can be seen that whenever, there is a profit, then MP ≥ SP > CP.

However, when there is a loss, then we can not have some definite relationship between the above written three quantities.

Margin

When we calculate profit percentage as a percentage of SP, then it is known as margin.

Calculating CP/SP Profit % or Loss %

If we say that there is a profit of 20%, then

CP × 1.2 = SP

So, if CP = Rs 120, then SP = Rs 120 × 1.2 = Rs 144

Or, if SP = Rs 144, then CP = 144/1.2 = Rs 120

(Needless to say that if there is a profit of 30%, then we will take 1.3 in the place of 1.2, and if there is a loss of 27%, then we will take 0.73 at the place of 1.2)

So, if there is a profit of R%, and CP = C, then,

$$SP = \frac{C}{100+R} \times 100$$

And in case of loss of R%, $SP = \frac{C}{100-R} \times 100$

Alternatively, we can use product stability ratio also to find out SP if CP is given.

Example 2 Two shopkeepers sell some article for Rs 4000 each. A shopkeeper calculates his profit per cent on his CP and another calculates his profit per cent wrongly on SP. What is the difference in their actual profit if both claim to have a profit of 20%?

Solution In 1st case, SP = Rs 4,000, Profit = 20% of CP, CP = Rs 3,333.33, Profit = Rs 666.66

In 2nd case SP = Rs 4,000, Profit = 20% of SP, CP = Rs 3,200, Profit = Rs 800

So, the difference in profit = Rs 133.33

PROFIT GENERATION

The purpose of studying Profit, Loss and Discount is to understand the mechanism of profit generation under different situations. Broadly it can be seen in two ways:

Honest Ways

Questions which you will find here will be based either in terms of Rs or in terms of goods.

Type 1: Questions in terms of goods

In these types of questions, CP of a fixed number of goods are compared with the SP of another field number of goods. Let us see through an example:

Example 3 The CP of 30 articles is equal to the SP of 40 articles. What is the profit or loss percentage?

Solution To obtain the same amount of money, which was needed to purchase 30 articles, we need to sell 40 articles, which is more than what we have got for the same sum. It means we need to arrange 10 more articles apart from the articles which we have purchased. So, there will be a loss.

Now, CP of 30 articles = SP of 40 articles

Or, CP/SP = 30/40 = ¾

Or, 1 – CP/SP = 1 – ¾ = ¼

So, Loss percentage

= (1 – CP/SP) × 100 = ¼ × 100 = 25%

Alternatively,

CP of 30 articles = SP of 40 articles

= Rs 120 (Assume)

So, CP of one article = Rs 4

SP of one article = Rs 3

Obviously, there is a loss of Re 1

Loss percentage = ¼ × 100 = 25%

Or, Otherwise we can use Profit/loss Percentage

$$= \frac{(\text{Goods left/added})}{(\text{Goods sold})} \times 100$$

Type 2: Questions in terms of money

Basically questions in terms of money relate to the CP or SP. However, sometimes the results given below also come handy in solving problems.

Some Important Results

When SPs of two articles are same

i. First one is sold at a profit of x% and second one is sold at a profit of y%.
   Ratio of $CP_1:CP_2 = (100 + y):(100 + x)$
ii. First one is sold at a profit of x% and second one is sold at a loss of y%.
   Ratio of $CP_1:CP_2 = (100 - y):(100 + x)$
iii. First one is sold at a loss of x% and second one is sold at a loss of y%.
   Ratio of $CP_1:CP_2 = (100 - y):(100 - x)$
iv. First one is sold at a loss of x% and second one is sold at a profit of y%.
   Ratio of $CP_1:CP_2 = (100 + y):(100 - x)$

Example 4 SPs of two articles are same. One is sold at a loss of 20% and another one at a profit of 20%. What is the net loss/profit in the whole transaction?

Solution Assume that SPs of each of the article = Rs 100

So, CP_1 (For the article which is sold at a loss of 20%) = 100/0.8 = Rs 125

CP_2 (For the article which is sold at a profit of 20%) = 100/1.2 = Rs 83.33

So, net CP = Rs 125 + Rs 83.33 = Rs 208.33

As we can observe now that SP<CP, so, there is a loss.

Loss % = (8.33/208.33) × 100 = (1/25) × 100 = 4%

Alternatively, if SPs of two articles are same, and one is sold at a profit of x% and another is sold at a loss of x%, then in that case there will be a loss always. And loss percentage

$$= \frac{x^2}{100}\%$$

Example 5 Two articles are sold at Rs 12,000 each. One is sold at a profit of 20% and another one at a loss of 20%. What is the net loss?

Solution SP_1 = Rs 12,000

CP_1 = Rs 12,000/ 1.2 = Rs 10,000

SP_2 = Rs 12,000

CP_2 = Rs 12,000/ 0.8 = Rs 15,000

So, total CP = Rs 25,000 and total SP = Rs 24,000

So, loss = Rs 1,000.

Alternatively, since there is a loss of 4%. So, for every Rs 100 invested, Rs 96 is coming back and Rs 4 is lost. In our case, 96% = Rs 24,000; so, 4% = Rs 1,000.

Example 6 Due North Inc. is the number 1 idea developer company worldwide. One day, they sold Idea 1 to CL and Idea 2 to AMS at a profit percentage of 20% and 30%, respectively. If the sum of developing Idea 1 and Idea 2 is Rs 25,000, what is the developing cost of Idea 1?

Solution The ratio of developing cost of Idea 1 and Idea 2 = 130:120

So, developing the cost of Idea 1 = $\frac{130}{250} \times 25{,}000$ = Rs 13,000

Dishonest Ways

Dishonest ways of earning profit includes adulterating or using faulty balance or both of these simultaneously.

Example 7 A shopkeeper sells his articles at his CP but uses a faulty balance which reads 1000 gm for 800 gm. What is his actual profit percentage?

Solution This question can be solved in several methods.

Method 1 Shopkeeper's net profit = 200 gm
CP of 1000 gm = SP of 800 gm
So, profit % = 200/800 × 100 = 25%

Method 2 Since while selling, 800 g = 1 kg
So, 200 gms = 1/4 kg
So, profit = 1/4 = 25%

Method 3 While selling, since the shopkeeper is branding 800 grams as 1 kg, so 1000 grams will be branded as 1000/800 = 1.25 kg and while purchasing, he has paid for just 1000 grams.
So, net profit = (1.25 – 1) = 0.25 kg
So, profit percentage = 0.25/1 × 100 = 25%

Example 8 A shopkeeper marks up his goods by 40% and gives a discount of 10%. Apart from this, he uses a faulty balance also, which reads 1000 gm for 800 gm. What is his net profit percentage?

Solution Let us assume his CP/1000 gm = Rs 100
So, his SP/kg (800 gm) = Rs 126
So, his CP/800 gm = Rs 80
So, profit = Rs 46
So, profit percentage = 46/80 × 100 = 57.5%

Some more examples

Example 9 A cloth store is offering "Buy 3, get 1 free." What is the net percentage discount being offered by the store?

Solution Suppose price of one article is Re 1. So, price of 4 articles = Rs 4. Now the whole scene can be understood as – We are paying only Rs 3 at the place of Rs 4. So, discount = Re 1. So, discount percentage = ¼ (discount/total price) = 25%

Example 10 A shopkeeper sells his goods at its CP only. But still he manages to gain a profit of 40% because he has manipulated his weights. Find how many grams he is actually selling at the place of 1000 gm.

Solution To earn a profit of 40%, shopkeeper needs to make 1.4 kg out of 1 kg. So, he will be selling $\frac{1000}{1.4}$ grams = 728 grams at the place of 1 kg.

Example 11 When an article is sold at 20% discount, SP is Rs 24. What will be SP when the discount is 30%?

Solution 20% discount = 0.8 of MP = Rs 24
So, MP = Rs 30
So, SP when discount is 30% = Rs 21

PRACTICE EXERCISES

WARM UP

Q.1. An article is bought for Rs 600 and sold for Rs 750. What is the profit percentage?
(a) 20% (b) 25%
(c) 30% (d) None of these

Q.2. By selling a VCD player for Rs 1,950, I got a profit of 30%. At what price should I have sold it in order to get a profit of 40%?
(a) Rs 2,000 (b) Rs 2,100
(c) Rs 2,500 (d) None of these

Q.3. The profit obtained by selling a book for Rs 56 is the same as the loss obtained by selling this book for Rs 42. What is the cost price of the book?
(a) Rs 40 (b) Rs 50
(c) Rs 49 (d) None of these

Q.4. A man bought a shirt at 3/4th of its list price and sold it at 50% above its list price. What is his percentage gain in the transaction?
(a) 50% (b) 70% (c) 80% (d) 100%

Q.5. By selling a CD for Rs 150, a shop owner lost 1/16 of what it costs. What is the cost price of CD?
(a) Rs 120 (b) Rs 140
(c) Rs 150 (d) Rs 160

Q.6. Saurabh sold 250 cycles and had a gain equal to the selling price of 50 cycles. What is his profit percentage?
(a) 25% (b) 20%
(c) 10% (d) None of these

Q.7. If profit as a percentage of the selling price is 25%, then what is profit as a percentage of the cost price?
(a) 25% (b) $33\frac{1}{3}\%$
(c) 20% (d) 15%

Q.8. A man sold an article at a profit of 10%. If he had charged Rs 45 more his profit percentage would have been 25%. What is the cost price?
(a) Rs 200 (b) Rs 300
(c) Rs 250 (c) None of these

Q.9. Richa purchased a pen drive at 4/5th of its list price and sold it at 20% above the original list price. What is Richa's profit percentage?
(a) 25% (b) 40%
(c) 50% (d) None of these

Q.10. A man buys 200 oranges for Rs 10. How many oranges a rupee can he sell so that his profit percentage is 25%?
(a) 10 (b) 14
(c) 16 (d) 20

Q.11. The CP of 24 apples is the same as the SP of 18 apples. Find the percentage gain.
(a) $12\frac{1}{2}\%$ (b) $14\frac{2}{7}\%$
(c) $16\frac{2}{3}\%$ (d) $33\frac{1}{3}\%$

Q.12. A manufacturer earns Rs 20 on selling his product at 10% above his cost. What is the cost price?
(a) Rs 125 (b) Rs 150
(c) Rs 200 (d) None of these

Q.13. By selling an article for Rs 360, the loss incurred is 10%. At what minimum price should he sell that article to avoid loss?
(a) Rs 320 (b) Rs 324
(c) Rs 396 (d) Rs 400

Q.14. Nitika buys a kinetic for Rs 16,000 and sells it for Rs 18,500. If she wants to gain 40%, how much more should she charge for the kinetic?
(a) Rs 3,800 (b) Rs 4,000
(c) Rs 4,200 (d) None of these

Q.15. The owner of Book Point does not get either profit or loss by selling 15 books for Rs 225. How many books should he sell for the same amount to gain 25% profit?
(a) 10 (b) 11 (c) 12 (d) 13

Q.16. A reduction 20% in the price of the rice enables to get 6 kg more of rice for Rs 120. What is the initial price of rice (Rs/kg)?
(a) 4 (b) 5
(c) 6 (d) None of these

Q.17. CP of 12 apples is equal to the SP of 9 apples and the discount on 10 apples is equal to the profit on 5 apples. What is the percentage difference between the CP and SP of apples?
(a) 20% (b) 25%
(c) 16.66% (d) None of these

Q.18. A shopkeeper professes to sell his articles at CP but gives only 800 gm in the place of 1000 gm. What is his profit percentage?
(a) 20% (b) 25%
(c) 16.66% (d) 33.33%

Q.19. In winters, a metre scale shrinks and starts reading 900 cm for every 100 cm. If the shopkeeper is selling the cloth at his CP, what is his loss percentage?
(a) 11.11% (b) 10%
(c) 12.5% (d) None of these

Q.20. A shopkeeper professes to sell his articles at CP but gives only N grams in the place of 1000 gm. If his profit percentage is 100%, what is the value of N?
(a) 750 (b) 500
(c) 0 (d) Not possible

FOUNDATION

Q.1. A car is sold for Rs 2,400 at a profit of 20%. What is the CP?
(a) Rs 2,100 (b) Rs 2,000
(c) Rs 1,800 (d) Rs 2,200

Q.2. A car is sold for Rs 2,400 at a profit of 20% over SP. What is the CP?
(a) Rs 2,000 (b) Rs 1,920
(c) Rs 1,980 (d) Rs 1,800

Q.3. A car is sold for Rs 2,400 at a profit of 20% over SP. What is the actual profit percentage?
(a) 16.66% (b) 25%
(c) 21.21% (d) 14.28%

Q.4. Apples are purchased @ 10 apples/Re. How many apples should be sold for one Rupee to obtain a profit of 25%?
(a) 6 (b) 8 (c) 12 (d) 4

Q.5. Sharat sells some articles to Chandra at a profit of 20%. Chandra now sells this article to Mayank at a loss of 30% and Mayank sells this article at a profit of 20%. If CP of Chandra is Rs 150, then what is the SP of Mayank?
(a) Rs 105 (b) Rs 87.5
(c) Rs 125 (d) Rs 126

Q.6. Petrol is purchased @ Rs 5/ litre and sold @ 5 litres/ Re. What is the profit/loss %?
(a) Loss of 96% (b) No profit, no loss
(c) Profit of 2400% (d) None of these

Q.7. Despite giving a discount of 11.11% on the mark-up price, profit obtained is equal to 14.28%. What is the per cent mark up over CP?
(a) 14.28% (b) 28.56%
(c) 25% (d) 50%

Q.8. A shopkeeper wants to earn a profit of 20% and at the same time, the minimum discount which he wants to offer is 25%. What should be the minimum percentage mark-up over CP?
(a) 60% (b) 42.5%
(c) 62.5% (d) 35%

Q.9. In Q. No. 8, what should be the maximum percentage mark-up over CP?
(a) 50% (b) 100%
(c) 71.4% (d) Cannot be determined

Q.10. An article when sold at 10% discount on the marked price gave a profit of Rs 70. What is the CP?
(a) Rs 700 (b) Rs 350
(c) Rs 125 (d) Cannot be determined

Q.11. 100 kgs of gold are purchased for Rs 1,100. It is sold in such a way that after selling the whole quantity, the quantum of loss is equal to the amount obtained by selling 20 kgs of gold. What is the selling price?
(a) Rs 9.16 (b) Rs 18.32
(c) Rs 11.11 (d) Rs 25

Q.12. A milkman buys two cows for Rs 750. He sells first cow at a profit of 22% and the second cow at a loss of 8%. What is the SP of second cow if in the whole transaction there is no profit no loss?
(a) Rs 312 (b) Rs 506
(c) Rs 484 (d) Rs 532

Q.13. Two cars are sold for Rs 24,000 each. One is sold at a profit of 20% and another at a loss of 20%. What is the net profit percentage/loss percentage, and amount of profit/loss??
(a) 4% profit, Rs 2,000 profit
(b) 4% loss, Rs 2,000 loss
(c) 1% profit, Rs 500 profit
(d) 1% loss, Rs 500 loss

Q.14. A supplier sells 20 pencils at the marked price of 16 pens to a retailer. The retailer, in turn, sells them at the marked price. What is the percentage profit or percentage loss of the retailer?
(a) Loss 25% (b) Profit 25%
(c) Loss 20% (d) Profit 20%

Q.15. Sum of the CP's of two cars is Rs 1,00,000. 1st car is sold at a profit of 20% and 2nd car is sold at a loss of 20%. However, their SPs are same. What is the CP of 1st car?
(a) Rs 40,000 (b) Rs 60,000
(c) Rs 52,400 (d) Rs 47,600

Q.16. Sum of CP's of two cows is Rs 13,000. Both the cows are sold at a profit of 20% and 40% respectively with their SP's being the same. What is the difference of CPs of both the cows?
(a) Rs 1,000 (b) Rs 2,000
(c) Rs 1,500 (d) Rs 2,500

Q.17. Buy three, get one free. What is the percentage discount being offered here?
(a) 33.33% (b) 25%
(c) 20% (d) 28.56%

Q.18. CP of 40 articles is equal to the SP of 30 articles. What is the profit/loss percentage?
(a) 25% Profit (b) 33.33% profit
(c) 25% loss (d) 33.33% loss

Q.19. Due to a price hike of 20%, 4 kgs less tea is available for Rs 120. What is the original price of tea?
(a) Rs 4/kg (b) Rs 5/kg
(c) Rs 6/kg (d) Rs 4.5/kg

Q.20. A shopkeeper sells his goods at its CP only. But he uses 750 grams weight at the place of 1000 gms weight for a kg. What is his net profit percentage?
(a) 25% (b) 20%
(c) 16.66% (d) 33.33%

Q.21. A shopkeeper sells his goods at its CP only. But still he manages to gain a profit of 30% because he has manipulated his weights. Find out how many grams he is actually selling at the place of 1000 gm.
(a) 700 (b) 769 (c) 800 (d) 820

Q.22. Which of the following discount options is better for a customer?
i. Successive discounts of 20%, 30% and then pay a service tax of 10%.

ii. Successive discounts of 30%, 20% and then pay a service tax of 10%.
iii. Pay service tax of 10% first, then successive discounts of 20% and 30%.
(a) i or ii
(b) iii
(c) Either of these three
(d) i and iii or ii and iii

Q.23. A wholeseller supplies few chips to a retailer every year. Each chip costs Rs 10,000 to the wholeseller. 5% of the chips are defective and they are to be replaced without charging anything extra. If the wholeseller still makes a profit of 20%, at what price is he selling it to retailer?
(a) Rs 12,300 (b) Rs 12,600
(c) Rs 13,200 (d) None of these

Q.24. A seller calculated his intended selling price at 6% profit on the cost of a product. However owing to some mistake while selling, the units and tens digits of the selling price got interchanged. This reduced the profit by Rs 9 and profit percentage to 2.4%. What is the cost price of the product?
(a) Rs 240 (b) Rs 250
(c) Rs 400 (d) Rs 480

Q.25. A shopkeeper wrongly calculates his profit on SP and finds it to be 25%. What is the actual profit percentage?
(a) 20% (b) 33.33%
(c) 40% (d) 25%

Q.26. There are two shopkeepers selling the same article at the same price for same quantity. One day, first shopkeeper offers a price discount of 25% for the same quantity whereas second shopkeeper offers 25% more quantity for the same price. From a customer's point of view, which deal is better?
(a) 1st shopkeeper's deal
(b) 2nd shopkeeper's deal
(c) Both are equal
(d) Cannot be determined

Q.27. A shopkeeper sells two cows in such a way that CP of first cow is equal to the SP of second cow and SP of first cow is equal to the CP of second cow. Which of the following is final result after the whole transaction?
(a) There is a loss (b) There is a profit
(c) No profit, no loss (d) Cannot be determined

Q.28. When an article is sold for Rs 180, a profit of 20% is made. At what price should the article be sold so that profit percentage is double?
(a) Rs 200 (b) Rs 210
(c) Rs 192 (d) Rs 240

Q.29. Vinit sells a car to Amit at a profit of 20%. Now Amit sells this car to Vicky at a profit of 12%. And finally Vicky sells this car to Nishu at a loss of 21%. What is the sum of CPs of Amit and Nishu?
(a) Rs 250 (b) Rs 475
(c) Rs 540 (d) Cannot be determined

Q.30. Anoop sells a book to Mayank at a profit of 20% and Mayank sells this book to Siddharth at a profit of 25%. Now Siddharth sells this book at a loss of 10% to Shishir. At what percentage loss should Shishir sell this book now so that his SP becomes equal to Anoop's CP?
(a) 36.68% (b) 25.92%
(c) 48.66 (d) Cannot be determined

Q.31. When an article is sold for Rs X, loss percentage is equal to L%. However, when the same article is sold for Rs Y, profit percentage is equal to P%. What is the CP of that article?
(a) 100 (P+L)/(Y−X) (b) (P+L)/(Y−X)
(c) 100 (Y−X)/(P+L) (d) (Y−X)/(P+L)

Q.32. A milkman professes to sell milk at its CP only. But still he is making a profit of 20% since he has mixed some amount of water in the milk. What is the percentage of milk in the mixture?
(a) 80% (b) 83.33%
(c) 75% (d) 66.66%

Q.33. There is some profit when an article is sold for Rs 720. However when the same article is sold for Rs 420, there is some loss. If the quantum of loss is two times the quantum of profit, find the cost price of the article.
(a) Rs 620 (b) Rs 700
(c) Rs 520 (d) None of these

Q.34. While I was shopping in Kamla Nagar market for my baby girl Tatto, I came across two shopkeepers who were giving me a discount of same amount for a top. While first shopkeeper was offering a discount of 14.28%, second shopkeeper was offering a discount of 12.5% over their respective mark-ups. Whose cost price is higher?
(a) 1st shopkeeper
(b) 2nd shopkeeper
(c) Either of them depending upon their respective MP
(d) Cannot be determined

Q.35. A seller offers discounts on the basis of the number of articles purchased. He gives a discount of 10% when 5 articles are bought and a discount of 12.5% when 8 articles are bought. If the profit he makes in each case is the same, then the ratio of the marked price to the cost price of the article is
(a) 4:3 (a) 3:4
(c) 6:5 (d) None of these

Q.36. A shopkeeper purchases his goods at a discount of 10% on the list price. He normally gives a discount of 6% to his customers. If during a special scheme, he gets an additional discount of 2% on his purchase price, find his approximate profit percentage.
(a) 8.8% (b) 7.8%
(c) 6.6% (d) None of these

Q.37. A shopkeeper purchases a packet of 50 pencils at Rs 10 per pencil. He sells a part of the packet at a profit of 30%. On the remaining part, he incurs a loss of 10%. If his overall profit on the whole packet is 10%, the number of pencils he sold at profit is
(a) 25 (b) 30 (c) 20 (d) 15

Q.38. The costs of a pen and a book are Rs 3 and Rs 4 respectively. A student has to spend a sum of exactly Rs 100 to buy pens and books, buying at least one pen and one book. How many combinations of books and pens can he buy?
(a) 6 (b) 8 (c) 12 (d) 10

Q.39. Anam bought a jute bag at 30% discount on the list price. He then sold it at a price which is 160% of the list price thereby making a profit of Rs 81. What is the list price of the bag?
(a) Rs 90 (b) Rs 100
(c) Rs 180 (d) Rs 200

Q.40. What is the profit percentage of a dishonest cloth merchant who professes to sell his articles at CP but uses a scale which weighs less by 16.67%?
(a) 20 (b) 15
(c) 16 (d) None of these

Q.41. A shopkeeper calculated his profit on the marked price and finds it to be 30%. He forgets the fact that he gave a discount of 20%. What is his actual profit percentage?
(a) $14^{2}/_{7}\%$ (b) 20% (c) 25% (d) $28^{4}/_{7}\%$

Q.42. Santa Singh, the local fruit vendor, buys a certain number of oranges at Rs 7 a dozen and equal number at Rs 6 a dozen. He sells the oranges at Rs 7.50 a dozen and makes a profit of Rs 80. How many oranges does he buy?
(a) 20 dozens (b) 30 dozens
(c) 40 dozens (d) 80 dozens

Q.43. A milkman who wants to make a fast buck by using a false measure, measures only 900 ml instead of a litre. If his cost price and selling price per litre of milk are Rs 10 and Rs 12 respectively, what is his daily profit and profit percentage if he sells 20 litres in one day?
(a) Rs 60, 20% (b) Rs 60, $33^{1}/_{3}\%$
(c) Rs 40,25% (d) Rs 40, $33^{1}/_{3}\%$

Q.44. Two lots of oranges with equal quantity, one costing Rs 20 per dozen and the other costing Rs 30 per dozen, are mixed together and the whole lot is sold at Rs 24 per dozen. Then what is the profit or loss?
(a) 4% profit (b) 5.6% profit
(c) 4% loss (d) 5.6% loss

Q.45. A shopkeeper marks up the price of his product by 40%. If he increases the discount from 5% to 10%, the profit would decrease by Rs 14. How much profit would he earn if he gives a discount of 20% on the marked price?
(a) Rs 56 (b) Rs 28
(c) Rs 32 (d) Rs 24

Q.46. Navneet is a bananawallah who buys a gross of bananas at Rs 10 a dozen. During transport and storage two dozen bananas get spoilt. If Navneet spends 10% of his outlay on transport costs and wants an overall profit of 20%, at what price must he sell a dozen bananas?
(a) Rs 14.64 (b) Rs 15.84
(c) Rs 16.24 (d) Rs 17.54

Q.47. Krishna sells his watch at a loss of 5%. If he had sold it for Rs 56.25 more he would have gained 10%. What will be his gain or loss percentage if he sells it for Rs 450?
(a) 10% (b) 15% (c) 20% (d) 30%

Q.48. A person sold his watch for Rs 24 and got a percentage of profit equal to the cost price; then the cost price is
(a) Rs 145 (b) Rs 18
(c) Rs 22 (d) Rs 20

Q.49. A merchant buying goods abroad at a discount of 33.33% on the catalogue price, has to pay import duty of 20% on the net cost of goods, realize a profit of 25% on his outlay, what must he charge for an article priced Rs 228 in the catalogue?
(a) 228
(b) 322
(c) Both of the above
(d) Cannot be determined

Q.50. A group of students prepared stuffed toys as part of their group activity. They spent Rs 100 on velvet, Rs 5 on thread and needle and Rs 27 on miscellaneous items. They made 30 toys, 50% of which were purchased by force by some senior students causing them a loss of 50%. At what % profit should they sell the remaining toys so as to gain 50% on their total cost?
(a) 100% (b) 150%
(c) 75% (d) None of these

MODERATE

Direction for questions 1 to 5: Read the passage below and solve the questions based on it.

Patna-mart, sister concern of Wal-mart, is planning to establish a mechanism to give discount. It is actually a Khul Ja Simsim kind of game where if you win any round, you will get some discount specific to that round, and it you lose, you will be losing all the discounts so far won.

Now, round 1 is having discount coupons of 2%.
Round 2 is having discount coupons of 4%.
Round 3 is having discount coupons of 6%.
And so on……

There are infinite rounds and you can participate in all the rounds but you can participate in any round only once and before participating in round N (any particular round), you should have participated in round 1 – round N - 1 (all the rounds before that particular round.)

All the discounts won will be successive, e.g., if you win Round 1 and then round 2 also, you will get a successive discount of 2% and then 4%. But again as the rule goes on, if you lose round 3, you will be losing all the discount coupons won so far. But despite losing round 3, you can participate in round 4 and hence.

Q.1. What is the minimum number of rounds to obtain maximum possible discount?
(a) 50 (b) 100
(c) Infinite (d) Cannot be determined

Q.2. What is the maximum possible discount?
(a) 1000% (b) 50%
(c) 100% (d) Infinite

Q.3. What is the minimum possible discount after 20 rounds provided that customer has won a minimum of 18 rounds?
(a) 100% (b) 50%
(c) 0% (d) None of these

Q.4. In the above question, what is the maximum possible discount?
(a) 100% (b) 50%
(c) 0% (d) None of these

Q.5. In a bid to attract more customers, Patna-mart is on a mass-compaigning move. With a new punch line "You can always check-out, but you can never leave", they are organizing several contact programs across the city. As a result, there is an increase in sale of 10% and 20% respectively on weekdays and weekends (Saturday and Sunday). If each of the articles sold by Patna-mart is sold at a profit of 40%, what is the maximum possible percentage growth in the profit of Patna-mart?
(a) 32% (b) 72%
(c) Minimum 50% (d) Cannot be determined

Direction for questions 6 to 8: Read the passage below and solve the questions based on it.

Had the CP been 10% less and SP been 10% more, profit % would have been double than that of earlier case.

Q.6. What is the actual CP?
(a) Rs 70 (b) Rs 100
(c) Rs 140 (d) Cannot be determined

Q.7. What is the profit % in original case?
(a) 20% (b) 28.56%
(c) 14.28% (d) Cannot be determined

Q.8. How many values for actual SP is/are possible?
(a) 1 (b) 0
(c) 3 (d) Infinite

Direction for questions 9 and 10: Read the passage below and solve the questions based on it.

King Nandan is the king of Divyagarh. One day he decides to sell all the animals living in his big kingdom. Now, there are five states in his kingdom and the only animals which his kingdom is having are horses, cows and sheeps. King Nandan sells all the animals to eight neighboring states. Each of the neighboring states purchases same number of animals at the following rate.

Horse – Rs 1,700 each
Cow – Rs 200 each
Sheep – Rs 200 each
King Nandan receives Rs 28,500 in all?

Q.9. What can be the maximum number of animals?
(a) 80 (b) 130
(c) 160 (d) None of these

Q.10. What are the respective numbers of horses, cows and sheeps in any state?
(a) (3, 109, 8) (b) (1, 37, 3)
(c) (2, 74, 6) (d) Cannot be determined

Q.11. Sunny marks up his goods by 40% and gives a discount of 10%. Apart from this, he uses a faulty balance which reads 800 gms for 1000 gms. What is his net profit/loss percentage?
(a) 8% (b) 57.2%
(c) 37.6% (d) None of these

Q.12. A shopkeeper marks up his goods by 20% and then gives a discount of 20%. Besides he cheats both his supplier and customer by 100 grams i.e., he takes 1100 gram from his supplier and sells only 900 grams to his customer. What is his net profit percentage?
(a) 24.5% (b) 17.33%
(c) 25% (d) 32.5%

Q.13. In question number 12, shopkeeper takes a discount of 20% from his supplier which he disregards while preparing his mark-up. What is his net profit percentage?
(a) 37.5% (b) 46.66%
(c) 52.33% (d) 58.33%

Q.14. In addition to all the data of question number 13, now he adulterates his goods in the ratio of 1:5 (adulterant: original). Assuming that adulterant comes free of cost, what is the net profit percentage?
(a) 62.5% (b) 66.66%
(c) 72.08% (d) None of these

Q.15. In question number 14, if the cost of adulterant is 20% the cost of original substance, what is the net profit percentage?
(a) 77.77% (b) 81.08%
(c) 86.75% (d) None of these

Q.16. Some mangoes are purchased at the rate of 8 mangoes/Re and some more mangoes at the rate of 6 mangoes/Re, investment being equal in both the cases. Now, the whole quantity is sold at the rate of 7 mangoes/Re What is the net percentage profit/loss?
(a) 0.6% profit (b) 0.6% loss
(c) 1.2% loss (d) No profit/no loss

Q.17. Some olives are purchased at the rate of 8 olives /Re and same number of olives at the rate of 6 olives/ Re. Now, the whole quantity is sold at the rate of 7 olives/Re. What is the net percentage profit/ loss?
(a) 0.6% profit (b) 0.6% loss
(c) 1.2% loss (d) No profit/ no loss

Direction for questions 18 to 22: Read the passage below and solve the questions based on it.

A shop has the following discount layers
1st layer – Buy 1, get 2 free.
2nd layer – Buy 2, get 3 free.
3rd layer – Buy 3, get 4 free.
And so on there are infinite layers on the same pattern.

Q.18. For how many of these discount layers discount percentage offered is more than 50%?
(a) 4 (b) 27
(c) 99 (d) None of these

Q.19. The shopkeeper wishes to lessen the percentage of discount by clubbing any two layers. Which two layers should be clubbed so that discount is minimized by maximum percentage?
(a) 1st + 2nd (b) 2nd + 3rd
(c) 3rd + 4th (d) None of these

Q.20. What is the net percentage discount if we club 1st three layers?
(a) 43.33% (b) 46.66%
(c) 53.33% (d) None of these

Q.21. A new condition has been introduced by the shop that anybody can avail one layer discount only once (i.e., if you are planning to purchase 6 articles, then you can not purchase 2 articles in 1st layer and get rest four free. Since you have availed 1st layer discount, you can now ask for only other layer discounts.) Sona is purchasing 8 articles from the store. What is the maximum possible discount that she can avail?
(a) 72% (b) 26%
(c) 38% (d) None of these

Q.22. In Q.21, what is the minimum percentage discount that she can avail?
(a) 0% (b) 22%
(c) 44% (d) None of these

Direction for questions 23 and 24: Read the passage below and solve the questions based on it.

To produce a certain type of engine, two components are imported— Component P is imported from USA and component Q is imported from Japan. P and Q account for 30% and 40% of the production cost of the engine. A profit of 25% is to be made but discount of 10% is to be offered to customers for promotional reasons.

Q.23. If the price of US Dollar appreciates by 20% and the price of Japanese Yen appreciates by 30%, what percentage above the original CP should the product be marked?
(a) 33.3% (b) 48.5%
(c) 63.88% (d) 71.4%

Q.24. US Dollar appreciates by 4% and Japanese Yen by 32%. If marked price remains same, then what is the change in profit percentage?
(a) 4.38% (b) 3.33%
(c) 6.25% (d) 7.69%

Direction for questions 25 and 26: Read the passage below and solve the questions based on it.

A shopkeeper sells two types of custom confiscated goods, A and B, both at same price of Rs 120 whereas they cost him Rs 90 and Rs 160 respectively. Sale for A started from 1 for the first day and then went on increasing every day by 4 units, whereas sale of article B was 450 on the first day and went on decreasing every day by 6 units.

Q.25. Shopkeeper has planned the things in such a way that he starts earning profit exactly on diwali which is after N days. What is the value of N?
(a) 103 (b) 105 (c) 107 (d) 109

Q.26. On which day will he earn profit for the first time on day-to-day basis?
(a) 49 (b) 51 (c) 53 (d) 55

Q.27. In the year 2002, the cost price of an item was 90% of the selling price while in 2003, the cost price was 95% of the selling price. If the gross profit remains the same for both the years, What is the percentage increase in the cost price of the item from 2002 to 2003?
(a) 60% (a) 80%
(c) 100% (d) None of these

Direction for questions 28 to 33: Read the passage below and solve the questions based on it.

Mittal has recently acquired four companies viz., Bank of Bozoland (BOB), My Own Bank (MOB), Zany Obliterated Bank (ZOB), and Dogmatically Obscure Bank (DOB). He noticed that the sales of DOB are half that of BOB whereas the profits of DOB are double that of BOB. The expenses of ZOB are Rs 3 crores less than that of DOB whereas the profits of MOB is Rs 1 crore less than that of ZOB. The expenses of BOB are three times that of DOB. It is also known that the sales of ZOB are Rs 15 crore or one-fourth that of MOB. All figures are for 1992-93. An insider further informs Mittal that the sales of DOB are Rs 10 crores more than that of ZOB and the expenses of BOB are 90% of its own sales. Sales – Expenses = Profit

Q.28. The total sales of all the four companies is (Rs crores)
(a) 200 (b) 150 (c) 125 (d) 160

Q.29. The expenses of BOB exceed that of ZOB by (Rs crores)
(a) 31 (b) 43 (c) 33 (d) 62

Q.30. Which bank had the maximum profit?
(a) BOB (b) MOB (c) ZOB (d) DOB

Q.31. Total profits of the four companies (Rs crores) is
(a) 51 (b) 52
(c) 53 (d) None of these

Q.32. What is the total expense (Rs crores) of all the four banks together?
(a) 110 (b) 130 (c) 150 (d) 160

Q.33. The profits of MOB form what percentage of the expenses of ZOB?
(a) 16% (b) 20% (c) 15% (d) 23%

Q.34. By selling a watch at a profit of 10 per cent, a man got Rs 15 more than half its price. What is the price of the watch?
(a) 10 (b) 15 (c) 25 (d) 5

Q.35. A man would gain 25% by selling a chair for Rs 47.5 and would gain 15% by selling a table for Rs 57.5. He sells the chair for Rs 36; what is the least price for which he must sell the table to avoid any loss on the two together?
(a) Rs 51.2 (b) Rs 58.5
(c) Rs 62.5 (d) None of these

Q.36. A bookseller marks his books at an advance of 69% on the actual cost of production. He allows a discount of 15% and also given a copy free for every dozen sold at a time. What rate per cent profit does the bookseller make, if books are sold in lots of 12?
(a) 32.6 (b) 47.5
(c) 24.9 (d) None of these

Q.37. While returning from Dubai festival, Mr Sanjay Singh purchased a number of articles, all at a discount of $33\frac{1}{3}$% of the list price. However he is required to pay a duty of 20% on the his cost of goods. If he realizes a profit of 25% on his outlay, what must he charge for an article priced Rs 228 as the list price?
(a) 228 (b) 322
(c) Neither (a) nor (b) (d) Data insufficient

Q.38. A certain manufacturer sells a product to the distributor at 10% profit. Then the distributor sells it to the dealer and the dealer sells to the retailer at a mark up of 10% and 20% respectively. The retailer marks up his cost by 20% and then offers a 10% discount to the customer. If the customer had bought it from the distributor directly at the distributor's selling price, then how much reduction in price would he have got with respect to buying it from the retailer?
(a) 29.6% (b) 18% (c) 32% (d) 22.8%

Direction for questions 39 and 40: Read the passage below and solve the questions based on it.

Machine X produces articles at the rate of 50 units/h. SP of each article is Rs 100 and the cost of production is Rs 40 each. However, 20% of the articles produced are defective and hence cannot be sold. The rate of production of machine can be increased, but with every increase of N units/h production cost would increase by $2x$% and the number of defectives would become $(20 + 1.5x)$%.

Q.39. Find the maximum increase in production that can be undertaken without incurring losses.
(a) 14 unit (b) 15 unit
(c) 16 unit (d) 17 units

Q.40. What will be the profit in 1 hour if the machine produces 58 units per hour?
(a) Rs 1,839 (b) Rs 1,178
(c) Rs 1,253 (d) Rs 1,624

Direction for questions 41 and 42: Read the passage below and solve the questions based on it.

There is a growing demand of platinum in the Indian market. A firm dealing in platinum wants to earn a profit to the tune of Rs 5 lacs per annum. Platinum can be produced from a mine. It is required to employ 25 people whose average salary is Rs 2,000 per month. Cost of the raw material i.e., platinum ore is Rs 200 per ton of pure platinum produced. However, the fixed cost associated with the plant is Rs 1,00,000 annually.

Q.41. If the production rate is 20 ton/month, then what should be the price per ton of platinum to earn the required profit?
(a) Rs 4,300 (b) Rs 5,200
(c) Rs 6,300 (d) None of these

Q.42. If the price per ton is Rs 3,000, then how much ton of platinum is required annually to fulfill the profit target?
(a) 300 (b) 350
(c) 400 (d) None of these

Q.43. Two pieces of the same cloth together cost Rs 910. On selling from the first piece as many metres as the second piece originally contained and from the second piece half as many metres as the first piece contained originally. The remaining of the first piece turned out to exceed the remaining of the second piece by 10 m. Each metre of the cloth costs Rs 14. How many metre of cloth did each piece originally contain?
(a) 45 m; 20 m (b) 45 m; 25 m
(c) 36 m; 29 m (d) 40 m; 25 m

Q.44. Rupesh marks up an article by p%, gives a discount of $\frac{p}{4}\%$ and gets a profit of $\frac{p}{4}\%$. Had he marked up by $\frac{p}{2}\%$ and given a discount $\frac{p}{6}\%$, what would be his profit percentage?
(a) 25% (b) $33\frac{1}{3}\%$
(c) 50% (d) $66\frac{2}{3}\%$

Q.45. Amit brought two cars. He then sold the first car at 10% profit and the second one at 25% profit. The selling price of the second car is 25% more than the selling price of the first car. What is the approximate profit per cent in both the cars together?
(a) 17.85% (b) 18.36%
(c) 16.19% (d) Cannot be determined

ADVANCED

Q.1. A publisher printed 3,000 copies of *'Future Shock'* at a cost of Rs 2,400. He gave 500 copies free to different philanthropic institutions. He allowed a discount of 25% on the published price and gave one copy free for every 25 copies bought at a time. He was able to sell all the copies in this manner. If the published price is Rs 3.25, then what is his overall gain or loss percentage in the whole transaction?
(a) 113% (b) 130% (c) 162% (d) 144%

Q.2. A car manufacturing company XYZ can make 300 cars per month at a cost of Rs 2 lacs per car and pay an excise duty of 30% per car. It has 500 cars in its storage room in a month and is able to sell 300 cars per month at a rate of Rs 3 lacs per car. In the budget in the last week of February, the excise duty was reduced by 10% and it will take effect from 1st April. But, the customers cannot wait till 1st April. So the company decides to give this rebate to every customer who buys a car in March as well, but for this particular month the company has to pay 30% duty. The company can reduce or increase manufacturing of car per month by a maximum of 25%. What is the profit or loss in the revenue to the company in the month of March?
(a) 15 lac (b) 60 lac
(c) 45 lac (d) 30 lac

Q.3. A shopkeeper uses a "point of sales" software which prepares the bill for customers based on the number of units per unit. Both of the above are 2-digit numbers. While entering the data, he erroneously swaps the digits of both quantity of units and price. As a result, the closing stock balance is lowered by 72 units, and the sales value shown is Rs 1,368. How many units have actually been sold?
(a) 27
(b) 91
(c) 83
(d) None of these

Direction for questions 4 and 5: Read the passage below and solve the questions based on it.

Given below is the production of fertilizers bags by ABC fertilizers during different days of the week.

Monday	*Tuesday*	*Wednesday*	*Thursday*	*Friday*	*Saturday*
650	750	700	800	750	600

There are two modes of transporting these bags to warehouse. A truck has a capacity of 800 bags and cost Rs 1,100 per trip. A Matador has a capacity of 500 bags and cost Rs 700 per trip. The cost of storing the material in case it is not transported is Re 1 per bag per day. The cost of transportation includes the storing cost.

Q.4. In order to minimize transportation cost, how many trips should the Matador and the Truck respectively make to the warehouse during the week?
(a) 3, 4 (b) 2, 4 (c) 1, 5 (d) 4, 3

Q.5. What should be the minimum cost of storing 1 bag/day, at which it is best to hire a truck everyday?
(a) Rs 0.4/day (b) Rs 1.4/day
(c) Rs 1.6/day (d) Rs 1.2/day

Q.6. Mohit goes to furniture shop to buy a sofa set and a center table. He bargains for a 10% discount on the center table and 25% discount on sofa set. However, the shopkeeper, by mistake, interchanged the discount percentage figures while making the bill and mohit paid accordingly. When compared to what he should pay for his purchases, what percentage did mohit pay extra given that the center table costs 40% as much as the sofa set.
(a) 12.3% (b) 7.2% (c) 8.1% (d) 6.3%

Q.7. In a laboratory experiment, a sample of air which is a mix of only oxygen and vapour is taken. Water contains hydrogen and oxygen gases. If air contains 70% oxygen (including that contained in the water vapour) by weight while water vapour contain 16.66% oxygen by weight, how many kilograms of water vapour is present in 1kilogram of water?
(a) 0.25 (b) 0.34 (c) 0.36 (d) 0.3

Q.8 A shopkeeper purchases a packet of 50 pens at Rs 10 per pen. He sells a part of the packet at a profit of 30%. On the remaining part, he incurs a loss of 10%. If his overall profit on the whole packet is 10%, the number of pens he sold at profit is
(a) 25 (b) 30 (c) 20 (d) 15

Q.9. Amar sold his moped to Bharat at 20% profit and Bharat sold it to Sridhar at 10% profit. Sridhar sold the same to a mechanic and received Rs 2,316. If Amar had sold the same moped to the mechanic and receive the same amount the mechanic paid to Sridhar, what profit percentage would Amar have made?
(a) 52% (b) 48%
(c) 33.3% (d) Cannot be determined

Direction for questions 10 to 13: Read the passage below and solve the questions based on it.

Newspaper major *The Asian Times* is trying to enter the city of Paperabad, where already two newspapers *The Deccan City* and *Good Morning Paperabad* are in circulation. The population of the city is 35 lakhs of which, according to the recent survey, 60% read newspaper. The two evening newspapers sell at Rs 1.75 per copy on weekdays and Rs 3.00 per copy on Sundays. *The Asian Times* promised its readers newspapers at Rs 1.50 per copy per day. Since people were aware of the quality of the paper, the management thought that it can capture a major chunk of the newspaper reading population. On weekdays, the operational cost of the *The Asian Times* were Rs 62,500 per day and the cost for the material and printing charges was Rs 2.75 per newspaper. For the Sunday edition the cost of the newspaper per copy was Rs 3.00 and the operational cost were Rs 75,000. The number of the copies of the newspaper *The Asian Times* sold on Sunday was 16.66% less than the number of copies sold on any of the week days. The number of copies sold on any weekday being the same. The amount collected through the advertisement on any weekday was the same and equaled Rs 2,50,000 per day while for Sunday it was 4,35,000.

Q.10. If *The Asian Times* wanted to break even, how many copies of the newspaper are required to be sold every week?
(a) 21,25,800 (b) 16,97,143
(c) 47,50,500 (d) 50,50,500

Q.11. What percentage of the newspaper readers should read *The Asian Times* so that the newspaper makes a profit of Rs 12,500 every weekday?
(a) 7.61% (b) 12.4 %
(c) 12.8 % (d) None of these

Q.12. If 50,000 of the readers of *The Deccan City* now read *The Asian Times* on all weekdays, how much profit would *The Asian Times* expect to make every week?
(a) 2,55,000 (b) 3,85,000
(c) 4,57,000 (d) Cannot be determined

Q.13. How many copies of the newspaper should *The Asian Times* sell on Sunday so that there is a profit of Rs 60,000 every Sunday?
(a) 1,88,000 (b) 2,40,000
(c) 3,20,000 (d) 3,76,000

Direction for questions 14 to 16: Read the passage below and solve the questions based on it.

In the "Get-One" supermarket, a new software was installed in the computerized billing section which made the calculations of each of the item sold very easy on the first day of the operations. To verify the workability of the software only one of the item was billed on the computer.

Due to some technical problem in the software, it reversed both the numbers of the item sold and price per piece as a result of which the closing of the stock items showed 63 items less and the value on the items still left was shown Rs 936. It was known that both the stocks left and the price per piece are two digit numbers.

Q.14. How many items were actually left at the end of the day?
(a) 29 (b) 81
(c) 92 (d) None of these

Q.15. What is the correct collection for the day if 44 pieces were sold?
(a) Rs 2,288 (b) Rs 1,744
(c) Rs 1,444 (d) None of these

Q.16. By how much would the incorrectly calculated value of the items sold be more/less than the correctly calculated value?
(a) Rs 887 more (b) Rs 887 less
(c) None of these (d) Cannot be determined

Q.17. Mahesh and Umesh purchased a radio each for the same price and both marked-up their respective radios by the same amount. Mahesh gave a discount of Rs 20 followed by another discount of 20% on the reduced price, while Umesh gave a discount of 20% followed by a discount of Rs 20. If Mahesh's profit percentage is equal to thrice of Umesh's loss percentage, what is the profit (in Rs) of Mahesh on his radio?
(a) Rs 2 (b) Rs 3 (c) Rs 4 (d) Rs 5

Q.18. Vinod is a very shrewd shopowner. He adjusted his electronic weighing scale in a typical way. Error percentage of his sale is directly proportional to the displayed weight of the commodity. But he gives a discount of 10% in every transaction. His weighing scale shows 5 kg for 4 kg. What is the actual profit % when the displayed weight is 10 kg?
(a) 35% (b) 30% (c) −5% (d) 50%

Direction for questions 19 and 20: Read the passage below and solve the questions based on it.

A shopkeeper believes in running his business on the sentiments of market i.e., quick response to the demand-supply relationship. He starts every morning by selling product X at Rs 500 for the first hour. If the number of customers for any hour is more than 10, he will increase the price by Rs 10 over the price of the previous hour. If less than 10, he will decrease the price by Rs 10 over the previous hour. If it is equal to 10, then he will maintain the same price level of the previous hour. CP of product X is Rs 440.

Q.19. If the customer for the first four hours are 15, 10, 6 and 12, then find the profit made by the shopkeeper.
(a) Rs 2,050 (b) Rs 2,270
(c) Rs 2,740 (d) None of these

Q.20. The shopkeeper never sells for a loss. If he shuts his shop after 10 hours due to this principle and there is exactly one hour when the number of customers was 10, then the minimum number of hour when the number of customers is more than 10 is
(a) 2 (b) 1 (c) 4 (d) 3

Q.21. In a joint venture, Ramesh invested Rs 6 lakhs as fixed cost. While Suresh had to pay the variable cost for every three months in advance (i.e., for the first three months at the start of first month and for the next three months at the start of the fourth month and so on).
Table for costs is given below

Quantity produced/month	*Variable cost per unit (Rs)*	*Fixed cost (Rs) for 6 months*
0-1,00,000	1.5	6,00,000
1,00,001-2,00,000	1.25	6,00,000
2,00,001-3,00,000	1	9,00,000

For the first three months they produced 1,00,000 units per month and for the next three months they produced 1,50,000 units per month.

All units produced were sold at Rs 4 each. After six months they sold their company for Rs 2,00,000 and divided total profit according to their investments. Find out the total profit.
(a) 15.875 lakhs (b) 15.125 lakhs
(c) 10.125 lakhs (d) None of these

Q.22. In the above question, what is the share of Suresh?
(a) 8.7 lakhs (b) 6.2 lakhs
(c) 4.3 lakhs (d) 5.6 lakhs

Q.23. In an orchard, there are only mango and apple trees. Fruits available on each tree are equal to the total number of trees of the same kind. Also, mango trees are twice in number as compared to the apple trees. Only 40% of total mangoes and 70% of total apples are in good condition. These have to be stored before selling. This year the farmer could sell only 75% of the stored mangoes and $\frac{6}{7}$ of the stored apples. The rest was wasted. Selling price per mango is Rs 10 and per apple is Rs 5. Investment to store (consider only this as the cost price) is Rs 20 per mango and Rs 10 per apple. In this transaction, the farmer lost Rs 15,000. What is the total number of trees in the garden?
(a) 75 (b) 80
(c) 25 (d) 60

Q.24. A spring balance reads 1 kg when actually 800 g is weighed on it. The shrewd grocer has already marked

up the price of his goods by 20%. One day, a police inspector detects this and orders him to repair his spring balance and revert to the original list price of the goods. Moreover, he asked him to give a discount of 10% to all the customers for a month from then onwards as a punishment. A day after, Ramu paid Rs 16 for 1 kg rice. Before the raid, how much he would have paid for the same quantity?

(a) Rs 26.66 (b) Rs 27.5
(c) Rs 24 (d) Rs 28

Q.25. The evergreen shrubs at Ravi's nursery are planted in rows on a square plot of land measuring 2,401 ft^2. The shrubs are planted in such a manner that the centers of the adjacent shrub are 7 ft apart and the outer shrubs are planted along the edges of the plot, with a shrub at each corner. Ravi spent $ 896 to cover all the costs necessary for raising this crop of the evergreen shrubs. If Ravi succeeds in selling each shrub for $ 35, his profit will be what percentage of his total cost?

(a) 50% (b) 90% (c) 150% (d) 200%

Q.26. A retailer keeps Reebok shoes, which are listed at a mark-up of 10% above their factory price. However, on Monday he realized the shortage of demand and hence, decreased the list price by 10%. On the very next day, i.e., Tuesday, he realized he is making a loss so, he increased the list price by 10% again. He continues this trend indefinitely. On which day he will suffer a loss of more than 10% per shoes for the first time?

(a) Monday (b) Tuesday
(c) Thursday (d) Sunday

Direction for questions 27 and 28: Read the passage below and solve the questions based on it.

A person purchased 50,000 cups of glasses at a rate of Rs 10 per cup. For transporting these cups to his shop, he has four options as follows:

Mode of transportation	*Rent for one trip*	*Capacity of number of glass in one trip*	*Breakages during transportation*
Truck	Rs 2,000	20,000	8%
Small truck	Rs 1,000	12,000	10%
Jeep	Rs 750	10,000	15%
Rickshaw	Rs 500	5,000	10%

Even if a transport vehicle is not full to its capacity, rent for the trip will remain the same.

Q.27. Which mode of transportation is most profitable?

(a) Truck (b) Small truck
(c) Jeep (d) Rickshaw

Q.28. If the most profitable mode of transportation is chosen, then what should be his selling price per cup so as to get the profit of 10%?

(a) Rs 12.45 (b) Rs 14.67
(c) Rs 13.80 (d) Rs 12.10

Direction for questions 29 to 31: Read the passage below and solve the questions based on it.

The air conditioning market is one of the few segments that have been stable since the last three-four years. The total air-conditioning industry is estimated to be around Rs 3,000 crores with central air-conditioning including packaged and ducted air-conditioning systems estimated to be around Rs 1,200 crore. Windows and mini-split air-conditioners are estimated at Rs 1,800 crores, with window ACs cornering 80% of the market.

Prior to 1997, corporations, institutions and the government dominated the consumer segment. They did around 85–90 per cent of the buying. It was only in the year 1997 that the home segment picked up. The institutional segment accounted for 65 per cent while the rest went to the home segment. The average growth for the last five years in the residential segment has been around 24 per cent.

Q.29. LG has priced their split-ACs at Rs 35,000. If they have estimated a market share of 70% in the split-AC market, then how many split-ACs would they be selling approximately?

(a) 72,000 (b) 68,000
(c) 71,000 (d) 70,000

Q.30. If prior to 1997, the institutional buyers comprised of Rs 2,700 crore worth of market, what is the approximate share of Samsung ACs in 1997 if they sold only to home buying segment? Their share prior to 1997 was 20% in the home AC segment, while it had increased by 15% in 1997 from its earlier levels. The overall market size for total AC market had remained the same.

(a) 300 crores (b) 69 crores
(c) 76 crores (d) 241.5 crores

Q.31. With respect to the previous question, if Samsung entered the institutional market in 1998 and had a share of 10% by 1999. What was the ratio of its share in the institutional market in 1998-1999 with respect to its share in the home market prior to 1997? The market size of institutional segment showed a 10% increase over its size in 1997. (Use the data from previous questions if required). (Ratio is required with respect to revenue).

(a) 3 : 1 (b) 2.75 : 1
(c) 3.57 : 1 (d) 2.8 : 1

Q.32. The cost price of a dozen mangoes is equal to the selling price of 10 mangoes. The selling price of apples per piece is the cost price of a mango (per piece). If the profit % made on selling an apple is half that of selling a mango, then what is the cost of an apple if the selling price of one mango is Rs X?

(a) Rs $\frac{X}{6}$ (b) Rs $\frac{5X}{6}$

(c) Rs $\frac{25X}{33}$ (d) Rs $\frac{33X}{25}$

Direction for questions 33 to 37: Read the passage below and solve the questions based on it.

Various costs involved in printing news papers

All the given cost are calculated in Rs

Now, the total cost of printing is calculated by the following formula:

Total cost = P × (Columns × Cost per column + Rows × Cost per row + Labour cost per print) + Plate cost + Setup cost

Where, P = The number of prints

Fixed cost = Plate Cost + Set up Cost

Total cost = Fixed cost + Variable Cost

Newspapers	*The Times of India*	*The Hindustan Times*	*The Pioneer*	*The Statesman*	*Dainik Jagran*
Columns	3	4	2	4	2
Cost/column	20	15	35	20	10
Rows	400	350	400	600	200
Cost/row	0.5	0.8	0.6	0.3	0.5
Plate	1000	800	750	800	600
Set-up cost	200	100	150	300	150
Labour cost per print	1.5	2	1.2	0.9	1

Q.33. If the number of prints of each newspaper (in thousands) is equal, which newspaper is more costly?

(a) *The Times of India*
(b) *Dainik Jagran*
(c) *The Hindustan Times*
(d) *The Pioneer*

Q.34. If one has to print hundred prints of *The Pioneer*, then what is the ratio of the variable cost to the fixed cost (approx)?

(a) 20 (b) 30
(c) 40 (d) 35

Q.35. Which is the costliest newspaper in terms of cost per ten prints?

(a) *The Times of India*
(b) *The Hindustan Times*
(c) *The Pioneer*
(d) None of these

Q.36. How many copies of the *Dainik Jagran* should be printed so that the total cost is equal to the cost of 100 prints in *The Times of India* (approx)?

(a) 150 (b) 175
(c) 225 (d) 200

Q.37. Which newspaper has the highest fixed cost?

(a) *The Pioneer*
(b) *The Statesman*
(c) *The Hindustan Times*
(d) None of these

Direction for questions 38 to 40: Read the passage below and solve the questions based on it.

The following table shows the relationship between production volumes, unit price and unit cost of a particular product. Refer to the following table to answer the questions that follow:

Contribution = volume x (price − cost)

Production Volumes (No. of Units)	*Unit Price (Rs)*	*Unit Cost (Rs)*
1000	140	100
2000	125	95
3000	110	90
4000	100	80
5000	90	75
6000	75	70
7000	65	60

Q.38. At what volume will the contribution be maximum?

(a) 2,000 units (b) 3,000 units
(c) 4,000 units (d) 5,000 units

Q.39. If the unit cost goes down by Rs 10 at all levels of production, what will be the maximum possible contribution (Rs)?

(a) Rs 75,000 (b) Rs 1,25,000
(c) Rs 1,20,000 (d) Rs 80,000

Q.40. A breakeven point is the volume of production at which for the first time the contribution equals fixed costs. If the fixed costs are Rs 60,000, then the production level at which the break even is reached, is

(a) 2,000 units (b) 4,000 units
(c) 5,000 units (d) 6,000 units

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(c)	(d)	(d)	(a)	(b)	(b)	(c)	(c)	(d)	(c)	(d)	(d)	(c)	(b)	(d)	(b)	(b)	(b)

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(b)	(b)	(d)	(a)	(b)	(a)	(d)	(d)	(a)	(b)	(b)	(b)	(a)	(a)	(b)	(b)	(b)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(b)	(c)	(b)	(b)	(b)	(a)	(c)	(b)	(d)	(b)	(c)	(b)	(c)	(d)	(c)	(d)	(a)	(a)	(a)	(a)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(a)	(d)	(b)	(c)	(d)	(b)	(c)	(d)	(a)	(b)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(c)	(c)	(d)	(d)	(b)	(b)	(d)	(d)	(d)	(a)	(b)	(b)	(c)	(d)	(d)	(b)	(d)	(a)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(a)	(c)	(a)	(c)	(b)	(d)	(b)	(c)	(d)	(d)	(b)	(a)	(c)	(d)	(a)	(a)	(d)	(d)	(c)
Q. No.	41	42	43	44	45															
Answer	(b)	(d)	(d)	(c)	(a)															

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(c)	(b)	(c)	(c)	(c)	(a)	(a)	(d)	(b)	(d)	(d)	(b)	(b)	(d)	(b)	(b)	(a)	(c)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(a)	(a)	(a)	(c)	(d)	(a)	(d)	(a)	(d)	(c)	(c)	(b)	(d)	(b)	(b)	(d)	(c)	(b)	(a)

HINTS AND SOLUTIONS

Warm up

1. Profit percentage $= \frac{150}{600} \times 100 = 25\%$

2. CP $= \frac{\text{Rs } 1950}{1.3} = \text{Rs } 1500$

   Hence required Selling Price = Rs 1500 × 1.4 = Rs 2100

3. Let us assume that the cost price = Rs X

   $56 - x = x - 42$

   $2x = 98$

   Hence, $x = 49$

4. MP = x

   CP = 3 x /4

   SP $= \frac{3}{2}x$

   Profit Percentage $= \frac{\frac{3}{2}x - \frac{3}{4}x}{\frac{3}{4}x} \times 100 = 100\%$

5. Assume cost price = Rs X

   $x - 150 = \frac{x}{16} \qquad \Rightarrow x = 160$

6. Profit Percentage $= \frac{50}{200} \times 100 = 25\%$

7. SP = x

   Profit Percentage $= \frac{\text{Profit}}{x} \times 100$

   $25 = \frac{\text{Profit}}{x} \times 100$

   Profit $= \frac{1}{4}x$

   CP $= x - \frac{1}{4}x = \frac{3}{4}x$

   Profit Percentage $= \frac{\frac{1}{4}x}{\frac{3}{4}x} \times 100 = 33\frac{1}{3}\%$

8. CP = Rs x

   Profit $= \frac{x}{10}$

   Now, $\frac{\frac{x}{10} + 45}{x} \times 100 = 25$

   $X = 300$

9. List Price = x

   CP $= \frac{4}{5}x$, SP $= \frac{6}{5}x$

   Profit Percentage $= \frac{\frac{6x}{5} - \frac{4x}{5}}{\frac{4}{5}x} \times 100 = 50\%$

10. Cost price of 1 orange = Rs $\frac{10}{200} = \text{Rs } \frac{1}{20}$

    SP of 1 orange = Rs $\left(\frac{1}{20} + \frac{1}{4} \cdot \frac{1}{20}\right) = \text{Rs } \frac{1}{16}$

    So, he should sell 16 oranges in a rupee to make a profit of 25%.

11. Profit Percentage $= \frac{6}{18} \times 100 = 33\frac{1}{3}\%$

12. $\frac{1}{10}x = 20$

    $x = 200$

13. CP $= 360 + 360 \cdot \frac{1}{9} = \text{Rs } 400$

14. Profit in order to gain 40% = Rs 6400

    More money charged = (Rs 6400–2500) = Rs 3900

15. CP of 1 book = Rs 15

    SP of 1 book in order to gain 25% profit = Rs 18.75

    Required No. of Books = $\frac{22.5}{18.75} = 12$

16. Suppose initially x kg of rice was bought.

    Now, $\frac{1}{4}x = 6$

    $x = 24$

    Initial price = Rs $\frac{120}{24} = \text{Rs } 5$

17. CP of 1 apple = Re 1

    CP or 12 apples = Rs 12, SP of 9 apples = Rs 12

    Required Difference = $\frac{3}{9} \times 100 = 33\frac{1}{3}\%$

18. Let CP of 1000 gm = Rs 1000

    CP of 800 gm = Rs 800

    SP of 800 gm = Rs 1000

    Profit Percentage $= \frac{200}{800} \times 100 = 25\%$

19. CP of 100 cm = Rs 100
CP of 110 cm = Rs 110
Loss = Rs 110

$$\text{Loss Percentage} = \frac{10}{100} \times 100 = 10\%$$

20. Go through options.
N = 500Fdfs

Foundation

1. $\text{CP} = 2400 - 2400 \times \frac{1}{6} - 2400 - 400 - 2000$

2. $$\text{Profit Percentage} = \frac{2400 \times 20}{100} = 480$$

CP = Rs (2400–480) = Rs 1920

3. $$\text{Actual Profit Percentage} = \frac{480}{1920} \times 100 = 25\%$$

4. CP of 10 Apples = Rs 1

CP of 1 Apple = Rs $\frac{1}{10}$

$$\text{SP of 1 Apple} = \text{Rs}\left(\frac{1}{10} + \frac{1}{4} \cdot \frac{1}{10}\right) = \frac{1}{8}$$

5. CP of Mayank = CP of Chandra × 0.7 = Rs 150 × 0.7 = Rs 105
SP of Mayank = Rs 126

6. CP of 1 liter petrol = Rs 5

SP of 1 liter Petrol = Rs $\frac{1}{5}$

$$\text{Loss \%} = \frac{5 - \frac{1}{5}}{5} \times 100 = \frac{24}{25} \times 100 = 96\%$$

7. MP = x

$$\text{S.P.} = x - \frac{1}{9}x = \frac{8}{9}x$$

$$\text{CP} = \frac{8}{9}x - \frac{8}{9}x \cdot \frac{1}{8} = \frac{8x}{9} - \frac{x}{9} = \frac{7x}{9}$$

$$\text{Required Percentage} = \frac{x - \frac{7x}{9}}{\frac{7}{9}x} \times 100 = \frac{\frac{2x}{9}}{\frac{7x}{9}} \times 100$$

=28.56%

8. Let CP = Rs 100
SP = Rs 120
MP = x

$$\frac{3}{4}x = 120$$

x = Rs 160
Hence, 60%

9. Maximum can be anything beyond 60%.

10. Without knowing the profit percentage, we cannot determine the CP.

11. Assume x is the SP per kg.
$1100 - 100x = 20x$
$120x = 1100$

$$x = \frac{110}{12} = 9.16$$

12. $$\frac{22}{100}x = (750 - x) \times \frac{8}{100}$$

x = 200
SP of 2nd Cow = (550-44) = Rs 506

13. $$\text{Net Loss Percentage} = \left(\frac{20}{10}\right)^2 = 4\%$$

For every Rs 100 invested, Rs 4 is the loss, or, for every Rs 96 realized through selling, Rs 4 is the loss.

$$\text{Loss amount} = \text{Rs}\left(48000 \times \frac{1}{24}\right) = \text{Rs}\,2000$$

14. MP of 1 Pencil = Re 1
For supplier, SP of 20 pencils = Rs 16
For retailer, SP of 20 pencils = Rs 20

$$\text{Profit Percentage} = \frac{4}{16} \times 100 = 25\%$$

15. $$x + \frac{1}{5}x = (100000 - x) - \frac{1}{5}(100000 - x)$$

$$x + \frac{1}{5}x = 100000 - x - 20000 + \frac{x}{5}$$

x = 40000

16. $$x + \frac{1}{5}x = (13000 - x) + \frac{2}{5}(13000 - x)$$

$$x + \frac{1}{5}x = 13000 - x + 5200 - \frac{2}{5}x$$

$$2x + \frac{3}{5}x = 18200$$

$$\text{Or, } \frac{13}{5}x = 18200$$

x = 7000

17. Required Difference = 7000 – 6000 = 1000

18. CP of 1 article = Re 1

CP of 40 articles = Rs 40

SP of 30 articles = Rs 40

$$\text{Profit Percentage} = \frac{10}{30} \times 100 = 33.33\%$$

19. Due to price hike of 20%, quantity of less tea available

$= \frac{1}{6}x = 4 \quad x = 24$ (where x = initial quantity)

$$\text{Original price} = \text{Rs}\frac{120}{24} = \text{Rs}\,5$$

20. $$\text{Profit Percentage} = \frac{250}{750} \times 100 = 33.33\%$$

21. Weight he is selling instead of 1000 grams = 1000/1.3 = 769 grams.

22. Let MP = Rs 100

(i) Customer has to pay = Rs 61.6

(ii) Customer has to pay = Rs 61.6

(iii) Customer has to pay = Rs 61.6

23. Let No. of chips supplied = 100

For whole seller, Net CP = (10000 × 100 + 5 × 10000) = Rs 1050000

$$\text{Profit} = \text{Rs}\left(1050000 \times \frac{1}{5}\right) = \text{Rs}\,210000$$

SP for whole seller = Rs 1260000

SP of 1 chip = Rs 12600

24. Go through options.

25. Let SP = Rs 100

Profit = Rs 25

CP = Rs 75

$$\text{Actual Profit Percentage} = \frac{25}{75} \times 100 = 33.33\%$$

26. Let SP of 1 kg article = Rs 100

For 1st Shopkeeper

SP of 1 kg Article = Rs 75

For 2nd Shopkeeper

SP of 1.25 kg Article = Rs 100

SP of 1 kg Article = Rs 80

27. Let CP of 1st Cow = Rs x

CP of 2nd Cow = Rs Y,

Hence net CP = Rs $(x + y)$

SP of 1st Cow = Rs y

SP of 2nd Cow = Rs X

Net SP = Rs $(x + y)$

Hence, No Profit No Loss

28. $$\text{CP} = \text{Rs}\left(180 - 180 \cdot \frac{1}{6}\right) = \text{Rs}\,150$$

29. In the whole question, there is nowhere the mention of any Rs figures. Since amount is not given, we cannot calculate any value (in Rs).

30. Let CP of book for Anoop = Rs 100

CP of book for Mayank = Rs 120

CP of book of Siddharth = Rs 150

CP of book of Shishir = Rs 135

$$\text{Required loss percentage} = \frac{35}{135} \times 100 = 25.92\%$$

31. Let CP = Rs 700

x = Rs 90

L = 10%

Y = Rs 110

P = 10%

Putting these values, we get (c) as the answer.

32. CP of 100% Pure Milk = Rs 100

$$\text{CP of the Mixture} = \text{Rs}\left(100 - 100 \times \frac{1}{6}\right) = \text{Rs}\,83.33$$

Hence, 83.33% milk is present 1 liter of mixture.

Alternatively, if he has initially 10 L of milk, he must have made it 12 L of mixture to get a profit of 20% (since SP per litre = CP per litre). Hence out of 12 litres of mixture, 10 litres is milk and 2 litres is water.

33. Let us assume that CP = Rs x

$720 - x = 2\,(x - 420)$

$720 - x = 2x - 840$

$3x = 1560$

$x = 520$

34. According to the question, 14.28% of Mark up of 1st shopkeeper = 12.5% of mark up of 2nd shopkeeper. With this we can compare the markup prices.

However nothing can be said about the cost prices of the shopkeepers.

35. Let MP = Rs 100 and CP = Rs X

then, $450 - 5x = 700 - 8x$

$$x = \frac{250}{3}$$

$$\text{Required ratio} = \frac{100}{250/3} = 6:5$$

36. Since, cost price and selling price are not known therefore, we cannot find profit percentage.

37. $13x + 9\,(50 - x) = 550$

$4x = 100$

$x = 25$

38. Go through options.

39. Assume that MP = Rs 100

then CP of Anam = Rs 70

and his SP = Rs 160

So, his Profit = Rs 90.

Now using unitary method, when profit of Anam = Rs 90, then his MP = 100. So when his MP = Rs 81, his MP = 90.

40. CP of 100 m = Rs 100

$$\text{CP of } \frac{500}{6}\text{m} = \text{Rs } \frac{500}{6}$$

$$\text{Profit Percentage } = \frac{100 - \frac{500}{6}}{\frac{500}{6}} \times 100 = 20\%$$

Alternatively, profit is 16.66% by selling 83.33% quantity.

$$\text{Hence profit percentage} = \frac{16.66 \times 100}{83.33}\% = 20\%$$

41. Let MP = Rs 100

CP = Rs 70 and SP = Rs 80

Hence, profit percentage = 14.28%

42. $(7.50 \times 2x) - (7x + 6x) = 80$

$2x = 80$

$x = 40$

43. CP of 1000 ml milk = Rs 100

CP of 900 ml milk = Rs 9

Profit = Rs (3×20) = Rs 60

$$\text{Profit Percentage } = \frac{60}{9 \times 20} \times 100 = 33.33\%$$

44. Net CP = Rs $50x$

Net SP = Rs $48x$

$$\text{Loss Percentage } = \frac{2x}{50x} \times 100 = 4\%$$

45. Let CP = Rs 100

MP = Rs 140

Profit, when discount is 5% = Rs 33

Profit, when discount is 10% = Rs 26

$$\text{Hence, CP} = \frac{100}{7} \times 14 = \text{Rs } 200$$

MP = Rs 280

SP = Rs 224

46. 1 Gross = 12 dozen

Net CP = Rs (120 + 12) = Rs 132

Net SP = Rs 158.4

$$\text{Required price } = \text{Rs } \frac{158.4}{10} = \text{Rs } 15.84$$

47. 15% of CP = Rs 56.25, Hence CP = Rs 375

New SP = Rs 450, Hence profit = Rs 75

Profit percentage = 20%

48. $x = \frac{24 - x}{x} \times 100$, So. $X = 20$

49. $$\text{Net CP} = \text{Rs}\left(228 - \frac{1}{3} \times 228 + \frac{1}{5} \times 152\right)$$

= Rs (152 + 30.4) = Rs 182.4

$$\text{SP} = \text{Rs}\left(182.4 + \frac{1}{4} \times 182.4\right) = \text{Rs } 228$$

50. Net CP = Rs 132

Money paid by seniors = Rs 33

$$\text{Required Percentage } = \frac{99}{66} \times 100 = 150\%$$

Moderate

1. Maximum possible discount in any case can be only 100% and any discount more than 100% is actually 100% only. Now, whatever is the discount, till the moment we get a discount of 100%, value of the article cannot be zero (even infinite discounts of 99.99% won't make it equal to a discount of 100%) So, number of rounds needed to obtain 100% discount = 50.

2. As discussed above, maximum possible discount can be only 100%.

3. Despite the fact that a customer has won 18 rounds out of the first 20 rounds, if he looses in the 20th round, net discount = 0%

6.

Actual CP	C
New CP	0.9C
Actual SP	S
New SP	1.1 S
Actual Profit %	100(S – C)/C
New Profit %	100 (1.1 S – 0.9C)/0.9C

So, [S/C] – 1 = 2/7

Profit percentage = 28.56%

And obviously CP cannot be calculated.

6. As done above, CP cannot be calculated.
Alternatively, since in the question, nowhere the term Rs has been used, the value of CP cannot be calculated.

7. Profit percentage as calculated above = 28.56%.

8. Since there can be any value of CP, the number of values of SP can also be anything, provided profit percentage = 28.56%.

11. Assume CP/1000 gm = Rs 1000
So, MP = Rs 1400
SP = Rs 1260
Now, when he sells 1000 gms, he actually obtains the money for only 800 grams (Go through the statement carefully).
So, when he sells articles worth Rs 1000, money obtained after selling will be equal to = Rs 1260 × 0.8 = Rs 1008
So, profit percentage = 8%

12. Assume his CP = Rs 1000/1100 gm
MP = Rs 1200 and SP = Rs 960/900 gm
So, SP/1100 gm = Rs 1173.33
So, profit = Rs 173.33
Profit percentage = 17.33%

18. Since in all these cases, more articles are given free than the purchased article, so the net discount is more than 50%. Hence in all these layers, the discount offered is more than 50%.
(If the articles given free is equal to the articles purchased, discount percentage will be equal to 50%).

20. If we club the first three layers, it is like: buy 6, get 9 free. So, net discount = 60%

	Sales	*Profit*	*Expenses*
BOB	$2x = 50$	$y = 5$	$3z = 45$
MOB	60	$t - 1$	$60 - 2 = 58$
ZOB	15	$t = 3$	$15 - 3 = 12$
DOB	$x = 25$	$2y = 10$	$25 - 10 = 25$

34. Price = Rs X
SP = Rs 1.1 x = 0.5 x + 15, So, 0.6 x = 15
So, x = 25
Alternatively, we can do this question very easily by using options.

36. CP = Rs X/dozen = 0.833/copy
So, MP = 1.69 x/dozen
SP = Rs 1.4365x /13 copies = 0.1105 x/copy
So, profit = 32.6%

37. List price = 228
CP (before duty) = Rs 152
CP (after duty) = Rs 152 × 1.2 =
SP = Rs 152 × 1.2 × 1.25 = Rs 228

38. This whole sequence goes like this:
100 →10% UP → 110 →10% UP → 121→20% UP→145.2→ UP by 20% and down by 10%→ 156.816
Distributors' SP = 121
So, percentage reduction = 22.8%

43. Go through the options.

44. This question can be done by forming equation by assuming the CP to be Rs X.
Alternatively, doing a bit of hit and trial gives the value of p = 200.
Using this, we get the answer as option (d) = 66.66%

Advanced

1. Cost Rs 2400
Published Price Rs 3.25
SP = 75/100 × 3.25 = Rs 2.4375
No. of free copy = (3000/25) = 120 + 500 = 620
So, total SP = 2380 × Rs 2.4375 = Rs 5801.25
Hence percentage gain = 5801.25 – 2400/2400 – 100 = 144%

3. $9(x - y) = 72$, $x - y = 8$
Therefore, only possibility is 19 or 91.

4. Matador's capacity is just 500, so it will be used minimum, i.e., 1 day.
So truck will be used for the rest 5 days.

6. Centre table – 40 sofa set – 100
According to Mohit, cost of centre table-
90/100 × 40 = 36
Cost of sofa set – 75/100 × 100
According to shopkeeper,
CT – 75/100 × 40 = 30
SS – 90/100 × 100 = $\frac{90}{120}$
Therefore, extra = 8.1%

8. Through the method of alligation
+30 –10
+10
20 20
1 : 1

9. Amar–100, Bharat–120, Sridhar–132
No profit or loss is mentioned about the deal between Sridhar and the mechanic. So the answer cannot be determined.

10.

	Weekdays	*Weekend*
Fixed Cost	Rs 3,75,000	Rs 75,000
Variable Cost	$2.75 \times 6x$	$(5x/6) \times 3$
Total Cost	Rs 3,75,000 + $2.75\,x$	Rs 75,000 + $(5x/6) \times 3$
Reveneue through advertisement	Rs 15 lacs	Rs 4,35,000

Where x is the number of copies sold any week day.
For breaking even,
$[\text{Rs } 375000 + 16.5x] + [\text{Rs } 75,000 + (5x/6) \times 3] = \text{Rs } 19,35,000 + 10.25x$
$\text{Rs } 450000 + 19x = \text{Rs } 1935000 + 10.25x$
So, $x = 1697142.28$
Hence, to obtain break-even point, minimum 1697143 copies are to be sold.

12. Since we are not having any information regarding the total number of copies sold (the information is talking about the reader of "The Deccan City"), we cannot determine the profit.

13. Rs 75, 000 + $3x$ = Rs 4,35000 + $1.5x$, where x is the number of copies sold on Sunday.

14. Number of items – Shown number of items = 63
$9(x - y) = 63$, $(x - y) = 7$ where x and y are its Ten's and Unit's digits
So, the number may be 81 and 18 or 92 and 29.
However value = Rs 936, which is divisible by 18 only.
So, actually left items = 81.

15. Rate = 936/18= 52
It is given that price/piece was also reversed.
So, price/piece = Rs 25
Collection = Rs 25 × 44 = Rs 1100

27.

Mode of Transportation	*Rent for one trip*	*Breakages During Transportation*	*Total Cost*
Truck	Rs 6,000	Rs 40,000	Rs 46,000
Small truck	Rs 5,000	Rs 50,000	Rs 55,000
Jeep	Rs, 3750	Rs 75,000	Rs 78,750
Rickshaw	Rs 5,000	Rs 50,000	Rs 55,000

Obviously, truck is the most profitable mode.

28. For 10% profit, he should obtain 1.1 × (Rs 5,00,000 + Rs 6,000) = Rs 5,56,600
Number of cups left = 46,000
So, SP per cup = Rs 12.1

38. Using the options,
(a) 2,000 units, C = 2,000 × 30 = Rs 60,000
(b) 3,000 units, C = 3,000 × 20 = Rs 60,000
(c) 4,000 units, C = 4,000 × 20 = Rs 80,000
(d) 5,000 units, C = 5,000 × 15 = Rs 75,000

39. At 5,000 units, C = 5,000 × 25 = Rs 1,25,000

40. From question number 38 At 2,000 units,
C = Rs 60,000.

5

CHAPTER

Average and Its Applications



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Average and its various interpretations
- Definitions and properties
- Applications of average in different cases such as alligation
- Different cases of mixtures and alligation
- Methods of solving questions

Introduction to the topic

Averages and their applications was once an important topic for CAT, but the number of questions asked from this topic has been reduced in the last couple of years. However, this concept is used vigorously in other exams. As a concept, average should be seen more as a logical concept of group formation than a pure mathematical concept. And this is true about its applications also.

AVERAGE

Traditionally, average is calculated by dividing the sum of all the numbers by the number of numbers.

$$\text{Average} = \frac{\text{Sum of numbers}}{\text{Number of numbers}}$$

For example, the average of the four numbers 214, 215, 219, 224 will be

$$\text{Average} = \frac{214+215+219+224}{4} = 218$$

Central Value Meaning of Average

Average can also be seen as the central value of all the given values.

Applying this definition for the above example let us assume the central value of all the given numbers = 214

Now, find the deviations of all the numbers from 214

214 215 219 224

When assumed central value is (214), the sum of the deviations

= 0 + 1 + 5 + 10

Now, finding the average of deviations gives us

$$\frac{0+1+5+10}{4} = \frac{16}{4} = 4$$

So, average = assumed central value + average of deviations = 214 + 4 = 218

Thus, we can assume any value to be the assumed average and then find the average of all the deviations; and when we add all the numbers and divide it by number of numbers, 0 is assumed to be the central value.

Example 1 Average age of A, B and C is 84 years. When D joins them the average age of A, B, C and D becomes 80 years. A new person E, whose age is 4 years more than D, replaces A and the average age of B, C, D and E becomes 78 years. What is the age of A?

Solution Since the average age of A, B and C is 84 years so, we can assume that age of A, B and C is 84 years.

A = 84 years

B = 84 years

C = 84 years

After D has joined them,

	Initially	*Finally*
A	84 years	80 years
B	84 years	80 years
C	84 years	80 years
D	.....	80 years

Decrease in the age of A, B and C can be attributed to the increase in the age of D. So, after getting 12 years in total (4 years each from A, B and C) D is at 80 years. The original age of D = 80 − 12 = 68 years.

So, age of E = 72 years

Now, the average age of A, B, C and D = 80 years; A + B + C + D = 320

And average of B, C, D and E = 78 years; B + C + D + E = 312

(Since the average difference between the age of A and E is 2 years,)

Difference (A − E) = 2 × 4 = 8 years

Since E = 72 years, so A = 80 years

By using central value method of averages every question of average can be done by mental calculation only.

Example 2 Average of ten two digit numbers is S. However when we reverse one of the numbers AB as BA from the given 10 numbers, then the average becomes S + 1.8. What is the value of B − A?

Solution Average of 10 numbers is increasing by 1.8, so it can be assumed that 1.8 has been added to all the numbers.

So, BA is 1.8 × 10 = 18 more than AB.

There are so many two-digit numbers which satisfy above condition. Using hit and trial method, the number can be 13, 24, 35, 46, 57, 68, 79. In every case, difference between the digits = 2

Otherwise, we can use the formula

(BA − AB) = 9 × (B − A)

Where BA and AB are two-digit numbers.

Example 3 The average score of Rahul Dravid after 25 innings is 46 runs per innings. If after the 26th innings, his average runs increased by 2 runs, then what is his score in the 26th inning?

Solution Runs in 26th inning = Total runs after 26th innings − Total runs after 25th innings

$$= 26 \times 48 - 25 \times 46 = 98$$

Alternatively, this question can be done by the above given central value meaning of average. Since the average increases by 2 runs per innings, we can assume that 2 runs have been added to his score in each of the first 25 innings. Now, the total runs added in these innings have been contributed by the runs scored in the 26th inning, which must be equal to 25 × 2 = 50 runs.

And after contributing 50 runs, his score in the 26th inning is 48 runs.

Hence, runs scored in the 26th inning = new average + old innings × change in average

$$= 48 + 25 \times 2 = 98.$$

To have a mental mapping, we can see the whole situation as:

Number of innings	*Average in the 1st 25 innings*	*Averate in the 1st 26 innings*	*Addition*
1	46	48	2
2	46	48	2
3	46	48	2
...	...	...	...
...	...	...	...
...	...	...	...
25	46	48	2
26		48	

Properties of Average

1. Average always lies in between the maximum and the minimum value. It can be equal to the maximum or minimum value if all the numbers are equal.
   For example—A1, A2, A3 and A4 are four numbers given where A1>A2>A3>A4.
   Average of these four numbers will always lie in between A1 and A4.
   However, if all the four numbers are equal (A1 = A2 = A3 = A4) then the average will be equal to each of these numbers.
   Average = A1 = A2 = A3 = A4.
2. Average is the resultant of net surplus and net deficit, as used in the central tendency method.
3. When weights of different quantities are same, then simple method is used to find the average. However, when different weights of different quantities are taken, then it is known as weighted average. Here the method of weighted average is used to find the average.
   For example, assume per capita income of India is USD 500 and per capita income of US is USD 200. Now if we merge India and US into one country then it is observed that per capita income of this new country will not be equal to $\frac{500+200}{2}$ = USD 350
4. If the value of each quantity is increased or decreased by the same value S, then the average will also increase or decrease respectively by S.
5. If the value of each quantity is multiplied by the same value S, then the average will also be multiplied by S.
6. If the value of each quantity is divided by the same value S ($S \neq 0$) then the average will also be divided by S.

Example 4 The average of 4 positive numbers is A and the average of all the possible triples formed out of these four positive numbers is B. Which of the following is true regarding A and B?

(a) A = B (b) A > B
(c) A < B (d) Cannot be determined

Solution Let us assume that the numbers are 1, 2, 3 and 4

Average of 1, 2, 3 and 4 : 1 + 2 + 3 + 4 = 10/4 = 2.5

The triplets are 1, 2 and 3; and the average = $\frac{6}{3}$

1, 2 and 4; and the average = $\frac{7}{3}$

1, 3 and 4; and the average = $\frac{8}{3}$

2, 3 and 4; and the average = $\frac{9}{3}$

Average of these four averages = $\frac{\frac{6}{3}+\frac{7}{3}+\frac{8}{3}+\frac{9}{3}}{4} = 2.5$

So, option a (A = B) is the answer.

Central value method

It should be observed here that when we find the average of all the possible triplets, all the numbers (1, 2, 3, 4) are added thrice. So effectively we are adding 12 numbers. Hence, the average should be equal to:

$$\frac{3(1+2+3+4)}{12} = 2.5$$

Thus, the average will be equal in all the cases.

Extension to this problem The average of four positive numbers is A and the average of all the possible pairs formed out of these four positive numbers is B. Which of the following is true?

(a) A = B (b) A > B
(c) A < B (d) Cannot be determined

Some Special Cases

1. Average involving time, speed and distance

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

However, while solving the questions involving time, speed and distance, we should assume some distance, preferably the LCM of all the given speeds.

Example 5 Lovely goes to Patna from New Delhi at a speed of 40 km/h and returns with a speed of 60 km/h. What is her average speed during the whole journey?

Solution Assuming that the total distance between Patna and New Delhi is 120 km (LCM of 40 and 60) the total time taken (Patna – New Delhi and New Delhi – Patna) = 3 + 2 = 5 h

So, average Speed = $\frac{240}{5}$ = 48 km/h

2. Average involving age

Average of a group of n persons given at any point of time can be calculated in the following way

5 years ago	*10 years ago*	*Now*	*10 years later*	*5 years later*
N – 5	N – 10	N	N + 10	N + 5

Example 6 The average age of the 5 members of a family is 20 years. The youngest member of the family is 4 years old. At the time of his birth the average age of the rest of the family was N years. What is the average age of the family (in terms of N) excluding the youngest member?

Solution Sum of ages of all the members of the family = 100

Sum of ages of all the members of the family excluding the youngest number = 100 – 4 = 96

So, average age of all the members of the family excluding the youngest number = 96/4 = 24 = N

What is the average age of the family(in terms of N) excluding the youngest member = N + 4

3. Average involving number system

1. Average of 1st n consecutive natural numbers $= \frac{n+1}{2}$
   For example, the average of 1st five natural numbers = 3
2. The average of 1st n consecutive even natural numbers $= n + 1$
   Sum of 1st n consecutive even natural numbers $= n(n + 1)$
   For example, the average of 1st five even natural numbers = 6
3. The average of 1st n consecutive odd natural numbers $= n$
   Sum of 1st n consecutive odd natural numbers $= n^2$
   For example, the average of 1st five odd natural numbers = 5

Weighted Average

It is observed that the average can be calculated only if the weights of all the factors, is same. So, the weighted average is a more generalized form of average. This can be further understood with the following illustration

	Class A	*Class B*
No. of students	10	10
Average age	12 years	16 years

Now, if we combine both these classes, then the average age of all the students $= \frac{12+16}{2} = \frac{28}{2} = 14$ years. This is one standard example of Average.

Let us see another example:

	Class A	*Class B*
No. of students	12	16
Average age	10 years	14 years

Now, if we combine these two classes, then the average can not be calculated by the above mentioned method, since the weights attached to different averages are different.

Some more cases of weighted average

1. As we have observed above in the case of average, if per capita income of India is USD 500 and per capita income of US is USD 200, and if we merge India and US into one country, then the per capita income of this new country (India + US) cannot be found by just adding the per capita income of both the countries and dividing it by 2.
   The weights, i.e., the population attached to the different averages, i.e., the per capita income would also have to the considered.
2. Average speed cannot be calculated by just adding the different speeds and then dividing it by 2. This can be understood by the following example:
   A person goes to A from B at a speed of 40 km/h and returns with a speed of 60 km/h, then the average speed for the whole journey can not be equal to 50 km/h.

   We know that average speed $= \frac{\text{Total distance}}{\text{Total time}}$

Finding expression for weighted average

Let us assume there are N groups with the following structure:

Group no.	*No. of members*	*Avg. age of the group*
G_1	N_1	A_1
G_2	N_2	A_2
G_3	N_3	A_3
...	...	...
...	...	...
G_N	N_N	A_N

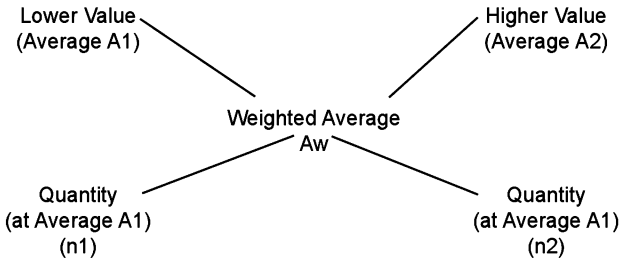
If we combine all these groups, then the average age of all the members = $(N_1 \times A_1 + N_2 \times A_2 \ldots + N_N \times A_N)/(N_1 + N_2 + N_3 + \ldots + N_N)$

$$= \frac{\sum AN}{\sum N}$$

Group no.	*No. of members*	*Avg. age of the group*
G_1	N_1	A_1
G_2	N_2	A_2

Considering that there are only two groups and both the groups are combined then the average age of all the members $= (N_1 \times A_1 + N_2 \times A_2)/(N_1 + N_2) = A_w$

Simplifying the above written expression, we get the conventional criss-cross method as given below



And we write this as: $\frac{n_1}{n_2} = \frac{A_2 - A_w}{A_w - A_1}$

i.e.,

$$\frac{\text{Quantity (Lower Priced)}}{\text{Quantity (Higher Priced)}} = \frac{\text{Higher Price} - \text{Averge Price}}{\text{Average Price} - \text{Lower Price}}$$

It is quite obvious that the ratio of the number of persons/items in different groups is proportionate to the deviations of their average from the average of all the people combined.

This average of all the members combined is known as weighted average, and is denoted by A_w. This process of mixing the two groups is also referred as alligation.

Elements of weighted average

As we can see from the above derivation there are five quantities

i. Number of members in 1st group (n_1)
ii. Number of members in 2nd group (n_1)
iii. Average of 1st group (A_1)
iv. Average of 2nd group (A_2)
v. Weighted average (A_w)

Normally, in the case of weighted average, we get questions in which one of these five elements is missing, and with the help of the remaining four quantities, the value of that missing quantity is found. Different possibility (situations) are given below (Y represents – data given, N represents – data not given):

Situations	n_1	n_2	A_1	A_2	A_w
First	Y	Y	Y	Y	N
Second	Y	Y	Y	N	Y
Third	Y	Y	N	Y	Y
Fourth	Y	N	Y	Y	Y
Fifth	N	Y	Y	Y	Y
Sixth	N	N	Y	Y	Y

First Situation

Example 7 10 kg of rice priced at Rs 12 per kg is mixed with 6 kg of rice priced at Rs 16 per kg. What is the average price of the whole mixture?

Solution Lower priced value = Rs 12 per kg and its quantity = 10 kg

Higher priced value = Rs 16 per kg and its quantity = 6 kg

Using Alligation,

$\frac{10}{6} = \frac{16 - A_w}{A_w - 12}$, or, A_w = Rs 13.5/kg

However, in my opinion, in this situation it is better to use the normal method rather than using the weighted average method of finding A_w.

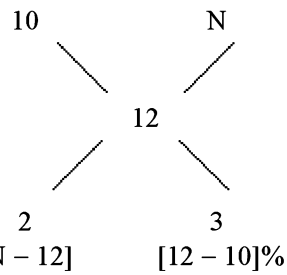
Normal method – Total value = $12 \times 10 + 16 \times 6 = 216$

So, average price = $\frac{216}{16}$ = Rs 13.5/kg

Second/Third Situation

Example 8 Two varieties of rice are mixed in the ratio 2:3. The price of the mixture is Rs 12 per kg and the price of the variety having lower weight is Rs 10 per kg. Find the price of the other variety.

Solution



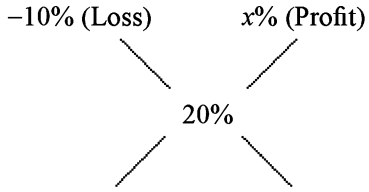
Now, $\frac{2}{3} = \frac{(N-12)}{(12-10)} = \frac{N-12}{2}$

So, N = Rs 13.33 per kg

Fourth/Fifth Situation

Example 9 Some articles are purchased for Rs 450. 1/3rd of the articles are sold at a loss of 10%. At what percentage profit should the remaining articles be sold to obtain a net profit of 20% on the whole transaction?

Solution



Now, $= [20-(-10)]/[x-20] = \frac{2}{1}$

So, $x = 35\%$

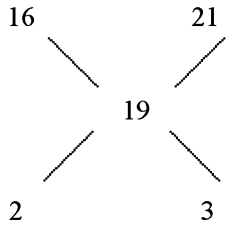
It is seen that the quantities are in the ratio of 1 : 2, so the deviation from mean percentage profit in the loss percentage and profit percentage will also be same.

2 corresponds to 30%, so 1 will correspond to 15%. So, $x = 35\%$

Sixth Situation

Example 10 Two different qualities of sugar are mixed in some ratio. The price of one quality of sugar is Rs 16/kg and that of another quality is Rs 21/kg. In what ratio have the sugar of two qualities been mixed if the price of the mixture is Rs 19/kg?

Solution



So, the ratio of quantity of sugar of different qualities = 2 : 3

MIXTURES

When two or more than two pure substances/mixtures are mixed in a certain ratio, they create a mixture. Here we shall confine ourselves to mostly homogenous mixtures in view of the questions commonly asked in CAT.

Mixing without Replacement

In this particular type of mixing, two or more than two substances are mixed without any part of any mixture being replaced.

Example 11 In a mixture of 420 litres the ratio of milk and water is 6 : 1. Now, 120 litres of the water is added to the mixture. What is the ratio of milk and water in the final mixture?

Solution Volume of milk = 360 litres and volume of water = 60 litres.

When 120 litres of water is added, volume of water = 180 litres

So, the ratio of milk water = 2 : 1

Example 12 A milkman mixes 20 litres of water with 80 litres of milk. After selling one-fourth of this mixture, he adds water to replenish the quantity that he had sold. What is the current proportion of water to milk?

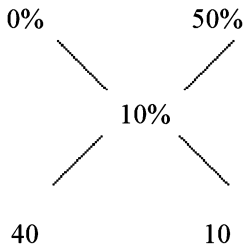
Solution Ratio of milk and water = 20 : 80

When 1/4th of this mixture is sold, total volume of mixture will be reduced by 25%, so 25% of milk and water both will reduce. So, volume of milk and water after selling out 1/4th of mixture = 60 litres and 15 litres respectively. Addition of 25 litres of water will finally give us the following: volume of milk = 60 litres and volume of water = 40 litres. Hence, the ratio of water and milk = 40 : 60 = 2 : 3.

Example 13 How many litres of fresh water should be mixed with 30 litres of 50% milk solution so that resultant solution is a 10% milk solution?

Solution

Method 1 *Using Alligation*



So, the ratio of fresh water added: milk solution = 4 : 1

Hence, 120 litres of fresh water should be added.

Method 2 *Principle of constant volume of one component*

Since we add fresh water, the volume of milk will be constant.

Now volume of milk = 15 litres = 10% of the new mixture.

So, 100% of the new mixture = 150 litres

So, volume of fresh water added = 150 − 30 = 120 litres.

Method 3 *Principle of inverse proportion*

We know that concentration is inversely proportional to the volume of solute added.

So, in this case $30 \times 50\% = 10\% \times (30 + x)$, where x is the volume of water added.

So, $x = 120$ litres

Method 4 *Using equation*

In the final mixture, $\frac{\text{Milk}}{\text{Total}} = 10\% = \frac{15}{30+x}$

So, x = 120 litres

Mixing with replacement In this particular type of mixing, two or more than two substances are mixed by replacing some part of a mixture. In these types of questions, total volume may or may not be the same and information regarding the same can be obtained from the question.

Case 1 When the quantity withdrawn and the quantity replaced are of the same volume.

Initially there are 40 litres of milk, and 4 litres of milk is replaced with 4 litres of water

Obviously, there will be 36 litres of milk and 4 litres of water.

Now, 4 litres of mixture is replaced with 4 litres of water

The quantity of milk and water being withdrawn here will be in the ratio of 9:1 (36:4). So, quantity of milk withdrawn = 9/10 × 4 = 3.6 l.

So, the volume of milk = 32.4

And the volume of water = 7.6

Now, again 4 litres of mixture is replaced with 4 litres of water

The quantity of milk and water being withdrawn here will be in the ratio of 81:19 (32.4:7.6). So, the quantity of milk withdrawn = (81/100) × 4 = 3.24 l

So, the volume of milk = 29.16

And the volume of water = 10.84

If we summarize the above values, then it looks like

	1st operation		*2nd operation*		*3rd operation*	
	Taken out	*Left*	*Taken out*	*Left*	*Taken out*	*Left*
Milk	4	36	3.6	32.4	3.24	29.16
Water	0	4	0.4	7.6	0.76	10.84

It can be seen that the quantity of water or milk withdrawn is 10% of the existing volume of milk or water because only 10% of the total volume of 40 litres taken out.

With this we can deduce a standard formula for these kinds of calculations.

If V is the initial volume of milk (or any liquid), and x litres of milk is always replaced by water, then quantity of milk left after n such operations = $V\left(1-\frac{x}{v}\right)^n$

This formula is very similar to the standard formula we have seen in the case of Compound Interest $\left[P(1+\frac{R}{100})n\right]$.

The only difference between the two formulae is that while the interest is being added every year (or for the given time-period), volume of milk gets reduced after every operation.

Using the values of the above example, quantity of milk left after 3 operations $= 40\times\frac{36}{40}\times\frac{36}{40}\times\frac{36}{40} = 19.16$ litres

The same problem can be solved with straight-line approach of percentage also Since 10% of existing volume is taken out every time, the percentage of milk in the final mixture after the third operation = 72.9%

$$\left(100\% \xrightarrow{10\%\downarrow} 90\% \xrightarrow{10\%\downarrow} 81\% \xrightarrow{10\%\downarrow} 72.9\%\right)$$

Since 100% = 40, so 72.9% = 29.16 litres

Case 2 When the quantity withdrawn and the quantity replaced are of the same volume, but the total volume before replacement does not remain the same.

Initially, there are 40 litres of milk, and 4 litres of milk is taken out and 4 litres of water is poured in

So, there will be 36 litres of milk and 4 litres of water.

Now, 5 litres of mixture is taken out and 5 litres of water is poured in.

The quantity of milk and water being withdrawn here will be in the ratio of 36:4. So, the quantity of milk withdrawn $= \frac{36}{40}\times 5$

Milk left = $40\times\frac{36}{40}\times\frac{35}{40}$

Again, if now 6 litres of mixture is taken out and 6 litres of water is poured in

Milk left = $40\times\frac{36}{40}\times\frac{35}{40}\times\frac{34}{40}$

Case 3 When the quantity withdrawn and the quantity replaced are not of the same volume.

Initially there are 40 litres of milk, and 4 litres of milk is taken out and 5 litres of water is poured in

Obviously, there will be 36 litres of milk and 5 litres of water.

Now, 5 litres of mixture is taken out and 6 litres of water is poured in then the quantity of milk and water being withdrawn will be in the ratio of 36:5. So, the quantity of milk withdrawn = $\frac{36}{41}\times 5$

Milk left = $40\times\frac{36}{40}\times\frac{36}{41}$

Again 6 litres of mixture is taken out and 7 litres of water is poured in.

Thus, the volume of milk in the final mixture

$= 40\times\frac{36}{40}\times\frac{36}{41}\times\frac{36}{42}$

Example 14 Two vessels A and B of equal capacities contain mixtures of milk and water in the ratio 4:1 and 3:1,

respectively. 25% of the mixture from A is taken out and added to B. After mixing it thoroughly, an equal amount is taken out from B and added back to A. The ratio of milk to water in vessel A after the second operation is

(a) 79:21 (b) 83:17
(c) 77:23 (d) 81:19

Solution Assume there is 20 litres of the mixture in both the vessels.

In vessel A, milk = 16 litres and water = 4 litres

25% from A to B = milk in B = 15 + 4 = 19 litres

= water in B = 5 + 1 = 6 litres

ratio = 19:6

Equal amount from vessel B to vessel A

$$= \text{milk in A} = 12 + \frac{19}{5} = \frac{79}{5}$$

$$= \text{water in A} = 3 + \frac{6}{5} = \frac{21}{5}$$

Hence, the ratio is 79:21

PRACTICE EXERCISES

WARM UP

Q.1. Find the average age of a family of five members, whose ages are 42, 49, 56, 63 and 35 years respectively.

(a) 50 years (b) 49 years
(c) 45 years (d) 48 years

Q.2. Rahul Ghosh walks from A to B at 8 km/h and comes back from B to A at 12 km/h. What is his average speed for the entire journey?

(a) 8.8 km/h (b) 9.6 km/h
(c) 10.2 km/h (d) 11.4 km/h

Q.3. The average of five numbers is 39.20 and the average of three of these numbers is 41. Find the average of the remaining two numbers.

(a) 35.5 (b) 36.5 (c) 37.5 (d) 38.5

Q.4. In a class of 60 students, the average height of 30 students is x cm and that of the remaining students is y cm. Find the average height of the whole class.

(a) $(x + y)$ cm (b) 30 cm
(c) $\frac{x+y}{2}$ cm (d) None of these

Q.5. What is the average of the first 15 multiples of 3?

(a) 21 (b) 27 (c) 24 (d) 15

Q.6. The average age of 7 members of a family is 25. If one of the members whose age is 28 years, is excluded, then others are in the ratio 1:2:3:4:5:6. Find the age of the eldest member of the family.

(a) 28 (b) 35 (c) 14 (d) 42

Q.7. The average of seven consecutive even numbers is 10. If the next three even numbers are included, then what is the new average?

(a) 10 (b) 12 (c) 13 (d) 14

Q.8. Average weight of three students is 40 kg. If the first student is twice the second and the second student is thrice the weight of the third student, then what is the weight of the student with lightest weight?

(a) 18 (b) 20
(c) 14 (d) None of these

Q.9. In a test, the average marks of five students was found to be 50. However during cross checking of the papers, it was discovered that the mark of one student was misread as 84 instead of 48. Find the correct average.

(a) 57.2 (b) 42.8 (c) 46.5 (d) 43.2

Q.10. The average weight of seven members of a family is 18 kg. If the head of the family is not considered, the average weight of the others would be 5 kg less. The weight of the head of the family is

(a) 42 kg (b) 48 kg
(c) 52 kg (d) 36 kg

Q.11. In a prep school, the average weight of 30 girls in a class of 50 students is 16 kg and that of the remaining students is 15.5 kg. What is the average weight of the all the students in the class?

(a) 15.2 kg (b) 15.8 kg
(c) 15.4 kg (d) None of these

Q.12. The average of the amount of money that 10 girls have is Rs 45. When two more girls join, the average increases by Rs 2. Find the average amount of the two new girls.

(a) Rs 57 (b) Rs 54
(c) Rs 48 (d) None of these

Q.13. In what ratio must 35% spirit solution be mixed with pure spirit to get a resultant solution of 56% spirit?

(a) 35:44 (b) 44:21 (c) 8:3 (d) 56:9

Q.14. A beggar has 50 coins with an amount of Rs 15 in the denominations of 25 paise and 50 paise. How many 25 paise coins are there with the beggar?
(a) 36 (b) 40 (c) 30 (d) 25

Q.15. Two litres of pure spirit is added to 6 litres of a spirit solution containing $16\frac{2}{3}\%$ spirit. What is the concentration of the spirit in the resultant solution?
(a) 12.5% (b) 25%
(c) 37.5% (d) None of these

Q.16. Three litres of pure milk is added to 10 litres of a milk solution containing 9% milk. Find the concentration of the resultant solution.
(a) $33\frac{1}{3}\%$ (b) 40%
(c) 50% (d) 30%

Q.17. In what ratio must a quality of sugar @ Rs 15.50/ kg be mixed with another quality of sugar @ Rs 21.80/ kg so that selling the mixture at Rs 20.02/ kg results in 10% profit?
(a) 5:4 (b) 3:2 (c) 2:1 (d) 4:3

Q.18. Parul Jaiswal has Rs 11,200 in Rs 100 and Rs 500 denomination. What is the ratio of the number of notes, of the two types, if she has a total of 40 notes with her?
(a) 3:2 (b) 7:3 (c) 8:7 (d) 11:9

Q.19. Ordinary sugar contains 32% impurities, while processed sugar has 18% impurities. How many kilograms of ordinary sugar must be processed to yield 102 kg of processed sugar?
(a) 185 (b) 246
(c) 123 (d) None of these

Q.20. A shrewd milkman mixes water and milk in the ratio of 2:3. What part of this mixture should be removed and replaced with water so that the solution contains water and milk in the ratio 1:1?
(a) 1/6 (b) 1/4 (c) 1/3 (d) 1/2

Q.21. A milkman sells his mixture of milk and water at the cost price of milk itself and thus he gains a profit of 20%. What is the ratio of milk and water in the mixture?
(a) 4:1 (b) 5:1
(c) 1:4 (d) None of these

FOUNDATION

Q.1. Neo is going to NDLS from PNBE by his car at a speed of 40 km/h. While coming back, he returns with a speed of x km/h. What should be the value of x so that his average speed during the intre journey is 80 km/h?
(a) 160 km/h (b) 40 km/h
(c) 120 km/h (d) It is not possible

Q.2. There are 125 middle level employees in Due North Inc. The average monthly salary of the middle level employees is Rs 5,500 and that of the senior level employees is Rs 14,000. If the average monthly salary of all these employees is Rs 8,687.5, find the total number of employees in the company, if middle and senior level employees of Due North Inc. form 80% of their total employees.
(a) 175 (b) 200 (c) 220 (d) 250

Q.3. I went to a hotel along with 12 friends. I paid Rs 145 and all the others paid an equal amount. In the end, we did some calculations and found that the average sum paid by all of us was Rs 5 more than what was originally paid by each of my friends. How much money did each of my friends pay?
(a) Rs 120 (b) Rs 100
(c) Rs 80 (d) Rs 70

Q.4. Saketprakash married ten years ago at the age of 27 years. His wife was 23 years old then. Six years after their marriage, the average age of Saketprakash, his wife and their son was 22 years. After how many years of Saketprakash's marriage was his son born?
(a) 6 years (b) 3 years
(c) 2 years (d) 4 years

Q.5. Mr Anant Roy, the renowned author, recently got his new novel released. To his utter dismay he found that for the 1,007 pages on an average there were 2 mistakes every page. While, in the first 612 pages there were only 434 mistakes, they seemed to increase for the latter pages. Find the average number of mistakes per page for the remaining pages.
(a) 6 (b) 4
(c) 2 (d) None of these

Q.6. The class X of Nav Yug Vidhyalaya has four sections—A, B, C and D. The average weight of the students of A, B, C together and A, C, D together are 45 kg and 55 kg respectively, while the average weight of the students of A, B, D together and B, C, D together are 50 kg and 60 kg respectively. Which of the following

could be the average weight of the students of all the four sections together?
(a) 47.6 kg (b) 52.5 kg
(c) 53.7 kg (d) 56.5 kg

Q.7. The average score of a cricketer in a certain number of innings is 44. He then played another eight innings in which he got out at 97, 3, 23, 0, 68, 40, 50 and 71 runs respectively. Then, which of the following statements is true regarding his new average score in all the innings together?
(a) The new average is more than the old average
(b) The new average is less than the old average
(c) The new average is the same as the old average
(d) Cannot conclude any of the above unless we know the number of innings played

Q.8. Average age of A, B and C is 84 years however when D joins them, then the average comes down to 80. Now, a new person E whose age is 3 years more than D replaces A and the new average is 79 years. What is the age of A?
(a) 75 (b) 79 (c) 81 (d) 77

Direction for questions 9 and 10: Read the passage below and solve the questions based on it.

The average age of the students in a class is 35 years.

Q.9. If a student whose age was 25 is absent from the class, the average of those present goes up by 1. How many students were there in the class originally?
(a) 9 (b) 10 (c) 11 (d) 12

Q.10. If 2 students whose ages were 30 years and 20 years left, by what value would the average of the remaining increase? (use the result from question 9.)
(a) $\frac{20}{9}$ (b) 2
(c) $\frac{20}{11}$ (d) None of these

Q.11. If the average of m numbers is a, and on adding x to the m numbers, the average of the $1 + m$ numbers is b. Then find the value of x.
(a) $m(b - a) + b$ (b) $m(b + a) + a$
(c) $m(a - b) + a$ (d) None of these

Q.12. Brian Lara, the famous batsman, scored 6,000 runs in certain number of innings. In the next five innings he was out of form and hence, could make only a total of 90 runs, as a result of which his average fell by 2 runs. How many innings did he play in all, if he gets out in all the innings?
(a) 105 (b) 95 (c) 115 (d) 104

Q.13. The average weight of A, B and C is x kg. A and C lose y kg each after dieting and B gains $\frac{y}{2}$ kg. After this their average weight decreases by 1 kg. Find y.
(a) 1 kg (b) 2 kg
(c) 3 kg (d) Cannot be determined

Q.14. One out of the five weights A, B, C, D and E is of a different weight. A person makes a measurement and finds the following: A + B = C + D + E
Which of the following should be known to find out which one is of a different weight?
(a) A + C > B + D + E
(b) A + D + E < B + C
(c) A + B + C > D + E
(d) A + C < B + D + E

Q.15. In a class, with a certain number of students, if another student weighing 50 kg joins, then the average weight of the class increases by 1. If one more student weighing 50 kg joins then the average weight of the class increases by 1.5 over the original average. What is the original average weight of the class?
(a) 46 (b) 4 (c) 2 (d) 47

Q.16. The average weight of the books carried by Kuku to school is 1.5 kg and 2.5 kg is the average weight of the exercise books. If Kuku is carrying only text books and exercise books in the ratio of 3 : 4, what is the total weight of his school bag? (Ignore any other weights such as that of school bag, stationary etc.)
(a) 10.5 kg (b) 1 kg
(c) 6 kg (d) Data insufficient

Q.17. The average of n numbers is 32. If 3/4 of the numbers are increased by 4 each and the remaining are decreased by 6 each, then what is the new average?
(a) 32 (b) 32.5 (c) 33.5 (d) 34.5

Q.18. The average of eight numbers is 25, that of the first two is 20 and of the next three is 26. The sixth number is less than the seventh by 4, and less than the eighth by 6. The last number is
(a) 30 (b) 32 (c) 40 (d) 36

Q.19. The average amount of sales per day in a shop for 5 days starting from Monday is Rs 1,800. The average amount of sales per day for 5 days from Tuesday is Rs 1,890. By how much does the sale on Saturday exceed that on Monday?
(a) Rs 450 (b) Rs 500
(c) Rs 600 (d) Rs 650

Q.20. Three years ago, the average age of a family of 5 members was 17 years. Inspite of the birth of a child in the family, the present average age of the family remains the same. The present age of the child is
(a) 3 years (b) 1 year
(c) 2 years (d) 1.5 years

Q.21. A batsman has a certain average of runs for 16 innings. In the 17th innng, he makes a score of 85 runs, thereby increasing his average by 3 runs. What is the average after the 17th inning?

(a) 58 (b) 37
(c) 35 (d) None of these

Q.22. If the average marks of 'a' students in a class is 'c' and that of the remaining 'b' students is 'd', what is the average marks of the class?
(a) $\frac{ac+bd}{b+d}$ (b) $\frac{ab+cd}{a+d}$
(c) $\frac{ac+bd}{a+b}$ (d) $\frac{ad+cd}{b+d}$

Q.23. The average marks of a class of 48 students is 35. Of them, two score zero, of the rest, the first 30 scored an average of 40, the next fourteen scored an average of 20. If the remaining two scored equal marks, what are their individual marks?
(a) 80 (b) 75 (c) 90 (d) 100

Q.24. The total salary of A, B and C is Rs 444. If they spend 80%, 85%, 75% of their salaries respectively, their savings are as 7:6:9. Then salary of B is:
(a) Rs 140 (b) Rs 160
(c) Rs 144 (d) None of these

Q.25. The average age of 2/3rd of the class is 17. What should be the average the age of the remaining 1/3rd students so that the average age of the entire class is 20?
(a) 21 (b) 22 (c) 24 (d) 26

Q.26. Sidharth Tripathi travels from Delhi to Lucknow at a speed of 75 km/h and returns to Delhi at a speed of 50 km/h. He again leaves for Lucknow immediately at a speed of 30 km/h and returns back to Delhi at a speed of 60 km/h. What is his average speed for the entire journey?
(a) 54 km/h (b) 48 km/h
(c) 56 km/h (d) Cannot be determined

Q.27. Average marks of a class of n students is 64. When eight new students with an average marks of 73 join the class, the new average of the entire class is a whole number. Find the number of students in the class given that n lies between 25 to 60.
(a) 44 (b) 32 (c) 36 (d) 72

Q.28. Average of 5 consecutive integers is N. What will be the new average when next two integers are also included?
(a) N + 2
(b) N + 2/7
(c) N + 1
(d) N + K / 7, where K depends upon the values taken.

Q.29. The average of 10 two-digit positive integers is Z. However, one number AB is taken as BA, then the average increases to Z + 2.7. What is the value of |B – A|?
(a) 1 (b) 2 (c) 3 (d) 4

Q.30. In the above question, how many set of values for A and B are possible?
(a) 5 (b) 6 (c) 7 (d) 8

Q.31. In a country, the average salary per month is calculated on the basis of 14 months and average expenses is calculated on the basis of 10 months. What will happen to the average savings per month with respect to actual savings (calculated on the basis of 12 months)?
(a) Decrease (b) Increase
(c) No change (d) Cannot be determined

Q.32. In the above question, what will be the net percentage change in the savings with respect to the actual savings?
(a) 11.11% (b) 22.22%
(c) 33.33% (d) Cannot be determined

Q.33. The average rainfall for the first 3 days out of five days was recorded to be 0.45 inches. The rainfall on the last two days was in the ratio 2:3. The average of five days was 0.40 inches. What was the rainfall on the last day?
(a) 0.45 inches (b) 0.39 inches
(c) 0.35 inches (d) 0.30 inches

Q.34. The average height of the girls of a class is 155 cm and the average height of the boys of the class is 4 cm more than the average of the class. If the number of girls is 205 less than the number of boys, what is the average height of the boys?
(a) 140 cm (b) 162 cm
(c) 152 cm (d) None of these

Q.35. The average price of a share is the average of 5 readings taken at regular intervals in a day. The index price is taken by a weighted arithmetic average price of a class A and class B stock. The respective weights are 1.1 and 0.9 for the two kinds of stocks. If the five readings of a class A stock were 19, 26, 31, 35, 39 and for a class B stock the readings were 7, 8, 17, 20, 23 then what was the index price that day?
(a) 46.5 (b) 25 (c) 23.25 (d) 45

Q.36. Ankit travels from New Delhi to Patna. He stops at equidistant places, i.e., Lucknow and Mirzapur during his journey. He covers the first 1/3rd distance at a speed of 50 km/h, the 2nd 1/3rd distance at a speed of 40 km/h and the last 1/3rd distance at a speed of 60 km/h. However, while returning, his uniform speed is 75 km/h and he stops only at Jaunpur, which is exactly at the mid-point between New Delhi and Patna. What is the average speed (in km/h) of Ankit during the whole journey, not considering his halt time?
(a) 59.8 (b) 59.01
(c) 60.8 (d) None of these

Q.37. The average age of 24 students and the class teacher is 16 years. If the age of the class teacher is excluded

the average reduces by 1 year. What is the age of the class teacher?
(a) 38 (b) 39
(c) 40 (d) None of these

Q.38. First ten multiples of 1, 2, ... 10 are taken. What is the average of all these 100 numbers?
(a) 25.5 (b) 50.5
(c) 75.5 (d) None of these

Q.39. There are 20 students in Mr Rahul Ghosh's class. He conducts an examination out of 100 and then arranged the marks in an ascending order. He found Chandan, the topper of the class, had slipped to the tenth position. When he was adding the scores of the last 11 students the average was 64 and that of the top 10 was 67. If the average marks obtained by all the students of his class was 65, how many marks did Chandan score?
(a) 64 (b) 67
(c) 74 (d) Cannot be determined

Q.40. The average of 38, 84, 63, 45, 32 and n lies between 50 and 60. If n is an integer greater than the average of the above numbers, then n lies between
(a) $53 < n < 97$ (b) $50 < n < 100$
(c) $52 < n < 98$ (d) None of these

Q.41. Milawati Sharma, the local grocer mixed three qualities of tea T_1, T_2 and T_3 priced at Rs 74 per kg, Rs 68 per kg and Rs 63 per kg in the ratio of 1:2:4. He added some more quantity of T_1 to 4 kg of this mixture. He sold this new mixture for Rs 84 per kg, thereby makmg a profit of 20%. How much of T_1 did he mix with the mixture?
(a) 4 kg (b) 6 kg
(c) 8 kg (d) None of these

Q.42. 60 litres of a 75% solution of wine is taken into a laboratory. 4 litres of the solution is flushed and 4 litres of water is added to compensate the loss. Then 5 litres of the solution is flushed again and 5 litres of water is added. Again 6 litres of the solution is flushed and 6 litres of water is added. The final concentration of wine in the solution is
(a) 255% (b) 57.75%
(c) 60.25% (d) 59%

Q.43. A and B are two alloys of gold and copper prepared by mixing the respective metals in the proportions of 7:2 and 7:11 respectively. If the alloys are melted and mixed to form a third alloy C, which has an equal proportion of gold and copper. What is the ratio of alloys A and B in the mixture?
(a) 2:5 (b) 5:2 (c) 11:2 (d) 1:2

Q.44. A vessel contains a mixture of two liquids A and B in the ratio of 4:1. When 10 litres of the mixture is replaced with liquid B, the ratio becomes 2:3. How many litres of liquid A was present in the jar initially?
(a) 12 litres (b) 14 litres
(c) 16 litres (d) 10 litres

Q.45. From a vessel containing pure milk, 9 litres is taken out. It is replaced with an equal quantity of water. This operation is repeated twice. The ratio of milk and water now in the vessel is 16:9. What was the volume of milk in the vessel initially?
(a) 54 litres (b) 36 litres
(c) 42 litres (d) 45 litres

Q.46. There are only deers and ducks in a zoo. When heads are counted, it is 204 and when the legs are counted, it is 512. How many deers are there in the zoo?
(a) 52 (b) 152 (c) 176 (d) 26

Q.47. Sharat invests a total of Rs 10,000 in two parts, some money @ 6% per annum and rest of the money @ 8% per annum of SI. After two years, he got a total of Rs 1,440 as interest. What is the amount put at 6% per annum interest?
(a) Rs 4,000 (b) Rs 6,000
(c) Rs 3,000 (d) Rs 7,000

Q.48. A vessel contains a mixture of two liquids A and B in the ratio of 7:5. When 9 litres of the mixture is taken out and the vessel is filled with B, the ratio of A and B is now 7:9. How many litres of liquid A was there in the vessel initially?
(a) 24 litres (b) 21 litres
(c) 18 litres (d) 27 litres

Q.49. A milkman dilutes 36 litres of pure milk with water. The percentage of milk in the solution is now 75%. How many litres of water did he add?
(a) 6 litres (b) 8 litres
(c) 12 litres (d) 16 litres

Q.50. Doodhimal, the local milkman was notorious for the quality of milk that he used to sell. Initially he had 60 liters of milk in his can. He removed one-third of it and replaced that with water. He then removed one-third of the mixture and again added water to it. How many times should this process be repeated if Doodhimal wants the concentration of milk to be just below 20%?
(a) 3 (b) 4 (c) 5 (d) 7

MODERATE

Q.1. The average marks of the girls in a class is equal to the number of boys and the average marks of boys is equal to the number of girls. If the class average is 4 less than the average of both the boys' and the girls' average marks, what will be the number of students in the class?
(a) 24 (b) 48
(c) 50 (d) 64

Q.2. In the above question, what can be the minimum number of students in the class?
(a) 0 (b) 2
(c) 8 (d) None of these

Q.3. A, B, C, D and E are five consecutive integers and the average of these five numbers is less than $\frac{1}{4}$th of A. Then A is
(a) Positive (b) Negative
(c) Even (d) Data Sufficient

Q.4. Find the difference between the average of the first n even numbers and the average of all the even numbers up to n (n is an even number).
(a) $\frac{n-1}{2}$ (b) $\frac{n+1}{2}$
(c) $\frac{n}{2}$ (d) None of these

Q.5. I was born 30 years after my father was born. My sister was born 25 years after my mother was born. The average age of my family is 26.25 years right now. My sister will get married 4 years from now and will leave the family. Then the average age of the family will be $\frac{107}{3}$ years. What is the age of my father?
(a) 30 year (b) 35 year
(c) 40 year (d) 45 years

Direction for questions 6 to 9: Read the passage below and solve the questions based on it.

India is supposed to send its boxing team for Olympics in each of the following 10 weight group divisions.

A (48 kg - 52 kg)	B (52 kg - 56 kg)
C (56 kg - 60 kg)	D (60 kg - 64 kg)
E (64 kg - 68 kg)	F (68 kg - 72 kg)
G (72 kg - 76 kg)	H (76 kg - 80 kg)
I (80 kg - 84 kg)	J (84 kg - 88 kg)

After selecting one player from each group, their average weight comes out to be 68 kg. If one of the players named X leaves the team, their average weight comes down to 66.5 kg.

Q.6. Player X is from group
(a) A (b) E
(c) I (d) None of these

Q.7. If X leaves the team, and two new players join the group, then their average weight increases to 68 kg. These players can be from group....
(a) A, C (b) D, J
(c) Both from G (d) Either 2 or 3

Q.8. What is the average weight (in kg) of all the players taken together?
(a) 56.7 (b) 58.8
(c) 61.4 (d) Cannot be determined

Q.9. In the average of all the groups together, which group contributes most in overall average?
(a) F (b) G
(c) H (d) Cannot be determined

Q.10. A shipping clerk has to weigh 6 distinct packets. He weighs them four at a time, weighing all the possible combinations of the packets from the six. The average weight of all the weighing combinations is found to be 500 gm. What is the combined weight of all the six packets?
(a) 375 gm (b) 1250 gm
(c) 750 gm (d) 1000/3 gm

Direction for questions 11 to 21: Read the passage below and solve the questions based on it.

There are three sections in class X at A. R. Academy. Below is the table indicating the number of students, and the range of marks obtained by the students of that particular section.

	Section A	*Section B*	*Section C*
No. of students	25	30	20
Minimum marks	22	24	20
Maximum marks	30	28	26

The minimum and the maximum marks provides us with only the range of marks which can be obtained by the students of that particular section.

Q.11. What is the average marks obtained by all the students of class X together?
(a) 24.25 (b) 26.33
(c) 26.75 (d) Cannot be determined

Q.12. If we shift 5 students from section A to section B and then 'x' students from section B to section C, then what

should be the value of 'x' so that the minimum possible average of section B and the maximum possible average of section C are the same?
(a) 6 (b) 7
(c) 9 (d) Not possible

Q.13. If we combine section A and section C, then what will be the ratio of the maximum possible average of this combined section and the average of section B?
(a) 2:1 (b) 3:2
(c) 1:2 (d) Cannot be determined

Q.14. What is the maximum number of students that can be transferred from section A to section C, so that the maximum possible average of section A is still more than the maximum possible average of section C?
(a) 20 (b) 22
(c) 24 (d) None of these

Q.15. What can be the maximum possible average of class X when all the three sections are combined?
(a) 26.33 (b) 28.13 (c) 27.28 (d) 28.5

Q.16. We take a sample size of the following numbers from different sections

	Section A	*Section B*	*Section C*
Total no. of students selected	10	10	10
No. of students of minimum marks	5	6	4
No. of students of maximum marks	5	4	6

What is the average of this sample size?
(a) 25.06 (b) 26.16 (c) 25.83 (d) 26.71

Q.17. In the above question, which section has contributed the maximum in the average of the whole sample size?
(a) A
(b) B
(c) C
(d) Two sections contributed equal

Q.18. If we shift 5 students from section A to section B and then 10 students from section B to section C, what will be the maximum average of section C?
(a) 27 (b) 26.66 (c) 26 (d) 27.33

Q.19. In the above question, what can be the minimum average of section C?
(a) 20 (b) 21 (c) 21.33 (d) 22

Q.20. 10 students are shifted from each of the sections A and C to section B and then 10 students each are shifted from section B to each of the section A and section C. Now what is the ratio of maximum possible average of section A and section C?
(a) 1:1 (b) 15:14
(c) 14:15 (d) Cannot be determined

Q.21. In the above question, what is the ratio of the minimum possible average of section A and section C?
(a) 11:12 (b) 12:11
(c) 1:1 (d) Cannot be determined

Q.22. The average of 5 positive numbers is A. The average of the average of all the possible distinct triplets of these 5 numbers is B. Which of the following is true?
(a) A>B (b) A<B
(c) A=B (d) Cannot be determined

Q.23. The average of 5 positive numbers is A. The average of the average of all the possible distinct pairs of these 5 numbers is B. Which of the following is true?
(a) A>B (b) A<B
(c) A=B (d) Cannot be determined

Direction for questions 24 to 26: Read the passage below and solve the questions based on it.

The following operators have been defined for two numbers A and B.
@(A, B) = Average of A and B
*(A, B) = Product of A and B
/(A, B) = A divided by B

Q.24. If A = 2 and B = 4, then the value of @ (/(*(A, B), B), A) would be
(a) 2 (b) 4 (c) 6 (d) 16

Q.25. The sum of A and B is given by
(a) *(@ (A, B), 2) (b) /(@ (A, B), 2)
(c) @(*(A, B), 2) (d) @ (/(A, B) 2)

Q.26. The sum of A, B and C is given by
(a) * (@ (* (@ (B, A), 2), C), 3)
(b) * (@ (* (@ (B, A), 2), C), 2)
(c) /(* (@ (* (B, A), 2), C), 3)
(d) None of these

Q.27. The average age of Mr and Mrs Sinha at the time of their marriage in 1972 was 23 years. On the occasion of their anniversary in 1976, they observed that the average age of their family had come down by 4 years compared to their average age at the time of their marriage. This was due to the fact that their son Vicky was born during that period. What was the age of Vicky in 1980?
(a) 6 (b) 7 (c) 8 (d) 5

Q.28. A man makes 60 articles in the 1st hour. His efficiency decreases by 25% in the 2nd hour, increases by 40% in the 3rd hour, decreases by 33% in the 4th hour and increases by 50% in the 5th hour. If he has to work

for more than 1 hour, then in which hour the average number of articles produced per hour then would be minimum?
(a) 2nd hour (b) After 5th hour
(c) 3rd hour (d) None of these

Q.29. Sachin Tendulkar has a certain batting average *N* (a whole number) in his career of 86 innings. In the 87th inning, he gets out after scoring 270 runs which increases his batting average by a whole number. The batting average is calculated by dividing the total number of runs scored by the total number of innings played by the player. How many values of his new average is/are possible?
(a) 0 (b) 1
(c) 2 (d) None of these

Q.30. In a school there are two sections for class X – section A and section B. The number of students in section A and section B are 60 and 70 respectively and their respective average weight is 35 kg and 38 kg. If one of the students from section A is shifted to section B then the average weight of both the sections decreases. What can be said about the weight of that student?
(a) His weight is less than 35 kg.
(b) His weight is more than 38 kg.
(c) His weight is more than 35 kg and less than 38 kg.
(d) Cannot be determined

Direction for questions 31 and 32: Read the passage below and solve the questions based on it.

The average score of a batsman for a certain number of innings was 21.75 per inning. He played 3 innings more and scored 28, 34 and 37 runs respectively, thus increasing his average by 1.25.

Q.31. How many innings did he play in all?
(a) 18 (b) 21
(c) 27 (d) None of these

Q.32. What was his total score?
(a) 520 (b) 601
(c) 621 (d) None of these

Q.33. A curious student of Statistics calculated the average height of all the students of his class as A. He also calculated the average of the average heights of all the possible pairs of students (two students taken at a time) as B. Further, he calculated the average of the average heights of all the possible triplets of students (three students taken at a time) as C. Which of the following is true of the relationship among A, B and C?
(a) A + 2B = C (b) A + B = 2C
(c) A = B = 3C (d) None of these

Q.34. A succession of numbers is said to be in the Arithmetic Progression (AP) if the difference between any given term and the previous term is constant throughout. In a AP $n_1, n_2, n_3, \ldots, n_{46}$, the sum of odd numbered terms $n_1, n_3, n_5, \ldots, n_{45}$, is 1272. What is the sum of all the 46 terms of this AP?
(a) 2,491 (b) 2,500
(c) 2,400 (d) Cannot be determined

Q.35. There are two vessels of milk of different prices with the volume of 220 litres and 180 litres respectively. Equal amounts of milk were poured off simultaneously from the two vessels and the milk poured off from the first vessel was poured into the second vessel and the milk poured off from the second vessel was poured into the first vessel. Then the price of milk in both the vessels becomes the same. How much milk was poured from one vessel into the other?
(a) 90 (b) 80
(c) 160 (d) 99

Q.36. There are two solutions of Sulphuric acid (acid + water) with concentration of 50% and 80% respectively. They are mixed in a certain ratio to get a 62% sulphuric acid solution. This solution is mixed with 6 litres of water to get back 50% solution. How much of the 80% solution has been used in the entire process?
(a) 15 litres (b) 12 litres
(c) 10 litres (d) None of these

Q.37. Two-fifths of the volume of the mixture of the milk and water is of the ratio of 4:3 is converted into a mixture of the ratio of 5:3 by the method of replacement with milk. The 4:3 mixture was prepared from a 4:1 mixture by the method of 'addition of the substances of the mixture'. If the replacement volume is 14 litres, what is the volume of the substance added?
(a) 12 litres (b) 60 litres
(c) 80 litres (d) 24 litres

Q.38. Rakesh has a stock of 1000 litres of pure milk. He decides to decrease the concentration to half, using the method of successive dilution, replacing 200 litres of contents by water, in each step of dilution. After the fourth dilution, he realizes that the concentration is less than the required value. Each of the contents removed during the dilution process is stored separately. He also has ample supply of water. Will he be able to make the mixture of desired concentration?
(a) Not possible
(b) Possible
(c) Cannot be determined
(d) None of these

Q.39. An industrial solvent of 90% strength is prepared and stored in a 150 litres capacity container. The container is filled to 80% of its capacity. It is required to decrease

the concentration of the solvent in the container to less than 60%. For doing this, either pure water can be added to it and/or solution can be removed. But removal can only be done if there is no place for adding water. At a given time, only 10 litres of the solution can be added or removed. How many times should water be added to the solution?
(a) 4 (b) 5 (c) 9 (d) 6

Q.40. Four gallons are drawn from a case full of wine. It is then filled with water. Four gallons of mixture are again drawn and the cask is re-filled with water. The ratio of the quantity of wine now left in the cask to that of the mixture in it is 36:49. How much does the cask hold?
(a) 30 gallons (b) 25 gallons
(c) 35 gallons (d) 28 gallons

Q.41. There is a vessel holding 40 litres of milk. 4 litres of milk is initially taken out from the vessel and 4 litres of water is then poured in. After this, 5 litres of mixture is replaced with 5 litres of water. And finally 6 litres of the mixture is replaced with 6 litres of water. How much milk (in litres) is there in the vessel?
(a) 26.775 (b) 29.16
(c) 24.72 (d) 27.42

Q.42. There is a vessel holding 40 litres of milk. 4 litres of milk is initially taken out from the vessel and 5 litres of water is poured in. After this, 5 litres of the mixture from this vessel is replaced with 6 litres of water. And finally 6 litres of mixture from this vessel is replaced with 7 litres of water. How much of the milk (in litres) is there in the vessel now?
(a) 22.42 (b) 27.09
(c) 24.72 (d) 29.42

Q.43. There is a vessel holding 40 litres of milk. 4 litres of milk initially is taken out from the vessel and 5 litres of water is poured in. After this, 6 litres of mixture from this vessel is replaced with 7 litres of water. And finally 8 litres of mixture from this vessel is replaced with 9 litres of water. How much of the milk (in litres) is there in the vessel now?
(a) 22.82 (b) 20.92
(c) 26.78 (d) 24.87

Q.44. There are three vessels A, B and C having a mixture of milk and water. Concentration of milk in the vessels is C_1, C_2 and C_3 respectively. When a part of vessel A is replaced with the mixture in vessel B, then the concentration of milk in the final mixture is less than C_1. However, when vessels B and C are mixed in the ratio of 1:1, then the quantity of milk in the final mixture is double than the quantity of milk in the vessel B. Which of the vessels is having the mixture of maximum concentration of milk?
(a) A (b) B
(c) C (d) Cannot be determined

Q.45. In the above question, which of the vessels is having the mixture of minimum concentration of milk?
(a) A (b) B
(c) C (d) Cannot be determined

Q.46. In question number 44, if we arrange the concentration of milk in different vessels in ascending order, then which of the following is true?
(a) ABC (b) BAC
(c) CAB (d) Cannot be determined

Q.47. In two alloys, the ratios of copper to zinc is 5:2 and 3:4. How many kilograms of the first alloy and the second alloy respectively should be melted together to obtain 28 kg of a new alloy with equal copper and zinc?
(a) 8 kg and 20 kg (b) 4 kg and 24 kg
(c) 3 kg and 25 kg (d) 7 kg and 21 kg

Q.48. From a 3:5 solution of milk and water, 20% is taken out and replaced by milk. How many times should this process be done to make the ratio of milk to water as 17:8?
(a) Once (b) Twice
(c) Thrice (d) Four times

ADVANCED

Direction for questions 1 and 2: Read the passage below and solve the questions based on it.

In the National Baseball League (NBL), the batting average (AB) of a hitter is computed from the home runs scored, innings played, completed innings and incomplete innings.

R1 = Number of home runs scored in complete innings.

N1 = Number of complete innings.

R2 = Number of home runs scored in incomplete innings.

N2 = Number of incomplete innings.

$$AB = \frac{R_1 + R_2}{N_1}$$

To accurately assess a hitter's accomplishment, the NBL has devised two other statistical computations called

the Quick Average (AQ) and the Comprehensive Average (AC), which are defined as follows:

$$AQ = \frac{R_1}{N_1} + \frac{N_2}{N_1}\max\left[0, \left(\frac{R_2}{N_2} - \frac{R_1}{N_1}\right)\right]$$

$$AC = \frac{R_1 + R_2}{N_1 + N_2}$$

Q.1. Based on the information, which of the following is true?
(a) AQ ≤ AB ≤ AC (b) AB ≤ AC ≤ AQ
(c) AC ≤ AB ≤ AQ (d) None of these

Q.2. An experienced hitter with no incomplete innings has AB of 50. The next time he bats, the inning is incomplete and he scores 45 runs. It can be inferred that
(a) AB and AQ will increase, while AC will decrease
(b) AB will increase, while AC will decrease
(c) AB will increase and we cannot comment on the changes in AQ and AC
(d) None of these

Q.3. If the average marks of 17 students in a class is A. The marks of the students when arranged in either an ascending or a descending order was found to be in arithmetic progression. The class teacher found that the students who were ranked 3rd, 7th, 9th, 11th, 15th had copied in the exam and hence got all of them rusticated. The average of the remainder of the class was B. Then
(a) A = B (b) A > B
(c) A < B (d) Data Insufficient

Q.4. The average weight of the students in four sections A, B, C and D is 60 kg. The average weights of the students of A, B, C and D individually are 45 kg, 50 kg, 72 kg and 80 kg respectively. If the average weight of the students of section A and B together is 48 kg and that of the students of B and C together is 60 kg, what is the ratio of the number of students in sections A and D?
(a) 4:3 (b) 3:2 (c) 8:5 (d) 12:7

Q.5. Illara, Puttu and Munchun had three bottles containing equal amounts of orange juice, mango juice and water respectively. Since Illara and Puttu wanted to dilute their drinks, Munchun gave one-fourth of the water that she had, to each of them. After mixing their bottles well, each of them gave to Munchun the same quantity of the solution that they had borrowed from her. Which of the following statements is true?
i. The concentration of water in Munchun's drink is more than the concentration of orange juice in Illara's drink.
ii. The concentration of orange juice in Illara's drink is more than the concentration of mango juice in Puttu's drink.
iii. The concentration of orange is more than the concentration of mango in Munchun's drink.
(a) i and ii (b) i, ii and iii
(c) i, ii and iii (d) None of these

Q.6. We write down all the digits from 1-9 side by side. Now we put '+' between as many digits as we wish to, so that the sum of numbers become 666. It is explained below
1 2 3 4 5 6 7 8 9 = 666
Now suppose we put plus signs at following places:
12 + 345 + 67 + 89 = 513
Since there are four numbers, so the average can be calculated by dividing the sum by 4. What is the average if the sum is 666?
(a) 166.5 (b) 111
(c) 133.2 (d) Cannot be determined

Q.7. In the above question, what is the average if the sum is 261?
(a) 54 (b) 52.2
(c) 55 (d) None of these

Direction for questions 8 to 11: Read the passage below and solve the questions based on it.

Go through the following definitions
A + B – All A are B.
A * B – Some A are B.
A @ B – No A are B.
A # B – Only A are B.

Following table gives the average age break-up of all the students at IIM-A (2005 batch)

Group of students	*Average age (years)*	*No. of persons in the group*
Males M	23	140
Females F	22	100
Engineers E	25	160
Fresher R	20	80
Working Professionals W	27	160

Assume that all the students mentioned above are either males or females. The minimum entry age in the IIM is 20 years (at the time of taking the entrance exam to qualify for IIM-A). No fresher can be a working professional and males/females/engineers can be either a fresher or a working professional. Everybody is either a fresher or a working professional.

Q.8. Following statement is definitely true: M + E. What is the average age (in years) of all the engineers who are not male?
(a) 39 (b) 40
(c) 38 (d) Cannot be determined

Q.9. Total number of working professionals is divided in the ratio of 3:2 among males and females. If A * B, then what is the range of the average age (x) of all the males who are engineers?
(a) $32 \le x \le 38$ (b) $39 \le x \le 43$
(c) $44 < x \le 49$ (d) Cannot be determined

Q.10. R + M and F + W are definitely true. What is the average age (in years) of all the males who are working professionals?
(a) 35 (b) 35.33
(c) 36 (d) None of these

Q.11. Consider the following statements – F @ E. What is the average age (in years) of all the working females?
(a) 23 (b) 24
(c) 24.5 (d) Data inconsistent

Direction for questions 12 to 14: Read the passage below and solve the questions based on it.

Ten different samples of milk solution— A, B, C, D, E, F, G, H, I and J are taken and the concentration of each sample (total quantity of milk as a percentage of total quantity of solution) was found out and the results were tabulated as follows

Sample	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*	*J*
Concentration (%)	78	57	82	84	98	66	34	87	79	71

Q.12. If two samples of milk are taken to form a sample whose concentration is more than 80%, then a maximum of how many distinct pairs of samples are there, which will never give the desired result for any ratio of their volumes?
(a) 30 (b) 48 (c) 12 (d) 15

Q.13. A triplet is a set of three distinct samples taken in some ratio of their volumes. How many such distinct triplets are there, which when mixed can give a sample with more than 80% concentration?
(a) 50 (b) 100 (c) 150 (d) 200

Q.14. There are how many distinct samples such that if two or mote than two such samples are mixed in any ratio, the resulting solution is always a sample of more than 80% concentration?
(a) 4 (b) 6 (c) 8 (d) 10

Q.15. There are three containers C_1, C_2 and C_3 having milk and water solutions in the ratio of a:b, where a ε (1, 2, 3) and b ε (4, 5, 6). Containers C_1, C_2 and C_3 can take any value of a and b. For how many different set of values of concentration of containers C_1, C_2 and C_3, the resultant mixture will have the ratio of milk and water as 1:1?
(a) 12 (b) 0
(c) Infinite (d) None of these

Q.16. In the above question, a ε (1, 2, 6) and b ε (4, 5, 3). Containers C_1, C_2 and C_3 can take any value of a and b. For how many different set of values of concentration of containers C_1, C_2 and C_3, the resultant mixture will have the ratio of milk and water as 1:1?
(a) 12 (b) 0
(c) Infinite (d) None of these

Q.17. The average weight of a class of 25 students is 30 kg. The average weight of the girls is 5 kg more than that of the boys. If the class teacher's weight, which is between 64 kg and 106 kg, is included, the average weight of the male members of the class equals that of the female members. If the average weight of the boys (in kg) is an integer, then what is the number of girls in the class?
(a) 20 (b) 15
(c) 10 (d) Cannot be determined

Q.18. A succession of numbers is said to be in Arithmetic Progression (AP) if the difference between any given term and the previous term is constant throughout. In an AP $n_1, n_2, n_3, \ldots, n_{47}$, the sum of odd numbered terms $n_1, n_3, n_5, \ldots, n_{47}$ is 1272. What is the sum of all the 47 terms of this AP?
(a) 2491 (b) 2500 (c) 2400 (d) 2591

Direction for questions 19 to 21: Read the passage below and solve the questions based on it.

In the entrance examination of IIMs, there were 200 questions, each of which carried the same marks. A correct answer gets 2 marks and there is 100% negative marking. A total of 70 candidates took the exam and it was later found that the average marks obtained by these 70 candidates was 240. The candidates were not required to attempt all the questions. None of the candidates got more number of incorrect answers than correct answers.

Q.19. Which of the following is always true?
(a) None of the applicants had a net score less than 240
(b) The net scores of the individuals would always be an even integer
(c) The net scores of any candidate would always be a positive integer
(d) The net scores of the topper could be exactly 24% more than the average of the entire group

Q.20. If the topper had a net score of 380 and the sixth ranker had a net score of 370 and the net scores of the top six

rankers were distinct, then what would be the average net score of the 64 students who got ranks from the seventh to the last?
(a) 225 (b) 245
(c) 275 (d) None of these

Q.21. When the scores of the top four students are deleted, the average score of the remaining 66 students falls by 6 marks. Assume that it is possible for two or more students to have the same net score. What is the minimum score possible for the fourth ranking student if no student got a net score of more then 352?
(a) 280 (b) 300 (c) 308 (d) 320

Q.22. Two alloys composed of gold and silver together weigh 20 kg. One lump contains 75% gold and 31.25 gm kg silver. Another alloy contains 85% gold and 30 gm per kg silver. The total quantity of silver in two lumps is 617.5 gm. If the two lumps are melted and formed into one, what percentage of gold will it contain?
(a) 50% (b) 89% (c) 78% (d) 67%

Q.23. Let N be a set of real numbers such that p is any real number in the set. There exists two numbers in N whose average is p, then
(a) N is a finite set
(b) N is a set containing all real numbers
(c) N is a set of all numbers in the interval (2, 3)
(d) None of these

Direction for questions 24 and 25: Read the passage below and solve the questions based on it.

There are two brands of TV, Onida and Videocon, in the market. Onida is an old brand launched in 1990 and Videocon is a new brand launched in 1997. For both these brands, 30% of the TV's purchased every year are disposed off as out-dated exactly two years later. It is known that 10 Onida TV's were disposed off in 1997. The following table gives us the number of TV's in operation at the end of the year from 1995 to 2000.

	1995	*1996*	*1997*	*1998*	*1999*	*2000*
Onida	120	162	182	222	236	236
Videocon	0	0	30	80	124	134

Q.24. What is the average number of Videocon TVs disposed every year by 2000?
(a) 4 (b) 12
(c) 8 (d) None of these

Q.25. What is the average number of Onida TVs purchased every year till 1995?
(a) 20 (b) 24
(c) 28 (d) Cannot be determined

TRUE/FALSE

1. For any given series of 100 terms, if we know the last term and the first term, we can always calculate the average of all the 100 terms.

   State whether True or False

2. A, B, C, D, E and F are six numbers, not necessarily natural numbers. They are divided into two groups—each group having three numbers each and we do not know which number belongs to which group. Final average of all the numbers can be calculated by calculating the average of individual average of the two groups.

   State whether True or False

3. In the above question, final average will depend upon the set of numbers forming the group.

   State whether True or False

4. 10 natural numbers are taken. Average of all the possible pairs of numbers formed out of these ten natural numbers are calculated. It is possible to determine the average of all the ten natural numbers using this information.

   State whether True or False

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(b)	(c)	(c)	(d)	(c)	(d)	(b)	(b)	(b)	(a)	(b)	(b)	(c)	(d)	(d)	(d)	(c)	(d)
Q. No.	21																			
Answer	(b)																			

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(d)	(c)	(c)	(b)	(b)	(c)	(a)	(c)	(a)	(a)	(a)	(b)	(d)	(d)	(d)	(c)	(a)	(a)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(b)	(c)	(d)	(b)	(d)	(b)	(c)	(c)	(c)	(b)	(a)	(d)	(b)	(d)	(c)	(b)	(c)	(b)	(c)	(a)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(d)	(b)	(a)	(c)	(d)	(a)	(a)	(b)	(c)	(b)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(d)	(a)	(d)	(d)	(c)	(d)	(d)	(d)	(c)	(d)	(d)	(d)	(c)	(b)	(a)	(a)	(a)	(b)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(c)	(c)	(a)	(a)	(b)	(b)	(d)	(d)	(c)	(c)	(c)	(d)	(d)	(d)	(c)	(c)	(b)	(d)	(d)
Q. No.	41	42	43	44	45	46	47	48												
Answer	(a)	(b)	(d)	(a)	(d)	(d)	(d)	(c)												

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(b)	(a)	(a)	(d)	(d)	(b)	(a)	(d)	(b)	(d)	(d)	(b)	(b)	(b)	(c)	(b)	(a)	(b)	(d)
Q. No.	21	22	23	24	25															
Answer	(b)	(c)	(b)	(a)	(d)															

■■■ True/False

1. False

   Formula to find out the average = (1st term + Last term)/2 can be used only if the series is in AP (i.e., the difference between any two consecutive terms is same).

2. True

   Average is only the central value of the numbers, and it does not matter what ways the numbers has been taken.

3. False

4. True

   Average is only the central value of the numbers, and it does not matter what ways the numbers has been taken – numbers are taken independently or they are first paired up and then taken together to find out the average.

HINTS AND SOLUTIONS

Warm up

1. $\frac{42+49+56+63+35}{5}=\frac{245}{5}=49$

2. Average speed = $\frac{2.8.12}{8+12}=\frac{192}{20}=9.6\,\text{km/h}$

3. Required Average = $\frac{39.2\times5-41\times3}{2}=\frac{196-123}{2}$

   $=\frac{73}{2}=36.5$

4. $\frac{30x+30y}{60}=\frac{x+y}{2}$

5. $\frac{15}{2}[2.3+14.3]=\frac{1}{2}[6+42]=24$

6. $25 \times 7 - 28 = 21x$

   $21x = 147$

   $x = 7$

   Age of eldest member = $6 \times 7 = 42$ years

7. $7x + 42 = 70$

   $7x = 28$

   $x = 4$

   Required average = $\frac{10x+90}{10}=x+9=13$

8. $6x + 3x + x = 120$

   $10x = 120$

   $x = 12$

9. Correct Average = $\frac{250-84+48}{5}=\frac{214}{5}=42.8$

10. Total weight of the whole family = $7\times 18 = 126$

    Weight of family excluding head of the family $=13 \times 6 = 78$ kg

    Weight of head of the family = 48 kg

11. $\frac{30.16+20\times15.5}{50}=\frac{480+310}{50}=\frac{790}{50}=15.8\,\text{kg}$

12. $\frac{47.12-45.10}{2}=\frac{564-450}{2}=\frac{114}{2}=57$

13. $\frac{x}{y}=\frac{100-56}{56-35}=\frac{44}{21}$

14. $x+y=50$

    $\frac{1}{4}x+\frac{1}{2}y=15$

    Solving both the equations, we get $x = 40$ and $y = 10$

15. $\frac{1+2}{6+2}=\frac{3}{8}=37.5\%$

16. $\frac{3+\frac{9}{100}\times10}{10+3}\times100=\frac{3.9}{13}\times100=30\%$

17. $\text{CP} = \text{Rs}\left(20.02-20.02\times\frac{1}{11}\right)=\text{Rs}\left(20.02-1.82\right)$

    Rs 18.20

18. $\frac{x}{y}=\frac{20000-11200}{11200-4000}=\frac{88000}{7200}=\frac{22}{18}=\frac{11}{9}$

20. Let x part of mixture is removed

    then, $\frac{2-\frac{2}{5}x+x}{3-\frac{3}{5}x}=1$

    $x=\frac{5}{6}$

21. Let CP of milk be Rs 100
then CP of mixture = Rs

$$\left(100-\frac{1}{6}\times 100\right)=\text{Rs}\,\frac{250}{3}$$

Using alligation, we get

$$\frac{\text{Milk}}{\text{Water}}=\frac{\frac{250}{3}-0}{100-\frac{40}{3}}=\frac{5}{1}$$

Foundation

1. $80=\frac{2\cdot x\cdot 40}{40+x}$

$40 + x = x$

Hence, not possible
Alternatively, the average speed for the whole journey has to be less than double the speed of one-side. In this case, speed in one direction = 40 kmph, hence average speed for the whole journey has to be less than 80 kmph.

2. Let x = no. of senior lever employees

then $\frac{125\times 5500+x\times 14000}{125+x}=8687.5$

$x = 75$

$\frac{4}{5}$ of total employees = (125 + 75) = 200

Hence no. of total employees = $\frac{5}{4}\cdot 200=250$

3. $\frac{140+12x}{13}=x+5$

$145 + 12x = 13x + 65$
$x = 80$

4. After 6 years from Marriage
Age of Satya Prakash = 33 years
Age of his wife = 29 years
Let the age of their son be x years.

Then, $\frac{33+29+x}{3}=22$

$62 + x = 66$
$x = 4$
Hence, option (c) is the answer.

5. Total mistakes = 1007 × 2 = 2014
Let x be average mistake per page for the remaining pages
$434 + 395x = 2014$
$395x = 1580$
$x = 4$

6. Let a, b, c, d the number of students in section A, B, C, D respectively
then,

$$=\frac{45(a+b+c)+55(a+c+d)+50(a+b+d)+60(b+c+d)}{3(a+b+c+d)}$$

$$=50+\frac{5b+10c+15d}{3(a+b+c+d)}$$

Clearly, a, b, c, d are natural no. put a = b = c = d = 1

Then, required average = $50+\frac{30}{12}=50+2.5$
= 52.5

7. Old average = 44
Average of 8 new innings

$$=\frac{97+3+23+0+68+40+50+71}{8}$$

$$=\frac{352}{8}=44$$

8. Age of D = 80 × 4 – 84 × 3 = 320 – 252 = 68
Age of E = 71 years
Age of A = 80 × 4 – 79 × 4 + 71
= 75

9. Let x students were in the class

$\frac{35x-25}{x-1}=36$

Or, $35x – 25 = 36x – 36$
$x = 11$

10. $\frac{11\times 35-30-20}{9}=\frac{385-50}{9}=\frac{335}{9}=37\frac{2}{9}$

Required No. $=37\frac{2}{9}-35=\frac{335-315}{9}=\frac{20}{9}$

11. $\frac{ma+x}{1+m}=b$ $\qquad ma+x=b+bm$

$x = b + m (b – a)$

12. Go through options
For (a)
He played 100 matches previously,

His old average = $\frac{6000}{100}=60$

His new average = 6090/105 = 58
Option (a) is satisfied. Hence option (a) is the answer.

13. $\frac{3x-y-y+\frac{y}{2}}{3}=x-1$

$3x-\frac{3y}{2}=3x-3$

$\frac{3y}{2}=3,\quad y=2$

14. Assign some numerical values, then proceed we will get (d) as answer.

16. No data regarding the numbers of books is given, hence we cannot determine the weight of school bag.
Hence, (d)

17. $$\frac{32\times n+\frac{3}{4}n.4-\frac{1}{4}n.6}{n}=\frac{70n-3n}{2n}=33.5$$

18. $40 + 78 + x - 6 + x - 2 + x = 200$
$3x = 90$
$x = 30$

19. Total sales from Monday to Friday = Rs 9000
Sales from Tuesday to Saturday = Rs 9450
Thus, a sale of Rs 450 more was on Saturday than on Monday.

20. Total age of family 3 years ago = 17 × 5 = 85 years
Total age of family now = 17 × 6 = 102 years
Total age of family excluding the child now = (85 + 15) = 100 years
Age of child = 2 years

21. Let a be the average till 16th inning.

$$\frac{16a+85}{17}=a+3$$

$16a + 85 = 17a + 51$
$a = 34$
Required average = 37

22. $$\frac{\text{Total Marks}}{\text{Total no. of Students}}=\frac{ac+bd}{a+b}$$

23. $$\frac{2\times 0+30\times 40+14.20+2x}{48}=35$$

$1200 + 280 + 2x = 1680$
$2x = 200$
$x = 100$

25. Let n = total no. of students

$$\frac{\frac{2}{3}n\cdot 14+\frac{1}{3}n\cdot x}{n}=20$$

$$\frac{34n+nx}{3n}=20$$

$34n + nx = 60n$
$nx = 26x$
$x = 26$

26. Average speed for 1st Journey

$$=\frac{2.75.50}{125}=60\,\text{km/h}$$

Average speed for 2nd Journey

$$=\frac{2.30.60}{90}=40\,\text{km/h}$$

Average speed for culture Journey

$$=\frac{2.60.40}{100}=48\,\text{km/h}$$

27. $$\frac{64n+8.73}{n+8}=\frac{64n+584}{n+8}$$

Put the values of n from the options to get the answer

28. Let the integers be 1,2,3,4,5

$$\text{N}=\frac{1+2+3+4+5}{5}=\frac{15}{5}=3$$

When next two integers are also included then new average $=\frac{28}{7}=4=\text{N}+1$

29. $10Z - (10.A + B) + (10B + A) = 10 (Z + 2.7)$
$10Z - 10A - B + 10B + A = 10Z + 27$
$9B - 9A = 27$
$B - A = 3$
$|B - A| = 3$

30. Clearly B > A, and none of them is zero

A	1	2	3	4	5	6
B	4	5	6	7	8	9

33. $$\frac{3\times 0.45+2x+3x}{5}=0.40$$

$1.35 + 5x = 2$
$5x = 0.65$
$x = 0.13$
Rainfall on last day = $3x = 3 \times 0.13 = 0.39$

35. $$\text{Average price of A}=\frac{19+26+31+35+39}{5}$$
$$=\frac{150}{5}=30$$

$$\text{Average price of B}=\frac{7+8+17+20+23}{5}=\frac{150}{5}=30$$

$$\text{Index price}=\frac{30\times 1.1+15\times 0.9}{2}=23.25$$

36. Average speed when Ankit travels from New Delhi to

$$\text{Patna}=\frac{d}{\frac{-d/3}{50}+\frac{d/3}{40}+\frac{d/3}{60}}$$

$$=\frac{d}{\frac{d}{150}+\frac{d}{120}+\frac{d}{180}}=\frac{1800d}{12d+15d+10d}$$

$$= \frac{1800d}{37d} = \frac{1800}{37} \text{km/h}$$

$$\text{Required Average speed} = \frac{2 \cdot \frac{1800}{37} \cdot 75}{\frac{1800}{37} + 75}$$

$$= \frac{270000}{4575} = 59.01$$

37. $\frac{25.16 - x}{24} = 15$, x = age of teacher

$x = 25.16 - 24.15 = 400 - 360 = 40$

38. Average of 1st 10 multiple of

$$1 = \frac{1 + 2 + 3 + ... + 10}{10} = 5.5$$

Average of 1st 10 multiple of

$$2 = \frac{2 + 4 + ... + 20}{10} = 11$$

Average of 1st 10 multiple of

$$3 = \frac{2 + 6 + ... + 30}{10} = 16.5$$

Hence, required Average

$$= \frac{5.5 + 11 + 16.5 + ...to\ 10\ \text{term}}{10}$$

$$= \frac{\frac{10}{2}[2 \times 5.5 + 9 \times 5.5]}{10}$$

$$= \frac{1}{2} \times 60.5 = 30.25$$

39. Score of Chandan = 64 × 11 + 10 × 67 – 20 × 65
= 704 + 670 – 1300
= 1374 –1300 = 74

40. $\frac{38 + 84 + 63 + 45 + 32 + n}{6} = \frac{262 + n}{6} = 43.66 + \frac{n}{6}$

only (c) satisfies the condition

41. Type of T_1 Tea in 4 kg mixture = $\frac{1}{7} \times 4\text{kg} = \frac{4}{7}$ kg

Type of T_2 tea in 4 kg mixture= $\frac{2}{7}.4$ kg = $\frac{8}{7}$kg

Type of T_3 tea in 4 kg mixture = $\frac{4}{7} \times 4 = \frac{16}{7}$kg

CP of 4 kg Tea mixture = Rs $\left(\frac{296 + 504 + 1008}{7}\right)$

= Rs $\frac{1808}{7}$

SP of New mixture = Rs 84

CP of New mixture = Rs $\left(84 - 84.\frac{1}{4}\right)$

= Rs 63 per kg

Suppose x kg of T_1 Tea is added to 4 kg mixture

CP of $(4 + x)$ kg tea mixture = $(4 + x)63$ = Rs $(252 + 63x)$

Also, CP of $(4 + x)$ kg Tea mixture

= Rs $\left(\frac{1808}{7} + \frac{x}{7}.74\right)$

= Rs $\left[\frac{1808 + 74x}{7}\right]$

Thus, $252 + 63x = \frac{1808 + 74x}{7}$

$1764 + 441x = 1808 + 74x$

$367x = 47$

$x = \frac{47}{367}$

42. Initial Concentration of wine = 75%

Concentration of wine after 1st flushing

$$= \frac{45 - 75\%\ \text{of}\ 4 \times 100}{60} = 70\%$$

Concentration of wine after 2nd flushing

$$= \frac{42 - 70\%\ \text{of}\ 5 \times 100}{60}$$

= 63.66%

Concentration of wine after 3rd flushing

$$= \frac{38.5 - 63.66\%\ \text{of}\ 38.5 \times 100}{60}$$

= 57.75%

43. x= amount of alloy B

y = Amount of Alloy A

In the Alloy

Take the quantity of gold in both the Alloy under consideration and then take alligation

$$\frac{x}{y} = \frac{7/9 \quad 1/2}{\frac{1}{2} - \frac{7}{18}} = \frac{\frac{14 - 9}{18}}{\frac{9 - 7}{18}} = \frac{5/18}{\frac{2}{18}} = 5:2$$

Hence, required ratio = 2 : 5

44. Suppose x litre was the initial volume of mixture. Then volume of liquid A $= \frac{4}{5}x$, and volume of liquid B = $x/5$

Now, $\frac{\frac{4}{5}x - \frac{4}{5}\cdot 10}{\frac{1}{5}x - \frac{1}{5}\cdot 10 + 10} = \frac{2}{3}$

$\frac{\frac{4}{5}x - 8}{\frac{1}{5}x - 2 + 10} = \frac{2}{3} \Rightarrow \frac{\frac{4}{5}n - 8}{\frac{1}{5}n + 8} = \frac{2}{3}$

$12x - 120 = 2x + 80$

$10x = 200$

$x = 20$

Volume of liquid = $\frac{4}{5}\cdot 20 = 16$ I

45. Let $25x$ be the volume of pure milk present in vessel. Applying the formula,

Amount left = Initial amount $\left(1 - \frac{\text{amount take out}}{\text{Initial Amount}}\right)^n$

n = no. of times operation carried out

$16x = 25x\left(1 - \frac{9}{25x}\right)^2$

$x = \frac{9.5}{25} = \frac{9}{5}$

Hence, volume of milk present initially = $25\cdot\frac{9}{5}$ 45I

46. Let x = no. of deer

y = no. of ducks

Total no. of heads are 204 i.e.,

$x + y = 204$ ..... (1)

total no. of legs = 512

i.e., $4x + 2y = 512$ ..... (2)

Solving 1 and 2

$x = 52$

$y = 152$

47. Net interest earned by him $= \frac{1}{2}\times\frac{1440}{1000}\times 100 = 7.2\%$

Let Rs x is invested at 6% p.a. and Rest invested at 8% p.a.

$\frac{x}{y} = \frac{8 - 7.2}{7.2 - 6} = \frac{.8}{1.2} = \frac{2}{3}$

Required Amount = $\frac{2}{5}\times 10000 =$ Rs 4000

48. Let $7x$ and $5x$ volume of liquids A and Liquid B are present in vessel then,

$\frac{7x - \frac{7}{12}\cdot 9}{5 - \frac{5}{12}\cdot + 9} = \frac{7}{9}$

$x = 3$

Hence, 7.3 = 21 l liquid A was present.

49. Let x = volume of water he added

$\frac{36}{36 + x} = \frac{3}{4}$

$144 = 108 + 3x$

$3x = 36$

$x = 12$

50. After 1st replacement concentration of milk = $\frac{40}{60}\times 100 = 66.66\%$

After 2nd replacement concentration of milk = $\frac{80/3}{60}\times 100 = 44.44\%$

A sequence is found - 66.66%, 44.44%, 22.22% us, this process is done for 4 times.

Moderate

1.

Number of Boys	B	Average of Girls
Number of Girls	G	Average of Boys

And class average = [BG + BG]/2

Average of Boys' average and Girls' average = [B + G]/2

Now, [BG + BG]/2 = [B + G]/2 – 4

Or, $[B - G]^2 = 8$ [B+G]

Since LHS is a perfect square, RHS should also be a perfect square. Now see the options that satisfy this.

2. Now different values of [B + G] possible are – $0(2\times 0^2)$, $2(2\times 1^2)$, $8(2\times 2^2)$, $18(2\times 3^2)$ etc.

We cannot take 0, since number of boys + number of girls cannot be equal to zero.

We cannot take 2, since in this case, B – G will exceed B + G.

We cannot take 8, since in this case, B – G will become equal to B + G, making G = 0. And this will result into making all the equations wrong.

So, the minimum possible value = 18

3. Take five consecutive integers such that it satisfies the condition.

4. Sum of 1st n even no. $= n(n + 1)$

Average of 1st n even number $= \frac{n(n+1)}{n} \equiv n+1$

Let $n = 2m$

Average of all the even no. upto n i.e., $2m$

$$= \frac{2m(2m+1)}{2m} = (2m+1)$$

$= (n + 1)$

Hence, required difference = 0.

5. Let present age of father = x year

and present age of mother = y year

therefore present age of son = $(x - 30)$

and present age of daughter = $(y - 25)$

Sum of their ages = $4 \times 26.25 = 105$ years

i.e., $x + y + x - 30 + y - 25 = 105$

$2x + 2y = 160$

$x + y = 80$ ...(1)

After 4 years, their total ages will be (excluding the daughter)

$x + 4 + y + 4 + x - 30 + 4 = 107$

$2x + y - 18 = 107$

$2x + y = 125$ ...(2)

Solving (1) and (2)

$x = 45$

6. Total weight of all players initially = $68 \times 10 = 680$ kg

Total weight of players when 1 player left the team = 598.5

Different in weight = (680 – 598.5) kg = 81.5

7. Total weight of 11 players (68×11) kg = 748 kg

Increase in weight = (748 – 598.5) kg = 149.5

8. Since the exact weight is not known, we cannot find out the average weight of all the players taken together.

9. Exact weight of players are not known hence (d)

10. If he is taking 4 packages at a time, then he is taking each of packet 15 times ($^6C_2$). So, total weight taken this way = 15×500. So, actual weight = $[7500 \times 3/2]/15$

11. What is given here is the range, and not the exact marks.

13. Since we do not have the exact average, we cannot calculate this.

14. Try with option. Also, max student that can shifted cannot be more than 24

when (c) then average of section A = 30

and average of section c

$$= \frac{24\times22+1\times20+19\times26}{44} = 23.6$$

15. Add in the existing solution

Max. possible average of all three sections

$$= \frac{1\times22+24+30+1\times24+29\times28+1\times20+19\times26}{77}$$

$= 28.13$

16. Required average

$$= \frac{5\times22+5\times30+6\times24+4\times28+20\times4+6\times26}{30}$$

25.06

17. Add in the existing solution

Marks contributed by Section

A = 5 × 22 + 5 × 30 = 260

Marks contributed by Section

B = 6 × 24 + 4 × 28 = 256

Marks contributed by Section

C = 20 × 4 + 6 × 26 = 236

18. First shift 5 students from A to B having the max. marks then, shift 5 students from B having max. marks and 5 those came from A.

Required Average

$$= \frac{5\times30+5\times28+1\times20+19\times26}{30} = 27$$

19. Now, shift 5 students from A having minimum marks and shift 5 from B having minimum marks and 5 those came from A

Required Average

$$= \frac{5\times22+5\times24+19\times20+1\times26}{30} = \frac{636}{30}$$

$= 21.2 \approx 21$

22. Take the values as 1, 2, 3 and 4 and check what happens.

Otherwise this question is very similar to the No. 10 of Moderate.

23. Let the numbers be 1,2,3,4 and 5 them solve

24. To calculate @ (/ (* (A, B), B), A); we will start with the innermost part.

* (A, B) = * (2, 4) = 8

/ (* (A, B), B = / 8, 4 = 2

@ (/ (* (A, B), B), A) = @ (2, A) = @ (2, 2) = 2

25. Check the options to find the answer as option (a).

27. Sum of ages of Mr and Mrs Sinha in 1972 = 46 years

Sum of age of their family in 1976 = 19 × 3 = 57 years

Sum of ages of Mr & Mrs Sinha in 1976 = (46 + 8) years = 54 years

their son's age in 1976 = (57 – 54) years = 3 years

28. Number of articles made in 1st hour = 60
Number of articles made in 2nd hour = 45
Number of articles made in 3rd hour = 63
Number of articles made in 4th hour = 42
Number of articles made in 5th hour = 63
So, obviously articles made in 4th hour is minimum.

29. Total number of runs scored till 86th inning = 86 N
Now, 86 N + 270 = 87 (N + S), where S is the increase in batting average.
Different values of S possible now S = 0, 1, 2 and 3.

30. Since, a boy in increased in Section B and the average is decreased ⇒ Hence, his weight is less than 38 kg. Again, the average weight is also decreased in both the sections. Hence, his weight is more than 35 kg. Hence, (c)

31. Let the no. of innings played be N

$$\text{then, } \frac{21.75N+28+34+37}{N+3}=N+1.25$$

Solving, we get N = 27

32. Total score = 21.75 × 24 + 28 + 34 + 37 = 621

33. Let the height of four students be 150, 160, 170, 180 cm then

$$A=\frac{150+160+170+180}{4}=\frac{660}{4}=165\,\text{cm}$$

$$B=\frac{\frac{150+160}{2}+\frac{160+170}{2}+\frac{170+180}{2}+\frac{150+180}{2}+\frac{150+170}{2}+\frac{160+180}{2}}{6}$$

$$=\frac{155+165+175+165+160+170}{6}=\frac{990}{6}=175\,\text{cm}$$

$$=\frac{670}{4}=167.5$$

Similarly, C = 167.5
Now going through the options,

34. Given $n_1+n_3.....+n_{45}=1273$
From above we cannot find the value of d.

35. Go through options.

36. Let x litres of 50% solution and y litres of 80% solutions are used

$$\frac{x}{y}=\frac{80-62}{62-50}=\frac{18}{12}=\frac{x}{y}=\frac{3}{2}$$

Solution get mixed in the ratio 3 : 2.
Now, suppose the value of acid is Z litres⇒

$$\frac{0.62z}{z+6}=\frac{1}{2}$$

$\Rightarrow 1.24\,Z = Z + 6 \quad \Rightarrow 0.24\,Z = 6$
$\Rightarrow Z = 25$

Hence, required rate = $\frac{2}{5}\times 25=10\,\text{litres}$

37. Solve with the help of options.

38. It is possible because each contents removed during the dilution process is stored separately.

39. Value of solvent in container = $\left(\frac{4}{5}\times 150\right)l=120l$

Concentration of solvent = $\left(\frac{9}{10}\times 120\right)l=108\,l$

When first time water is added the concentration of solvent

$$=\frac{108}{130}\times 100=83\%$$

When 2nd time water is added the concentration of solvent

$$=\frac{108}{140}\times 100=77\%$$

When 3rd the water is added the concentration of

Solvent $=\frac{108}{150}\times 100=72\%$

Now, 10 l solution should be removed
Solvent present in container = (108 – 7.2) l = 100.8 l
Again 10 l water is added and 10 l solvent is removed. To get the required solution, water is added for 6 times.

40. Let the initial volume be $85x$

$$36x=49x\left(1-\frac{4}{49x}\right)^2$$

$$\left(\frac{6}{7}\right)^2=\left(1-\frac{4}{49x}\right)^2$$

$$1-\frac{4}{49x}=\frac{6}{7}$$

$\Rightarrow x = 4/7$

Required volume = 49 × 4/7 = 28 g

41. After 1st operation milk left =

$$40\left(1-\frac{4}{40}\right)=40\cdot\left(1-\frac{1}{10}\right)=40\cdot\frac{9}{10}=36\,L$$

After 2nd operation milk left = 36 [1– (5/40)] = 31.5

After 3rd operation milk left $\left(36.5-6\times\frac{31.5}{40}\right)$
= (26.775L)

42. When 4 litres of milk is taken out, volume of milk left in the vessel = 36 litres [40{1−(4/40)}]
When 5 litres of mixture is taken out, volume of milk left in the vessel = [40{1−(4/40)}{1−(5/41)}
When 6 litres of mixture is taken out, volume of milk left in the vessel = [40{1−(4/40)}{1−(5/41){1−(6/42)}]

43. Do as given in 42.

44. When a part of vessel A is replaced with the mixture in vessel B, then the concentration of milk in the final mixture is less than C_1 – It implies that $C_1 > C_2$
When vessels B and C are mixed in the ratio 1:1, then the quantity of milk in the final mixture is double than the quantity of milk in the vessel B – It implies that $C_2 = C_3$
Hence C_1 is greatest.

45. Obviously, $C_2 = C_3$ is the right answer.

46. When a part of vessel A is replaced with the mixture in vessel B, then the concentration of milk in the final mixture is less than C_1 – It implies that $C_1 > C_2$
When vessels B and C are mixed in the ratio 1 : 1, then the concentration of milk in the final mixture is double than the concentration of milk in the vessel B. It implies that $C_3 > C_2$. Now we don't know which of C_1 or C_3 is greater. So answer is cannot be determined.

47. Let x and y kg of 1st Alloy and 2nd Alloy are taken

$$\frac{x}{y} = \frac{\frac{5}{7} - \frac{1}{2}}{\frac{1}{2} - \frac{3}{7}} = \frac{\frac{10-7}{14}}{\frac{7-6}{14}} = \frac{3}{1}, \quad \frac{x}{y} = \frac{3}{1}$$

Hence, (d)

48. Let the value of milk = $3x$ litres and volume of water = $5x$ litres
When 20% of mixture is taken out

$$\text{Volume of milk left} = \left(3x - \frac{1}{5} \times 3x\right) = \frac{12x}{5} \text{ litres}$$

= 2.4 x litres

$$\text{Volume of water left} = \left(5x - \frac{1}{5} \times 5x\right) = 4x \text{ litres}$$

And 1.6 x litres of milk is added, New ratio = 1 : 1
Similarly, proceeding we get
(c) as answer

Advanced

1. Obviously, in this case, AC ≤ AB. Now we are required to compare AB and AQ.

$$AB = \frac{R_1}{N_1} + \frac{R_2}{N_1} \text{ and } AQ = \frac{R_1}{N_1} + \frac{R_2}{N_1} \max\left[0, \left(\frac{R_2}{N_2} - \frac{R_1}{N_1}\right)\right]$$

So, basically we have to compare $\frac{R_2}{N_1}$ and $\frac{R_2}{N_1} \max\left[0, \left(\frac{R_2}{N_2} - \frac{R_1}{N_1}\right)\right]$

Since we don't have any information regarding values of R_2, R_1, N_2 and N_1, we cannot determine anything. So, Answer is option (d) None of these.

2. Since total number of runs is increasing without increase in the number of completed innings, AB will increase. However, since runs scored is less than 50 in the last inning, so AC will decrease.

3. This question is one standard example of the definition of average in terms of surplus and deficit of the values. (Refer to the definition given)
Now, what we were expected to mark here was the fact that whatever is the average (in this case, it is A), surplus generated by the marks of 3rd student will be same as deficit incurred due to 15th student. So, rusticating both of them is not going to create any difference on average marks of the class (remember marks are in AP). And similar will be the impact of rusticating 7th student and 11th student and then finally 9th student.
So, A = B

5. Basically in these types of situations, we should start with an attempt of refuting the statements given. The statement that cannot be refuted will be true, otherwise false.
Checking statement i, since Munchun is donating ½ of water from her bottle and Illara is donating only ¼ of mixture from her bottle, concentration of water in Munchun's drink will be less than the concentration of orange juice in Illara's drink. So, statement i. is not true.
Checking statement ii, since Illara and Puttu are receiving and then donating the same amount of liquid, so concentration of orange juice in Illara's drink will be equal to the concentration of mango juice in Puttu's drink. So, statement ii. is also not true.
Checking statement iii, concentration of orange will be equal to the concentration of mango in Munchun's drink. So, statement iii is also wrong.
Hence option (d). None of these is the answer.

6. We can get the sum 666 in two ways:
1 + 2 + 3 + 4 + 567 + 89 = 666 or
123 + 456 + 78 + 9 = 666
So, average cannot be uniquely determined.

7. 261 is possible only if we take 123 + 45 + 6 + 78 + 9
So, the average will be 261/5 = 52.2

8. It is equal to (Sum of ages of all the Engineers – Sum of ages of all the males)/(Number of engineers – Number of males) = 780/20 = 39
9. Since the relationship here is: Some A are B, so it cannot be determined.
10. Required average = (Sum of ages of all working professional – sum of ages of all females) / (No. of working professional – No. of females)
$$= \frac{27.160 - 22.100}{160 - 100} = \frac{4320 - 2200}{40} = 35.33$$
11. If no females (F) are engineers (E), then it is not possible to have 160 engineers from 140 males. So, F @ E is not possible.
12. The required result will be produce when we choose sample of A, B, F, G, I, J

    No. of pairs = ${}^{6}C_{2} = \frac{6.5}{1.2} = 15$
14. The required result well be produced by taking the samples C, D, E and H.
    Hence, required = ${}^{4}C_{2} = 6$
15. All the ratios formed from a and b will be resulting into a ratio less than 1 (like 1 : 4, 2 : 5 etc). To obtain a resultant ratio of 1:1, we should have atleast one of the ratios greater than 1. Hence we cannot get the desired ratio of 1 : 1.
16. When three solutions are taken together we can have infinite values.
18. Average of the sum of n_1, n_3, n_5,..... n_{47} terms
$$= \frac{1272}{24} = 53$$
    Required sum = average of odd terms × no. of terms
    = 53 × 47 = 2491
22. Eliminating the option, we get (c) as answer because average always lies between greatest and lowest.
23. A set of all real number can give the above conditions. Hence, (b)

[redacted]

19. All the additions in the net score would be in the multiple of 2 and similarly all the subtractions will also be in the multiple of 2. Hence, net scores will also be a multiple of 2.
21. The sum of scores of top four students = 1356
    Maximum possible score possible for top three students = 352 × 3 = 1056
    Hence, the minimum possible score for the 4th student = 300
24. Total number of TVs disposed off in 1999 = 20% of 30 = 6
    Total number of TVs disposed off in 2000 = 20% of 50 = 10
    So, Total number of TVs disposed off by 2000 = 16
    So, average number of TVs disposed off = $\frac{16}{4} = 4$
25. Since we do not have the figure of disposed off TVs every year, we cannot determine the average number of TVs purchased every year till 1995.

6

CHAPTER

Ratio, Proportion and Variation



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Ratio, proportion and variation
- Definitions and properties
- Different applications of ratio, proportion and variation
- Kinds of questions asked in the CAT
- Methods of solving questions

Introduction to the topic

Ratio and proportion is important not only for the QA paper but also for DI, especially for calculations. Questions from this topic are based on conceptual clarity and their different applications are often tested in the CAT. However, this is a favourite topic for all the non-CAT and non-XAT level examinations. While going through this chapter, students should try to internalize the mathematical as well as the logical method of solving the problems.

RATIO, PROPORTION AND VARIATION

Ratio is a quantity which expresses the relationship between two similar quantities. It expresses a magnitude by which one quantity is a part of or a multiple of another quantity.

If the value of A and B are 8 and 6 respectively, then they are in the ratio 8:6 (read as 8 is to 6). Ratio can be understood also as the relationship which one quantity bears with the other of the same kind. Due to this reason, we cannot compare salary of one person with the percentage expenditure of another person.

The ratio of two quantities A and B is written as A:B. Here, A is known as an antecedent and B is known as a consequent. It can also be said that A:B = kA:kB, where k is any constant known as constant of proportionality, $k \neq 0$.

If the antecedent is more than the consequent (or, the numerator is more than the denominator), then the ratio is known as an improper ratio.

For example, 5/3, 55/29 etc.

If the antecedent is less than the consequent (or, the numerator is less than the denominator), then the ratio is known as a proper ratio.

For example, 3/7, 7/18, etc.

Since Ratio compares two similar quantities, it cannot have any units.

Example 1 Consider any ratio $\frac{a}{b}$. Now x is added to the numerator and the denominator of this fraction. Which of the following is greater: $\frac{a+x}{b+x}$ or $\frac{a}{b}$?

Solution It depends upon two factors:

i. If the ratio is proper or improper
ii. x is positive or negative.

If $\frac{a}{b} > 1$ and $x > 0$, or, $\frac{a}{b} < 1$ and $x < 0$

$$\frac{a}{b} > \frac{a+x}{b+x}$$

and if $\frac{a}{b} > 1$ and $x < 0$, or, $\frac{a}{b} < 1$ and $x > 0$

$$\frac{a}{b} < \frac{a+x}{b+x}$$

RATIO

Ratio can be understood in two ways:

1. Ratio as a bridging element
2. Ratio as a multiplier

1. Ratio as a Bridging Element

Ratio as a bridging element helps us in establishing the relationship between more than two quantities. This can be further understood with the following example

Suppose conversion rate of our currency Rupee is given with respect to US dollar and also with respect to Pound sterling. If we have to find the conversion ratio of US dollar with respect to pound sterling, we can do it by making rupee as the bridge between US dollar and pound sterling.

Example 2 The ratio of the age of A and B is 2:5 and ratio of the age of B and C is 3:4. What is the ratio of the age of A, B and C?

Solution Since B is the common platform which associates A and C, so we will try to make B equal in both the cases.

Age of A : Age of B = 2:5] × 3
Age of B : Age of C = 3:4] × 5
Or, Age of A : Age of B = 6:15 ...(1)
Age of B : Age of C = 15:20 ...(2)

Since ratio of B is same in both the cases, hence age of A:Age of B:Age of C = 6:15:20.

Example 3 Given that

Salary of A : Salary of B = 1:2
Salary of B: Salary of C = 3:4
Salary of C: Salary of D = 5:6
Salary of D: Salary of E = 7:8
Salary of E: Salary of F = 9:10

What is the ratio of the salaries of A, B, C, D, E, and F?

Solution Salary of A : Salary of B:Salary of C:Salary of D:Salary of E:Salary of F = (1 × 3 × 5 × 7 × 9):(2 ×3 × 5 × 7 × 9):(2 × 4 × 5 × 7 × 9):(2 × 4 × 6 × 7 × 9): (2 × 4 × 6 × 8 × 9):(2 × 4 × 6 × 8 × 10).

(Understand the above mechanism with the help of the method given in Example 2. In these cases, this method can be used as a shortcut to find the ratios in the following way: For A, take all the leftmost digits, and now keep shifting towards the right digits by quitting one by one all the leftmost digits. So, B = Right digit of 1st ratio and so on for C, D, E and F.)

Example 4 If A : B = 3:4,

B : C = 5:7
C:D = 10:11

What is the ratio of A:D?

Solution A = 3 × 5 × 10 and D = 4 × 7 × 11.

So, the ratio = 150 : 308

Alternatively, (A/B) × (B/C) × (C/D) = (3/4) × (5/7) × (10/11) = (3 × 5 × 10)/(4 × 7 × 11) = 150:308

Example 5 A, B, C and D purchase a gift worth Rs 60. A pays 1/2 of what others are paying, B pays 1/3rd of what others are paying and C pays 1/4th of what others are paying. What is the amount paid by D?

Solution Since A is paying 1/2 of what others are paying, so A is paying 1/3rd of the total amount.

(To understand this, let us assume that B, C and D are paying Rs $2x$. So A is paying Rs x. The total amount being paid by A, B, C and D = $3x$ = Rs 60, hence, the amount paid by A = $x/3x$ = 1/3rd of the total.)

So, the amount paid by A = 60/3 = Rs 20

Similarly, B is paying 1/4th of the total and C is paying 1/5th of the total.

Hence, the amount paid by B and C are Rs 15 and Rs 12 respectively.

So, the amount paid by D = Rs 13

2. Ratio as a Multiplier

The moment we say that the ratio of two numbers A and B is 5 : 1, what we mean to say that A is 5 times of B.

It can also be seen that A:B:C in A/2:B/3:C/4 = K is not same as A:B:C = 1/2:1/3:1/4 since multiplier of A, B and C are not the same in both the cases.

Ratio of A:B:C in A/2:B/3:C/4 = K can be calculated in the following way

Since A/2 = B/3 = C/4 = K, so A = 2K, B = 3K and C = 4 K

Hence, the ratio of A:B:C = 2:3:4.

While calculating the ratio of A, B and C in A:B:C = 1/2:1/3:1/4, we will multiply each of A, B and C by the LCM of the denominator of all the ratios i.e., 12.

So, A:B:C = 6:4:3

Example 6 10 persons can cut 8 trees in 12 days. How many days will 8 persons take to cut 6 trees?

Solution Let us see this question in a changed perspective

Suppose if the question is—10 persons can cut 8 trees in 12 days. How many days will 10 persons take to cut 4 trees?

Answer to this question is:Since the amount of work is getting halved, so the number of days will also get halved.

There are three factors, namely, the number of men, the number of days and the number of trees, which are responsible for the final answer.

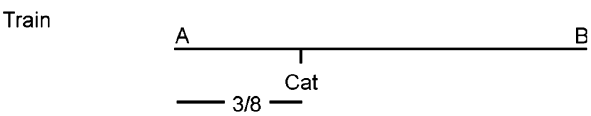
Since the number of men are less in the final situation, so more number of days will be required. Hence, multiplier = 10/8 (had there been 12 persons, multiplier would have been 10/12.)

The number of trees are less in the final situation, so less number days will be required. So, multiplier = 6/8

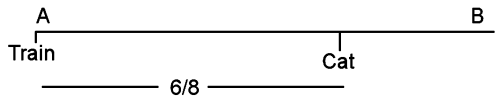
Hence, the total number of days = 12 × 10/8 × 6/8 = 90/8 = 11.25 days

Example 7 A train approaches a tunnel AB. Inside the tunnel, a cat is sitting at a point that is 3/8th of the distance of AB measured from the entrance A. When the train whistles, the cat runs. If the cat moves to the entrance of the tunnel A, the train catches the cat exactly at the entrance. If the cat moves to the exit B, the train catches the cat exactly at the exit. What is the ratio of the speed of the train and the speed of the cat?

Solution



Initially, this was the position of the train and the cat. Now, let us assume that the cat is moving towards exit B. The moment the cat covers 3/8th of AB distance in the direction of exit B, the train will be at the entrance A.



Now, if the cat moves in the direction of exit B, the train is catching up with the cat at the exit B. So, in the time cat covers 2/8th distance, the train is covering the whole distance from A to B.

So, the ratio of the distance covered by train and the distance covered by the cat = 4 : 1

So, the ratio of speed = 4:1

Example 8 Pranesh can do a work in 15 days. In how many days will the work be completed by his brother Saket if efficiency of Saket is 60% more than that of Pranesh?

Solution Since the ratio of efficiency of Pranesh and Saket = 100:160 = 5:8 the number of days taken by Pranesh and Saket will be in the ratio of 8:5

Since Pranesh takes 15 days to do this work, Saket will take 15 × 5/8 = 9.37 days

Comparison of Ratio

This is one of the most important calculations and is extensively used in DI. On an average, if somebody does 100 calculations in DI at least 8–10 calculations will be from comparing the ratios. Normally, there are two methods to compare two or more than two ratios:

1. Cross multiplication method

E.g., Let us compare 11/15 and 13/18.

$$\frac{11}{15} \times \frac{13}{18}$$

Cross multiplying numerator of 1st fraction with the denominator of 2nd fraction and denominator of 1st fraction with the numerator of 2nd fraction,

11×18 13×15

198 195

Since, 198 is greater than 195 the 1st fraction (11/15) is greater than the 2nd fraction (13/18).

2. Decimal calculation

$\frac{11}{15} = 0.733$ $\frac{13}{18} = 0.722$

Obviously, here the 1st fraction $\left(\frac{11}{15}\right)$ is greater than the 2nd fraction $\left(\frac{13}{18}\right)$.

However, if we have to compare 3156/5438 and 3423/5822, then using any of the above two methods becomes cumbersome and time-consuming.

Here, we will compare ratios with the help of percentage.

3. Percentage comparison

Let us first understand this with the help of the following ratios

1st Case $\frac{10}{15} \xrightarrow{100\%} \xrightarrow{100\%} \frac{20}{30}$

2nd Case $\frac{10}{15} \xrightarrow{100\%} \xrightarrow{200\%} \frac{30}{30}$

3rd Case $\frac{10}{15} \xrightarrow{300\%} \xrightarrow{200\%} \frac{30}{60}$

In the 1st case, percentage change in numerator (100%↑) = percentage change in denominator (100%↑), so ratios are equal.

In the 2nd case, percentage change in the numerator (200%↑)>percentage change in the denominator (100%↑), so the 2nd ratio is greater than the 1st ratio.

In the 3rd case, percentage change in the numerator (200%↑)<percentage change in the denominator (300%↑), so the 1st ratio is greater than the 2nd ratio.

This particular example can also be seen as a general rule for determining the order of ratios.

PROPORTION

In case of an analogy, two quantities share same kind of relationship. For example, what Macbeth is to William Shakespeare, Dr Zivago is to Boris Pasternak.

In QA, the same is true for proportion. It is basically the equality of the two ratios.

$$\frac{A}{B} = \frac{C}{D}$$

When A, B, C and D are in proportion, then A and D are known as 'extremes' and B and C are known as 'means'.

Thus, we can say,

Product of Extremes = Product of Means

Example 9 What is the value of x in the following expression?

$$\frac{5}{8} = \frac{x}{12}$$

Solution $\frac{5}{8} = \frac{x}{12}$

$\Rightarrow x = \frac{60}{8} = 7.5$

It can be calculated with the help of percentages also. In this question, the percentage increase in the denominator is 50%, so the numerator will also increase by 50%.

Standard Results/Definitions on Ratio/ Proportion

1. Continued proportion

   a, b and c are said to be in continued proportion if $\frac{a}{b} = \frac{b}{c}$

   So, $b^2 = ac$. Here b is known as the mean proportion. Similarly, if a, b, c and d are in continued proportion, then we get:

   $$\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$$

2. Componendo

   If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{b} = \frac{c+d}{d}$

3. Dividendo

   If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a-b}{b} = \frac{c-d}{d}$

4. Componendo and Dividendo

   If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

5. Invertendo

   If $\frac{a}{b} = \frac{c}{d}$, then $\frac{b}{a} = \frac{d}{c}$

6. Alterando

   If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{c} = \frac{b}{d}$

7. $\frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$

   In general, if $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \ldots = K$

   Then, $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \ldots\ldots = K = \frac{a+c+e+\ldots..}{b+d+f+\ldots..}$

   = (any combination of the numerator/any combination of the corresponding denominator)

   For example, $1/2 = 3/6 = 4/8 = \ldots = (1+3+4)/(2+6+8) = (3+4)/(6+8)$

8. If we multiply the numerator and the denominator of a ratio by any number N ($N \neq 0$), then the ratio remains same.

   A/B = NA/NB

9. If we divide the numerator and the denominator of a ratio by any number N ($N \neq 0$), then the ratio remains same.

   A/B = (A/N)/(B/N)

10. If a/b, c/d, e/f ... etc. are all unequal ratios, then the value of $(a+c+e+\ldots)/(b+d+f+\ldots)$ lies in between the minimum and the maximum of all these ratios.

VARIATION

Two quantities A and B are said to be varying with each other if there exists some relationship between A and B such that the change in A and B is uniform and guided by some rule.

Some typical examples of variation

- Area (A) of a circle = πR^2, where R is the radius of the circle.

Area of a circle depends upon the value of the radius of a circle, or, in other words we can say that the area of a circle varies as the square of the radius of a circle.

- At a constant temperature, pressure is inversely proportional to the volume.
- If the speed of any vehicle is constant, then the distance traversed is proportional to the time taken to cover the distance.

Direct Proportion

If A varies directly to B, then A is said to be in direct proportion to B.

It is written as – A α B.

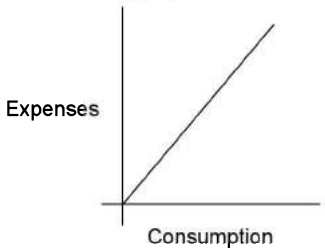
It can be understood with the typical example of percentage relating to expenses, consumption and price of the article.

If the price of a article is constant, then Consumption α Expenses.

$\Rightarrow$ Consumption = K. Expenses, where K is proportionality constant.

If we increase consumption by 20%, then the expenses will also increase by 20%.

At a constant price, if a graph is drawn between consumption and expenses by taking them at X-axis and Y-axis respectively, then this graph will be a straight line.



This is what we mean to say with direct proportion.

Inverse Proportion

If A varies inversely to B, then A is said to be in inverse proportion to B.

It is written as – A α 1/B

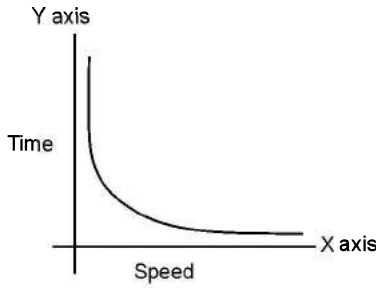
It can be understood with a time-speed-distance example, where if the distance is constant, then speed α 1/time.

Assuming the distance between New Delhi to Patna is 1000 km then consider the following table:

Speed (km/h)	*Time (in h)*
500	2
2 × 500	1/2 × 2
3 × 500	1/3 × 2
4/3 × 500	3/4 × 2

It can be seen here that the multiplier of time is reciprocal of the multiplier of speed.

For any fixed distance, if we draw a graph between speed and time by taking them at X-axis and Y-axis respectively, then this graph will be a curve.



Thus, when the speed is minimum, the time is maximum and when the speed is maximum, the time is minimum.

Example 10 The height of a tree varies as the square root of its age (between 5 and 17 years). When the age of a tree is 9 years, its height is 4 feet. What will be the height of the tree at the age of 16?

Solution Let us assume the height of the tree is H and its age is A years.

So, H α $\sqrt{A}$, or, H = K $\times$ $\sqrt{A}$

Now, 4 = K $\times$ $\sqrt{9}$

$\Rightarrow$ K = 4/3

So, height at the age of 16 years = H = K $\times$ $\sqrt{A}$ = 4/3 $\times$ 4 = 16/3 = 5 feet 4 inches.

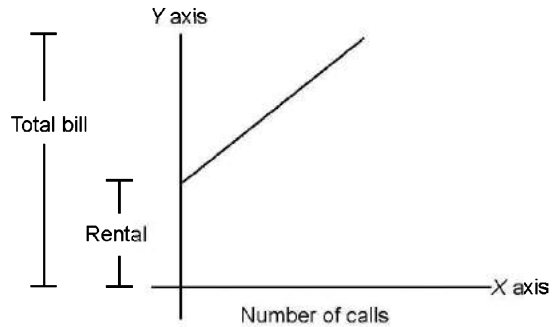
Direct Relation

A is directly related to B and as B changes, A also changes but not proportionally.

It is written as – A = C + K. B, where C and K are constants.

One classical example of direct relation can be seen as the telephone connection in a house. In a telephone connection we pay some money as the rent along with the phone charges according to the rate and number of calls made. So, the total bill = rental + K. (number of calls).

If we draw a graph between the number of calls and the total bill by taking them at X-axis and Y-axis respectively, this graph will be a straight line in the following way:



Example 11 Total expenses at a hostel is partly fixed and partly variable. When the number of students is 20, total expense is Rs 15,000 and when the number of students is 30, total expense is Rs 20,000. What will be the expense when the number of students is 40?

Solution Expenses = F + K. V; where F is the fixed cost and V is the number of students.

Rs 15,000 = F + K.20 .....(1)

Rs 20,000 = F + K.30 .....(2)

Solving (1) and (2),

Rs 5,000 = 10. K ⇒ K = Rs 500

So, F = Rs 5000

So, F + 40 K = Rs 5,000 + 40 × 500 = Rs 25,000

APPLICATION OF RATIO, PROPORTION AND VARIATION

1. Income–expense ratio

When the ratio of incomes and expenses of two persons are given and their savings is being asked.

Example 12 The ratio of the incomes of Mr Vinay Singh and Mr Arun Sharma is 3 : 5 and the ratio of their expenses is 1:3. Who is saving more?

Solution Let us assume the values of income and expenses of A and B,

	Income	*Expense*	*Savings*
Vinay Singh	Rs 3	Rs 1.5	Rs 1.5
Arun Sharma	Rs 5	Rs 4.5	Rs 0.5

So, Vinay saves more than Arun.

In the other case,

	Income	*Expense*	*Savings*
Vinay Singh	Rs 3	Rs 1	Rs 2
Arun Sharma	Rs 5	Rs 3	Rs 2

So, savings of both of them is equal.

	Income	*Expense*	*Savings*
Vinay Singh	Rs 3,000	Rs 1,600	Rs 1,400
Arun Sharma	Rs 5,000	Rs 4,800	Rs 200

So, in this case Mr Singh is saving less than Mr Sharma.

Thus, it is difficult to determine who is saving more.

The concept tells us: If the value of the ratio of income is more than the value of the ratio of expenses, then we cannot determine who is saving more. If the value of the ratio of expenses is more than the value of the ratio of income, then we can determine who is saving more. (Ratio should be taken in such a way that the value of ratio is less than 1, i.e., the numerator should be less than the denominator.)

In the above case, the value of the ratio of income = 3/5 = 0.6 and value of ratio of expenses = 1/3 = 0.33

Since the value of ratio of expenses < value of the ratio of income, we cannot determine who is saving more.

However, in the above question, if we take the ratio of income of Vinay and Arun as 3 : 5 and the ratio of their expenses as 3:1, then Arun is saving more.

2. Partnership

If two or more than two persons are investing their money in a joint venture, then the profit or loss incurred is distributed among everybody in the ratio of the time period for which the money has been invested by each of them.

It can be observed here that the partnership is a typical case of direct variation.

Example 13 A, B, C, … , K, L invested money in the ratio of 12:11:10:… : 2:1. The duration for which they invested the money is in the ratio of 1:2:3 :… : 11:12. Who will obtain the maximum profit at the end of year?

Solution Profit will be distributed in the ratio of product of the time and money of every individual.

So, the ratio of profit = 12 × 1:11 × 2:10 × 3:9 × 4:8 × 5:7 × 6:6 × 7:5 × 8:4 × 9:3 × 10:2 × 11:1 × 12.

So, the maximum profit will be obtained by F and G both.

PRACTICE EXERCISES

WARM UP

Q.1. The speed of three buses are in the ratio 2 : 3 : 4. The ratio between the time taken by these buses to travel the same distance is
(a) 2 : 3 : 4 (b) 4 : 3 : 2
(c) 4 : 3 : 6 (d) 6 : 4 : 3

Q.2. The difference between the two positive numbers is 10 and the ratio between them is 5:3. Find the product of the two numbers.
(a) 375 (b) 325 (c) 275 (d) 125

Q.3. If 30 oxen can plough 1/7th of a field in 4 hours, in how many hours will 18 oxen take to do the remaining work?
(a) 30 hours (b) 20 hours
(c) 15 hours (d) 40 hours

Q.4. Four numbers in the ratio of 1:3:4:7 add up to give a sum of 75. Find the value of the biggest number.
(a) 42 (b) 35 (c) 49 (d) 63

Q.5. In the above question, what is the difference between the biggest and the smallest number?
(a) 42 (b) 30 (c) 49 (d) 63

Q.6. When we increase both the numerator and the denominator by 7, a fraction changes to 3/4. Find the original fraction.
(a) 5/12 (b) 7/9 (c) 2/5 (d) 3/8

Q.7. According to the Boyle's law, at a constant temperature, pressure of a definite mass of gas is inversely proportional to the volume. If the pressure is reduced by 20%, find the respective change in volume.
(a) −33.33% (b) +25%
(c) −25% (d) +33.33%

Q.8. What number must be subtracted from both the numerator and the denominator 27 : 35, so that it becomes equal to 2 : 3?
(a) 8 (b) 12
(c) 6 (d) None of these

Q.9. P is directly proportional to Q, and Q = 7 when P = 15, find P when Q = 14.
(a) 36 (b) 54 (c) 30 (d) 60

Q.10. The concentration of three milk solutions A, B and C are 10%, 20% and 30% respectively. They are mixed in the ratio 2 : 3 : N resulting in a 23% concentration solution. Find N.
(a) 7 (b) 6 (c) 5 (d) 4

Q.11. Which of the following will have the maximum change in their values if 10 is added to both the numerator and the denominator of all the fractions?
(a) 3 : 4 (b) 2 : 3 (c) 4 : 7 (d) 5 : 7

Q.12. If $p/q = 3/4$, then find the value of the expression $(5p - 3q)/(7p + 2q)$.
(a) 3/29 (b) 5/27 (c) 7/31 (d) 5/58

Q.13. In a mixture of milk and water of volume 30 litres, the ratio of milk and water is 7:3. Find the quantity of water to be added to the mixture to make the ratio of milk and water 1:2.
(a) 30 (b) 32 (c) 33 (d) 35

Q.14. A and B invested Rs 12,000 and Rs 18,000 respectively in a business for the whole year. At the year-end, there was a total profit of Rs 2,000. What is the share of A in the profit?
(a) Rs 800 (b) Rs 1,200
(c) Rs 1,600 (d) None of these

Q.15. A and B invested the same capital in a business. At the year-end they share the profit in the ratio of 3:2. If A has invested his capital for the whole year, for how many months B has invested his capital?
(a) 6 months (b) 8 months
(c) 9 months (d) None of these

Q.16. A and B invest Rs 12,000 and Rs 16,000 respectively in a business. At the year-end, they share the profit in the ratio of 3:1. If A has invested his capital for the whole year, for how many months B has invested his capital?
(a) 4 months (b) 3 months
(c) 6 months (d) 8 months

FOUNDATION

Q.1. King Dashrath decided to distribute gold coins to his three queens in the following way: 2nd queen would get 5/7th of what the 1st queen would get and the 3rd queen would get 3/5th of what the 2nd queen would get. 1st queen got 60 gold coins more than the 3rd queen. How many gold coins were distributed to the three queens?
(a) 120 (b) 175 (c) 225 (d) 250

Q.2. The sum of the ages of the 4 members of Sinha family is 140 years. 5 years ago the ages of the 4 members Nishu, Vicky, Mrs Sinha and Mr Sinha were in the ratio of 2:3:7:8. After how many years would Nishu be as old as the present age of his mother?
(a) 10 yrs (b) 17 yrs
(c) 30 yrs (d) 32 yrs

Q.3. In the prime minister office, one-third of the workers are women, half of the women are married and one-third of the married women have children. If three-fourth of the men are married and two-third of the married men have children, what part of the persons working at the PMO are without children?
(a) 5/18 (b) 4/9 (c) 11/18 (d) 17/36

Q.4. Two-fifth of Anil's salary is equal of Bhuvan's salary and seven-ninth of Bhuvan's salary is equal to Chandra's salary. The sum of the salary of all of them is Rs 770. Which of the following is the salary of each?
(a) 300, 225, 250 (b) 500, 425, 375
(c) 450, 180, 140 (d) 520, 610, 475

Q.5. Inspector Vijay Shankar spends one-fourth of his salary on the house rent, one-third on food and one-sixth on travel. After spending one-tenth of the remaining amount on clothes, he is left with Rs 13,500. What is the difference between the amounts spent on the house rent and food?
(a) Rs 7,500 (b) Rs 10,000
(c) Rs 4,000 (d) Rs 5,000

Q.6. Bidhan is planning to buy a bike worth Rs 35,000, provided his brother agrees to lend him 3/2 times the money. Bidhan contributes and the financer provides 3 times the brother's contribution. How much does Bidhan's brother contribute?
(a) 5,000 (b) 7,500
(c) 10,000 (d) 7,000

Q.7. A group of students decided to buy a book jointly which costs between Rs 170 to 195. However at the last moment two students decided not to contribute and so, each of the remaining students had to pay one rupee each more. What was the price of the book if the students paid equal shares?
(a) Rs 182 (b) Rs 188.12
(c) Rs 192.4 (d) Rs 180

Q.8. An alloy of manganese, tin and bronze contains 90% bronze, 7% manganese and 3% tin. A second alloy of bronze and tin is melted with the first and the mixture contains 85% of bronze, 5% of manganese and 10% of tin. What is the percentage of bronze in the second alloy?
(a) 67.5% (b) 72.5%
(c) 77.5% (d) 82.5%

Q.9. Vinit is carrying some money with him. He gives 25% of the money to his friend, Sachin. He spends $\frac{1}{3}$ rd of the remaining, on a movie ticket. With 40% of the remaining money he does shopping. After all this he meets his friend Gita who gives him 66.66% of the money which he was having at that time. Finally, Vinit gives $\frac{4}{5}$th of the total money to his mother. What is left is the square root of the initial amount he was carrying. How much money does Gita give him?
(a) 40 (b) 25 (c) 10 (d) 20

Q.10. Two cogged wheels of which one has 32 cogs and other 54 cogs, work into each other. If the latter turns 80 times in three quarter of a minute, how often does the other turn in 8 seconds? (Assume equal size cogs and equi-spaced).
(a) 24 (b) 16 (c) 32 (d) 8

Q.11. Kunal, Saurav and Gopi assemble for a dutch party. Kunal brings 3 guavas while Saurav brings 5. Since Gopi did not have any guavas, he contributed Rs 8. How many rupees should Kunal and Saurav respectively get, assuming each of the three consumes an equal portion of the guavas?
(a) 1, 6 (b) 2, 5
(c) 5, 7 (d) None of these

Q.12. A bag contains Rs 600 in the form of one rupee, 50 paise and 25 paise coins in the ratio of 3:4:12. Find the total number of 25 paise coins in the bag.
(a) 75 (b) 200 (c) 300 (d) 900

Q.13. A certain sum of money was divided among Anand, Bidhan and Chandra in the following way: Chandra got half as much as Anand and Bidhan together got. Anand got one-third of what Bidhan and Chandra

together got. What is the ratio of Anand's share to that of Chandra's share?
(a) 1:2 (b) 3:5
(c) 4:7 (d) None of these

Q.14. Anil started a manufacturing unit with a certain amount of money. After a few months, Dheeraj became his partner, contributing three times of what Anil had contributed. At the end of the year, each was entitled to half the total profit. If Anil started the unit in January, then when did Dheeraj join as a partner?
(a) August (b) September
(c) July (d) October

Q.15. Pawan, Qureshi and Ravi entered into partnership, and provided capital of Rs 22,000, Rs 26,000 and Rs 34,000 respectively. Some months later Rs 10,000 extra capital was supplied by Qureshi. At the end of 12 months, the total profit was Rs 50,274, and Pawan's share was Rs 12,474. When did Qureshi supply the extra capital?
(a) After 4 months (b) After 5 months
(c) After 6 months (d) None of these

Q.16. A's salary is $\frac{5}{4}$ of B's salary, and B's salary is $\frac{4}{3}$ of C's salary, while M's salary is equal to the sum of salaries of A, B and C. If salary of C is Rs 1,500, What is the salary of M?
(a) 5,500 (b) 6,500
(c) 7,000 (d) 6,000

Q.17. Four years ago, the ratio of the ages of A and B was 13:9 and eight years hence it would be 4:3. The difference of their present ages is
(a) 56 years (b) 40 years
(c) 16 years (d) 24 years

Q.18. The work done by a man, a woman and a child is in the ratio of 3:2:1. There are 20 men, 30 women and 36 children in a factory. Their weekly wages amount to Rs 780, which is divided in the ratio of work done by the men, women and children. What will be the wages of 15 men, 21 women and 30 children for 2 weeks?
(a) Rs 585 (b) Rs 292.5
(c) Rs 1,170 (d) Rs 900

Q.19. On the eve of the foundation day party of Due North Inc., the ratio of males to females was 5:3. However, when 10 males left the party, this ratio got changed to 1:1. How many people were there originally at the party?
(a) 48 (b) 32 (c) 64 (d) 40

Q.20. What is the angle in degrees made by a sector, the ratio of whose area with the area of the semicircle is equal to 1:10?
(a) 36 (b) 18 (c) 24 (d) 9

Q.21. Two candles of the same length are lighted at 12 noon. The first is consumed in 6 hours and the second in 4 hours. Assuming that each candle burns at a constant rate, in how many hours after being lighted, was the first candle twice the length of the second?
(a) 3 p.m. (b) 2 p.m.
(c) 1.30 p.m. (d) 2.30 p.m.

Q.22. The value of a diamond varies directly as the square of its weight. If a diamond worth Rs 10,000 is divided into 2 pieces in the ratio of 4:6, what is the loss in value?
(a) 52% (b) 48%
(c) 36% (d) None of these

Q.23. The monthly income of A and B are in the ratio of 3:2, and their expenditures are in the ratio of 5:3. If each of them saves Rs 1,000 a month, then what is the income of A?
(a) Rs 4,000 (b) Rs 6,000
(c) Rs 9,000 (d) None of these

Q.24. If ab, bc, x and c^2 are in proportion, then find x, if none of these are equal to zero.
(a) a^2c (b) c^2a
(c) ac (d) None of these

Q.25. The first, second and third class fares between two railway stations, Patna and Lucknow were 10:8:3 and the number of first, second and third class passengers between the two stations was is 3:4:10. If total sales of the ticket is Rs 16,100, find the money obtained by the sales of second class tickets.
(a) Rs 5,250 (b) Rs 5,600
(c) Rs 6,400 (d) Rs 6,650

Q.26. If A:B = 3:4. What will be the value of $(3A^2 + 4B)/(3A - 4B^2)$?
(a) 43:55
(b) −43:55
(c) Either of (1) or (2)
(d) Cannot be determined

Q.27. The sum total of the salary of Atul, Binod and Charu is Rs 350. If they spend 60%, 80% and 75% of their respective salaries, then their savings are in the ratio of 8:8:5. Find the salary of Atul.
(a) Rs 50 (b) Rs 100
(c) Rs 200 (d) Rs 250

Q.28. Two horses cost as much as 5 dogs, 6 dogs as much as 8 oxen, 10 oxen as much as 50 sheep, 14 sheep as much as 9 goats. If the price of one goat is Rs 70, how much will one horse cost?
(a) Rs 670 (b) Rs 730
(c) Rs 830 (d) Rs 750

Q.29. A pond has a rare breed of lotus that triples in number every minute. If $\frac{1}{3}$rd of the pond is full of lotus in

30 min, what is the total time taken for the whole pond to be full?
(a) 1 min (b) 31 min
(c) 90 min (d) 60 min

Q.30. Satish used 6 litres of oil paint to paint a map of India, 6 m high. How many litres of paint would Santosh need for painting a proportionally scaled map of the height of 18 m?
(a) 54 (b) 18
(c) 30 (d) Cannot be determined

Q.31. A is proportional to B. B is inversely proportional to C. C is proportional to the square of D. D is directly proportional to the cube root of E. Assuming positive integers, if A increases then E
(a) Increases
(b) Decreases
(c) Cannot say
(d) Could increase or decrease

Q.32. Measurement of the temperature is carried out using thermometers in which the mercury expands linearly with the change in temperature. If the temperature range from the melting point of water to the boiling point of water is divided into 0 to 100 in Celsius scale and 32 to 212 in Fahrenheit scale, what is the temperature at which both the scales indicate the same value?
(a) 0 (b) 40 (c) 32 (d) – 40

Q.33. The sum of four numbers is 255. The ratio of the first number to the second number is 2:3. The ratio of the second number to the third number is 5:6. The ratio of the third number to the fourth number is 8:9. What is the average of the second number and the third number?
(a) 72 (b) 132
(c) 60 (d) None of these

Q.34. The population of the Balearic Island situated in Mediterranean Sea is 18,000. Balearic Island has three ports A, B and C. Every year, the entire population of each port moves to the other two ports, half going to one and the remaining half going to the other. The current populations of A, B and C are 2,000, 6,000 and 10,000 respectively. Then four years from now, the population of port A will be
(a) 5,000 (b) 6,500
(c) 6,000 (d) 5,750

Q.35. Arun has a certain amount of money in the denomination of 1 rupee and 10 rupee notes. The number of 1 rupee notes multiplied by the number of 10 rupee notes is equal to the total money (in rupees) that he has. What is the total number of ten rupee notes that he can have?
(a) 11 (b) 13
(c) 15 (d) None of these

Q.36. Divya and Nandan go to a bakery and buy some black currant pastries. Divya buys 9 black currants more than Nandan. Later in the day, they go to the bakery again and buy some butter scotch pastries. In the entire day, Divya and Nandan buy a total of 12 and 21 pastries respectively but spend the same amount of money. Find the ratio of the price of a black current to that of a butter scotch.
(a) 3:2 (b) 2:1 (c) 5:2 (d) 3:1

Q.37. Rohan purchases some pastries of two varieties— one variety costing Rs 3 per piece and the other Rs 7 per piece. He wanted to spend exactly 50 rupees for the same. What is the maximum number of pastries he can purchase of the costlier variety?
(a) 6 (b) 5 (c) 4 (d) 2

Q.38. Three friends A, B and C started a venture with capitals in the ratio of 4:1:15. At the end of every quarter A halves his capital, while B doubles his capital and C leaves his capital untouched. This process is repeated till the end of the year. If at the end of the year B's share of the profit is Rs 22,000, what is the total profit?
(a) Rs 88,000 (b) Rs 1,10,000
(c) Rs 1,21,000 (d) None of these

Q.39. Rakesh, Shyam, and Prashant have some money with them. If Rakesh gives Rs 15 to Shyam then he would have the same amount. If Prashant takes half of Shyam's money, he would have 5 times as much money left with Shyam. After this, the ratio of money with Rakesh, Shyam and Prashant is 5:2:4. What is the total amount with them?
(a) Rs 55 (b) Rs 110
(c) Rs 165 (d) Rs 231

Q.40. King Fahd of Saudi Arabia distributed his property equally among his 6 sons as follows— to his eldest son he gave 1 lakh gold coins and 1/7th of the remaining coins. To the second son, he gave 2 lakh gold coins and 1/7th of the remaining gold coins and so on and to his youngest son he gave 6 lakh gold coins. The entire number of gold coins was distributed without any remainder. What was the total number of gold coins with the king?
(a) 22 lakh (b) 50 lakh
(c) 71 lakh (d) 36 lakh

Q.41. Abhishek bought a total of 'N' dozen fruits consisting of apples and bananas in the ratio of 3:7. The number of dozen of apples he bought is equal to the cost per dozen of bananas and vice versa. Total amount spent by him is equal to Rs 1,050. Find the number of dozen of apples and the number of dozen of bananas he bought respectively.
(a) 12 dozen and 38 dozen
(b) 14 dozen and 36 dozen

(c) 15 dozen and 35 dozen
(d) 18 dozen and 32 dozen

Q.42. If $xy = r, yz = r^3, xz = r^2$, also $x + y + z = 13$ and $x^2 + y^2 + z^2 = 91$, then what is the value of $z:y$?
(a) 3 (b) $\frac{7}{3}$ (c) 13 (d) $\frac{13}{3}$

Q.43. Gandhiji owns cows, some black and some white. He finds that 4 black cows and 3 white cows gave the same amount of milk in 5 days as 3 black cows and 5 white cows gave in 4 days. What is the ratio of milk given by a black cow in a day to that given by a white cow in a day?
(a) 8 : 5 (b) 5 : 8 (c) 3 : 5 (d) 5 : 3

Q.44. Two software developers Vinit and Harpal share a mainframe and the final cost is Rs 74,750. Vinit uses 5 type I terminals for 7 weeks and 22 type II terminals for 4 weeks. Harpal uses 10 type I terminals for 6 week and 8 type II terminals for 6 weeks. If cost of operating each of type I and type II terminal is in a ratio of 2:3, find the share of Vinit and Harpal in the final cost.
(a) Rs 41,750, Rs 33,000
(b) Rs 40,000, Rs 34,000
(c) Rs 41,000, Rs 3,80,000
(d) Rs 50,000, Rs 24,750

Q.45. A metal trader buys 2 kinds of silver foils, the ratio of their prices being 1:4. He sells the alloy at Rs 90 per kg so that he can make a profit of 20%. If the ratio of their quantities present in a alloy is 6:1 respectively, find the purchase price of the foil present in lesser quantity.
(a) Rs 52.5 (b) Rs 55
(c) Rs 47.5 (d) Rs 45

Q.46. A heap of topaz, rubies and emeralds is required to be formed. A topaz is worth Rs 5 crores, a ruby Rs 2 crores and an emerald Rs 3 crores. The weight of a topaz is 0.7 kg, weight of a ruby is 0.2 kg and that of an emerald is 0.3 kg. If the total weight allowed is 8 kgs, then which combination maximizes the worth of the entire heap?
(a) 10 topaz, 2 rubies 2 emeralds
(b) 6 topaz, 1 ruby, 12 emeralds
(c) 11 topaz, 1 emerald
(d) 26 emeralds, 1 ruby

Direction for questions 47 and 48: Read the passage below and solve the questions based on it.

A book having pages between 4,000 and 5,000 is divided into four parts, each part being divided into chapters. The total number of pages in each of the four parts is the same. The ratio of the chapters across all the parts is 6:5:10:14. The number of chapters in the fourth part is 70.

Q.47. What is the total number of pages in the book?
(a) 4,000 (b) 4,800 (c) 4,200 (d) 4,600

Q.48. What is the total number of chapters in the book?
(a) 113 (b) 226
(c) 127 (d) None of these

Q.49. A rat takes 5 leaps for every 4 leaps of a cat, but 3 leaps of the cat are equal to 4 leaps of the rat. What is the ratio of the speed of the rat to that of the cat?
(a) 11 : 15 (b) 15 : 11 (c) 16 : 15 (d) 15 : 16

Q.50. The ratio of the age of Sita and her mother is 2:3. N years from now, ratio of their ages will become 3:5. What is the value of N?
(a) Infinite (b) 1
(c) More than 1 but finite (d) Not possible

MODERATE

Direction for questions 1 to 3: Read the passage below and solve the questions based on it.

Two of my friends Nishant and Rajesh, have the same number of children. Each son of Nishant has 5 times as many sisters as the number of brothers, while each son of Rajesh has as many sisters as the number of brothers. Each daughter of Nishant has twice the number of the brothers but each daughter of Rajesh has twice the number of brothers as the number of sisters.

Q.1. How many children did Rajesh have?
(a) 3 (b) 4 (c) 6 (d) 7

Q.2. How many sons does Nishant have?
(a) 1 (b) 2
(c) 4 (d) Cannot be determined

Q.3. What is the ratio of the number of sons to the number of daughters in the 2 families?
(a) 1:3 (b) 3:1
(c) 7:2 (d) None of these

Direction for questions 4 to 6: Read the passage below and solve the questions based on it.

Three persons, Anny, Berry and Cherry, started playing a game. They had money in the ratio of 3:2:1. It was decided

that if Anny lost the game he would distribute 1/4th of the money that he had prior to the round, equally between Berry and Cherry. If Berry lost the game he would distribute half of his money equally between Anny and Cherry and if Cherry lost the game he would distribute 2/3rd of his money prior to that round equally between Anny and Berry. Berry lost the game first, and then Cherry lost the next game followed by Anny and then followed by Berry again. In the process Anny gained Rs 10.

Q.4. What was the total sum of money they had initially?
(a) Rs 60 (b) Rs 120
(c) Rs 150 (d) Rs 240

Q.5. After which round the game would have been stopped for Anny to gain the maximum profit?
(a) Round 1 (b) Round 2
(c) Round 3 (d) Round 4

Q.6. Which of these did not have any chance of making profit in any of the rounds?
(a) Both Berry and Cherry
(b) Both Anny and Cherry
(c) Only Cherry
(d) Only Berry

Direction for questions 7 to 9: Read the passage below and solve the questions based on it.

Anu, Bunty, Christy and Dorthy are 4 salesmen. In the 1st year they received a commission of Rs 4,200 from their company and divided it in the ratio of 2 : 3 : 4 : 5. In the 2nd year the commission doubled, the amount was divided in the ratio of 3 : 4 : 5 : 2. In the 3rd year the commission was tripled when compared to the first year and they shared in the ratio of 4 : 5 : 3 : 2 and in the 4th year the commission was half compared to the previous year and they shared in the ratio of 4 : 3 : 5 : 2.

Q.7. What was the average yearly income of Christie over the period?
(a) Rs 2,287.5 (b) Rs 2,252.5
(c) Rs 1,680.25 (d) None of these

Q.8. How much should Bunty have earned so that his average yearly income would have been Rs 3,500?
(a) 2,450 (b) 3,250
(c) 4,850 (d) 5,750

Q.9. The ratio of the rate of tomatoes to the rate of brinjals is
(a) 1:2 (b) 1:3
(c) 1:5 (d) None of these

Direction for questions 10 and 11: Read the passage below and solve the questions based on it.

Narendrajee is the office assistant at LC's North Campus centre. One day he realized that (i) among all the visitors the ratio of the number of men and women was the same as that between women and children, (ii) maximum number of the visitors to the centre were children. At the end of the day, he finds that 7 women visited the centre.

Q.10. How many children visited the centre that day?
(a) 44 (b) 49 (c) 52 (d) 57

Q.11. How many men visited the centre that day?
(a) 25 (b) 68
(c) 49 (d) None of these

Q.12. P works twice as fast as Q, whereas P and Q together can work three times as fast as R. If P, Q and R together work on a job, in what ratio should they share the earnings?
(a) 2:1:1 (b) 4:2:1
(c) 4:3:2 (d) 4:2:3

Direction for questions 13 to 15: Read the passage below and solve the questions based on it.

Sachin has 3 children Ten, Dul and Kar. One day he brought 117 sweets for his children and planned to divide the sweets in such a manner that two times the number of sweets that Ten would have got should be equal to 2/3 times that Dul got and 4 times that Kar got. Since Sachin was poor at counting, he divided the sweets in the ratio of 2:3:4.

Q.13. How many sweets did Kar get as a result of wrong sharing?
(a) 16 (b) 20
(c) 22 (d) None of these

Q.14. If Kar were to get the same number of sweets that Dul got now after the wrong distribution had the distribution being done correctly, how much should the total number of sweets be?
(a) 150 (b) 169
(c) 216 (d) None of these

Q.15. If X bulbs consume X^2 units of electricity in X^3 days, find the time in which Y bulbs will consume Y units of electricity.
(a) X^1 days (b) X^2 days
(c) X^3 days (d) X^0 days

Direction for questions 16 and 17: Read the passage below and solve the questions based on it.

There are certain number of apples, guavas and oranges in a basket. The number of each variety is more than one. The ratio of the number of apples to the number of guavas is equal to the ratio of the number of guavas to the number of oranges.

Q.16. If the total number of fruits is 61, then find the number of guavas.

(a) 16 (b) 20
(c) 25 (d) Cannot be determined

Q.17. If the number of guavas is 21, then which of the following can be the total number of fruits in the basket?
(a) 63 (b) 89 (c) 101 (d) 117

Q.18. The time taken by a jet, to reach a certain place while flying at a speed, slower by 300 m/h than the normal, is double the time taken by it while flying at a speed faster by 300 m/h than normal. What is the ratio of the 'delay time' while flying slower to the 'early time' while flying faster?
(a) 3:2 (b) 1:2
(c) 2:1 (d) Cannot be determined

Q.19. Anand and Bhavana have a few one rupee, 50 paise and 25 paise coins each. The total amount with each of them is Rs 5. The number of coins of two of the three types of coins that Anand has are equal and same is the case for the coins that Bhavana has. If Anand has the maximum possible total number of coins, then find the maximum possible total number of coins that Bhavana can have given that she has fewer coins than Anand.
(a) 16 (b) 12 (c) 10 (d) 9

Q.20. In the above question, what is the number of 50 paise coins with Anand?
(a) 1 (b) 2 (c) 3 (d) 4

Q.21. In a flood-affected village called "Waterland", there are 500 adult residents. They planned to construct a dam to overcome the situation. Each of the men contributed a sum equal to the number of women in the village and each woman a sum equal to the number of men. The village-chief came to know about this and contributed an amount equal to the contribution of the villagers. The total contribution then became Rs 2.4 lakhs. Find the number of men in "Waterland".
(a) 250 (b) 300
(c) 350 (d) Cannot be determined

Q.22. If the man initially had Rs 82, instead of Rs 100, when he went to the market, then the ratio of the rate of tomatoes to the rate of cucumber was
(a) 19:4 (b) 2:3
(c) 4:19 (d) None of these

Q.23. The sides of a right angled triangle are a, $a + x$ and $a + 2x$ with a and x both positive. The ratio of a to x is
(a) 3:1 (b) 5:2 (c) 2:1 (d) 3:4

Q.24. If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b}$, then each fraction is equal to

i. −1 ii. 1/2 iii. 1
(a) i and ii only (b) ii and iii only
(c) i, ii and iii only (d) None of these

Direction for questions 25 and 26: Read the passage below and solve the questions based on it.

Ujala Jewellers is a well known diamond jewellery shop. Owner of the shop has priced his diamond necklaces such that the cost varied as the square of the number of diamonds, for example the necklace placed at the first right top corner has 8 diamonds and costs Rs 1,60,000. There are lots of necklaces and the price of all necklaces have been found by the same method.

Q.25. What would be the price of the bracelet with 10 diamonds?
(a) 1,75,000 (b) 2,25,000
(c) 2,50,000 (d) 4,00,000

Q.26. Vijay Shanker asked for diamonds in a necklace to be divided in the ratio of 5:3:2. Owner of Ujala Jewellers knew that he would suffer some loss. What is the % decrease in the price that he would incur, if he has to oblige Vijay Shanker?
(a) 22% (b) 38% (c) 62% (d) 78%

Q.27. Between two stations Kanpur and Mumbai the first, second and third class fares were fixed in the ratio of 8:6:3. However, when Mr L Prasad, the railway minister, presented the railway budget, he reduced the first class fares by 1/6 and the second class by 1/12 in order to rationalize the fares vis-à-vis the second-class fares. In a year, ratio of the number of first, second and third class passengers were respectively 9:12:26 and the money at the booking offices was Rs 1,08,800. How much was paid by the first class passengers?
(a) Rs 32,080 (b) Rs 16,800
(c) Rs 24,400 (d) None of these

Direction for questions 28 and 29: Read the passage below and solve the questions based on it.

In the famous Hawthorne studies by Elton Mayo, employees of the organization were given the options of going to several places for entertainment. One-seventh of the employees preferred to go to the polo ground, a number equal to the square root of the remaining employees went to the museum. One-fifth of the remaining employees went to the fair. One-third of the remaining to the science exhibition and the rest to watch movie ASHOKA, which was running house full in the nearby theatre. The total number of employees who went to the polo ground and the museum is twice that of the employees who went to the fair.

Q.28. What is the strength of the organization?
(a) 14 (b) 35 (c) 42 (d) 49

Q.29. The entry tickets for museum, fair, exhibition and the movie were Rs 5, Rs 4, Rs 2 and Rs 25 respectively. How much money did the employees pay for all these on that Sunday?

(a) Rs 275 (b) Rs 320
(c) Rs 388 (d) Rs 470

Direction for questions 30 and 31: Read the passage given and solve the questions based on it.

In the last year's "Tycoon Contest", nine teams from different colleges participated. Navneet Anand, the quiz master, asked the nine teams— "I have 9 coins, the ratio of the weight of the coins are same, except one. I give the different numbers to these 9 coins starting from 11 to 19. The team who can find the coin with different weights in the least number of weighings can take that coin and will be the winner."

Q.30. Team number 5 said that they knew the answer and if they had won the coin, what was their answer?
(a) One weighing (b) Two weighings
(c) Three weighings (d) Four weighings

Q.31. Navneet Anand now said that the coin whose weight was different from the rest was the heaviest of the all. What is the minimum number of weighings required to find that coin? Contestant 8 gives the correct answer and his answer is
(a) One weighing (b) Two weighings
(c) Three weighings (d) Four weighings

Q.32. The cost of building a wall is Rs 1,347. Wages of workmen is increased by 1/8 of the former wages and working hours per day have been increased by 1/20 of the former duration. What is the new cost (approximately) of building a wall, the length of which is two times the length of this wall and the other dimensions of this wall are same?
(a) Rs 2,692 (b) Rs 2,724
(c) Rs 2,886 (d) Rs 2,484

Q.33. If $a^2 = by + cz$, $b^2 = cz + ax$, $c^2 = ax + by$, then $\frac{x}{a+x}+\frac{y}{b+y}+\frac{z}{c+z}=?$
(a) abc (b) 1 (c) $\frac{abc}{xyz}$ (d) $\frac{xyz}{abc}$

Q.34. If 29 goats can feed on a field of uniformly grown grass in 7 days or 25 goats can feed on the same field in 9 days, how many goats will feed in 6 days?
(a) 36 (b) 44
(c) 42 (d) 32

Q.35. Given, $x:y = a:b = 1:5$. Then, the value of $\frac{x^2+a^2}{y^2+b^2}$ is
(a) $\frac{1}{5}$ (b) $\frac{1}{50}$
(c) $\frac{1}{250}$ (d) None of these

Q.36. a,b and c are three positive numbers and $s=\frac{(a+b+c)}{2}$. If $(s-a):(s-b):(s-c) = 1:7:4$, the ratio of $a:b:c =$
(a) 8 : 10 : 6 (b) 9 : 4 : 12
(c) 7 : 8 : 9 (d) 11 : 5 : 8

Q.37. In an election, each voter may vote for two candidates. Half of the voters vote for A, but divide their votes between B, C and D in the ratio of 3:2:1. Of the remaining, half vote for B and divides their votes between C and D in the ratio of 2:1. Of the remaining, half vote for C and D and the remaining 840 do not vote for anyone. How many votes were received by A, B, C and D respectively?
(a) 3,360, 3,360, 3,080, 1,960
(b) 3,360, 3,360, 1,960, 3,080
(c) 3,450, 3,210, 4,120, 1,210
(d) 5,000, 5,000, 2,500, 1,250.

Direction for questions 38 and 39: Read the passage below and solve the questions based on it.

Two friends Pankaj and Pranav decide to play a game where the person losing that round gives half of his money to the one who wins. They start playing with Rs 100 each.

Q.38. At the end of 3 rounds which of the following cannot be the amount that Pankaj has?
(a) Rs 137.5 (b) Rs 162.5
(c) Rs 112.5 (d) Rs 125

Q.39. What is the maximum amount that changes hands in the game in 3 rounds?
(a) Rs 100 (b) Rs 75
(c) Rs 11.5 (d) Rs 87.5

Q.40. The square of the time taken by a planet to complete one revolution around the sun varies as the cube of its mean distance from the sun. If the ratio of the mean distances of Mars and Saturn from the sun is 16:3. Find the number of days on Saturn that are equal to a year on Mars.
(a) 4,496 days (b) 4,294 days
(c) 5,598 days (d) 4,696 days

Q.41. In a forest reserve, the ratio of the number of deers, bear and fox is 3:7:5. If the difference between the number of deers and bears is a multiple of 3 as well as 7, what is the minimum number of animals in the park?
(a) 315 (b) 310
(c) 45 (d) Cannot be determined

Q.42. A watermelon is cut into two pieces in the ratio of 3:5 by weight. The bigger of the two is further cut in the ratio of 5:7 by weight. Find the ratio of each of the three pieces.

(a) 3:5:7 (b) 36:25:35
(c) 15:25:56 (d) None of these

Q.43. Shalaj and Associates, a shoe-manufacturing company, has three machines producing shoes. If machine X produces 1/4th as many as machine Y produces in the same time, and machine Y produces twice as many as machine Z produces in the same time, then during a fixed time, what fraction of the total number of shoes is produced by machine Z?

(a) $\frac{1}{14}$ (b) $\frac{1}{3}$
(c) $\frac{2}{7}$ (d) $\frac{1}{2}$

Direction for questions 44 and 45: Read the passage below and solve the questions based on it.

A man went to the market with Rs 100 in his pocket. He bought 3 kgs of brinjals, y kgs of tomatoes and 4 kgs of cucumber. The amount spent on tomatoes was double than that spent on cucumbers and also is same as one third of that spent on brinjals. After the shopping the man was left with Rs 10 in his pocket.

Q.44. Which of the following statements is true?
(a) The ratio of the rate of tomatoes to the rate of brinjals is 1:4
(b) The ratio of the rate of tomatoes to the rate of cucumber is 1:8
(c) The ratio of the rate of cucumber to the rate of brinjals is 8:1
(d) None of these

Q.45. If the man was left with Rs 19 instead of Rs 10 and the total weight of vegetables bought is 10 kgs then, which of the following statement is true?
(a) The value of y is 4 kg.
(b) The value of y is one more than the rate of tomatoes.
(c) The value of y is one-sixth of the amount spent on tomatoes.
(d) None of these

ADVANCED

Q.1. The Road Transport Corporation has hired a consultancy firm to determine the standard time that should be taken for a journey between two local destinations. The consultant after a detailed study and discussion submitted in his report that the time of journey is proportional to the product of the square root of number of passengers and the distance to be covered. For two distinct trips, the times taken are in the ratio of 2:1 and the distances covered are 80 km and 30 km respectively. How many passengers are there in the first bus, if the second bus has 64 passengers?
(a) 9 (b) 16 (c) 36 (d) 24

Q.2. The inhabitants of Hawaii Island are taxed in a peculiar manner. They have to pay a fixed sum irrespective of their income strata. In addition to this they have to pay an amount which is proportional to the excess of their salary over Rs 50,000. Mr Kalyan pays a total tax of Rs 6,200 when his annual salary is Rs 6,0000 and his wife Mrs. Kalyan pays a total tax of Rs 7,700 when her annual salary is Rs 75,000. What is the yearly salary of Mr Ajay who pays an annual tax of Rs 8,200?
(a) Rs 80,000 (b) Rs 82,000
(c) Rs 90,000 (d) None of these

Q.3. Rohit finds that the distance he covers is equal to the sum of two quantities, one of which varies with time 'T' and the second varies with the square of the time 't' that has elapsed from the start. After 5 seconds Rohit has covered a distance of 100 metres and after 6 seconds, he has covered a distance of 138 metres. How far is he from the starting point after a time of 20 seconds?
(a) 450 m (b) 880 m
(c) 1,220 m (d) None of these

Q.4. While developing Tejas, the Light Combat Aircraft developed by DRDO to replace MIG series aircraft, it was found that the distance a man falls under gravity is $5t^2$ m in t s and the distance a man falls with an open parachute under gravity is $2t^2 - 3t$ m in t s. Squadron leader Ajay Rathod jumps out of a Tejas at a height of 2,300 m with a closed parachute. Find the height at which he can open the parachute so as to touch the ground in exactly 40 s.
(a) 1,050 m (b) 1,775 m
(c) 1,175 m (d) 1,250 m

Q.5. A manufacturing unit is to be set up at Geneva. The cost of setting up a unit is proportional to the square of the raw material it handles. If for all the processes four separate units are deployed, then the raw material that the individual unit is going to handle is in the ratio of 6:3:3:1. But if a single unit is deployed, then the total raw material that is to be handled is summation of

raw material that is to be handled by an individual unit. By installing separate units, the company saved Rs 57 lakh. What is the cost of installing a single unit?
(a) Rs 84.5 lakh (b) Rs 285 lakh
(c) Rs 338 lakh (d) None of these

Direction for questions 6 and 7: Read the passage given and solve the questions based on it.

At 70° F, 100 litres of milk can dissolve a maximum of 1.5 kg of sugar. For every increase in temperature by 1° F, the solubility of milk increases by 0.1 litre. For example, at 71° F, 100 litre of milk can dissolve 1.6 kg of sugar. The boiling point of milk is 100° F i.e., when the temperature of milk reaches 100° F, it stops increasing further and the milk starts evaporating @ 200 gm/min.

Q.6. If 20 litres of milk at 80° F with 180 gm sugar is heated till it starts to evaporate, then what is the maximum amount of sugar that can be added so that it does not sediment?
(a) 900 g (b) 520 g (c) 720 g (d) 770 g

Q.7. At 75° F, 75 litres of milk with 1 kg of sugar is heated till it starts to evaporate, and appropriate quantity of sugar is added so that it becomes fully saturated. Thereafter, it is heated for 10 min. How much sugar would have deposited at the bottom after 10 min?
(a) 120 g (b) 90 g (c) 100 g (d) 60 g

Q.8. Pranav, Raj and Santosh entered into a partnership with a sum of Rs 6,200, Rs 9,300 and Rs 12,400 respectively. It was decided that Pranav will manage the business for the whole year. But after some months due to some personal problems Pranav handed over his responsibility of managing the business to Raj for the rest of the year. The person who manages the business gets 10% of the profit as commission. If the total income of Pranav at the end of the year is Rs 290 and the total profit is Rs 1,200, for how many months did Pranav manage the business?
(a) 2 (b) 4
(c) 6 (d) None of these

Q.9. The total profit made by running a train between New Delhi to Lucknow consists of two parts: (i) a constant amount of Rs 24 lakh per trip and (ii) it varies as the square of the number of coaches attached to the engine that amounts to Rs $7P^2$, where P is the number of coaches attached to the engine in the trip. If the average profit per coach per trip should not fall below Rs 169 lakh, then what is the minimum number of coaches that have to be attached to the engine?
(a) 23 (b) 24
(c) 25 (d) None of these

Direction for questions 10 and 11: Answer the questions based on the following information.

A simple weighing balance having two pans is in perfect balance when no weights are kept in the two pans even though both the pans are not of the same weight. Further even when 10 kg is kept in the first pan and 5 kg is kept in the second pan, it remains in balance.

Q.10. What is the ratio of the weight of the first pan to the second pan?
(a) 1:2 (b) 2:1
(c) 3:1 (d) Cannot be determined

Q.11. What weight should be kept in the second pan if a weight of 5 kg is kept in the first pan so that the weighing scale is in balance?
(a) 2.5 kg (b) 1.25 kg
(c) 7.5 kg (d) None of these

Q.12. There are two alloys of gold and copper. In the first alloy there is twice as much gold as copper and in the second alloy there is 5 times less gold than copper. How many times of the second alloy should be taken than the first, in order to obtain a new alloy in which there would be twice as much copper as gold?
(a) Two times (b) Three Times
(c) Four Times (d) None of these

Q.13. The average age of the students in a class of 50 is 13. The weight of each student is directly proportional to the height. A 165 cm tall student has a weight of 33 kg. The average weight of the class is:
(a) 33 kg (b) 39 kg
(c) 36 kg (d) Data insufficient

Q.14. Nishit Transport Corporation (NTC) operates buses between Kolkata and Patna. To increase the profit, NTC has sought the services of Dipanjan Das and Sons consultants. After a thorough study of the mechanism of the whole business, Dipanjan Das and Sons suggested that— (i) profit in running a bus is directly proportional to the distance when the number of passengers is constant, and (ii) directly proportional to the number of passengers in excess of a certain fixed number when the distance is constant. The profit is Rs 80 when 30 passengers are carried over a distance of 40 km, and Rs 180 when 35 passengers are carried over 60 km. What should be the advice, furnished by Dipanjan Das and Sons consultants, regarding the minimum number of passengers to be carried, so that there is no loss?
(a) 19 (b) 18 (c) 20 (d) 25

Q.15. During the acceleration and breaking test by two men for a new bike launched by Kajaj, the 1st man starts the scooter and accelerates upto 40 km/h for a distance of 30.2 m and drives the remaining distance at 40 km/h. Second man starts and accelerates upto 60 km/h for a

distance of 80.9 m and travels with a speed of 60 km/h and then applies brakes to 0 km/h at the distance of 24.2 m. The total distance travelled is 200 m. If both of them started at the same time, then who will reach the 200 m mark first?
(a) Ist man (b) 2nd
(c) Same time (d) Data insufficient

Q.16. Two alloys of chromium have different percentage of chromium in them. The first one weighs 6 kg and second one weighs 12 kg. One piece each of equal weight was cut off from both the alloys and the first piece was alloyed with the second alloy and the second piece was alloyed with the first one. As a result, the percentage of chromium became the same in the resulting two new alloys. What was the weight of each cut-off piece?
(a) 4 kg (b) 2 kg (c) 3 kg (d) 1 kg

Q.17. The cost C of manufacturing a brick can be estimated by the formula $C = 0.03\ ABt^2$, where A and B are the amounts in Rs of the two major ingredients sand and clay and t is the production time in hours. If A is increased by 50%, B is increased by 20% and t is decreased by 30% by what percentage will the estimated cost of manufacturing the brick change?
(a) 12.4% (b) 11.8%
(c) 14.6% (d) 18.2%

Q.18. During the bank wiring experiment to understand the impact of pattern of informal relationship among employees in any organization, it is found that the quantity of work done by a man in an hour is directly proportional to his pay per hour and inversely as the square root of the number of hours he works per day. He can finish a piece of work in six days when he works 9 hours a day at Re 1 per hour. How many days will he take to finish the same work when working 16 hours a day at the rate of Rs 1.50 per hour?
(a) 10 days (b) 3 days
(c) 8 days (d) 4 days

Q.19. According to the personnel administration prevalent in India, people are awarded pensions in proportion to the square root of the number of years they have served. One has served 9 years longer than the other and receives a pension greater by Rs 250. If the length of service of the first had exceeded that of the second by $4\frac{1}{4}$ years their pensions would have been in the ratio of 9:8. How long have they served respectively?
(a) 16 years and 25 years
(b) 9 years and 16 years
(c) 9 years and 25 years
(d) None of these

Q.20. The speed of man while walking, running and driving are in the ratio of 1:3:9. If he travels equal distance while walking, running and driving, what is the ratio of $\frac{\text{time taken}}{\text{velocity}}$ in the above three cases?
(a) 9 : 3 : 1 (b) 1 : 1 : 1
(c) 81 : 9 : 1 (d) 27 : 9 : 1

Q.21. Given that $2^x = 4^y = 8^z$ and $xyz = 288$. What is the value of $\frac{1}{2x} + \frac{1}{4y} + \frac{1}{8z}$?
(a) $\frac{18}{74}$ (b) $\frac{18}{96}$ (c) $\frac{11}{74}$ (d) $\frac{11}{96}$

Q.22. According to census 2001 of India, population of 5 states: state 1, state 2, state 3, state 4, state 5, showed that their population figures are in the ratio of 1 : 3 : 2 : 3 : 2. Further, the ratio of the percentage of women in the population in these states is 1 : 3 : 2 : 5 : 4. If the total population of women in the five states is 3.7 million, then what is the excess number of women in state 2 over that of state 5?
(a) 10 lakh (b) 1 lakh
(c) 2.2 lakh (d) Cannot be determined

Q.23. The time period of oscillation T for a pendulum is given by $T = k\sqrt{\frac{\ell}{g}}$, where ℓ is the length of the pendulum and g is the acceleration due to gravity, 'k' is any const. If 'k' and 'g' remain unchanged under any condition, what should be the percentage change in ℓ such that the time period T increases by 10%?
(a) 10% increase (b) 10% decrease
(c) 20% increase (d) 21% increase

Q.24. A tap A can fill an acid into a vessel in 30 min. Another tap B can fill alkali into a vessel in 20 min and a leak C at the bottom of the vessel can empty the vessel in 12 min. Initially, both the taps A and B are opened for 6 min. Then tap B is closed and leak C is opened. For how many more minutes should both these taps A and C be kept opened so that the proportion of acid and alkali is same? (Assume tap C removes acid and alkali in exactly the same ratio as they were present in the solution when tap C was just opened and no reaction takes place).
(a) 1 min (b) 2 min
(c) 3 min (d) 4 min

Q.25. If after the operation of question 24 is performed (till the ratio is 1:1), tap C is closed and B is opened again and then taps A and B are kept open till the vessel is full, what will the ratio of acid and alkali be in the full vessel?
(a) 19:11 (b) 11:19 (c) 14:11 (d) 11:14

Direction for questions 26 to 29: Read the passage below and solve the questions based on it.

On 13th of November, I kept some money in a magical purse in which on every alternate day starting from the 13th, money increases with respect to the closing amount on the evening of the previous day, that the ratio of money becomes 2:1 but on every alternate day starting from the 14th, the money decreases by Rs 10, when compared to the closing amount on the evening of the previous day. I had Rs 500 on the 19th and the magical purse kept on multiplying money for me till the 25th of that November.

Q.26. What is the initial amount of money that I kept into the purse?
(a) 50 (b) 80
(c) 140 (d) Cannot be determined

Q.27. How much money was there on the 22nd of November?
(a) 440 (b) 360
(c) 240 (d) None of these

Q.28. Had I stopped on the 21st, how much less money would I have received with respect to the money which I would have obtained on the 25th?
(a) 2,880 (b) 3,260
(c) 4,270 (d) None of these

Q.29. Had I started with Rs 100, how much more/less money would I have received than I am receiving now?
(a) 400 (b) 760 (c) 1,280 (d) 1,640

Q.30. The square of the ratio of the number of sweets with Aldrin's sons and daughters is equal to the cube of the ratio of the number of sweets with Bladimir's sons and daughters. The total number of chocolates with all the children of Aldrin and Bladimir is a prime number The number of sweets with Bladimir's sons equals the number of sweets with Aldrin's daughters. The number of sweets with Bladimir's daughters is 1/32 of the number of sweets with Aldrin's sons. What is the total number of sweets with all the children?
(a) 31 (b) 41 (c) 61 (d) 101

Direction for questions 31 and 33: Read the passage below and solve the questions based on it.

Surya Pratap Singh has 4 children (two daughters who are older to the two sons). He decides to divide his property among his children such that both the daughters together get the same share as the two sons together. Elder son gets more than the younger son and the elder daughter gets more than the younger daughter. His entire property consists of cash, gold coin and copper plate. Each gold coin is worth Rs 5,000 and copper plate in worth Rs 1,000. The eldest daughter got 1/3rd of the gold and 1/5th of the copper while elder son got Rs 1,50,000 in cash, 50% of his wealth as gold and 250 copper plates. The younger daughter got 18% of the copper and 15% of the cash. The younger son got cash equal to 90% of the cash the elder daughter got and 1/6th of the gold and 12% of the copper. Surya Pratap Singh has gold worth 3 times that of copper.

Q.31. Who got the maximum wealth?
(a) Elder son (b) Elder daughter
(c) Younger son (d) Both (1) and (2)

Q.32. How much cash did the younger daughter get?
(a) Rs 50,000
(b) Rs 60,000
(c) Rs 75,000
(d) Cannot be determined

Q.33. If the younger son wants to sell all the copper he has, who is capable of buying it by paying cash?
(a) Elder son (b) Elder daughter
(c) Younger son (d) All the three

Q.34. Two items having the same cost price and marked price are given two different discounts, which differ by 20 percentage points. If their profits differ by 32 percentage points, what is the ratio of their cost price to the marked price?
(a) 5:8 (b) 4:7
(c) 2:3 (d) Cannot be determined

Q.35. A, B and C run around a circular track of the length 300 m. A and B run with speeds of 10 m/s and 12 m/s in the same direction respectively and C runs in the opposite direction with a speed of 15 m/s. If all the three start at the same time, which of the following is true?
(a) For a given period of time, the ratio of the number of meeting points of A and C to the number of meeting points of B and C is more than 1.
(b) For a given period of time, the ratio of the number of meeting points of B and C to the number of meeting points of A and C is more than 1.
(c) For a given period of time, the ratio of the number of meeting points of A and C to the number of meeting points of B and C is equal to 1.
(d) None of the above can be concluded.

Direction for questions 36 to 38: Read the passage below and solve the questions based on it.

In a class, every student plays exactly one of the following games—squash, table tennis, hockey, handball, cricket. Exactly 1/12th of the girls and 1/8th of the boys play squash, 1/15th of the girls and 1/6th of the boys play hockey, 1/4th of the girls and 1/12th of the boys play hand-ball, 1/5th of the girls and 3/8th of the boys plays table tennis and the remaining boys and girls play cricket. The total strength of the class is less than 120.

Q.36. How many girls are there in the class?
(a) 60 (b) 48
(c) 90 (d) Cannot be determined

Q.37. Which of the following cannot be the ratio of boys and girls in the class?
(a) 2:5 (b) 4:5
(c) 3:4 (d) Cannot be determined

Q.38. What is the total strength of the class?
(a) 84 (b) 108
(c) 132 (d) Either (1) or (2)

Direction for questions 39 and 40: Read the passage below and solve the questions based on it.

There are four friends Sharat, Chandra, Mayank and Sid living in the same village of Bhagalpur. Once Sharat had nothing to eat, so, Chandra, Mayank and Sid collected their rotis and gave a part of their rotis to Sharat. It was later found that the number of rotis given by Chandra is one more than the rotis given by Mayank, and the number of rotis given by Mayank is one more than the rotis given by Sid. Now Sharat was required to pay the cost of rotis in the ratio of the rotis contributed by all three of them. It was found that Chandra should get twice as much money as Mayank. Price of one roti = Rs 5.

(All the values, number of rotis and money are in integers.)

Q.39. How much money did Chandra get?
(a) Rs 8 (b) Rs 12
(c) Rs 16 (d) None of these

Q.40. How many rotis were consumed by all of them?
(a) 6 (b) 9
(c) 12 (d) 15

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Answer	(d)	(a)	(d)	(b)	(b)	(c)	(b)	(d)	(c)	(c)	(b)	(a)	(c)	(a)	(b)	(b)

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(c)	(c)	(c)	(d)	(b)	(d)	(b)	(d)	(a)	(d)	(d)	(d)	(b)	(d)	(d)	(c)	(c)	(d)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(a)	(b)	(b)	(c)	(b)	(d)	(a)	(d)	(b)	(a)	(b)	(d)	(d)	(d)	(d)	(b)	(b)	(d)	(b)	(d)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(c)	(a)	(b)	(a)	(a)	(d)	(c)	(d)	(d)	(d)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(d)	(d)	(b)	(b)	(d)	(d)	(c)	(d)	(b)	(d)	(a)	(d)	(d)	(c)	(b)	(a)	(c)	(b)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(b)	(a)	(a)	(a)	(c)	(b)	(d)	(c)	(d)	(c)	(b)	(c)	(b)	(d)	(d)	(d)	(a)	(d)	(d)	(a)
Q. No.	41	42	43	44	45															
Answer	(a)	(b)	(c)	(d)	(c)															

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(a)	(d)	(c)	(d)	(c)	(b)	(d)	(b)	(b)	(a)	(a)	(b)	(c)	(b)	(a)	(b)	(b)	(a)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(b)	(d)	(b)	(d)	(b)	(d)	(a)	(c)	(b)	(a)	(b)	(d)	(a)	(b)	(a)	(d)	(d)	(d)	(c)

HINTS AND SOLUTIONS

Warm up

1. Required ratio = 1/2 : 1/3 : 1/4
   To find the simple ratio, we will multiply each of these by the LCM of 2, 3 and 4.
   = 12 (1/2 : 1/3 : 1/4) = 6 : 4 : 3
2. Let the nos. be $5x$ and $3x$.
   then, $5x - 3x = 10$
   $2x = 10$, $x = 5$
   Required product = $5x \times 3x = 5 \times 5 \times 3 \times 5 = 375$
3. | No. of oxen | work | time |
   |---|---|---|
   | 30 | 1/7 | 4 |
   | 18 | $\frac{6}{7}$ | x |

   Applying the variation, we get
   $30 \times 4 \times 7 = \frac{18 \times x \times 7}{6}$ $x = 40$ hrs.
4. Let the number be x, $3x$, $4x$ and $7x$.
   then, $x + 3x + 4x + 7x = 75$
   $15x = 75 \Rightarrow x = 5$
   Required value = $7 \times 5 = 35$
5. Required difference = $35 - 5 = 30$
6. Go through options. (c) is the answer.
7. p = pressure, v = volume

   $p = \frac{k}{v}$ k = constant term

   $k = pv$

   p is reduced by 20% i.e., by $\frac{1}{5}$

   then V is increased $\frac{1}{4}$ i.e., 25%
8. $\frac{27-x}{35-x} = \frac{2}{3}$

   $81 - 3x = 70 - 2x$, $x = 11$
9. $p \propto q \Rightarrow p = k\,q$

   $15 = k \times 7$, $k = \frac{15}{7}$

   $P = kq = 14 \times 15/7 = 30$
10. Go through options
11. Go through options
12. $\frac{5p-3q}{7p+2q} = \frac{5.\frac{3}{4}q-3q}{7.\frac{3}{4}q+2q} = \frac{15q-12q}{21q+8q} = 3/29$
13. Volume of milk in mixture = $\frac{7}{10} \times 30 = 21\ l$

    Volume of water = 9 L
    Let x litres of water is added

    $\frac{21}{9+x} = \frac{1}{2}$

    $42 = 9 + x$

    $x = 33$
14. Ratio of their investment = 12000 : 18000 = 2: 3
    A's share in profit = (2/5) × 2000 = Rs 800
15. Suppose B invested for n months

    Hence, $\frac{12 \times x}{n \times x} = \frac{3}{2}$, x = their investment

    $\frac{12}{n} = \frac{3}{2}$, $n = 8$
16. Let B invested for n months

    then, $\frac{12000 \times 12}{16000 \times n} = \frac{3}{1}$

    $n = 3$

Foundation

1. Let x gold coins were given to 1st queen. Then no. of gold coins given to 2nd queen = $\frac{5}{7}x$

   And no. of gold coins given to 3rd queen = $\frac{3}{7}x$

   then, $x - \frac{3}{7}x = 60 \Rightarrow \frac{4x}{7} = 60$

   $x = 105$

   Required no. = $105 + \frac{5}{7} \times 105 + \frac{3}{7} \times 105 = 225$

2. Let their ages 5 years ago be $2x$, $3x$, $7x$ and $8x$.
Their ages now $2x + 5$, $3x + 5$, $7x + 5$, $8x + 5$.
Or, $20x + 20 = 140$
$\Rightarrow 20x = 120 \Rightarrow x = 6$
Present age of Nishu = $2 \times 6 + 5 = 17$ years
Present age of mother = $7 \times 6 + 5 = 47$ years
Hence, required years (47 –17) years = 30 years

3. Let x = total no. of workers

No. of women workers = $\frac{1}{3}x$

No. of men workers = $\frac{2}{3}x$

No. of married women = $\frac{1}{6}x$

No. of women having children = $\frac{1}{3}.\frac{1}{6}x = \frac{1}{18}x$

No. of women without children = $5x/18$

No. of married men = $\frac{3}{4}.\frac{2}{3}x = \frac{1}{2}x$

No. of men having children = $x/3$

No. of men without children = $\left(\frac{2}{3}x - \frac{1}{3}x\right) = \frac{x}{3}$

Required fraction = $\frac{\frac{5x}{18} + x/3}{x} = \frac{\frac{5x+6x}{18}}{4} = \frac{11}{18}$

4. Let salary of Anil = Rs x

Bhuvan's salary $= \frac{2}{5}x$

Chandra's Salary = $\frac{7}{9}.\frac{2}{5}x = \frac{14x}{45}$

$x + \frac{2}{5}x + \frac{14x}{45} = 770$

$\Rightarrow \frac{77x}{45} = 770$

$x = 450$

5. Let his salary be Rs x

Money spent on house rent = $\frac{1}{4}x$

Money spent on food = $\frac{1}{3}x$

Money spent on travel = $\frac{1}{6}x$

Money spent = $\frac{1}{4}x + \frac{1}{3}x + \frac{1}{6}x = \frac{6x+8x+4x}{24}$

$= \frac{18x}{24} = \frac{3}{4}x$

Money left = $\frac{x}{4}$

Money spent on clothes = $\frac{1}{10}.\frac{x}{4} = \frac{x}{40}$

Money left = $\frac{x}{4} - \frac{x}{40} = \frac{10x-x}{40} = \frac{9x}{40}$

i.e., $\frac{9x}{40} = 13,500$

$x = 1500 \times 40 = 60000$

Required amount = 5000

6. Let Rs x be Bidhan's contribution. Then his brother's

contribution = $\frac{3}{2}x$

Financers contribution = $3.\frac{3}{2}x = \frac{9x}{2}$

then, $x + \frac{3}{2}x + \frac{9x}{2} = 35000$

then, $\frac{14x}{2} = 35000 \Rightarrow x = \frac{35000 \times 2}{14} = 5000$

His brother's contribution = Rs 7500

7. Go through options.

8. Let x and y be the mass of 1st alloy and 2nd alloy respectively

Bronze in 1st alloy = $\frac{9}{10}x$

Manganese in 1st alloy = $\frac{7}{100}x$

Tin in 1st alloy = $\frac{3}{100}x$

For manganese, we have

$\frac{\frac{7}{100}x}{x+y} \times 100 = 5 \Rightarrow 7x = 5x + 5y \Rightarrow 2x = 5y.$

Hence, $x/y = 5/2$

Applying alligation,

t = % of bronze in 2nd Alloy

$\frac{y}{x} = \frac{5}{85-t}$

$\frac{2}{5} = \frac{S}{85-t}$, $t = 72.5$

9. Let Rs x was initially with Vineet

Money given to Sachin = $\frac{1}{4}x$

Money spent on movie ticket = $\frac{1}{3}.\frac{3}{4}x = \frac{1}{4}x$

Money left = $\frac{1}{2}x$

Money spent on shopping = $\frac{2}{5}.\frac{1}{2}x = \frac{1}{5}x$

Money given by Gita = $\frac{2}{3}.\frac{3x}{10} = \frac{x}{5}$

Total money he had = $\frac{3x}{10}+\frac{x}{5} = \frac{5x}{10} = \frac{1}{2}x$

Money given to mother = $\frac{4}{5}.\frac{1}{2}x = \frac{2x}{5}$

Money left = $\frac{1}{2}x - \frac{2}{5}x = \frac{x}{10}$

Given that,

$\sqrt{x} = \frac{x}{10}$ $\qquad$ $x = \frac{x^2}{100}$

$100x - x^2 = 0 \Rightarrow x(100 - x) = 0 \Rightarrow x = 100$

Money given by Gita $= \frac{100}{5} =$ Rs 20

10. No. of cogs remains inversely proportional to the number of turns.

No. of turns $= \frac{80}{45}\times\frac{54}{32}\times 8$

$= 24$

12. Let the no. of coins denomination Rs 1, 50 paise and 25 paise be $3x$, $4x$ and $12x$ respectively.

Then, $3x.1 + 4x.\frac{1}{2} + 12x.\frac{1}{4} = 600$

$3x + 2x + 3x = 600 \Rightarrow 8x = 600 \Rightarrow x = 75$

No. of 25 paise coins = $12 \times 75 = 900$

13. Let Anand, Bidhan and Chandra got Rs x, y and z respectively then, Chandra's Share = $\frac{1}{2}(x + y)$ and

Anand's Share = $\frac{1}{3}(y+z)$

We donot know the values of x, y, z, therefore (d)

14. Let Anil invested Rs X, then Dheeraj's investment = Rs $3x$

Let Dheeraj join him for n month

then, $\frac{x\times 12}{3x\times n} = 1 \Rightarrow x \times 12 = 3x \times n \Rightarrow n = 4$

Hence, Dheeraj joined him is September.

15. Since, we don't know the share of either of them, we cannot find the required time.

16. Let salary of C = Rs x = Rs 1500

Salary of B = $\frac{4}{3}x = \frac{4}{3}\times 1500 =$ Rs 2000

Salary of A = $\frac{5}{4}.\frac{4}{3}x = \frac{5}{3}x =$ Rs $\frac{5}{3}\times 1500 = Rs\ 2500$

Salary of M = (Rs (1500 + 2000 + 2500) = Rs 6000

17. Four years ago, let ages of A and B be $13x$ and $9x$ years respectively.

then, $\frac{13x+12}{9x+12} = \frac{4}{3} \Rightarrow 39x + 36 = 36x + 48$

Or, $3x = 12 \Rightarrow x = 4$

Hence, present age of A = $13x + 4 = 13 \times 4 + 4 = 56$

And present age of B = $9 \times 4 + 4 = 40$

18. Let the wages of a man, a woman and a child he $3x$, $2x$ and x respectively,

then, $20 \times 3x + 30 \times 2x + x \times 36 = 780$

$60x + 60x + 36x = 780$

$\Rightarrow 156x = 780 \Rightarrow x = 5$

Required Amount = $2(15 \times 3 \times 5 + 21 \times 2 \times 5 + 30 \times 5)$

$= 2\ (225 + 210 + 150)$, = Rs 1170

19. Let $5x$ and $3x$ be the no. of males and females originally at the party.

Now, $\frac{5x-10}{3x} = 1$, $5x - 10 = 3x$

$\Rightarrow 2x = 10 \Rightarrow x = 5$

No. of people originally at the party = $8x = 40$

20. Area of semicircle = $\frac{1}{2}\pi r^2$

Area of sector = $\frac{\theta}{360}\times\pi r^2$

$\frac{\frac{\theta}{360}}{\frac{1}{2}\pi r^2}\times\pi r^2 = \frac{1}{10}$, $\frac{\theta}{180} = \frac{1}{10} = \theta = 18^\circ$

22. $V \propto W^2$

$V = KW^2$, V = Value of diamond and W = Weight of diamond

K = constant

$10000 = K\ .\ (10)^2 \Rightarrow K = 100$

Hence, $V = 100 \times 4^2 = 1600$

$V = 100.6^2 = 3600$

Net loss = (Rs (10000 – 5200) = Rs 4800 i.e., 48%.

23. Monthly income of A and B be $3x$ and $2x$ respectively.

Monthly Expenditure of A and B be $5y$ and $3y$ respectively.

$3x - 5y = 1000$, $2x - 3y - 1000$

Solving, we get, $y = 1000$, $x = 2000$

Income of A = Rs 6000

24. $\frac{ab}{bc}=\frac{x}{c^2}$ and $x=\frac{ab.c^2}{bc}=ac$

25. Let the fare of 1st, 2nd and 3rd class be $10x$, $8x$, and $3x$ respectively and let the no. of passengers of 1st, 2nd and 3rd class be $3y$, $4y$ and $10y$ respectively.
then $30xy + 32xy + 30xy = 16100 \Rightarrow 92xy = 16100$
Hence $xy = 175$
Required amount $= 32xy = 32 \times 175 = 5600$

26. $$\frac{3A^2+4B}{3A-4B^2}=\frac{3.\left(\frac{3}{4}B\right)^2+4B}{4.\frac{3}{4}B-4B^2}=\frac{\frac{27B^2}{16}+4B}{3B-4B^2}$$

$$=\frac{27B^2+64B}{16(3B-4B^2)}=\frac{27B+64}{16(3-4B)}$$

Further it cannot be solved. Hence (d) is the correct answer
Note: Had there been no square terms, we could have calculated the final value by just putting A = 3 and B = 4.

28. Price of one goat = Rs 70 $\Rightarrow$ Price of 9 goats = Rs 9 × 70 = Rs 630
Price of 14 sheeps = Rs 630 $\Rightarrow$ Price of 1 sheep = Rs $\frac{630}{14}$ = Rs 45
Price of 50 sheeps = Rs 45 × 50 = Rs 2250
Price of 10 oxen = Rs 2250
Price of 1 oxen = Rs 225 $\Rightarrow$ Price of 8 oxen = Rs 1800
Price of 6 dogs = Rs 1800 $\Rightarrow$ Price of 1 dog = 300
Price of 5 dogs = Rs 1500 $\Rightarrow$ Price of two horses = Rs 1500
Price of one horse = Rs 750

29. Since in 30 minutes pond is $\frac{1}{3}$ full, in next 1 minute it will triple i.e., the pond well be full. Hence (b) is the correct answer

31. $A \alpha B$
$B \alpha (1/C)$
$C \alpha D^2$
$D \alpha E^{1/3}$
When A increase
$\Rightarrow$ B also increase $\Rightarrow$ C decreases.
Decrease in C results in decrease in D and decrease in D results decrease in E.

32. Using the formula $\frac{C}{5}=\frac{F-32}{9}$
We well get [– 40] as answer

33. Let the 1st no. be $2x$ and 2nd no = $3x$.

3rd no. $=\frac{18}{5}x$
4th no. = $81x/20$

So, $x+3x+\frac{18}{5}x+\frac{81x}{20}=233$

$$\frac{20x+60x+72x+81x}{20}=233$$

$$x=\frac{233\times 20}{233} \Rightarrow x=20$$

2nd no. $= 3x = 3 \times 20 = 60$

3rd no. $=\frac{18.20}{2}=72$

Average $=\frac{60+72}{2}=\frac{132}{2}=66$

34. After one year
Population of port A = 8000
Population of port B = 6000
Population of Port C = 4000
After 2nd year
Population of A = 5000
Population of B = 6000
Population of C = 7000
After 3rd year
Population of A = 6500
Population of B = 6000
Population of C = 5500
After 4th year
Population of A = 5750

35. Let x = no. of Re 1 notes and y = no. of Rs 10 notes
So, $xy = x + 10y \Rightarrow xy = x + 10y$
Now, go through options

36. At first let Nandan buys x black currents, then no. of pastries bought by Divya = $x + 9$
Now, let Divya bought y butter scotches, then no. of butter scotches bought by Nandan = $y + 18$

$$\frac{x+y+9}{x+y+18}=\frac{12}{21} \Rightarrow x+y=3$$

x and y cannot be zeroes.
When $x = 1, y = 2$ …… A
When $x = 2, y = 1$ …… B
Let the price of black current be Rs A/unit and price of butter scotch be Rs B/unit.
For Case (A)
$10a + 2b = a + 20b \Rightarrow 9a = 18b \Rightarrow a : b = 2 : 1$

37. Go through options.

38. Ratio of their investments for 1 year
$= 4x \times 3 + 2x \times 3 + x \times 3 + \frac{x}{2} \times 3 : x \times 3 + 2x \times 3 +$
$4x \times 3 + 8x \times 3 : 15x \times 12$

$$=\frac{45x}{2}:45x:180x = 45x:90x:360x = 1:2:4$$

B's share in profit $=\frac{2}{7}$ of profit

Let y be profit, then $\frac{2}{7}y = 22000$, $y = 77000$

42. $y=\frac{r}{x}$, $z=\frac{r^2}{x^2}$

Then, $x + y + z = 13$

$$x + \frac{r}{x} + \frac{r^2}{x^2} = 13$$

$$x^2 + xr + r^2 = 13x^2 \quad \ldots(1)$$

And, $x^2 + \frac{r^2}{x^2} + \frac{r^4}{x^4} = 91 \quad \ldots(2)$

With the help of (1) and (2), we get (a) as answer.

43. Let 1 black cow gives x litres milk in 1 day and 1 white cow gives y litres milk in 1 day. Then
$5(4x + 3y) = 4(3x + 5y) \Rightarrow 20x + 15y = 12x + 20y$
$\Rightarrow 8x = 5y$. Hence, $\frac{x}{y} = \frac{5}{8}$

44. Let the cost of type I terminal be Rs x per week and cost of type II terminal be Rs y per week.
$5 \times x \times 7 + 22 \times 4 \times y + 10 \times 6x + 8 \times 6x = 74750$ ...(1)
Also, given $\frac{x}{y} = \frac{2}{3} \quad \ldots(2)$
Using (2) and (1), we get option (a) as answer.

45. S.P. of alloy = Rs 90
Profit = 20%= $\frac{1}{5}$
C.P. = Rs $\left(90 - \frac{1}{6}.90\right)$ = Rs 75
Prices of foils are in ratio 1 : 4.
Hence, price of cheaper quality = $\frac{1}{5} \times 75$ = Rs 15
and price of dearer quality = $\frac{4}{5} \times 75$ = Rs 60

46. Go through options

47. Let the no. of chapters be $6x$, $5x$, $10x$ and $14x$.
Also, $14x = 70$, $x = 5$
Thus, chapters are 30, 25, 50 and 70.
We can easily eliminate (a) as no. of pages has to be in between 4000 and 5000.
For (b) Each parts well have 1200 pages which is not possible for part having 70 chapters. Only option (c) satisfies the conditions.

48. Total no. of chapters = $(6x + 5x + 10x + 14x) = 35x = 35 \times 5 = 175$

49. Required ratio = 15/16

50. Let their present ages be $2x$ and $3x$.
Given that
$$\frac{2x + N}{3x + N} = \frac{3}{5}$$
$10x + 5N = 9x + 3N$
$X = -2N$ which 18 not possible.
Hence, (d)

Moderate

	Brother	*Sister*
Mr Nishant's son	X	5X
Mr Rajesh' son	Y	Y
Mr Nishant's daughter	2Z	Z
Mr Rajesh' daughter	2A	A

Each son of Mr Rajesh has as many sisters as the number of brothers and each daughter of Mr Rajesh has twice the number of brothers as the number of sisters. It is possible only if the number of sons = 4 and number of daughters = 3. However, nothing concrete can be said about the children of Mr Nishant.

Number of men : number of women = Number of women : number of children

10. Only the second option is satisfying the values.

11. Since this ratio is less than 1, the number of men should be less than the number of females visited. So, number of males should be less than 7. So, the answer is option d. None of these.

12. If P is taking 3 days to do some work, then Q takes 6 days to do the same work. Now, both of them will take 2 days to complete the work. So, R takes 6 days to complete the same work.
Hence, earning should be distributed in the ratio of their efficiency, i.e., 2 : 1 : 1.

Questions 13 to 15

Assume that Ten, Dul and Kar are getting T, D and K sweets respectively.
2 T = 2/3 D = 4 K = 4P (assume)
So, T = 2 P, D = 6 P and K = P
So, T : D : K = 2:6:1

13. Kar would get – 117(4/9). So, the answer is option (d). None of these.

15. x bulbs consume $1/x$ units everyday.
So, 1 bulb will consume $1/x^2$ units of electricity everyday
Y bulbs consume y/x^2 units everyday
Now, y/x^2 units is consume in 1 day.
Hence, y units will be consumed in x^2 days.

16. Number of apples/Number of guavas = Number of guavas/Number of oranges
(Number of guavas)^2 = Number of apples × Number of oranges
This gives us a hint that Number of apples × Number of oranges should be a prefect square.
Now start using the options.
Option (b) satisfies the relationship (16/20 = 20/25)

17. We have to check all the possibilities in this question. If we take the ratio to be equal Number of apples/Number of guavas = Number of guavas/Number of oranges = 1, then Number of apples = Number of guavas = Number of oranges. So, one possible number of total fruits = 63
In 16 and 17, if we start using the concept of GP, then it will be difficult to find the number of fruits by GP as this questions contains a lot many variables.

19. Village-chief contribution = Rs 1.2 lakhs, so villagers' contribution = Rs 1.2 lakhs
Now, Go through the options.

28. Go through the options.

29.

No. of people	6	6	8	16
Event	Museum	Fair	Exhibition	Movie
Rate	5	4	2	25

36. Using options is the best way of solving this question.

37. Best way of solving this question is—Go through the options.

38. Its all about taking different values and checking which option cannot satisfy the money with Pankaj.

41. No. of deers, bears and foxes = $3x:7x:5x$
Difference between the no. of deers and bears = $4x$
Now, $4x$ has to be a multiple of 21. So, x has to be a multiple of 21.
So, Total number of animals = $15x = 15 \times 21 = 315$

42. Weight of two pieces of Watermelon
$= 3 \times 12y : 5 \times 12y$
(We are assuming the weight to be the multiple of 12 because the bigger of two pieces has to be cut in the ratio 5 : 7.)
Now $5 \times 12y$ has to be cut in the ratio 5 : 7. So, the two parts will be $25y$ and $35y$.
So, finally the ratio is 36:25:35.

43. Assume that Y produces 4P units in unit time. Then X and Z produce P and 2P units in the same time. So, Total shoes produced = 7P, Fraction of the total number of shoes produced by machine Z = 2/7

48. Use options.

49. $$\frac{a}{a+b} = \frac{b}{c+a} = \frac{c}{a+b} = \frac{a+b+c}{2(a+b+c)}$$

Now if $a + b + c = 0$, then $a + b = -c$

So, $\frac{a+b+c}{2(a+b+c)} = -1$

And if $a + b + c \neq 0$, then $\frac{a+b+c}{2(a+b+c)} = \frac{1}{2}$

Advanced

1. Time a (√Number of passengers) × Distance
Expressing this in symbols, T α (√N) × D
Or, T1 / T2 = (√N1/N2) × D1/D2
Or, 2 / 1 = (√N1/64) × 80/30

2. Tax = Fixed + Variable
Since Mr Kalyan pays Rs 6200 and Mrs Kalyan pays Rs 7700, so it can be inferred that for Rs 10,000 (Difference between their salaries), one has to pay Rs 1500 as the variable tax. So, Fixed component of tax = Rs 5200. The point worth to be noted here is the fact that tax is to be paid over an above Rs 50,000. So, the variable tax percentage above Rs 50,000 = 10%. Now the answer can be calculated using options.

Questions 6 and 7

Generate the equations first and then use options for faster calculations.

9. Total profit = Rs 24 lakh per trip + Rs 7P^2, where P is the number of coaches.
Now go through the options.

10. When 10 kg is kept in first pan and 5 kg is kept in second pan, it remains in balance – It means that 1st pan is heavier than the second pan (or, more difficult to pull down), so the ratio of weight of pans = 2 : 1.

11. Do as given in the solution of 10.

12.

	Gold	*Copper*
First Alloy	2	1
Second Alloy	2	7
Finally	1	2

It is possible if we mix first alloy and second alloy in the ratio 1 : 3. So, the second alloy will be mixed two times more than the first alloy.

13. Weight α Height and Height α Age.

19. Go through the options.

20. Assume the distance = 9 units. So, the ratio of the time taken = 9 : 3 : 1
So, the answer = 9/1 : 3/3 : 1/9

22. Population state 1, state 2, state 3, state 4, state 5 = x:3x:2x:3x:2x. The ratio of the percentage of women in the population = y: 3y :2y: 5y :4y.
Now use the above values to establish the relationship between xy and the total population of women = 3.7 million.

23. It is possible if new length of the pendulum become 1.21 times of the old length of the pendulum.

Questions 26 to 29

Using the options given in question 26, we can see that it is possible to have Rs 500 on 19th only if he starts with Rs 80 on 13th of November.

Date	*Money*
13th	Rs 80
14th	Rs 70
15th	Rs 140
16th	Rs 130
17th	Rs 260
18th	Rs 250
19th	Rs 500
20th	Rs 490
21st	Rs 980
22nd	Rs 970
23rd	Rs 1940
24th	Rs 1930
25th	Rs 3860

27. It is Rs 970 on 22nd.

28. It is obvious from the table given above.

29. The new table would have been like this:

Date	*Money*
13th	Rs 100
14th	Rs 90
15th	Rs 180
16th	Rs 170
17th	Rs 340
18th	Rs 330
19th	Rs 660
20th	Rs 650
21st	Rs 1300
22nd	Rs 1290
23rd	Rs 2580
24th	Rs 2570
25th	Rs 5140

Now it can be calculated.

35. Total number of distinct meeting points for B and C = 9 and Total number of distinct meeting points for A and C = 5. Now it can be checked using the options.

Questions 36 to 38

Whatever is the total number of boys, it should be divisible by 8, 6, 12 and total number of girls should be divisible by 12, 15 and 4. Now use options.

7

CHAPTER

Time and Work



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Time and work equivalence
- Relationship between time and efficiency
- Different types of work
- Kinds of questions asked at the CAT
- Methods of solving questions

Introduction to the topic

A look at the CAT papers of the last 15 years shows that time and work is an important topic for this exam. There have been questions from this topic almost every year. The number of questions, however, have varied between 1 and 3. A basic understanding of the concept of time and work will enable you to solve problems from this topic with considerable ease.

TIME AND WORK

There is a definite relationship between the amount of time taken, the number of persons doing the work and the unit of work done. Under this topic we study the phenomenon of accomplishment of a work with relation to the time. There are certain formulae that demonstrate the above-mentioned relationship and exhibit how each of these factors influence the other two.

TIME–WORK EQUIVALENCE

The essence of time-work equivalence lies in the fact that it exhibits the most fundamental relationship between the three factors as mentioned above viz., work, time and the agent which is completing the work. That is,

Work done = Number of days × Number of men

$W = M \times D$

This gives us an important concept of man-days.

Suppose there are 20 persons working for 10 days to complete a job, then the total work done is equal to 200 man-days. Now, if we change the number of days in which the work is to be completed, then the other factor i.e., the number of persons will change accordingly, so that, the product of the factors becomes equal to 200 man-days. Product-stability-ratio (Chapter 3) is a very effective tool to calculate this.

Example 1 Seven persons can clean 7 floors by 7 mops in 7 days. In how many days can 5 persons clean 5 floors by 5 mops?

Solution This problem can be solved through several methods.

Method 1 To clean 7 floors, we need to have $7 \times 7 = 49$ man-days.

So, to clean 5 floors, we need to have 35 man-days.

So, $35 = D \times 5$. So, $D = 7$ days

Method 2 Using ratio proportion, less work and less men are involved here.

So, the number of days $= 7 \times \frac{5}{7} \times \frac{7}{5} = 7$ days

Method 3 Let us try to have a mental image of this situation: There is a building with seven floors namely $F_1, F_2, \ldots, F_7$ and seven persons $P_1, P_2, \ldots P_7$ are cleaning this building in such a way that one floor is being cleaned by each one of them. Since it takes 7 days to complete the whole work, it can be inferred that everybody is taking 7 days to clean his respective floor. So, if there are just floors and five persons are cleaning these five floors, then it will take them seven days (assuming that the top two floors have been demolished.)

Now, depending upon different situations, three conditions are possible, in the relationship W = MD (where W = quantity of work, M = number of persons, D = number of days)

Condition 1: W is constant

$M \times D = \text{Constant}$

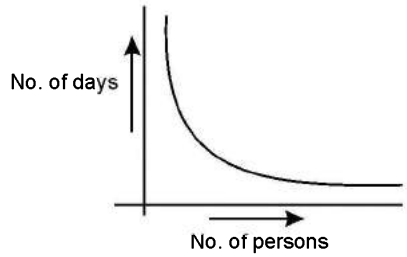
$M \propto 1/D$

It can be observed that if the work done is constant, then the number of persons is inversely proportional to the number of days, which means that the multipliers of M and D will be reciprocal. Extending this situation, if 10 persons can do a work in 20 days, then 5 persons can do the same work in 40 days, or, 20 persons will do the same work in 10 days. Further it can be summarized as

W	=	M	×	D
200	=	10	×	20
200	=	$5(10 \times 1/2)$	×	$40(20 \times 2)$
200	=	$20(10 \times 2)$	×	$10(20 \times 1/2)$

Thus, it can be said that the multiplier of M and multiplier of D are reciprocal to each other.

It can be seen with the help of the graph given below



Example 2 Yadavjee contractor undertakes to get a work done in 50 days by 50 labourers. After 40 days, he realizes that only 50% of the work is done. How many more men should be employed so that the work is complete on time?

Solution

W	=	M	×	D	
50%		50		40	...(i)
Rest 50%		(50 + M)		10	...(ii)

Since work is constant in both the cases, so, the number of men and the number of days will be reciprocal to each other. As the number of days left in (ii) is 1/4th of initial period (i), so the number of persons will become 4 times of the initial number of persons.

Hence, the number of persons = 50 × 4 = 200. So, M = 150 men

Example 3 In the above question, if the schedule can go behind by 10 days, then how many extra men are required to complete the work?

Solution So, now we will have to complete the work in 60 days.

W	=	M	×	D	
50%		50		40	..... (i)
Rest 50%		(50 + M)	20		(ii)

Since work is constant in both the cases, so the number of men and the number of days will be reciprocal to each other. As, the number of days left in (ii) is 1/2 of initial period (i), so the number of persons will become 2 times of the initial number of persons.

Hence, the number of persons = 50 × 2 = 100. So, M = 50 men

Example 4 In question 2, he realizes after 40 days that work is only 20% complete. How many extra men should be employed now so that the work is completed on time?

Solution

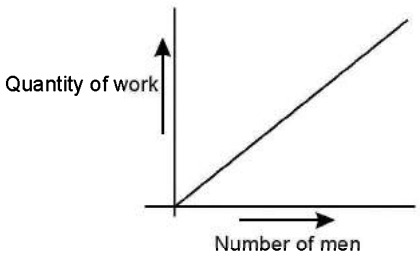
W	=	M	×	D	
20%		50		40	.....(i)
20%		200		10	.....(ii)
So, 80%		800		10	.....(iii)

Hence, 750 more persons are needed to complete the job on time.

Condition 2: D is constant

$W \alpha M$

More work will be done if we employ more men and vice-versa. It means that multiplier of W and M will be same. It can be seen with the help of the graph given below



Condition 3: M is constant

$W \alpha D$

In general, we can summarize that

$$\frac{M_1D_1}{W_1} = \frac{M_2D_2}{W_2}$$

Example 5 12 persons can cut 10 trees in 16 days. In how many days can 8 persons cut 12 trees?

Solution Here $W_1 = 10\ W_2 = 12$
$M_1 = 12\ M_2 = 8$
$D_1 = 16\ D_2 = ?$

Putting the values in the equation $W_1/W_2 = (M_1/M_2) \times (D_1/D_2)$

We get $\frac{10}{12} = \frac{12}{8} \times \frac{16}{D_2} \Rightarrow D_2 = 28.8$ days

INDIVIDUAL WORK AND INDIVIDUAL EFFICIENCY

Individual Work

If Amit can do a certain work in 10 days, then he will finish 1/10th of the work in one day.

Example 6 Amit can do some work in 12 days and Vinit can do the same work in 15 days. In how many days will both of them do the work when working together?

Solution Assume total work = 1 unit

Work done by Amit in one day = 1/12 unit

Work done by Vinit in one day = 1/15 unit

Work done by both of them in one day when working together = (1/12) + (1/15) = 9/60 unit

Hence, they will be doing the whole work in 60/9 days $= 6\frac{2}{3}$ days

LCM method of solving time and work questions

This can be understood in terms of the above written example in the following way:

Let us assume total work to be equal to the LCM of the days taken by Amit and Vinit (i.e., of 10 and 15)

Assume work = 60 units

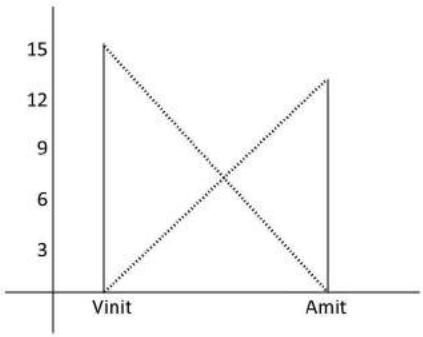
Work done by Amit in one day = 5 units

Work done by Vinit in one day = 4 units

Work done by both of them in one day when working together = 9 units

So, the number of days taken by both of them when working together = 60/9 = $6\frac{2}{3}$ days

Alternatively, we can do this problem with the help of a graph also.



We have created the pillars of the number of days of Amit and Vinit. Now the point of intersection of the top of the first pillar to the bottom of the second pillar and the top of second pillar and the bottom of the first pillar is the number of days taken by both of them when working together. In the above drawn graph, it can be clearly seen that the point of intersection of both the straight lines is a bit more than 6.

Despite graphical method appearing easier than the earlier two methods, the usage of this method should be avoided due to the complexity of denoting the points on the graph paper.

Example 7 A, B and C can do a piece of work individually in 8, 12 and 15 days respectively. A and B start working, but A quits after working for 2 days. After this, C joins B till the completion of work. In how many days will the work be complete?

Solution Let us assume that the Work = LCM (8, 12, 15) = 120 units

So, the work done by A in one day = 15 units

Work done by B in one day = 10 units

Work done by C in one day = 8 units

Work done by A and B in two days = 2 × 25 = 50 units
Remaining work = 70 units
Work done by C and B in one day = 18 units

Time taken to complete the remaining work by C and B = $\frac{70}{18} = 3\frac{16}{18}$ days

So, the total number of days = $5\frac{16}{18}$ days

Example 8 A and B together can do a work in 12 days, B and C together can do the same work in 10 days and A and C together can do the same work in 8 days. In how many days will the work be complete if A, B and C are working together?

Solution Let us assume work = LCM of (12, 10, 8) = 120 units

So, A and B are doing 10 units in one day, B and C are doing 12 units in a day and A and C are doing 15 units in a day.

Adding all these, 2 (A + B + C) are doing 37 units in a day.

$\Rightarrow$ (A + B + C) are doing $\frac{37}{2}$ = 18.5 units in a day

So, time taken to complete the work = $\frac{120}{18.5}$ days = 6.48 days

Individual Efficiency

Efficiency is also known as work-rate.

If A is taking less number of days with respect to B to complete the same work, we can say that the efficiency of A is more than the efficiency of B.

So, more the efficiency, less will be the number of days and less the efficiency, more will be the number of days to do a certain work. We have observed individual efficiency in case of percentage also (Product Stability ratio).

Now, assume A takes 20 days to complete a work and B takes 25 days to complete the same work. It means A is doing 5% (100%/20) work in one day and B is doing 4% (100%/25) work in a day. So, efficiency of A is 25% more than efficiency of B.

General expression correlating time taken and efficiency

If efficiency of A is x per cent more than the efficiency of B and B takes 'B' days to complete the work, then A will take $\left(\frac{B}{100+x}\times 100\right)$ days to complete the same work.

If efficiency of A is x% less than the efficiency of B and B takes 'B' days to complete the work, then A will take $\left(\frac{B}{100-x}\times 100\right)$ days to complete the same work.

So, if A is 20% more efficient than B and B takes 'B' days to complete the work, then A will take $\frac{B}{1.2}$ days to do the same work.

With this, it can also be observed that if work is constant then time taken is inversely proportional to efficiency.

Example 9 John is thrice as efficient as Abraham and hence completes a work in 60 days less than the number of days taken by Abraham. What will be the number of days taken by both of them when working together?

Solution Since John is thrice as efficient as Abraham, so the number of days taken by him will be 1/3rd the number of days taken by Abraham. If John is taking x days, then Abraham will take $3x$ days to complete the same work.

Now, $3x - x = 2x = 60$ days

So, $x = 30$ days and $3x = 90$ days

Let us assume that the total work = 90 units (LCM of 30 and 90)

So, the total work done by both of them in one day = 3 + 1 = 4 units of work.

So, the total number of days = $\frac{90}{4}$ days = 22.5 days

Example 10 A can do a work in 4 days. Efficiency of B is half the efficiency of A, efficiency of C is half the efficiency of B and efficiency of D is half the efficiency of C. After they have been grouped in two pairs it is found that the total number of days taken by one group is 2/3rd the time taken by the other group. Which of the following is a possible group? (CAT 2001)
(a) AB (b) BC (c) CD (d) AC

Solution Number of days taken by A = 4 days
Number of days taken by B = 8 days
Number of days taken by C = 16 days
Number of days taken by D = 32 days
Assume that the total work = 32 units
So, the work done by A = 8 units
Work done by B = 4 units
Work done by C = 2 units
Work done by D = 1 units

It can be observed that the work done by B and C together in one day is 2/3rd the work done by A and D. So the groups are—AD and BC.

Example 11 In a nuts and bolts factory, one machine produces only nuts at the rate of 100 nuts per minute and needs to be cleaned for 5 min after production of every 1,000 nuts. Another machine produces only bolts at the rate of 75 bolts per minute and needs to be cleaned for 10 min after production of every 1,500 bolts. If both the machines start their production at the same time, what is the minimum duration required for producing 9,000 pairs of nuts and bolts?
(a) 130 min (b) 135 min (c) 170 min (d) 180 min

Solution

Machine I
Number of nuts produced in one min = 100
Time required to produce 1,000 nuts = 10 min
Cleaning time for nuts = 5 min
Overall time to produce 1,000 nuts = 15 min.

Over all time to produce 9,000 = 138 min − 5 min = 133 min ... (1)

Machine II

Time required to produce 75 bolts = 1 min

Time required to produce 1,500 bolts = 20 min

Cleaning time for bolts = 10 in

Effective time to produce 1,500 bolts = 30 min

Effective time to produce 9,000 bolts = 30 × 6 − 10 = 170 min ... (2)

From (1) and (2)

Minimum time = 170 min

Example 12 A, B and C are assigned a piece of work which they can complete by working together in 15 days. Their efficiencies (measured in terms of rate of doing work) are in the ratio of 1:2:3. After $\frac{1}{3}$rd of the work is completed, one of them has to be withdrawn due to budget constraint. Their wages per day are in the ratio of 3:5:6. The number of days in which the remaining two persons can complete the work (at optimal cost) is

(a) 18 (b) 20 (c) 15 (d) 12

Solution A, B and C together in 15 days

= A alone in 90 days, B alone in 45 days, C alone in 30 days

Wages per day per unit work for A, B and C are $\frac{3}{1}:\frac{5}{2}:\frac{6}{3}$

Hence, A is the least efficient and hence, must be done away with.

For B and C, the whole work can be finished in 18 days and hence, remaining 2/3rd of the work can be finished in 12 days only.

COLLECTIVE WORK AND COLLECTIVE EFFICIENCY

When people of different efficiencies start working together, the method of time-work equivalence to find the time or the amount of work done cannot be used. In those cases, we will be required to relate time with the efficiency of a group.

For example, if 5 men and 8 women can do a piece of work in 10 days, from this information we can not find out that in how many days a man or a woman can do the work individually, since we are not aware of their individual efficiencies. However, if we get a similar equation like x men and y women can do the same work in 'p' days, then we can correlate these two equations to find the number of days taken by one man and one woman to do the work.

Example 13 10 men and 9 women can do a piece of work in 20 days. The same work can be done by 6 men and 12 women in 30 days. In how many days can the same work be done by 1 man and 1 woman?

Solution 10 men and 9 women can do a work in 20 days ⇒ 20 × (10 men and 9 women) can do the same work in 1 day.

Similarly, 6 men and 9 women can do the same work in 30 days ⇒ 30 × (6 men and 12 women) can do the same work in 1 day.

So, efficiency of 20 × (10 men and 9 women) = efficiency of 30 × (6 men and 12 women)

Or, 200 men + 180 women = 180 men + 360 women

Or, 20 men = 180 women ⇒ 1 man = 9 women

So, the total work = 20 × (10 men and 9 women) = 20 × (90 + 9) women = 99 × 20 = 1980 units

Total number of persons employed = 1 man + 1 women = 10 women

So, the time taken = $\frac{1980}{10}$ = 198 days

EXTENSION OF THE CONCEPT OF TIME AND WORK

1. Pipes and cisterns

Pipes and cisterns is just another application of the concept of time and work. While we see only +ve work being done in normal cases of time and work, in case of pipes and cisterns, −ve work is also possible.

Given that pipes A and B can fill a tank in 20 min and 25 min working individually ⇒ this statement is similar to "A can do a work in 20 min and B can do the same work in 25 min."

Again, given that pipe C can empty a tank in 40 min we can say this statement is similar to "C can demolish a wall in 40 min (assuming that the work is building or demolishing the wall)

Let us understand this with the help of an example.

Example 14 A and B are two taps which can fill a tank individually in 10 min and 20 min respectively. However, there is a leakage at the bottom, which can empty a filled tank in 40 min. If the tank is empty initially, how much time will both the taps take to fill the tank (leakage is still there)?

Solution Let us assume the units of work = LCM of (10, 20, 40) = 40 units

Work done by Tap A/min = 4 units/min (Positive work)

Work done by Tap B/min = 2 units/min (Positive work)

Work done by leakage/min = 1 unit/min (Negative work)

Net work done/min = 5 units/min

Hence time taken = 8 min

Example 15 Pipe A can fill a tank in 3 hour. But there is a leakage also, due to which it takes 3.5 hour for the tank to be filled. How much time will the leakage take in emptying the tank if the tank is filled initially?

Solution Assume the total units of work = 10.5 units

Work done by Tap A/h = 3.5 units/h (Positive work)

Work done by leakage/h = 3 units/h (Negative work)

Net work done/h = 0.5 units/h

So, the time taken = $\frac{10.5}{0.5}$ = 21 hour

Alternatively, due to the leakage, the pipe is required to work for an extra half an hour. So, the quantity filled by pipe in half an hour is being emptied by the leakage in 3.5 hour. Hence, the quantity filled by pipe in 3 hour will be emptied by the leakage in 21 hour.

2. Variable work

The concept of variable work comes from the possibility

⇒ that the rate of working can be different or

⇒ can be dependent upon some external agent.

In these cases, the rate of work will be proportional to some external factor.

Understand this with the help of a simple statement: The rate of the flow of water from a pipe is directly proportional to the area of the cross section of the pipe.

Example 16 There are three inlet taps whose diameters are 1 cm, 2 cm and 3 cm respectively. The rate of flow of the water is directly proportional to the square of the diameter. It takes 9 min for the smallest pipe to fill an empty tank. Find the time taken to fill an empty tank when all the three taps are opened.

Solution The rate of flow of a diameter^2, or, rate of flow = K × diameter^2 (where K is a constant)

For 1st tap, rate of flow = K × 1

For 2nd tap, rate of flow = K × 4

For 3rd tap, rate of flow = K × 9

We know, the quantity filled will be equal to the product of the rate of flow and time.

So, the quantity filled by the smallest pipe = K × 1 × 9 = 9K = Capacity of tank

Quantity of water filled by all the taps together in 1 min = 9K + 4K + 1K = 14K

Assume that all the taps working together take 't' min.

So, 14 K × t = 9 K

So, the time taken $t = \frac{9K}{14K} = \frac{9}{14}$ min

3. Alternate work

The concept of alternate work is analogous to the concept of man-days. As we have seen in the concept of man-hour that if 20 men can do a work in 10 days, then this work is equivalent to 200 man-days. However, in the case of alternate work, two or more than two people of different efficiencies work alternately or in some particular pattern.

Example 17 Navneet can build a wall in 30 days and Rakesh can demolish the same wall in 40 days. If they work on alternate days with Navneet starting the job on the 1st days, then in how many days will the wall be built for the first time?

Solution Let us assume the total units of work = 120 units

So, the wall built by Navneet in one day = 4 units

And wall demolished by Rakesh in one day = 3 units

So, effectively in two days, total wall built = 1 unit

Now, they work on alternate days, so days taken to built 116 units = 116 days

On 117th day Navneet will add another 4 units and so completing the construction of wall in 117 days.

(This problem can be understood well with another very traditional problem—A frog climbs up a pole 4 inches in 1 hour and slips 3 inches next hour. If height of the pole is 120 inches, then what is the time taken by the frog to reach the top of the pole?)

PRACTICE EXERCISES

WARM UP

Q.1. Anil can do a piece of work in 18 days and Nishit can do the same work in 12 days (Nishit is more efficient than Anil). In how many days both of them will complete the work when working together?

(a) 7.2 days (b) 6 days

(c) 5.2 days (d) None of these

Q.2. A can do a piece of work in 'a' days and B can do the same piece of work in 'b' days, then the number of days in which both of them will complete the work working together is N. Which of the following can never be true?

(a) $2N$ is the Arithmetic Mean of a and b

(b) $2N$ is the Geometric Mean of a and b

(c) $2N$ is the Harmonic Mean of a and b

(d) None of these

Q.3. In the context of the above question, if $a \neq b$, then which of the following is true?

(a) $2N$ is the Arithmetic Mean of a and b

(b) $2N$ is the Geometric Mean of a and b

(c) $2N$ is the Harmonic Mean of a and b

(d) None of these

Q.4. Aftab can do a piece of work in 12 days and Ravi can do the same work in 18 days. If Aftab works for three days and then quits, in how many days will the remaining work be completed by B alone?
(a) 13½ (b) 9 (c) 16½ (d) 12

Q.5. Efficiency of Ravi and Mayank in doing the same work is 3:2. If together they can complete a work in 18 days, then in how many days Ravi can complete the work alone?
(a) 45 (b) 30 (c) 24 (d) 40

Q.6. A tap can fill an empty tank in 12 hours and a leakage can empty half the tank in 10 hours. If the tap and the leakage are working simultaneously, how long would it take for the empty tank to be filled to half its capacity?
(a) 60 hour (b) 30 hour
(c) 15 hour (d) 12 hour

Q.7. Pipe A can fill an empty tank in eight hours and pipe B can empty the full tank in four more hours. If both the pipes are opened simultaneously, then how long would it take to fill an empty tank?
(a) 12 hours (b) 24 hours
(c) 18 hours (d) 30 hours

Q.8. Sanjay can wash a truck in 45 min and he works for six hours a day. How many trucks can he wash in three days?
(a) 24 (b) 12 (c) 21 (d) 15

Q.9. Sumit Niyogi can build a wall measuring 20 ft × 3 ft × 15 ft in 6 days. Vartika Dikshit can build a wall measuring 30 ft × 4 ft × 10 ft in 5 days. What is the ratio of the work done by Sumit Niyogi and Vartika Dikshit in one day?
(a) 5:8 (b) 6:5 (c) 6:7 (d) 7:5

Q.10. Nandan is twice as good a workman as Divya and together they complete the work in 15 days. In how many days can Divya alone complete the work?
(a) 45 days (b) 40 days
(c) 35 days (d) None of these

Q.11. N men can do a piece of work in 8 days. The same work can be done by (N + 4) men in 6 days. Find the value of N.
(a) 10 (b) 12
(c) 14 (d) None of these

Q.12. Harveen can do a piece of work in 18 days. He worked for 12 days and left. Deepak finished the remaining work in 8 days. In how many days can Deepak alone complete the work?
(a) 21 (b) 22 (c) 23 (d) 24

Q.13. Munchun can do a work in 20 days and Chunmun can do it in 10 days. Both worked together and were paid Rs 1,800. The share of Chunmun is:
(a) Rs 1,200 (b) Rs 1,500
(c) Rs 900 (d) None of these

Q.14. Shahid Bhai can do 2/3rd of the work in 18 days. In how many days can he finish 1/4th of the work?
(a) 6½ (b) 6.75 (c) 6 (d) 13.5

Q.15. A can do half as much work as B in one day. B alone can do a certain work in 12 days. In how many days can A and B together finish that work?
(a) 6 days (b) 7 days
(c) 8 days (d) None of these

Q.16. If 35 persons can do a piece of work in 6 days, in how many days can 15 persons do it?
(a) 12 (b) 14
(c) 16 (d) 18

Q.17. A and B can complete a work in 15 days. A is 50% more efficient than B. How long would A take to complete the work alone?
(a) 20 days (b) 22 days
(c) 24 days (d) None of these

Q.18. Forty persons can finish a work in 60 days. If 40 persons start the work and at the end of every 10 days 40 more persons of the same efficiency join them, then how long will it take to complete the work?
(a) 30 days (b) 20 days
(c) 40 days (d) None of these

Q.19. A group of 18 men and 12 women can do a work in 18 days. A woman takes twice as much time as a man to do the work. How many days will 8 men take to finish the same work?
(a) 45 (b) 48
(c) 54 (d) None of these

Q.20. A team has a food stock for N number of days. After 20 days, 1/4th of the team quits and the food lasted for another N number of days for the remaining persons in the team. What is the value of N?
(a) 50 (b) 60 (c) 70 (d) 80

FOUNDATION

Q.1. To complete a work, A and B take 30 days, B and C take 24 days, and C and A take 20 days. They all start together and work for 5 days after which B and C leave. How many more days will A take to finish the remaining work?
(a) 11 (b) 28 (c) 33 (d) 18

Q.2. A water tank normally takes 7 hours to be filled by a tap but because of the leak, it takes another 2 hours. In how many hours will the leak empty a full water tank?
(a) 20.5 hours (b) 24.5 hours
(c) 30 hours (d) 31.5 hours

Q.3 The wages of 10 men and 8 boys amount to Rs 37. If 4 men together receive Re 1 more than 6 boys, what are the wages of each man and boy respectively?
(a) 2.5 and 1.5 (b) 11 and 9
(c) 3 and 4 (d) 6 and 5

Q.4. Fourteen machine can produce 500 units in 12 hours. After 4 hours of production, two more machines were started. How many more hours will it take to complete the production of 500 units?
(a) 4 (b) 5 (c) 6 (d) 7

Q.5 Pramod can do 2/5th of the work in 8 days and Binod can finish the remainder in 18 days. How long would they take to finish it together?
(a) 8 (b) 10 (c) 12 (d) 14

Q.6 A cistern can be filled separately by two pipes P and Q in 45 min and 35 min respectively. A tap R at the bottom can empty the full cistern in 30 min. If the tap R is opened 7 min after the two pipes P and Q are opened then after what time from the opening of pipe P and Q the cistern becomes full?
(a) 40th min (b) 61st min
(c) 37th min (d) None of these

Q.7. A can do a job in 20 days, B in 30 days and C in 60 days. If A is helped by B and C every 3rd day. How long will it take for them to complete the job?
(a) 12 days (b) 4 days
(c) 15 days (d) 18 days

Q.8. Saket and Prakash take 7.5 days working together. However, if Saket and Prakash work individually to complete the equal amount of work, it takes them 20 days to complete the job. How long will Prakash alone take to do the job if Saket is more efficient than Prakash?
(a) 18 days (b) 24 days
(c) 30 days (d) 27 days

Q.9. Four machines A, B, C and D are to be used for producing 10,000 units. Machine A alone can produce 50 units in a day, B is half as efficient as A, C is as efficient as A and D is twice as efficient as A. In how many days would the work be complete if all the four machines start working together?
(a) 19 days (b) 29 days
(c) 46 days (d) None of these

Q.10. In an organization, regular working day is of 8 hours and regular week is of 5 working days. A man is paid Rs 3.20 per hours overtime. If he has earned Rs 432 in 4 weeks, what is the total number of hours he worked in those 4 weeks?
(a) 180 (b) 175
(c) 160 (d) None of these

Q.11. Tap A fills a tank in 20 min while C empties it at 1/3rd the rate at which A fills it. At 12.00 noon, A and C are simultaneously started and when the tank is 50% full, tap A is turned off. At what time will the tank be empty?
(a) 12.35 P.M. (b) 12.45 P.M.
(c) 12.30 P.M. (d) 12.55 P.M.

Q.12. Two pipes X and Y can fill a tank in 6 hours and 9 hours respectively. If they are turned on alternately for one hour each starting with X, then what is the time taken to fill the tank?
(a) 6.25 hours (b) 5.66 hours
(c) 7 hours (d) 8 hours

Q.13. A 10 hectare field is reaped by 2 men, 3 women and 4 boys in 10 days. If a man, a woman and a boy work in the ratio of 5:4:2, then in how much time 6 men, 4 women and 7 boys reap a 16 hectare field?
(a) 5 days (b) 6 days
(c) 8 days (d) 7 days

Q.14. Four plumbers had to repair 364 taps. At the end of 3 days working 7 hours per day they had repaired 168 taps. In order to speed up the work, 3 more plumbers were added who also worked 7 hours daily. If efficiency of each of the plumbers is same, how many additional days would it take to complete the task?
(a) 2 days (b) 3 days
(c) 4 days (d) 1½ day

Q.15. Three factories are capable of manufacturing a particular part of the bike. Each of the two factories can produce 1,00,000 parts in 15 days. The third factory can produce them 30% faster. How many days (approx) would it take to produce a million parts with all the three working simultaneously?
(a) 43 (b) 45 (c) 48 (d) 52

Q.16. Five persons working 8 hours daily can complete a wall in 10 days. When they have worked for 5 and 1/2 days, 5 more persons are brought to work. In how many more days will the wall be complete?
(a) One more day (b) More than 2 days
(c) More than 4 days (d) More than 3 days

Q.17. Two filling pipes can fill a tank in 15 hours and 12 hours respectively while a third pipe C can empty it in 20 hours when filled. If the tank is empty and all the three pipes are opened, in how much time will the tank be full?
(a) 14 hours (b) 10 hours
(c) 6 hours (d) 8 horus

Q.18. A work can be finished by Arnold, Bill and Kennedy in 20 days, 24 days and 30 days respectively provided the working hours are 10 hours per day. If all of them are engaged to complete this work and they finished the work in 10 days, how many hours per day did they work?
(a) 7 hours (b) 10 hours
(c) 8 hours (d) 9 hours

Q.19. If A completes a work in t days, B in 1/r days and C in s days then in how many days will they complete the work together?
(a) st/(s+t+rst) (b) rt/(r+t+rst)
(c) rst/(rs+st+tr) (d) rs/(r+s+rst)

Q.20. A certain job was assigned to a group of people to complete it in 20 days. But 12 of them did not turn up for the job and the remaining persons did the job in 32 days. What is the original number of men in the group?
(a) 32 (b) 34
(c) 36 (d) None of these

Q.21. A machine P can print one lakh books in 8 hours, machine Q can print the same number of books in 10 hours while machine R can print them in 12 hours. If all of them work together for 6 hours, how many books will be printed?
(a) 60,000 (b) 1,80,000
(c) 1,85,000 (d) 2,85,000

Q.22. A pipe can fill a cistern in 6 hours. Due to a leak in its bottom, it is filled in 7 hours. When cistern is full, in how much time will it be emptied by the leakage?
(a) 42 hours (b) 40 hours
(c) 43 hours (d) 45 hours

Q.23. Tap A can fill a tank in 16 min and tap B can empty it in 8 min. If the tank is already 1/2 full and both the taps are opened together, the tank will be either filled or emptied? How long will it take before the tank is either filled or emptied?
(a) Emptied; 16 min (b) Filled; 8 min
(c) Emptied; 8 min (d) Filled; 12 min

Q.24. Three-fourth of a tank is full of water. If 5 liters are added to it. Then four-fifth of the tank becomes full. What is capacity of the tank?
(a) 75 litres (b) 80 litres
(c) 100 litres (c) 120 litres

Q.25. A 6-cm long cigarette will burn itself up in 15 min if no puff is taken. For every puff, it burns three times as fast as during the duration of the puff. If the cigarette burns itself in 13 min, then how many puffs has the smoker taken if his average puff lasted 3 s?
(a) 30 (b) 20
(c) 15 (d) None of these

Q.26. Two taps A and B can fill a vessel in 12 min and 15 min respectively, but a third tap C can empty the full tank in 6 min. A and B is kept open for 5 min in the beginning and then C is also opened. In what time will the vessel be emptied?
(a) 30 min (b) 33 min
(c) 45 min (d) None of these

Q.27. Two pipes A and B can separately fill a cistern in 15 and 10 min respectively and a waste pipe C can carry off 7 litres per min. If all the pipes are opened when the cistern is full, it is emptied in 2 hours. How many litres does the cistern hold?
(a) 30 litres (b) 40 litres
(c) 35 litres (d) 45 litres

Q.28. One pipe can fill a tank three times as fast as another pipe. If together the two pipes can fill the tank in 36 min, then how much time will the slower pipe take to fill the tank working alone?
(a) 81 min (b) 108 min
(c) 144 min (d) 192 min

Q.29. Two pipes A and B can separately fill a cistern in 10 min and 5 min respectively and a waste pipe C can carry off 5 litre per min. If all the pipes are opened when the cistern is full, it is emptied in 1 hour. What is the capacity of the cistern?
(a) 40 litres (b) 35 litres
(c) 95 litres (d) None of these

Q.30. Sanjay can do as much work in 16 hours as Balram can do in 24 hours. Pranesh can do the same work in 32 hours. If all of them work together, they can complete the same work in 24 hours. In how many hours can Pranesh alone complete the work?
(a) 80 (b) 88 (c) 104 (d) 112

Q.31. One fill-pipe A is 10 times faster than the second fill-pipe B. If B can fill a cistern in 55 min, then find the time when the cistern will be full if both the fill-pipes are opened together?
(a) 5 min (b) 6 min
(c) 7 min (d) 3 min

Q.32. A tank can be filled by one tap in 20 min and by another in 25 min. Both the taps are kept open for 5 min and then the second tap is turned off. In how much more time will the tank be filled?
(a) 11 min (b) 13 min
(c) 15.5 min (d) 18 min

Q.33. Fifteen men take 21 days working 8 hours each to do a piece of work. How many days of 6 hours each would 21 women take, if 3 women can do as much work as 2 men can do?
(a) 20 (b) 30
(c) 21 (d) 28

Q.34. Three hundred workers are set to build a dam. In 21 weeks they have done 60% of the work, but subsequent rains lasting 4 weeks washed away 25% of what they had done. The job is resumed by 200 workers. What is the additional time it will take to complete the work?
(a) 12 (b) 24
(c) 36 (d) None of these

Q.35. Tap A can fill a tank completely in 4 hours, while tap B can empty it in 8 hours. However the operator forgot to close the tap B, and as a result both the taps remained open. After 4 hours, the person realized his mistake and immediately closed the tap B. How much more time would the tank take to get full now?
(a) 1 hour (b) 2 hours
(c) 3 hours (d) 4 hours

Q.36. If work done by 2 men are equivalent to work done by 5 women and work done by 2 women are equivalent to work done by 5 children, then in how many days can the work be completed by one man alone if the same work can be completed by 100 children in 10 days?
(a) 120 days (b) 80 days
(c) 100 days (d) 160 days

Q.37. A certain number of people completed a job in 24 days. If there were 2 persons less, the work would have taken 6 more days. How many persons were there initially?
(a) 12 (b) 16
(c) 18 (d) 10

Q.38. Twelve men complete a work in 9 days. After they have worked for 6 days, 6 more men join them. How many days will they take to complete the remaining work?
(a) 2 days (b) 3 days
(c) 4 days (d) 5 days

Q.39. A certain number of men can finish a piece of work in 100 days. If however there were 10 men less, it would take another 10 days to finish the work. How many men were there originally?
(a) 50 (b) 75
(c) 82 (d) 110

Q.40. A Zamindar hired a certain number of men to plough his 480 acres of fields. Some of the men did not turn up at the job, and consequently each of those who reported had to plough 20 acres more of the field each than originally planned by the zamindar, assuming equal distribution of labour. How many men were initially appointed and how many abstained from the job?
(a) 15, 5 (b) 12, 8
(c) 10, 3 (d) 16,10

Q.41. Anoop takes 5 hours more than Mayank to build a wall. However, Anoop takes 9 hours more than Manoj. Anoop and Mayank together can do the same work in the same time as Manoj. How many hours will Anoop take to do it?
(a) 12 hours (b) 15 hours
(c) 18 hours (d) 24 hours

Q.42. A tank can be filled in by two taps A and B in 3 min. However, due to a leakage at the bottom of the tank, it takes 3.5 min to fill the tank. If tank is full, then how much time will the leakage take to empty the tank?
(a) 21 min (b) 12 min
(c) 24 min (d) 18 min

Q.43. A road construction agency takes up the work of making a road 189 km long in 15 months. He engages 256 men, but after 1/5th of the time, agency finds that only 1/6th of the work has been completed. How many more men should be now employed to complete the work in time?
(a) 57 (b) 59
(c) 64 (d) 67

Q.44. Twenty-five men are employed to do a piece of work in 24 days. After 15 days, 10 more men are engaged and the work was finished a day earlier than actually planned. In what time could they finish the work if extra men were not employed?
(a) 29.5 days (b) 24.6 days
(c) 26.2 days (d) 21.7 days

Q.45. Twelve men or 18 women can do some work in 7 days. In what time can 4 men and 8 women do the same work?
(a) 4 days (b) 8 days
(c) 7 days (d) 9 days

Q.46. A man works for 10 hours per day to finish a work in 3 days. A boy finishes the same in 5 days working 18 hours per day. How many boys are required to do the job in the time taken by one man?
(a) 1 (b) 2
(c) 3 (d) 4

Q.47. A tap can fill a tank in 60 min. Tap is turned on to fill the tank. When the tank is half full, it springs a leak at the bottom and the water level falls to 1/4 of the tank in 10 min. The tap is now closed. What is the time taken by the leakage to empty the tank now?
(a) 7 min (b) 4 min
(c) 8 min (d) None of these

Q.48. A piece of work can be done by A and B in 10 days, by B and C in 15 days and by C and A in 20 days. All three worked for 6 days and then A withdrew. B and C worked for another 4 days and then B stopped working. How many more days will C take to complete the remaining work?
(a) 12 (b) 12.5
(c) 10.33 (d) 10

Direction for questions 49 and 50: Read the passage given below and solve the questions based on it.

A group of 30 people can complete a job by working for 10 hours a day in 15 days. The group starts the work. But at the end of every day, starting from the first day, one person leaves the group and the remaining people work for 20 min less on the next day.

Q.49. On which day will the work be completed?
(a) 28th (b) 29th
(c) 30th (d) None of these

Q.50. What part of the work, approximately remains to be done at the end of 15 days?
(a) 0.51 (b) 0.61
(c) 0.49 (d) 0.39

MODERATE

Q.1. Mayank can do 50% more work than Shishu in the same time. Shishu alone can do a piece of work in 30 hours. Shishu starts working and he had already worked for 12 hours when Mayank joins him. How many hours should Shishu and Mayank work together to complete the remaining work?
(a) 6 (b) 12 (c) 4.8 (d) 7.2

Q.2. A man works twice as fast as a woman. A woman works twice as fast as a child. If 16 men can complete a job in 12 days, how many days would be required for 32 women, and 64 boys together to complete the same job?
(a) 2 day (b) 3 days
(c) 4 days (d) 6 days

Q.3. A tank with capacity T litres is empty. If water flows into the tank from pipe X at the rate of X litre per minute and water is pumped out by pipe Y at the rate of Y litres per minute and X>Y, then in how many minutes will the tank be filled?
(a) T/(Y-X) (b) T/(X-Y)
(c) (T-X) Y (d) (X-Y)/60T

Q.4. A can do a piece of work in 90 days, B in 40 days and C in 12 days. They work for a day each in turn, i.e., first day A does it alone, second day B does it alone and 3rd day C does it alone. After that the cycle is repeated till the work is finished. They get Rs 240 for this job. If the wages are divided in proportion to the work each had done. Find the amount A will get?
(a) 14 (b) 24 (c) 34 (d) 36

Q.5. There are 12 pipes connected to a tank. Some of them are fill pipes and the others are drain pipes. Each of the fill pipes can fill the tank in 8 h and each of the drain pipes can drain completely in 6 h. If all the pipes are kept open, an empty tank gets filled in 24 h. How many of the 12 pipes are fills pipes?
(a) 5 (b) 6 (c) 7 (d) 8

Direction for questions 6 and 7: Read the passage below and solve the questions based on it.

The tank at a water supply station is filled with water by several pumps. At first, three pumps of the same capacity are turned on; 2.5 hours later, two more pumps (both the same) of a different capacity are set into operation. After 1 hour, the additional pumps were set into operation; the tank was almost filled to its capacity (15 m^3 were still lacking); in another hour the tank was full. One of the two additional pumps could have filled the tank in 40 hours.

Q.6. What is the volume of the tank?
(a) 60 m^3 (b) 80 m^3 (c) 75 m^3 (d) 90 m^3

Q.7. How much water does one of the first three pumps fill in an hour?
(a) 5 m^3 (b) 4 m^3 (c) 3 m^3 (d) 2 m^3

Direction for questions 8 and 9: Read the passage below and solve the questions based on it.

A person can dig a trench 40 metres in depth in 4 days working 8 hours a day. However, after every day he finds that 1/5th of the depth got filled up with mud again.

Q.8. In how many days can he actually dig a trench if it is of 30 metres in depth?
(a) Exactly 4 days (b) On the 5th day
(c) Exactly 5 days (d) On the 6th day

Q.9. What was the depth of the trench at the beginning of the fourth day?
(a) 14.4 metres (b) 19.52 metres
(c) 23.616 metres (d) None of these

Direction for questions 10 to 12: Read the passage below and solve the questions based on it.

There are three taps A,B, C and an outlet pipe D. A, B and C can fill the tank in the Panikam locality in 10, 20 and 25 h respectively. The outlet pipe can empty the same tank in 100 h. There are 2,000 houses in the locality. The tank has a capacity of 50,000 litres.

Q.10. If all the taps and the outlet pipe are opened simultaneously, how much water is thrown into the tank every hour?
(a) 8000 litres (b) 9000 litres
(c) 1000 litres (d) Cannot be determined

Q.11. All the three taps were open and the emptying pipe is closed. At the time when the tank was supposed to be full, it was found that only 2/5th of the tank was full. It was discovered that all the residents had kept their water taps open during this period. At what rate were the residents of each house getting water? (Consider that each house has only one tap.)
(a) 1.1 litres/h (b) 2.22 litres/h
(c) 2.85 litres/h (d) 4.46 litres/h

Q.12. If the average requirement of every house in the area per day is 40 litres, for how many h, for the second time should the taps be opened? (Consider that all the three tapes were opened on both occasions).
(a) $3\frac{3}{19}$ h (b) $4\frac{1}{17}$ h
(c) 5 h (d) None of these

Direction for questions 13 and 14: Refer to the data below and answer the questions that follow

Fifty cows can graze a field for 40 days and 40 Cows can graze the same field for 80 days. Some grass is already there and the grass grows at a constant rate.

Q.13. For how many days can 60 cows graze the same field? (growth of grass is also there.)
(a) 20.66 (b) 50.66 (c) 60.66 (d) 26.66

Q.14. What is the maximum number of cows that can graze the same field for an infinite time?
(a) 30 (b) 25 (c) 8 (d) 35

Q.15. Anand got an order from a boutique for 480 shirts. He brought 12 sewing machines and appointed some expert tailors to do the job. However, many did not report to duty. As a result, each of those who reported had to stitch 32 more shirts than was originally planned by Anand, with equal distribution of work. How many tailors had been appointed earlier and how many had not reported to work?
(a) 12, 4 (b) 10, 3
(c) 10, 4 (d) None of these

Direction for questions 16 to 18: Refer to the data below and answer the questions that follow

Anoop and Sandeep can dig a well each in 10 and 5 days respectively. But on a particularly difficult terrain the work is such that due to fatigue every subsequent day the efficiency of a worker falls by 10%.

Q.16. If Anoop is given a task of digging one such well in the difficult terrain, then in how many days will he finish the work?
(a) 12th day (b) 15th day
(c) 11th day (d) never

Q.17. If Sandeep is given the task of digging one such well in the difficult terrain, then in how many days will he finish the work?
(a) 9th day (b) 7th day
(c) 8th day (d) Never

Q.18. If both Anoop and Sandeep work together to finish two such wells then, in how many days will the work finish?
(a) 11th day (b) 12th day
(c) 8th day (d) Never

Direction for questions 19 and 20: Refer to the data below and answer the questions that follow.

Anoop was writing the reading comprehension sections in the DOG entrance examinations. There were four passages of exactly equal length in terms of number of words and the four passages had 5, 8, 8 and 6 questions following each of them respectively. It is known that Anoop can answer exactly 12 questions in the time he takes to read any one of the four passages. Assume that his rate of reading and answering questions remains the same throughout the section.

Q.19. Anoop took 13 min more to finish the first three passages than the time he took to finish the last passage. Assuming that Anoop answered all the questions in each passage, What percentage of the total time did he spend on the first passage?
(a) 24.5% (b) 25.4%
(c) 22.6% (d) 26.2%

Q.20. By what per cent should Anoop increase his reading speed if he has to cut down on his total time spent on the section by 20%? Assume that the time spent on

answering the questions is constant and as given in the directions.
(a) 36.36% (b) 54.54%
(c) 50.50% (d) 45.45%

Q.21. The kingdom of Charles had a circular wall around it and four-city gates pointing north, south, east and west respectively, Now the Shire of Nottingham and the Shire of Birmingham decided to race on the shortest route touching all the four gates. They start at the west gate and one of the shires rides five times as fast as the second. They will meet for the 19th time after they start, at the
(a) The north gate
(b) The south gate
(c) Exactly midway between the north and the west gate
(d) Exactly midway between the south and the east gate

Q.22. The packing process in a garment factory involves processing by 3 machines. The first machine, A, inserts the garment in a packet, after which the second machine, B, seals the packet and finally the third machine, C, attaches a price tag to the packet. Machine A takes at least 10 per unit. Machine B takes at least 15 per unit. Machine C takes at least 12 per unit. The factory starts at 8 a.m. and runs non-stop till 6 p.m., What is the maximum number of packets that can be processed in a one-hour period?
(a) 360 (b) 240 (c) 300 (d) 900

Q.23. P, Q and R can each complete a certain work in 16, 20 and 30 days respectively. The three of them start the work together. P leaves after 4 days; Q leave 4 days before the work is finished. How long did the work last?
(a) 12.6 days (b) 9.2 days
(c) 16.4 days (d) 11.4 days

Q.24. A certain work was started by 4 men and 10 women who completed 50% of the work in 6 days. Then another 2 men and 2 women joined them and they could complete 2/3rd of the remaining work in three days. How many men along with 6 women are required to complete the remaining work in two more days?
(a) 3 (b) 5 (c) 6 4.9

Q.25. A can work half as fast as B and C together. A and B together can work three times as fast as C. If all three of them complete a job in 15 days, how long would C take to complete the same work?
(a) 45 days (b) 54 days
(c) 60 days (d) 65 days

Q.26. A cistern can be filled by two taps A and B coming from the same main pipeline in 20 and 30 min respectively. Both the taps are opened at the same time but due to partial closing of a valve in the main pipeline, tap A was supplying only 4/5th of its capacity and tap B, 5/6th of its capacity. After some time, the valve in the main pipeline was opened fully thereby enabling the two taps to supply at full capacity. It took another 5 min to fill the cistern completely. How long (approx.) was it before the valve on the main pipeline opened?
(a) 7 minutes (b) 9 minutes
(c) 11 minutes (d) 13 minutes

Q.27. In what time would a cistern be filled by three pipes of diameter of 1 cm, 2 cm and 3 cm if the largest pipe alone can fill the cistern in 49 minutes, the amount of water flowing through each pipe being proportional to the square of its diameter?
(a) 31.5 minutes (b) 63 minutes
(c) 126 minutes (d) None of these

Q.28. Three workers— A, B and C— working simultaneously, produced 144 dyes in four hours. In one hour, A can produce as many dyes more than B as B can produce more than C. During a period of five hours, A can produce as many dyes as C can in seven hours. What is the difference between the number of dyes produced by B and C per hour?
(a) 1 (b) 2 (c) 3 (d) 4

Q.29. A can do a piece of work in 36 days, B in 54 days and C in 72 days. All three began the work together but A left after 8 days and B 12 days before the completion of the work. In all for how many days did C work?
(a) 24 days (b) 28 days
(c) 32 days (d) 36 days

Q.30. Three men with 5 boys can do a piece of work in 2 days and 4 men and 16 boys can complete the job in one day. How much time will it take for 1 boy together with a woman who can work twice as fast as the boy to complete a job that is 3 times as time consuming?
(a) 24 days (b) 28 days
(c) 32 days (d) 36 days

Q.31. Tatto, Bubby and Tappo are doing a work in 10 days. However, Tappo only worked for the first 3 days till 37/100th of total work is done. The remaining work was done by Tatto and Bubby alone. Efficiency of Tatto is 25% more than the efficiency of Bubby. What is the time taken by the fastest worker to finish the work alone?
(a) 16.66 days (b) 30 days
(c) 25 days (d) 20 days

Q.32. P works twice as fast as Q, whereas P and Q together can work three times as fast as R. If P, Q and R work together on a job, in what ratio should they share the earnings?
(a) 2:1:1 (b) 4:2:1
(c) 4:3:2 (d) 4:2:3

Q.33. Rakesh and Sid can complete a piece of work in 12 and 24 days respectively. Rakesh started the work and after a few days Sid joined him and the work was completed in 10 days. After how many days, did Sid join Rakesh?
(a) 6 (b) 7 (c) 8 (d) 10

Q.34. Sagar is 20% more efficient than Pranesh. If Pranesh can complete a piece of work in 30 days, then in how many days can both Sagar and Pranesh complete the work?
(a) (11 + 5/11) days (b) (13+ 7/11) days
(c) (15+ 5/11) days (d) (17+ 7/11) days

Q.35. Manoj and Harvinder can complete a piece of work in 16 days, which Harvinder can complete in 32 days. If Manoj and Harvinder work on alternate days, starting with Harvinder, in how much time would the work be completed?
(a) 17.25 days (b) 17.66 days
(c) 20 days (d) 32 days

Q.36. A factory manufactures dyes. It has 12 men and two machines which can be operated by all of its men. It takes 4 hours to manufacture one dye on the machine with the operator. The machines can work continuously without a break. Without the machine each of the men can manufacture a dye in 8 hours. The policy is such that the production is maximized and the men are ready to work in three shifts of 8 hours per day. What will be the average cost incurred per dye if 1 man hour costs Rs 20 and 1 machine hour costs Rs 15?
(a) Rs 140 (b) Rs 160
(c) Rs 147 (d) Cannot be determined

Q.37. Ramesh and Suresh could finish the garden cleaning in 'p' days. They worked together for 'q' days; then Ramesh fell sick and the remaining cleaning was completed by Suresh in 'r' days. In how many days could each of them clean the garden?

(a) Ramesh = $\frac{pr}{r-p+q}$ days; Suresh = $\frac{pr}{p-q}$ days

(b) Ramesh = $\frac{p}{r-p+q}$ days; Suresh $\frac{p}{p-q}$ days

(c) Ramesh = $\frac{pr}{r+p-q}$ days; Suresh $\frac{pr}{q-p}$ days

(d) Ramesh = $\frac{pr^2}{r-p+q}$ days; Suresh $\frac{pr^2}{q-p}$ days

Q.38. A, B, C, D and E are five taps. The capacity of B is 2 times that of A, the capacity of C is 3 times that of A. Capacities of D and E are 4 and 5 times that of A respectively. In the first case A, C and E act as input pipes and B and D act as output pipes.
In the second case, C, D, E act as input pipes and A and B act as output pipes.

If A and B working together as input pipes can fill the tank in 4 hours, then what is the difference in the time required to fill the tank in the first and second case stated above?
(a) 4.44 hours (b) 3.33 hours
(c) 2.22 hours (d) 1.11 hours

Q.39. A tank has 5 input pipes, A, B, C, D and E. Among them A takes the maximum time to fill the tank. Their capacities to fill the tank are in "Geometric Progression (GP)" in the given order with common ratio 2. The tank also has 5 output pipes F, G, H, I and J. Among them F takes the maximum time to empty the tank. G's capacity to empty the tank is 2 times that of F. H's capacity? is times that of F and so on. Working together A and F can fill the tank in 100 h. and all the taps working together can fill the tank in $\left(\frac{100}{63}\right)$ hr. How much time will E and J take to fill the tank when working together?

(a) $\frac{100}{62}$ (b) $\frac{100}{38}$

(c) $\frac{100}{48}$ (d) Cannot be determined

Q.40. A man makes 60 articles in the 1st hour. His efficiency decreases by 25% in the 2nd hour, increases by 40% in the 3rd hour, decreases by 33% in the 4th hour and increases by 50% in the 5th hour. If he has to work for more than 1 hour, then in which hour the average number of articles produced per hour till then would be minimum?
(a) The 2nd hour
(b) After the 5th hour
(c) The 2nd and the 4th hour
(d) None of these

Q.41. A worm climbs a vertical bamboo stick 32 cm tall. The worm climbs 3 cm in the day and slips 2 cm during the night each time. However, there are cavities in the stick at 2 cm, 4 cm,16 cm from the bottom that would not let the worm slip further. If the worm climbs only during the daytime and slips only during the night, on which day from the start will it reach the top?
(a) 29 days (b) 28 days
(c) 27 days (d) 26 days

Q.42. Anil, Bhuvan and Chandan take 10, 20 and 25 days to complete a job. What is the minimum time required to finish the job if not more than 2 of them work together on a single day and no two consecutive days have the same pair of people working?

(a) 6.5 days (b) $6\frac{13}{15}$ days

(c) 7.5 days (d) $7\frac{21}{28}$ days

Q.43. A swimming pool is fitted with three pipes. The first two pipes working simultaneously fill the pool in the same time as the third pipe alone. The second pipe alone fills the pool 5 hours faster than the first pipe and 4 hours slower than the third pipe. In what time will the second and the third pipe together fill the pool?
(a) 3.25 h (b) 6.5 h
(c) 5 h (d) None of these

Q.44. A contractor needs to complete a given work in 24 days. For every day that he finishes ahead of time, he gets an incentive 'I' and for every day that he finishes behind time, he needs to pay a penalty 'P'. He has a choice of 2 workers who work as follows: A completes the work in 24 days for 75% of the time and in 30 days for 25% of the time. B completes the work in 20 days for 60% of the time and in 30 days for 40% of the time. At what ratio of P to I will the contractor be indifferent of using A or B?
(a) $\frac{8}{3}$ (b) 2
(c) $\frac{7}{3}$ (d) $\frac{7}{4}$

Q.45. Anuj can do a piece of work in a certain number of days. To do the same piece of work, Bhanu takes thrice the number of days as Anuj takes whereas Chandu takes thrice as many days as Bhanu does and Dodo takes thrice as many days as Chandu does. Now, they are paired and two groups are formed. The first pair takes one-third the time taken by the second pair to complete the work. which is the first pair?
(a) Anuj and Dodo
(b) Anuj and Chandu
(c) Chandu and Bhanu
(d) Bhanu and Dodo

ADVANCED

Direction for questions 1 to 5: Read the passage below and solve the questions based on it.

There are infinite pipes attached to a very big tank. Pipes are numbered like P_1, P_2, P_3... and so on. Its also known that efficiency of every subsequent pipe is half the efficiency of earlier pipe i.e., the efficiency of P_2 pipe is half the efficiency of P_1 pipe and so on. However, in case of P_1 pipe, this rule is not true since there is no earlier pipe. When all the pipes are working together, tank gets filled in 2 h.

Q.1. What time will P_4 take to fill the tank working alone?
(a) 8 (b) 16
(c) 32 (d) Cannot be determined

Q.2. What is the difference in the time taken by P_5 and P_6?
(a) 32 (b) 64
(c) 128 (d) Cannot be determined

Q.3. How many pipes can fill the tank within 100 h working alone?
(a) 4 (b) 5 (c) 6 (d) Infinite

Q.4. Due to some technical problem, only four pipes P_1, P_2, P_3 and P_4 are in working condition. These pipes are now grouped to obtain two pairs of pipes. It is found that one pair of the pipe is taking 2/3rd of the time taken by the other pair to fill the tank independently. Which of the following is one of the two pairs?
(a) P_1, P_2 (b) P_1, P_3
(c) P_1, P_4 (d) Cannot be determined

Q.5. What time will P_2 take to fill the tank working alone?
(a) 4 (b) 8 (c) 16 (d) 32

Direction for questions 6 and 7: Read the passage below and solve the questions based on it.

There are 2 filling taps A and B and an emptying tap C which are used in a laboratory to fill or empty bottles. The three taps are attached to a computer and they function according to the commands given by the computer. The following are the commands that the computer gives:

Command	*Effect*
F(A, x, t_1)	Fill the bottle with tap A at *x* cc/min for t_1 min
F(B, y, t_2)	Fill the bottle with tap B at *y* cc/min for t_2 min
E(C, z, t_3)	Empty the bottle with tap C at *z* cc/min for t_3 min

Q.6. The following commands, in the given order, are used to fill an empty bottle and each command can be executed only after the previous command is completed. What is the final volume after these commands?
F (A, 10, 600), E (C, 50, 20)
F (A, 10, 400), F (B, 20, 200)
F (A, 50, 100), F (B, 20, 50)
E (C, 10, 50), F (B, 20, 75)

(a) 15 (b) 20
(c) 22 (d) None of these

Q.7. The commands F (A, 10, 500) and (B, 40, 500) can fill an empty bottle. Which of the following commands can now empty the bottle?
(a) E (C, 20, 500) (b) E (C, 50, 500)
(c) E (C, 100, 200) (d) None of these

Q.8. A group of workers can complete a certain job in 9 days. But it so happens that every alternate day starting from the second day, 2 workers are withdrawn from the job and every alternate day starting from the third day, one worker is added to the group. In such a way, the job is finished by the time, there is no worker left. If it takes the double time to finish the job now, find the number of workers who started the job?
(a) 5 (b) 10 (c) 15 (d) 20

Q.9. A student studying the weather for d days observed that (i) it rained on 7 days, morning or afternoon; (ii) when it rained in the afternoon, it was clear in the morning; (iii) there were five clear afternoons and (iv) there were six clear mornings. Then d equals
(a) 3 (b) 7 (c) 11 (d) 9

Q.10. Tanveer constructs a wall working in a special way and takes 12 days to complete it. If S_n is the length of the wall (in m) that he constructs on the nth day, then
$S_n = 2n$, $0 \le n \le 4$
$S_n = 8$, for $n = 5$
$S_n = 3n - 7$, $6 \le n \le 12$
Find the total length of the wall he constructs in the first 10 days.
(a) 31 m (b) 35 m
(c) 113 m (d) None of these

Direction for questions 11 and 12: Read the passage below and solve the questions based on it.

There are infinite number of taps that are filling a certain tank. The taps are numbered 1, 2, 3 and so on. Any tap numbered p ($p \ge 2$) can fill the tank in $(p^2 - 1)$ min while the first tap does not work at all.

Q.11. If all the taps from the second tap onwards are opened simultaneously, then what is the shortest time in which the tank can be filled?
(a) 11/3 min (b) 22/3 min
(c) 2/3 min (d) 11/5 min

Q.12. The minimum number of taps that must be opened so that the tank is filled within 106 seconds is
(a) 6 (b) 3 (c) 5 (d) 4

Q.13. Alok, Mithilesh and Bimlesh started a work and after completing 1/5th of the work Bimlesh left. Alok and Mithilesh then worked for 20 days. Bimlesh then took over from Alok and Mithilesh and completed the remaining portion of the work in 12 days. If Bimlesh takes 40 days to complete the work, in how many days would Alok alone or Mithilesh alone complete the work if the efficiencies with which they work is the same?
(a) 40 (b) 60
(c) 100 (d) None of these

Q.14. Sixty-four men working 8 h a day plan to complete a piece of work in 9 days. However, 5 days later they found that they had completed only 40% of the work. They now wanted to finish the remaining portion of the work in 4 more days. How many hours per day should they need to work in order to achieve the target?
(a) 11 (b) 12
(c) 13 (d) 15

Direction for questions 15 and 16: Read the passage below and solve the questions based on it.

The employees of ABC Corporation put forward demands for a shorter working week and a better pay. They were paid by the hour and worked 5 days a week. Their demands were:

(i) the working days to be shorter by 1 h and
(ii) in the course of the shorter working day that they are demanding, payment for a working day be the same as the number of h needed to earn Rs 49.

At present, a full day's work brings in Re 1 less than that and as many rupees as it is necessary to work and earn Rs 40.

Q.15. For how many hours were the employee proposing to work?
(a) $8\frac{1}{2}$ (b) 10 (c) $10\frac{1}{2}$ (d) 9

Q.16. What was the revised rate demanded per hour?
(a) Rs $1\frac{1}{4}$ (b) Rs $2\frac{1}{3}$
(c) Rs $2\frac{3}{4}$ (d) Rs $2\frac{1}{2}$

Direction for questions 17 to 19: Read the passage given below and solve the questions based on it.

Mr Bose can do a job in two days if his four secretaries A, B, C and D are all present. His efficiency got reduced to $\frac{1}{6}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}$ respectively as A, B, C and D remain absent. In any day only one secretary remains absent. Also Mr Bose finishes his work exactly in integral number of days.

Q.17. Which of the following secretaries cannot be absent for an odd number of days?
(a) A (b) B (c) C (d) D

Q.18. If each secretary remained absent for at least one day, then in how many days will Mr Bose finish the work?
(a) 5, 6 or 7 days (b) 6, 7 or 8 days
(c) 6, 8 or 9 days (d) 7, 8 or 9 days

Q.19. If Mr Bose finished his work in 6 days with one secretary being present every day, then the secretary was
(a) A (b) B
(c) C (d) Cannot be determined

Direction for questions 20 to 22: Read the passage given below and solve the questions based on it.

A team of N workers are selected to do a certain job. Each worker can individually complete the job in 10 days. The team follows the following procedure for any job:

The work is started by one person on the first day. Then, from the next day onwards, till the work is completed, everyday one new person joins the work.

The work is such that, on any day it starts at 9.00 am and stops at 5.30 pm with a 30-min break for lunch from 1.00 pm to 1.30 pm. At whatever stage a worker joins the work, he does not leave till the job is over.

Answer each question in this set independently of the other questions.

Q.20. If the entire team works at only 50% efficiency, and the worker who is supposed to join on the last day does not report for work, then the work is completed
(a) Exactly at the end of the 6th day
(b) Before lunch, on the 6th day
(c) After lunch, before the end of the 6th day
(d) Before lunch, on the 7th day

Q.21. If the team takes up a task which is twice as time consuming as the original work, then the work will be completed on the
(a) 5th day (b) 6th day
(c) 7th day (d) 8th day

Q.22. If the team takes up a task which is twice as time consuming as the original work, then at what time on the concluding day will the work be completed ?
(a) 3.40 pm (b) 4.20 pm
(c) 5.30 pm (d) 4.10 pm

Q.23. The rate of flow of water (in litre per min) of three pipes are 2, N and 3, where $2 < N < 3$. The lowest and the highest flow rates are both decreased by a certain quantity x, while the intermediate rate is left unchanged. If the reciprocals of the three flow rates, in the order given above, are in arithmetic progression both before and after the change, then what is the quantity x (in litre per min)? (Negative flow rates indicate that the pipes act as emptying pipes instead of filling pipes.
(a) 2.6 (b) −3.2 (c) 3.8 (d) −2.6

Direction for questions 24 to 26: Read the passage below and solve the questions based on it.

Two products A and B can be made on either of the two extrusion machines M_1 or M_2. The time in min, taken by these products according to the machine used, are as given below. Assume a day has 8 working hours.

	M_1	M_2
A	8	10
B	6	6

Q.24. If A and B are produced in the ratio of 3 : 1, then what is the maximum total of the products A and B that can be produced in a day?
(a) 120 (b) 160
(c) 116 (d) None of these

Q.25. If A is produced only on Machine M_1 and B produced only on machine M_2, then what is the unutilized time on M_2 in a day, if A and B are produced in 1 : 1 ratio?
(a) 20 min (b) 60 min
(c) 150 min (d) None of these

Q.26. If each product requires working on both the machines and the time required on each machine is as indicated in the table, then what is the maximum production possible in a day, assuming only A is produced?
(a) 80 (b) 180
(c) 48 (d) None of these

Direction for questions 27 to 30: Read the passage below and solve the questions based on it.

There are three bottles of water- A, B and C whose capacities are 5 litres, 3 litres and 2 litres respectively. For transferring water from one bottle to another and to drain out the bottles, there exists a programe which is computer-controlled. The computer that controls the flow through these pipes can be fed with three types of instructions, as explained below:

Instruction type	*Explanation of the instruction*
Fill (x, y)	Fill bottle labelled x from the water in bottle labeled y, where the remaining capacity of x is less than or equal to the amount of water in y.
Empty (x, y)	Empty out the water in bottle labelled x into bottle labelled y, where the amount of water in x is less than or equal to remaining capacity of y.
Drain (x)	Drain out all the water contained in bottle labelled x.

Initially, A full of water and B and C are empty.

Q.27. First instruction: Fill (B, A), Second instruction: Fill (C, B)
What is the quantity of water in B after executing all the instructions?
(a) 1 litre (b) 2 litres
(c) 3 litres (d) None of these

Q.28. After executing a sequence of three instructions, bottle A contains one litre of water. The first and the third of these instructions are shown below
First instruction: Fill (C, A)
Third instruction: Fill (C, A)
Then, which of the following statements about the instructions is true?
(a) The second instruction is Fill (B, A)
(b) The second instruction is Empty (C, B)
(c) The second instruction transfers water from B to C
(d) The second instruction involves using the water in bottle A

Q.29. Consider the same sequence of three instructions and the same initial state mentioned in the last question. Three more instructions are added at the end of the above sequence to have A contain 4 litres of water. In this total sequence of six instructions, the fourth one is Drain (A). This is the only Drain instruction in the entire sequence. At the end of the execution of the above sequence, how much water is contained in C?
(a) 1 litres (b) 2 litres
(c) 0 (d) None of these

Q.30. Initially, A had 5 litres, B had 2 litres and C had 1 litre. What is the minimum number of times Fill (x, y) is required to be used so that A is empty, provided we can not use Empty (x, y) instruction?
(a) 1 (b) 2
(c) 3 (d) None of these

Direction for questions 31 to 35: Read the passage below and solve the questions based on it.

Modern Boutique received a large order for stitching uniforms from A. R. Academy and T. M. High School. He has two cutters who will cut the fabric, five tailors who will do the stitching, and two assistants to stitch the buttons and button-holes. Each of these 9 persons will work for exactly 10 hours a day. Each of the A. R. Academy uniform requires 20 min for cutting the fabric, 1 hour for doing the stitching and 15 min for stitching the buttons and the button holes. The T. M. High School uniform requires 30 min, 1 hour and 30 min respectively for the same activities.

Q.31. Find the maximum number of T. M. High School uniforms that Modern Boutique can complete in a day.
(a) 30 (b) 35 (c) 40 (d) 36

Q.32. On a particular day, Modern Boutique decided to complete 20 T. M. High School uniforms. How many A. R. Academy uniforms can it complete on that day?
(a) 30 (b) 35 (c) 40 (d) 36

Q.33. If Modern Boutique decided to complete 30 T. M. High School uniforms only and no other uniform on that particular day, how many total man-hours will go idle?
(a) 30 (b) 60 (c) 45 (d) 36

Q.34. If Modern Boutique hires one more assistant, what is the maximum number of A. R. Academy uniforms that can be completed in a day?
(a) 60 (b) 48 (c) 120 (d) 50

Q.35. Modern Boutique has the option to hire one more employee of any category. Whom should it hire to get the maximum increase in the production capacity, assuming that it needs to stitch only A. R. Academy uniforms on that day?
(a) One cutter
(b) One tailor
(c) One assistant
(d) Either of the tailor or the cutter

Q.36. Two poles, one 78 m in height above the ground and the other 91 m in height above the ground, are at some distance from each other. Two strings are tied, one from the top of one pole to the bottom of the other and the other from the top of the second pole to the bottom of the first. What is the height above the ground at which the string meets?
(a) 52 m (b) 86 m (c) 42 m (d) 48 m

Q.37. The daily work of 2 men is equal to that of 3 women or that of 4 youngsters. By employing 14 men, 12 women, and 12 youngsters a certain work can be finished in 24 days. If it is required to finish it in 14 days and as an additional labour, only men are available, how many of them will be required?
(a) 20 men (b) 18 men
(c) 48 men (d) 28 men

Q.38. Two Pipes A and B can fill a cistern in $12\frac{1}{2}$ and 15 hours respectively. If the pipes can be opened or closed only after every 30 min (i.e., at 6.00, 6.30 etc.) and the tank is to be filled by using both the pipes without any overflow then,
(a) Pipe B should be turned on later
(b) A should be turned on later
(c) A should be turned off after some time.
(d) Both the pipes should remain on till the tank is full

Q.39. A, B and C are assigned a piece of job which they can complete by working together in 15 days. Their efficiencies (measured in terms of rate of doing job) are in the ratio of 1:2:3. After 1/3rd of the job is

completed, one of them has to be withdrawn due to budget constraint. Their wages per day are in the ratio of 3 : 5 : 6. The number of days in which the remaining two persons can finish the job (at optimal cost) is
(a) 9 (b) 12 (c) 15 (d) 18

Direction for questions 40 and 41: Read the passage below and solve the questions based on it.

A group of 10 people, each with a different capacity to work, is assigned to complete a certain work. They work in such a way that only two people work on any given day, with no group of the same two people working on more than one day. By working in this manner, they could finish only 3/4th of the work by the time all the possible pairs of people had worked once. The remaining part of the work was then finished by all of them working together.

Q.40. What is the number of days for which all of them worked together?
(a) 135 days (b) 120 days
(c) 45 days (d) 3 days

Q.41. If on the last day before the end of the work, the work assigned to the group is doubled, then find the total number of days from the start that the group took to complete the work, assuming that all the people continue to work together?
(a) 240 days (b) 160 days
(c) 90 days (d) 60 days

Direction for questions 42 to 46: Read the passage below and solve the questions based on it.

ABC is a firm that deals with furniture. Manufacturing of a table requires three levels of assembly. The finished table is at the first level. The leg assembly and the tabletop are at the second level. The pieces that go into the leg assembly are at the third level which consist of shortrails, longrails and legs. One unit of the table requires one unit of the tabletop and one unit of the leg assembly. One unit of the leg assembly requires 2 units of shortrails, 2 units of longrails and 4 units of legs. The orders are placed just in time to minimize storage.

The lead time for activities are (Lead time is the waiting time required to complete one activity.):

Parts	*Weeks*
Assemble table	1
Finished leg assembly	1
Purchase legs	1
Purchase shortrails	1
Purchase longrails	1
Purchase table top	2

The availability of the parts at the present time

Parts	*Units*
Table	50
Leg assembly	100
Legs	150
Shortrails	50
Longrails	0
Table top	50

Demand of the finished tables:

Details	*Week 4*	*Week 5*	*Week 6*
Demand (units)	200	150	100

Q.42. For meeting the demand of 200 units of the finished table of week 4, when should the first order of the tabletops be placed?
(a) Week 1 (b) Week 3
(c) Week 5 (d) Week 5

Q.43. What is the net requirement of legs for meeting the demand of the finished table of week 4?
(a) 200 (b) 50 (c) 400 (d) 800

Q.44. When and how many units of shortrail would be placed for meeting the demand of the finished table of week 6?
(a) 100 units in week 1
(b) 200 units in week 3
(c) 300 units in week 6
(d) Data insufficient

Q.45. Due North Inc. needs a new kind of table in which 8 legs and 4 short rails would be needed for manufacturing a single table. What would be the net requirement of legs and shortrails for meeting the total demand of the finished table (including the additional demand of 200 tables for due North inc.) in week 5?
(a) 1800, 900
(b) 2200, 1100
(c) 1600, 800
(d) 800, 400

Q.46. The supplier of longrails has shifted its manufacturing units to a new location and owing to this, the delivery time of the long rail has been increased by 1 week. When should the order of longrails be placed to meet the week 5 demand of the finished table?
(a) Week 0 (b) Week 1
(c) Week 2 (d) Week 3

Q.47. Labour allocation is a very important process. A particular weaving section has 20 looms and with five labourers, loom efficiency is 75%. The production of a loom at 100% efficiency is 10 m/h. Salary of a labourer is Rs 11,000 per month. I removed one labourer due to which the efficiency came down to 70%. How much do I gain or loose due to this action? (Assume that the profit on 1 m cloth is Rs 4 and the looms are working for 30 days in a month and 10 hours per day.)

(a) Rs 1,000 profit

(b) Rs 1,500 loss

(c) Rs 1,500 profit

(d) Rs 1,000 loss

Q.48. To make an article, it takes 40 hour for a workman who is paid Rs 1.80 per hour. 20% the material of is wasted in the course of working, which costs Rs 22.5 per kilogram. At what price must the article be sold so as to yield a profit of 33.33%, if its weight was measured to be 8 kg?

(a) Rs 360 (b) Rs 404

(c) Rs 396 (d) None of these

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(d)	(c)	(a)	(b)	(c)	(b)	(a)	(a)	(a)	(b)	(d)	(a)	(b)	(c)	(b)	(d)	(a)	(c)	(d)

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(d)	(a)	(d)	(c)	(c)	(c)	(c)	(d)	(d)	(b)	(c)	(c)	(a)	(b)	(b)	(b)	(c)	(a)	(a)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(c)	(a)	(c)	(c)	(d)	(c)	(b)	(c)	(c)	(c)	(a)	(a)	(b)	(b)	(d)	(d)	(d)	(a)	(d)	(b)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(b)	(a)	(c)	(c)	(d)	(c)	(d)	(d)	(b)	(d)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(d)	(b)	(b)	(c)	(a)	(b)	(b)	(b)	(b)	(c)	(a)	(d)	(a)	(c)	(d)	(b)	(c)	(c)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(c)	(b)	(d)	(c)	(c)	(b)	(a)	(b)	(a)	(b)	(d)	(a)	(a)	(b)	(d)	(c)	(a)	(a)	(b)	(c)
Q. No.	41	42	43	44	45															
Answer	(d)	(b)	(d)	(a)	(b)															

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(b)	(b)	(c)	(b)	(d)	(b)	(a)	(d)	(c)	(a)	(c)	(c)	(d)	(d)	(b)	(b)	(b)	(d)	(a)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(b)	(d)	(a)	(a)	(c)	(d)	(a)	(b)	(c)	(b)	(c)	(a)	(a)	(d)	(b)	(c)	(a)	(a)	(b)	(d)
Q. No.	41	42	43	44	45	46	47	48												
Answer	(d)	(a)	(b)	(b)	(b)	(b)	(d)	(c)												

HINTS AND SOLUTIONS

Moderate

1. Ratio of efficiency of Mayank and Shishu = 3/2
So ratio of time taken by Mayank and Shishu = 2/3
So if Shishu takens 30 hours, then Mayank will take 20 hours
Shishu in 6 hours = 1/5th work.
Remaining work = 1 – 1/5 = 4/5th work, 1 1 1

Shishu and Mayank together $= \frac{1}{20}+\frac{1}{30}=\frac{1}{12}$

So required time $= \frac{4/5}{1/12} = 9.6$ hours

2. 1 men = 2 women and 1 women = 2 boys.
So 1 men = 2 women = 4 boys.
Alternatively, assume work done by a child = x units/day, then work done by a woman and a man will be equal to $2x$ units per day and $4x$ units per day.
Now 16 men are completing the work in 12 days, so total work done = $4x \times 16 \times 12$. Equate this with the supplied work force to obtain the answer.

4. Let the total amount of work = 180 unit
A does = 2 units/day
B does = 4.5 units/day
C does = 15 units/day
To finish 180 units of work, every body has to work for 9 days.
In 9 days A will do = $9 \times 2 = 18$ units work.

So A's contribution $= \ldots = \frac{18}{180}\ldots\frac{1}{10}$

So out of Rs 240, A's share = $240 \times 1/10 = 24$

6. First three pumps are on for 4.5 hours and two additional pumps for 2 hours. Let any one of the first three pipes takes x hours to fill the tank and it is given that any one of the two additional pumps take 40 hours to fill the tank
So according to question,

$$\frac{4.5}{x}+\frac{4.5}{x}+\frac{4.5}{x}+\frac{2}{40}+\frac{2}{40}=1$$

10. Water filled into the tank by all pipes/hour

$$\frac{1}{10}+\frac{1}{20}+\frac{1}{25}-\frac{1}{100}$$

11. If emptying pipe is closed, then it takes 100/19 hours to fill the tank.
But, only 2/5th of 50,000 = 20,000 litres is filled.
So in 100/19 hours, 30,000 litres water drained
Hence in 1 hour, 5700 litres water drained.
So rate at which each house gets water.

$$= \frac{5700}{2000} = 2.85 \text{ litres/hours}$$

12. For the second time, the total amount of water needed.
$= 40 \times 2000 - 50{,}000 = 30{,}000$ liters, this amount in 3/4th of total
Let pipes A, B and C are opened for x hours.
So, $x\,(1/10 + 1/20 + 1/25) = 3/5 => x = 60/19$

15. Solve using options.

16. Lets assume that digging one well = 40 units work.
So, A does = 4 units/day, B does = 8 units/day.
Now efficiency is falling by 10%, so it is a case of Geometric progression.

$4, \frac{18}{5}, \frac{81}{25}, \ldots$ Where common ratio = 9/10

Now sum of an infinite GP = a/(1−r)

$$= \frac{4}{1-9/10} = 40$$

Now it is clear that 'A' takes infinite time, so he will never finish the work.

17. B does = 8 units/day

$$\text{Using} \frac{4\left[1-(9/10)^2\right]}{1-9/10} = 40 \Rightarrow n \cong 7$$

18. A and B together do = 4 + 18 = 12 units/day
Now proceed the same way, as in the previous question.

19. Let's assume that time to read one passage = x
So according to the question, $3x + 21x / 12 - 13 = 6x + x/12 \Rightarrow x = 4$
So total time taken for answering all questions = $4x\ 4 + 4x\ 27/12 = 25$ minutes
Time spent on first passage = $4 + 4x\ 5/12 = 17/3$ minutes
Required percentage = 22.6%

20. As we have found in the previous question that total time spent = 25 minutes.
If total time spent is cut down by 20%, then time = 20 minutes.
Now, let's assume increased reading speed = x
So $20 = 4x + 4x\ 27/12 \Rightarrow x = 11/4$
Hence percentage increase = 45.45%

25. Let us assume that
A finishes the work alone in a days
B finishes the work alone in b days
C finishes the work alone in c days
So according to the question,

$$\frac{1}{a} = \frac{1}{2}\left(\frac{1}{b} + \frac{1}{c}\right) \quad \text{(i)}$$

$$\frac{1}{c} = \frac{1}{3}\left(\frac{1}{c} + \frac{1}{b}\right) \quad \text{(ii)}$$

$$\text{nd } \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{15} \quad \text{(iii)}$$

Using these options, $C = 60$ days

26. Let's assume the cistern's capacity = 60 litres
So A supplies = 3 litres/min, B supplies = 2 litres/min.
Now, due to closing of valve,
A supplies = = 3 × 4/5 = 12/5 litres/min.
B supplies = = 2 × 5/6 = 5/3 litres/min.
In last 5 minutes cistern is filled at full capacity of pipes, and it is filled by = 5 × 5 = 25 litres.
Remaining capacity = 60 – 25 = 35 litres
So required time ≈ 9 minutes

27. Since the amount of water flowing through each pipe is proportional to square of its diameter so if efficiency of longest pipe (3 cm) = 1/49
Then efficiency of pipe (2 cm) = 4/(49 × 9)
and efficiency of pipe (1 cm) = 1/ (49 × 9)
Now let cistern is filled by all three pipes in x minutes.

$$\text{So, } \frac{1}{49} + \frac{4}{49\times 9} + \frac{1}{49\times 9} = \frac{1}{x}$$

$=> x = 31.5$ minutes

28. In four hours, A + B + C = 144 – (i)
And given, A – B = B – C => A + C – 2B = 0 – (ii)
From (i) and (ii), B = 48, Hence A = 56 and C = 40
So B–C = 48 – 40 = 8 hours.
It is for 4 hours so per hours = 8/4 = 2 hours

32. Ratio of efficiency, P = 2Q – (1)
And P + Q = 3R – (2)
From (1) and (2), Q = R
P = 2Q = 2R
Hence P : Q : R = 2:1:1

34. Pranesh takes 30 days to finish the work.
Since Sagar is 20% more efficient than Pranesh, he will take 16.66% less time.
So Pranesh takes = 25 days
So work done by both of them in one day = (1/30) + (1/25)

35. Let's assume the total amount of work = 32 units
So Manoj and Harvinder does = 2 unit/day
Harvinder does = 1 unit/day
So Manoj does = 1 unit/day
So 32 units of work will be completed in 32 days.

36. To maximize the production, four men will work in each shift.
2 men will work with machines and 2 men work alone.
Total cost incurred in one hour

$$= \frac{2\times 20 + (2\times 20 + 2\times 15)}{6} = 110/6$$

So total cost incurred in 8 hour (per die) = (110/6) × 8 = 146.66

37. Let Suresh take x days to complete the work. So in one day Suresh does = $1/x$.
Given total efficiency of Ramesh and Suresh in one day = 1/P

$$\text{So Ramesh} = \frac{1}{P} - \frac{1}{x}$$

$$\text{According to question} = \frac{q}{P} + r(1/x) = 1$$

$$\text{So, } x = \frac{Pr}{P-q}$$

$$\text{So, Suresh takes} = \frac{Pr}{P-q}$$

$$\text{So, Ramesh takes} = \frac{Pr}{r-P-q}$$

38. Let B take x hours to fill the tank,
So according to question, time required by each
A = $2x$
B = x
C = $2x/3$
D = $2x/4$
E = $2x/5$
A and B together fills the tank in 4 hours
$\frac{1}{2x}+\frac{1}{x}+=...\frac{1}{4}\Rightarrow x=6$
In first case, when A, C and E act as input pipes and B and D as output pipes,
Total time = 1/12 + 1/4 + 5/12 – 1/6 – 1/3 = 4 hours
In 2nd case, when C, D and E act as input pipes and A and B as output pipes,
Total time = 1/4 + 1/3 + 5/12 – 1/12 – 1/6 = 4/3 hours

40. Solve using options.

42. Assume total amount of work = 100 units
A does = 10 units/day, B does = 5 units/day and C does = 4 units/day
Possible Pairs:
A + B = 15 units/day, A + C = 14 units/day, B + C = 9 units/day
To minimize time, we will use the first two pairs.
So, 15 + 14 + 15 + 14 + 15 + 14 + 15 = 102 Units
So, 7 days are required.

45. Let Anuj do the work in x days,
So, Bhanu = $3x$ days, Chandu = $9x$ days and Dodo = $27x$ days
Now, using options,
Anuj + Chandu = $1/x + 1/9x = 10/9x$, So, they take $9x/10$ days
Bhanu + Dodo = $1/3x + 1/27x = 10/27x$, So, they take $27x/10$ days
Now, 1/3 of $27x/10 = 9x/10$, So, Anuj and Chandu is first pair.

Advanced

[redacted]

Let efficiency of Pipe P1 – e
Let efficiency of Pipe P2 – e/2
Let efficiency of Pipe P3 – e/4
Let efficiency of Pipe P4 – e/8 and so on.
Now e, e/2, e/4, e/8… are in Geometric progression
So total efficiency = $\frac{e}{2}=e\frac{e}{4}+...\frac{e}{8}+...+...+...\infty$
All pipes together means can fill the tank in 2 hours.
So pipe P1 with efficiency e will take = 2 × 2 =4 hours

Similarly P2 = 8 hours
P3 = 16 hours
P4 = 32 hours
P5 = 64 hours
P6 = 128 hours

[redacted]

6.
F(A,10,600), E(C,50,20) = + 6000 – 1000
F(A,10,400), E(B,20,200) = + 4000 + 4000
F(A,50,100), F(B,20,50) = + 5000 + 1000
E(C,10,50), F(B,20,75) = – 500 + 1500
Final Volume = 20,000

7. F(A, 10, 500), F(B,40,500) = 5000 + 20,000 = 25000
Now option (b) E(C, 50,500) gives same amount 25000 to be emptied.

9. Let x = Number of days it rained in the morning and had clear afternoons.
y = Number of days it rained in the afternoon and had clear mornings.
z = Number of days it rained in the morning or afternoon
So according to question, $x+y=7$
$x+z=5$
$y+z=6$
Adding all three equations, $x+y+z=9$
So, d = 9 days

10. Let us make a table of the units of work everyday:

Day	1	2	3	4	5	6	7	8	9	10
Length	2	4	6	8	8	11	14	17	20	23

Total length = 113 m

13. Ratio of efficiency of Sanjay : Balram : Pranesh = 1/16 : 1/24 : 1/32 = 6 : 4 : 3
Work finished in 24 hours, so in 1 hour = 1/24 part of work
Pranesh's contribution = 3/13 × 1/24 = 1/104 Part of work
So Pranesh will take 104 days
Answer: Option (c)

14.

Men	*Working Hours*	*Day*	*Work done*
64	8	5	40%
64	n (say)	4	60%

40% work in 40 hours → 60% work in 60 hours.
Hence, working hours = 60/4 = 15 hours.

[redacted]

17. Combined efficiency of four secretaries = 1/2

$$\frac{1}{A}+\frac{1}{B}+\frac{1}{C}+\frac{1}{D}=\frac{1}{2}$$

As it is given in the question that boss's efficiency reduces respectively according as A, B, C and D remain absent, it means their efficiencies are in descending order by trial and error.

[redacted]

20. A worker in a day completes 10% work.
Since entire team works at their 50%

Day 1	*Day 2*	*Day 3*	*Day 4*	*Day 5*	*Day 6*
5%	10%	15%	20%	25%	25%

(Day 6 = 25% because that day worker is absent)
Total = 5 + 10 + 15 + 20 + 25 + 25 = 100%

21. Amount of work = 200%

Day 1	*Day 2*	*Day 3*	*Day 4*	*Day 5*	*Day 6*
10%	20%	30%	40%	50%	60%

Total = 10 + 20 + 30 + 40 + 50 + 60 = 210%
So work is finished on 6th day

22. We can use the data of previous question. On the last day 50% work is required but efficiency to finish 60% work is there so 60% work in 8 hours.
50% work in (8/60) × 50 = 6 hours 40 minutes
They start working at 9.00 am and work till 1.00 pm
So 4 hours passed till 1.00 pm.
Now 2 hours 40 minutes is left, they again started working at 1.30
So 1.30 + 2 hours 40 minutes

23. According to the question, 1/2, 1/a, 1/3 are in A.P.
So 'a' can be calculated.
Now, $\frac{1}{2-x}, \frac{P}{a}, \frac{1}{3-x}$
So x can be calculated.
$x = 2.6$.

36. Let the height above the ground = EF = h.
ΔECF ~ ΔACB

$$\text{So, } \frac{h}{78}=\frac{FC}{BF} \quad \ldots (1)$$

ΔBEF ~ ΔBCD

$$\text{So, } \frac{h}{91}=\frac{FC}{BF+FC} \quad \ldots (2)$$

ΔECF ~ ΔACB

$$(1) \div (2) \ldots = \frac{FC}{BF} \Rightarrow \frac{91}{78}\ 7BF = 6FC \quad \ldots (3)$$

By putting values of equation (3) in equation (1), h = 42 m.

[redacted]

40. Total numbers of possible pairs = $^{10}C_2 = 45$
So each is working 9 times to finish 3/4th of the work.
In one day, all of them working together will do = 1/9 of 3/4 = 1/12th work
Remaining work = 1/4
So required days = 3 days

41. As we have found in a previous problem that working together, they do 1/12th work in a day i.e., they take 12 days to complete the work working together.
Now till the last day they have worked for 45 + 3 = 48 days.
Now for the same work again they will take 12 more days working together.
Hence, total number of days required from start = 48 + 12 = 60 days.

42. Fifty table and 50 tabletops are available so we need to purchase 100 more tabletops. Lead time for purchasing table top in 2 week and lead time for assemble table is 1 week so order should be placed 3 weeks in advance, i.e., Week 1.

47. At 75% efficiency loom works 7.5 m/hour
So, 20 looms @ 10 h/day in 30 days will make = 7.5 × 20 × 10 × 30 = 45000 m cloths
At 70% efficiency loom works = 7 m/hour
So, 20 looms @ 10 h/day in 30 days will make = 7 × 10 × 20 × 30 = 42000 m cloths
So, production of cloths decreases by 45000 – 42000 = 3000 m
So, Loss = 3000 × 4 = Rs 1200000
Saving on a labourer by removing him = 11000

8

CHAPTER

Time, Speed and Distance



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Basic relationship between time, speed and distance
- Different situations like Motion in a straight line, Trains, Races, Boats and Streams, Circular Track, Clocks
- Concepts attached to these motions
- Kinds of questions asked in the CAT
- Methods of solving questions

Introduction to the topic

Time, speed and distance (TSD) is one of the most popular topics in the CAT. Almost 5%–10% questions in the paper are asked from this chapter every year. The concept of time, speed and distance is also used extensively for questions relating to different areas in various other examinations. Some of these areas are: motion in a straight line, boats and streams/escalators, races, circular motion, trains, clocks. Students should try to understand the inter-relationship between the factors time, speed and distance.

RELATIONSHIP BETWEEN TIME, SPEED AND DISTANCE

As we know, Distance = Speed × Time

It means that if a person is running at a speed of 20 km/h and he runs for 2 h, he will be covering a total distance of 40 km.

Distance = 20 × 2 = 40 km

Distance

When an object is moving with a certain speed in a particular time, the displacement made by an object is called the distance.

Unit of Distance Kilometre (km) and metre (m) is usually taken as the unit of distance. Sometimes, mile or feet, etc., can be found as the unit.

Time

Time is defined as a quantity, which governs the order or sequence of an occurrence. In the absence of time, the actual sequence of any occurrence or incident would be lost. If we did not have the concept of time, we would not be able know in what period or in what order something took place.

So, time could be seen as a big building with a number of floors where all the floors are designated according to the occurrence of incidents/events on the respective floors. In our case time shall be seen as the duration of happening of any event.

Unit of Time Hour and second are mostly taken as the unit of time. However, day or minute are also taken as units.

Speed

Speed is defined as the distance covered per unit time. In other words, it is the rate at which the distance is covered.

Unit of Speed Though we commonly take km/hour and metre/sec as the units of speed. Yet, any unit of distance upon any unit of time can be taken as the unit of speed. For example, mile/h, feet/s, mile/s, feet/h, etc.

Conversion from m/s to km/h and vice versa

If speed is given in m/s and it is required to convert it into km/h, then we multiply it by $\frac{18}{5}$ and when speed is given in km/h and we have to convert it into m/s, we multiply it by $\frac{5}{18}$.

$$36 \text{ km/h} = 36 \times \frac{5}{18} = 10 \text{ m/s}$$

$$20 \text{ m/s} = 20 \times \frac{18}{5} = 72 \text{ km/h}$$

MOTION IN A STRAIGHT LINE

This is one of the primary areas of application of time, speed and distance. By using the basic relationship between time, speed and distance the following three different cases are possible:

Case 1 When S (Distance) is constant

$$V \propto \frac{1}{T}$$

So, $V_1/V_2 = T_2/T_1$

It is read as V is inversely proportional to T.

It can be understood in terms of the reciprocal-ratio-multiplication relationship between V and T.

Suppose distance = 1000 km and Speed = 100 km/h

Speed = 100 km/h	Time = 10 h
Speed = 100 × 2	Time = ½ × 10
Speed = 100 × 3	Time = 1/3 × 10
Speed = 100 × ½	Time = 2 × 10

To simplify it, the product stability ratio can be further used. So, the more the speed, the lesser is the time taken and the lesser the speed, more will be the time taken.

Example 1 A man cycles with a speed of 10 km/h and reaches his office at 1 p.m. However, when he cycles with a speed of 15 km/h, he reaches his office at 11 a.m. At what speed should he cycle so that he reaches his office at 12 noon? **(CAT 2004)**

Solution Using the product stability ratio,

The speed is being increased by 50%, so the time taken will reduce by 33.33%.

So, 33.33 % of Time = 2 h

Hence, total time = 6 h

So, distance = 10 × 6 = 60 km

This distance is to be covered in 5 h.

So, speed = 60/5 = 12 km/h

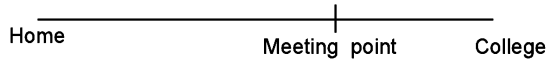
Alternatively, it can be seen that time taken in three (Given) situations are in AP. Hence speeds will be in HP.

Required Speed = Harmonic Mean of two speeds.

So, Required Speed = $\frac{2 \times 15 \times 10}{15 + 10}$ = 12 km/h

Example 2 Siddharth goes by a bike to pick up his girlfriend everyday from college and then drops her at her house. College timings are till 5 p.m. daily, but today the college at 4 p.m. His girlfriend, not finding Siddharth at the college gate, starts walking towards her house. Siddharth, unaware of this fact, leaves his house as usual meets his girlfriend on way, picks her up and drops her at her house. At night, Siddharth realizes that he had saved 40 min that day. What is the ratio of the speed of Siddharth to that of his girlfriend? (Both of them live in the same building).

Solution Let us see the following schematic representation:



The usual route of Siddharth is home-college-home

His route today is – home – meeting point – home.

And, in this way, 40 min are saved. So, he takes 20 min to cover the distance between the meeting point and the college. It can be further concluded that he usually reaches college at

5 P.M., but today he reached at 4:40 (20 min. are saved) and his girlfriend took 40 min. (she starts at 4 P.M.) to cover the distance between her college to the meeting point.

The ratio of time of Siddharth and his girlfriend = 20:40 = 1:2

The ratio of the speed of Siddharth and his girlfriend = 2:1.

Case 2 When T (Time) is constant

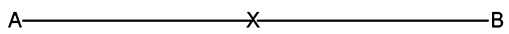
$S \alpha V$

So, $S_1/S_2 = V_1/V_2$

The higher is the speed, the more will be the distance covered and the lower the speed, the lesser will be the distance covered.

We'll see that T constant is a situation specific to meeting point cases.

Example 3 Distance between two points AB = 110 km. Manoj starts running from A at a speed of 60 km/h and Ravi starts running from B at a speed of 40 km/h at the same time. They meet at a point X, somewhere on the line AB. What is ratio of A*X* to B*X*?



Solution Since both Manoj and Ravi are running for the same time, T is constant. Hence, the ratio of the distance covered by them will be the same as the ratio of their speed.

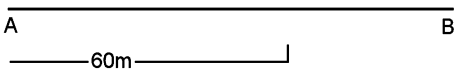
So, AX/BX = Speed of Manoj/Speed of Ravi = 60/40 = 3:2

Some typical meeting point cases

When two persons are running between the ends of a linear track for infinite time:

Example 4 Two robots Mango and Bango start from the opposite ends A and B of a linear track respectively and keep running between the ends for infinite time. They meet for the first time at a point 60 m from A. If AB = 100 m, which point is their point of 4th meeting?

Solution 1st meeting point



The ratio of the speed of Mango and Bango is 60:40 = 3:2. Now Mango is moving towards B and Bango is moving towards A.

For the 2nd meeting, Bango reaches A after covering 60 m, Mango must have covered 90 m in the same time. So, he is at the mid-point of A and B. Now, the distance between Mango and Bango is 50 m. They will cover this distance of 50 m in the ratio of 3:2.

So, the point of their 2nd meeting is



It can be seen here that the sum of the distance covered by both Mango and Bango from the starting till now is 300 m, and the difference between the distance covered between the 1st meeting and the 2nd meeting is 200 m.

So, again they will meet for the 3rd time when they have covered a total distance of 200 m together. Mango and Bango will cover this distance of 200 m in the ratio of 3:2.

Distance covered by Mango = 120 m and distance covered by Bango = 80 m.

So, the 3rd meeting point is point B.

To have a 4th meeting, they will again have to cover a total distance of 200 m. So, the point of their 4th meeting is 20 m from A.

So, we can now generalize the above situation as:

Distance covered by both of them for the 1st meeting = 100 m

Distance covered by both of them for the 2nd meeting = 100 m + 200 m = 300 m

Distance covered by both of them for the 3rd meeting = 300 m + 200 m = 500 m

Distance covered by both of them for the 4th meeting = 500 m + 200 m = 700 m

The ratio of the speed of Mango and Bango is 3:2, so the distance covered by Mango = 420 m

Hence they will meet at 20 m from A.

Example 5 Two persons, Ram and Mohan, start from the same end A of a linear track AB and keep running to and fro for infinite time. They meet for the first time at a point 20 m from B. If AB = 100 m, which point is their point of 4th meeting?

Solution Using the above generalization, distance covered by both of them for the 1st meeting = 200 m

Distance covered by both of them for the 2nd meeting = 200 m + 200 m = 400 m

Distance covered by both of them for the 3rd meeting = 400 m + 200 m = 600 m

Distance covered by both of them for the 4th meeting = 600 m + 200 m = 800 m

The ratio of the speed of Ram and Mohan is 3:2, so the distance covered by Ram = 480 m

Hence they will meet at 80 m from A.

Limitation of above generalization For the meeting to occur after every 200 m, the ratio of the speed of the two runners should be less than 2. If it is more than or equal to 2, then the problems can only be evaluated on the basis of actual calculation.

Case 3 When V (Speed) is constant

$S \alpha T$

So, $S_1/S_2 = T_1/T_2$

In layman terms, if a person is running with a speed of 20 km/h, then the ratio of the distance covered in one hour to the distance covered in two hours will be 1:2.

BOATS AND STREAMS/ESCALATOR



Boats and streams should be ideally seen as just a logical extension of the motion in a straight Line, with distance being constant.

As we know, if the distance is constant then $V \alpha 1/T$.

Basic Terminology

Downstream movement When the direction of the movement of a river and a boat is the same, their collective movement is known as the downstream movement. And the distance covered by boat is known as downstream distance.

If the speed of the River = R and the speed of the boat = B, then Downstream Speed = B + R

Upstream movement When the direction of the movement of the river and a boat is opposite, they are said to be in upstream movement. The distance covered in this case is known as upstream distance.

If the speed of the river = R and the speed of the boat = B, then upstream speed = B – R (Conventionally the speed of one boat is taken more the than speed of the river otherwise the boat would not be able to go back.)

Now, speed of boat = ½ (downstream speed + upstream speed) = ½ (B + R + B – R) = B

And speed of river = ½ (downstream speed – upstream speed) = ½ (B + R – B + R) = R

Hence, if downstream speed and upstream speed are given as 20 km/h and 10 km/h respestively, then the speed of the boat = 15 km/h and speed of the river = 5 km/h.

In most of the cases of boats and streams, the distances covered downstream and upstream are the same. In those cases, the ratio of the time taken becomes inverse of the ratio of the speeds.

Time taken downstream: Time taken upstream = upstream speed: downstream speed

Example 6 The speed of the boat in still water is 6 km/h and the speed of the river is 1.2 km/h. Boat takes a total of 10 h to go to a place and come back. What is the total distance covered in the whole process?

Solution Let us assume D is the distance.

Upstream Speed = 4.8 km/h

Downstream speed = 7.2 km/h

According to the question, D/4.8 + D/7.2 = 10

So, D = 28.8 km and hence the total distance = 57.6 km

Alternatively, the ratio of downstream speed: upstream speed = 3:2

Ratio of the downstream time: upstream time = 2:3

The time taken in the downstream movement = 4 h and the time taken in the upstream movement = 6 h

So, the distance covered = $4 \times 7.2 = 6 \times 4.8 = 28.8$ km

Hence the total distance = 57.6 km

In the case of escalators, moving staircase works like an external agent as the river works for boats and streams. The speed of an escalator and the person will be added when the staircase is going up and the person walking up with it have the same direction of the movement.

Now if the direction of the movement of an escalator and the person are opposite, then the resultant speed (or, the relative speed) will be equal to the speed of the person – to the speed of an escalator.

Example 7 A man can walk up in a moving escalator (upwards) in 30 s. The same man can walk down this moving 'up' escalator in 90 s. Assume that this walking speed is the same both upwards and downwards. How much time will he take to walk up the escalator when it is not moving?

Solution Let us assume that the speed of the man = m steps/s and the speed of the escalator = e steps/s

Distance covered while going up = 30 m + 30 e

Distance covered while going down = 90 m – 90 e

Now, these two are equal.

So, 30 m + 30 e = 90 m – 90 e

Or, 60 m = 120 e, Hence, 1 m = 2 e

So, the total length of escalator = 45 m

So, the time taken by the man to cover the whole escalator = Distance/Speed = 45m/m = 45 s

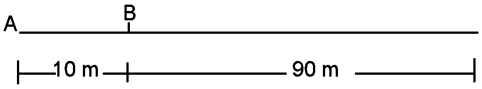
Alternatively, Answer would be Harmonic Mean of the given time = $\frac{2 \times 30 \times 90}{30 + 90} = 45$ secs.

[See the solution of Example 1 on last page].

RACES

Basic Statements

1. A gives a start of 10 m to B → When B has already run 10 m, then A starts running.

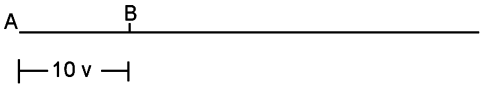


Example 8 In a race of 100 m, A gives a start of 10 m to B. Despite this, A wins the race by 20 m. What is the ratio of the speed of A and B?

Solution Time taken by A to cover 70 m = Time taken by B to cover 100 m

Since the distance is constant, the ratio of speed of A and B = 10:7

2. A gives a start of 10 secs to B → B has already run for 10 secs, now A starts running.



Where v m/s is the speed of B.

Example 9 In a 100 m race, Tom runs at a speed of 1.66 m/s. If Tom gives a start of 4 m to Jerry and still beats him by 12 s, what is the speed of Jerry?

Solution Time taken by Tom to cover 100 m = 60 s

Now, since Tom beats Jerry by 12 s, time taken by Jerry = 72 s

And the distance covered by Jerry = 96 m

So, speed = 96/72 = 1.33 m/s

Example 10 Karan and Arjun run a 100-m race where Karan beats Arjun by 10 m. To do a favour to Arjun, Karan starts 10 m behind the starting line in a second 100 m race. They both run at their earlier speeds. Which of the following is true in connection with the second race?

1. Karan and Arjun reach the finishing line simultaneously.
2. Arjun beats Karan by 1 m.
3. Arjun beats Karan by 11 m.
4. Karan beats Arjun by 1 m.

Solution Situation (I)

In whatever time Karan covers a distance of 100 m, Arjun covers 90 m in the same time.

Situation (II)

Karan is 10 m behind the starting point. Once again to cover 100 m from this new point Karan will take the same time as before. In the same time Arjun will cover only 90 m. This means that both of them now will be at the same point, which is 10 m away from the finish point. Since both of them are required to cover the same distance of 10 m and Karan has a higher speed, he will beat Arjun. There is no need for calculations as option (4) is the only such option.

CIRCULAR MOTION

In the case of races and motions in straight line, we have observed that if the two bodies or persons are moving with different speeds in a straight line in one direction, then they will never meet. This is due to the fact that with the passage of time, the distance between them is increasing constantly.

Circular motion should be seen as a logical extension of Races where runners are running on a circular track. Since it an enclosed track (by virtue of it being circular), runners are bound to meet at some point or the other.

Case 1 When two or more than two persons are running around a circular track in the same direction.

Example 11 When will they meet for the first time anywhere on the track?

Solution To understand the situation completely, let us assume that there are two persons A and B. Speed of A = 20 m/s, speed of B = 10 m/s, length of the track is 1000 m and they are running in the same direction. It can be seen in figure 1 that initially both of them are at the same point i.e., the starting point.

They will be meeting for the first time only if the faster runner A has taken one more round of the track than the slower runner B. This can be interpreted as—A will have to cover 1000 m more than B.

It is understood with the figures given above that the distance will keep on increasing between them with the passage of time. And the moment distance between them becomes equal to 1000 m, they will be at the same point.

So, the time taken = distance/relative speed = $\frac{1000}{10}$

= 100 s

Or, this can be done by using unitary method also:

Distance of 10 m is created in 1 s

So, the distance of 1000 m will be created in 100 s

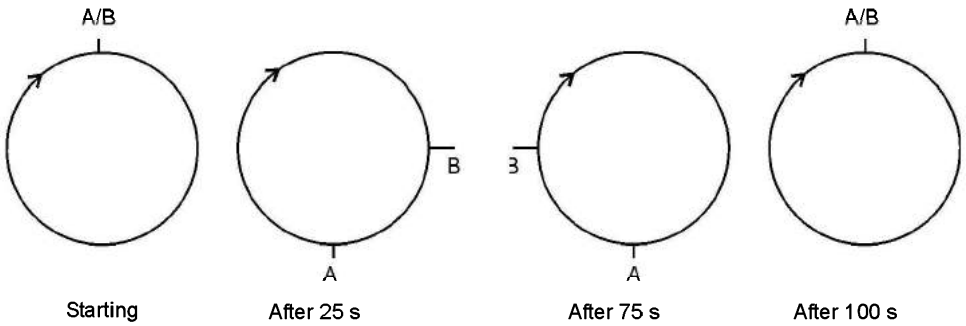
Now, let us assume that there are three persons A, B and C running with following speeds in the same direction:

Speed of A = 30 m/s

Speed of B = 20 m/s

Speed of C = 10 m/s

To calculate when will they meet for the first time, we are required to find the time taken by the fastest runner to take one round over the other runners.



Time taken by A to take one round over B = t_{A-B} = 1000/10 = 100 s

Time taken by A to take one round over C = t_{A-c} =1000/20 = 50 s

Now, the LCM of these two values t_{A-B} and t_{A-c} will give us the time after which all of them will be meeting at the same place.

LCM = (100, 50) = 100 s

It can also be seen that they will be meeting after every 100 s.

Example 12 When will they meet for the first time at the starting point?

To calculate this, we will use the concept of LCM (Usage of LCM and HCF, chapter 2, case 2)

Find the time taken by each individual to take one round and then calculate LCM of these values.

Assume there are three persons A, B and C with a respective speed of 30 m/s, 20 m/s and 10 m/s running in the same direction. Length of the circular track is 1000 m.

Time taken by A to take one round = t_1 = 1000/30 = 33.33 s

Solution Time taken by B to take one round = t_2 = 1000/20 = 50 s

Time taken by C to take one round = t_3 =1000/10 = 100 s

LCM of t_1, t_2, t_3 = 100 sec.

Example 13 At how many different points of the track will they be meeting?

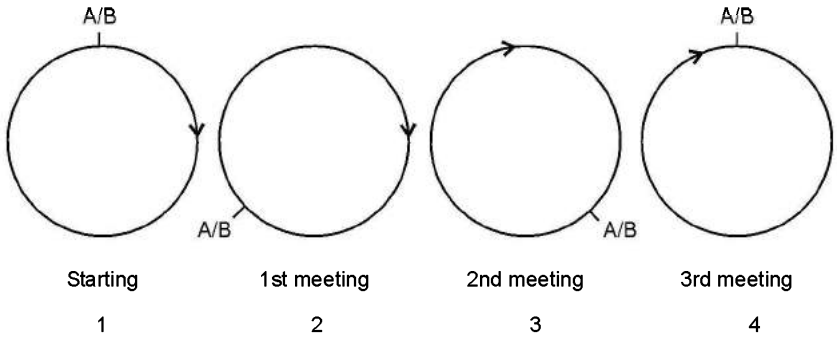
Solution Let us assume that the speed of A = 25 m/s and the speed of B = 10 m/s and the length of the track =1000 m

They will be meeting for the first time after a time-gap of 1000/15 = 66.66 s

Till this time, A has covered 1666.66 m and B has covered 666.66 m. This point is 666.66 m from the starting point. Now, this point can be assumed to be the starting point.

So, they will meet at a distance of 666.66 m from here. This is the second meeting point, at a distance of 333.33 m from the starting point. Next meeting point will be 666.66 m from here. This point will be nothing but the starting point again (Fig.1 and Fig.4 are same).

This can be seen through the figures given below:



So, there are a total of 3 distinct meeting points on the track.

In general, number of meeting points = difference of ratio of the speed the of A and B in its simplest form.

Ratio of speed of A and B = 5:2

So, the number of different meeting points = 5 – 2 = 3 points

Case 2 When two or more than two persons are running around a circular track in the opposite direction.

Here again there are two persons A and B with a speed of 20 m/s and 10 m/s respectively, and length of track is 1000 m.

Example 14 When will they meet for the first time anywhere on the track?

Solution Since they are running in the opposite direction, relative speed =10 + 20 = 30 m/s

So, time taken = distance/relative speed = 1000/30 = 33.33 s

Example 15 When will they meet for the first time at the starting point?

First, we will calculate the time taken by each individual to take one round and then calculate the LCM of those values.

Time taken by A to take one round = t_1 = 1000/20 = 50 s

Time taken by B to take one round = t_2 = 1000/10 = 100 s

LCM of (t_1, t_2) = 100 s

Example 16 At how many different points of the track will they be meeting?

Solution They are meeting after 33.33 s for the first time. Till this time, A has covered 666.66 m and B has covered 333.33 m. So, obviously they are meeting at a distance of 666.66 m from starting point in the direction of A. Next point will be again 666.66 m ahead of this point. And, the next point will be another 666.66 m ahead of this point, which will be the starting point.

So, a total of 3 points will be there.

In general, number of distinct meeting points = addition of the ratio of the speed of A and B in its simplest form.

The ratio of speed of A and B = 2:1

So, the number of different meeting points = 2 + 1 = 3 points

Example 17 Anup and Shishir start running from the same point of a circular track at the same time. After how much time will Anup and Shishir, who are running with a speed of 35 m/s and 40 m/s, respectively, meet at diametrically opposite point?

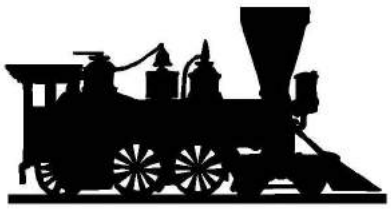
Solution The simplest ratio of speed of Anup and Shishir = 7:8

So, if they are running in the same direction, they will meet at 1 point and if they are running in the opposite direction, they will meet at 15 different points.

Now, for them to meet at a diametrically opposite point, there should be atleast two meeting points or the number of meeting points should be a multiple of 2.

Since, they would meet either at 1 point or at 15 different points, depending on the direction of their movement, they will, therefore, not meet at a diametrically opposite point.

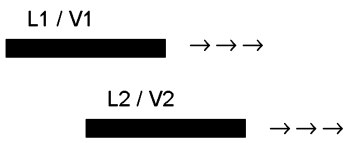
TRAINS



We know that when the direction of the movement of a boat and a river is the same, the relative speed is obtained by adding the speeds of both, the boat and the river. But if two trains are moving in the same direction, then what is the relative speed?

Let us see some cases:

1. When two trains of length L_1 and L_2 and speed V_1 m/s and V_2 m/s respectively are crossing each other:
   i. The direction of the movement of both the trains are the same:



   Relative speed = $|V_1 - V_2|$
   Total distance covered = $L_1 + L_2$
   ii. The direction of the movement of both the trains are opposite:



   Relative speed = $|V_1 + V_2|$
   Total distance covered = $L_1 + L_2$

2. When a train is crossing a stationary object:
   i. When the train is crossing a pole or a stationary human being:



(Fig. 1)



(Fig. 2)

Let us assume that A is a pole. In figure 1, the front of the train is about to cross the pole and in figure 2, the tail of the train has just crossed the pole. It is understood here that the train has crossed its whole length with respect to the pole. So, when the train is crossing any stationary object of negligible width, total distance covered is its own length.

Relative speed = $V_1 + V_2$, since $V_2 = 0$, then, the relative speed = V_2

Total distance covered = $L_1 + L_2$, since L_2 (width of the pole) is negligible with respect to L_2 (Length of the train), so we do not consider it while calculating the quantities. Thus, distance = L_1

However, it should be remembered that this is mathematically not correct and all the solutions are on the assumption that the width of the pole is zero, which is obviously not true.

ii. When the train is crossing a platform or a standing train:

Relative speed = $V_1 + V_2$,

Where V_1 is the speed of the moving train and V_2 is the speed of the standing train or the platform.

Since $V_2 = 0$, so the relative speed = V_1

Total distance covered = $L_1 + L_2$

Where L_1 is the length of the moving train and L_2 is the length of the standing train or the platform.

Example 18 A train takes 10 s to cross a pole and 20 s to cross a platform of length 200 m. What is the length of train?

Solution The train takes 10 s to cross its own length and 20 s to cross its own length and length of the platform. So, it is infered that the train takes 10 s to cross the platform and 10 s to cross its own length.

Since the time taken to cross the platform = time taken to cross its own length

So, length of the platform=length of the train=200 m

Example 19 Speed of a train is 36 km/h. It takes 25 s to cross a pole. What is the length of this train?

Solution Speed of train = 10 m/s (36 × 5/18)

Distance covered = 10 × 25 = 250 m

So, length of train = 250 m

Some Special Cases

Case 1 Two trains are moving in an opposite direction with a speed of V_1 and V_2. Their lengths are L_1 and L_2. Now, see the whole situation from the point of view of a person sitting on the window seat of the 1st train.

Relative speed = $V_1 + V_2$ (This person can be assumed to be running with a speed of V_2)

Relative distance = L_2

Case 2 A train is running with a speed V_1 and a person X is running inside the train with a speed of V_2 in the direction of the movement of train. Now if a person Y is watching this from the outside of the train, then the relative speed of Y with respect to $X = V_1 + V_2$.

Speed of person X with respect of another person Z who is sitting in the train = V_2

There is also a person P who is outside the train and is moving with a speed of V_3 in the opposite direction of train.

Relative speed of P with respect to person

$X = V_1 + V_2 + V_3$

Had this person P been running in the same direction as that of the train, then the relative speed of P with respect to person $X = |V_1 + V_2 - V_3|$

CLOCKS



A clock is a typical example of a circular motion where length of the track is equal to 60 km (Assume 1 min = 1 km). Now on this track, two runners i.e., hour hand and minute hand are running with a speed of 5 km/h and 60 km/h respectively. Since the direction of their movement is the same, so the relative speed = 55 km/h.

Example 20 When will the hour hand and the minute hand of a clock be together between 1 and 2?

Solution Hands have to be together in between 1 o'clock to 2 o'clock.

At 1 o'clock, the distance between hour and minute = 5 km

And the relative speed = 55 km/h

Time = 5/55 h = 1/11 h = 60/11 min = 5 5/11 min. = 5300/11 s = 5:27.27s

So, the hour hand and the minute hands will be together at 1:05:27.27 s

Students can learn this value 5 min 27.27 s as a standard result.

So, both the hands will meet at

1:05:27.27 – Between 1 o'clock and 2 o'clock

2:10:54.54 – Between 2 o'clock and 3 o'clock

3:16:21.81 – Between 3 o'clock and 4 o'clock

and so on.

Example 21 How many times in a day will the hands of a clock be together?

Solution Using the data from the above question, hands of a clock meet at a regular interval of 5 min 27.27 s. So, the number of times they will the meet = $\frac{60}{5:27.27}$ = 11.3 times.

So, the hands will meet for a total of 11 times.

However, it can also be observed that the hands of a clock meet once every hour except in between 11–1. They meet just once in between 11–1. So, they are meeting for 11 times.

Degree Concept of Clocks

Total angle subtended at the centre of a clock = 360°.

Angle made by an hour hand at the centre per hour = 30° per hour, or, 0.5° per min

Angle made by the minute hand at the centre per hour = 360° per hour, or, 6° per min

Example 21 solving 20 by this method,

Angle between an hour hand and the minute hand at 1 o'clock = 30°

Relative speed (in terms of angle) = 5.5°/h

So, time taken = 30°/5.5° = 60/11 min

Example 22 Mr Binod Kumar Roy goes to a market between 4 P.M. and 5 P.M. When he comes back, he finds that the h hand and the min hand have interchanged their positions. For how much time was he out of his house?

Solution Since hands are interchanging their position, minute hand is taking the place of an hour hand and an hour hand is taking the place of min hand. So sum of the angles formed by h hand and min hand = 360°

Let us assume that he was out of house for 't' min.

So, the angle formed by min hand = $6 \times t$ and by hour hand = $0.5 \times t$

So, $0.5 \times t + 6 \times t = 360$

Or, $6.5 \times t = 360$

So, $t = 55.38$ min

Important Derivations

→The number of times hands of a clock are in a straight line (either at 0° or at 180°) in 24 h = 44

→The number of times hands of a clock are at a right angle (at 90°) in 24 h = 44

→Both the hands of clock are together after every $65\frac{5}{11}$ min

(So if both the hands of the clock are meeting after every 65 min or anything less than $65\frac{5}{11}$ mins, then the clock is running fast and if both the hands of the clock are meeting after every 66 min or anything more than $65\frac{5}{11}$ min, then clock is running slow.)

PRACTICE EXERCISES

WARM UP

Q.1. A man is running at the speed of 36 km/h, then how many m does he cover in 40 s?
(a) 400 m (b) 300 m
(c) 2,400 m (d) 100 m

Q.2. The speed of a train is 20 m/s. The time it takes to cross a telephone pole is 10 s. What is the length of the train?
(a) 100 m (b) 1.5 km
(c) 150 m (d) 200 m

Q.3. A train having a length of 400 m, travelling at a speed of 15 m/s crosses a platform in one min. What is the length of the platform?
(a) 2000 m (b) 500 m
(c) 400 m (d) Cannot be determined

Q.4. Speed of a train is 88 km/h. What time will it take to cross a platform 180 m long if the length of the train is 150 m?
(a) 11½ s (b) 10½ s (c) 13½ s (d) 14½ s

Q.5. A certain distance is covered at a speed V km/h. If half of the same distance is covered in double the time, then the ratio for the former speed to that of the latter is
(a) 4:1 (b) 1:4 (c) 2:1 (d) 1:2

Q.6. A person can row a distance of one km upstream in ten min and downstream in four min. What is the speed of the stream?
(a) 4 km/h (b) 9 km/h
(c) 5.6 km/h (d) 4.5 km/h

Q.7. Anil calculated that it will take 45 min to cover a distance of 60 km by his car. How long will it take to cover the same distance if the speed of his car is reduced by 15 km/h (approximate)?
(a) 36 min (b) 55 min
(c) 48 min (d) 40 min

Q.8. A car travels from Patna to Jehanabad at a speed of 65 km/h in one hour. If the speed is reduced by 15 km/h then, how much more time will the car take to cover the same distance?
(a) 12 min (b) 16 min
(c) 18 min (d) 44 min

Q.9. In a kilometer race, A beats B by 10 m. In a two kilometers race, A beats B by
(a) 10 m (b) 20 m (c) 40 m (d) 25 m

Q.10. A train, 225 m long, is running at 45 km/h. In what time does it cross a man running at 18 km/h in the same direction?
(a) 20 s (b) 30 s (c) 25 s (d) 15 s

Q.11. Two trains started from a station, at the same time in the same direction, at speeds of 70 km/h and 60 km/h. What will be the distance between the two trains after 6 h?
(a) 42 km (b) 54 km
(c) 60 km (d) 84 km

Q.12. A man can row half a km against the current in 12 min and returns in 6 min. Find the speed of the current.
(a) 1.25 km/h (b) 1.5 km/h
(c) 2.5 km/h (d) 3 km/h

Q.13. In a kilometer race, Amit beats Bahadur by 100 m and Bahadur beats Chandra by 200 m. By how many metres does Amit beat Chandra in the same race?
(a) 100 m (b) $83^1/_3$ m
(c) 68 m (d) 280 m

Q.14. On a river, Q is the mid-point between two points P and R on the same bank of the river. A boat can go from P to Q and back in 12 hours, and from P to R in 16 hours 40 min. How long would it take to go from R to P?
(a) $3\frac{1}{3}$ h (b) 5 h
(c) $6\frac{2}{3}$ h (d) None of these

Q.15. Two trains are running on a parallel line in the same direction at a speed of 50 km/h and 30 km/h respectively. Given that the faster train crosses a man in the slower train in 18 s, what is the length of the faster train?
(a) 170 m (b) 100 m
(c) 98 m (d) 85 m

Q.16. In a kilometre race, Ajay beats Bijay by 100 m and Bijay beats Chand by 100 m. By how many m does Ajay beat Chand in the same race?
(a) 100 m (b) 200 m
(c) 190 m (d) 119 m

Q.17. Two persons are walking with the speed A and B respectively. If the first person takes ten min less to cover

a distance, what is the time taken by the second person to cover the same distance provided A : B = 3:2?
(a) 20 min (b) 30 min
(c) 10 min (d) 50 min

Q.18. A train of length 200 m crosses a pole in 15 s and crosses a platform in 60 s. What is the length of the platform?
(a) 450 m (b) 600 m
(c) 800 m (d) None of these

Q.19. Hands of a clock are meeting after every 65 min. We can conclude that the clock is running
(a) Fast
(b) Slow
(c) Cannot be determined

Q.20. In 12 hours, how many times the hour and the minutes hands of a clock will be at 0° to each other?
(a) 11 (b) 12
(c) 13 (d) None of these

FOUNDATION

Q.1. When the speed of a train is increased by 20%, it takes 20 min less to cover the same distance. What is the time taken to cover the same distance with the actual speed?
(a) 2 h (b) 1 h (c) 1.5 h (d) 2.5 h

Q.2. Difference between the time taken in covering a certain journey decreases by 45 minutes when the speed is increased from 10 km/h to 20 km/h. What is the difference between the time taken when the same distance is covered at a speed of 40 km/h and 25 km/h?
(a) 12.5 min (b) 13.5 min
(c) 18 mins (d) None of these

Q.3. Shramjeevi express goes to Patna from New Delhi at a speed of 60 km/h. At what speed (in km/h) should Shramjeevi express return from Patna to New Delhi so that the average speed during the whole journey is 120 km/h?
(a) 180 (b) 240
(c) 120 (d) This is not possible

Q.4. Sona travels 600 km to his house partly by train and partly by car. He takes 8 hours if he travels 120 km by train and the rest by car. He takes 20 min more if he travels 200 km by train and the rest by car. Find the speed of the train in km/h.
(a) 45 (b) 60
(c) 75 (d) None of these

Q.5. Sama traveled the first half of the distance from her house to school at a speed of 10 km/h and for exactly half of the remaining time she travelled at 15 km/h and the rest of the time at 20 km/h. Find her average speed.
(a) 13 km/h (b) $12\frac{8}{11}$ km/h
(c) 15 km/h (d) None of these

Q.6. Shilu can go 30 km upstream and 44 km downstream in 10 h. Also, she can go 40 km upstream and 55 km downstream in 13 h. What is her speed in still water?
(a) 6 km/h (b) 12 km/h
(c) 8 km/h (d) 16 km/h

Q.7. A train travels a distance of 192 km at an average speed of 80 km/h from station P to another station Q. It then travels back to P taking 5 hours 36 min. Find the average speed of the train on the return journey to P.
(a) 72 km/h (b) 36 km/h
(c) 20 km/h (d) 60 km/h

Q.8. Two persons, 27 km apart, start at the same time and are together in 9 h if they walk in the same direction. But, if they walk in the opposite directions they are together in 3 h. What are their speeds (in km/h)?
(a) 5, 4 (b) 4, 3 (c) 6, 3 (d) 8, 7

Q.9. In a race of 300 m, Abhishek beats Bijay by 30 m while Bijay beats Chandan by 50 m. By what distance should Abhishek beat Chandan in the same 300 m race?
(a) 80 m (b) 40 m (c) 160 m (d) 75 m

Q.10. What is the speed of a train, which overtakes a cow walking at a speed of 5 km/h in 30 s, if the train is 274 m long?
(a) 51.88 km/h (b) 27.88 km/h
(c) 37.88 km/h (d) 21.67 km/h

Q.11. In eleven hours C walks 12.5 km less than what D does in twelve h; and in five h D walks 3.25 km less than what C does in seven h. How many km does each walk per hour?
(a) 3 and 4 (b) 3.5 and 4.25
(c) 2.33 and 3.4 (d) None of these

Q.12. The ship Vikrant starts from a point P towards a point Q at noon and at 1.00 pm ship, Viraat starts from Q towards P. If ship Vikrant is expected to complete the voyage in 6 h and ship Viraat is moving at a speed of

2/3rd of that of ship Vikrant, at what time are the two ships expected to meet?
(a) 4 pm (b) 4.30 pm
(c) 3 pm (d) 2.30 pm

Q.13. Amit Kumar starts driving from Patna to Quillon at 9 a.m and Bhartendu starts driving from Quillon to Patna at 10 a.m Bhartendu is 50% faster than Amit Kumar. At what time will they meet if Patna and Quillon are 300 km apart and Amit Kumar's speed is 50 km/h?
(a) 12.30 pm (b) 12 noon
(c) 11 am (d) 11.30 am

Q.14. A ship, 156 km away from the seacoast springs a leak, which admits $2\frac{1}{3}$ metric tons of water in $6\frac{1}{2}$ minutes, but the pumps throw out 15 metric tons of water in 1 h. 70 metric tons would be sufficient to sink the ship. What should be the average speed (in km/h) of the ship so that it may just reach the shore as she begins to sink?
(a) 14.5 (b) 15 (c) 18 (d) 20

Q.15. A man rows a boat a certain distance upstream and then returns to the same place. If the time taken by him in going upstream is twice the time taken in rowing downstream, find the ratio of the speed of the boat in still water and the speed of the stream.
(a) 2:1 (b) 3:2 (c) 5:3 (d) 3:1

Q.16. Excluding stoppages, the speed of a bus is 54 km/h and including stoppages it is 45 km/h. What is the stoppage time of the bus (in minutes) per h?
(a) 10 (b) 12 (c) 9 (d) 20

Q.17. For covering a distance of P km, a man takes t h. He takes 40 min less if he moves 3 km/h faster, But if he moves slower by 2 km/h, he takes 40 min more. Then P is equal to.
(a) 22.5 km (b) 32.5 km
(c) 42.5 km (d) None of these

Q.18. Two trains start at the same time from A and B and proceed towards B and A at 36 km/h and 42 km/h respectively. When they meet, it is found that one train has moved 48 km more than the other. Then the distance between A and B (in km) is
(a) 624 (b) 636 (c) 544 (d) 460

Q.19. Anand can beat Bidhan by 5 m in a 100 m race and Bidhan can beat Chandan by 10 m in a 200 m race. Then in a race of 400 m, Anand can beat Chandan by
(a) 40 m (b) 39 m (c) 15 m (d) 10 m

Q.20. A man can swim with the stream at the rate of 3 km/h and against the stream at the rate of 2 km/h. How long will he take to swim 7 km in still water?
(a) 3 h (b) 2.8 h (c) 2.6 h (d) 3.2 h

Q.21. Points P and Q are 120 km apart. A motorcyclist starts from P to Q along a straight road P–Q with a speed of 30 km/h. At the same time a cyclist starts along a road perpendicular to road PQ, with a speed of 10 km/h. After how much time will the distance between them be the least?
(a) 3 h (b) 3.4 h
(c) 4 h (d) Cannot be determined

Q.22. A and B start at the same time from P and Q (55 km apart) to Q and P at 3 and 2½ km/h respectively. They meet at R and then reach Q and P to return immediately and meet again at S. The distance from R to S is
(a) 10 km (b) 12 km
(c) 15 km (d) 16 km

Q.23. A runs 1.75 times as fast as B. If A gives B a start of 60 m, how far should the winning post be so that A and B reach there at the same time?
(a) 140 m (b) 70 m
(c) 210 m (d) 175 m

Q.24. X, Y and Z walk at 6, 12, and 18 km/h respectively. They start from Jhumritalaiya towards Delhi at 2, 5, and 7 p.m. respectively, when Y catches up with X, Y sends X back with a message to Z, when will Z get the message?
(a) 9.45 pm (b) 9 pm
(c) 8.45 pm (d) 9.15 pm

Q.25. A train 75 meters long overtook a person who was walking at the speed of 6 km/h in the opposite direction and passed him in $7\frac{1}{2}$ s. Subsequently, it overtook a second person, walking in the same direction as the first person and passed him in $6\frac{3}{4}$ s. At what rate was the second person travelling?
(a) $11\frac{3}{4}$ km/h (b) 10 km/h
(c) 8 km/h (d) $4\frac{1}{2}$ km/h

Q.26. Alok rows a boat against a stream flowing at 2 km/h for a distance of 9 km, and then turns around and rows back with the current. If the whole trip lasts 6 h, find Alok's rowing speed in still water.
(a) 4 km/h (b) 3 km/h
(c) 2 km/h (d) 5 km/h

Q.27. A man walked 12 km at a certain rate, and then 6 km farther at a rate of $\frac{1}{2}$ km/h faster. If he had walked the whole distance at a faster rate, his time would have been 20 min less. How long did he really take to walk the distance of 18 km?
(a) 4.5 h (b) 5.33 h
(c) 5.66 h (d) None of these

Q.28. Shramjeevi Express leaves Patna for New Delhi at 10.10 am and reaches New Delhi at 3.30 pm. Magadh

Express leaves New Delhi at 9.48 am and reaches Patna at 2.36 pm. If routes of both the trains are the same, then at what time will both the trains meet each other?
(a) 12.52 pm (b) 1.12 pm
(c) 1.32 pm (d) None of these

Q.29. A can give a start of 20 m to B and B can give a start of 10 m to C in a race of 200 m. By how many metres can A beat C in the same race?
(a) 21 m (b) 28 m (c) 29 m (d) 31 m

Q.30. Find the distance between 2 towns when by increasing the speed to 7 km/h a train finishes the journey in 1 h less, and when the speed is reduced to 5 km/h, the train finishes the journey in 1 h more.
(a) 210 km (b) 180 km
(c) 240 km (d) 164.4 km

Q.31. Preeti and Uma stay next to each other and study in the same school. Preeti and Uma start walking towards their school at the speed of 2 km/h and 3 km/h respectively. The faster of the two reaches the school first, turns around and starts walking back. If Preeti and Uma meet 200 m away from the home, what is the distance between their residences and the school?
(a) 260 m (b) 240 m (c) 200 m (d) 250 m

Q.32. Vinit has to cover a certain distance of 7 km in 1 h 12 min. If he walks some distance at 4 km/h and rides a horse for some distance at 10 km/h, he reaches his destination in time. How far should he walk on foot?
(a) 14/3 km (b) 11/3 km
(c) 13/3 km (d) 10/3 km

Q.33. A starts driving from P to Q at 9 a.m. and B starts driving from Q to P at 10 a.m. B is 50% faster than A. What is the time when they meet if P and Q are 300 km apart and A's speed is 50 km/h.
(a) 12.30 pm (b) 12 noon
(c) 11.00 am (d) 11.30 am

Q.34. Binod and Neeraj start from one end of a 1000 m track while Deleep starts from the other end. Binod is 50% faster than Neeraj and Deleep is 150% faster than Neeraj. If Binod meets Deleep in 25 seconds, how long does Neeraj take to meet Deleep?
(a) 30 s (b) 35 s
(c) $28^4/_7$ s (d) $26^3/_7$ s

Q.35. Vinod and Kaurvaki start travelling in the same direction at 8 km/h and 13 km/h respectively. After 4 hours, Vinod doubled his speed and Kaurvaki reduced his speed by 1 km/h and reached the destination together. How long did the entire journey last?
(a) 3 h (b) 6 h (c) 9 h (d) 12 h

Q.36. A bus B1, left a town T1 for another town T2 at 6.00 am at 7.30 am. Another bus B2 also left T1 for T2 and the speed of this bus was 5 km/h more than that of B1. At 10.30 pm on the same day the buses were 21 km apart. Find the speed of bus B1.
i. 36 ii. 64 iii. 72
(a) i or ii only (b) i or iii only
(c) Either i or ii or iii (d) None of these

Q.37. Two men leave places A and B simultaneously. One of them left A for B and the other B for A. Both travel at their own uniform velocity. Having arrived at their destination, they turn back without stopping and travell to their starting points. For the first time they met at 12 km from B, the second time at 6 km from A. Find the distance AB.
(a) 72 km (b) 30 km
(c) 45 km (d) 60 km

Q.38. P and Q start running simultaneously – one from point A to B and the second from point B to A. P's speed is 6/5th of Q's speed. If after crossing Q, P takes 21/2 hours to reach B, how much time does Q take to reach A after crossing P?
(a) 3.36 min (b) 3.48 min
(c) 4.12 min (d) None of these

Q.39. A beats B by 20 m in a 100 m race and B beats C by 20 m in a 100 m race. How much start should A give to C in a 100 m race so that both of them reach the winning post at the same time?
(a) 30 m (b) 44 m (c) 32 m (d) 36 m

Q.40. Mahima beats Kareena in 1000 m race by 50 s and Bhumika by 450 m. If Kareena and Bhumika run 1000 m race, Kareena wins by 40 s. How much time does Bhumika take to run 1000 m?
(a) 210 sec (b) 225 sec
(c) 180 sec (d) 200 sec

Q.41. Two trains have respective lengths as 230 m and 190 m. They cross each other completely in 21 s, if they are travelling in the opposite direction and in 42 s, if they are travelling in the same direction. Find the ratio of the speeds of the two trains.
(a) 3:1 (b) 4:1
(c) 3:2 (d) None of these

Q.42. A car after travelling a distance of 110 km develops a problem in the engine and proceeds at 3/4th of its former speed and arrives at the destination 60 min late. Had the problem developed 30 km further on, the car would have arrived 12 min sooner. Find the original speed of the car.
(a) 45 km/h (b) 60 km/h
(c) 50 km/h (d) None of these

Q.43. A stone, allowed to fall under gravity falls $5t^2$ m in 't' seconds. When a stone is dropped into a well,

the sound of the stone striking the water was heard 65/9 sec after it was dropped. Find the depth of the well if sound travels 324 metres in a second?
(a) 360 m (b) 180 m
(c) 210 m (d) 240 m

Q.44. A watch which gains uniformly is 5 min slow at 8.00 am on sunday and is 5 min 48 s fast at 8.00 pm on the following sunday. When did it show the correct time?
(a) 7.20 pm Thursday
(b) 7.20 am Wednesday
(c) 7.20 pm Wednesday
(d) 7.20 am Thursday

Q.45. Anil and Dheeraj run a 4 km race on a circular course 400 m long in the same direction. If their speeds are in the ratio of 3:2, how many times does Anil pass Dheeraj?
(a) 2 times (b) 3 times
(c) 4 times (d) 5 times

Q.46. At what time between 8 and 9 O'clock will the hours and min hand make an angle of 72° with each other?
(a) 8 h 306/11 (b) 8 h 568/11
(c) 1 or 2 (d) None of these

Q.47. At what time between 2 and 3 O'clock will the hands of a clock be in a straight line facing away from each other?
(a) $2:42:\frac{9}{11}$ (b) $2:42:\frac{49}{77}$
(c) $2:43:\frac{9}{11}$ (d) $2:43:\frac{49}{77}$

Q.48. A man can row a distance of 2 km upstream in 15 min and returns the same distance in 10 min. How much time will he take to row the same distance upstream if due to a tide the speed of the current gets doubled?
(a) 25 min (b) 45 min
(c) 20 min (d) None of these

Q.49. Two points A and B are diametrically opposite points on a 6 km long circular road. A cyclist started from A and made two rounds. He made the first round with a certain uniform speed and then decreased his speed by 3 km/h. The interval between his two passages through point B is 50 min. Find the speed with which the cyclist made the first round.
(a) 6 km/h (b) 9 km/h
(c) 12 km/h (c) 16 km/h

Q.50. Minute hand of a clock overtakes the hour hand at the intervals of 63 min of the correct time. How much does the clock lose or gain in a day?
(a) $56\frac{24}{77}$ min (b) $55\frac{8}{77}$ min
(c) $56\frac{8}{77}$ min (d) None of these

MODERATE

Q.1. A train of 300 m is travelling with the speed of 45 km/h when it passes point A completely. At the same time, a motorbike starts from point A with the speed of 70 km/h. When it exactly reaches the middle point of the train, the train increases its speed to 60 km/h and motorbike reduces its speed to 65 km/h. How much distance will the motorbike travel while passing the train completely?
(a) 2.52 km (b) 2.37 km
(c) 2 km (d) None of these

Q.2. Two persons A and B are standing diametrically opposite on a circular track of with a radius of 3.5 units. They start running in an opposite direction on the circular track and their speeds are 4 units/min and 7 units/min respectively. On meeting each other they take rest for half a min and then start running with twice the original speed. In five min how many times will they take rest together?
(a) 6 times (b) 7 times
(c) 10 times (d) None of these

Q.3. Two riders on the horseback with a gun and a bullet proof shield were moving towards each other at a constant speed of 20 km/h and 5 km/h respectively. When they were 100 km apart, they started firing bullets at each other at the speed of 10 km/h. When a bullet of rider 1 hits the shield of rider 2, rider 2 fires a bullet and the process continues vice versa. Neglecting the time lag at the instant when the bullet hits the shield and the rider fires the shot, find the total distance covered by all the bullets shot by both the riders.
(a) 50 km (b) 40 km
(c) 25 km (d) None of these

Q.4. A group of soldiers are marching with a speed of 5 m/s. The distance between the first and the last row of soldiers is 100 m. A dog starts running from the last row and moves towards the first row, turns and comes back to the last row. If the dog has travelled 400 m, the speed of the dog is
(a) $5\sqrt{2}$ m/s (b) $3\sqrt{5}$ m/s
(c) $6\sqrt{5}$ m/s (d) $6\sqrt{2}$ m/s

Q.5. A train approaches a tunnel PQ which is 16 m long. Two rabbits A and B are standing at points which are 12 m and 8 m inside the tunnel with respect to the entrance P. When the train is x m away from P, A starts running towards P and B towards Q. Difference between the ratios of the speed of A to that of the train and the ratio of the speed of B to that of train is 1/8. How much can the distance x be, if both of them get caught at the ends of the tunnel?
(a) 30 m (b) 16 m
(c) 48 m (d) None of these

Q.6. Pranesh wants to travel to Pune. He has three options.
Route-1 Travel by a train from Chiplun to Mumbai and then from Mumbai to Pune. The trains take 4.5 h and 3 h respectively.
Route-2 Travel by a bus via Bhor. The distance covered by this route is 300 km at 40 km/h and there is a half an hour break in between.
Route-3 Travel by a bus via Koyana. The distance covered by this route is 500 km at 60 km/h. If Pranesh goes by train there is a time span of 30 min between the respective trains. Which route should he take to save his time?
(a) Route 1 (b) Route 2
(c) Route 3 (d) [1] or [2]

Q.7. The numbers expressing (in metres) the length of a fast and a slow trains are three times the numbers expressing (in km/h) the speed of the slow and the fast train respectively. The speed of the slow train and that of the fast train consists of the same two digits, but in a reverse order. The time taken by the two trains to pass each other, when traveling in an opposite direction on parallel lines will be
(a) 10.8 s (b) 5.4 s
(c) 9.77 s (d) 5/18 s

Q.8. Michael Johnson is a great sprinter. He can climb 12 stairs of a staircase in 3 s. But after climbing 480 stairs his speed reduces to 8 stairs in 3 s. The empire state building in NY has 90 floors and the lift takes 4 s per floor. To get to the next floor one has to climb 12 stairs. If Michael is on the ground floor how should he reach the 90th floor to be the first?
(a) By lift
(b) Running
(c) Run till 50th floor and then take the lift
(d) None of these

Q.9. A car goes 20 miles on a gallon of gas driving at 60 miles/h. If the car is driven at 75 miles/h it goes as far as 70%. How many gallons of gas will be required to travel 210 miles/h.
(a) 6.17 (b) 22.5 (c) 12 (d) 15

Q.10. Rakesh and Siddharth row on a river simultaneously from the same point. Rakesh rows downstream and Siddharth rows upstream. In 15 min, they are 2.25 km apart. Rakesh then turns to follow Siddharth and after 30 min from the beginning, the boats have rowed together 3.5 km. If the speeds of Rakesh and Siddharth and the stream are constant at how many km/hour does the stream flow?
(a) 3 km (b) 2.5 km
(c) 2 km (d) 5 km

Direction for questions 11 and 12: Refer to the data below and answer the questions that follow.

There is a race between Sagar and Sapna. Both of them Delhi. Sagar started on a bike with a speed of 40 km/h and Sapna has started in a car with a speed of 60 km/h from Mumbai to Delhi. After 3½ h of the journey there was a snag in the car. She tried to repair the car but in vain. After half an hour she got a lift for another 500 km in a truck, which was travelling with a speed of 45 km/h. From there Delhi was at a distance of 200 km on road, instead, Sapna took a shorter route which was only 100 km away from Delhi. She started running at the speed of 30 km/h on the shorter route to reach Delhi.

Q.11. How much time did Sapna take to reach Delhi?
(a) 15.5 h (b) 16.5 h
(c) 17.5 h (d) None of these

Q.12. If there was no snag in the car, by how much distance Sapna would have defeated Sagar?
(a) 252 km (b) 264 km
(c) 303 km (d) 321 km

Q.13. A 200 m-long train passes a 350 m long platform in 5 s. If a man is walking at a speed of 4 m/s along the track and the train is 100 m away from him, how much time will it take to reach the man?
(a) Less than 1 s
(b) 1.04 s
(c) More than 2 s
(d) Data insufficient

Q.14. A man waiting at a point P has to reach point Q at some distance from P. He is considering three plans of action.
A. Wait at P itself and catch the first taxi going towards Q.
B. Start walking towards Q and catch the first taxi going towards Q.
C. Start walking away from Q and catch the first taxi going towards Q.
He walks at a constant rate of 5 km/h and taxi travel at a constant rate of 50 km/h. The arrival time of the taxi is totally random. Which of the following statement is then true? (Assume that he is the only person who is looking for a taxi on that road.)
(a) He will always reach Q first if he chooses plan A.
(b) He will always reach Q at the same time regardless of which plan he chooses.

(c) He will sometimes reach Q at the earliest if he chooses plan B.
(d) He will always reach Q at the earliest if he chooses plan C.

Q.15. Three cockroaches John, Mac and Roe decide to take a race from one corner of the room to the diagonally opposite corner of the room. John can fly, Mac can walk and Roe can walk only along the edges. If all of them reach the destination at the same time by taking the smaller paths, what is the ratio of the speeds of John, Mac and Roe assuming that the room is cubical in shape?
(a) $\sqrt{3}+1:3:\sqrt{3}$ (b) $\sqrt{3}:5:3$
(c) $\sqrt{3}:\sqrt{2}:3$ (d) None of these

Q.16. Four men start to move from a particular point. Two of them go in the direction opposite to each other at an equal speed. The other two also move in the opposite directions at an equal speed, but in a direction that is perpendicular to that of the original two. All of them stop at the same time, if the final positions of the men are joined, then it forms.
(a) A parallelogram (b) A rectangle
(c) A rhombus (d) A square

Q.17. Saket runs twice on a circular track at a uniform speed. Bidhan starts at the same time as Saket and after Bidhan has completed one lap, he stops and Sunny starts and completes the second lap at a uniform speed. It is given that they start the race exactly at 2.30 pm and the track is 500 m in circumference. Sunny runs at a speed of 5 km/h greater than that of Bidhan and Saket's speed is 2 km/h greater than that of Bidhan. When does Saket finish the 2nd lap if Sunny also finishes at the same time?
(a) 2.32 pm (b) 2.35 pm
(c) 2.33 pm (d) 2.36 pm

Q.18. Two persons A and B are running between two points P and Q to and from infinitely. A is starting from P and B is starting from Q. They meet for the first time at 0.6 D from point P, where D is the distance between the points AB. At how many distinct points will they meet till their 10th meeting?
(a) 10 (b) 3 (c) 4 (d) 6

Direction for questions 19 and 20: Refer to the data below and answer the questions that follow.

When BMW introduced its most advanced engine, it was found that the engine of the car followed certain mileage characteristics in a speed limit of 25 km/h to 80 km/h. The car can travel a distance of 15 km in one litre of petrol while speeding at 50 km/h. From this point onwards for every increase of 5 km/h in the speed of the car the mileage is increased by 1 km per litre. Similarly, for every decrease of 5 km/h in the the speed of the car, the mileage is decreased by 1 km per litre. Assume that the cars operate in this speed limit only.

Q.19. Rahul Ghosh has filled exactly 10 litres of petrol in his car. He drives at a speed of 45 km/h for first two hours. What is the maximum distance that he can cover before his tank goes dry?
(a) 80 km (b) 67.37 km
(c) 75 km (d) None of these

Q.20. In the previous question, what is the maximum span of time for which Rahul Ghosh can enjoy the drive?
(a) 24/7 h (b) 15/4 h
(c) 37/15 h (d) None of these

Q.21. Two rabbits A and B are running a race in which they have to go upto 50 m mark and then come back to the starting point. A runs in a sequence of 3 jumps in which the distance traveled in the first one is twice that of the other two and B runs in a sequence of 3 jumps in which the distance travelled in the first jump is 1.5 that of the distance travelled in the 2nd jump and distance travelled in the second jump is twice the distance travelled in the third jump. Distance travelled by A in 2 jumps and by B in 3 jumps equal to 6 m. In the 2nd race they interchange their individual sequence of jumps. Who will win the 1st and the 2nd match if on completing the 50 m mark they turn back and start from the 1st jump?
(a) A, A (b) B,B
(c) A,B (d) None of these

Q.22. There are two clocks. One of them gains 2 min in 12 h and another one loses 2 min in 36 h. Both are set right at 12 noon on Tuesday. What will be the correct time when both of them show the same time for the next time?
(a) 12 night (b) 1.30 am
(c) 10.30 pm (d) 12 noon

Q.23. A train's journey is disrupted due to an accident on its track after it has travelled 30 km. Its speed then comes down to 4/5th of its original and consequently it runs 45 min late. Had the accident taken place 18 km farther away, it would have been 36 min late. Find the original speed of the train.
(a) 25 km/h (b) 36 km/h
(c) 30 km/h (d) 20 km/h

Q.24. A passenger train departs from Ahmadabad at 6 pm for Bombay. At 9 p.m. an express train, whose average speed exceeds that of the passenger train by 15 km/h, leaves Bombay for Ahmedabad. Two trains meet each other mid-route. At what time do they meet, given that the distance between the cities is 1080 km?
(a) 4 pm (b) 2 pm
(c) 12 midnight (d) 6 am

Q.25. In a race on a circular track, there were 12 laps of 150 m each. Akhil and Bharat were very close till the end of the third lap, but thereafter Bharat went on to

win the race by 81 m. If Bharat gained uniformly over Akhil in all the remaining laps, the ratio of the speed of Akhil and Bharat for each of other laps was
(a) 129:150 (b) 150:141
(c) 150:129 (d) 141:150

Q.26. Magadh Express and Vikramshila Express run between Patna and New Delhi. Every night each of these train leaves Patna for Delhi. Likewise every night each of the trains leaves Delhi for Patna. The train journey lasts 36 h and the trains are identical in all aspects. If the train that leaves in the night is like the one that arrived in the morning, what is the minimum number of trains required?
(a) 4 (b) 8
(c) 16 (d) None of these

Q.27. An athlete runs to and fro between points A and B at a speed of 10 km/h. A second athlete simultaneously runs from point B to A and back at a speed of 15 km/h. If they cross each other 12 min after the start, after how much time will they cross each other?
(a) 18 min (b) 24 min
(c) 36 min (d) 48 min

Q.28. There are two swimmers A and B who start swimming towards each other from opposite banks of the lake. They meet at a point 900 ft from one shore for the first time. They cross each other, touch the opposite bank and return. They meet each other again at 300 ft from the other shore. What is the width of the lake?
(a) 2400 ft (b) 1800 ft
(c) 2700 ft (d) 3600 ft

Q.29. DMRC is plying trains between Vishwavidyalaya and Central Secretariat, which are at a distance of 120 km. There are three more stations in between Vishwavidyalaya and Central Secretariat and each train stops at these two stations for 2 minutes. What is the minimum numbers of trains DMRC should put in this route, if the trains are travelling at a speed of 40 km/h and it is desirable to have a regular interval of 6 minutes between the departures of two trains from both the stations?
(a) 32 (b) 63
(c) 64 (d) None of these

Q.30. The front wheels of a tractor make 36 revolutions per min while the rear wheels make 18 revolutions per min. After how many seconds will the two wheels be in the same position relative to each other as they were when they started moving?
(a) 2.22 (b) 3.33 (c) 1.66 (d) 4.44

Q.31. A train and a car start at the same time, parallel to each other in the same direction. The car starts its journey from the rear end of the train. The car reaches the front end of the train and comes back to the back end of the train. In the meantime, the mail train travels a distance of 1 km. If the speed and the length of the mail train is 1 km/h and 1 km respectively then what is the distance covered by the car?
(a) 2 km (b) $1+\sqrt{2}$ km
(c) $2+\sqrt{2}$ km (d) Cannot be determined

Direction for questions 32 to 34: Refer to the data below and answer the questions that follow.

A, B and C participate in a triathlon. The triathlon consists of three parts. The first part consists of running, the second part cycling and the third part swimming. A can cycle thrice as fast as he can run and swim one-third as fast as he can run. B can cycle four times as fast as he can run and swim one-fourth as fast as he can run. C can swim half as fast as he can run and cycle four times as fast he can run. A can run 10 percent faster than B and 20% faster than C. The ratio of the length of the running track to that of the cycling is ½ and to that of swimming tracks is 2.

Q.32. Who is leading after the 2nd part is over?
(a) A (b) B
(c) C (d) Cannot be determined

Q.33. Who will win the triathlon?
(a) A (b) B
(c) C (d) Cannot be determined

Q.34. C beats B by 100 m finally. What is the length of the cycling tracks given that the length of the swimming track is more than 100 m?
(a) 580 m (b) 622 m
(c) 518 m (d) 576 m

Q.35. Pramod can row at 3 km/h in still water. He wants to cross a river that is flowing at the speed of 4 km/h in the shortest possible time. Find the ratio of the distance between the point he leaves the bank on one side and the point that he reaches on the other side to the distance between the shortest possible distance to cross the river?
(a) 1:1 (b) 1:2
(c) 2:3 (d) None of these

Q.36. Vinit's rowing at a speed of 4 km/h in still water. He has to reach a destination that is 130 km downstream of a river. The river flows at a speed of 2 km/h but due to tides every two h the velocity doubles and then after another two h it halves. If Vinit starts at the change of the tide when the river was flowing at 2 km/h, then after how much time will he reach his destination?
(a) 8¼ h (b) 18 ½ h
(c) 9 ½ h (d) None of these

Q.37. Ramesh and Somesh are competing in a 100 m race. Initially, Ramesh runs at twice the speed of Somesh for the first fifty m. After the 50 m mark, Ramesh runs at 1/4th his initial speed while Somesh continues to run at his original speed. If Somesh catches up with

Ramesh at a distance of 'N' m from the finish line, then N is equal to
(a) 35 (b) 10
(c) 45 (d) None of these

Direction for questions 38 and 39: Read the passage given below and solve the questions based on it.

Two cars A and B start simultaneously from two different cities P and Q and head for the cities Q and P respectively. As soon as car A reaches the city Q, it turns and starts for city P and as soon as it reaches P again, it leaves for city Q and so on. Similarly car B, travels from Q to P, from P to Q and so on. The speeds of the car A and B are in the ratio of 3 : 4.

Q.38. If car B travelled 1040 m when the two cars meet for the seventh time, what is the distance between the two cities P and Q?
(a) 140 m (b) 280 m
(c) 182 m (d) Cannot be determined

Q.39. If car A travelled 1860 m, how many times did the cars A and B meet? (Take the distance PQ as calculated in the previous question.)
(a) 14 (b) 15
(c) 16 (d) Cannot be determined

Q.40. The Howrah Mail leaves Trivandrum everyday at 12:00 noon and reaches Howrah exactly three days later. Also, the Trivandrum Mail leaves Howrah at 12:00 noon everyday and reaches Trivandrum exactly three days later. Not counting the trains just starting or just leaving the platform when a train leaves or arrives, how many Howrah Mails will one cross when travelling by Trivandrum Mail?
(a) 2 (b) 3 (c) 5 (d) 9

Q.41. A person starts writing natural numbers, starting from 1, on a blackboard, at the rate of 60 digits per minute. At the same time, another person begins to erase the digits from 1 onwards, at the rate of 40 digits per minute. Find the difference between the leftmost digit and the rightmost digit on the blackboard immediately after one hour.
(a) 3 (b) 5
(c) 2 (d) None of these

Q.42. Two persons Prabhat and Vinay are walking around a circular park of the length 960 m. Prabhat walks at the rate of 80 m/min while Vinay walks at the rate of 60 m/min. If both of them start from the same starting point at the same time in the same direction, when will they be together?
(a) 24 min (b) 48 min
(c) 96 min (d) 120 min

Q.43. A 100 m-long train is approaching an unmanned railway crossing. The train is running at a uniform speed of 90 km/h and is 1 km away from the crossing. At the same time, a packed jeep, also approaching the crossing, is 700 m away from it. Assuming that the jeep is also moving at a uniform speed of N, what is the range of N, for which the jeep will not collide with the train?
(a) 59 km/h < N < 61 km/h
(b) 57 km/h < N < 63 km/h
(c) 55 km/h < N < 65 km/h
(d) 53 km/h < N < 67 km/h

Q.44. A road from A to B is 11½ km long, first goes uphill, then crosses a plain, and then goes downhill. A person walking from A to B covered this road in 2 h 54 min, and the return journey took him 3 hours 6 min. His speed uphill is 3 km/h, on the plain is 4 km/h and downhill is 5 km/h. What is the length of the plain part of the journey?
(a) 4 km (b) 6 km
(c) 4½ km (d) None of these

Q.45. Two cyclists start from the same place to ride in the same direction. Aflatoon starts at noon with a speed of 8 km/h and Bablajoon starts at 2 pm with a speed of 10 km/h. At what times Aflatoon and Bablajoon will be 5 km apart?
(a) 7.30 pm same day and 1.30 am on the next day
(b) 7.30 pm same day and 12.30 am on the next day
(c) 8.30 pm same day and 1.30 am on the next day
(d) 8.30 pm same day and 12.30 am on the next day

Q.46. Mrs and Mr Sinha travel from Patna to Lucknow and break their journey at Buxer. Somewhere between Patna and Lucknow, Mrs Sinha asks, "How far have we travelled?" Mr Sinha replies, "Half as far as the distance from here to Lucknow." Somewhere between Buxer and Lucknow, exactly 300 km from the point where she asked the first question, Mrs Sinha asks, "How far have we to go?" Mr Sinha replies, "Half as far as the distance from Buxer to here." The distance between Patna to Lucknow is
(a) 750 km (b) 600 km
(c) 450 km (d) None of these

Q.47. At his normal speed, Rishu can go 18 km downstream in a flowing river in 9 h less than what he takes to go the same distance upstream. The downstream journey would take 1 h less than what the upstream journey would take provided he doubles his rate of rowing. Find the speed of the river (in km/h)?
(a) $6\frac{2}{3}$ km (b) 8 km
(c) $9\frac{1}{2}$ km (d) 12 km

Q.48. Umesh and Harish are running on a circular track of length 1200 m in opposite direction. Everytime they meet, they exchange their speed as well as their directions. What is the shorter distance along the track between their 13th and 23rd meeting points, if the ratio of their oringinal speeds is 1:5?
(a) 200 m (b) 400 m
(c) 800 m (d) 1000 m

Q.49. Two friends Manoj and Sagar start running simultaneously in an opposite direction on a circular track of the length 2100 m with speed of 7 m/s and 3 m/s respectively. With every meeting they exchange their speed and they meet for a total of ten times. How much distance did Sagar cover in total?
(a) 1400 m (b) 700 m
(c) 1050 m (d) None of these

Q.50. A and B start simultaneously at one end of a swimming pool whose length is 50 m. The swimming race is a race of 1000 m. If A beats B and meets him 17 times during the course and A's speed is 5 m/s then the speed of B could be
(a) 1 m/s (b) 3 m/s
(c) 4 m/s (d) 6 m/s

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Direction for questions 1 to 5: Read the passage given below and solve the questions based on it.

There are 8 days in a week from Sunday to Saturday and another day called Funday on planet North. There are 36 h in a day and each hour has 90 min while each minute has 60 s.

Q.1. Find the approximate angle between the hands of a clock on North when the time is 16.50 am?
(a) 189 (b) 131
(c) 320 (d) 165

Q.2. At what time between 14 o'clock and 15 o'clock will the two hands of the clock be at an angle of 60°?
(a) $14:89\ ^{16}/_{17}$ (b) $14:37\ ^{13}/_{17}$
(c) $14:58\ ^{14}/_{17}$ (d) $14:55\ ^{5}/_{17}$

Q.3. Bubby, an inhabitant of the planet North leaves home between 7 am and 8 am and returns between 1 pm and 2 pm to find that the minutes and hours hands have interchanged their positions. How long was bubby out of the house?
(a) 6 h 28 min (b) 12 h 39 min
(c) 12 h 28 min (d) 12 h 22 min

Q.4. A watch which is correctly set at 16 o'clock in the morning shows 10 min after 13 o' clock in the evening when the correct time is 14 o'clock. What is the time when the watch indicates 15 o'clock in the evening?
(a) 16:00 (b) $16:32\ ^{5}/_{7}$
(c) $16:16:^{16}/_{17}$ (d) $16:33\ ^{6}/_{17}$

Q.5. A watch on planet North which loses time uniformly is 15 min fast at 14 am on Sunday and is 45 min slow at 8 pm on the next Sunday. When did the clock show the correct time?
(a) 6 am on Saturday
(b) 8 am on Funday
(c) 17 am on Tuesday
(d) 11 am on Funday

Q.6. In the shooting of the movie 'AKS', Manu, the actor, is running towards a vertical mirror with a speed of 10 m/s on a line which is perpendicular to the plane of the mirror. The mirror is moving towards Manu with the speed of 6 m/s along the same line. Find the speed of Manu's image with respect to Manu (the image of Manu forms on the other side of the mirror such that the distance between the image of Manu and the mirror is equal to the distance between Manu and the mirror).
(a) 15 m/s (b) 25 m/s
(c) 30 m/s (d) None of these

Q.7. A rabbit, a tortoise and a mongoose decide to run a closed circuit race. If the rabbit overtakes the mongoose 2 times per round, with the 2nd overtake being at the starting point and the mongoose overtakes the tortoise 4 times per round, the 4th overtake being at the starting point, then find the ratio of the speed of the rabbit to that of the tortoise.
(a) 15:1 (b) 8:1 (c) 6:1 (d) 10:1

Q.8. Anil left his house for school S min late but he traveled at 4/3th of his usual speed and reached school S min early. Next day he left home S min early than the previous day and travelled at 8/11th of his usual speed. He reached his office
(a) S min late (b) 2S min late
(c) Exactly on time (d) Cannot be determined

Direction for questions 9 and 10: Read the passage given below and solve the questions based on it.

There are two cities Nagpur and Pune along a straight line 120 km apart. A and B start at the same time from Nagpur and Pune respectively with the speed of 40 km/h and 60 km/h respectively. They travel towards each other, and after they meet for the first time, they reverse directions and also interchange their speeds. After reaching their respective starting points, they reverse their directions and start proceeding towards each other again.

Q.9. How many hours from the beginning do they meet for the second time?
(a) 2 h (b) $2^1/_5$ h
(c) $2^1/_3$ h (d) $3^3/_5$ h

Q.10. How much distance had B covered till the time he met A for the second time?
(a) 216 km (b) 180 km
(c) 174 km (d) 168 km

Direction for questions 11 and 12: Read the passage below and solve the questions based on it.

A number of runners, numbered 1, 2, 3,…, N and so on, start simultaneously at the same point on a circular track and keep on running continuously in the same direction, around the track. They run in such a way that the speed of the runner numbered N (N>1) is N times that of the runner numbered 1.

Q.11. If there are exactly six runners, then at how many distinct points on the track is the runner numbered 1 overtaken by any of the other five runners?
(a) 15 (b) 11 (c) 9 (d) 10

Q.12. If there are exactly four runners, then at how many distinct points on the track do two or more runners meet?
(a) 3 (b) 4 (c) 5 (d) 6

Direction for questions 13 and 14: Read the passage given below and solve the questions based on it.

Amit intended to travel a certain distance at a certain uniform speed. But after one hour, he increased his speed by 25%. As a result, in the remaining part of the time that he originally planned for the journey, he could now cover as much distance as he initially thought he would be able to cover.

Q.13. What is the total time taken for the journey?
(a) 4 hrs (b) 5 hrs
(c) 6 hrs (d) None of these

Q.14. After Amit increased his speed, if he decided to terminate his journey after covering the distance he initially intended to cover and not cover the extra distance as given in the data, what is the total time taken for the journey?
(a) 4 h 12 min (b) 5 h 24 min
(c) 3 h 36 min (d) None of these

Direction for questions 15 and 16: Read the passage given below and solve the questions based on it.

A train started at 9.00 am from station X with a speed of 72 km/h. After two hours, another train started from station Y towards X with a speed 90 km/h. The two trains are expected to cross each other at 1.30 pm. Owing to a signal problem arising at 12 noon, the speed of each of them was reduced by the same quantity and they crossed each other at 4.30 pm.

Q.15. What is the new speed of the train that started from station X?
(a) 18 km/h (b) 36 km/h
(c) 45 km/h (d) 54 km/h

Q.16. If the signal problem had occurred at 1.00 pm instead of 12 noon, at what time would the two trains cross each other?
(a) 2.30 pm (b) 3.00 pm
(c) 2.00 pm (d) 2.30 pm

Direction for questions 17 to 19: Read the passage given below and solve the questions based on it.

Bihar Couriers (BC) sends a person on motorcycle every day to the airport to collect the courier. This person reaches the airport exactly when the plane lands at its scheduled time. One day BC came to know that the flight would be late by 50 min. So, the BC person also started from his office 50 min later than the usual time. However, the plane landed earlier than was anticipated and thus the airport authorities dispatched the courier through a cyclist to BC. The cyclist meets the BC personnel after 20 min and hands over the courier to the motorcyclist. As a result, the BC person reached office 10 min earlier than expected. (Assume that everybody travels at their own uniform speeds.)

Q.17. What was the delay in the landing of the plane when compared to the normal schedule?
(a) 30 min (b) 35 min
(c) 20 min (d) 25 min

Q.18. If the cyclist covered a distance of 2.5 km before he met the motorcyclist, what is the speed of the motorcyclist?
(a) 60 km/h (b) 45 km/h
(c) 30 km/h (d) Cannot be determined

Q.19. If on that particular day when the cyclist covered 2.5 km before he met the motorcyclist, the motorcyclist spent a total of 30 min for his complete journey, what is the distance from BC's office to the airport?
(a) 20 km (b) 15 km
(c) 10 km (c) Cannot be determined

Q.20. My digital clock is peculiar. It counts 10 s of a normal clock as 1 min and 60 such min of itself as 1 h. It has also a display which shows the day. It was at par with the normal clock at 12 noon on Monday. At 3 pm of the same day (Actual time), I just read the time display of my weird clock. I read
(a) 6.00 pm Monday
(b) 6.00 pm Tuesday
(c) 6.00 am Tuesday
(d) 4.00 am Tuesday

Direction for questions 21 and 22: Read the passage below and solve the questions based on it.

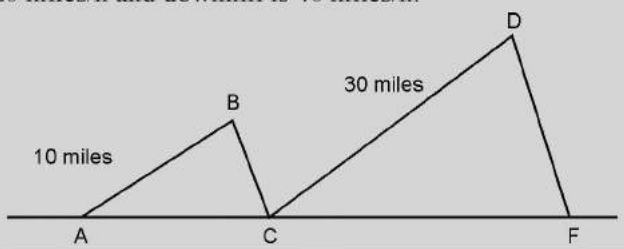
Every morning Ravi and Beena start walking from A and B respectively towards each other. They meet at C. Ravi always starts at 5.30 am and Beena starts at 4.45 am. Ravi's speed is 6 km/h, whereas Beena's speed is 5 km/h.

Q.21. One fine morning, Beena started late by a few min and hence met Ravi who started on time at D in between CB. CD = 2.5 km. By how many min was Beena late in starting on that day?
(a) 25 min (b) 30 min
(c) 55 min (d) Data insufficient

Q.22. One morning Ravi got delayed but Beena was on time and that day the two met at 7.00 am at E which was 2.5 km away from C. What is the distance between A and B?
(a) 20.25 km (b) 17.25 km
(c) 14.75 km (d) Data insufficient

Direction for questions 23 and 24: Read the following passage and solve the questions based on it.

Given below is the structure of roads from A to F. ABC and CDF are two similar mountains with ideal straight lines AB and CD as their slant heights. $\angle BAC = \angle DCF = 30°$ and $\angle BCA = \angle DFC = 60°$. Speed of Manoj Kumar uphill is 20 miles/h and downhill is 40 miles/h.



Q.23. What is the total time taken by him to travel from A to F ?
(a) 3 h 12 min (b) 2 h 21 min
(c) 2 h 35 min (d) Cannot be determined

Q.24. Jonathan Livingston Seagull is flying between the points B and D. What is the distance traveled by Jonathan?
(a) $10\sqrt{\frac{28}{3}}$ miles (b) $10\sqrt{\frac{7}{3}}$ miles
(c) $\frac{10}{3}$ miles (d) None of these

Q.25. Ram and Rahim left Bombay simultaneously towards Pune. Speed of Ram is 15 km/h and the speed of Rahim is 12 km/h. Half an hour later, Mohan started from Bombay towards Pune on the same road in the same direction. After some time he overtook Rahim and 90 min further, he overtook Ram. What is Mohan's speed?
(a) 18 km/h (b) 24 km/h
(c) 32 km/h (d) 36 km/h

Direction for questions 26 and 27: Read the passage below and solve the questions based on it.

Ankit and Sanket are traveling by bus from Ahmedabad to Kolkatta via Bangalore in a bus which leaves Ahmedabad at 8.00 am. The bus reaches Bangalore at 10.00 am, just in time for them to catch the last bus to Kolkata. However, Ankit decides to get off earlier (at point P_1) and travels to Bangalore by car. Because of this he reaches 15 min late. If he had got off at a distance 16 km ahead (at point P_2), he would have reached only 7 min late. Sanket also gets off at a point 16 kms from Bangalore, where as the bus continues at its normal speed of 40 km/h and the ratio of speeds of Ankit and Sanket is 15:16.

Q.26. If the bus takes 45 minutes to reach Bangalore from point P_1, Sanket will reach
(a) Early by 6 min (b) Late by 6 min
(c) On time (d) Cannot be determined

Q.27. Had Sanket got off at P_2, at what time would he reach Bangalore?
(a) 10.05.15 (b) 10.05.05
(c) 9.54.45 (d) Cannot be determined

Q.28. A candidate misinterpreted the timings of an interview and reached the venue 35 min before the scheduled time. After 25 min, he presumed that the interview might have been cancelled and so he started walking back home at the speed of 1 km/h. But then he realized that he had left his bag there. So he comes back with the same speed and enters the office along with the interviewer, who is 5 min late. How much distance had he moved away from the office?

(a) 7.5 km (b) 250 m
(c) 125 m (d) 750 m

Q.29. Two cyclists are travelling at a speed of 3 km/h and 10 km/h around a circullar track. What is the ratio of number of meeting points that thye would meet on that track if they travel in the same direction first and in the opposite direction next?
(a) 3:13 (b) 7:13
(c) 3:7 (c) Cannot be determined

Direction for questions 30 and 31: Read the passage below and solve the questions based on it.

A jetfighter moving 3 km above the ground with a speed of 360 km/h tries to cross the border. A radar station notices it at distance of 6 km from the Jet. But it can fire anti-craft missile only when the plane enters in its border. The border is at a distance of $\sqrt{3}$ km from radar station. (Assume that the fighter moves in a straight line in the same plane with the same speed).

Q.30. If it fires the missile at the fighter as soon as it notices the jetfighter such that it will hit the fighter as soon as it enters in the borderline, then at what speed should the be missile fired?
(a) 360 km/h (b) 240 km/h
(c) $200\sqrt{3}$ km/h (d) None of these

Q.31. The radar station can fire 10 missiles simultaneously in such a way that the first missile will meet the fighter exactly at the border if the fighter moves with its original speed. The other missiles will reach up to the line of movement of the jet at same time but at a distance of 100 m from each other such that one of them will hit the fighter even if the fighter increases its speed. If jetfighter has to escape the attack, at what increased speed it should move?
(a) 400 km/h (b) 545 km/h
(c) 464 km/h (d) 380 km/h

Q.32. Two trains going on a parallel line in opposite directions take 10 s to cross each other. But if they are going in the same direction the longer train crosses the shorter train in 30 s. If the length of the longer train is decreased by 50%, the time taken to cross the shorter train while going in the same direction decreases by 8 s. Find the time taken by the longer train to cross a tunnel twice its length, if the difference between the length of the trains is 25 m?
(a) 30 s (b) 24 s (c) 40 s (d) 36 s

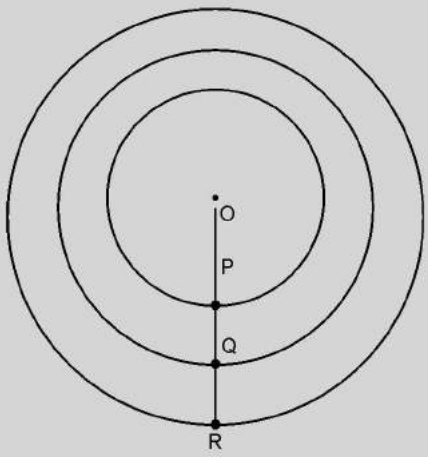
Q.33. A runner, Ajay runs twice on a circular track at a uniform speed. Vijay start at the same time as Ajay and after Vijay has completed one lap, he stops and Sanjay starts and completes the second lap at a uniform speed. It is given that they start the race exactly at 2.30 pm and the track is 500 m is circumference. Sanjay runs at a speed of 5 km/h greater than that of Vijay and Ajay's speed is 2 km/h greater than that of Vijay. When does Ajay finish the 2nd lap, if Sanjay also finishes at the same time?
(a) 2.32 pm (b) 2.33 pm
(c) 2.35 pm (d) 2.36 pm

Q.34. A man travels 1200 km from A to B by air, train and ship. The distance travelled by ship is twice the distance travelled by train. The average speed of the aircraft is twice the overall speed and the time he was on the ship was thrice the time he was on the train. If the man travels 10 h by train out of the total travel time of 50 h, then what is the total distance covered by the ship?
(a) 720 km (b) 630 km
(c) 480 km (d) 570 km

Q.35. Three friends Rupesh, Nishit and Vijay planned to race with each other on their bikes. Rupesh has a lot of riding experience, while Nishit had just learned how to ride. The race seemed unfair. Therefore, they decided that the excellent biker Rupesh will form a team with Nishit. They would bike against the other rider Vijay, each riding half the total distance of 50 kms. Also Nishit can ride only half as fast as Vijay and Rupesh travels four times as faster as Nishit. What is the ratio of the time taken by the winner to that of the looser?
(a) 4:5 (b) 5:2 (c) 2:5 (d) 4:3

Direction for questions 36 and 37: Read the passage below and solve the questions based on it.

Three persons A, B and C start running simultaneously on three different concentric circular tracks from three collinear points P, Q and R respectively which are collinear with the centre O and are on the same side of the centre as shown. The speeds of A, B and C are 5 m/s, 9 m/s and 8 m/s respectively. The lengths of the tracks on which A, B and C are running are 400 m, 600 m and 800 m respectively.



Q.36. If A and B run in a clockwise direction and C in an anti-clockwise direction, after how much time will the positions of A, B and C be collinear with the centre (and on the same side of the centre), for the first time after they start?
(a) 200 s (b) 400 s
(c) 600 s (d) 800 s

Q.37. If A runs in the clock wise direction and C in the anti-clockwise direction, after how much time will the positions of A and C be collinear with the centre, (and on the same side), for the fourth time after they start?
(a) $177\frac{7}{9}$s (b) 160 s
(c) $145\frac{5}{11}$s (d) 1

Direction for questions 38 and 39: Read the passage below and solve the questions based on it.

A train started at 9.00 am from station X with a speed of 72 km/h. After two hours, another train started from station Y towards X with a speed of 90 km/h. The two trains are expected to cross each other at 1.30 pm. Owing to a signal problem arising at 12 noon, the speed of each of them was reduced by the same quantity and they crossed each other at 4.30 pm.

Q.38. What is the new speed of the train that started from station X?
(a) 18 km/h (b) 36 km/h
(c) 45 km/h (d) 54 km/h

Q.39. If the signal problem had occurred at 1.00 pm instead of 12 noon, at what time would the two trains have crossed each other?
(a) 3.30 pm (b) 3.00 pm
(c) 2.00 pm (d) 2.30 pm

Direction for questions 40 to 42: Read the passage below and solve the questions based on it.

P, Q and R are three villages, located in the same order, along a highway. An absent minded person, Andrews, intending to reach R, starts from Q and travels at a speed of 60 km/h in the direction of QP instead of QR. At the same time as Andrews starts, Bill leaves R to reach Q and travels at the speed of 90 km/h in the direction of RQ. After travelling for 1 h in the direction of QP, Andrews realizes his mistake and reverses his direction and increases his speed to 90 km/h. After reaching Q, Bill reverses his direction and proceeds back to R at a reduced speed of 60 km/h. After reaching R, Andrews starts back towards Q with no change in speed. The distance PQ and QR are 162 km and 108 km respectively.

Q.40. At what distance from Q, do Andrews and Bill meet for the first time?
(a) 72 km (b) 90 km
(c) 84 km (d) 96 km

Q.41. At what distance from R, do the two meet for the second time?
(a) 9.6 km (b) 2.4 km
(c) 4.8 km (d) 7.2 km

Q.42. Which of the two persons travelled more distance up to the point of their second meeting, and by how much?
(a) Bill by 33.6 km
(b) Bill by 31.2 km
(c) Andrews by 16.8 km
(d) Andrews by 21.6 km

Q.43. In a certain race, the winner beats the other 6 contestants by 10, 20, 30, 40, 50, 60 m respectively. Find by what distance the 3rd runner-up beats the 4th runner-up, if he was 11.11 m ahead of the 4th runner-up, when the fist runner-up finished the race.
(a) 14.14 m (b) 14.28 m
(c) 16.66 m (d) None of these

Q.44. Distance between Lucknow and Patna is 300 km. Mayank leaves at a speed of x km/h from Lucknow towards Patna. After three hours Sharat leaves at the speed of $(x + 10)$ km/h from Lucknow towards Patna. If x and the number of hours taken to meet after Sharat starts are integers, how much distance can Mayank cover before they meet?
(a) 174 km (b) 60 km (c) 150 km (d) 180 km

Q.45. Two champion swimmers start a two-length swimming race at the same time, but from the opposite ends of the pool. They swim at constant but different speeds. They first pass at a point 18.5 m from the deep end. Having completed one length, each swimmer takes a rest at the edge of the pool for 45s. After setting off on the return length, the swimmers pass for the second time just 10.5 m from the shallow end. Thus, the length of the pool is
(a) 90 m (b) 45 m
(c) 26.5 m (d) Data insufficient

Q.46. A tortoise lays eggs on the beach and goes back to the sea where its mate is waiting. One way to reach the mate is going down 12 km on a straight line perpendicular to the sea, turning 90° and swimming for 5 dm on a straight line. On its way down to the sea, the tortoise can cut the water at any point. The speed of the tortoise on land is 1 km/h and in water is 2 km/h. If the tortoise takes the described route, at what distance from the point where it had laid the eggs will the tortoise cut the water, if the total time taken is 12 h?
(a) 7 km (b) 8 km (c) 5 km (d) 4 km

Direction for questions 47 and 48: Read the passage below and solve the questions based on it.

Trains X and Y start towards each other at the same time from points A and B for destinations B and A respectively that are 300 km apart. They meet each other 120 km from point A.

Q.47. Another express train Z starts from point A to point B, 1 h after X starts. This express train meets train Y 1.5 h after Z starts. If this express train is at least 20 km/h faster than the faster of X and Y, which of the following is true? (Ignore train lengths in calculations.)
(a) The minimum possible speed of Y is 60 km/h
(b) The minimum possible speed of X is 45 km/h
(c) The maximum possible speed of Y is 60 km/h
(d) The maximum possible speed of X is 45 km/h

Q.48. What is the minimum speed for train Z to overtake train X, before it meets train Y? (use the data from previous question, if necessary)
(a) 45 km/h
(b) 50 km/h
(c) 60 km/h
(d) Train Z will always overtake train X, before it meets train Y

Q.49. A, B and C start running a race from the same starting point at the same time in the same direction. A's speed around a path which is an equilateral triangle. B's path is a square and C's path is a regular hexagon. One edge of the triangular path, square path and hexagonal path completely overlaps with each other. If all of them complete one round at the same time then which of the following is true?
(a) Speed of C is twice that of the speed of B.
(b) Speed of A is half that of C.
(c) Speed of B is 50% more than that of A.
(d) None of these

Q.50. On a river there is no current form A to B but a current from B to C. A man rows down a stream from A to C in 3 h, and up stream from C to A in $3\frac{1}{2}$ h; had there been the same current in all the ways as from B to C his journey down stream would have occupied $2\frac{3}{4}$ h; find the length of the time, his return journey would have taken under the same circumstances.
(a) 4 h (b) 4.2 h
(c) $3\frac{17}{20}$ h (d) 2 h 24 mins

TRUE/FALSE

1. If the ratio of speed of two trains and ratio of their lengths are given, we can find out the time taken by them to cross each other while passing in the same direction.

   State whether true or false.

2. Sum of lengths of two trains are given. Speed of the longer train is less than the speed of the smaller train and both the speeds are given. We can find out the time taken by these trains to cross each other while passing in the opposite direction.

   State whether true or false.

3. Two runners are running in opposite direction on a circular track starting from the same point in the same direction same time. Length of track is given. We can find out the distinct meeting points of the runners if the ratio of their speeds are given.

   State whether true or false.

4. Two runners A and B are running on a linear track. Length of linear track is given. B gives a start of 10 seconds to A and both of them finish the race in dead heat. With the information given we can find out the ratio of speed of A and B.

   State whether true or false.

5. There are 5 runners running on a circular track starting from the same point in the same direction same time. If we know the speeds of each of the runners and the distances between each of the possible distinct meeting points between any two of them is given, we can find out the length of track.

   State whether true or false.

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(d)	(b)	(c)	(a)	(d)	(b)	(c)	(b)	(b)	(c)	(a)	(d)	(d)	(b)	(c)	(b)	(b)	(a)	(a)

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(b)	(d)	(b)	(b)	(c)	(d)	(c)	(d)	(c)	(b)	(a)	(b)	(b)	(d)	(a)	(d)	(a)	(b)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(a)	(a)	(c)	(b)	(a)	(d)	(d)	(c)	(a)	(d)	(d)	(b)	(c)	(c)	(a)	(b)	(a)	(d)	(d)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(a)	(c)	(b)	(c)	(b)	(c)	(d)	(d)	(b)	(c)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(d)	(b)	(a)	(c)	(d)	(a)	(d)	(d)	(c)	(d)	(c)	(a)	(c)	(d)	(a)	(b)	(b)	(c)	(a)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(d)	(c)	(c)	(d)	(b)	(c)	(a)	(c)	(b)	(b)	(b)	(c)	(c)	(d)	(d)	(d)	(a)	(c)	(c)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(d)	(b)	(b)	(a)	(b)	(d)	(a)	(b)	(d)	(b)										

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(c)	(c)	(a)	(b)	(d)	(a)	(b)	(a)	(a)	(d)	(b)	(b)	(a)	(a)	(a)	(d)	(c)	(d)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(c)	(c)	(c)	(a)	(a)	(b)	(a)	(c)	(b)	(a)	(c)	(b)	(c)	(c)	(a)	(b)	(a)	(a)	(d)	(c)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(c)	(d)	(d)	(b)	(b)	(a)	(d)	(d)	(b)	(c)										

■■■ True/False

1. False 2. True 3. True 4. False 5. True

HINTS AND SOLUTIONS

Moderate

1. Speed of train while passing point
A = 70 × (5/18) m/s = V1
Speed of bike initially = 70 × (5/18) m/s = V2
Time taken by the bike to reach at the mid-point of the train = 150/(V2 – V1)
Again find out the new speeds of train and bike, and calculate the time taken by the bike to cover the rest 150 m distance relative to the train.

3. This question has actually got nothing to do with bullets initially. We can see that it takes them 4 hours to reach each other. And this is the same time for which bullets will cover some distance.
So, the total distance covered by the bullet = 4 × 10 = 40 km

4. Form the equations first and then use the options.

5. Go through the actual calculation method.

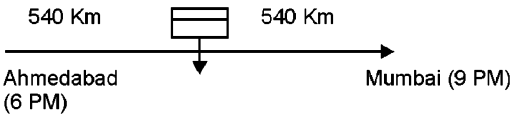
12. Total distance from Mumbai to Delhi = 910 km
Time taken by Sapna to cover this distance, had there been no snag = 910/60
Since their speeds are in the ratio 3 : 2, Manoj would have covered [910 – (910 × 40)/60] km less than Sapna.

13. The train can cover (200 + 350) m distance in five seconds which means the speed of the train is 110 m/s. Relative speed of man and train is 114 m/s. To cover the distance of 100 metre, it will take less than one second.

14. Go through the options.

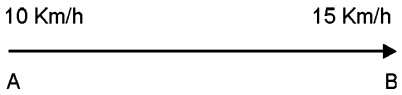
23. Let the original speed be X km/h
According to the question, 18/(4/5x) – 18/x = 9/60 hr
x = 30 km/h

24.



Now using options can get us the result.
Take the option 6 A.M. which means the train from Ahmedabad takes 12 hours to cover 540 km. In this way, the speed will be 45 km/h and train from Mumbai takes 9 hours to cover 540 km which means the speed is 60 km/h. It is written in the question that the difference between the speed of the train from Ahmedabad and that from Mumbai is 15 km/h. Hence, this is the answer.

27.



Both the trains are crossing each other after 12 minutes which means the distance between them is 5 km. It will be easy to go through the ratio of the speed which is 2 : 3. The answer is 36 minutes.

28. Let us assume that the width of the lake = x. So, when one of the runners A covers 900 m, the other one B is covering (x – 900) m. To meet next time, A will be covering (x – 900 + 300) m whereas B will be covering (900 + x – 300) m.
Now, 900/(x – 900) = (x – 900 + 300)/(x + 900 – 300)
Now use options to find the answer.

36. Go through the options.

37. This question gives us the freedom to assume any value of speeds of Ramesh and Somesh. Let us assume the initial speed of Somesh = 20 m/s, then the initial speed of Ramesh = 40 m/s. Till 50 m they are running with this speed only. Time taken by Ramesh in covering 50 m = 1.25 sec. In the same time Somesh is covering 25 m.
After this stage, speed of Somesh is 20 m/s, whereas speed of Ramesh = 10 m/s. Now relative speed = 10 m/s and distance = 25 m. At 75 m from the starting, both of them will be meeting.

[redacted]

38. For seventh meeting, both of them together must have covered a total of 13 D distance, where D is the distance between P and Q. Now, ratio of their speeds are 3 : 4, so distance covered will also be in the same ratio.

Hence, distance covered by B = $13D \times \frac{4}{7} = 1040$

Hence, D = 140 m

39. If A covers 1860 m, then in the same time B covers 2480 m (ratio of speed 3 : 4). So, total distance covered = 4340 m = 140 m + 30 × 140 m = 31 × 140 m, this is the distance covered for 16th meeting.

40. Assume that the Howrah mail is leaving on 13th November at 12 noon. It will be reaching its destination on 16th November at 12 noon. Now any train which leaves Trivandrum between the 10th and 16th November (excluding extremes) will be meeting Howrah mail on the track. These are five trains.

42. Relative speed of Vinay and Prabhat will be 20 m/minute to cover the track of 960 m. It will take 48 minutes.

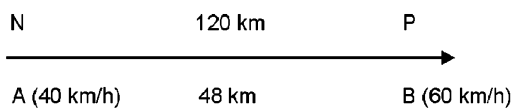
43. The train has to cover 1100 m @ 90 km/h and to cover that it will take 44 seconds to cover the distance 700 m by jeep and not to collide with each other the jeep has to take either less than 44 seconds or more than that.

45. At 2 O' clock Aflatoon has already covered 16 km @ 8 km/h, Bablajoon starts running in the same direction @ 10 km/h. The relative speed is 2 km/h. They will be 5 km apart at 7.30 p.m. the same day and 12.30 a.m. on the next day.

Advanced

1. Since there are 36 hours in a day, 18 hours dial will be there in the clock.
At 16:50, the hour hand will be in between 10 and 11 of the actual clock and the minute hand will be in between 6 and 7 of the actual clock. Now we can eliminate the options.

8. Assume any value of S and solve.



Ratio of relative speed 2:3
Distance travelled by A [2/5]×120= 48 km
Total distance travelled by B [3/5]120 = 72 km

9. According to the question:
First time meet at 48 km and reverse the direction with interchanging speed
T = [48 km/40] × 60 = 72 minutes (Reverse T)
And second time = [7260]/60 = 72

10. First time B covers 72 km. In reverse B covers (72 + 72) km. Hence, the total distance is 216 km.

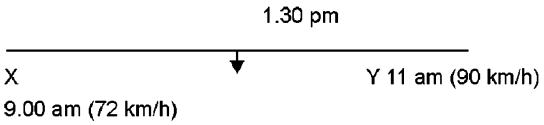
11. Assume that the track length is 1000 m.
Now, runner 1 and runner 2 will meet at one point, i.e., the starting point.
Runner 1 and runner 3 will meet at two points, at 500 m and at the starting point.
Runner 1 and runner 4 will meet at three points, at 333.33 m, at 666.66 m and at the starting point.
Runner 1 and runner 5 will meet at four points, at 250 m, 500 m, 750 m and at starting point.
Runner 1 and runner 6 will meet at five points, at 200 m, at 400 m, at 600 m, at 800 m and at the starting point.
These are 10 distinct points.

12. Do as done in 11.

13. If Amit would have increased his speed by 25% in the beginning, he would have saved one hour in covering the actual planned distance. So, 1/5 T = 1 hr. (Where T is the actual planned time). Hence, T = 5 hours.

14. Saving in time would be in the later 4 hours part. Since speed is increase by 25%, time taken to cover the same distance would reduce by 20%. Hence, he would save 48 minutes.

15.



They cross each other at 1.30 p.m. which means the distance is 549 km.
At 12 noon, a signal was down, distance travelled by x and y at 12 noon is 306 km. Rest of the distance will be covered in 4.5 hours which mean the relative speed is 54 km. After reducing an equal speed from the speed of both the trains, the equation is like this:
72 – p + 90 – p = 54, so, p = 54
Hence, the answer is 18 km/h.

16. Suppose that the signal problem occured at 1 p.m., the distance travelled by them is (72 × 4 + 19 × 2) km. Rest of the distance will be covered by reduced speed.

36. It does not create any difference whether they are running in the clockwise direction or anti-clockwise direction.

44. One of the ways of solving this question is going through equations. But after a certain stages we will be required to start assuming the values because all the data are not given.
Another way of doing this problem is: Start working by assuming some values. Let us assume the speed of Mayank = 10 km/h. In three hours he has covered 30 km. Now Sharat starts with a speed of 20 km/h. He will take 3 hours to meet Mayank. Till that time, the total distance covered by Mayank = 60 km.

BENCHMARKING TEST 2

This test paper contains 30 questions of 4 marks each. One-third of the marks allotted to a particular question will be deducted in case of wrong answer.

Time Given: 70 Minutes **Total Marks: 120** **Qualifying Marks: 55 Marks**

Break-up of the Expected Percentile
(assuming this to be a CAT paper)

Marks	*Expected Percentile*
88+	99+
83	98.5+
78	98
55	92
40	85
30	80

Direction for questions 1 and 2: Read the passage given below and solve the questions based on it.

A pedestrian having travelled from A to B found that if his rate had been one km/h quicker, he would have completed his journey in $1\frac{1}{2}$ h less and if it had been one km/h slower, he would have taken 2 hours more to complete it.

Q.1. The distance from A to B is
(a) 12 km (b) 20 km
(c) 26 km (d) 30 km

Q.2. The rate at which, the pedestrian covered the distance is
(a) 2 km/h (b) 3 km/h
(c) 4 km/h (d) 6 km/h

Q.3. There were 500 seats in PVR placed in similar rows. After the reconstruction of the hall the total number of seats became 450. The number of rows was reduced by 5 but each row contained 5 seats more than the earlier setting. What is the total number of rows and total numbers of seats in a row initially in the hall?
(a) 30 and 15 (b) 25 and 20
(c) 20 and 25 (d) 50 and 10

Q.4. Krishna owns several cows, some black and some white. He finds that 4 black cows and 3 white cows provide the same amount of milk in 5 days as 3 black cows and 5 white cows provide in 4 days. What is the ratio of milk provided by a black cow in a day to that by a white cow in a day?
(a) 8 : 5 (b) 5 : 8 (c) 3 : 5 (d) 5 : 3

Q.5. Two cities P and Q are 60 km apart. A college is to be built to serve 150 students in city P and 50 students in city Q.If the total distance to be travelled by all 200 students is to be as small as possible, then the college should be built at
(a) City P
(b) City Q
(c) 45 km from City P
(d) 45 km from City Q

Q.6. When ACP Rathod was going to Wahed from New Delhi to arrest Mirchi Seth, he observed few surprising things through his rear-view mirror

i. In nth hour he saw inscribed on a milestone—New Delhi—xy kms, where x and y are the tens and units place digit of the distance from that place to New Delhi (remember, he was observing it in rear-view mirror)

ii. In $(n + 1)$th hour, he saw inscribed on a milestone—New Delhi—yx kms, where x and y are representing the same digits.

What is the speed of ACP Rathod?

(a) 9 km/h (b) 12 km/h
(c) 18 km/h (d) Cannot be determined

Q.7. A and B start from the same point and in the same direction at 7 a.m. to walk around a rectangular field 400 m × 300 m. A and B walk at the rate of 3 km/h and 2.5 km/h respectively. How many times shall they cross each other, if they continue to walk till 12.30 p.m.?

(a) Not even once (b) Once
(c) Twice (d) Thrice

Q.8. Left pan of a faulty balance weighs 100 g more than its right pan. A shopkeeper keeps the weight measure in the left pan while buying goods but keeps it in the right pan while selling his goods. He uses only 1 kg weight measure. If he sells his goods at cost price, what is his gain percentage?

(a) $1\frac{1}{2}$ (b) $\frac{100}{11}\%$
(c) $\frac{1000}{9}\%$ (d) $\frac{200}{9}\%$

Q.9. Four different candles, which can last for 5 h, 4 h, 3 h and 2 h of burning, respectively, are lit in a room at the same instance and allowed to burn till the time the three candles in the room get extinguished. The cost of burning each one of these candles is 75 paise per h. What is the total cost involved?

(a) Rs 2.75 (b) Rs 3.75
(c) Rs 9.75 (d) None of these

Q.10. Two ladies simultaneously leave cities A and B connected by a straight road and travel towards each other. The first lady travels 2 km/h faster than the second lady and reaches B one hour before the second lady reaches A. The two cities A and B are 24 km. apart. What is the difference in their speeds/h?

(a) 4 km/h (b) 3 km/h
(c) 2 km/h (d) 1 km/h

Q.11. In a company, 605 of the employees are men. Of these 40% are drawing more than Rs 50,000 per year. If 36% of the total employees of the company draw more than Rs 50,000 per year, what is the percentage of women who are drawing less than Rs 50,000 per year?

(a) 70 (b) 60
(c) 40 (d) None of these

Q.12. The following table shows the percentage change in the consumption of electricity by five towns P, Q, R, S and T from 1986 to 1988:

Town	*Percentage change*	
	From 1986 to 1987	*From 1987 to 1988*
P	+8	–18
Q	–15	+11
R	+6	+9
S	–7	–5
T	+13	–6

In town T were 50,000 units consumed in 1986, how much did it consume in 1988?

(a) 37100 units (b) 53110 units
(c) 55110 units (d) 57100 units

Q.13. Water is filled in a container in such a manner that its volume doubles after every five minutes. If it takes 30 minutes for the container to be full, in how much time will it be one-fourth full?

(a) 7.30 min (b) 10 min
(c) 20 min (d) 25 min

Q.14. Amar, Akbar and Anthony are friends, being looked after by a matron Farah, Amar weighs 50% more than Akbar and Anthony weighs 25% less than Amar. Farah weighs a third of the combined weight of the three boys. All four together weigh 232 kg. What is the correct sequence of the persons in the ascending order of their weights?

(a) Anthony, Akbar, Farah, Amar
(b) Anthony, Akbar, Amar, Farah
(c) Akbar, Anthony, Amar, Farah
(d) Akbar, Anthony, Farah, Amar

Q.15. In a family, a couple has a son and a daughter. The age of the father is three times that of his daughter and the age of the son is half of his mother. The wife is nine years younger to her husband and the brother is seven years older than his sister. What is the average age of the mother and son?

(a) 40 years (b) 45 Years
(c) 50 years (d) 60 years

Direction for questions 16 and 17: Read the passage below and solve the questions based on it.

Several runners, numbered 1, 2, 3,... and so on, start running simultaneously from the same point on a circular track and run continuously, in the same direction. They run such that the speed of the runner numbered n ($n > 1$) is n times that of the runner numbered 1 and so on.

Q.16. If there are exactly six runners, then at N distinct points on the track is the runner numbered 1 overtaken by any of the other five runners. What is the value of N?
(a) 15 (b) 11 (c) 9 (d) 10

Q.17. If there are exactly four runners, then at how many distinct points on the track do two or more runners meet?
(a) 3 (b) 4 (c) 5 (d) 6

Q.18. Sharat is walking down a descending escalator and takes 40 steps to reach the bottom. Chandra starts simultaneously from the bottom, taking two steps for every one step taken by Sharat. Time taken by Sharat to reach the bottom from the top is the same as the time taken by Chandra to reach the top from the bottom. How many more steps did Chandra take than Sharat before they crossed each other on the escalator?
(a) 20 steps (b) 25 steps
(c) 40 steps (d) 60 steps

Q.19. A hare and a tortoise decide to run around a pond, in opposite directions from the same point at the same time. The hare meets the tortoise at a certain pine tree for the first time. The second time, they meet at a banyan tree and the third time at a mango tree. The fourth time, the hare meets the tortoise exactly at the first pine tree. How many of the following statements is/are true?
i. One of them runs three times as fast as the other.
ii. One of them is twice as fast as the other.
iii. One of them is six times as fast as the other.
iv. The starting points and starting times are required to comment about the ratio of their speeds.
(a) 0 (b) 1
(c) 2 (d) Cannot be determined

Q.20. Chunmun is running at a uniform speed from Rajender Nagar to Patna Junction along the railway tracks. After every 12 min, a train moving from Rajender Nagar to Patna Junction overtakes him. Also, after every 4 min, he meets a train travelling from Patna Junction to Rajender Nagar. Trains leave from Rajender Nagar for Patna Junction and Patna Junction for Rajender Nagar at a regular interval of 'S' min. What is the value of 'S'?
(a) 4 (b) 6 (c) 8 (d) 12

Q.21. Two brothers Kumar Kalyan and Amar Kumar are walking around a circular track of length 1800 m, with a speed of 40 m/min and 50 m/min respectively. Both of them start in the same direction simultaneously, but Amar Kumar reverses his direction every time he completes one round. After how much time (in min) from the start will they meet for the first time?
(a) 120 (b) 40 (c) 80 (d) 160

Q.22. Trains are coming from Patna to New Delhi and New Delhi to Patna at a regular interval of 1 h. They take 5 h to complete the journey. How many trains coming from New Delhi will cross a train coming from Patna that started at 10 p.m.?
(Given that the trains start from both the stations at the same time.)
(a) 5 (b) 6 (c) 10 (d) 11

Q.23. Two runners are running in a clockwise direction on a circular race-track of n ($n > 1$) km long. They meet for the second time at a point C which is 200 m from their starting point in the anticlockwise direction on the circular track. If both of the runners started at the same time, then the location of their first meeting point is
i. 100 m from the starting point in the anti-clockwise direction.
ii. 100 m from the point diagonally opposite to the starting point in anti-clockwise direction.
(a) Definitely 1 (b) Definitely 1
(c) Either 1 or 2 (d) Neither 1 nor 2

Q.24. In a 4000 m race around a circular stadium having a circumference of 1000 m, the fastest runner and the slowest runner reach the same point at the end of the 5th min, for the first time after the start of the race. All the runners have the same starting point and each runner maintains a uniform speed throughout the race. If the fastest runner runs at twice the speed of the slowest runner, what is the time taken by the fastest runner to finish the race?
(a) 5 min (b) 10 min
(c) 15 min (d) None of these

Q.25. In a race of 1000 m, A beats B by 100 m or 10 sec. If they start a race of 1000 m simultaneously from the same point and if B gets injured, after running 50 m less than half the race length and due to which his speed gets halved, then by how much time will A beat B?
(a) 45 s (b) 50 s
(c) 60 s (d) 65 s

Q.26. A train, 100 m long is approaching an unmanned railway crossing. The train is travelling at a uniform speed of 90 km/h and is 1 km away from the crossing.

At the same time, a bus, also approaching the crossing, is 700 m away from it. Assuming the bus is also travelling at a uniform speed of B, for what range of values of N will the bus collide with the train?
(a) 59 km/h < N < 61 km/h
(b) 57 km/h < N < 63 km/h
(c) 55 km/h < N < 65 km/h
(d) 53 km/h < N < 67 km/h

Q.27. Two friends Karan and Arjun start running around a circular track in the same direction. Karan and Arjun take one full round in 7 min and 4 min respectively. After how much time will they meet at diametrically opposite point of their starting point?
(a) 120 s
(b) 240 s
(c) They will never meet
(d) None of these

Q.28. The analysis of the recently held CBSE exams shows that Chennai zone, Delhi zone and Ajmer zone have performed the best. The average marks of the Ajmer zone and the Delhi zone put together is 71%. The average marks of the Chennai zone and the Delhi zone put together is 76% and the average marks of the Ajmer zone and the Chennai zone put together is 79%. What is the range of the average marks of all the three zones (Assume N) put together?
(a) 71% < N < 78%
(b) 70.5% < N < 77.5%
(c) 73.5% < N < 77.5%
(d) 73% < N < 79%

Q.29. At the end of the year 1998, a shepherd bought nine dozen goats. Henceforth, every year he added p% of the goats at the beginning of the year and sold q% of the goats at the end of the year, where $p > 0$ and $q > 0$. If shepherd had nine dozen goats at the end of the year 2002, after making the sales for that year, which of the following is true?
(a) $p = q$ (b) $p < q$ (c) $p > q$ (d) $p = q/2$

Q.30. Two fuel tanks, one shaped like a cylinder and other like a cone, contain fuel. The cylindrical tank holds 500 litres more fuel than the conical fuel. After 200 litres of fuel have been taken out from each tank, the cylindrical tank contains twice the quantity of fuel in the conical tank. How many litres of fuel did the cylindrical tank have when it was full?
(a) 700 (b) 1000 (c) 1100 (d) 1200

ANSWERS

■■■ Benchmarking Test 2

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(c)	(b)	(b)	(a)	(a)	(b)	(a)	(c)	(c)	(a)	(b)	(c)	(b)	(a)	(d)	(d)	(a)	(b)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30										
Answer	(b)	(d)	(c)	(b)	(d)	(b)	(c)	(c)	(c)	(d)										

Module 3

X+2 Maths

- Basics of Algebra
- Equations
- Squence and Series
- Function
- Graphs and Maxima Minima
- Logarithm
- Permutation and Combination
- Probability

9

CHAPTER

Basics of Algebra



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Basic terms involved in Algebra
- Different types of solving the equations
- System of equations
- Kinds of questions in the CAT
- Methods of solving questions

Introduction to the topic

Algebra is important for CAT in terms of weightage of the questions asked from this area. In this chapter, I have tried to brush up the basics of Algebra of those students who have 'passed' through it at some point of time in their life, but have not much of idea currently. Initially students are advised to see this chapter without any connection with the other chapters/concepts of the algebra. Once one has completed this chapter thoroughly, one will start seeing the connection between the basics given in this chapter with the concepts given ahead.

WHAT IS ALGEBRA?

Algebra is a branch of mathematics that substitutes letters for numbers Þ It simply means that while in Arithmetic we were having the statements like–"Ram is having 5 apples", in Algebra we will be having the statements like–"Ram is having x apples" where value of x can be deduced from the given conditions.

Moving from Arithmetic to Algebra will look something like this: Arithmetic: $3 + 4 = 3 + 4$. In Algebra it would look like: $x + y = y + x$

While in Arithmetic, we mostly deal with the numbers (which have a constant fixed value always), in case of Algebra, we mostly deal with the variables (which don't have a fixed value and can take different values according to the conditions governing.) For example, if we say that "Ram is having Rs 100", we get a clear picture that how much money Ram is having. And even if next day Ram is having 100 apples instead of Rs 100, the number "100" has the same meaning in this different situation.

Now replace "100" by "x" ⇒ "Ram is having Rs x", this does not give us a clear picture that how much money Ram is having. And if next day, Ram is having x apples, the letter "x" may not have the same value in a different situation.

ALGEBRAIC NOTATIONS AND OPERATIONS

i. Notations

Algebraic Notations	Meaning	Example
$x = y$	x is equal to y	$1 = 1$
$x \neq y$	x is not equal to y	$1 + 1 \neq 1$
$x < y$	x is less than y, or, y is greater than x	$2 < 5$
$x > y$	x is greater than y, or, y is less than x	$4 > 3$
$x \leq y$	y is greater than or equal to x, or, x is less than or equal to y	
$x \geq y$	x is greater than or equal to y, or, y is less than or equal to x	
$x \approx y$	x is approximately equal to y	$2.99999 \approx 3$

ii. Additions/Subtractions

Statement 1 – Ram is having 5 more apples than Shyam.

⇒ Number of apples with Ram = Number of apples with Shyam + 5

Assume that number of apples with Shyam = x, then number of apples with Ram = $x + 5$

Alternatively, if we assume that the number of apples with Ram = y, then number of apples with Shyam = $y - 5$

iii. Multiplication

When there are two or more equal numbers to be added together, the expression of their sum may be abridged. For example,

$x + x = 2 \times x = 2x$
$x + x + x = 3 \times x = 3x$
$x + x + x + x = 4 \times x = 4x$

In this manner, we may form an idea of multiplication; and it is to be observed that,

$2 \times x$ signifies 2 times x or twice x
$3 \times x$ signifies 3 times x or thrice x

Further, we can multiply such products again by other numbers; for example:

$2y \times 5 = 10y$
$2y \times 5z = 10yz$
$2y \times 6y = 12y^2$

iv. Linear Equation

An algebraic equation, such as $y = 2x + 7$ or $3x + 2y - z = 4$, in which the highest degree term in the variable or variables is of the first degree. The graph of such an equation is a straight line if there are two variables.

The general form of linear equation in two variables x and y is $ax + by + c = 0$, $a \neq 0$, $b \neq 0$, and a, b and c real numbers. Here a and b are known as coefficient of x and y respectively and c is a constant. A solution of such an equation is a pair of values, one for x and the other for y, which makes LHS and RHS of the equation equal.

For a linear equation in two variables x and y:

$$ax + by + c = 0 \Rightarrow by = -ax - c \Rightarrow y = \frac{-ax - c}{b}$$

In the above equation, for every real x, there exists a real number y corresponding to x.

Therefore every linear equation in two variables has infinitely many solutions, i.e., infinitely many pairs (x,y). For example, the equation $2x + 3y = 10$ will have infinite solution.

All these solutions are represented by points on a certain line. Due to this fact only this equation is called LINEAR because the graph of the equation on the x–y Cartesian plane is a straight line.

For example, consider the equation

$$3x + 2y = 8 \Rightarrow 2y = 8 - 3x \Rightarrow y = \frac{8 - 3x}{2} \qquad (1)$$

To find a solution of linear equation in two variables, we assign any value of one of the two variables and determine the value of the other variable, from the given equation (1).

Thus, taking $x = 1$, we get corresponding value of $y = \frac{5}{2}$.

Similarly, taking $x = 0$, we get $y = 4$; and so on.

The following table lists six possible values for x and the corresponding values for y, i.e. six solutions of the equation:

x	–2	–1	0	1	2	8/3
y	7	11/2	4	5/2	1	0

Taking any two pairs of given equation, we plot corresponding points, say, P and Q. The line PQ through these points is related with the given equation in the following manner:

(i) Every solution $x = p$ and $y = q$ of the given equation determines a point (p, q) that lies on this line.

(ii) Every point (x_i, y_i) lying on the line PQ, determines a solution $x = x_p\, y = y_i$ of the given equation.

The line PQ is said to be the graph of the given equation. It is worth noting that:

(a) We can add or subtract any number on both sides of the equation without affecting the equation and its solution.

(b) We can multiply or divide both sides of an equation by a non-zero number without affecting the equation and its solution.

If we plot the solutions of the equation $3x + 2y = 8$, which is represented in the table above then we notice that they all lie on the same line. We call this line the graph of the equation since it corresponds precisely to the solution set of the equation.

SYSTEM OF TWO LINEAR EQUATIONS IN TWO UNKNOWNS

A system of two linear equations in two unknowns is a system of two equations of the form

$$a_1 x + b_1 y = c_1$$
$$a_2 x + b_2 y = c_2$$

where $a_1, a_2, b_1, b_2, c_1, c_2$ are arbitrary real numbers.

The solution set of the system of linear equation in two unknowns is a pair of real numbers (x_0, y_0) which satisfies each of the equations of the system. In general,

$$\Rightarrow x = \frac{c_1 b_2 - c_2 b_1}{a_1 b_2 - a_2 b_1}$$

x is defined if $a_1 b_2 - a_2 b_1 \neq 0$ (i.e., for the existence of a unique solution of the system of linear equations)

In other words, the system of linear equations has a unique solution if $(a_1 b_2 - a_2 b_1) \neq 0$

$$\Rightarrow \frac{a_1}{a_2} \neq \frac{b_1}{b_2} or \Rightarrow \frac{a_1}{b_1} \neq \frac{a_2}{b_2}$$

Now let us consider the geometric interpretation of the discussing given above. We know that the set of points in a plane whose coordinates satisfy an equation of the form $ax + by = c$, where either a or b are non-zero, constitutes a straight line. And to solve a system in which every equation has at least one unknown means to find the common point of two straight lines. Therefore, a system of linear equations has a unique solution if the lines intersect.

E.g., 1. Consider the system of linear equations.

$4x + 6y = 10$ and $4x - 2y = 2$

Point (1,1) is the intersection of two straight lines. Hence $x = 1$, $y = 1$ is the unique solution of system of linear equations.

We can say that the system of linear equations has a unique solution if $a_1 b_2 - a_2 b_1 \neq 0$.

What if $a_1 b_2 - a_2 b_1 = 0$?

If $a_1 b_2 - a_2 b_1 = 0$

or $a_1 b_2 = a_2 b_1$

or $\frac{a_1}{a_2} = \frac{b_1}{b_2}$, then there are two situations:

Situation (i)

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

If situation (i) exists, then the system of linear equations has an infinite number of solutions.

Situation (ii)

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

If condition (ii) exists, then the system of linear equations has no solutions.

For example, consider following equations:

$$x + y = 10 \quad (1)$$
$$2x + 2y = 25 \quad (2)$$

In the given equations, $a_1/a_2 = b_1/b_2$, hence unique solution is not possible. To have a better understanding, it can be seen that LHS of equation (1) × 2 will give us the LHS of equation (2). Hence there will not be any point of intersection of these two graphs drawn on X–Y axis. In other words lines will be parallel.

Summarizing the whole discussion:

(i) The lines intersect	System of linear equations has a unique solution, i.e. $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (known as system is determinate)
(ii) The lines are	The system is inconsistent, i.e., parallel the system of linear equations has no solution, i.e.,
(iii) The lines are	System of linear equations has coincident infinitely many solutions, i.e., $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ (known as system is indeterminate)

SOLVING A SYSTEM OF LINEAR EQUATIONS

(i) Graphical Method

We plot the straight lines correspondence to each of the given linear equations in two variables. The point of intersection is the solution for x and y (i.e., two variables) in the system of linear equation.

Example 1 Solve the two equations $2x - y = 1$ and $2x + 3y = 5$.

Solution We will draw the graph of both the equations. We can see that the straight lines intersect at the point (1, 1). Hence the set of solutions of the given equations is (1, 1).

(ii) Elimination Method

Step 1 Multiply the coefficients of the equations by suitable numbers so that the coefficients of one of the variables becomes same in both the equations.

Step 2 Add or subtract to get one of the variables (which has become same in step 1) cancelled. This will lead to obtaining the value of other variable.

Step 3 Substitute this value in one of the equations to obtain the value of the other variable.

Example 2 Solve $3x - 5y = 1$
$x + 4y = 6$

Solution

Given equations are $3x - 5y = 1$ (1)
$x + 4y = 6$ (2)

Step 1 – If we multiply equation (2) by 3, then co-efficient of x will become same in both the equations. Alternatively, if we multiply equation (1) by 4 and equation (2) by 5, co-efficient of y will become same in both the equations. We are making co-efficient of x same in both the equations.

$3x - 5y = 1$ (3)
$3x + 12y = 18$ (4)

Step 2- Subtracting (3) from (4), we get
$(3x + 12y) - (3x - 5y) = 18 - 1$
$\Rightarrow 17y = 17 \Rightarrow y = 1$

Step 3 – Substitute value of y in equation 1.
$3x - 5 \times 1 = 1 \Rightarrow x = 2$

Alternative Method:

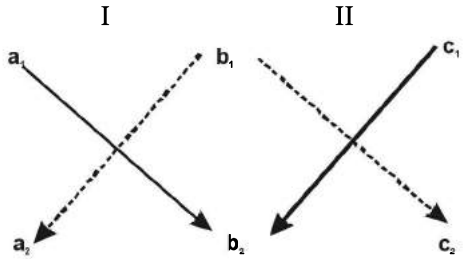
Suppose we have to solve the system of equations

$a_1x + b_1y = c_1$
$a_2x + b_2y = c_2$

Write the coefficients of x, y and the constants in the following manner (taking notice of the signs, i.e., positive and negative values, of the coefficients).

I		II
a_1	b_1	c_1
a_2	b_2	c_2

Now, do the cross-multiplication in the part I and part Ii as given below:



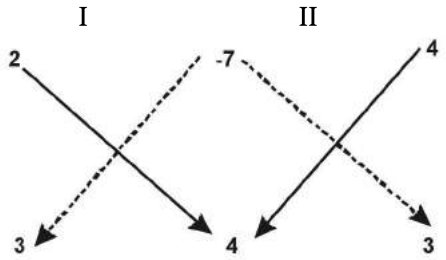
Note that there are two types of cross-multiplication in each of the Part I and Part II - expressed by bold lines and the expressed by dotted lines.

Now, subtract the value obtained by the cross-multiplication of dotted lines from that by the cross-multiplication of bold lines in each of the parts. Suppose the values obtained from part II is m and the value obtained from part I is n; then the value of the variables corresponding to coefficient a_1 and a_2 is $\frac{m}{n}$;

i.e., $x = \frac{m}{n}$.

Let us see an example:

Given equations are
$2x - 7y = 4$
$3x + 4y = 3$



$8 - (-21) = 29$ $16 - (-21) = 37$

can be obtained by putting this value of x in any of the equations [$y = (2x - 4)/7$]

(iii) Substitution Method

Assume that equations are in terms of x and y.

Step 1 – Express x in terms of y in one of the equations.

Step 2 – Put the value of x obtained from step 1 in the 2nd equation. Calculate the value of 'y' now.

Step 3 – Substitute the value of 'y' in any of the equations to find the value of x.

Example 3 Solve $x + 7y = 15$
$7x - 3y = 1$

Solution We have $x + 7y = 15$ (1)
$7x - 3y = 7$ (2)
From equation (1), $x = (15 - 7y)$
Substituting this value of x in equation (2),
$7 \times (15 - 7y) - 3y = 1$
$\Rightarrow y = 2$ and corresponding value of $x = 1$.

REMAINDER AND FACTOR THEOREM

(i) Remainder Theorem

Remainder theorem gives us a method for finding the remainder without actual division.

Let us understand this with the examples:

Example 4 Let $p(x) = x^4 - 3x^2 + 2x + 5$. Find the remainder when $p(x)$ is divided by $(x-1)$.

Solution Rather going for the actual division, Let us compute $p(1)$, ie value of $p(x)$ when x is replaced by 1, we have, $p(1) = 1 - 3 + 2 + 5 = 5$.

We find that the remainder when $p(x)$ is divided by $(x-1)$ is equal to $p(1)$, i.e., the value of $p(x)$ at $x = 1$.

Example 5 Find the remainder when
$p(y) = y^3 + y^2 + 2y + 3$ is divided by $y+2$

Solution Compute $p(-2)$ = Value of $p(y)$ when y is replaced by -2, we have,
$P(-2) = (-2)^3 + (2)^2 + 2(-2) + 3 = -8 + 4 - 4 + 3 = -5$

It follows from these two examples that the remainder obtained when $p(x)$ is divided by $(x-a)$ is equal to $p(a)$ i.e. the value of $p(x)$ at $x = a$.

The above result is stated as remainder theorem (Look in chapter 2 – Number system for more reference.)

Note:

1. If a polynomial $p(x)$ is divided by $(x + a)$, the remainder is the value of $p(x)$ at $x = -a$, i.e., $p(-a)$.
2. If a polynomial $p(x)$ is divided by $(ax-b)$, the remainder is the value of $p(x)$ at $x = b/a$, i.e., $P(b/a)$.
3. If a polynomial $p(x)$ is divided by $(ax + b)$, the remainder is the value of $p(x)$ at $x = -b/a$, i.e., $p(-b/a)$.
4. If a polynomial $p(x)$ is divided by $(b - ax)$, the remainder is equal to the value of $p(x)$ at $x = b/a$, i.e., $p(b/a)$.

Example 6

Determine the remainder when the polynomial $p(x) = x^4 - 3x^2 + 1$ is divided by $x-1$.

Solution
By remainder theorem, the required remainder is equal to $p(1)$.
Now, $p(x) = x^4 - 3x^2 + 2x + 1$
$\Rightarrow p(1) = (1)^4 - 3 \times 1^2 + 2 \times 1 + 1 = 1 - 3 + 2 + 1 = 1$
Hence, required remainder = $p(1) = 1$.

(ii) Factor Theorem

If $g(x)$ divides $f(x)$, we say that $f(x)$ is divisible by $g(x)$ or $g(x)$ is a factor of $f(x)$.

Factor theorem gives us a method to determine whether a polynomial $g(x)$ is a factor of a polynomial $f(x)$ or not without actual division.

Let $f(x)$ be a polynomial of degree greater than or equal to 1 and 'a' be a real number such that $f(a) = 0$, then $(x - a)$ is a factor of $f(x)$, Conversely, if $(x - a)$ is a factor of $f(x)$ then $f(a) = 0$.

Note:

1. $(x + a)$ is a factor of a polynomial $f(x)$ if $f(-a) = 0$
2. $(ax-b)$ is a factor of a polynomial $f(x)$ if $f(b/a) = 0$.
3. $ax + b$ is a factor of polynomial $f(x)$ if $f(-b/a)=0$
4. $(x-a)$ $(x-b)$ is a factor of a polynomial $f(x)$ if $f(a) = 0$ and $f(b) = 0$.

Example 7

Show that $(x - 3)$ is a factor of the polynomial
$x^3 - 3x^2 + 4x - 12$

Solution
Let $f(x) = x^3 - 3x^2 + 4x - 12$ be the given polynomial. By factor theorem, $(x-a)$ is a factor of a polynomial $f(x)$ if $f(a) = 0$. Therefore, in order to prove that $(x-3)$ is a factor of $f(x)$, it is sufficient to show that $f(3) = 0$.

Now, $f(x) = x^3-3x^2+4x-12$

$\Rightarrow f(3) = 3^3-3\times3^2+4\times3-12 = 27-27+12-12=0$

Hence, $(x-3)$ is a factor of $f(x)$.

Example 8

Find out if $(x-1)$ is a factor of (i) and (ii) .

Solution

Let $f(x) = x^{10}-1$ and $g(x)=x^{11}-1$.

To find out that $(x-1)$ is a factor of both $f(x)$ and $g(x)$, it is sufficient to show that $f(1) = 0$ and $g(1) = 0$.
Now, $f(x) = x^{10}-1$ and $g(x)=x^{11}-1$

$\Rightarrow f(1)=1^{10}-1=0$ and $g(1)=1^{11}-1=0$

$\Rightarrow (x-1)$ is a factor of both $f(x)$ and $g(x)$

More worked out problems:

Example 9

Solve the equation $3x - 2y = 7$.

Solution

Let us first find one particular solution and then we will extend that. Its not too difficult to find one solution using hit and trial. One such solution is (3, 1). Alternatively, we can find it using $[x = (7 + 2y)/3]$

To find the next solution, let us re-write the equation as $3x = 2y + 7$.

Using the concepts of remainder, above written equation can be explained as:

LHS = A multiple of 3

RHS = A number that gives remainder 7 when divided by 2 which means a number that gives remainder 1 when divided by 2.

Obviously the next number will be obtained after the LCM (2,3) = 6. To move ahead by 6, we will cover 2 multiples of 3 (2×3) and 3 multiples of 2 (3×2). Hence the next solutions are (5, 4) (7,7) (9, 10) (11, 13) etc.

Note– Since it is a linear equation, there will be obviously infinite solution.

Example 10

Find all integral roots of the equation $21x + 48y = 5$.

Solution

LHS is divisible by 3, but RHS is not divisible by 3. Hence no integral solution is possible.

Statement I

(i) $1990x - 173y = 11$ has no solution in integers for x and y.

(ii) $3x - 12y = 7$ has no solution in integers for x and y.

Which of the following is true?

(1) Statement I is wrong and statement II is right.
(2) Statement I is right and statement II is wrong.
(3) Statement I and II both are right.
(4) Statement I and II both are wrong.

Statement II

The coefficients in the equation are large enough to find a particular solution. However, we can see that the numbers 1990 and 173 are relatively prime and this helps us reach the conclusion.

Hence we can find the solution of $1990x - 173y = 11$ in integers for x and y.

Hence statement I is wrong.

Statement II

In the linear equation $3x - 12y = 7$; coefficients of x and y, i.e. 3 and 12, are not relatively prime. HCF of (3, 12) = 3. Also, constant 7 is not exactly divisible by 3. In other words, LHS is divisible by 3, but RHS is not divisible by 3.

Hence $3x - 12y = 7$ has no solution for x and y in integers.

Hence, statement II is wrong.

Therefore option (4) is true.

PRACTICE EXERCISES

Q.1. If $x + y + z = 0$, then $x^3 + y^3 + z^3$ is equal to

(a) 0 (b) $3xyz$

(c) $\frac{xy+yz+zx}{xyz}$ (d) $xyz\,(xy + yz + zx)$

Q.2. If $x-\frac{1}{x}=2$, then the value of $x^4+\frac{1}{x^4}$ is

(a) 4 (b) 8

(c) 12 (d) 34

Q.3. If $\left(x+\frac{1}{x}\right)=3$, then the value of $\left(x^6+\frac{1}{x^6}\right)$ is
(a) 927 (b) 414
(c) 364 (d) 322

Q.4. Let $f(x)=x^2-27x+196$. If $(x)=x$, then what is the value of x?
(a) 28 (b) 14
(c) 7 (d) 4

Q.5. Consider the following statement: The equations $ax + by = 1$ and $ax - by = 1$ have a unique solution for
(a) $a = 1, b = 0$ (b) $a = 0, b = 1$
(c) $a = 0, b = 0$ (d) $a = 1, b = 1$

Q.6. Find the numerical value of 'c', if the expression $xy - 3x + 5y + c$ can be factorized.
(a) 9 (b) –15
(c) 15 (d) –9

Q.7. If x is less than 2, then which of the following statements is always true?
(a) x is negative
(b) $2x$ is greater than or equal to x
(c) x^2 is greater than or equal to x
(d) None of these

Q.8. If $x - y = 8$, then which of the following must be true?
I Both x and y are positive.
II if x is positive, y must be negative
III If x is negative, y must be negative
(a) I only (b) II only
(c) I and II (d) III only

Q.9. Five tables and eight chairs cost Rs 7,350. Three tables and five chairs cost Rs 4,475. What is the price of one table?
(a) Rs 950 (b) Rs 325
(c) Rs 925 (d) Rs 350

Q.10. The cost of 3 apples and 2 guavas is Rs 23 and that of one apple and four guavas of the same type is Rs 21. What is the cost of 5 guavas?
(a) Rs 4 (b) Rs 20
(c) Rs 12 (d) Rs 25

Q.11. If the two digits of the age of Mr Manoj Sagar are reversed then the new age so obtained is the age of his wife. If the sum of their ages is equal to the difference between their ages. If Mr Manoj Sagar is elder than his wife then find the difference between their ages.
(a) 9 years (b) 10 years
(c) 8 years (d) Cannot be determined

Q.12. Find the remainder when the polynomial $f(x)=2x^4-6-x^3+2x^2-x+2$ is divided by $x+2$.
(a) 102 (b) –102
(c) 92 (d) –92

Q.13. Find the remainder when $f(x) = 4x^3 - 12x^2 + 14x - 3$ is divided by $g(x) = x - 1/2$
(a) 2/3 (b) –2/3
(c) 3/2 (d) –3/2

Q.14. Find the remainder when $f(x) = x^3 - 6x^2 + 2x - 4$ is divided by $g(x) = 3x - 1$.
(a) 107/27 (b) – 107/27
(c) – 4 (d) 4

Q.15. Find the remainder when $f(x) = x^3 - 6x^2 + 2x - 4$ is divided by $g(x) = 1 - 3x$.
(a) 107/27 (b) – 107/27
(c) – 4 (d) 4

Q.16. If the polynomials $ax^3 + 4x^2 + 3x - 4$ and $x^3 - 4x + a$ leave the same remainder when divided by $(x - 3)$, find the value of a.
(a) 1 (b) 0
(c) – 1 (d) None of these

Q.17. The polynomials $ax^3 + 3x^2 - 13$ and $2x^3 - 5x + a$ are divided by $x + 2$. If the remainder in each case is the same, find the value of a.
(a) 5/9 (b) – 5/9
(c) 2/7 (d) – 2/7

Q.18. What is the value of a, if $(x - a)$ is a factor of $(x^3 - a^2x + x + 2)$?
(a) 2 (b) – 2
(c) 0 (d) 1

Q.19. Find the values of $(a + b)$ if the polynomial $x^3 - ax^2 - 13x + b$ has $(x - 1)$ and $(x + 3)$ as factors.
(a) 2 (b) 15
(c) 18 (d) None of these

Q.20. What is the value of p, if $x+3$ is a factor of $3x^2 + px + 6$?
(a) 9 (b) – 9
(c) 11 (d) – 11

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(d)	(d)	(b)	(d)	(b)	(d)	(d)	(a)	(b)	(a)	(c)	(c)	(b)	(b)	(c)	(a)	(b)	(c)	(c)

HINTS AND SOLUTIONS

1. $x + y = -z \Rightarrow (x + y)^3 = (-z)^3$

   $\Rightarrow x^3 + y^3 + 3xy(x + y) + z^3 = 0.$

   So, $x^3 + y^3 + z^3 = 3xyz$

2. $x - \frac{1}{x} = 2 \Rightarrow \left[x - \frac{1}{x}\right]^2 = 4 \Rightarrow x^2 + 1/x^2 = 6$

   Or, $[x^2 + 1/x^2]^2 = 36$

   Hence $x^4 + \frac{1}{x^4} = 34 = 34$

4. $f(x) = x^2 - 27x + 196 = x$, or, $x^2 - 28x + 196 = 0$

   $\Rightarrow (x - 14)(x - 14) = 0$

   So, $x = 14$

5. $a_1x + b_1y = 1$

   $a_2x - b_2y = 1$

   For unique solution we know $\frac{a_1}{a_2} \neq \frac{b_1}{-b_2}$

   Now go through the options a and b cannot be zero hence options (a) (b) (c) rejected. Only option (4) is answer.

6. $xy - 3x + 5y + c$ can be factorized (given) as

   $x(y - 3) + 5(y - 3) \Rightarrow xy - 3x + 5y - 15$

   After comparing we get ($c = -15$)

7. Statement 1 is obviously not always true.

   Statement 2 is not true for negative values of x.

   Statement 3 is not true for $0 < x < 1$.

8. If $x = 10$ and $y = 2 \therefore x - y = 8$ so II rejected.

   If $x = 6$ and $y = -2 \therefore x - (-2) = 6 + 2 = 8$ so (I) rejected

   If $x =$ –ve, for $x - y =$ +ve, y must be –ve as

   $x = -2$ and $y = -10 \Rightarrow x - y = -2 - (-10) = -2 + 10 = 8$

   So III is true.

9. 5T (Table) + 8C (chair) = 7350 (given) –(1)

   3T + 5C = 4475 (given) –(2)

   $5 \times (1) - 8 \times (2)$

   $\Rightarrow$ 25T + 40C = 36,750

   <u>24T + 40C = 35, 800</u>

   T = 950 Rs.

10. Assume that price of an apple = Rs x and price of a guava = Rs y

    Given is $3x + 2y = 23$ and $x + 4y = 21$

    Solving we get $y = 4$, Hence cost of 5 guavas = Rs 20

12. We have, $x + 2 = x - (-2)$. So, by remainder theorem, when $f(x)$ is divided by $\{x - (-2)\}$ the remainder is equal to $f(-2)$.

    Now, $f(x) = 2x^4 - 6x^3 + 2x^2 - x + 2$

    $\Rightarrow f(-2) = 2(-2)^4 - 6(-2)^3 + 2(-2)^2 - (-2) + 2$

    $\Rightarrow f(-2) = 2 \times 16 - 6 \times -8 + 2 \times 4 + 2 + 2$

    $\Rightarrow f(-2) = 32 + 48 + 8 + 2 + 2 = 92$

    Hence, required remainder = 92.

13. Using remainder theorem, we know that $f(x)$ when divided by $g(x) =$ gives a remainder equal to $f\left(\frac{1}{2}\right)$.

    Now, $f(x) = 4x^3 - 12x^2 + 14x - 3$

    $\Rightarrow f\left(\frac{1}{2}\right) = \frac{3}{2}$

    Hence, required remainder $= f\left(\frac{1}{2}\right) = \frac{3}{2}$

14. We have, $g(x) = 3x - 1 = 3\left(x - \frac{1}{3}\right)$. Therefore, by remainder theorem, when $f(x)$ is divided by $g(x) = 3\left(x - \frac{1}{3}\right)$, the remainder is equal to $f\left(\frac{1}{3}\right)$.

Now, $f(x) = x^3 - 6x^2 - 2x - 4$

$\Rightarrow f\left(\frac{1}{3}\right) = \left(\frac{1}{3}\right)^3 - 6\left(\frac{1}{3}\right)^2 + 2\left(\frac{1}{3}\right) - 4$

$= \frac{1}{27} - \frac{6}{9} + \frac{2}{3} - 4 = -\frac{107}{27}$

Remainder $= -\frac{107}{27}$

15. Using remainder theorem, when $f(x)$ is divided by $g(x) = 1 - 3x$, the remainder is equal to $f(1/3)$.

Now, $f(x) = x^3 - 6x^2 + 2x - 4$

$f\left(\frac{1}{3}\right) = \frac{1}{27} - \frac{2}{3} + \frac{2}{3} - 4 = \frac{-107}{27}$

Hence, required remainder

16. Let $f(x) = ax^3 + 4x^2 + 3x - 4$ and $g(x) = x^3 - 4x + a$ be the given polynomials. The remainders when $f(x)$ and $g(x)$ are divided by $(x-3)$ are $f(3)$ and $g(3)$ respectively.

According to the question, $f(3) = g(3)$

$\Rightarrow 27a + 36 + 9 - 4 = 27 - 12 + a$

$\Rightarrow 26a + 26 = 0 \Rightarrow a = -1$

17. Let $f(x) = ax^3 + 3x^2 - 13$ and $g(x) = 2x^3 - 5x + a$ be the given polynomials. The remainder when $p(x)$ and $q(x)$ are divided by $(x+2)$ are $f(-2)$ and $g(-2)$ respectively.

Given that $f(-2) = g(-2)$

$\Rightarrow a(-2)^3 + 3(-2)^2 - 13 = 2(-2)^3 - 5(-2) + a$

$\Rightarrow -8a + 12 - 13 = -16 + 10 + a$

$\Rightarrow -8a - 1 = a - 6$

$\Rightarrow a = \frac{5}{9}$

18. Let $f(x) = x^3 - a^2x + x + 2$ be the given polynomial. By factor theorem, $(x - a)$ is a factor of $f(x)$ if $f(a) = 0$.

Now, $f(a) = 0 \Rightarrow a^3 - a^2 \times a + a + 2 = 0$

$\Rightarrow a + 2 = 0 \Rightarrow a = -2$.

Hence, $(x-a)$ is a factor of the given polynomial, if $a = -2$

19. Let $f(x) = x^3 - ax^2 - 13x + b$ be the given polynomial. If $(x-1)$ and $(x-3)$ are factors of $f(x)$, then

$f(1) = 0$ and $f(-3) = 0$

$\Rightarrow 1^2 - a \times 1^2 - 13 \times 1 + b = 0$

and $(-3)^2 - 13 \times (-3) + b = 0$

$\Rightarrow 1 - a - 13 + b = 0$

and $-27 - 9a + 39 + b = 0$

$\Rightarrow -12 - a + b = 0$

and $-9a - b = 12$

Solving the equations, we get $\Rightarrow a = 3$

Putting $a = 3$ in $a - b = -12$, we get $b = 15$

Hence, $a = 3$ and $b = 15$

20. $f(-3) = 0$

$\Rightarrow 3(-3)^2 + p(-3) + 6 = 0$

$\Rightarrow 27 - 3p + 6 = 0 \Rightarrow p = 11$

10

CHAPTER

Equations



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Polynomials and different equations
- Definitions and properties of those equations
- Concepts attached to the these equations and their maxima-minima
- Inequations
- Kinds of questions asked in the CAT
- Methods of solving problems

Introduction to the topic

Quadratic equations is one of the popular topics for the CAT. Almost every year, 1–3 questions are asked from this topic. Normally, the questions are asked using the simple concepts of quadratic equations only, but it is desirable to have a good knowledge of the cubic equation and other polynomials also. Most of the questions asked from this topic are logical in nature, i.e., they can be solved with the help of alternative methods such as using options, or assuming some value.

DEFINITIONS

Polynomials

Assume that a_1, a_2, a_3, a_4, ... are real numbers and x is a real variable. Then $f(x) = a_1x^n + a_2x^{n-1} + a_3x^{n-2} + \ldots + a_{n-1}x + a_nx$ is called a polynomial.

For example, $5x^5 + 3x^4 + \ldots + x$ is a polynomial in x, where x is a real variable.

Degree of a Polynomial

Polynomial $f(x) = a_1x^n + a_2x^{n-1} + a_3x^{n-2} + \ldots + a_{n-1}x + a_nx$ is a polynomial of degree n, where $a_1 \neq 0$.

For example, $5x^5 + 3x^4 + \ldots + x$ is a polynomial of degree 5.

$2x^4 + x^3 + 4x^2 + 2x + 10$ is a polynomial of degree 4.

$4x^3 + 4x^2 + 2x + 10$ is a polynomial of degree 3.

$4x^2 + 2x + 10$ is a polynomial of degree 2.

Remember that

- The degree of a polynomial is defined for real as well as complex polynomials.
- The degree of a polynomial cannot be a fraction.

Polynomial Equation

If $f(x)$ is a real or complex polynomial, then $f(x) = 0$ is the corresponding equation. Simply put, when we equate any polynomial with zero, then it becomes equation.

For example, $5x^5 + 3x^4 + \ldots + x = 0$ is an equation.

$2x^4 + x^3 + 4x^2 + 2x + 10 = 0$ is an equation.

Polynomial Inequation

If $f(x)$ is a real or complex polynomial, then $f(x) \neq 0$ is the corresponding inequation. Simply put, if any polynomial is not equal to zero, i.e., either greater than zero or less than zero, then it is known as an inequation.

For example, $2x^4 + x^3 + 4x^2 + 2x + 10 > 0$ is an inequation.

$4x^2 + 2x + 10 < 0$ is an inequation.

Roots of an Equation

All the values of a variable satisfying the equation are known as the roots of the equation. It is known that an equation of degree n will have n roots, real or imaginary. Roots can be the same or distinct. When the roots are the same, they are known as repeated roots.

Quadratic Equation

Any equation of degree 2 is known as a quadratic equation.

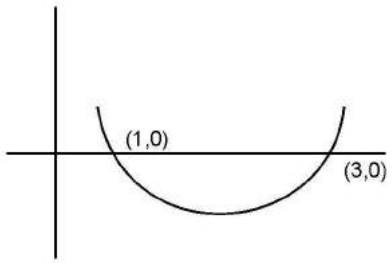
$ax^2 + bx + c = 0$ is known to be the standard equation of quadratic equation. This equation will have two roots either real or imaginary.

Geometrical Meaning of Roots

For any given equation $y = f(x) = 0$, the number of times the graph of this equation cuts X-axis is equal to the distinct real roots of this equation.

For example, $y = (x-1)(x+2)(x-2) = 0$ will intersect X-axis at three distinct points, namely $x = 1, -2$ and 2.

For any quadratic equation $f(x) = (x-1)(x-3)$, the graph of this equation (as can be seen below) will intersect X-axis at two distinct points, namely $x = 1$ and $x = 3$.



So, the equation $f(x) = x^2 - 4x + 3 = 0$ will have two roots, i.e., two values of $x = 1$ and $x = 3$, which will satisfy this equation.

If we say that the two roots 'p' and 'q' of the quadratic equation $f(x) = ax^2 + bx + c = 0$ are such that $f(2) \times f(-2) > 0$, then either both the roots 'p' and 'q' of this equation will lie inside -2 and 2 or both the roots 'p' and 'q' will lie outside the range of -2 and 2.

Similarly, if we say that the two roots 'p' and 'q' of the quadratic equation $f(x) = ax^2 + bx + c = 0$ are such that $f(2) \times f(-2) < 0$, then one of the two roots of this equation will lie inside this range and another will lie outside the range of -2 and 2.

FACTORIZATION OF A QUADRATIC EQUATION

When factorizing quadratic equations, there is generally nothing that is common in every term. In this case, we cannot pull a constant value out and rewrite the problem.

You should know that your answer would look like. $(y+a) \times (y+b)$. The trick is knowing how to find a and b.

For example, $y^2 + 4y - 12$

Start with the constant term. In this particular case, the constant term is -12. Factorize -12 into groups. Each group is a set of two numbers that when multiplied together will give -12. With the number -12, these can be the following groups:

$\{-1, 12\}, \{-2, 6\}, (-3, 4\}, \{1, -12\}, \{2, -6\}, \{3, -4\}$

Now you have six different groups. The next step is to find out the coefficient of the 'y' term and the number to be multiplied by 'y' in the quadratic equation. In this case, that number is 4. Now, we have to find out which one of the group will give us 4, when the two numbers of each group are added together.

If you add -1 and 12, you get 11. This is not correct, So, try again.

If you add −2 and 6, then you get 4. This is the required group.

The numbers that you choose in the group (in this case −2 and 6) will be the a and the b that we have been looking for. Which one you chose for a and which you chose for b does not matter. I will just use $a = -2$ and $b = 6$,

$[y + (-2)] \times (y + 6)$

$= (y - 2)(y + 6)$

Properties of the Roots

1. A polynomial equation of degree n will have n roots, real or imaginary.
2. Complex roots or surds always occur in pair.
   For example, if $2 + \sqrt{5}$ is a root of any equation, then $2 - \sqrt{5}$ will also be a root of that equation.
3. Every equation of an even degree with coefficient of the highest degree term positive and the constant term negative will have at least two real roots. Out of these two real roots, one will be positive and the other one will be odd.
4. Any equation with the sum of all of its coefficients equal to zero will have 1 as one of its roots.
   For example, $x^2 - 5x + 4 = 0$ will have 1 as one of its roots.
5. If all the terms of an equation are +ve and it does not involve any odd powers of x, then this equation will not have any real root.
   For example, $f(x) = x^4 + x^2 + 1 = 0$ will not have any real root. As $x^4 \geq 0$, $x^2 \geq 0$, either of the numbers can become the a or b, here we will take 1 and 1 is positive. And we know that sum of three positive number. can never be equal to zero for any real value of x.

Example 1 How many real roots will be there of the quadratic equation $f(x) = x^2 + 5|x| + 6 = 0$?

Solution If $x > 0$, then $f(x) = x^2 + 5x + 6 = 0$

Or, $x^2 + 5x + 6 = (x + 2)(x + 3) = 0$

So, $x = -2$ and $x = -3$

But as we have taken $x > 0$, the negative values of x are not admissible.

Now, if $x < 0$, then $f(x) = x^2 - 5x + 6 = 0$

Or, $x^2 - 5x + 6 = (x - 2)(x - 3) = 0$, So, $x = 2$ and $x = 3$

Now as we have taken $x < 0$, so the positive values of x are not admissible.

And $x = 0$ is not possible.

So, no real value of x can be obtained.

Alternatively, it is worth observing that all the terms of $f(x)$, viz., x^2 and $5|x|$ and 6 are positive. So, the sum of these three terms cannot be equal to zero for any real value. Hence, no real value of x can be obtained.

Descartes' Rule of Signs of Roots

Descartes' rule tells us how to find the maximum number of positive roots in any equation. The maximum number of positive roots in $f(x)$ is equal to the change of signs from positive (+ve) to negative (–ve) and from negative (–ve) to positive (+ve).

With the help of this, we can also find out the maximum number of real roots. To find out the negative roots, we apply the same procedure with $f(-x)$.

For example, let us see the roots of the quadratic equation $x^2 + x - 2 = 0$

Signs of different terms: x^2 has a positive sign, x has a '+' sign, the constant term (2) has a negative sign.

$f(x) = x^2 + x - 2$, and their signs are

\+ + −

As there is only one sign change, we can conclude that this equation will have one positive real root.

Now, $f(-x) = (-x)^2 + (-x) - 2 = x^2 - x - 2$

Again, there is just one sign change, so there will be one negative root. Also there are no positive or negative imaginary roots and only real numbers can be negative or positive.

Example 2 How many real roots will be there for the equation $x^4 + x^2 + 1 = 0$?

Solution To find out the positive roots:

$f(x) = x^4 + x^2 + 1$, and signs are

\+ + +

There is no sign change between any two terms, so, no roots will be positive.

To find out the negative roots:

$f(-x) = (-x)^4 + (-x)^2 + 1 = x^4 + x^2 + 1$

Again, there is no sign change, so, the equation will have no negative roots.

And $x = 0$ is also not a root of this equation.

So, no real roots are possible which can satisfy this equation.

It can also be inferred, that all the four roots of this equation will be negative.

Alternatively, we have done this question while seeing the properties of roots (v) also.

ROOTS OF A QUADRATIC EQUATION AND THEIR NATURE

Given is the quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$.

Assuming that a and b are the roots of this equation.

Then $\alpha = \dfrac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $\beta = \dfrac{-b - \sqrt{b^2 - 4ac}}{2a}$

It can be seen that the nature of both these two roots depends upon the value of $b^2 - 4ac$.

$(b^2 - 4ac)$ is also known as **Discriminant (D)** of the quadratic equation $ax^2 + bx + c = 0$.

If D = 0, then $\sqrt{b^2 - 4ac} = 0$.

So, the roots will be real and equal.

If D > 0, then $\sqrt{b^2 - 4ac} > 0$.

So, the roots will be real and distinct.

If D < 0, then $\sqrt{b^2 - 4ac}$ is not real.

So, the roots will not be real.

If D is a perfect square (including D = 0) and a, b and c are rational, then the roots will also be rational.

Example 3 Given is the quadratic equation $ax^2 + bx + 1 = 0$, where a, b (1, 2, 3, 4). For how many set of values of (a, b), the quadratic equation $ax^2 + bx + 1 = 0$ will have real roots? **(CAT 2003)**

Solution For the roots to be real, D ≥ 0.

$D = b^2 - 4a \geq 0$.

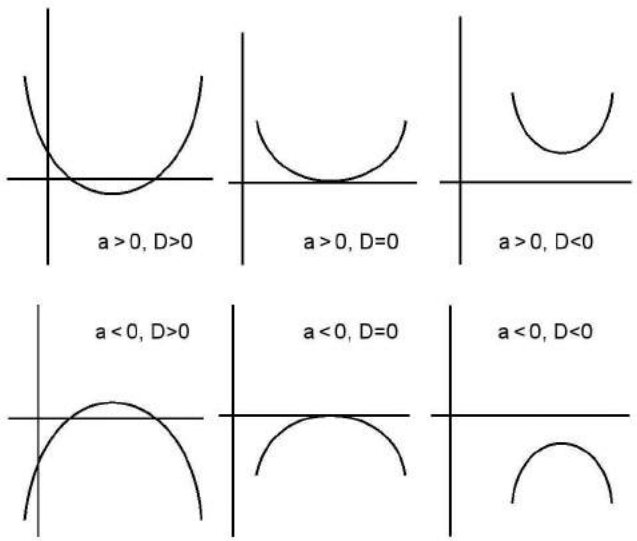
Forming the table for the above-written condition:

b	*a*
1	Not possible
2	1
3	1
3	2
4	1
4	2
4	3
4	4

So, there are seven set of values for (a, b).

GRAPH OF A QUADRATIC EQUATION

$f(x) = ax^2 + bx + c = 0$, where $a \neq 0$, and b and c are real coefficients.



As is obvious from the above graphs, when D > 0, there are two points at which this curve is intersecting X-axis. And we also know that there are two distinct values of x which satisfy $f(x)$.

When D = 0, there is only one point at which this curve is intersecting X-axis, and there is only one distinct value of x, which satisfies $f(x)$.

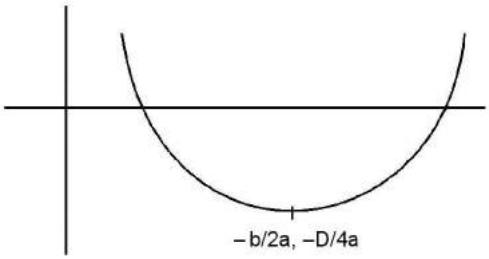
When D < 0, there is no point at which this curve is intersecting X-axis. So, there are no real values of x which can satisfy $f(x)$.

MAXIMA AND MINIMA OF A QUADRATIC EQUATION

$f(x) = ax^2 + bx + c$, if $a > 0$.

Finding Minima

Graphical method



At $x = -b/2a$, $f(x)$ will have the minimum value $= -D/4a$

Equation method

In this method, the objective is to break the equation into a format of $(P)^2 \pm Q$. Now, as we know that any perfect square can have a minimum value of 0, so the minimum value of $f(x) = \pm Q$

Let us assume $f(x) = x^2 - 5x + 6$

$$f(x) = \left[x^2 - 2\times\frac{5}{2}\times x + \left(\frac{5}{2}\right)^2\right] - 0.25 = \left(x - \frac{5}{2}\right)^2 - 0.25$$

Since $\left(x - \frac{5}{2}\right)^2$ can have a minimum value of zero, so the minimum value of $f(x)$ here is (– 0.25).

Using the graphical method, at $x = -b/2a = 5/2$, the minimum value of $f(x) = -D/4a = -1/4 = (-0.25)$

Finding maxima As the value of x is increased, the value of y also increases. So, the maximum value of y is $+\infty$.

It is obvious from the graph that it is going upwards till infinity.

$f(x) = ax^2 + bx + c$, if $a < 0$.

Finding minima As the value of x is increased, the value of y decreases (since $a < 0$). So, the minimum value of y is $-\infty$.

It is obvious from the graph that it is going downward till negative of infinity.

Finding maxima The same method of finding minima is applied as in the earlier case ($a > 0$).

ALGEBRAIC CALCULATIONS ON ROOTS

$f(x) = ax^n + bx^{n-1} + cx^{n-2} + \ldots + K$

The sum of the roots taking one at a time = (Coefficient of x^{n-1}/Coefficient of x^n) × $(-1)^n$, where n is the number of roots taken at a time.

The sum of the roots taking two at a time = (Coefficient of x^{n-2}/Coefficient of x^n) × $(-1)^n$, where n is the number of roots taken at a time.

The sum of the roots taking three at a time = (Coefficient of x^{n-3}/Coefficient of x^n) × $(-1)^n$, where n is the number of roots taken at a time.

The product of roots, taking all at a time = (Constant term/Coefficient of x^n) × $(-1)^n$, where n is the total number of roots.

Quadratic Equation

$f(x) = ax^2 + bx + c = 0$

Assume that the roots are α and β.

$\alpha + \beta$ = (Coefficient of x^{n-1}/Coefficient of x^n) × $(-1)^n = \frac{b}{a}(-1)^n = -\frac{b}{a}$

$\alpha\ \beta$ = (Constant term/Coefficient of x^n) × $(-1)^n = \frac{c}{a}(-1)^2 = \frac{c}{a}$

Cubic Equation

$f(x) = ax^3 + bx^2 + cx + d = 0$

Assume that the roots are α, β and γ.

$\alpha + \beta + \gamma$ = (Coefficient of x^{n-1}/Coefficient of x^n) × $(-1)^n = \frac{b}{a}(-1)^1 = -\frac{b}{a}$

$\alpha\beta + \gamma\alpha + \beta\gamma$ = (Coefficient of x^{n-2}/Coefficient of x^n) × $(-1)^n = \frac{c}{a}(-1)^2 = \frac{c}{a}$

$\alpha\ \beta\ \gamma$ = (Constant term/Coefficient of x^n) × $(-1)^n = \frac{d}{a}(-1)^3 = -\frac{d}{a}$

Bi-quadratic Equation

$f(x) = ax^4 + bx^3 + cx^2 + dx + e = 0$

Assume that the roots are α, β, γ and δ.

$\alpha + \beta + \gamma + \delta$ = (Coefficient of x^{n-1}/Coefficient of x^n) × $(-1)^n = \frac{b}{a}(-1)^1 = -\frac{b}{a}$

$\alpha\beta + \gamma\delta + \alpha\delta + \beta\gamma + \alpha\gamma + \delta\beta$ = (Coefficient of x^{n-2}/Coefficient of x^n) × $(-1)^n = \frac{e}{a}(-1)^4 = \frac{e}{a}$

$\alpha\beta\gamma + \delta\alpha\beta + \gamma\delta\alpha + \beta\gamma\delta$ = (Coefficient of x^{n-3}/ Coefficient of x^n) × $(-1)^n = \frac{d}{a}(-1)^3 = -\frac{d}{a}$

$\alpha\ \beta\ \gamma\ \delta$ = (Constant term/Coefficient of x^n) × $(-1)^n = \frac{e}{a}(-1)^4 = \frac{e}{a}$

Example 4 If the polynomial $ax^4 + bx^3 + cx^2 + dx + e$ has the property that the product of all the roots, taken at a time, is 1/3rd of the sum of the product of the roots, taking two at a time. What is the relationship between e and c?

Solution The product of all the roots, taken at a time = $\frac{e}{a}$

The sum of the product of the roots, taking two at a time = $\frac{c}{a}$

Now, $\frac{e}{a} = 1/3\ \frac{c}{a}$

So, $c = 3 \times e$

Example 5 If a, b and c are the roots of the equation $x^3 - 3x^2 + 2x + 1 = 0$, then what is the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$?

Solution $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{ab + bc + ca}{abc}$

$ab + bc + ca = 2$ and $abc = -1$

$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{ab + bc + ca}{abc} = -2$

FORMATION OF QUADRATIC EQUATION FROM ITS ROOTS

Case 1 If α and β are the roots of the equation $ax^2 + bx + c = 0$, then we can write

$ax^2 + bx + c = x^2 - (\alpha + \beta)x + \alpha\beta = x^2 -$ (sum of the roots) x + product of the roots = 0

Or, $ax^2 + bx + c = a(x - \alpha)(x - \beta) = 0$

In general, if $x_1, x_2, x_3, \ldots, x_n$ are the roots of the equation $f(x) = ax^n + bx^{n-1} + cx^{n-2} + \ldots + K = 0$

Then $f(x) = ax^n + bx^{n-1} + cx^{n-2} + \ldots + K = a(x - x_1)(x - x_2)(x - x_3)\ldots(x - x_n)$

Case 2 p and q are the roots of the equation $ax^2 + bx + c = 0$, and the quadratic equation with $(1/p)$ and $(1/q)$ as the roots is to be found:

Put $x = 1/p$

$a(1/p)^2 + b(1/p) + c = 0$

or, $cp^2 + bp + a = 0$

So, the quadratic equation is: $cx^2 + bx + a = 0$

Case 3 p and q are the roots of the equation $ax^2 + bx + c = 0$, and the quadratic equation with $(-p)$ and $(-q)$ as the roots is to be found:

Put $x = -p$

$a(-p)^2 + b(-p) + c = 0$

Or, $cp^2 - bp + a = 0$

So, the quadratic equation is: $cx^2 - bx + a = 0$

Example 6 What will be the quadratic equation having the roots opposite in sign as that of the quadratic equation $x^2 + 5x + 6 = 0$?

Solution Putting $(-x)$ at the place of x will give us the solution.

Desired equation is $(-x)^2 + 5(-x) + 6 = 0$

So, equation is: $x^2 - 5x + 6 = 0$

Alternatively, the roots of the quadratic equation $x^2 + 5x + 6 = 0$ are -2 and -3.

So, the roots of the desired equation should be 2 and 3.

So, the quadratic equation is $(x-2)(x-3) = 0$, or, $x^2 - 5x + 6 = 0$

Case 4 p and q are the roots of the equation $ax^2 + bx + c = 0$, and the quadratic equation with $(p)^2$ and $(q)^2$ as the roots is to be found:

Let x be a root of the original equation and y be a root of the transformed equation.

$y = x^2$, so we will put $x = \sqrt{y}$

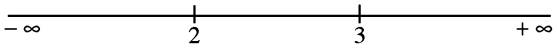
$a(\sqrt{y})^2 + b(\sqrt{y}) + c = 0$,

Solving this will give us the desired equation.

Solution of Inequations

Case 1 $f(x) = x^2 - 5x + 6 \geq 0$

$f(x) = x^2 - 5x + 6 = (x-2)(x-3) \geq 0$



We have got three intervals of values here: $-\infty$ to 2, 2 to 3 and 3 to $+\infty$

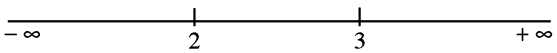
To check the values of x satisfying the above written inequation, any one value of x is taken from any of the intervals. If this value satisfies the inequation, then inequality will be satisfied in the alternate intervals.

Taking $x = 10$, $f(x) = (x-2)(x-3) = (10-2)(10-3) = +ve$

So, inequality is satisfied in this interval. Hence, it will not be satisfied in the interval $2 \leq x \leq 3$ and again will be satisfied in the interval $-\infty < x \leq 2$

So, the values of x satisfying the above written inequality lies outside 2 and 3 i.e., $x \leq 2$ or $x \geq 3$

Case 2 $f(x) = x^2 - 5x + 6 \leq 0$



It can similarly be observed that the values of x are lying in between 2 and 3

Hence, $2 \leq x \leq 3$

In general, if $(x-a)(x-b) \geq 0$, then $a \leq x \leq b$

And if $(x-a)(x-b) \leq 0$ and $a < b$, then $x \leq a$ or $x \geq b$

Example 7 What values of x will satisfy the quadratic inequation $f(x) = -x^2 + 3x + 4 > 0$?

Solution $f(x) = -x^2 + 3x + 4 > 0$, or, $x^2 - 3x - 4 < 0$

Or, $(x+1)(x-4) < 0$

So the values of x satisfying $f(x) = -1 < x < 4$

Worked-out Problems

Example 8 If the roots of the quadratic equation $x^2 - 4x - \log_3 N = 0$ are real, then find the minimum value of N?

Solution Since the roots of the given equation are real, so $D \geq 0$

Or, $16 + 4\log_3 N \geq 0$, or, $\log_3 N \geq -4$

Or, $N \geq 3^{-4}$

So, the least value of $N = 1/81$

Example 9 Find the real values of N for which the quadratic equation $2x^2 - (N^3 + 8N - 1)x + N^2 - 4N = 0$ will have the roots of an opposite sign.

Solution The roots of the given equation will be of opposite signs only if they are real and the product of the roots is negative.

Or, $D \geq 0$, and the product of the roots < 0

Or, $(N^3 + 8N - 1)^2 - 8(N^2 - 4N) \geq 0$ and $\frac{N^2 - 4N}{2} < 0$

Or, $N^2 - 4N < 0$. Hence $0 < N < 4$

Example 10 If p and q are the roots of the quadratic equation $2x^2 + 6x + N = 0$ $(N < 0)$, then what is the maximum value of $\frac{p}{q} + \frac{q}{p}$?

Solution $p + q = -3$, and $pq = N/2$

Now, $N < 0$, hence $D = 36 - 4N > 0$. So, p and q are real.

Again, $\frac{p}{q} + \frac{q}{p} = \frac{(p+q)^2}{pq} - 2 = \frac{18}{N} - 2$

So, the maximum value of $\frac{p}{q} + \frac{q}{p} = -2$

Example 11 For which value of k does the following pair of equators yield a unique solution of x such that the solution is positive? (CAT 2005)

$x^2 - y^2 = 0$

$(x-k)^2 + y^2 = 1$

(a) 2 (b) 0 (c) $\sqrt{2}$ (d) $-\sqrt{2}$

Solution (3) $y^2 = x^2$

$2x^2 - 2kx + k^2 - 1 = 0$

$D = 0$

$\rightarrow 4k^2 = 8k^2 - 8$

$\rightarrow 4k^2 = 8$

$\rightarrow k = \sqrt{2}$

Example 12 Let

$x = \sqrt{4+\sqrt{4-\sqrt{4+\sqrt{4-...\text{to infinity.}}}}}$ Then x equals

(a) 3 (b) $\left(\frac{\sqrt{13}-1}{2}\right)$

(c) $\left(\frac{\sqrt{13}+1}{2}\right)$ (d) $\sqrt{3}$ **(CAT 2005)**

Solution (c) $x = \sqrt{4+\sqrt{4-x}} \Rightarrow x^2 - 4 + \sqrt{4-x}$

$$\left(x^2-4\right) = \sqrt{4=x}$$

Now, put the values from the options.

Only the 3rd option satisfies the condition.

Alternatively, the value of x will be more than 2, which is given only in option 1 and option 3. Since it is only slightly more than 3, option 2 will be the answer.

Example 13 In a quadratic equation $ax^2 + bx + c = 0$, a, b and c are positive. Graph of this equation cuts X-axis at two distinct points. If both the roots of this equation are real, what can be said about the signs of both the roots?

Solution **Method 1**

Let us assume p and q are the roots of the equation.

Product of the roots = $p\,q = c/a$ = + ve [Both c and a are positive]

Hence, p and q are both positive, or both are negative.

Sum of the roots = $p + q = -b/a$ = – ve

Hence, p and q are both negative.

Method 2

Since the roots are real and distinct, D > 0

$$p = \left[-b - \sqrt{b^2-4ac}\right]/2a$$

Obviously p is negative.

$$q = \left[-b + \sqrt{b^2-4ac}\right]/2a$$

In this case, we need to compare b and $\left[\sqrt{b^2-4ac}\right]$.

$b > \left[\sqrt{b^2-4ac}\right]$, so q will also be negative.

Method 3

Using Descartes' Theorem, sign of $f(x)$ = + + +, so there will be no positive roots and $x = 0$ is not a solution of this equation.

So, it can be concluded that both the roots will be negative.

PRACTICE EXERCISES

WARM UP

Q.1. What is the value of k when the equation $x^2 - (k+6)x + 2(2k-1) = 0$ has the sum of the roots equal to half of their product?
(a) −2 (b) 7 (c) 9 (d) 12

Q.2. What is the value of P when the sum of the squares of the roots of the equation $x^2-(p-2)x-p-1=0$? Assume the least value.
(a) −2 (b) 7 (c) 9 (d) 12

Q.3. If the equation $x^2 + 2(p+1)x + 9p - 5 = 0$ has only positive roots, then which of the following is true?
(a) $p \geq 6$ (b) $p \leq 0$
(c) $p \leq -6$ (d) None of these

Q.4. The quadratic equations $x^2 - ax - 21 = 0$ and $x^2 - 3ax + 35 = 0$, where $a > 0$ have a common root. What is the value of a?
(a) 0 (b) 1 (c) 4 (d) 6

Q.5. A, B and C are real values such that $A + B + C = 2$, $A^2 + B^2 + C^2 = 6$ and $A^3 + B^3 + C^3 = 8$, then find the value of $A^4 + B^4 + C^4$?
(a) 6 (b) 12 (c) 18 (d) 24

Q.6. What is the sum of the roots of the quadratic equation $3x^2 + 2x - 1 = 0$?
(a) 1/3 (b) 2/3 (c) 3 (d) −2/3

Q.7. What is the product of the roots of the quadratic equation $3x^2 + 2x - 1 = 0$?
(a) −1/3 (b) 2/3 (c) 3 (d) −2/3

Q.8. What is the sum of the roots of the cubic equation $x^3 + 2x^2 + x - 1 = 0$?
(a) −2 (b) 2/3 (c) −1 (d) 2

Q.9. Which of the following quadratic equations will not have real roots?

(a) $3x^2 + 2x - 1 = 0$ (b) $x^2 + 2x + 1 = 0$
(c) $3x^2 + 2x + 5 = 0$ (d) $2x^2 + 3x + 1 = 0$

Q.10. Roots of the quadratic equation $ax^2 + bx + c = 0$ are 1 and 2, for some certain value of a, b and c. What will be the sum of the roots of the equation having the roots opposite in sign of the equation $ax^2 + bx + c = 0$?
(a) 3 (b) 6
(c) −3 (d) None of these

Q.11. The roots of the equation $a^2x^2 + abx = b^2$, where $a \neq 0$, are
(a) Real but unequal
(b) Real but equal
(c) Not real
(d) None of these

Q.12. Which of the following satisfies the equation: $-x^2 + 6x - 8 > 0$
(a) $3 < x < 5$ (b) $2 < x < 4$
(c) $x < 2$ and $x > 4$ (d) $x < 3$ and $x > 5$

Q.13. For what value of K, the equations $2x + 3y = 5$ and $4x + Ky = A$ will not have any solution?
(a) 2
(b) −2
(c) 6
(d) More than one value is possible

Q.14. Five burgers, six pizzas and seven cold drinks cost Rs 178 and six burgers, four pizzas and two cold drinks cost Rs 124. What is the cost of 3 (pizzas + burgers + cold drinks)?
(a) Rs 60 (b) Rs 62.5
(c) Rs 90 (d) Cannot be determined

Q.15. A and B are the roots of the equation $x^2 - 13x + N = 0$. If $A - B = 3$, what is the value of N?
(a) 20 (b) 27
(c) 36 (d) None of these

Q.16. If $x < 0$. then what is the maximum value of $\frac{9}{x} + \frac{x}{9}$?
(a) 2 (b) −2
(b) $+\infty$ (d) None of these

Q.17. If the coefficient of x^2 and the constant term of a quadratic equation are interchanged, then which of the following will not get changed?
(a) Sum of the roots
(b) Product of the roots
(c) Roots of the equation
(d) None of these

Q.18. If $(x + 2)^2 = 9$ and $(y + 3)^2 = 25$, then what is the maximum value of $\frac{x}{y}$?
(a) −5/8 (b) 5/8
(c) 3/7 (d) −3/7

Q.19. If one of the roots of the quadratic equation $2x^2 - 7x + q = 0$ is 3, what is the other root of this equation?
(a) 1/3 (b) 1/2 (c) 1 (d) 2

Q.20. What is the remainder when the polynomial $x^4 - 3x^2 + 7x - 10$ is divided by $x - 2$?
(a) 12 (b) 1 (c) −1 (d) 8

FOUNDATION

Q.1. How many real roots will be there of the equation $x^4 + x^2 = 0$
(a) 0 (b) 2
(c) 4 (d) None of these

Q.2. If $(x) + (1/x) = 2$, then what is the value of $(x)^3 + (1/x)^3$?
(a) 1 (b) −2 (c) −1 (d) 2

Q.3. Given that $as^4 + bs^3 + cs^2 + ds + e = 0$ is a bi-quadratic equation in s and $a \neq 0$. What is the value of $(1 - s_1)(1 - s_2)(1 - s_3)(1 - s_4)$?
(a) 1 (b) $(a+b+c+d+e)/a$
(c) 0 (d) $a + b + c + d + e$

Q.4. If $(x + 2)$ is a factor of $x^4 - 4x^2 + 2ax + 3 = 0$, then what is the value of a?
(a) −2 (b) 3 (c) $^3/_4$ (d) −3

Q.5. $x^2 - 5x + 6 = 0$ and $x^2 + mx + 3 = 0$ have a root in common. What is the value of m?
(a) 4 (b) −7/2
(c) Both 1 and 2 (d) None of these

Q.6. Find the number of integral roots of the equation $x3 - x^2 + 2x - 17 = 0$.
(a) 1 (b) 2
(c) 3 (d) None of these

Q.7. The number of real solutions of $x - \frac{1}{x^2 - 4} = -2\frac{1}{x^2 - 4}$ is
(a) 0 (b) 1 (c) 2 (d) Infinite

Q.8. The number of real solutions of the equation $e^x = x$ is

(a) 1 (b) 2
(c) 0 (d) None of these

Q.9. The sum of the real roots of the equation $x^2 + |x| - 6 = 0$ is
(a) 4 (b) 0
(c) −1 (d) None of these

Q.10. If one root of the equation $(k^2 + 1)\,x^2 + 13x + 4k = 0$ is reciprocal of the other then k has the value
(a) $-2 + \sqrt{3}$ (b) $2 - \sqrt{3}$
(c) 1 (d) None of these

Q.11. If α, β are roots of the equation $(x - a)(x - b) = c$, $c \neq 0$, then the roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are
(a) a, c (b) b, c
(c) a, b (d) $a + c, b + c$

Q.12. If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals then $\frac{b^2}{ac} + \frac{bc}{a^2}$ is equal to
(a) 2 (b) −2 (c) 1 (d) −1

Q.13. If a, b are roots of $x^2 + px + q = 0$ and g, d are the roots of $x^2 + px - r = 0$ then $(a - g)(a - d)$ is equal to
(a) $q + r$ (b) $q - r$
(c) $-(q + r)$ (d) $-(p + q + r)$

Q.14. If α, β, γ be the roots of the equation $x(1 + x^2) + x^2 (6 + x) + 2 = 0$ then the value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ is
(a) −3 (b) $\frac{1}{2}$
(c) $-\frac{1}{2}$ (d) None of these

Q.15. If the equations $2x^2 - 7x + 1 = 0$ and $ax^2 + bx + 2 = 0$ have a common root then
(a) $a = 2, b - 7$ (b) $a = \frac{7}{2}, b = 1$
(c) $a = 4, b = -14$ (d) None of these

Q.16. If $(x - 1)^3$ is a factor of $x^4 + ax^3 + bx^2 + cx - 1$ then the other factor is
(a) $x - 3$ (b) $x + 1$
(c) $x + 2$ (d) None of these

Q.17. If $x^2 - bx + c = 0$ has equal integral roots then
(a) b and c are integers
(b) b and c are even integers
(c) b is an even integer and c is a perfect square of a positive integer
(d) 1 and 3 both

Q.18. Find the number which exceeds its reciprocal by 35/6.
(a) 4 (b) 5 (c) 6 (c) 7

Q.19. If a and b are the roots of the equation $2x^2 - 3x - 2 = 0$. Find the value of $(a^3 + b^3)$ without finding the roots of the equation.
(a) $\frac{8}{3}$ (b) $\frac{64}{3}$ (c) $\frac{63}{8}$ (d) 16

Q.20. If the roots of the equation $3ax^2 + 2bx + c = 0$ are in the ratio of 2:3, then
(a) $8ac = 25b$ (b) $8ac = 9b^2$
(c) $8b^2 = 9ac$ (d) $8b^2 = 25ac$

Q.21. In writing a quadratic equation of the form $x^2 + bx + c = 0$, a student writes the coefficient of x incorrectly and finds the roots as 7 and 8. Another student makes a mistake in writing the constant term and gets the roots as 8 and −3. Find the correct quadratic equation.
(a) $x^2 - 5x + 56 = 0$ (b) $x^2 - 5x + 21 = 0$
(c) $x^2 - 21x + 56 = 0$ (d) $x^2 - 7x + 21 = 0$

Q.22. The quadratic equation, whose roots are reciprocal of the roots of $1 - 3x - 2x^2 = 0$, is
(a) $2 - 3x - 2x^2 = 0$ (b) $2x^2 + 3x - 1 = 0$
(c) $x^2 - 3x - 2 = 0$ (d) $x^2 - 2x - 3 = 0$

Q.23. Find the value of K in the quadratic equations $Kx^2 - 3x + 2 = 0$, such that one root is double the other.
(a) −1 (b) 1 (c) −2 (d) 2

Q.24. The denominator of a positive fraction is 1 more than the square of the numerator. If 4 is added to both the numerator and the denominator, the fraction becomes ½. Find the original fraction.
(a) 1/3 (b) 2/5 (c) 4/17 (c) 3/10

Q.25. Ramesh bought a certain number of chairs for Rs 2,400. If the price of each chair is reduced by Rs 20, then ten more chairs can be purchased for the same amount. Find the number of chairs he purchased initially.
(a) 20 (b) 30 (c) 40 (d) 50

Q.26. If $x^2 - kx + 5 = 0$ has 3 as a root, then the value of K is
(a) 17/5 (b) 13/2 (c) 14/3 (d) 16/5

Q.27. If a, b and c are real numbers such that $a + b + c = 0$, then the value of $a^3 + b^3 + c^3$ is
(a) 0 (b) $ab^2 + bc^2 + ca^2$
(c) $3abc$ (d) None of these

Q.28. One root of the equation $x^3 - ax^2 + 11x - 6 = 0$ is 2. Find other two roots.
(a) 0, 1 (b) 1, 3 (c) 2, 3 (d) 1, 2

Q.29. If $x^4 + \frac{1}{x^4} = 47$, then find the value of $x^3 + \frac{1}{x^3}$.
(a) 9 (b) 18
(c) 12 (d) 27

Q.30. If the roots of the equation $px^2 + rx + r = 0$ are in the ratio of $a : b$, then the value of $\sqrt{b/a} + \sqrt{a/b}$ is

(a) $\sqrt{r/p}$ (b) $-\sqrt{r/p}$
(c) $\sqrt{1/p}$ (d) $\sqrt{1/r}$

Q.31. How many numbers in the set $\{-4, -3, 0, 2\}$ satisfy the conditions $|y-4| \leq 6$ and $|y+4|<5$?
(a) 3 (b) 1
(c) 2 (d) None of these

Q.32. If $x^3 - ax^2 + bx + 10$ is perfectly divisible by $(x+5)$ and $x^4 + x^3 + bx^2 - ax + 42$ is perfectly divisible by $(x-3)$, find the value of $(a-3b)$.
(a) $\frac{450}{7}$ (b) 50
(c) 100 (d) None of these

Q.33. If a and b are the roots of $x^2 + x + 2 = 0$, then $\frac{\alpha^{10} + \beta^{10}}{\alpha^{-10} + \beta^{-10}}$ is
(a) 4096 (b) 2048
(c) 1024 (d) 512

Q.34. A man born in 1900s realized that in 1980 his age was the square root of the year of his birth. What is his birth year?
(a) 1929 (b) 1949 (c) 1936 (d) 1946

Q.35. In solving a problem on quadratic equation, one student makes a mistake in the constant term of the equation and gets -3 and -2 for the roots. Another student makes a mistake in the coefficient of the first degree term and finds -1 and -2 for the roots. The correct equation was.
(a) $x^2 - 5x + 2 = 0$ (b) $x^2 + 5x + 2 = 0$
(c) $x^2 + 5x - 2 = 0$ (d) $x^2 - 5x - 2 = 0$

Q.36. What is the solution set of the inequation $x^3 - x^2 + x - 1 > 0$?
(a) $x<-1$ (b) $-1<x<1$
(c) $x>1$ (d) $x>0$

Q.37. The number of distinct real roots of the equation $|V^2 + V - 6| - 3V + 7 = 0$ is
(a) 0 (b) 4 (c) 2 (d) 3

Q.38. In solving an equation of the form $ax - b = 0$ (a and b having only 1 as the common factor), A made a mistake of copying 'b' and got $\frac{7}{3}$ as the root whereas B made a mistake of copying 'a' and got $\frac{8}{5}$ as the root. The correct root is
(a) $\frac{7}{5}$ (b) $\frac{3}{8}$
(c) $\frac{8}{3}$ (d) Cannot be determined

Q.39. If one of the roots of the cubic expression $x^3 - ax^2 + 11x - 6$ is 3, what are the other roots?
(a) 6 and 2 (b) -1 and 2
(c) 1 and 2 (d) -2 and -1

Q.40. If m and n are natural numbers, then
(a) $m^{100} + n^{100} > (m+n)^{100}$
(b) $m^{100} + n^{100} \geq (m+n)^{100}$
(c) $m^{100} + n^{100} < (m+n)^{100}$
(d) $m^{100} + n^{100} \leq (m+n)^{100}$

MODERATE

Q.1. How many real roots will be there of the equation $= x^2 + 5|x| + 6 = 0$
(a) 0 (b) 2
(c) 4 (d) None of these

Q.2. If $s = 2 + 2^{1/3} + 2^{2/3}$, then what is the value of $s^3 - 6s^2 + 6s$?
(a) (b) 2 (c) (d)

Direction for questions 3 and 4: Read the passage below and solve the questions based on it.

x_1, x_2 and x_3 are the roots of two distinct quadratic equations (with coefficient of x^2 being 1 in both the equations) in which x_1 is a common root.

Q.3. If x_1, x_2 and x_3 are in AP and the coefficient of x in the equation with x_1 and x_2 as its root is -4, then one of the roots is
(a) 1 (b) 3
(c) 5 (d) Cannot be determined

Q.4. If x_1, x_2 and x_3 are in AP and the coefficient of x in the equation with x_1 and x_3 as its root is -6, then one of the roots is
(a) 1 (b) 3
(c) 5 (d) Cannot be determined

Direction for questions 5 and 6: Read the passage below and solve the questions based on it.

One of the +ve roots of a quadratic equation $ax^2 + mx + c = 0$, where $a = 1$, is twice that of the other. The sum of the coefficient of x and constant term is 2.

Q.5. What is the value of m in the equation?
(a) -4 (b) 4 (c) -6 (d) 6

Q.6. If x_1 and x_2 are the roots of this equation, then find the value of $|x_1 - x_2|$?
(a) 2 (b) 4
(c) 6 (d) None of these

Q.7. The two roots of the quadratic equation $ax^2 + bx + c = 0$ are both two-digit integers with the same units digit, but D/a^2 (where D is the discriminant given as $b^2 - 4ac$) is not a multiple of 100. Which of the following is a possible value of the unit digit of $b/2a$?
(a) 4 (b) 5 (c) 6 (d) 7

Q.8. The polynomial $(ax^2+bx+c)\ (ax^2-dx-c)$, $ac \neq 0$, has
(a) Four real zeros
(b) At least two real zeros
(c) At most two real zeros
(d) No real zeros

Q.9. The number of real solutions of $1 + |e^x - 1| = e^x\ (e^x - 2)$ is
(a) 0 (b) 1 (c) 2 (d) 4

Q.10. If $y \neq 0$ then the number of values of the pair (x, y) such that $x + y + \frac{x}{y} = \frac{1}{2}$ and $(x+y)\frac{x}{y} = -\frac{1}{2}$, is:
(a) 1 (b) 2
(c) 0 (d) None of these

Q.11. The equation has $\sqrt{x+1} - \sqrt{x-1} = \sqrt{4x-1}$
(a) No solution (b) One solution
(c) Two solutions (d) More than two solutions

Q.12. The number of real solutions of the equation $\frac{6-x}{x^2-4} = 2 + \frac{x}{x+2}$ is
(a) Two (b) One
(c) Zero (d) None of these

Q.13. If $x \in Z$ (the set of integers) such that $x^2 - 3x < 4$ then the number of possible values of x is
(a) 3 (b) 4
(c) 6 (d) None of these

Q.14. If x is an integral satisfying $x^2-6x+5 \leq 0$ and $x^2-2x>0$ then the number of possible value of x is
(a) 3 (b) 4 (c) 2 (d) Infinite

Q.15. If l, m are real and $l \neq m$ then the roots of the equation $(1 - m)\ x^2 - 5\ (1 + m)\ x - 2\ (10\ m) = 0$ are
(a) Real and equal (b) Nonreal complex
(c) Real and unequal (d) None of these

Q.16. The number of solutions of $\log_4(x-1) = \log_2(x-3)$ is
(a) 2 (b) 1 (c) 3 (d) 0

Q.17. If $a > 1$, roots of the equation $(1 - a)\ x^2 + 3ax - 1 = 0$ are
(a) One positive and one negative
(b) Both negative
(c) Both positive
(d) Both nonreal complex

Q.18. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are less than 3 then
(a) $a < 2$ (b) $2 \leq a \leq 3$
(c) $3 < a \leq 4$ (d) $a > 4$

Q.19. The number of integral values for which $x^2-(a-1)\ x+3 = 0$ has both the roots positive and $x^2+3x+6-a = 0$ has both the roots negative, is
(a) 0 (b) 1 (c) 2 (d) Infinite

Q.20. If the equations $ax^2 + bx + c = 0$, $cx^2 + bx + a = 0$, $a \neq c$ have a negative common root then the value of $a - b + c$ is
(a) 0 (b) 2
(c) 1 (d) None of these

Q.21. The number of values of k for which $(x^2 - (k - 2)\ x + k^2)\ (x^2 + kx + (2k - 1))$ is a perfect square, is
(a) 1 (b) 2
(c) 0 (d) None of these

Q.22. If $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$, $a \neq b$, have a common root a then, which of the following is true?
(a) $a + b = 1$ (b) $a + 1 = 0$
(c) $a = 0$ (d) $a + b + 1 = 0$

Q.23. If 2 and 4 are two roots of the expression $x^4 + 2x^3 + mx^2 + nx + 3 = 0$ (m and n are constants) then what is the value of $2m + 3n$?
(a) 106 (b) −106 (c) 53 (d) −53

Q.24. If $p+q+r = 1$, $p^2+q^2+r^2 = 9$ and $p^3+q^3+r^3 = 1$, then what is the value of $\frac{1}{p} + \frac{1}{q} + \frac{1}{r}$?
(a) 1 (b) −4 (c) −1 (d) −8

Q.25. If the roots of the equation $x^2 + (3k - 36)\ x + k^2 - 24\ k + 144 = 0$ are reciprocal to each other, then find the value of k.
(a) K = 11 or K = 13 (b) K = −11 or K = −13
(c) K = 12 (d) K = −12

Q.26. The number of common roots of $x^2 + 3x - 4 = 0$ and $x^3 - 4x = 0$ is
(a) 0 (b) 1 (c) 2 (d) 3

Q.27. If $Z + \frac{1}{Z} = 1, Z^{64} + \frac{1}{Z^{64}}$ is equal to
(a) 0 (b) 1 (c) −1 (d) −2

Q.28. Find the maximum and the minimum values of the function $\frac{x^2 - x + 1}{x^2 + x + 1}$ for real values of x.
(a) 3 and −3 (b) $\frac{1}{3}$ and $-\frac{1}{3}$
(c) 3 and $\frac{1}{3}$ (d) None of these

Q.29. Consider the equation of the form $x^2 + bx + c = 0$. The number of such equations that have real roots and have

coefficients b and c in the set {1, 2, 3, 4, 5, 6}, (b may be equal to c), is
(a) 20 (b) 18 (c) 17 (d) 19

Q.30. If a, b, c, d are four consecutive terms of an increasing AP then the roots of the equation $(x - a)(x - c) + 2(x - b)(x - d) = 0$ are
(a) Real and distinct
(b) Nonreal complex
(c) Real and equal
(d) Integers

Q.31. If $2a^{-1/3} + 2a^{1/3} = 5$, then find the value of a.
(a) 8 or $\frac{1}{8}$ (b) 2 or $\frac{1}{2}$
(c) 3 or $\frac{1}{3}$ (d) None of these

Q.32. If α, β and γ are the roots of the equation $9x^3 - 7x + 6 = 0$, then the equation whose roots are $3a + 2$, $3b + 2$ and $3\gamma + 2$ is
(a) $x^3 - 6x^2 + 5x + 24 = 0$
(b) $9x^3 - x + 16 = 0$
(c) $2x^3 - 27x - 8 = 0$
(d) None of these

Q.33. If the roots of equation $(q - r)x^2 + (r - p)x + p - q = 0$ are equal then p, q, r are in
(a) AP (b) GP
(c) HP (d) None of these

Q.34. Both the roots of the equation $(x-b)(x-c) + (x-a)(x-c) + (x-a)(x-b) = 0$ are always
(a) +ve (b) −ve
(c) Real (d) None of these

Q.35. If $a(b-c)x^2 + b(c-a)xy + c(a-b)y^2$ is a perfect square, then, a, b, c are in
(a) AP (b) GP
(c) HP (d) None of these

Q.36. $x^{3^n} + y^{3^n}$ is divisible by $x + y$ if
(a) n is any integer ≥ 0
(b) n is an odd positive integer
(c) n is an even positive integer
(d) n is a rational number

Q.37. The condition that both the roots of quadratic equation $ax^2 + bx + c = 0$ are positive is
(a) a and c have an opposite sign that of b
(b) b and c have an opposite sign that of a
(c) a and b have an opposite sign that of c
(d) None of these

Q.38. The condition that both the roots of the equation $ax^2 + bx + c = 0$ are zero is
(a) $b \neq 0$
(b) $c \neq 0$
(c) Either b or c (not both) = 0
(d) $b = c = 0$

Q.39. There is a quadratic equation of the form $ax^2 + bx + c = 0$ along with 3 conditions.
I. If only the value of a is correct in the equation, the roots are $\frac{1}{3}$ and 1.
II. If only the value of b is correct, the roots are 6 and 5.
III. If only the value of c is correct, the roots are −3 and −2.
What is the equation of a quadratic equation opposite in sign to the roots of the given quadratic equation?
(a) $3x^2 + 11x + 6 = 0$ (b) $3x^2 - 7x - 6 = 0$
(c) $x^2 - 11x + 30 = 0$ (d) $x^2 - 6x - 1 = 0$

Q.40. Quadratic equations $(2p - 1)z^2 + (2p + 1)z + c = 0$ and $(q + 1)y^2 + (4q + 1)y + 3c = 0$ have the same pair of roots. Given that $c \neq 0$, what is the value of $(p + q)$?
(a) 3 (b) 4
(c) 2 (d) Cannot be determined

ADVANCED

Q.1. Quadratic equation $x^2 + bx + c =$ has roots a and b, such that LCM of $(a, b) = 24$ and HCF of $(a, b) = 2$. What is the total number of such quadratic equations if a and b are natural numbers?
(a) 1 (b) 2 (c) 3 (d) 4

Q.2. $(p^2 + q^2)x^2 - \sqrt{2}(p + q)x + 1 = 0$. What must be the relation between q and p if the equation has equal and real roots?
(a) $p = q$ (b) $p > q$
(c) $p < q$ (d) None of these

Q.3. How many positive integers p, q and r exist such that $p/q + q/r + r/p = 2$?
(a) 1 (b) 3
(c) 15 (d) None of these

Q.4. If $x^2 + y^2 + z^2 = 1$ then the value of $xy + yz + zx$ lies in the interval of
(a) $\left[\frac{1}{2}, 2\right]$ (b) $[-1, 2]$
(c) $\left[-\frac{1}{2}, 1\right]$ (d) $\left[-1, \frac{1}{2}\right]$

Q.5. If $a \in R$, $b \in R$ then the factors of the expression $a(x^2 - y^2) - bxy$ are
(a) Real and different (b) Real and identical
(c) Complex (d) None of these

Q.6. If the sum of the roots of the quadratic equations $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then $\frac{b^2}{ac} + \frac{bc}{a^2} =$
(a) 0 (b) −1 (c) 1 (d) 2

Q.7. If equations $x^2 + 3x + 10 = 0$ and $ax^2 + bx + c = 0$ have a common root, then $a:b:c =$
(a) 2:3:10 (b) 1:3:5
(c) 1:3:10 (d) None of these

Q.8. Sum of all the real roots of the equation $|x - 2|^2 + |x - 2| - 2 = 0$ is...
(a) 0 (b) −4 (c) 4 (d) 2

Q.9. If $f(x) = \sum_{r=1}^{x} \frac{1}{r}, f(10) = \frac{a}{b}, f(20) = \frac{c}{d}$ and $f(30) = \frac{m}{n}$, where HCF of a and b = HCF of c and d = HCF of m and $n = 1$, then the roots of the equation $ax^2 + cx + m = 0$ are
(a) Imaginary (b) Rational
(c) Cannot be rational (d) Integers

Q.10. If pth, qth and rth term of a GP be 27, 8 and 12 respectively, then the equation $px^2 + 2qx - 2r = 0$ has
(a) Only one root in (0, 1)
(b) No root in (0, 1)
(c) Both roots in (0, 1)
(d) Imaginary roots

Q.11. If a, b be the roots of equation $x^2 + ax + b = 0$ and m, n, p, q are distinct integers in AP such that $a < m < n < p < q < b$ and $q = m^2 + n^2 + p^2$, then which of the following is true?
(a) $b + 1 + |a| > 0$ (b) $b + 1 + |a| = 0$
(c) $b + |a| = 0$ (d) $1 + b + |a| < 0$

Q.12. For $a, b, c \in R$ if $ax^2 + bx + c$ is real for real values of x and imaginary for imaginary values of x, then
(a) $a > 0$ (b) $a < 0$
(c) $a = 0$ (d) Nothing can be said

Q.13. If $x + \frac{1}{x} = 1$ and $p = x^{4000} + \frac{1}{x^{4000}}$ and q be the digit at units place in the number $2^{2^n} + 1, n$ being a natural number greater than 1, then $p + q =$
(a) 2 (b) 4
(c) 6 (d) None of these

Q.14. If the roots of $ax^2 + bx + b = 0$ be in the ratio $p:q$, then $\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{b}{a}} =$
(a) 1 (b) −1 (c) 2 (d) 0

Q.15. There are two quadratic expressions $a_1x^2 + b_1x + c_1$ and $a_2x^2 + b_2x + c_2$. Both of them have the same roots. If the ratio of a_1 to a_2 is 1:2, what is the ratio of the maximum values of the two quadratic expressions? (a_1, $a_2 < 0$)
(a) 1:3 (b) 3:1
(c) 1:5 (d) None of these

Q.16. If the two equations $ax^2 + bx + c = 0$ and $cx^2 + bx + a = 0$ have a common root, then how many of the following statements can be true?
I. $a = c$
II The common root is 1
III. $a - b + c = 0$
(a) 0 (b) 1 (c) 2 (d) 3

Q.17. Given that P and Q are the roots of $Ax^2 - 4x + 1 = 0$ and R and S are the roots of the equation $Bx^2 - 6x + 1 = 0$. What is the value of A so that P, Q, R and S are in HP?
(a) 1 (b) 2 (c) 3 (d) 4

Q.18. If $a < b < c < d$, then which of the following is true about the roots of the quadratic equation $(x-a)(x-c) + K(x-b)(x-d) = 0$, for the real value of K?
(a) Both the roots of this equation will have a positive sign
(b) Both the roots of this equation will have a negative sign
(c) Both the roots of this equation will be real
(d) Cannot be determined

Q.19. If $a + b + c = 0$ and a, b, c are rational, then which of the following is true about the roots of the quadratic equation $(b + c - a)x^2 + (c + a - b)x + (a + b - c) = 0$?
(a) Both the roots of the equation are rational
(b) Both the roots of the equation are irrational
(c) One of the roots of the equation is rational and the other is irrational
(d) None of these

Q.20. Find all the values of p for which one root of the equation $x^2 - (p + 1)x + p^2 + p - 8 = 0$, is greater than 2 and of the other root is smaller than 2.
(a) $-2 < p < 3$ (b) $-3 < p < 2$
(c) $-3 < p$ (d) $p < 2$

Q.21. For how many values of a, $a > 0$, both the roots of $ax^2 - (a+1)x + (a - 2) = 0$ are greater than 3?
(a) 0 (b) 2 (c) 5 (d) Infinite

Q.22. The number of real values of the parameter k for which $(\log_{16}x)^2 - \log_{16} x + \log_{16} k = 0$ with real coefficients will have exactly one solution is:
(a) 2 (b) 1
(c) 4 (d) None of these

Q.23. If the roots of the equation $(x-a)(x-b)+(x-b)(x-c)+(x-c)(x-a)=0$, (where a, b, c are real numbers) are equal, then:
(a) $b^2-4ac=0$ (b) $a=b=c$
(c) $a+b+c=0$ (d) None of these

Q.24. If the equation $ax^2+bx+c=0$ has a root less than -2 and a root greater than 2, and $a>0$, then which of the following is true?
(a) $4a+2|b|+c<0$
(b) $4a+2|b|+c>0$
(c) $4a+2|b|+c=0$
(d) None of these

Q.25. The equations $x^2+x+a=0$ and $x^2+ax+1=0$.
(a) Cannot have a common real root for any value of a
(b) Have a common real root for exactly one value of a
(c) Have a common real root for exactly two values of a
(d) Have a common real root for exactly three values of a

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(a)	(c)	(c)	(d)	(a)	(a)	(c)	(c)	(d)	(b)	(c)	(c)	(d)	(b)	(d)	(b)	(b)	(d)

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(d)	(b)	(c)	(b)	(d)	(a)	(c)	(b)	(b)	(c)	(a)	(c)	(c)	(c)	(b)	(d)	(c)	(c)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(a)	(c)	(b)	(d)	(b)	(c)	(a)	(d)	(b)	(b)	(b)	(b)	(c)	(c)	(b)	(c)	(a)	(d)	(c)	(c)

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(b)	(d)	(d)	(c)	(d)	(b)	(b)	(b)	(b)	(a)	(b)	(b)	(a)	(c)	(b)	(c)	(a)	(c)	(a)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(a)	(d)	(a)	(a)	(a)	(a)	(c)	(c)	(d)	(a)	(a)	(d)	(a)	(c)	(c)	(a)	(a)	(d)	(a)	(a)

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(a)	(b)	(c)	(a)	(d)	(c)	(c)	(d)	(a)	(d)	(c)	(c)	(d)	(d)	(d)	(c)	(c)	(a)	(a)
Q. No.	21	22	23	24	25															
Answer	(a)	(b)	(b)	(a)	(b)															

HINTS AND SOLUTIONS

Warm up

1. $K + 6 = 2K - 1$
   $K = 7$

2. Let roots = α, β
   So, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (P-2)^2 + 2(p+1)$
   $= p^2 - 4p + 4 + 2p + 2 = (P-1)^2 + 5$
   So, value of p for the least value = 1

3. $\alpha + \beta = -2(P+1)$; $\alpha\beta = 9p - 5$
   $\therefore\ \alpha < 0$ and $\beta < 0$
   $\therefore\ \alpha + \beta < 0 \Rightarrow -2(p+1) < 0 \Rightarrow p > -1$
   $\therefore\ \alpha\beta > 0 \Rightarrow 9P - 5 > 0 \Rightarrow p > 5/9$
   Hence option (*d*) is the answer.

4. Let common root = α
   $\therefore\ \alpha^2 - a\alpha - 21 = 0$
   $\alpha^2 - 3a\alpha + 35 = 0$
   Solving the two equations, we get $a = 4$

5. Taking the values of A, B and C as 1, 2 and –1.
   We get $A^4 + B^4 + C^4 = 18$

6. Sum of roots = $-2/3(-b/a)$.

7. Product of roots = $-1/3(c/a)$

8. In a cubic equation $ax^3 + bx^2 + cx + d = 0$, Sum of roots = $-b/a$
   $\therefore$ Here sum of roots = –2

9. Calculating $D = b^2 - 4ac$ for each of the given option and for a non–real root, D should be less than 0.

10. For roots with opposite sign replace x with $-x$ in the given equation.
    We get, $ax^2 - bx + c = 0$
    Now the sum of roots: b/a.
    For the new equation sum of roots = –3

11. $a^2x^2 + abx = b^2$
    $$\left(ax + \frac{b}{2}\right)^2 = \frac{5b^2}{4}$$
    $$ax + b/2 = \pm\frac{\sqrt{5}}{4}b$$
    If $b = 0$, then real and equal roots, else real and unequal.
    Hence option *d* is the answer.

13. $2x + 3y = 5$
    $4x + ky = A$
    For no solution
    $$\frac{2}{4} = \frac{3}{k} \neq \frac{5}{A}$$
    $\therefore\ K = 6$

14. $5b + 6p + 7c = 178$ ---(1)
    $6b + 4p + 2c = 124$ ---(2)
    Multiplying equation (1) by 2 and adding to equation (2):
    $16b + 16p + 16c = 480$
    $b + p + c = 30$
    $\therefore\ 3b + 3p + 3c = 90$

15. $x^2 - 13x + N = 0$ with roots A and B
    $\therefore\ A + B = 13$
    and $A - B = 3$
    $\therefore\ A = 8$ and $B = 5$
    $\therefore\ N = AB = 40$

16. let $x = -a$ where $a > 0$
    For any 2 positive numbers $\frac{9}{a}, \frac{a}{9}$

AM ≥ GM

$$\Rightarrow \frac{\frac{9}{a}+\frac{a}{9}}{2} \geq \sqrt{\frac{9}{a}\times\frac{a}{9}}$$

$$\Rightarrow \frac{9}{a}+\frac{a}{9} \geq 2$$

$$\Rightarrow -\left(\frac{a}{9}+\frac{9}{a}\right) \leq -2$$

17. Take a quadratic equation: $x^2 + 3x + 2 = 0$

$\therefore$ sum of roots = –3

products of roots = 2

roots = –1, –2

Now new quadratic equation: $2x^2 + 3x + 1 = 0$

$\therefore$ sum of roots = –3/2

product of roots: 1/2

roots: –1, –1/2

18. $(x + 2)^2 = 9$, $(y + 3)^2 = 25$

$\therefore x = 1, -5; y = 2, -8$

$\therefore$ maximum value of $\frac{x}{y} = \frac{5}{8}$

19. Sum of roots $= \frac{7}{2}$ and one of the roots = 3

So other root $= \frac{7}{2} - 3 = 1/2$

20. Using remainder theorem; put $x = 2$ in the main polynomial:

$\therefore p(2) = 16 - 12 + 14 - 10 = 8$

$\therefore$ Remainder = 8

Foundation

1. $x^4 + x^2 = 0$ can be written as $\Rightarrow x^2(x^2 + 1) = 0$

So, either $x^2 = 0$ or $(x^2 + 1) = 0$

Case 1 - If $x^2 = 0$, then $x = 0, 0$ (two solutions)

Case 2 - $(x^2 + 1) = 0 \Rightarrow x^2 = -1$

Minimum value of $x^2 = 0$, and $x^2 = -1$ is not possible.

Hence a total of 2 real solutions are possible.

2. Method 1 - Using the formula $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$

$(x^3 + 1/x^3) = (x + 1/x)^3 - 3(x).(1/x)(x + 1/x) = 2^3 - 3 \times 1 \times (2) = 2$

Method 2 - $(x + 1/x) = 2$

Or, $(x + 1/x) - 2 = 0$, or, $x^2 - 2x + 1 = 0$

$\Rightarrow (x - 1)^2 = 0 \Rightarrow x = 1$

So, $(x^3 + 1/x^3) = 2$

Method 3:

$x + \frac{1}{x} = 2$. Going by hit and trial, we can see that $x = 1$.

$\therefore \quad x^3 + \frac{1}{x^3} = 1 + 1 = 2$

3. $(1 - S_1)(1 - S_2)(1 - S_3)(1 - S_4)$

$= 1 - (S_1 + S_2 + S_3 + S_4) + (S_1S_2 + S_2S_3 + S_3S_4 + S_4S_1)$

$- (S_1S_2S_3 + S_2S_3S_4 + S_3S_4S_1 + S_4S_1S_2) + S_1S_2S_3S_4$

$$= 1 + \frac{b}{a} + \frac{c}{a} + \frac{d}{a} + \frac{e}{a}$$

$$= \frac{a + b + c + d + e}{a}$$

Alternate solution:

$as^4 + bs^3 + cs^2 + ds + e = 0$

$\Rightarrow a(s - s_1)(s - s_2)(s - s_3)(s - s_4) = 0$ (because S_1, S_2, S_3 and S_4 are roots)

Now, putting $s = 1$ in L.H.S

$\Rightarrow P(1) = a(1 - s_1)(1 - s_2)(1 - s_3)(1 - s_4)$

$\Rightarrow (1 - s_1)(1 - s_2)(1 - s_3)(1 - s_4) = (a + b + c + d + e)/a$

4. Since $(x + 2)$ is a factor of $P(X)$, so $p(-2) = 0$

$\Rightarrow 16 - 16 - 4a + 3 = 0 \Rightarrow a = \frac{3}{4}$.

5. $x^2 - 5x + 6 = 0 \Rightarrow$ Roots = 2, 3

When 2 is the common root then $p(2) = 0 \Rightarrow 2^2 + 2m + 3 = 0$

$\Rightarrow m = -7/2$

When 3 is the common root then $P(3) = 0 \Rightarrow 3^2 + 3m + 3 = 0$

$\Rightarrow m = -4$

6. $x^3 - x^2 + 2x - 17 = 0$

let roots be α, β and γ

$\therefore \alpha\beta\gamma = 17$

Now, 17 is having only two factors 1 and 17, so the only possible integral roots = 1 and 17. Checking for these two values we find no integral root possible.

7. $$x - \frac{1}{x^2 - 4} = 2 - \frac{1}{x^2 - 4}$$

$\Rightarrow x = 2$. Although we can see that for $x = 2$, $\frac{1}{x^2 - 4}$ will not hold.

Hence no value of x is possible.

8. Draw the graphs of both L.H.S and R.H.S to find there are no intersection points.

9. $x^2 + |x| - 6 = 0 \Rightarrow x^2 + x - 6 = 0$ where ≥ 0 $\therefore$ root : 2
Else $x^2 - x - 6 = 0$ if $x < 0$ $\therefore$ root: -2.
So, sum of roots = 0

10. Product of roots = 1
$\Rightarrow \frac{4k}{k^2+1} = 1 \Rightarrow k = 2 \pm \sqrt{3}$

11. $(x - a)(x - b) = c$
$\Rightarrow x^2 - (a + b)x + ab - c = 0$; roots: α, β
$(x - \alpha)(x - \beta) + c = 0$
$\Rightarrow x^2 - (a + \beta)x + \alpha\beta + c = 0$
$\Rightarrow x^2 - (a + b) + ab - c + c = 0$
(putting values of $(\alpha + \beta)$ and $\alpha\beta$ from previous equation)
Hence new equation $= x^2 - (a + b) + ab = 0$. So the roots are a and b.

12. $\alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$
$\Rightarrow \frac{-b}{a} = \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} \Rightarrow \frac{-b}{a} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha^2\beta^2}$
$\Rightarrow \frac{-b}{a} = \frac{(b^2/a^2) - \frac{2c}{a}}{c^2/a^2}$
$\Rightarrow \frac{b^2}{ac} + \frac{bc}{a^2} = 2$

14. $x(1 + x^2) + x^2(6 + x) + 2 = 0$
$\Rightarrow 2x^3 + 6x^2 + x + 2 = 0$
For roots: α, β, γ
$\alpha + \beta + \gamma = -\left(\frac{6}{2}\right) = -3$
$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{1}{2}$
$\alpha\beta\gamma = -\left(\frac{2}{2}\right) = -1$
$\therefore \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{1/2}{-1} = -1/2$

15. let 'x' be the common root
$\therefore$ $2x^2 - 7x + 1 = 0$ and $ax^2 + bx + 2 = 0$
$\Rightarrow 4x^2 - 7x + 1 = 0$ and $ax^2 + bx + 2 = 0$
$\therefore$ Subtracting the two equation
$(a - 4)x^2 + (b + 14)x = 0 \Rightarrow (a - 4)x^2 + (b + 14)x = 0$
{because $x \neq 0$}
Now go through the options.
Alternatively, for common roots between the equations $2x^2 - 7x + 1 = 0$ and $ax^2 + bx + 2 = 0$, value of constant term in 2nd equation is twice the constant term in 1st equation. Hence value of a and b in 2nd equation also should be twice than that of 1st equation.
Hence $a = 4$, $b = -14$.

16. As, $(x - 1)^3$ is a factor of the polynomial, so, 1 is a repeated root (3 times) of the given equation.
Let the fourth root be x, so, 1.1.1. $x = -1$
$\Rightarrow x = -1$. $\Rightarrow -1$ is another root
Hence $(x + 1)$ is a factor.

17. Let 'α' be the equal integral root,
$\therefore x^2 - bx + c = 0$
$(x - \alpha)^2 = 0$
$\Rightarrow x^2 - 2\alpha x + \alpha^2 = 0$
Now comparing the two equations
$\therefore 2\alpha = b$ and $c = \alpha^2$
$\therefore$ b is even integer and c is an integer as well as a perfect square

18. $x - \frac{1}{x} = \frac{35}{6}$
$\therefore x = 6$

19. $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
$= \left(\frac{3}{2}\right)^2 + 3 \times 1 \times \frac{3}{2} = \frac{63}{8}$

20. Assume that roots of the equation $3ax^2 + 2bx + c = 0$ are α, β.
$\alpha + \beta = -\frac{2b}{3a}$, $\alpha\beta = \frac{c}{3a}$ and $\frac{\alpha}{\beta} = \frac{2}{3}$ (given)
$\therefore \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{2}{3} + \frac{3}{2}$
$\frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{13}{6} \Rightarrow \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{13}{6}$
Now putting the values of $(\alpha + \beta)$ and $\alpha\beta$ and then solving,
We get, $8b^2 = 25ac$.

21. Coefficient of $x^2 = 2$
In first case the product of the roots = $8 \times 7 = 56$ which is the constant term. So $c = 56$.
In the second case the sum of the roots = $-b = 8 - 3 = 5$
$\Rightarrow b = -5$
$\therefore$ The correct quadratic equation is $x^2 - 5x + 56 = 0$

22. For reciprocal roots, replace x with $1/x$ in L.H.S of
$1 - 3x - 2x^2 = 0$
$\Rightarrow x^2 - 3x - 2 = 0$

23. Let the roots be α and 2α

So, Sum of roots = $\alpha + 2\alpha = 3\alpha = \frac{3}{k} \Rightarrow \alpha = \frac{1}{k}$

and product of roots $\Rightarrow 2\alpha^2 = \frac{2}{k} \Rightarrow \frac{2}{k^2} = \frac{2}{k}$

Hence $k = 1$.

24. Go through options.

25. Let the number of chairs bought initially = n

$\therefore \quad \frac{2400}{n} - \frac{2400}{n+10} = 20$

$\Rightarrow \quad 120\left[\frac{10}{n(n+10)}\right] = 1$

$\therefore \quad n(n + 10) = 1200 \Rightarrow n = 30$

26. Here sum of roots = k and product of roots = 5

One root is 3 the other root is $\frac{5}{3}$

$\therefore k = 3 + \frac{5}{3} = \frac{14}{3}$

27. We know

$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

As, $a + b + c = 0$

$\therefore \quad a^3 + b^3 + c^3 = 3abc$

28. Let the roots be 3, α and β

$\therefore 3 + \alpha + \beta = a$; $3\alpha + 3\beta + \alpha\beta = 11$; $3\alpha\beta = 6$

So, $\alpha\beta = 2$ and $3(\alpha + \beta) + 2 = 11$

$\Rightarrow 3(\alpha + \beta) + 2 = 11 \Rightarrow \alpha + \beta = 3$

$\therefore \alpha, \beta = (1,2)$

29. $x^4 + \frac{1}{x^4} = 47$

$\Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 = 47 \Rightarrow x^2 + \frac{1}{x^2} = 7$

$\Rightarrow \left(x + \frac{1}{x}\right)^2 - 2 = 7$

$\Rightarrow \left(x + \frac{1}{x}\right)^2 = 9$

$\Rightarrow x + \frac{1}{x} = 3 \Rightarrow \left(x + \frac{1}{x}\right)^3 = 27$

$\Rightarrow x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right) = 27$

$\Rightarrow \quad x^3 + \frac{1}{x^3} + 3 \times 3 = 27 \Rightarrow x^3 + \frac{1}{x^3} = 18$

30. Let roots be ak and bk.

$\therefore (a + b)k = -r/p$

$\therefore abk^2 = r/p$

now, $\sqrt{\frac{b}{a}} + \sqrt{\frac{a}{b}} = \frac{a+b}{\sqrt{ab}} = \frac{(a+b)k}{\sqrt{abk^2}} = -\frac{r/p}{\sqrt{r/p}}$

$= -\sqrt{r/p}$

31. $|y - 4| \le 6 \Rightarrow -2 \le y \le 10$

$|y + 4| < 5 \Rightarrow -9 < y < 1$

$\therefore$ Combining the two region we get.

$-2 \le y < 1$

$\therefore$ Only one element satisfies the condition i.e. = 0

Alternative solution:

Go through the options.

32. $p(x) = x^3 - ax^2 + bx + 10$; since it is divisible by $(x + 5)$

$\therefore \quad p(-5) = 0$

$\Rightarrow (-5)^3 - 25a - 5b + 10 = 0$

$\Rightarrow \quad 5a + b = -23 \qquad$ ---- (1)

$\Rightarrow \quad Q(x) = x^4 + x^3 + bx^2 - ax + 42 = 0$

$\therefore \quad Q(3) = 0$

$\Rightarrow \quad 81 + 27 + 9b - 3a + 42 = 0$

$\Rightarrow \quad a - 3b = 50$

33. $x^2 + x + 2 = 0$

$\therefore \alpha + \beta = -1$

$\alpha\beta = 2$

Now, $\frac{\alpha^{10} + \beta^{10}}{\alpha^{-10} + \beta^{-10}} = \frac{\alpha^{10} + \beta^{10}}{\frac{\alpha^{10} + \beta^{10}}{(\alpha\beta)^{10}}} = (\alpha\beta)^{10} = 2^{10} = 1024$

34. Go through the options to find

For option (c) $44^2 = 1936$ and $1980 - 44 = 1936$.

35. In the correct equation;

Sum of roots = $-3 - 2 = -5$ and the product of roots = $(-1) \times (-2) = 2$

So, correct equation = $x^2 + 5x + 2 = 0$

36. $x^3 - x^2 + x - 1 > 0$

$x^2(x - 1) + (x - 1) > 0$

$(x^2 + 1)(x - 1) > 0$

As, $x^2 + 1 > 0$ for all x, so, $x > 1$.

38. $ax - b = 0$

$\therefore \quad x = \frac{b}{a}$

Since 'b' and a have only 'a' have only '1' as a common factor

$\therefore$ we can directly equate with a rational number in its empirical form.

$\therefore$ For A; $X = \frac{7}{3}$ with incorrect 'b'

$\therefore$ Correct value of 'a' = 3 or any multiple of 3.

Similarly from 2nd statement, $x = \frac{8}{5} \Rightarrow$ we get correct value of 'b' = 8 or any multiple of 8.

Since we don't get any unique value of a and b, we cannot find unique value of $\frac{b}{a}$.

39. Let the roots be 3, α, β

$\therefore 3\alpha\beta = 6 \Rightarrow \alpha\beta = 2$

and $3x + 3\beta + \alpha\beta = 11$

$\Rightarrow 3(\alpha + \beta) + 2 = 11$

$\therefore \alpha + \beta = 3$

$\therefore \alpha = 1, \beta = 2$

40. $(m + n)^{100} = m^{100} + {}^{100}c_1 m^{99}.n + {}^{100}c_2 m^{98}.n + \ldots\ldots + n^{100}$

$= m^{100} + n^{100} + k$

[where $k = {}^{100}C_1 m^{99}n + {}^{100}C_2 m^{98}.n + \ldots\ldots + {}^{100}C_{99} mn^{99}$]

$\therefore \quad k > 0$ for m and n belonging to natural no.

$\therefore \quad (m + n)^{100} > m^{100} + n^{100}$

Moderate

1. $x^2 + 5|x| + 6 = 0$

$\because$ All the terms in LHS are positive

Hence no real roots possible.

2. $S - 2 = 2\frac{1}{3} + 2^{2/3} \Rightarrow (S-2)^2 = 2^{2/3} + 2.2^{1/3} + 2.2$

Now, required $= 2^{2/3} - 2.2^{1/3} + 2.2 - (2.2^{1/3} + 2.2^{2/3}) - 2$

$= 2$

5. Let the roots be α, 2α. Where $\alpha > 0$

$\therefore \alpha + 2\alpha = -m \Rightarrow m = -3\alpha$

and, $2\alpha^2 = C$

Now, since $m + c = 2 \Rightarrow 2\alpha^2 - 3\alpha = 2$

$\alpha = -1/2, 2$

$\because \alpha > 0$

$\therefore \alpha = 2 \therefore m = -3\alpha = -6$

6. In the above question the roots are α and 2α i.e. (2, 4)

$\therefore |x_1 - x_2| = |\alpha - 2\alpha| = 2$

7. Method 1–

Let p, q be the roots with same unit digit.

$\therefore \quad \frac{D}{a^2} = \frac{b^2 - 4ac}{a^2} = \left(\frac{b}{a}\right)^2 - \frac{4c}{a} = (p+q)^2 - 4pq$

$= (p - q)^2$

Now, given that 'p' and 'q' have same unit digit still $(p - q)^2$ is not divisible by 100.

Which means p and q have to be of opposite signs.

Now $\frac{b}{2a}$ is nothing but $= \frac{-1}{2}(p + a)$

Now, since p and q are of opposite signs with same unit digit.

Thus unit digit of $p + q = 0$

Thus, unit digits of $\frac{1}{2}(p + q)$ will be either '0' or '5'.

Out of the given options, 2nd option is the answer.

Method 2 -

Let the roots be –21 and 11

Equation is $X^2 + 10X - 231 = 0$

$(X + 21)(X - 11) = 0$

$\frac{b^2 - 4ac}{a^2} = \frac{100 + 924}{1} = 1024$ (not multiple of 100)

$b/2a = 10/2 = 5$

8. $(ax^2 + bx + c)(ax^2 - dx - c) = 0$

$\therefore$ either $ax^2 + bx + c = 0$ or, $ax^2 - dx + c = 0$ or both.

$\therefore$ roots of $ax^2 + bx + c = 0$ will be real if

$b^2 - 4ac > 0$

Similarly for $ax^2 - dx - c$ roots will be real if

$d^2 + 4ac > 0$

Now, at least one of the two conditions will hold true since either $4ac$ will be greater than zero or less than zero or equal to zero.

$\therefore$ At least 2 real zeroes will be there.

9. $1 + |e^x - 1| = e^x(e^x - 2)$

When $x \geq 0$

$1 + e^x - 1 = e^x(e^x - 2) \Rightarrow e^x - e^x(e^x - 2) = 0$

$\Rightarrow e^x(1 - e^x + 2) = 0$

$\Rightarrow e^x = 0 \; e^x = 3$

for $x \geq 0$ $e^x = 0$ is not possible

$\therefore e^x = 3$ is one root

When $x < 0$

$1 - e^x + 1 = e^x(e^x - 2)$

$\Rightarrow 2 - e^x + (2 - e^x)e^x = 0$

$(2 - e^x)(1 + e^x) = 0 \Rightarrow e^x = 2 \qquad e^x = -1$

for $x < 0$ $\quad e^x = 2$ is not possible and e^x can never be equal to -1

$\therefore$ no. of roots = 1

10. $(x+y)+\left(\frac{x}{y}\right)=\frac{1}{2}$ and $(x+y)\frac{x}{y}=\frac{-1}{2}$

Solving these two equations the values of

$(x+y)$ and $\left(\frac{x}{y}\right)$ will be $(1, -1/2)$

When $x + y = 1$ and $\frac{x}{y} = -1/2$

$(x, y) = (2, -1)$

When $x + y = -1/2$ and $\frac{x}{y} = 1$

$(x_1 y) = \left(\frac{-1}{4}, -\frac{1}{4}\right)$.

$\therefore$ No. of possible pairs = 2.

11. $\sqrt{x+1}-\sqrt{x-1}=\sqrt{4x-1}$

$\Rightarrow (x + 1) + (x - 1) - 2\sqrt{x^2 - 1} = 4x - 1$

$\Rightarrow -2\sqrt{x^2 - 1} = 2x - 1$

$\Rightarrow 4(x^2 - 1) = 4x^2 + 1 - 4x$

$\Rightarrow x = \frac{5}{4}$ which when put in the main equation does not satisfy it.

Hence no solution is possible.

12. $\frac{6-x}{x^2-4}=\frac{2(x+2)+x}{x+2}$ with $x + 2 \neq 0$ and $x - 2 \neq 0$

$\Rightarrow \frac{6-x}{x-2} = 3x + 4$

$\Rightarrow 6 - x = (3x + 4)(x - 2)$

$\Rightarrow 3x^2 - x - 14 = 0$

$\Rightarrow x = -2$ or $x = 7/3$

But $(x + 2) \neq 0$, hence only one root is possible.

13. $x^2 - 3x - 4 < 0$

$(x - 4)(x + 1) < 0$

$\therefore$ Possible values = (0, 1, 2, 3)

Hence there are 4 values.

14. $x^2 - 6x + 5 \leq 0 \Rightarrow (x - 5)(x - 1) \leq 0 \Rightarrow x \in (1, 5)$

$x(x - 2) > 0 \Rightarrow (-\infty, 0) \cup (2, \infty)$

Combining these two regions will give following integers $-3, 4, 5$.

Hence there are three values.

16. $\text{Log}_4(x - 1) = \log_2(x - 3)$

$\Rightarrow \frac{1}{2}\log_2(x-1) = \log_2(x-3)$

$\Rightarrow (x - 1) = (x - 3)^2$

$\therefore x = 5, 2$

Now $x = 2$ is not possible as $\log(x - 3) = \log(-1)$ is not possible

17. $\alpha+\beta=\frac{3a}{a-1}, \alpha\beta=\frac{1}{a-1}$ where α and β are the roots.

$\because a > 1$

$\therefore \alpha + \beta > 0$ and $\alpha\beta > 0$

$\therefore \alpha > 0$ and $\beta > 0$. Hence both the roots are positive.

18. $x^2 - 2ax + a^2 + a - 3 = 0$

$\because$ Roots are less than 3.

$\therefore$ Their sum will be less than 6.

$\therefore 2a < 6 \Rightarrow a < 3$. Out of the given options only option 1 satisfies it.

20. $ax^2 + bx + c = 0$ and $cx^2 + bx + a = 0$ have roots which are reciprocal to each other.

i.e. if the roots of $ax^2 + bx + c = 0$ are α, β then the roots of

$cx^2 + bx + c = 0$ are $\frac{1}{\alpha}, \frac{1}{\beta}$.

Now it is given that $\alpha < 0$ and $\alpha = \frac{1}{\alpha}$

$\therefore \quad a = -1$

$\therefore$ The common root is -1.

$\therefore$ putting $x = -1$ in any of the equations we get, $a - b + c = 0$.

21. $(x^2 - (K - 2) + K^2)(x^2 + Kx + (2K - 1))$

for the expression to be a perfect square there can two possible ways:

i) when both the quadratic expression are perfect square for a particular value

For that to happen, $K - 2 = 2K \Rightarrow K = -2$

Now for $K = -2$ in the 2nd expression we get, $x^2 - 2x - 5$

which is not a perfect square.

ii) The other is when both the quadratic equations are same.

$\therefore -(K-2) = K \Rightarrow K = 1$

$+K^2 = 2K - 1 \Rightarrow K = 1$

$\therefore$ at $K = 1$ the expression is a perfect square.

22. Let the common root be x

$\therefore \alpha^2 + \alpha x + b = 0$

and $\alpha^2 + b\alpha + a = 0$

$\Rightarrow (a-b)\alpha - (a-b) = 0$

$\Rightarrow (a-b)(\alpha - 1) = 0 \Rightarrow \alpha = 1 \ \{\because a \neq b\}$

$\therefore$ putting $\alpha = 1$ is any of the equations.

$a + b + 1 = 0$

23. $P(x) = x^4 + 2x^3 + mx^2 + nx + 3$

Now, $P(2) = 0$

$\Rightarrow 16 + 16 + 4m + 2n + 3 = 0$ ..... (1)

$\Rightarrow 4m + 2n + 35 = 0$

and, $P(4) = 0$

$\Rightarrow 256 + 128 + 16m + 4n + 3 = 0$

$\Rightarrow 16m + 4n + 387 = 0$ ..... (2)

Multiplying 5 in equation (1) and then subtracting from eq. (2)

$4m + 6n - 212 = 0$

$\therefore 2m + 3n = 106$

24. Putting $p = 2$, $q = -2$, $r = 1$; we can get the value of $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$

25. $x^2 + (3K - 36)x + K^2 - 24K + 144 = 0$

Since, roots are reciprocal, product of roots = 1

$\Rightarrow K^2 - 24K + 144 = 1$

$\therefore K = 11, 13.$

27. $Z + \frac{1}{Z} = 1$

$\Rightarrow Z^2 - Z + 1 = 0$

$\Rightarrow (Z+1)(Z^2 - Z + 1) = 0$ {if $z \neq -1$}

$\Rightarrow (Z^3 + 1) = 0$

$\therefore Z^3 = -1$ and $Z \neq -1$

Now, $Z^{64} + \frac{1}{Z^{64}} = Z^{63}.Z + \frac{1}{Z^{63}.Z}$

$\Rightarrow (Z^3)^{21}.Z + \frac{1}{(Z^3)^{21}.Z} = -Z - \frac{1}{Z} = -1$

29. $X^2 + bx + c = 0$

for real roots $b^2 - 4c \geq 0$

$\Rightarrow b^2 \geq 4c$

Now for each values of 'b' from the given set checking the no. of values of 'c' possible in the same set.

$b = 1$ no. of values of $c = 0$

$b = 2$ no. of values of $c = 1(1)$

$b = 3$ no. of values of $c = 2(1, 2)$

$b = 4$ no. of values of $c = 4(1, 2, 3, 4)$

$b = 5$ no. of values of $c = 6(1, 2, 3, 4, 5, 6)$

$b = 6$ no. of values of $c = 6\ (1, 2, 3, 4, 5, 6)$

Therefore, total possible sets = 1 + 2 + 4 + 6 + 6 = 19

30. Let $a, b, c, d, = -1, 0, 1, 2$ (respectively)

$\therefore$ Given expression comes out to be $(x-1)(x+1) + 2(x-2)x = 0$

$\Rightarrow 3x^2 - 4x - 1 = 0$; $\therefore D = 4^2 + 12 = 28$

Hence it will have Real and distinct roots.

31. $2\left[a^{1/3} + \frac{1}{a^{1/3}}\right] = 5$

$\Rightarrow 2a^{2/3} - 5a^{1/3} + 2 = 0$

$\Rightarrow (a^{1/3} - 2)(2a^{1/3} - 1) = 0$

$\therefore a^{1/3} = 2, a^{1/3} = 1/2$

$\Rightarrow a = 8, a = 1/8$

32. Since each of the roots are changing symmetrically from

$\alpha \rightarrow 3\alpha + 2$

Thus, to get the required equation, just replace x with $\frac{x-2}{3}$ in the given equation.

Thus we get

$9\left(\frac{x-2}{3}\right)^3 - 7\left(\frac{x-2}{3}\right) + 6 = 0$

$\Rightarrow x^3 - 6x^2 + 5x + 24 = 0$

33. Since the roots are equal, so, $D = 0$

$\Rightarrow (r-p)^2 - 4(q-r)(p-q) = 0$

$\Rightarrow [(r-q) + (q-p)]^2 - 4(r-q)(q-p) = 0$

$\Rightarrow [(r-q) - (q-p)]^2 = 0$

$\Rightarrow r - q = q - p$

$\Rightarrow p + r = 2q$ i.e. p, q, r are in A.P.

34. $(x-b)(x-c) + (x-c) + (x-a)(x-b) = 0$

$\Rightarrow 3x^2 - 2(a+b+c)x + ab + bc + ca = 0$

$\therefore X = 4(a+b+c)^2 - 4(ab + bc + ca) \times 3$

$\Rightarrow D = 4[a^2 + b^2 + c^2 - (ab + bc + ca)]$

Now we know $(a-b)^2 > 0$

$\Rightarrow a^2 + b^2 > 2ab$

Similarly, $b^2 + c^2 > 2bc$

and, $c^2 + a^2 > 2ca$

$\therefore a^2 + b^2 + c^2 > ab + bc + ca$

$\Rightarrow a^2 + b^2 + c^2 - (ab + bc + ca) > 0$

$\Rightarrow 4(a^2 + b^2 + c^2 - (ab + bc + ca)) > 0$

$\Rightarrow D > 0$

$\therefore$ The roots are Real.

35. Applying $D = 0$ in the given equation

$b^2(c-a)^2 = 4ac(a-b)(b-c)$

$$\Rightarrow \frac{b^2(c-a)^2}{ac} = 4(a-b)(b-c)$$

$$\Rightarrow b^2 ac\left(\frac{c-a}{ac}\right)^2 = 4\left(\frac{a-b}{ab}\right)\left(\frac{b-c}{bc}\right).ab^2c$$

$$\Rightarrow \left(\frac{1}{a}-\frac{1}{c}\right)^2 = 4\left(\frac{1}{b}-\frac{1}{a}\right)\left(\frac{1}{c}-\frac{1}{b}\right)$$

$$\Rightarrow \left[\left(\frac{1}{b}-\frac{1}{b}\right)+\left(\frac{1}{b}+\frac{1}{c}\right)\right]^2 - 4\left(\frac{1}{a}-\frac{1}{b}\right)\left(\frac{1}{b}-\frac{1}{c}\right) = 0$$

$$\Rightarrow \left[\left(\frac{1}{a}-\frac{1}{b}\right)-\left(\frac{1}{b}-\frac{1}{c}\right)\right]^2 = 0$$

$$\Rightarrow \frac{1}{a}-\frac{1}{b} = \frac{1}{b}-\frac{1}{c}$$

$\Rightarrow \frac{2}{b} = \frac{1}{c}+\frac{1}{a}$. So, a, b, c are in H.P.

36. We know $(x + y)$ is a factor of $x^p + y^p$ if p is odd.

Thus here $p = 3^n$ which is odd integer when 'n' is an integer ≥ 0

37. For both then roots : (α, β) to be positive

$\alpha + \beta > 0$ and $\alpha\beta > 0$

$$\Rightarrow \frac{-b}{a} > 0 \text{ and } \frac{c}{a} > 0$$

i.e. b and a are of opposite sign and c and a are of same sign.

38. $ax^2 + bx + c = 0$

for both roots to be zero, sum of roots $= \frac{-b}{a} = 0$

$\therefore b = 0$

and product of roots $= \frac{c}{a} = 0 \ \therefore c = 0$

39. $ax^2 + bx + c = 0$

Now in first case when root is 1/3 and 1

The equation is $(3x - 1)(x - 1) = 0$

$\therefore$ correct value of $a = x\ 3 \times 1 = 3$

Similarly in second case the equation is $(x - 6)(x - 5) = 0$

$\therefore$ correct value of $b = -(5 + 6) = -11$

and in third case the equation is $(x + 3)(x + 2) = 0$

$\therefore$ Correct equation is $3x^2 + 11x + 6 = 0$

40. For the equations to have same pair of roots

$$\frac{2p-1}{q+1} = \frac{2p+1}{4q+1} = \frac{c}{3c}$$

$\therefore 3(2p - 1) = q + 1 \Rightarrow 6p - q = 4$

and, $3(2p + 1) = 4q + 1 \Rightarrow 6p - 4q = -1$

Solving two equation $q = 2$ and $p = 1$

$\therefore (p + q) = 3$

Advanced

1. HCF of $\alpha, \beta = 2$, so we can assume $\alpha = 2x$ and $\beta = 2y$.

So, $2x \times 2y = 24 \times 2$, or, $xy = 12$

Different values of x and y possible are (12, 1), (4, 3). and hence different values of roots of the equation will be (24, 2), (8, 6).

Now, since we are talking about LCM and HCF of α, β, roots of the equation has to be non-negative and real.

So, 2 such quadratic equations are possible.

This is a question on Number System, and not on Quadratic Equation.

2. If roots are real and equal, then $D = 0$

$D = [\sqrt{2}(p + q)]^2 - 4(p^2 + q^2) \times 1 = 2(p^2 + q^2 + 2pq) - 4(p^2 + q^2) = -2(p^2 + q^2 - 2pq) = -2[(p - q)^2] = 0$

So, $p = q$

4. Due to symmetry, we can say that the maximum value of $xy + yz + zx$ will be at $x = y = x$

Now $x^2 + y^2 + z^2 = 1$

$\Rightarrow x = y = z = 1/\sqrt{3}$

$\therefore xy + yz + zx \leq 1$ which is present in only one option.

5. $ax^2 - bxy - ay^2 = 0$

$D = b^2y^2 + 4a^2y^2 = y^2\{b^2 + (2a)^2\} > 0$ but not a perfect square. So, factors are real and different.

6. Let the roots be α and β.

$$\therefore \alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

$$\Rightarrow \frac{-b}{a} = \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} \Rightarrow \frac{-b}{a} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha^2\beta^2}$$

$$\Rightarrow \frac{-b}{a} = \frac{(b/a)^2 - \frac{2c}{a}}{c^2/a^2}$$

$$\Rightarrow \frac{-b}{a} = \frac{b^2 - 2ca}{c^2} \Rightarrow b^2a + bc^2 = 2ca^2$$

$$\Rightarrow \frac{b^2}{ac} + \frac{bc}{a^2} = 2$$

7. $ax^2 + bx + c = 0$

$x^3 + 3x + 10 = 0$

For common roots $a = 1$, $b = 3$, $c = 10$ (ratio)

$a : b : c = 1 : 3 : 10$

8. Taking $(x - 2) > 0$ we get $x = 0, 3$. So, $x = 3$

Taking $(x - 2) < 0$ we get $x = 1, 4$. So, $x = 1$

Hence sum of roots = 4

Alternatively,

Assuming $|x - 2| = Z$, this equation is now $z^2 + z - 2 = 0$

Or, $(z + 2)(z - 1) = 0$, Hence $z = -2$ or 1.

Taking z = –2 is not possible since $z = |x - 2|$

Taking $z = 1$, or, $|x - 2| = 1$, so $x = 3$ and $x = 1$.

And these are the only real roots of this equation. So sum of real roots = 1 + 3 = 4

12. It is possible only if the given quadratic equation becomes linear equation. So, $a = 0$

13. $x + \frac{1}{x} = 1 \Rightarrow x^2 - x + 1 = 0$

Now as $x \neq -1 \Rightarrow (x + 1)(x^2 - x + 1) = 0$

$\Rightarrow x^3 + 1 = 0$

$\therefore x^3 = -1 \Rightarrow x^{4000} = (x^3)^{1333}. x = -x$

$\therefore P = x^{4000} + \frac{1}{x^{4000}}$

$= -x - \frac{1}{x} = -1 \Rightarrow P = -1$

Now let $n = 2$

$\therefore$ p = unit digit of 17 i.e. 7. So, $p + q = 7 - 1 = 6$

14. Let the roots of the equation $ax^2 + bx + b = 0$ be pk and qk.

$\therefore (p + q) K = \frac{-b}{a}$ and $pqk^2 = \frac{b}{a}$

$\therefore \sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{b}{a}} = \frac{p+q}{\sqrt{pq}} + \sqrt{\frac{b}{a}}$

$= \frac{(p+q)k}{\sqrt{pak^2}} + \sqrt{\frac{b}{a}} = \frac{-b/a}{\sqrt{b/a}} + \sqrt{b/a} = 0$

15. $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$

For same pair of roots $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{1}{2}$ (given)

Expression (2) $= a_2x^2 + b_2x + c_2 = 2(a_1x^2 + b_1x + c_1)$

$= 2$(expression (1))

Then the ratio of the maximum value is 1:2

16. We have to check that if the statements can be true.

Checking Statement I, if $a = c$, then obviously both the equation will have common root.

Checking Statement II, if the common root is 1, then both the equations are giving $a + b + c = 0$.

Checking Statement III, If the common root is –1, then the given condition is true.

So, all the three statements are true.

18. Solving equation we get $(1 + k)X^2 - \{a + c + (b + d)k\}x + (ac + bdk) = 0$

Here $D \geq 0$, so roots of equation will be real.

19. Any operation of rational nos. will always result in rational numbers.

20. Let the roots of the given equation be α, β

Now for roots $(\alpha - \beta)$, $(\beta - 2)$ the equation can be deduced by replacing x with $(x + 2)$

$\therefore$ The deduced equation would be

$\Rightarrow (x + 2)^2 - (p + 1)(x + 2) + p^2 + p - 8 = 0$

$\Rightarrow x^2 + (3 - p)x + p^2 - p - 6 = 0$

$\Rightarrow x^2 + (p - 3)x + (p + 2)(p - 3) = 0$

Now, $\alpha > 2$ and $\beta < 2$

$\therefore (\alpha - 2) > 0$ and $(\beta - 2) < 0$

$\therefore (\alpha - 2)(\beta - 2) < 0 \Rightarrow (p + 2)(p - 3) < 0$

$\therefore (\alpha - 2)(\beta - 2) < 0 \Rightarrow (p + 2)(p - 3) < 0$

$\therefore -2 < p < 3$.

21. Solving eqn we get $D = -5a^2 + 10a + 1$

Hence we can conclude that no value of a fulfills the equation.

22. If there is only one solution of any quadratic equation, then it means both the roots of the equation are equal or D = 0.

Alternatively, Put $x = 4$ and $k = 2$.

23. Solving eqn we get $3x^2 - 2(a + b + c)x + ab + bc + ca = 0$

For roots to be equal, $a = b = c$.

24. Let the eqn be $x^2 - x - 12 = 0$ $(x = -3, 4)$

Here $a = 1$, $b = -1$, $c = -12$

25. Putting $a = 1$ we get one common real roots for both equation.

11

CHAPTER

Sequence and Series



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- The concept of sequences and series
- Progression and their types
- Definitions and properties of progressions
- Kinds of questions asked in the CAT
- Methods of solving questions

Introduction to the topic

Sequence and series is a mathematical concept that draws majorly from the basic number system and the simple concepts of arithmetic. This is the reason that makes it an important topic for this exam. On an average, 1–3 questions have been asked from this topic in the CAT almost every year in the last 12 years. Besides the CAT, this topic is important for other exams like IIFT, SNAP, XAT, MAT. One good thing about the problems from this chapter is that they can be solved simply by the application of logic or some very simple concepts of calculation.

SEQUENCE AND SERIES

Let us consider the following series:

- 1, 4, 9, 16, ...
- 2, 6, 12, 20,...

It can be observed here that each of these two series shares some or the other common property:

Series (i) is → $1^2, 2^2, 3^2, 4^2$...

Series (ii) is → $1^2 + 1, 2^2 + 2, 3^2 + 3, 4^2 + 4$...

With this, any term or in general t_n, for either of the two series can be very easily found out.

For series (i), $t_{10} = 10^2$

For series (i), $t_{10} = 10^2 + 10$.

If the terms of a sequence are written under some specific conditions, then the sequence is called a *progression*.

With respect to preparation for the CAT, we will confine ourselves only to the following standard series of progression:

1. Arithmetic Progression
2. Geometric Progression
3. Harmonic Progression

1. Arithmetic Progression

A succession of numbers is said to be in arithmetic progression (AP) if the difference between any term and the previous term is constant throughout. In other words, the difference between any of the two consecutive terms should be the same. This difference which is common between any two consecutive terms is known as common difference of this AP and is denoted by *'d'*.

For example → Series (i) 1, 2, 3, ...

Series (ii) → 2, 5, 7, 10,...

Series(iii) → $a, a + d, a + 2d$,...

Common Difference(d) of series(i) = 1

Common Difference(d) of series(ii) = 3

Common Difference(d) of series(iii) = d

nth term of an arithmetic progression

First term $t_1 = a = a + (1 - 1)d$

Second term $t_2 = a + d = a + (2 - 1)d$

Third term $t_3 = (a + d) + d = a + 2d = a + (3 - 1)d$

Fourth term $t_4 = a + 3d = a + (4 - 1)d$

nth term $t_n = a + (n - 1)d$, where a is the first term, d is the common difference and n is the number of terms.

Important points

- t_n is also known as the general term of AP.
- If in any question, some particular term is given (like t_4 or t_{10}), then we should assume those terms in the form of t_n. However, if the total number of terms are given then we should assume the terms in the following way:

If three terms or any odd number of terms are involved, then we should assume these terms as $a - d, a, a + d$ and so on.

If four terms or any even number of terms are involved, then we should assume these terms as $a - 3d, a - d, a + d, a + 3d$ and so on.

Example 1 The sum of three numbers in an AP is 27 and the sum of their squares is 293. Find the numbers.

Solution Let the numbers be $a - d, a, a + d$.

Given is $(a - d + a + a + d) = 27$

So, $a = 9$

Also, $(a - d)^2 + a^2 + (a + d)^2 = 293$

$\Rightarrow d^2 = 25$

$\Rightarrow d = \pm 5$

When $d = +5$, then the terms are 4, 9, 14.

When $d = -5$, then the terms are 14, 9, 4.

Alternatively, this question can be worked out very easily with the help of options.

Example 2 If the sum of the first 11 terms of an arithmetic progression equals that of the first 19 terms, then what is the sum of the first 30 terms?

(a) 0 (b) −1

(c) 1 (d) Not unique

Solution Given

$t_1 + t_2 + \ldots + t_{11} = t_1 + t_2 + \ldots + t_{19}$ (for an AP)

$$\Rightarrow \frac{11}{2}[2a + (11-1)d] = \frac{19}{2}[2a + (19-1)d]$$

$\Rightarrow 22a + 110d = 38a + 342d$

$\Rightarrow 16a + 232d = 0$

$\Rightarrow 2a + 29d = 0$

$$\Rightarrow \frac{30}{2}[2a + (30-1)d] = 0$$

$\Rightarrow S_{30 \text{ terms}} = 0$

Properties of AP

If $a, b, c, d, \ldots$ are in AP, then

1. $a + k, b + k, c + k, d + k \ldots$ will be in AP, where k is any constant.
2. $a - k, b - k, c - k, d - k \ldots$ will be in AP, where k is any constant.
   In the above two cases, the common difference will be the same as earlier.
3. $ak, bk, ck, dk \ldots$ will be in AP, where k is any constant.
   In this case, new common difference will be k times the earlier common difference.
4. $\frac{a}{k}, \frac{b}{k}, \frac{c}{k}, \frac{d}{k}$ will be in AP, where $k \neq 0$.
   In this case, new common difference will be $\frac{1}{k}$ times the earlier common difference.

Example 3 If a, b, c are in AP, then $b + c, c + a, a + b$ will be in

(a) AP
(b) GP
(c) HP
(d) Cannot be determined uniquely

Solution a, b, c are in AP, then $a-(a+b+c)$, $b-(a+b+c)$, $c-(a+b+c)$ will be in AP.

$\Rightarrow -(b+c)$, $-(a+c)$ and $-(a+b)$ will be in AP.

$\Rightarrow (b+c)$, $(a+c)$ and $(a+b)$ will also be in AP.

Alternatively, let us assume a, b, c to be 1, 2, 3. Then $(b+c) = 5$, $(a+c) = 4$ and $(a+b) = 3$, which are obviously in AP.

Example 4 If x, y, z are in GP, then $1/(1+\log_{10}x)$, $1/(1+\log_{10}y)$ and $1/(1+\log_{10}z)$ will be in
(a) AP
(b) GP
(c) HP
(d) Cannot be determined uniquely

Solution Let us go through the options

Checking option (a), the three will be in AP if the 2nd expression is the average of the 1st and the 3rd expressions. This can be mathematically written as

$$2/(1+\log_{10}y) = 1/(1+\log_{10}x) + 1/(1+\log_{10}z)$$

$$= \frac{[1+(1+\log_{10}x)+1+(1+\log_{10}z)]}{[(1+\log_{10}x)(1+\log_{10}z)]}$$

$$= [2+\log_{10}xz]/(1+\log_{10}x)(1+\log_{10}z)$$

Obviously this will not give us the answer.

Checking option (b),

$$[1/(1+\log_{10}y)]^2 = [1/(1+\log_{10}x)]\,[1/(1+\log_{10}z)]$$

$$= [1/(1+\log_{10}(x+z)+\log_{10}xz)]$$

Again, no solution is found.

Checking option (c),

$1/(1+\log_{10}x)$, $1/(1+\log_{10}y)$ and $1/(1+\log_{10}z)$ are in HP then $1+\log_{10}x$, $1+\log_{10}y$ and $1+\log_{10}z$ will be in AP.

So, $\log_{10}x$, $\log_{10}y$ and $\log_{10}z$ will also be in AP.

Hence, $2\log_{10}y = \log_{10}x + \log_{10}z$

$\Rightarrow y^2 = xz$ which is given.

So, (c) is the answer.

Alternatively, we can also apply the following process:

Assume $x = 1$, $y = 10$ and $z = 100$ as x, y, z are in GP.

So, $1+\log_{10}x = 1$, $1+\log_{10}y = 2$ and $1+\log_{10}z = 3$.

$\Rightarrow$ Thus we find that since 1, 2 and 3 are in AP, we can assume that

$1+\log_{10}x$, $1+\log_{10}y$ and $1+\log_{10}z$ are in AP.

Hence, by definition of an AP we have that

$1/(1+\log_{10}x)$, $1/(1+\log_{10}y)$ and $1/(1+\log_{10}z)$ are in AP.

So, option (c) is the answer.

Sum of n terms of an arithmetic progression

$S_n = \frac{n}{2}[2a+(n-1)d]$, where n = number of terms, a = 1st term and d = common difference.

Example 5 Find the sum of AP 3, 5, 7, ... 50 terms and find its sum.

Solution Here $n = 50$, $d = 2$ and $a = 3$.

Using formula $S_n = \frac{n}{2}[2a+(n-1)d]$

$= 25 \times [2\times 3 + (50-1)2)] = 25 \times 104 = 2600$

However, we can find out the sum of any AP in a better way through average also.

The last term of this series = 101, so the average

$$= \frac{3+101}{2} = 52$$

So, sum = average × number of numbers

$$= 52 \times 50 = 2600$$

Hence, $S_n = \frac{F+L}{2} \times n$, where F is the first term, L is the last term and n is the number of terms of that AP.

Example 6 What is the sum of all the two-digit numbers which when divided by 7 gives a remainder of 3? (CAT 2003)

Solution This series is like → 10, 17, 21, ... 94.

Here $n = 13$, $d = 7$ and $a = 10$

Using the formula for the sum

$S_n = \frac{n}{2}[2a+(n-1)d]$, sum = 676

Alternatively, using the average method,

average = (1st number + last number)/2

$$\text{Average} = \frac{10+94}{2} = 52$$

So, the sum = average × number of numbers

$$= 52 \times 13 = 676$$

Example 7 Find the value of the expression

$1-4+2-5+3-6+\ldots$ to 100 terms.

(a) −250 (b) −500
(c) −450 (d) −300

Solution We can write the given expression $(1-4+2-5+3-6+\ldots$ to 100 terms) as:

(1 + 2 + 3 + ... to 50 terms) − (4 + 5 + 6 + ... to 50 terms)

Both of these are AP's with different values of 'a' and 'd'.

$a = 1$, $n = 50$ and $d = 1$ for 1st series and $a = 4$, $n = 50$ and $d = 1$ respectively.

Using the formula for the sum of AP

$= \frac{n}{2}[2a + (n-1)d)]$, we get

$\frac{50}{2}[2.1 + (50-1).1] - \frac{50}{2}[2.4 + (50-1).1]$

$= 25(2 + 49) - 25(8 + 49)$

$= 25(51 - 57) = -150$

Alternatively, logically this question can be done a bit faster by assuming (1 – 4), (2 – 5), etc as one unit.

$1 - 4 = 2 - 5 = \ldots = -3$.

Thus the above series is equivalent to a series of fifty (–3)'s added to each other.

So, (1 – 4) + (2 – 5) + (3 – 6) + … 50 terms = $-3 \times 50 = -150$

2. Geometric Progression

A succession of numbers is said to be in a geometric progression if the ratio of any term and the previous term is constant. This constant ratio which is common to any of the two terms is known as the common ratio and is denoted by '*r*'.

Example i. 1, 2, 4, 8,.,.,.,.,.,.,.,.,.,.,

ii. 20, 10, 5,.,.,.,.,.,.,

iii. $a, ar, ar^2, \ldots\ldots$

Common ratio of series (i) is 2.
Common ratio of series (ii) is 0.5.
Common ratio of series (iii) is *r*.

nth term of a geometric progression

First term $t_1 = a = ar^{1-1}$
Second term $t_2 = ar = ar^{2-1}$
Third term $t_3 = ar^2 = ar^{3-1}$
Fourth term $t_4 = ar^3 = ar^{4-1}$
*n*th term $t_n = ar^{n-1}$

where *a* is the first term, *r* is the common ratio and *n* is the number of terms.

Important points

- t_n is also known as the general term of GP.
- In any question, if some particular term is given like t_4 or t_{10}, then we should assume those terms in the form of t_n. However, if the total number of terms are given then we should assume the terms in the following way:

If three terms or any odd number of terms are involved, then we should assume these terms as $\frac{a}{r}$, *a*, *ar* and so on.

Example 8 The seventh term of a GP is 8 times the fourth term and the 5th term of the same GP is 48. Find the 6th term of this GP.

Solution Given $t_7 = 8 \times t_4$,

Or, $ar^6 = 8 \times ar^3$

$\Rightarrow r^3 = 8$, or, $r = 2$

Now, $ar^4 = 48$, So, $a = 3$

So, 6th term = $ar^5 = 3 \times 2^5 = 96$

Properties of GP

If *a, b, c, d,* …are in GP, then

1. *a*k, *b*k, *c*k, *d*k … will be in GP, where k is any non-zero constant.
2. $\frac{a}{k}, \frac{b}{k}, \frac{c}{k}, \frac{d}{k}$ will be in GP, where k is any non-zero constant.

   In the above two cases, the common ratio will be the same as the earlier.
3. If a GP of any even number of terms is given, then its common ratio will be the same as the ratio of the sum of all the even terms and the sum of all the odd terms.

Sum of n terms of a geometric progression

$$S_n = \frac{a(1-r^n)}{1-r} \text{ when } r \neq 1$$

$S_n = na$ when $r = 1$

where *n* = number of terms, *a* = 1st term and *d* = common difference.

Sum of infinite geometric progression

So far, we have done the summation of *n* terms of a GP. Now there is also a need of a separate expression for the sum of infinite GP. In case of AP, since either of the terms are always decreasing and going till $-\infty$ or are always increasing and going till $+\infty$. So, the summation of the infinite terms in AP will be either $-\infty$ or $+\infty$. However, the case is not the same in GP.

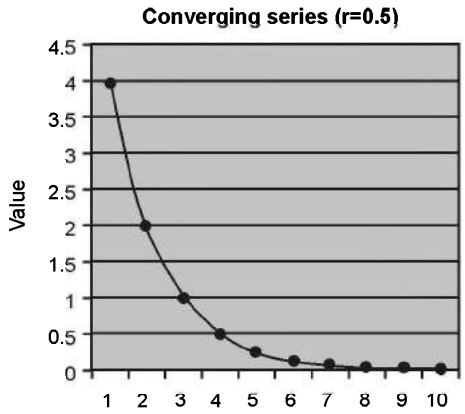
The need of a formula for infinite GP can be seen with the following example:

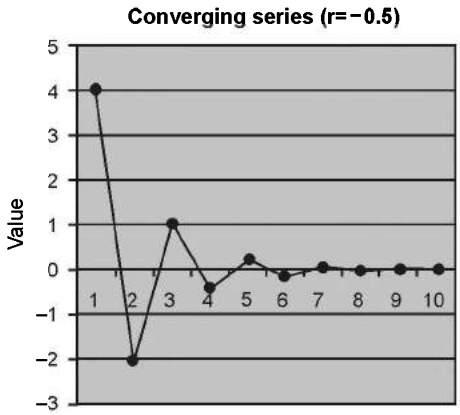
Find the sum of the series $\rightarrow 2 + 1 + \frac{1}{2} + \frac{1}{4} + \ldots$ till infinite terms.

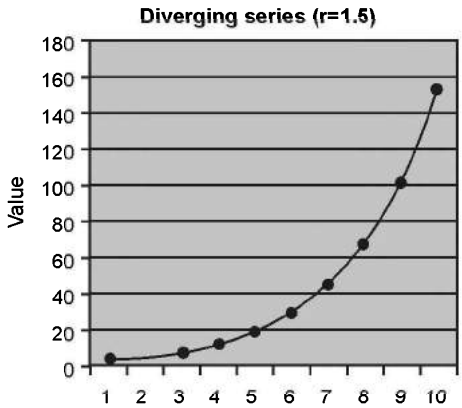
In the above-written expression, the number of terms is not given, so we cannot find out the sum using the formula for finding the sum of *n* terms of a GP.

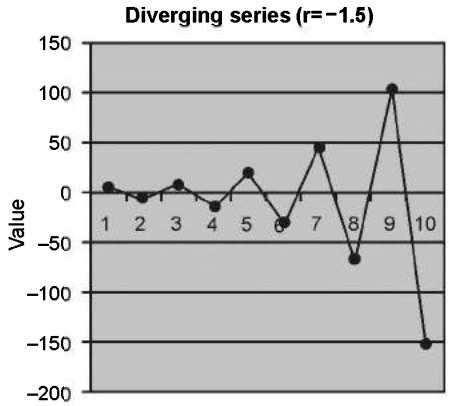
This can be further seen with the help of the following graphs:

If ***r*** is outside the range of –1 to 1, the terms of the series get bigger and bigger (even if they change the sign), and the series ***diverges***. If ***r*** is within the range of –1 to 1, the terms get smaller and smaller (closer to 0) and the series ***converges***.









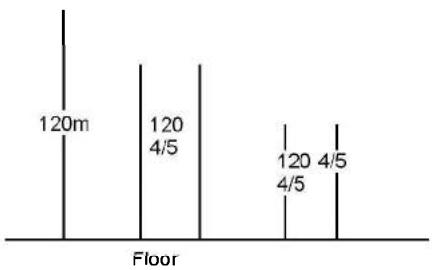
Sum of Infinite GP = $\frac{a}{1-r}$, where $-1 < r < 1$.

Example 9 What is the sum of the following series: $1 + 2 + 4 + \ldots$ till infinity.

Solution It is very obvious that the sum is going to be $+\infty$. This formula is applicable only for $-1 < r < 1$.

Example 10 After striking the floor, a ball rebounds to 4/5th of the height from which it has fallen. What is the total distance that it travels before coming to rest if it is gently dropped from a height of 120 m?

Solution The distance covered before the first rebound = 120 m



And then the ball bounces back to a height of $120 \times \frac{4}{5}$ and then falls from the same height. Next time the ball will go up by $120 \times \frac{4}{5} \times \frac{4}{5}$ m and then it will fall from the same height.

So, the total distance covered

$$= 120 + 2 \times 120 \times \frac{4}{5} + 2 \times 120 \times \frac{4}{5} \times \frac{4}{5} + \ldots \infty.$$

$$= 120 + 2 \times 120 \left(\frac{4}{5} + \frac{4}{5} \times \frac{4}{5} + \ldots \infty \right)$$

$$= 120 + 240 \left[\frac{\frac{4}{5}}{1 - \frac{4}{5}} \right] = 1080 \text{ m}$$

Alternatively, if the ball rebounds to $\frac{a}{b}$ th of the original height H, then the total distance covered = $H \times \frac{a + b}{a - b}$.

In the above example, the total distance covered $= 120 \times \frac{4+5}{5-4} = 1080$ m

Example 11 On 1 January 2004, two new societies S_1 and S_2 are formed, each of n numbers. On the first day of each subsequent month, S_1 adds b members while S_2 multiples its current numbers by a constant factor r. Both the

societies have the same number of members on 2 July 2004. If $b = 10.5n$, what is the value of r? **(CAT 2004)**

(a) 2.0 (b) 1.9 (c) 1.8 (d) 1.7

Solution There will be an increase of 6 times. The number of members in S_1 will be in an AP.

On 2 July 2004, S_1 will have $n + 6b$ members

$= n + 6 \times 10.5\, n, = 64n$

The number of members in S_2 will be in a GP.

On 2 July 2004, the number of members in $S_2 = nr^6$

They are equal, hence $64\, n = nr^6$

$\Rightarrow 64 = r^6 \Rightarrow r = 2$

3. Harmonic Progression

Unequal numbers $a, b, c, \ldots$ are said to be in a harmonic progression HP if reciprocals of these terms i.e., $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \ldots$ are in an AP. It is noteworthy, that no term of a HP can be equal to zero.

nth term of a harmonic progression

*n*th term of HP = 1/(*n*th term of the corresponding AP)

Example 12 If a, b and c are in HP, then $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in

(a) AP
(b) GP
(c) HP
(d) Cannot be determined uniquely

Solution a, b, c are in HP, so, $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ will be in AP.

Or, $\frac{a+b+c}{a}, \frac{a+b+c}{b}$ and $\frac{a+b+c}{c}$ will be in AP.

Or, $1+\frac{b+c}{a}, 1\frac{a+c}{b}$ and $1+\frac{a+b}{c}$ will be in AP.

Hence $\frac{b+c}{a}, \frac{a+c}{b}$ and $\frac{a+b}{c}$ are in AP.

So, $\frac{a}{b+c}, \frac{b}{c+a}$ and $\frac{c}{a+b}$ will be in HP.

Alternatively, these kinds of problems can also be done by assuming values.

Let us take 1, ½ , 1/3 (which are in HP)

$\frac{a}{b+c} = \frac{6}{5}, \frac{b}{a+c} = \frac{3}{8}, \frac{c}{a+b} = \frac{2}{9}$

Now, when we check these values for AP, GP and HP, we find that $\frac{8}{3}$ is the AM of $\frac{5}{6}$ and $\frac{9}{2}$.

So, obviously the given terms are in HP.

Properties of HP

If a, b, c and d are in HP, then

1. $a + d > b + c$
2. $ad > bc$

Sum of n terms of a harmonic progression

There is no standard formula for finding the sum of n terms of a HP.

ARITHMETICO GEOMETRIC SERIES

A series is said to be in arithmetico geometric series if each of its term is the product of the corresponding terms of an AP and a GP.

For example, $1, 2x, 3x^2, 4x^3, \ldots$

In the above series, the first part of this series is in an AP $(1, 2, 3, 4, \ldots)$ and the 2nd part is $(x^0, x^1, x^2, x^3, \ldots.)$ in a GP.

Sum of *n* terms of any arithmetico geometric series (AGS)

The sum of n terms of any AGS a, $(a + d)$r, $(a + 2d)$r^2, ... is given by

$$S_n = \frac{a}{1-r} + dr\frac{(1-r^{n-1})}{1-r} - \frac{[a+(n-1)d]}{1-r} \cdot r^n, \text{ if } r \neq 1.$$

$$S_n = \frac{n}{2}[2a + (n-1)d], \text{ if } r = 1$$

Sum of infinite terms of any arithmetico geometric series (AGS)

$$S_\infty = \frac{a}{1-r} + \frac{d \times r}{(1-r)^2}, |r| < 1$$

However, I would suggest students to desist from using these formulae. They should use the standard process to find out the sum of any AGS which is given below:

Let N be the sum of the arithmetico geometric series. Then each term of the series is multiplied by r (the common ratio of GP) and is written by shifting each term one step rightward, and then by subtracting rN from N to get (1−r) N. Thus N is finally obtained.

Example 13 What is the sum of the following series till infinity: $1 + 2x + 3x^2 + 4x^3 + \ldots, |x| < 1$

Solution Assume $S = 1 + 2x + 3x^2 + 4x^3 +$ ... (1)

Multiplying S by x, x. $S = x + 2x^2 + 3x^3 + 4x^4 +$ ... (2)

Subtracting (2) from (1)

$S - xS = 1 + (x + x^2 + x^3 + \ldots \infty)$

$S(1 - x) = 1 + (x + x^2 + x^3 + \ldots \infty) = 1 + \frac{x}{1-x}$

$S = \frac{1}{1-x} + \frac{x}{(1-x)^2} = \frac{1}{(1-x)^2}$

MEANS

Arithmetic Mean

It two numbers a and b are in AP then their arithmetic mean (AM) $= \frac{a+b}{2}$

In general, if $a, b, c, \ldots n$ terms are in AP, then their AM $= \frac{a+b+c+\ldots n \text{ terms}}{n}$

Its imperative to mention here is that, in simple terms, AM of n terms is nothing but the average of n terms.

Hence, the sum of n AMs between P and Q

$= \left(\frac{P+Q}{2}\right)n$

Example 14 If a, b, c, d, e and f are the AMs between 2 and 12, then find the value of $a + b + c + d + e + f$?

Solution The Sum of n AMs between P and Q

$= \left(\frac{P+Q}{2}\right)n$

Hence the sum $= \left(\frac{2+12}{2}\right)6 = 42$

Inserting a given number of arithmetic means between two given quantities

Let p and q be the given quantities and n be the number of means which is to be inserted. After inserting n means, the total number of terms including the extremes will be equal to $n + 2$. Now we have to find a series of $n + 2$ terms in AP, of which p is the first, and q is the last term.

Let d be the common difference;

then q = the $(n + 2)$th term $= p + (n + 1)\,d$;

Hence, $d = \frac{(q-p)}{(n+1)}$

And the inserted means are:

$p + \frac{(q-p)}{(n+1)}, p + 2\frac{(q-p)}{(n+1)}, \ldots, p + n\frac{(q-p)}{(n+1)}$.

Geometric Mean

If two numbers a and b are in GP, then their geometric mean (GM) $= \sqrt{ab}$.

In general, if $a, b, c, \ldots n$ terms are in GP, then their GM $= \sqrt[n]{a \times b \times c \times \ldots n \text{ terms}}$.

So, if three terms a, b and c are in GP, then their GM $= b = \sqrt[3]{a \times b \times c}$.

Inserting a given number of geometric means between two given quantities

Let p and q be the given quantities and n be the required number of geometric means to be inserted between p and q. In all, there will be $n + 2$ terms, so we have to find a series of $n + 2$ terms in a GP of which p is the first term and q is the last term.

Let us assume that r is the common ratio.

Then q is the $(n + 2)$th term.

$q = pr^{n+1}$;

So, $r^{n+1} = \frac{q}{p}$; $r = \frac{q}{p}^{\frac{1}{n+1}}$

Hence, the required means are $p\left(\frac{q}{p}\right)^{\frac{1}{n+1}}, p\left(\frac{q}{p}\right)^{2\frac{1}{n+1}}, \ldots$

$p\left(\frac{q}{p}\right)^{n\frac{1}{n+1}}$.

The above written means are nothing but $pr, pr^2, \ldots, pr^n$, where $r = \frac{q}{p}^{\frac{1}{n+1}}$

Harmonic Mean

If two numbers a and b are in HP then their harmonic mean (HM) $= \frac{2ab}{a+b}$

The process of inserting n HMs between two given numbers is quite similar to the process of inserting AM's between two given numbers.

Relationship among AM, GM and HM

Now we know that for any two given numbers a and b,

$AM = \frac{a+b}{2}$

$GM = \sqrt{ab}$

$HM = \frac{2ab}{a+b}$

1. AM, GM and HM will be in a GP.
   So, GM is the geometric mean of this series.
   And, GM $= \sqrt{AM \times HM}$ [True only for two terms]
2. AM ≥ GM ≥ HM [Always true for any no. of terms]
3. The equation having a and b as its roots is $x^2 - 2Ax + G^2 = 0$.

Sum of *n* terms of some special series

In this part of progression, we shall see the sum of some other special sequences.

1. The sum of the 1st n natural numbers:

   $S_n = \frac{n(n+1)}{2}$

   Hence $\sum_{S=1}^{n} S = 1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$

2. The sum of the squares of the 1st n natural numbers:

   $S_n = \frac{n(n+1)(2n+1)}{6}$

   Hence $\sum_{S=1}^{n} S^2 = 1^2 + 2^2 + 3^2 + \ldots + n^2$

   $= \frac{n(n+1)(2n+1)}{6}$

3. The sum of the cubes of the 1st n natural numbers:

   S_n = (Sum of the 1st n natural numbers)^2

   $= \left(\frac{n(n+1)}{2}\right)^2$

   It can be seen that $(1 + 2 + 3)^2 = 1^3 + 2^3 + 3^3$

   Hence, $\sum_{S=1}^{n} S^3 = 1^3 + 2^3 + 3^3 + \ldots + n^3$

   $= \left(\frac{n(n+1)}{2}\right)^2$

4. $2^2 + 4^2 + 6^2 + \ldots + n$ terms

   $= 1/4 \ \frac{n(n+1)(2n+1)}{6}$

Worked-out Problems

Example 15 An investor starts with $ 500 in an investment account, and each month it earns a constant interest of $ 32. After how many months will the sum exceed $ 700 in the account?

(a) 2 (b) 3 (c) 6 (d) 8

Solution Here, we know that $a + (n-1)\,d = 700$ where $a = 500$ and $d = 32$. The reason for not using the sum formula is that the amount in the account, after each successive month, represents each term of the series and not the added up terms.

The equation becomes $500 + (n - 1) \times 32 = 700$. This solution gives $n = 7.25$, which is rounded up to 8. After eight months the account will hold in excess of $ 700.

So, option (d) is the answer.

Example 16 The sum of an AP, consisting of 12 terms, is 354. The ratio of the sum of the odd terms to the sum of the even terms is 27:32. What is the common difference of this AP?

(a) 2 (b) 3 (c) 4 (d) 5

Solution Let the common difference be 'd' and the terms are $T_1, T_2, T_3 \ldots$

So, $T_2 = T_1 + d$

And $T_4 = T_3 + d$ and so on.

So, $T_2 + T_4 + T_6 + T_8 + T_{10} + T_{12} = T_1 + T_3 + T_5 + T_7 + T_9 + T_{11} + 6d$ ...(1)

Sum of the even terms : Sum of the odd terms = 32 : 27 and their sum = 354

So, the sum of the even terms = 192 and of the odd terms = 162, Using (1), d = 5

Example 17 Find the sum of n terms of the series $11 + 103 + 1005 + \ldots$

(a) $10/9(10^n - 1) - 1$ (b) $100/99(10^n - 1) + n^2$

(c) $10/9(10^n - 1) + n^2$ (d) None of these

Solution Ideally, in these kinds of problems, instead of going by the mathematical process of solving, we should use options.

Checking option (1), Put $n = 1$.

$10/9\,(10^n - 1) - 1 = 9$, so it is not correct.

Checking option (2), Put $n = 1$.

$100/99\,(10^n - 1) + n^2$ is not equal to 11, so this is also not correct.

Checking option (c), Put $n = 1$.

$10/9(10^n - 1) + n^2 = 11$. But just because this option satisfies $n = 1$, it should not be assumed to be correct. Let us check it for $n = 2$.

Option (c) gives us 104. So, this is the answer.

Normally, in these cases, checking the options till $n = 2$ guarantees the answer, but sometimes we need to check it till $n = 3$.

PRACTICE EXERCISES

WARM UP

Q.1. Find the sum of the first 15 terms of the series whose nth term is $(4n + 1)$.
(a) 485 (b) 495
(c) 505 (d) 630

Q.2. What is the sum of the first 7 terms of the series 1/3, 1/2, 3/4, ...?
(a) 2059/164 (b) 2050/192
(c) 2059/164 (d) None of these

Q.3. In a GP, the first term is 7 and nth term is 448, and the sum of the n terms is 889. What is the common ratio of this GP?
(a) 2 (b) 4 (c) 1 (d) 3

Q.4. The sum of 15 terms of an AP is 600, and the common difference is 5. Find the first term.
(a) 4 (b) 5
(c) 3 (d) None of these

Q.5. A man arranges to pay off a debt of Rs 3,600 by 40 annual installments which form an AP. When 30 of the installments were repaid he died leaving a third of the debt unpaid. What is the value of the first installment?
(a) 50 (b) 51 (c) 52 (d) 53

Q.6. An AP has 23 terms, the sum of the middle three terms is 144, the sum of the last three terms is 264. What is the 16th term?
(a) 102 (b) 64
(c) 82 (d) None of these

Q.7. Find A, B and C between 2 and 18 such that (i) their sum is 25, (ii) 2, A and B are consecutive terms of an AP and (iii) B, C and 18 are the consecutive terms of a GP.
(a) 5, 8, 14 (b) 4, 8, 12
(c) 5, 8, 12 (d) None of these

Q.8. If the sum of an AP is the same for p terms as for the q terms, find the sum for $(p + q)$ terms.
(a) 2 (b) 0
(c) 4 (d) None of these

Q.9. The sum of the three numbers in GP is 70. If the two extreme terms are multiplied by 4, and the middle term by 5, the resultants are in an AP. Find the numbers.
(a) 10, 25, 35 (b) 10, 20, 40
(c) 15, 30, 35 (d) 12, 24, 34

Q.10. Find the first term of the infinite GP whose first two terms add up to 5, and each term is three times the sum of the all terms that follow it.
(a) 1 (b) 2
(c) 3 (d) 4

Q.11. If $x = 1 + a + a^2 + a^3 + \ldots$ to ∞ $(|a| < 1)$,
$y = 1 + b + b^2 + b^3 + \ldots$ to ∞ $(|b| < 1)$,
Then find $1 + ab + a^2b^2 + a^3b^3 + \ldots$ to ∞
(a) $x + y/x + y - 1$ (b) $xy/x + y$
(c) $xy/x + y - 1$ (d) None of these

Q.12. If the $(n + 1)$th term of a harmonic progression is twice the $(3n + 1)$th term, find the ratio of the first term to the $(n + 1)$th term.
(a) 1 (b) 2 (c) 3 (d) 4

Q.13. If the sum of the reciprocals of the first seven terms of a harmonic progression is 70, find the 4th term of the HP.
(a) 2/15 (b) 1/10 (c) 3/7 (d) 5/12

Q.14. If the mth term of a HP is n and the nth term is m, what is the value of the $(m + n)$th term?
(a) $m/(m + n)$ (b) $mn/m + n$
(c) $n/m + n$ (d) None of these

Q.15. If $(x^{n+1} + y^{n+1})/(x^n + y^n)$ is the harmonic mean of x and y, find the value of n.
(a) 1 (b) -1 (c) 2 (d) 4

Q.16. A series is simultaneously in an AP and a HP. Which of the following is true?
(a) The set of common ratios of all the possible GPs can have only one element.
(b) The set of common ratios of all the possible GPs can have more than one element but there will be a finite value.
(c) The set of common ratios of all the possible GPs can have infinite elements.
(d) None of these

Q.17. A series is simultaneously in an AP and a HP. Which of the following is true?

(a) The set of common differences of all the possible APs can have only one element.
(b) The set of common differences of all the possible APs can have more than one element but there will be a finite value.
(c) The set of common differences of all the possible APs can have infinite elements.
(d) None of these

Q.18. Find the sum of all the numbers divisible by 6 in between 100 to 400.
(a) 12,550
(b) 12,450
(c) 11,450
(d) 11,550

FOUNDATION

Q.1. Let t_r denote the rth term of an AP. If $t_m = \frac{1}{n}$ and $t_n = \frac{1}{m}$ then t_{mn} equals
(a) $\frac{1}{mn}$ (b) $\frac{1}{m}+\frac{1}{n}$
(c) 1 (d) 0

Q.2. Let x, y, z be three positive prime numbers. The progression in which $\sqrt{x}, \sqrt{y}, \sqrt{z}$ can be three terms (not necessarily consecutive) is
(a) AP (b) GP
(c) HP (d) None of these

Q.3. Let $f(x) = 2x + 1$. Then the number of real values of x for which the three unequal numbers $f(x), f(2x)$ and $f(4x)$ are in a GP is
(a) 1 (b) 2
(c) 0 (d) None of these

Q.4. If a, b and c are in a GP then $a + b, 2b, b + c$ are in
(a) AP (b) GP
(c) HP (d) None of these

Q.5. If a, b and c are in an AP then $\frac{a}{bc}, \frac{1}{c}, \frac{2}{b}$ are in
(a) AP (b) GP
(c) HP (d) None of these

Q.6. In the sequence 1, 2, 2, 4, 4, 4, 4, 8, 8, 8, 8, 8, 8, 8, 8, ..., where n consecutive terms have the value n, the 1025th term is
(a) 2^9 (b) 2^{10} (c) 2^{11} (d) 2^8

Q.7. If the pth, qth and rth terms of an AP are in a GP then the common ratio of the GP is
(a) $\frac{p+q}{r+q}$ (b) $\frac{r-1}{q-p}$
(c) $\frac{p-r}{p-1}$ (d) None of these

Q.8. The sum of an infinite geometric series is 4 and the sum of the cubes of the terms of the same GP is 192. The common ratio of the original geometric series is
(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{6}$

Q.9. The 10th common term between the series 3 + 7 + 11 + ... and 1 + 6 + 11 + ... is
(a) 191 (b) 193
(c) 211 (d) None of these

Q.10. If a, b, c and d are four numbers such that the first three are in an AP while the last three are in a HP then
(a) $bc = ad$ (b) $ac = bd$
(c) $ab = cd$ (d) None of these

Q.11. The ratio of the sum of n terms of two AP's is $(3n - 13) : (5n + 21)$.
Find the ratio of their 24th terms.
(a) 1:2 (b) 1:3 (c) 2:3 (d) 2:5

Q.12. In a GP of even number of terms, the sum of all the terms is 5 times the sum of the odd terms. The common ratio of the GP is
(a) $\frac{-4}{5}$ (b) $\frac{1}{5}$
(c) 4 (d) None of these

Q.13. If a, b and c are in an AP then $a+\frac{1}{b}, b+\frac{1}{ca}, c+\frac{1}{ab}$ are in
(a) AP (b) GP
(c) HP (d) None of these

Q.14. The ratio of HM and GM for two positive numbers is 4:5. What is the ratio of the numbers?
(a) 4:1 (b) 3:2 (c) 3:4 (d) 2:3

Q.15. In a GP of alternatively positive and negative terms, any term is the AM of the next two terms. Then the common ratio is
(a) −1 (b) −3 (c) −2 (d) $-\frac{1}{2}$

Q.16. The 288th term of the series $a, b, b, c, c, c, d, d, d, d, e, e, e, e, e, \ldots$ is
(a) u (b) v (c) w (d) x

Q.17. The product of n positive numbers is 1. Their sum is
(a) A positive integer
(b) Divisible by n
(c) Equal to $n+\frac{1}{n}$
(d) Greater than or equal to n

Q.18. If t_n denotes the nth term of the series 2 + 3 + 6 + 11 + 18 + ... then t_{50} is
(a) $49^2 - 1$ (b) 49^2
(c) $50^2 + 1$ (d) $49^2 + 2$

Q.19. If $a^x = b^y = c^z$ and x, y, z are in GP then $\log_c b$ is equal to
(a) $\log_b a$ (b) $\log_a b$
(c) $\frac{z}{y}$ (d) None of these

Q.20. The numbers 1, 4 and 16 can be three terms (not necessarily consecutive) of
(a) No AP
(b) Only one GP
(c) Infinite number of APs
(d) More than one but finite number of GPs

Q.21. If a, b, c and d are four positive numbers then which of the following is not true?
(a) $\left(\frac{a}{b}+\frac{b}{c}\right)\left(\frac{c}{d}+\frac{d}{e}\right) \geq 4.\sqrt{\frac{a}{e}}$
(b) $\left(\frac{a}{b}+\frac{c}{d}\right)\left(\frac{b}{c}+\frac{d}{e}\right) \geq 4.\sqrt{\frac{a}{e}}$
(c) $\frac{a}{b}+\frac{b}{c}+\frac{c}{d}+\frac{d}{e}+\frac{e}{a} \geq 5$
(d) $\frac{b}{a}+\frac{c}{b}+\frac{d}{c}+\frac{e}{d}+\frac{a}{e} \geq \frac{1}{5}$

Q.22. Find the 45th term of the series 12, 13½, 15
(a) 57 (b) 77.5 (c) 74 (d) 78

Q.23. The 5th term of an AP is 15 and the 9th term is 23. Find the 14th term.
(a) 31 (b) 33 (c) 35 (d) 37

Q.24. If 9 times the 9th term in an AP is equal to 15 times the 15th term in the AP what is the 24th term?
(a) −1 (b) −3 (c) 0 (d) 1

Q.25. Find the sum of an AP of 14 terms whose first and the last terms are 8 and 99 respectively.
(a) 707 (b) 749 (c) 789 (d) 747

Q.26. A gets as many marks less than B as B gets less than C. If the total marks secured together by them is 270 and C got 90 marks more than A, find the marks secured by them individually.
(a) 30, 60, 180 (b) 45, 90, 135
(c) 40, 80, 150 (d) 35, 70, 165

Q.27. Find the sum of all the 2-digit numbers that leave remainder 1 when divided by 5.
(a) 963 (b) 968 (c) 967 (d) 960

Q.28. Find the common ratio of GP whose first term is 3, the last is 3072 and the sum of the series is 4095.
(a) 2 (b) 3 (c) 4 (d) 6

Q.29. The least value of n, for which $1 + 4^2 + 4^4 + \ldots$ to n terms is greater than 1000, is
(a) 5 (b) 4 (c) 3 (d) 2

Q.30. The number of terms in an AP is odd. If the sum of all the terms is 45 and the number of terms is 4 more than the middle term, find the middle term.
(a) 9 (b) 4 (c) 6 (d) 5

Q.31. If $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is the arithmetic mean between a and b then n is
(a) 0 (b) 1
(c) 1/2 (d) None of these

Q.32. In an infinite GP, every term is equal to the sum of all the terms that follow. Find the common ratio.
(a) 1/2 (b) 1/3 (c) 1/4 (d) 1/6

Q.33. The difference between the two numbers is four and the AM between them is six. The product of the numbers is
(a) 24 (b) 12 (c) 32 (d) 48

Q.34. Which of the following statements is always true? (statements are related to AP and the terms are having their general notations.)
(a) $t_5 + t_8 = t_1 + t_{12}$ (b) $t_8 = a + 8d$
(c) $t_1 + t_2 = t_3$ (d) Only a and b

Q.35. The number of terms common between the series 1 + 2 + 4 + 8 + ... to 100 terms and 1 + 4 + 7 + 10 + ... to 100 terms is
(a) 6 (b) 4
(c) 5 (d) None of these

Q.36. If the geometric mean of two non-negative numbers is equal to their harmonic mean, then which of the following is necessarily true?
I. One of the numbers is zero
II. Both the numbers are equal
III. One of the numbers is one
(a) I only (b) Either I or II
(c) II only (d) I or III

Q.37. Consider three numbers in an AP. Which of the following will definitely divide the sum of the cubes of all the three numbers?
(a) Middle number
(b) Difference of the first and last numbers
(c) 3
(d) (a) and (c)

Q.38. My neighbour, Mr Makrant Chatpatia, is a nice person. Last night he hosted a party for me. He told me that he would retire next year. He also added that when he joined his firm he was 19 years old and his basic salary was Rs 100 and after every two years there was an increase of Rs 60. What must be the basic salary of Mr Chatpatia now, if the age of retirement is 60 years?
(a) Rs 1,990 (b) Rs 1,240
(c) Rs 1,200 (d) Rs 1,300

Q.39. Given is 54 + 51 + 48 + ... till n terms = 513. How many values of 'n' are possible?
(a) 1 (b) 2
(c) 0 (d) None of these

Q.40. In a decreasing AP the sum of all its terms, except the first term, is equal to –36, the sum of all its terms, except the last term, is zero, and the difference of the tenth and the sixth term is equal to –16. What will be first term of this series?
(a) 16 (b) 20 (c) –16 (d) –20

MODERATE

Q.1. A group of friends have some money which was in an increasing GP. The total money with the first and the last friend was Rs 66 and the product of the money that the second friend had and that the last but one friend had was Rs 128. If the total money with all of them together was Rs 126, then how many friends were there?
(a) 6 (b) 5
(c) 3 (d) Cannot be determined

Q.2. If the sum of the 10th, 20th and 30th terms of an AP is equal to the 58th term, what is the ratio of the sum of the 10th, 20th and 30th terms to the sum of the 5th, 10th and 15th terms; when the common difference ≠ 0?
(a) 2 : 1 (b) 20:9
(c) 19:9 (d) Cannot be determined

Q.3. Let $a_1, a_2, a_3, \ldots$ be in an AP and a_p, a_q, a_r, be in a GP. Then $a_q : a_p$ is equal to
(a) $\frac{r-p}{q-p}$ (b) $\frac{q-p}{r-q}$
(c) $\frac{r-1}{q-q}$ (d) None of these

Q.4. If a, b, c and d are non-zero real numbers such that $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) \le (ab + bc + cd)^2$, then a, b and c and d are in
(a) AP (b) GP
(c) HP (d) None of these

Q.5. If $\log\left(\frac{5c}{a}\right), \log\left(\frac{3b}{5c}\right)$ and $\log\left(\frac{a}{3b}\right)$ are in an AP, where a, b and c are in a GP, then a, b and c, are the lengths of sides of
(a) An isosceles triangle
(b) An equilateral triangle
(c) A scalene triangle
(d) None of these

Q.6. The coefficient of x^{15} in the product $(1-x)(1-2x)(1-2^2.x)\ldots(1-2^{15}.x)$ is equal to
(a) $2^{105} - 2^{121}$ (b) $2^{121} - 2^{105}$
(c) $2^{120} - 2^{104}$ (d) None of these

Q.7. The coefficient of x^{49} in the product $(x-1)(x-3)\ldots(x-99)$ is
(a) -99^2 (b) 1
(c) -2500 (d) None of these

Q.8. The AM of two given positive numbers is 2. If the larger number is increased by 1, the GM of the numbers becomes equal to the AM of the given numbers. Then the HM of the given numbers is
(a) $\frac{3}{2}$ (b) $\frac{2}{3}$
(c) $\frac{1}{2}$ (d) None of these

Q.9. Let a and b be two positive numbers, where $a > b$ and 4 × GM = 5 × HM for the numbers. Then a is
(a) $4b$ (b) $\frac{1}{4}b$
(c) $2b$ (d) b

Q.10. The sequence P_n is in a GP with P_2/P_4 = ¼ and $P_1 + P_4 = 108$. What will be the value of P_3?
(a) 42 (b) 48 (c) 44 (d) 56

Q.11. Let $t_n = n.(n!)$. Then $\sum_{n=1}^{15} t_n$ is equal to
(a) 15! – 1 (b) 15! + 1
(c) 16! – 1 (d) None of these

Q.12. In a GP the product of the first four terms is 4 and the second term is the reciprocal of the fourth term. The sum of the GP upto infinite terms is
(i) 8 (ii) –8 (iii) $\frac{8}{3}$ (iv) $-\frac{8}{3}$
(a) Only i and ii (b) Only ii and iii
(c) Only i, ii and iii (d) i, ii, iii and iv

Q.13. Let a_n = product of the first n natural numbers. Then for all $n \in N$

(i) $n^n \geq a_n$ (ii) $\left(\frac{n+1}{2}\right)^n \geq n!$

(iii) $n^n \geq a_{n+1}$ (iv) None of these

(a) i and ii only (b) ii and iii only

(c) i, ii and iii only (d) None of these

Q.14. A GP consists of 1000 terms. The sum of the terms occupying the odd places is P_1 and the sum of the terms occupying the even places is P_2. Find the common ratio of this GP.

(a) P2/P1 (b) P_1/P_2

(c) $(P_2 - P_1)/P_1$ (d) $(P_2 + P_1)/P_2$

Q.15. The harmonic mean of the roots of the equation $(5 + \sqrt{2})x^2 - (4 + \sqrt{5})x + 8 + 2\sqrt{5} = 0$ is

(a) 2 (b) 4 (c) 6 (d) 8

Q.16. Let A, G and H be the AM, GM and HM of two positive numbers a and b. The quadratic equation whose roots are A and H is

(a) $Ax^2 - (A^2 + G^2)x + AG^2 = 0$

(b) $Ax^2 - (A^2 + H^2)x + AH^2 = 0$

(c) $Hx^2 - (H^2 + G^2)x + HG^2 = 0$

(d) (a) and (c) both

Q.17. If the roots of $x^3 - 12x^2 + 39x - 28 = 0$ are in an AP then their common difference is

(a) ± 1 (b) ± 2 (c) ± 3 (d) ± 4

Q.18. In an infinite GP, each term is equal to four times the sum of all the terms that follow. Find the common ratio.

(a) 1/4 (b) 1/5 (c) 1/8 (d) 1/7

Q.19. If the terms of one geometric progression are multiplied by the corresponding terms of another geometric progression, the sequence obtained will be in

(a) AP (b) HP

(c) GP (d) Cannot be determined

Q.20. In a network system, each person has to include four more persons under him and such a chain should continue. The person at any level would get Re 1 commission per person below him in his group. If a person earns Rs 84, find the number of persons under him, in his group earning zero amount.

(a) 32 (b) 64 (c) 128 (d) 256

Q.21. If the nth term of AP is p and the mth term of the same AP is q, then $(m + n)$th term of AP is

(a) $p + q$ (b) $p + q - a + d$

(c) $p + q - d - a$ (d) $p + q - d + a$

Q.22. Two sequences of numbers {1, 4, 16, 64 ... } and {3, 12, 48, 192,... } are mixed as follows {1, 3, 4, 12, 16, 48, 64, 192, ... }. One of the numbers in the mixed series is 1048576. Then the number immediately preceding it is

(a) 7,86,432 (b) 2,62,144

(c) 8,14,572 (d) 7,86,516

Q.23. For each positive integer n consider the set S_n defined as follows

$S_1 = \{1\}$, $S_2 = \{2, 3\}$, $S_3 = \{4, 5, 6\}$ ..., and in general, S_{n+1} consists of $n+1$ consecutive integers the smallest of which is one more than the largest integer in S_n. Then the sum of all the integers in S_{21} equals.

(a) 1,113 (b) 5,336 (c) 5,082 (d) 4641

Q.24. If $a_1, a_2, a_3, a_4 \ldots a_{24}$ are in an arithmetic progression and $a_1 + a_5 + a_{10} + a_{20} + a_{24} = 225$, then the sum of the series $a_1 + a_2 + a_3 + a_4 + \ldots + a_{24}$ is

(a) 909 (b) 75 (c) 750 (d) 900

Q.25. The harmonic mean of two positive real numbers is 4. Their arithmetic mean A and their geometric mean G satisfy the relation $2A + G^2 = 27$. Find the two numbers.

(a) 4 and 4 (b) 2 and 6

(c) 3 and 6 (d) 5 and 10/3

Q.26. Find the sum of the 37th bracket of the following series.

$(1) + (7 + 7^2 + 7^3) + (7^4 + 7^5 + 7^6 + 7^7 + 7^8) + (7^9 + 7^{10} + \ldots + 7^{15}) \ldots$.

(a) $\frac{7^{37}}{6}\left(7^{73} - 1\right)$ (b) $\frac{\left(7^{73} - 1\right)}{6}$

(c) $\frac{7^{71}}{6}\left(7^{73} - 1\right)$ (d) None of these

Q.27. If a, b and c are positive integers, then find the product of $(a + b)$ $(b + c)$ $(c + a)$.

(a) $> 8\ abc$ (b) $< 8\ abc$

(c) $= 8\ abc$ (d) None of these

Q.28. The sum the of 'n^{th}' terms of two different arithmetic progression is zero. The sum of a certain number of terms (beginning with the first) of one of the progression is 386. What is the sum of the same number of terms (from the beginning) of the second progression?

(a) 386 (b) + 193 (c) − 193 (d) − 386

Q.29. An infinite GP has the first term 'X' and sum '5', then X belongs to

(a) $X < -10$ (b) $-10 < X < 0$

(c) $0 < X < 10$ (d) $X > 10$

Q.30. $f(x) = 2x + 1$, where $x = 1, 2, \ldots, 100$. $g(x) = 3x - 2$, where $x = 1, 2, \ldots, 100$. For how many values of x, $f(x) = g(x)$?

(a) 31 (b) 32 (c) 33 (d) 34

Q.31. Find the sum of the series:
$1.2 + 2.2 + 3.2^2 + 4.2^3 + \ldots + 100.2^{99}$.
(a) $99 \times 2100 - 1$ (b) $99 \times 2101 + 1$
(c) $99 \times 2101 - 1$ (d) $99 \times 2100 + 1$

Q.32. The mid-points of the adjacent sides of a square are joined. Again the mid-points of the adjacent sides of the newly formed figure are connected and this process is repeated again and again. Calculate the sum of the areas of all such figures given that the diagonal of outermost square is $6\sqrt{2}$cm.
(a) 35 cm^2 (b) 44 cm^2
(c) 72 cm^2 (d) 58 cm^2

Q.33. The sum of the first n terms of an AP is $n(n-1)$. Then the sum of the squares of these terms is
(a) $n^2(n-1)^2$ (b) $\frac{n}{6}(n-1)(2n-1)$
(c) $\frac{2}{3}n(n-1)(2n-1)$ (d) $\frac{2}{3}n(n-1)(2n+1)$

Q.34. What is the sum of the following series?
$7 + 26 + 63 + 124 + \ldots + 999$
(a) 3014 (b) 3013
(c) 3015 (d) None of these

Q.35. In ΔABC, points P_1, Q_1 and R_1 divide the lines AB, BC and AC respectively in the ratio of 2 : 1. In $\Delta P_1Q_1R_1$, the points P_2, Q_2 and R_2 divide the sides P_1Q_1, Q_1R_1 and P_1R_1 in the ratio of 2: 1. In every such new triangle, a new triangle is generated by joining the points on the sides that divide these sides in the ratio of 2 : 1. Find the sum of the areas of all such triangles formed till infinity. (Area of ΔABC = 'A' sq. units.)
(a) $\frac{A}{2}$ sq. units (b) 2A sq. units
(c) $\frac{A}{3}$ sq. units (d) 3A sq. units

Q.36. Find the sum of the series $-1 + 1^2 - 2 + 2^2 - 3 + 3^2 + \ldots n + n^2$.
(a) $\frac{-n(n+1)}{3}$ (b) $\frac{n(n+1)(n-1)}{3}$
(c) $\frac{n(n-1)}{3}$ (d) None of these

Q.37. There are three numbers in an arithmetic progression. If the two larger numbers are increased by one, then the resulting numbers are prime. The product of these two primes and the smallest of the original numbers is 598. Find the sum of the three numbers.
(a) 45 (b) 29 (c) 42 (d) 36

Q.38. If three successive terms of a GP with the common ratio r > 1 form the sides of a triangle and [r] denotes the integral part of x, then [r] + [−r] =
(a) 0 (b) 1
(c) −1 (d) None of these

Q.39. In how many ways can we select three natural numbers out of the first 10 natural numbers so that they are in a geometric progression with the common ratio greater than 1?
(a) 2 ways (b) 3 ways
(c) 4 ways (d) 5 ways

Q.40. If m times the mth term of an AP is equal to p times the pth term, find the $(m+p)$th term.
(a) 0 (b) $a^2 - b^2$
(c) $a - b$ (d) 1

ADVANCED

Q.1. If $a_1, a_2, a_3, \ldots a_n$ $(n \geq 3)$ are in an AP then $\left(\frac{1}{a_1a_2} + \frac{1}{a_2a_3} + \ldots + \frac{1}{a_{n-1}a_{n-2}}\right)$ will be equal to
(a) $\frac{n^2}{a_1a_n}$ (b) $\frac{(n-1)}{a_1a_n}$
(c) $\frac{2n}{a_1a_n}$ (d) None of these

Q.2. Three distinct numbers x, y, z, form a GP in that order and the numbers $x + y, y + z, z + x$ form an AP in that order. Find the common ratio of the GP.
(a) 1 (b) −2
(c) 2 (d) Either (a) or (b)

Direction for questions 3 and 4: Answer the questions based on the following information.

A series of positive integers have the following terms. First term: 1; Second term: (2 + 3); Third term: (4 + 5 + 6), ... so on.

Q.3. The last number in the nth term of the series is
(a) $\frac{(n^2 - n + 2)}{2}$ (b) $\frac{(n^2 + n)}{2}$
(c) $\frac{(2n^2 + 3)}{4n}$ (d) None of these

Q.4. The sum of the numbers of the nth term is

(a) n^2+n (b) $\frac{n\left(n^2+1\right)}{2}$

(c) $\frac{(n+n)^2}{3n}$ (d) None of these

Q.5. An arithmetic series consists of $2n$ terms, and the first term equals the value of the common difference. If a new series is formed taking the 1st, 3rd, 5th, ... $(2n-1)$ th term of the old series, find the ratio of the sum of the new series to that of the sum of the terms of the old series.

(a) $\frac{n+1}{2(2n+1)}$ (b) $\frac{n}{2n+1}$

(c) $\frac{1}{2}$ (d) Cannot be determined

Q.6. If the sum of the first $2n$ terms of the AP 2, 5, 8 ... is equal to the sum of first n terms of the AP 57, 59, 6 ..., then what is the value of n?

(a) 7 (b) 9 (c) 11 (d) 13

Q.7. Let the positive numbers a, b, c and d be in AP. Then abc, abd, acd, bcd are

(a) Not in AP or GP or HP
(b) In AP
(c) In GP
(d) In HP

Q.8. Given that $(m+1)$th, $(n+1)$th and $(r+1)$th term of an AP are in GP and m, n, r are in HP, then find the ratio of the first term of the AP to its common difference in terms of n.

(a) $2:n$ (b) $n:2$ (c) $2n:3$ (d) $3:4n$

Q.9. Let x be the arithmetic mean and y and z be the two geometric means between any two positive numbers then the value of $\frac{y^3+z^3}{xyz}=\ldots$

(a) xy (b) 2 (c) 0 (d) xyz

Q.10. If rational numbers a, b, c and d are in GP then the roots of the equation $(a-c)^2x^2+(b-c)^2x+(b-d)^2=(a-d)^2$ are necessarily

(a) Imaginary (b) Irrational
(c) Rational (d) Real and unequal

Q.11. Let $a = 111\ldots1$ (55 digits),
$b = 1 + 10 + 10^2 + 10^3 + 10^4$,
$c = 1 + 10^5 + 10^{10} + 10^{15} + \ldots + 10^{50}$, then

(a) $a = b + c$ (b) $a = bc$
(c) $b = ac$ (d) $c = ab$

Q.12. If b be the pth term of a GP where $(p+q)$th and $(p-q)$th terms are a and c respectively, then for all $x \in R$

(a) $af(x) = 0$ (b) $af(x) > 0$
(c) $af(x) \leq 0$ (d) $af(x) > 0$

Q.13. If a geometric mean of two non-negative numbers is equal to their harmonic mean, then which of the following is necessarily true?

I. One of the numbers is zero.
II. Both the numbers are equal.
III. One of the numbers is one.

(a) I and III only (b) Either I or III
(c) III only (d) None of these

Q.14. The middle points of the sides of a triangle are joined forming a second triangle. Again a third triangle is formed by joining the middle points of this second triangle and this process is repeated infinitely. If the perimeter and the area of the outer triangle are P and A respectively, find the sum of the areas of all the triangles.

(a) 4/5A (b) 4/3 A (c) 3/4A (d) 5/4A

Q.15. A series in which any term is the sum of the preceding two terms is called a Fibonacci series. The first two terms are given initially and together they determine the entire series. If the difference of the squares of the ninth and the eighth terms of a Fibonacci series is 715 then, what is the 12th term of that series?

(a) 157 (b) 142
(c) 144 (d) Cannot be determined

Q.16. In a sequence of terms, a_1, a_2, a_3 ..., the nth term, a_n (for $n>1$), is given by $a_n = a_{n-1} + a_{n+1}$. If $a_7 = 6$ and $a_{14} = 7$, find S_{26}/S_{13}, where S_i is the sum of all the terms from a_1 to a_i in the sequence.

(a) $\frac{1}{6}$ (b) $-\frac{1}{6}$

(c) $\frac{13}{6}$ (d) Cannot be determined

Q.17. The sum of an infinite GP is 162 and the sum of its first n terms is 160. If the inverse of its common ratio is an integer, then how many values of common ratio is/are possible, common ratio is greater than 0?

(a) 0 (b) 1 (c) 2 (d) 3

Q.18. In the above question, which of the following is the value of n?

(a) 4 (b) −2
(c) −1 (d) None of these

Q.19. The first term of an AP = the common ratio of a GP and the first term of the GP = common difference of the AP. If the sum of the first two terms of the GP is equal to the sum of the first 2 terms of the AP, then the ratio of the first term of the GP to the first term of an AP is

(a) > 1
(b) Independent of first term of AP
(c) Independent of first term of GP
(d) < 1

Direction for questions 20 to 23: Read the passage below and solve the questions based on it.

Let there be a series 'S' with it's nth term be equal to $n(x)^n$. Also S_n denotes the sum of the first n terms of the series S.

Q.20. What is $S_5 - S_4$ equal to?
(a) $3x^3$ (b) $4x^4$
(c) $5x5$ (d) None of these

Additional information for questions 21 to 23:

If the $(n + 1)$th term of S is equal to 64 and the nth term is equal to 24 then:

Q.21. What is the value of n?
(a) 2 (b) 3
(c) 4 (d) Can't say

Q.22. What is the value of x?
(a) 2 (b) 3
(c) 4 (d) None of these

Q.23. What is the value of S_3?
(a) 20 (b) 14
(c) 34 (d) None of these

Q.24. Let S_n be defined as the sum of n terms of the following series: $S_n = 2^{66} - 2^{65} - 2^{64}$. What is the sum of S_{20}?
(a) 2^{47} (b) 2^{12}
(c) 2^{19} (d) None of these

Q.25. The odd natural numbers are arranged in a pyramidical shape given below:

1
3 5
7 9 11
13 15 17 19
… … … … …

And so on.
What is the sum of the numbers in the nth row of this structure?
(a) $2n^2 - 1$ (b) n^3
(c) $(n - 1)^2 + 1$ (d) $2n^2 - 1$

Q.26. Set X is having 2000 elements in AP with the first term = 1, and the common difference = 1. What is the maximum number of elements Set Y can have such that sum of no two elements of Y is equal to 9 or a multiple of 9?
(a) 890 (b) 891
(c) 892 (d) None of these

Q.27. In an AP of even number of terms, the sum of the odd terms is 24 while that of the even terms is 30. Find the number of terms if the last term exceeds the first by 10.5.
(a) 7 (b) 9 (c) 6 (d) 8

Q.28. Find the sum of the products of the integers, 1, 2, 3,... n taking two at a time.
(a) $1/12\ n(n^2 + 1)\ (n + 1)$
(b) $1/6\ n(n^2 + 1)\ (n + 1)$
(c) $\frac{1}{12} n\ (n2 - 1)\ (3n + 2)$
(d) None of these

Q.29. The sum of the first 10 terms of an AP is equal to 155 and the sum of the first two terms of a GP is 9. How many values for the first term of an AP is possible if the first term of an AP is equal to the common ratio of a GP and the first term of a GP is equal to the common difference of a AP.
(a) 0 (b) 1 (c) 2 (d) Infinite

Q.30. The sum upto the n terms of the following series:
$\frac{1}{1.3.5} + \frac{1}{3.5.7} + \frac{1}{5.7.9} \cdots$
(a) $\frac{(n+2)(n+4)}{n(n+1)(n+3)}$ (b) $\frac{(n+2)(n+4)}{(2n+1)(n-2)}$
(c) $\frac{n(n+2)(n+4)}{3(2n+1)(n+3)}$ (d) $\frac{n(n+2)}{3(2n+1)(2n+3)}$

Direction for questions 31 and 32: Read the passage below and solve the questions based on it.

30 Alphonso mangoes are to be distributed among Tom, Dick and Harry. Harry's share is less than that of Dick. Further, they have agreed to divide the mangoes such that the number of mangoes with the three of them are in an arithmetic progression.

Q.31. If none of them is to receive more than 15 mangoes, in who many ways can the mangoes be distributed?
(a) 10 (b) 20 (c) 5 (d) 15

Q.32. Which of the following statements must be true?
(a) The number of mangoes Dick receives is at least 11.
(b) The number of mangoes Tom receives is at most 9.
(c) One person gets exactly 10 mangoes.
(d) More than one of the above.

Q.33. The sum of the first n terms $(n > 1)$ of an AP is 153 and the common difference is 2. If the first term is an integer, then the number of possible values of n is
(a) 2 (b) 3 (c) 4 (d) 5

Q.34. The positive numbers a, b, c and d are in a HP and $a \neq b$, then
(a) $a + d > b + c$ is always true
(b) $a + b > d + c$ is always true
(c) $a + c > b + d$ is always true
(d) None of these

Q.35. The number of ways in which three distinct numbers in an AP can be selected from 1, 2, … , 24 is
(a) 112 (b) 124
(c) 132 (d) 146

Q.36. Numbers are placed on the vertices of a polygon with n (>4) sides in such a way that each of them equals the arithmetic mean of its neighbours. Then it can be concluded that all the numbers are equal
(a) For every n (>4)
(b) Only for even n (>4)
(c) Only for odd n (>4)
(d) Only for odd prime n (>4)

Q.37. In an infinite geometric progression, the nth term is equal to three times the sum of all the terms that follow it and the sum of the first two terms is 15. What is the sum of the whole series?
(a) 32 (b) 16
(c) $+\infty$ (d) None of these

Direction for questions 38 and 39: Read the passage below and solve the questions based on it.

$N_1 + N_2 + N_3 + \ldots + N_N = P$, where $N_1, N_2, N_3 \ldots$ and N_N are $n(n>1)$ consecutive natural numbers such that $N_1 < N_2 < N_3 < N_4 \ldots < N_N$.

Q.38. If P = 100, then how many values of N is/are possible?
(a) 0 (b) 1 (c) 2 (d) Infinite

Q.39. If P = 100, then what is the highest value of N?
(a) 5 (b) 10 (c) 20 (d) 8

Q.40. An employee joined a company on 01.04.2004 in the salary grade of Rs 8,000-500-9,500-750-12,500 with a basic salary of 9,000. He is due to retire on 31.03.2007. He contributes 10% of his basic salary to an EPF scheme. His employer contributes an equal amount. If, on retirement, he gets the full amount of his share of the EPF and 50% of the employer's share, what is the amount of EPF received by him, ignoring any interest on the deposits?
(a) 34,500 (b) 17,250
(c) 51,750 (d) 62,500

TRUE/FALSE

1. If we know any two terms of an AP, we can definitely find out the remaining terms of the same AP.
State whether true or false.

2. If we know any two terms of a GP, we can definitely find out the remaining terms of the same GP.
State whether true or false.

3. If we know any two terms of a GP, we can definitely find out the common ratio of the same GP.
State whether true or false.

4. If we know any two terms of an AP, we can definitely find out the common difference of the same AP.
State whether true or false.

5. Sum of all the odd terms (viz., 1st term, 3rd term etc.) and sum of all the even terms (viz., 2nd term, 4thterm etc.) of a GP with 'n' terms are given. We can find out the common ratio of this GP with this information.
State whether true or false.

6. If a series is in both AP and GP, then sum of 1st 10 terms will be equal to sum of any other set of consecutive 10 terms.
State whether true or false.

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Answer	(b)	(d)	(a)	(b)	(b)	(b)	(c)	(b)	(b)	(d)	(c)	(b)	(b)	(b)	(b)	(a)	(a)	(b)

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(d)	(a)	(c)	(d)	(b)	(b)	(b)	(a)	(a)	(a)	(c)	(a)	(a)	(c)	(c)	(d)	(d)	(a)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(d)	(b)	(c)	(b)	(b)	(a)	(c)	(b)	(d)	(a)	(a)	(c)	(a)	(c)	(b)	(d)	(d)	(b)	(a)

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(c)	(c)	(b)	(d)	(a)	(c)	(a)	(a)	(b)	(c)	(d)	(a)	(a)	(b)	(d)	(c)	(b)	(c)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(b)	(a)	(d)	(d)	(c)	(d)	(a)	(d)	(c)	(c)	(d)	(c)	(c)	(c)	(b)	(d)	(c)	(c)	(b)	(a)

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(d)	(b)	(b)	(b)	(c)	(d)	(b)	(b)	(c)	(b)	(d)	(d)	(b)	(c)	(c)	(c)	(a)	(c)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(b)	(a)	(c)	(a)	(b)	(b)	(d)	(c)	(c)	(d)	(d)	(c)	(d)	(d)	(c)	(a)	(b)	(c)	(d)	(c)

■■■ True/False

1. True 2. True 3. True 4. True 5. False 6. True

HINTS AND SOLUTIONS

Moderate

1. The sum of money with the first and the last friend = 66. This can be used as a hint. Let us assume the first friend was having Rs 2 and the last friend was having Rs 64. So, the money can be in the sequence 2, 4, 8, 16, 32, 64. It satisfies the given conditions.
   Alternatively, this can be done by using the formula for t_n of GP also.
2. Sum of the 10th, 20th, 30th terms of an AP is equal to the 58th term. Hence, $3a + 57d = a + 57d$
   So, $a = 0$ and d can be anything not equal to zero.
   Now, we have to find out the ratio $3a + 57d$ and $3a + 27d$, or, we have to find out the ratio $57d : 27d = 19 : 9$.
5. For a, b, c to be the length of the sides of the triangle, it's AM should not be 0.
8. Take two numbers such that their AM is 2 like 1 and 3. And go according to the conditions given in the question.
9. Using option (a), let $a = 2$, $b = 8$ and solve. If it does not satisfy, use the other options.
11. Solve through options.
14. Take any small sample of 4 terms of any GP viz. 1, 2, 4 and 8. Its common ratio = 2 = (8+2)/(1 +4)
    And this will be true for any number of terms as long as we are taking total number of terms as even(in this question, it is given as 1000). However, if the total number of terms is given as odd, we cannot find the common ratio in this way, because the number of odd terms ≠ number of even terms.
15. We can easily find out the values : $2ab$ and $a + b$, where a and b are the roots of the equation given.
16. Solve through options.
17. Factorize the equation and we get $(x-1)(x-4)(x-7)$
19. Take two series and solve.
20. Make the three and solve though option.
21. $T_n = p = a + (n-1)d$
    $T_m = q = a + (m-1)d$
    Add the two and solve through the options given.
22. 1048576 is divisible by 4
    so, previous number is divisible by 3.
23. Sum = $n(n^2 + 1)/2$, where n is the number of terms in S.
25. Solve through options.
26. Solve through options.
27. Put $a = 1$, $b = 2$ $c = 3$ and check through options.
28. Assume the 1st AP as 1, 1, 1,... or 2, 2, 2,... and the 2nd AP as −1, −1, −1 or −2, −2, −2
30. Get the terms of sets $f(x)$ and $g(x)$ and check how many terms are common.
31. Sum = $(n-1)2^n + 1$.
32. The side of the outermost square is 6 cm. Area of the square one in G.P with first term 36 and common ratio = ½
36. Break the series in two sequence 1, 2, 3, 4,... n and 1^2, 2^2, 3^2, ... n^2 and solve
37. Factorize 598 as the product of two prime numbers and an even number.
    $598 = 2 \times 13 \times 23$.
38. Take any value of $r > 1$ and then solve.
39. Make pair of three numbers. Only sequence with 2 and 3 as the common ratio is possible.
40. Take any series for which the given condition is true. One such series can be: 4, 3, 2, 1, 0, ...

Advanced

1. Since a_1, a_2, a_3 are in AP, So $a_2 - a_1 = a_3 - a_2 = d$

$$\left[\frac{1}{a_1a_2} + \frac{1}{a_2a_3} + \ldots \frac{1}{a_{n-1}a_{n-2}}\right]$$

$$= \frac{1}{d}\left[\frac{1}{a_1} - \frac{1}{a_2} + \frac{1}{a_2} - \frac{1}{a_3} + \ldots\right]$$

2. Since x, y, z are in GP so $y/x = z/y$ and also $x + y$, $y + z$, $z + x$ are in AP so $2(y + z) = 2x + y + z$ i.e., $z = 2x - y$.
   So, y/x $(2x - y)/y \Rightarrow y/x = (2x/y) - 1$
   Since y/x is the ratio, assume $y/x = r$
   and solve the equation $r^2 + r - 2 = 0$
3. Go through options.
4. Go through options.
5. The series consist of $2n$ terms,
   first term = a ,common diff = a, no of terms = $2n$

$$\text{Sum of all terms} = \frac{2n}{2}[2a + (2n-1)a] \quad \ldots(\text{i})$$

   for the new series taking 1st, 3rd, 5th, $(2n-1)$th term of old series.
   First term = a, common difference = $2a$, number of terms = n

$$\text{Sum of all terms} = \frac{n}{2}[2a + (2n-1)2a] \quad \ldots(\text{ii})$$

   Dividing (i) by (ii) gives the required ratio.

6. The equation can be written as:

$$\frac{2n}{2}[4+(2n-1)3]=\frac{n}{2}[114+(n-1)2]$$

Solving this equation gives us the value of n.

7. Divide each term by $abcd$ will give $1/d$, $1/c$, $1/b$ and $1/a$ respectively which are in HP (since a, b, c, d are in AP).
Alternatively, assume values of a, b, c and d such that they are in AP.

8. Since the $(m + 1)$th, $(n + 1)$th and $(r + 1)$th term of an A.P. are in G.P. so,

$(a + nd)^2 = (a + md)(a + rd)$ ... (i)

[assume a and d as the first term and common difference of an AP]. Also, m, n, r are in HP so,

$$\frac{2n}{2}=\frac{m+r}{mr} \quad \text{... (ii)}$$

By solving the equation you will get

$\frac{a}{d}=\frac{n^2-mr}{m+r-2n}$, put the $m + r = 2mr/n$ from equation (iii)

You will get $a/d = -n/2 \approx n{:}2$ [–ve sign indicates that either common difference or first term is –ve]

9. Assume any two positive numbers a, b. So, according to the question a, x, b are in AP

So, $x=\frac{a+b}{2}$, also, a, y, z, b are in GP so, $y^2 = az$ i.e., $y^3 = ayz$ and $z^3 = byz$

Now solve $\frac{y^3+z^3}{xyz}$.

10. In a quadratic equation, $ax^2 + bx + c = 0$. If $a + b + c = 0$ then, roots will be rational. Use this property. Also a, b, c, d are in GP i.e., $b/a = c/b = d/c$.

11. Since, $a = 1111 \ldots 1$ (55 digits)

$$b = 1+10+10^2+10^3+10^4 = \frac{1(10^5-1)}{10-1}=\frac{10^5-1}{9} \text{ and}$$

$$c = 1 + 10^5 + 10^{10} + 10^{15} + \ldots 10^{50} = 1\ \frac{\left[(10^5)^{11}-1\right]}{10^5-1}$$

$bc = (10^5 - 1)/9 \times (10^{55} - 1)/(10^5 - 1) = (9999\ldots$ 55 digits $)/9 = a$

13. Assuming two non-negative a and b, then according to the question $\sqrt{ab}=\frac{2ab}{a+b}$.
It is true only if $a = b$

14. Area of the triangle formed by joining the middle point of the triangle will be 1/4th of the original triangle so, the series will be
A+1/4 A + 1/16 A + 1/64 A + ... ∞.

15. The series is like 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 ...
The difference is 715 and the 12th term is 144.

17. $a/(1 - r) = 162$ and $a(1 - r^n)/1 - r = 160$
$\Rightarrow 1 - r^n = 160/162 \Rightarrow r^n = 1/81$
Hence, there will be only two values of r, i.e., 2 and 4.

18. See the hints of the above question.

19. $(\alpha - \beta)\ (\alpha)\ (\alpha - \beta)$ AP
(a, ar, ar^2) GP
$\alpha - \beta = r + a = \beta$
$a + ar = 2\alpha - \beta$
$a + ar = 2\alpha - 2\beta + \beta$
$a(1 + r) = 2(r + a) + r + a$
$a(1 + r) = 3(r + a)$
$a + r = a(1 + r)/3$
$\frac{a}{\alpha-\beta} = ?$
$a/(a + r) = (a.3)/a(1 + r)$
$= 3/1 + r$
This is independent of the first term of GP.

20. $S_5 = x + 2x^2 + 3x^3 + 4x^4 + 5x^5$ and $S_4 = x + 2x^2 + 3x^3 + 4x^4$

27. $a = 0.75$, $d = 1.75$

28. Go through the options.

29 Put values like (3, 6)
Alternatively this question can be done by going through options also.

30. Put $n = 1$ and check the options.

31.

D	H	T	T	H	D	T	H	D
10	5	15	5	10	15	10	5	15

.....
.....
.....
Total = 15

32. Assume that they are getting $a - d$, a and $a + d$ mangoes irrespective of the fact who gets what numbers.
Now, $a - d + a + a + d = 30$, So, $a = 10$
Regarding who gets how many mangoes, we have only one information, i.e., D > H. So, we cannot decide about the first two options.

35. ... = 1 × 12 × 13 = Number of ways = 132.

36. Go through the options.

37. The series is: 12, 3, 3/4, 3/16 ...

38. $N_1 + N_2 + N_3 + \ldots + N_N = 100$ is possible only in two ways:
23 + 24 + 25 + 26 + 27 = 100, or, 9 + 10 + 11 + 12 + 13 + 14 + 15 + 16 = 100

12

CHAPTER

Function



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Functions and its types
- Definitions and properties
- Nature of functions
- Ways of representing a function and questions associated with it
- Methods of solving questions

Introduction to the topic

Function is an important chapter for CAT and other exams. The number of questions being asked from this topic is almost constant. This chapter is almost a nightmare for the non-math students, all due to its technical language and interpretations. But most of the questions asked in the CAT from this area can be solved with the knowledge of the basics of functions, and a bit of logic.

FUNCTION

A function is a rule which associates two or more than two variables. Functions can be understood with the help of some very basic examples:

i. Area of a circle (A) = πr^2, where r is the radius.
So, the area of a circle is dependent upon the value of the radius of a circle. We can write this mathematically as $A = f(r)$

ii. If the distance (d) is constant, then the time taken (t) to cover that distance (d) will be dependent on the value of the speed (v). This can be written mathematically as $t = f(v)$
Normally any function is represented in the format of $y = f(x)$

Domain and range of $y = f(x)$ In case of $y = f(x)$, values of y are dependent upon the corresponding values of x. Here y is known as the dependent variable and x is known as the independent variable.

All the real values of x for which atleast one real value of y exists, are known as the domain of this function $y = f(x)$

All the possible real values of y are known as the range of this function $y = f(x)$.

While finding out the domain, the variable generally takes a range of values unlike the case of equations where the variable will, mostly, take one value or a discrete set of values in interval notation. Hence, it is important to understand the standard notations which are used to represent the solution set of the variable as well as the interval notation.

- (a, b) read as "open interval a, b" means all real numbers between a and b excluding a and b; $a < b$.
- $[a, b]$ read as "closed interval a, b" means all real numbers between a and b including a and b; $a < b$.
- $[a, b)$ means all numbers between a and b, with a being included and b excluded; $a < b$.
- $(a, b]$ means all numbers between a and b, with a being excluded and b included; $a < b$.

Example 1 Represent all real numbers between 1 and 10 in the interval from where

(i) 1 and 10 are included,
(ii) 1 and 10 are excluded
(iii) 1 is included and 10 is excluded and
(iv) 1 is excluded and 10 is included

Solution

(i) $[1, 10]$ i.e., $1 \le x \le 10$ and x is a real number.
(ii) $(1, 10)$ i.e., $1 < x < 10$ and x is a real number.
(iii) $[1, 10)$ i.e., $1 \le x < 10$ and x is a real number.
(iv) $[1, 7]$ i.e., $1 \le x \le 7$ and x is a real number.

Process to find out the domain of $y = f(x)$

1. Denominator should not be equal to zero.
2. Expression under square root should be a non-negative.

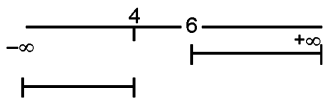
Example 2 Find the domain of the definition of function $y = f(x) = \sqrt{4-x} + \sqrt{x-6}$.

Solution We know that the expression under square root should be non-negative.

So, $4 - x \ge 0$, or, $x \le 4$

And $x - 6 \ge 0$, or, $x \ge 6$.

Arranging these values on the numberline:



So, no value of x is common. Hence the domain of $y = f(x)$ is not defined.

Example 3 The domain of $y = f(x) = \dfrac{1}{\sqrt{|x| - x}}$ is

Solution For $y = f(x)$ to be defined, $|x| - x > 0$, or, $|x| > x$.

It is true only if $x < 0$. So, the domain is $-\infty < x < 0$.

Example 4 Are the following functions identical?

$y = f(x) = x$

$y = h(x) = \dfrac{x^2}{x}$

Solution Domain of $f(x)$ = all the real values

Domain of $h(x)$ = all the real values except at $x = 0$.

Because the domains are not the same for $f(x)$ and $h(x)$, so $f(x)$ and $h(x)$ are not identical.

SOME SPECIAL FUNCTIONS

Modulus Function

$y = |x|$

It is defined as

$y = x$; if $x > 0$

$y = -x$; if $x < 0$

$y = 0$; if $x = 0$

Despite that in the above equations, we are finding a negative value of x if $x < 0$, its absolute value can never be negative. This can be seen through the following example:

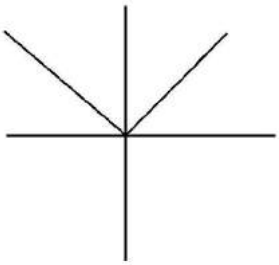
We are finding out the value of $y = |-5|$.

Assuming that $-5 = x$, so $y = |x|$

Now, since $x < 0$, so $y = -x = -(-5) = 5$

Graphical representation of modulus function

This is the graph of $y = |x|$



It is observed that for every value of x, be it +ve or –ve, the value of y cannot be negative.

Example 5 What is the value of x if $|2x + 3| = 9$?

Solution In the questions involving modulus, first the value of expression under the modulus is taken as a positive and then as a negative.

Case 1 When $(2x + 3) > 0$, or, $x > \frac{-3}{2}$, then $|2x + 3|$ $= 2x + 3$

So, $2x + 3 = 9$ or, $2x = 6$.

So, $x = 3$

Case 2 When $(2x + 3) < 0$, or, $x < \frac{-3}{2}$, then $|2x + 3|$ $= -(2x + 3)$

So, $-(2x + 3) = 9$, or, $-2x = 12$

So, $x = -6$

Example 6 What is the value of 'x' if $x^2 + 5|x| + 6 = 0$?

Solution Taking $x > 0$, $x^2 + 5|x| + 6 = x^2 + 5x + 6$ $= (x + 2)(x + 3) = 0$

Or, $x = -2$ and $x = -3$

But as we have assumed that $x > 0$, so $x = -2$ and -3 are not admissible.

Taking $x < 0$, $x^2 + 5|x| + 6 = x^2 - 5x + 6 = (x - 2)$ $(x - 3) = 0$

Or, $x = 2$ and $x = 3$

But as we have assumed that $x < 0$, so $x = 2$ and 3 are not admissible.

So, there is no real value of x which can satisfy this equation.

Alternatively, it can be seen that x^2 and $5|x|$ and 6, are all positive values. So the sum of these three can never be equal to 0, so no real value of x is possible.

Example 7 If $|x^2 - 5x + 6| > x^2 - 5x + 6$, then find the values of x?

Solution If $|N| > N$, then $N < 0$. (It can be understood by assuming the values)

So, $x^2 - 5x + 6 < 0$

Or, $(x - 2)(x - 3) < 0$

So, $2 < x < 3$

Greatest Integer Value Function

$y = [x]$

It is defined as the largest integral value of x which is less than or equal to x.

It is given $y = [3.23]$ and we have to find the greatest integer value of y.

Taking the second part of the definition, i.e., the integer less than or equal to 3.23, we get a set of integers less than or equal to $3.23 \Rightarrow 3, 2, 1, 0, -1, \ldots$ and so on.

The largest integer among all these integers = 3. So the greatest integer value of $[3.23] = 3$

Similarly, if we find the greatest integer value of $y = [-2.76]$, then all the integers less than this value $(-2.76) = \{-3, -4, -5, -6, \ldots\}$.

Now the greatest integer among all these integers given in the above set $= -3$

It can also be seen through the tabular presentation:

x	y
0–1 (excluding at $x = 1$)	0
1–2 (excluding at $x = 2$)	1
2–3 (excluding at $x = 3$)	2

And so on …

Example 8 What is the value of x in the following expression?

$[x]^2 \leq 16$?

Solution $4 \leq [x] \leq 4$

$-4 \leq [x]$, or, $-4 \leq x$

And $[x] \leq 4$, or, $x < 5$.

So, the value of x is: $-4 \leq x < 5$.

Logarithmic Function

$y = \log_a x$

It is known that the value of x has to be positive here. However, y can have negative values.

Since with an increase in the value of x, y always increases; so $y = \log_a x$ is an increasing function.

We will discuss more about this function in the logarithm chapter.

Exponential Function

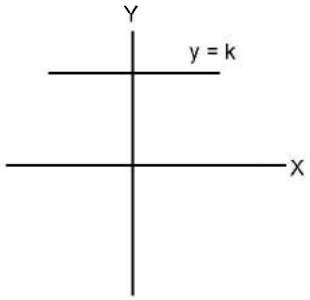
$y = e^x$

Exponential function is the inverse of a logarithmic function.

Again it can be observed that the value of y cannot be negative, whatever be the value of x in $y = e^x$

Constant Function

$f(x)$ = k, (where k is any constant) is known as a constant function.



WAYS OF REPRESENTING FUNCTIONS

There are three standard methods of representing functions:

1. Equation method
2. Tabular method
3. Graphical method

1. Equation method

It is the most commonly used form of writing any function. This form of representation involves a simple equation written in terms of the variable.

Example $y = f(x) = 5x^2 + 8$.

2. Tabular method

In this form of representation a table of the values of x and y is formed, with or without mentioning the equation of functions.

Example A function $y = f(x)$ is represented by the table given below:

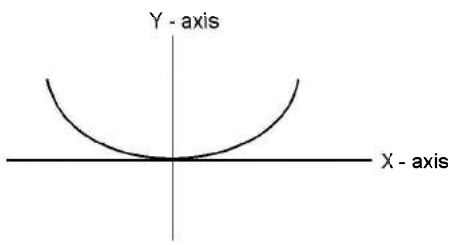
x	1	2	–1	–2
y	1	4	1	4

Obviously, the function which will satisfy the above written table is $y = f(x) = x^2$

3. Graphical method

A graph is considered to be the most-revealing-way of representing a function. Apart from giving an inkling about the function $y = f(x)$, its nature and shape, it gives us the information about the maximum and minimum value of the function also.

For example,



It is quite evident here that for both the negative and the positive values of x, y is having the same value and the shape of the graph is parabolic. So this graph is being represented by the function $y = f(x) = x^2$.

NATURE OF A FUNCTION (IN TERMS OF EQUATION)

On the basis of the symmetric nature of functions, we define the graphs to be of three natures:

Even Function

For $y = f(x)$,

Mathematically, even functions are defined as $y = f(x) = f(-x)$

In layman terms, if we replace 'x' from the given equation by '$-x$', then the equation should be the same.

For example, $y = f(x) = x^2 + 8$.

Here, $f(-x) = (-x)^2 + 8 = x^2 + 8 = f(x)$

So, $y = f(x) = x^2 + 8$ is an even function.

Properties of even function

1. The sum or the difference or the product or the division of any two even functions is an even function.
2. For $y = f(x)$, the graph of the even functions are symmetrical to Y-axis.

Odd Function

For $y = f(x)$,

Mathematically odd functions are defined as

$y = f(x) = -f(-x)$

In layman terms, if we replace 'x' from the given equation by '$-x$', then the equation should be the same with an opposite sign.

For example, $y = f(x) = x^3$.

Here $f(-x) = (-x)^3 = -(x^3) = -f(x)$

So, $y = f(x) = x^3$ is an odd function.

Properties of odd function

1. The sum or the difference of any two odd functions is an odd function.
2. For $y = f(x)$, the graph of the odd functions are symmetrical to the origin.

Neither Odd Nor Even Function

For $y = f(x)$,

If any equation is not satisfying the conditions of either an odd or an even function, then it is said to be neither odd nor even function.

In layman terms, if we replace 'x' from the given equation by '$-x$', then the equation should not be the same, either with the same sign or with an opposite sign.

For example, $y = f(x) = x^3 + 1$

Here, $f(-x) = (-x)^3 + 1 = -x^3 + 1$

It is quite clear now that $f(-x)$ is neither equal to $f(x)$ or $-f(x)$.

So, $y = f(x) = x^3 + 1$ is neither an even function nor an odd function.

Properties of neither odd nor even function

1. The sum of the difference of odd and even functions is neither an odd nor an even function.
2. To understand the mechanism of the sum/difference/product of two functions, we can take an even function as a positive number and an odd function as a negative number. For example, x^2 (Even function) $\times$ x^3 (Odd function) = Odd function
   x^2 (Even function) + x^3 (Odd function) = Neither odd nor even function
   Remember that no function can be both an even and an odd function simultaneously.

Example 9 Find the nature of the following functions:

(a) $y = f(x) = e^{-x}$

(b) $y = f(x) = x^2 + x^4$

(c) $y = f(x) = \frac{1}{x} + x^3$

(d) $y = f(x) = x^2 + x^3$

(e) $y = f(x) = \log x^2$

(f) $y = f(x) = \log_a x$

Solution (a) Neither odd nor even
(b) Even
(c) Odd
(d) Neither odd nor even
(e) Even
(f) Neither odd nor even

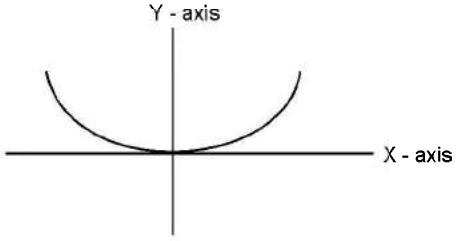
NATURE OF FUNCTION (IN TERMS OF A GRAPH)

For $y = f(x)$

Even Function

The graph of an even function will be symmetrical to Y-axis. It simply means that the graph on the left side of the Y-axis will be the same as the graph on the right side of the Y-axis.

As we have seen in the case of $y = f(x) = x^2$, the graph on the left hand side of the Y-axis is the same in shape and size with respect to the graph on the right hand side of the Y-axis.



Properties of the graph of an even function

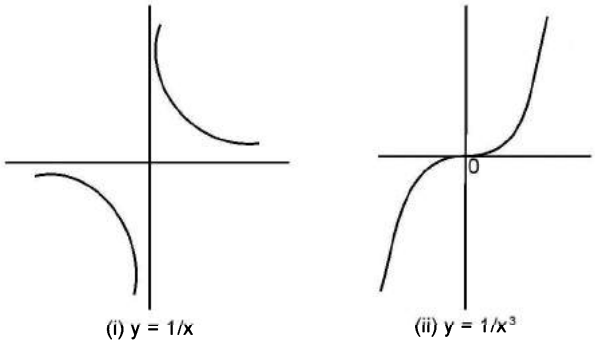
1. Even function graphs are one-fold graphs. i.e., we can make both the parts of the graphs overlap on each other, by just folding the graph paper once.
2. Whenever we fold the paper, we fold it along the axis, either on the X-axis or on the Y-axis.

Odd Function

For $y = f(x)$

The graph of an odd function will be symmetrical to the origin. It simply means that the graph on one side of the origin will be same as the graph on the other side of the origin.

As we have seen in the case of $y = f(x) = 1/x$, the graph on one side of the origin is the same in shape and size with respect to the graph on the other side of the origin.



Properties of the graph of an odd function

1. Odd function graphs are two-fold graphs, i.e., we can make both the parts of the graph overlap on each other, by folding the graph paper twice, once along the X-axis and then along the Y-axis.
2. Whenever we fold the paper to overlap the graph, we fold it along the axis, either on the X-axis or on the Y-axis.

Composite Function

Composite function is the kind of function which is composed into one by two or more than two functions.

For example, if $y = f(x) = 5x + 3$ and $y = g(x) = x^2$ is given, then $f(g(x))$ or $g(f(x))$ is the composite function.

In the above example, $f(g(x)) = 5\,g(x) + 3 = 5\,x^2 + 3$

Similarly, $g(f(x)) = [f(x)]^2 = (5x + 3)^2$

Example 10 If $g(x) = \frac{1}{f(x)}$ and $f(x) = x$, $x \neq 0$ is given. Which of the following is true?

(a) $f(f(f(g(g(f(x)))))) = g(g(f(g(f(x)))))$

(b) $f(f(g(g(g(f(x)))))) = g(g(f(g(f(x)))))$

(c) $f(g(f(g(g(f(g(x))))))) = g(g(f(g(f(x)))))$

(d) $f(f(f(g(g(f(x)))))) = f(f(f(g(f(x)))))$

Solution $f(x) = x$ and $g(x) = 1/x$

Since applying 'f' any number of times is not changing the final result, but applying 'g' is making the value reciprocal. So, LHS can be equal to RHS only if the number of 'g' on both the sides is either even or odd.

So, option (b) is the answer.

Inverse Function

If two functions f and g satisfy $g(f(x)) = x$ for every x in the domain of f and similarly $f(g((x)) = x$ for every x in the domain of g, then f is said to be the inverse of g and vice-versa. It is written as f^{-1} and g^{-1}. For example, $y = \log_e x$ and $y = e^x$ are inverse to each other.

Example 11 $y = f(x) = \dfrac{x+1}{x+2}$. Find $f^{-1}(x)$.

Solution $y = \dfrac{x+1}{x+2}$, or, $y - 1 = \dfrac{x+1}{x+2} - 1 = \dfrac{-1}{x+2}$

Or, $x + 2 = \dfrac{1}{1-y}$, or, $x = \dfrac{1}{1-y} - 2 = \dfrac{2y-1}{1-y}$

$f^{-1}(x) = \dfrac{2x-1}{1-x}$

There lies an important relationship between function and graphs. We will see more application of this in the Chapter: Graphs and Maxima Minima

Example 12 A polynomial $f(x)$ with a real coefficient satisfies the functional equation $f(x).f\left(\dfrac{1}{x}\right) = f(x) + f\left(\dfrac{1}{x}\right)$. If $f(2) = 9$, then $f(4)$ is:

(a) 82 (b) 17
(c) 65 (d) None of these

Solution Assume $f(x) = x^n + 1$ and $f\left(\dfrac{1}{x}\right) = \dfrac{1}{x^n} + 1$

So, $f(x).\ f\left(\dfrac{1}{x}\right) = \left(x^n + 1\right)\left(\dfrac{1}{x^n}\right) + 1$

$= \left(1 + x^n\right) + \left(\dfrac{1}{x^n} + 1\right)$

Now $f(x) = x^n + 1 = 9$

Or, $x^n = 8$, or, $n = 3$

$f(x) = x^3 + 1$

Hence, $f(x) = 4^3 + 1 = 65$

Example 13 Let $f(x)$ be a function such that $f(x + 1) + f(x - 1) = f(x)$ for every real x. Then for what value of p is the relation $f(x + p) = f(x)$ necessarily true for every real x?

(a) 3 (b) 4 (c) 5 (d) 6

Solution $f(x + 1) + f(x - 1) = f(x)$ ...1

Putting $(x + 1)$ at the place of x

or, $f(x + 2) + f(x) = f(x + 1)$ ...2

Adding 1 and 2, we get $f(x + 2) + f(x - 1) = 0$

Similarly, $f(x + 3) + f(x) = 0$

Or, $f(x + 4) + f(x + 1) = 0$

Or, $f(x + 5) + f(x + 2) = 0$

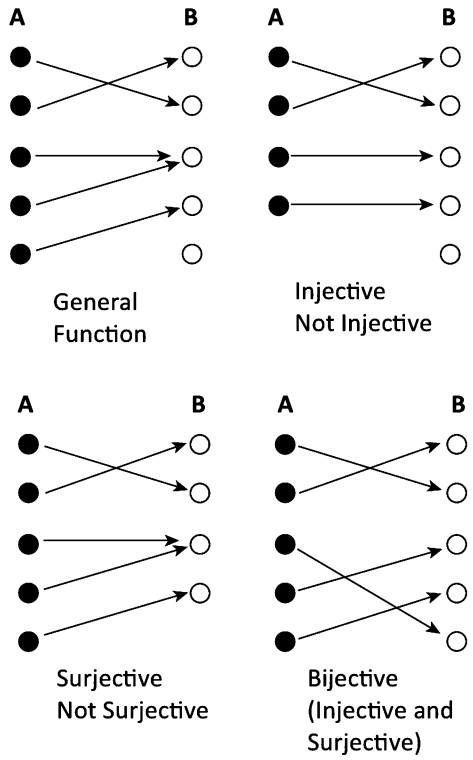
Or, $f(x + 6) + f(x + 3) = 0$

Now replacing $f(x + 3)$ by $-f(x)$, $f(x + 6) + f(x + 3) = f(x + 6) - f(x) = 0$

So, $p = 6$

INJECTIVE, SURJECTIVE AND BIJECTIVE FUNCTIONS

Let us consider a function from A to B and understand the types of function:



(a) General Function – We have already discussed this. Two or more elements of A can have same mapping in B. For example: $y = f(x) = x^2$. In this function, two values of x will point to same value of y. ($x = 2$ or $x = -2$ both will give $y = 4$).

(b) Injective – In this type of functions, each element in A has its unique corresponding mapping in B. B may have some elements without any correspondence in A. For example: $y = f(x) = 3x + 2$

In this case, you won't have two values of A having same mapping in B. So "many to one" is not

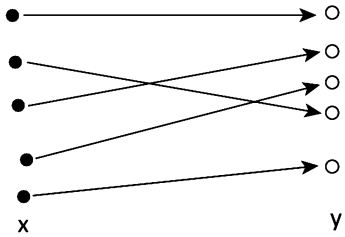
possible. (One to many is obviously not possible because it's a function).

(c) Surjective – In this type of functions, each element in B should be mapped by one or more than one element in A.
There won't be an element in "B" left out.
Surjective Functions are also known as Onto function.

(d) Bijective – It means both Injective and Surjective.
So each element of B will be mapped by a unique element in A.

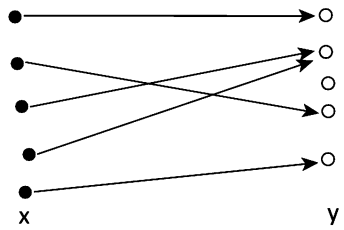
Example 14 Indentify what type of function is this (x to y)?

(a)



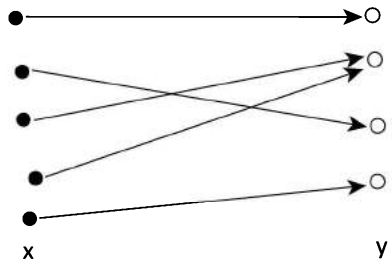
(i) A general function (ii) Injective
(iii) Surjective (iv) Bijective

(b)



(i) A general function
(ii) Injective
(iii) Surjective
(iv) Bijective

(c)



(i) A general function
(ii) Injective
(iii) Surjective
(iv) Bijective

Solution

(a) Bijective is the answer.

(b) A general function is the answer because many to one mapping is there.

(c) It cannot be Injective because two X are pointing to single Y. Since no Y is left out, hence Surjective is the answer.

PRACTICE EXERCISES

WARM UP

Q.1. The domain of the function $f(x) = \sqrt{|x|}$ is
(a) $x \in (-\infty, \infty)$
(b) $x \in (0, \infty)$
(c) $x \in (-\infty, 0) \cup (0, \infty)$
(d) $x \in [0, \infty)$

Q.2. The domain of the function $f(x) = |\sqrt{x}|$ is
(a) $x \in (-\infty, \infty)$
(b) $x \in (0, \infty)$
(c) $x \in [0, \infty)$
(d) None of these

Q.3. The domain of the function $f(x) = \frac{1}{\sqrt{x}}$ is
(a) $-\infty < x < \infty$
(b) $0 < x < \infty$
(c) $-\infty < x < \infty$ excluding $x = 0$
(d) $0 \le x < \infty$

Q.4. The domain of the function $f(x) = \sqrt{x^2}$ is
(a) $-\infty < x < \infty$
(b) $0 < x < \infty$
(c) $-\infty < x < \infty$ excluding $x = 0$
(d) None of these

Q.5. The domain of the function $f(x) = \sqrt{4x-3} + \sqrt{2x-6}$ is

(a) $[0, \infty)$ (b) $\left[\frac{3}{4}, \infty\right]$

(c) $\left[\frac{4}{3}, \infty\right]$ (d) $[3, \infty)$

Q.6. The domain of the function $f(x) = \log(5x - 6 - x^2)$ is

(a) (5, 6) (b) (2, 3)

(c) $[2, \infty)$ (d) None of these

Q.7. The domain of $y = \log_x 5$ is

(a) $x > 5$

(b) $(0, \infty)$

(c) $(0, \infty)$ excluding $x = 1$

(d) $[5, \infty)$

Q.8. If $f(x) = \frac{x}{\sqrt{1+x^2}}$, then $f(f(f(x)))$ is

(a) $\frac{3}{\sqrt{1+x^2}}$ (b) $\frac{x}{\sqrt{1+3x^2}}$

(c) $\frac{x^3}{\sqrt{1+x^2}}$ (d) $\frac{x}{\sqrt{3+3x^2}}$

Q.9. Let $f(x) = \frac{x}{x+3}$, then $f(x + 1) =$

(a) $\frac{3x+2}{x+2}$ (b) $\frac{x+1}{x+4}$ (c) $\frac{x+1}{x+3}$ (d) $\frac{2x+3}{x+3}$

Q.10. If $f(x) = \frac{x+1}{x-1}$, $x \neq 1$, find $f(f(f(f(f(2)))))$.

(a) 1 (b) 2 (c) 3 (d) 4

Q.11. Let $(x) = \sqrt{x^5}$, then $f(5x)$ is equal to

(a) $5x\sqrt{5x^5}$ (b) $25x\sqrt{5x^5}$

(c) $5x$ (d) None of these

Q.12. If $f(x) = 4x - 5$, $g(x) = x^2$ and $h(x) = \frac{1}{x}$, then $f(g(h(x)))$ is

(a) $\frac{4}{x-5}$ (b) $\frac{1}{(4x-5)^2}$

(c) $\left(\frac{x}{4} - 5\right)$ (d) None of these

Q.13. $f(x) = \text{In}\left(x + \sqrt{x^2+1}\right)$ is

(a) An even function

(b) An odd function

(c) Neither even nor odd

(d) None of these

Q.14. Which of the following is an even function?

(a) $\text{In}\left(\frac{1+x^3}{1-x^3}\right)$ (b) $x\,\frac{e^{ax}-1}{e^{ax}+1}$

(c) $\frac{(1+2^x)^2}{2^x}$ (d) None of these

Q.15. Which of the following function is an even function?

(a) $f(x) = \log\frac{1-x}{1+x}$

(b) $f(x) = x\,\frac{a^x+1}{a^x-1}$

(c) $f(x) = \log\left[x + \sqrt{(1+x^2)}\right]$

(d) $f(x) = \sqrt{(1+x+x^2)} - \sqrt{(1-x+x^2)}$

Q.16. The function $f(x) = \frac{x}{e^x-1} + \frac{x}{2} + 1$ is

(a) Even

(b) Odd

(c) Neither even nor odd

(d) Cannot be determined

Q.17. Given $f(x) = \frac{1}{(1-x)}$, $g(x) = f(f(x))$ and $h(x) = f(f(f(x)))$, then the value of $f(x).\ g(x).\ h(x)$ is:

(a) 0 (b) −1 (c) −2 (d) 2

Q.18. If $f(x) = x^2 + 2$ then $f^{-1}(x)$ is

(a) x^5 (b) $\sqrt{x+2}$

(c) $\sqrt{x} + 2$ (d) None of these

Q.19. If $f(x) = \frac{x}{x+1}$, then $f^{-1}(x)$ is

(a) $\frac{x+1}{x}$ (b) $\frac{x}{1-x}$

(c) $\frac{x-1}{x}$ (d) None of these

Q.20. If $f(x) = \frac{x-1}{x+1}$, then $f^{-1}(x)$ is

(a) $\frac{x+1}{1-x}$ (b) $\frac{x+1}{1-x}$

(c) $\frac{1}{x+1}$ (d) None of these

FOUNDATION

Q.1. Given is $f(x) = x^2 + 4x + 4$ and $g(x) = x^2 + 4x + 3$, then find the values of x such that $f(g(x)) = g(f(x))$.
(a) $x = -1$ (b) $x = -2$
(c) $x = -3$ (d) Not possible

Q.2. For $y = f(x)$
Even functions are defined as $f(-x) = f(x)$ and
Odd functions are defined as $f(-x) = -f(x)$ and
$f(x) = \log_{10}\left(x + \sqrt{x^2+1}\right)$ is
(a) An odd function
(b) An even function
(c) Both even and odd function
(d) Neither odd nor even function

Q.3. A function $f(x)$ is such that $f(x) + f(y) = f(xy)$. Which of the following could be $f(x)$?
(a) a^x (b) $\sqrt{x}$
(c) x^2 (d) $\log ax$

Q.4. $f(x) = 3x^2 - 2x + 4$ and $g(x) = x + 1$. Find $f(g(f(1)))$.
(a) 69 (b) 100
(c) 96 (d) None of these

Q.5. If $f(x) = x^2$ and $f[g(x)] = g[f(x)]$, then which of the following can never be equal to $g(x)$?
(a) $\sqrt{x}$ (b) x^2 (c) x (d) $2x$

Q.6. Let $f(x) = |x - 1|$, then
(a) $f(x^2) = [f(x)]^2$ (b) $f(x+y) = f(x) + f(y)$
(c) $f(|x|) = |f(x)|$ (d) None of these

Direction for questions 7 to 9: Read the passage below and solve the questions based on it.

For any $y = f(x)$
If $f(-x) = f(x)$, then this function is known as an even function.
And if $f(-x) = -f(x)$, then this function is known as an odd function.

Q.7. $f(x) = |x|^3$ is
(a) An odd function
(b) An even function
(c) Neither odd nor even function
(d) Cannot be determined

Q.8. The sum of the two odd functions is always
(a) An odd function
(b) An even function
(c) Neither odd nor even function
(d) Nothing can be said

Q.9. The sum of an odd function and an even function is always
(a) An odd function
(b) An even function
(c) Neither odd nor even function
(d) Nothing can be said

Q.10. $f(x) = c.x + 1$ and $g(x) = 3x + 2$. If $f(g(x)) = g(f(x))$, then what is the value of c?
(a) 1 (b) 2 (c) 3 (d) 4

Q.11. If $f(x) = 2^x - 2^{-x}$, then the value of $2f(x) - 5f(x-1) + 2f(x-2)$ is
(a) 1 (b) −3
(c) 15 (d) None of these

Direction for questions 12 to 14: Read the passage below and solve the questions based on it.

If $(x, y) = x^{y^2}$
$g(x, y) = y^{y^2}$
$h(x, y)$ = Remainder when x is divided by y.
$i(x, y)$ = Quotient when x is divided by y.

Q.12. The value of g[h {f {i (16, 3), 2}, 7}, 3] is
(a) 16 (b) 25 (c) 49 (d) 81

Q.13. The value of $\frac{h[g\{i(76,28),2\},3]}{h[f\{i(16,3),2\},7]}$ is
(a) 1 (b) $\frac{1}{4}$ (c) $\frac{1}{3}$ (d) 1/2

Q.14. The value of f[h {g (2,2), 3}, 2] −g[h[f (5,2), 7}, 3] is
(a) −100 (b) −90 (c) −85 (d) −80

Q.15. If $(x) = 2x + 3$ then the value of $f[f\{f(x)\}]$ is
(a) $-6x + 7$ (b) $7x + 8$
(c) $8x + 21$ (d) $9x + 10$

Q.16. If $f(x) = e^x + e^{-x}$, then $f(x)$ is
(a) An odd function
(b) An even function
(c) Neither odd nor even
(d) None of these

Q.17. If $f(x) = \frac{x^2 - 2x}{1+x}$, then the value of $f\left(\frac{1}{-x}\right)$ is
(a) $\frac{x^2-2}{1+2x}$ (b) $\frac{1-2x^2}{1+x}$
(c) $\frac{1+2x}{x(x-1)}$ (d) None of these

Q.18. If one root of the equation $x^2 - 3ax + f(a) = 0$ is double of the other, then which of the following is equal to $f(x)$?

(a) x^2 (b) $2x$
(c) $2x^2$ (d) $\frac{x^2}{2}$

Q.19. If $f(x, y) = 3x^2-2xy-y^2 + 4$, then $f(f(2, 3), f(-1, 1) = ?$
(a) −68 (b) 95 (c) 251 (d) 232

Q.20. If $y = f(x) = \frac{x+2}{x-1}$, then it is incorrect to say
(a) $x = \frac{y+2}{y-1}$ (b) $f(0) = -2$
(c) $f(1) = 0$ (d) $f(-2) = 0$

Direction for questions 21 and 22: Read the passage below and solve the questions based on it.

A function $f(x)$ is defined as $f(x) = \log (g(x))$ where $g(x)$ is any function of x. Then

Q.21. For which of the values of $g(x)$, can $f(x)$ be expressed as $f1(x) + f2(x)$, where $f1(x)$ and $f2(x)$ are any two functions of x?
(a) $g(x) = e^x$ (b) $g(x) = x^2$
(c) $g(x) = \log (x)$ (d) None of these

Q.22. For what value of $g(x)$, $g(x) = g(f(x))$?
(a) e (b) $\log x$ (c) e^x (d) None of these

Q.23. For what value of $g(x)$ is the function $f(x) = g(x)$?
(a) e^{-1} (b) $\log x^2$
(c) e^{2x} (d) None of these

Q.24. If $f(t) = \frac{t^3+2t}{t-1}$, then $\frac{f(t) + f(-t)}{f(t)-f(-t)}$ is equal to
(a) t^3 (b) $\frac{1}{t}$ (c) t (d) t^2

Q.25. If $f(x) = 2x^2 + 6x-1$ and $g(x) = |x + 5|$ then the value of $f[g\{f(g(-6))\}]$ is
(a) 225 (b) 249 (c) 341 (d) 359

Q.26. If $f(x) = x^3 + 1$ and the value of the expression $f(2x) -2 f(x) = 383$, the value of x is
(a) 1 (b) 0 (c) (d) (d) 6

Direction for questions 27 and 28: Read the passage below and solve the questions based on it.

$F(x) = 0$ when $x = 1$
$= 1$ when $x = 0$
$= 0.5$ when $x = 0.5$
$g(x) = 1$ when $x = 1$
$= 0.5$ when $x = 0$
$= 0$ when $x = 0.5$

Q.27. If $f[g[f[g(t)]]]] = 0$, the value of t is
(a) 1 (b) 0
(c) 0.5 (d) None of these

Q.28. If $f[g(t) + f(t)] + g[g(t)] = 1$, the value of t is
(a) 1 (b) 0
(c) 0.5 (d) None of these

Q.29. If $f(x) = x^2$ and $h(x) = \sqrt{x}$, then
i. h of (−2) = 4
ii. h of (4) = 4
iii. h of (3) = 6
(a) i is correct but other two are incorrect
(b) ii is correct but other two are incorrect
(c) i and iii are correct but ii is incorrect
(d) None of the above is true

Q.30. Let $h(x) = \frac{1-x}{1+x}$, find the domain of $h^{-1}(x)$.
i. $(-\infty, +\infty)$ ii. $(-\infty, -1)$
iii. $(1, +\infty)$ iv. $[-1, +1]$
(a) i alone is correct
(b) ii ∪ iii are correct, where ∪ represents union of the sets.
(c) ii ∪ iii ∪ iv are correct, where ∪ represents union of the sets.
(d) None of these

Q.31. If $f(n) = 2n^2 - n$ for $n > 5$, $= n^2 + n$ for n 5, then $f(1) + f(2) + \ldots + f(10)$ equals
(a) 705 (b) 440 (c) 690 (d) 715

Q.32. If $y = f(x) = (ax + b)/(cx - a)$, then which of the following is equal to $f(y)$?
(a) x (b) $2x$ (c) $x/2$ (d) x^2

Q.33. If $y = f(x) = (x + 2)/(x -1)$, then which of the following is equal to $f(y)$?
(a) x (b) $2x$ (c) $x/2$ (d) x^2

Q.34. If $f(x) = \frac{x}{x-1}$, then find the value of $\frac{f\left(\frac{a}{b}\right)}{f\left(\frac{b}{a}\right)}$.
(a) a/b (b) $-b/a$ (c) $-a/b$ (d) b/a

Q.35. Find the domain of the function $y = f(x)$, which is defined as $f(x) \frac{1}{\sqrt{|x|+x}}$.
(a) $(-\infty, +\infty)$ (b) $(0,+\infty)$
(c) $[0, +\infty)$ (d) $[1, +\infty)$

Q.36. Find the domain of the function $y = f(x)$, which is defined as $f(x) = \frac{1}{\sqrt{\{x-[x]\}}}$ $[x]$ is the greatest integer function.
(a) x is any real number other than integers
(b) Any real value of x
(c) All the natural numbers
(d) None of these

Q.37. Are these two functions identical?
$f(x) = \log (x-2) + \log (x-3)$
and $g(x) = \log (x-2) (x-3)$

(a) Yes
(b) No
(c) Cannot be determined
(d) None of these

Q.38. Are these two functions identical?
$f(x) = 2\log x$ and $g(x) = \log x^2$.
(a) Yes
(b) No
(c) Cannot be determined
(d) None of these

Q.39. The domain of the function $f(x) = \log_{10}\log_{10}(1+x^3)$ is
(a) $(-1,+\infty)$ (b) $(0,+\infty)$
(c) $(0,+\infty)$ (d) $(-1, 0)$

Q.40. The set of real values of x satisfying $|x-1| \le 3$ and $|x-1| \ge 1$ is
(a) [2, 4]
(b) $(-\infty, 2) \cup [4, +\infty]$
(c) $[-2, 0] \cup [2, 4]$
(d) None of these

MODERATE

Q.1. If $b = f(a)$ and $f(a) = \dfrac{a-1}{a+1}$ which of the following is true?
(a) $f(2a) = f(a) + 1$
(b) $f\left(\dfrac{1}{a}\right) = -f(a)$
(c) $a = f(b) + f\left(\dfrac{1}{a}\right)$
(d) $a = f(b)$

Q.2. Let $f(x)$ be defined on $[-1, 1]$ in such a way that the area of the equilateral D with two of its vertices at (0, 0) and $\{x, f(x)\}$ is $\dfrac{\sqrt{3}}{4}$ sq. units. Which of the following can be $f(x)$?

1. $\sqrt{1-x^2}$ **2.** $-\sqrt{1-x^2}$

(a) 1 is correct but 2 is incorrect
(b) 2 is correct but 1 is incorrect
(c) Both 1 and 2 are correct
(d) Neither 1 nor 2 is correct

Direction for questions 3 and 4: Read the passage below and solve the questions based on it.

$[x]$ = Greatest integer less than or equal to x.
$\{x\}$ = Smallest integer more than or equal to x.

Q.3. Given that 'x' is not an integer, then what is the value of $[x] - \{x\}$?
(a) -1
(b) 1
(c) 0
(d) Depends upon the value of x

Q.4. Given that 'x' is not an integer, then $[x] + \{x\}$ is
(a) An even integer (b) An odd integer
(c) Positive number (d) Non positive integer

Q.5. Let $f(n) = \left[\dfrac{1}{2} + \dfrac{n}{100}\right]$ where $[x]$ denotes the integral part of x. Then the value of $\sum_{n=1}^{100} f(n)$ is
(a) 50 (b) 51
(c) 1 (d) None of these

Q.6. If $0 < x < 1000$ and $\left[\dfrac{x}{2}\right] + \left[\dfrac{x}{3}\right] + \left[\dfrac{x}{5}\right] = \dfrac{31}{30}x$, where $[x]$ is the greatest integer less than or equal to x, the number of possible values of x is
(a) 34 (b) 32
(c) 33 (d) None of these

Q.7. If $f(x) = 1/x, x \neq 0; f^{n+1}(x) = f^n(f(x))$, find the value of the product $f^{13}(1) \times f^{15}(1) \times f^{19}(1) f^{21}(1)$.
(a) 34 (b) 16 (c) 1 (d) 17

Q.8. The set of real values of x satisfying $||x-1|-1| \le 1$ is:
(a) $[-1, 3]$ (b) $[0, 2]$
(c) $[-1, 1]$ (d) None of these

Q.9. A function is defined as $f^n(x) = f\{f^{n-k}(x)\}$
If n is odd, k = 1
If n is even, k = 2 and $f^0(x) = 1$.
Find $f^7(x)$ at $x = 1$, if $f(x) = x^2 + 1$.
(a) 676 (b) 677 (c) 577 (d) 1076

Q.10. $(x^2 - 1)$ is a factor of $f(x) = (x^5 + ax^4 + bx^3 + cx^2 + x + d)$. The graph of $f(x)$ intersects Y axis at $(0, -3)$. Find the value of $(a + c)$.
(a) 0 (b) 3 (c) -3 (d) -1

Q.11. Let $f(x) = ax^2 + bx + c$, where a, b and c are rational, and $f: z \to z$, where z is the set of integers. Which of the following best describes the value of $a + b$?
(a) A negative integer
(b) An integer
(c) Non integral rational number
(d) None of these

Q.12. If $f(x+1)+f(x-1)=2f(x)$ and $f(0)=0$, then $f(n)$, $n \in N$ is
(a) $nf(1)$ (b) $\{f(1)\}^n$
(c) 0 (d) None of these

Q.13. Let $f(x)=1+|x|$, $x<-1$
$[x]$, $x \geq -1$, where [.] denotes the greatest integer function. Then
$f\{f(-2.3)\}$ is equal to
(a) 4 (b) 2 (c) −3 (d) 3

Q.14. The largest set of real values of x for which $f(x) = \sqrt{(x+2)(5-x)} - \frac{1}{\sqrt{x^2-4}}$ is a real function is
(a) $[1, 2] \cup [2, 5]$ (b) $[2, 5]$
(c) $[3, 4]$ (d) None of these

Q.15. Let a, b and c be fixed positive real numbers. Let $f(x) = \frac{ax}{b+cx}$ for $x \geq 1$. Then as x increases,
(a) $f(x)$ increases
(b) $f(x)$ decreases
(c) $f(x)$ increases first and then decreases
(d) None of these

Q.16. For any real number x, let (x) denote the largest integer less than or equal to x and $\langle x \rangle = x - [x]$, that is, the fractional part of x. For arbitrary real numbers x, y and z, only one of the following statements is correct. Which one is it?
(a) $[x+y+z] = [x] + [y] + [z]$
(b) $[x+y+z] = [x+y] + [z] = [x] + [y+z] + [y]$
(c) $\langle x+y+z \rangle = y + z - [z+z] + \langle x \rangle$
(d) $[x+y+z] = [x+y] + [z + \langle y+x \rangle]$

Q.17. The domain of the function $f(x) = {}^{16-x}C_{2x-1} + {}^{20-3x}P_{4x-5}$, where the symbols have their usual meanings, is the set
(a) [1, 2, 3, 4, 5] (b) {2, 3, 4}
(c) {2, 3} (d) None of these

Q.18. The domain of the real valued function $f(x) = \log_e |\log_e x|$ is
(a) $(1, +\infty)$ (b) $(0, +\infty)$
(c) $(e, +\infty)$ (d) None of these

Q.19. The domain of the function $f(x) = \log_e(x-[x])$, where [.] denotes the greatest integer function, is
(a) R (b) R−Z
(c) $(0, +\infty)$ (d) None of these

Q.20. Let $f(x) = \log_{x^2} 25$ and $g(x) = \log_x 5$, then $f(x) = g(x)$ holds for x belonging to
(a) All the real values
(b) $(0, 1) \cup (2, +\infty)$
(c) ϕ
(d) None of these

Q.21. Let $f(x) = |x-2|+|x-3|+|x-4|$ and $g(x) = f(x+1)$. Then
(a) $g(x)$ is an even function
(b) $g(x)$ is an odd function
(c) $g(x)$ is neither even nor odd
(d) None of these

Q.22. If $f(x) = x^n$, $n \in N$ and g of $(x) = n\,g(x)$, then $g(x)$ can be
(a) $n|x|$ (b) $3\sqrt[3]{x}$ (c) e^x (d) $\log |x|$

Q.23. The inverse function of the function $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ is
(a) $\frac{1}{2}\log\frac{1+x}{1-x}$ (b) $\frac{1}{2}\log\frac{2+x}{2-x}$
(c) $\frac{1}{2}\log\frac{1-x}{1+x}$ (d) None of these

Q.24. The graph of the function $y=f(x)$ is symmetrical about the line $x = 2$. Then
(a) $f(x+2) = f(x-2)$
(b) $f(2+x) = f(2-x)$
(c) $f(x) = f(-x)$
(d) None of these

Q.25. If $f(x) = \log\frac{(1+x)}{1-x}$ and $g(x) = \frac{3x+x^3}{1+3x^2}$, then $f[g(x)] =$
(a) $[f(x)]^3$ (b) $-f(x)$
(c) $[f^3(x)]$ (d) $3[f(x)]$

Q.26. If $f(x) = 2x^2 + 3x + 4$ and $g(x) = 5 - x^2$, the min $[f(x)]$ − max $[g(x)]$ is
(a) 0 (b) 5 (c) $\frac{17}{8}$ (d) $-\frac{17}{8}$

Q.27. If $f(x) = x^3 - x^2 + x - f(x)$, where x is a whole number less than or equal to 15. For how many values of x, $f(x)$ will not be a whole number?
(a) 7 (b) 8
(c) 5 (d) None of these

Q.28. For any real number x, define Mag (x) = Maximum $(x, -x)$ and Nag (x)=Minimum $(x, -x)$. What is the value of Mag (8, (Nag (4, (Nag (7,−7))))) − (Mag (6,(Nag (7,−7)))?
(a) 7 (b) 0 (c) 2 (d) − 14

Q.29. If $f(x) = \frac{4^x}{4^x+2}$; find the value of
$f\left(\frac{1}{1999}\right) + f\left(\frac{2}{1999}\right) + \ldots + f\left(\frac{1998}{1999}\right)$.
(a) 1998 (b) 1999 (c) 998 (d) 999

Q.30. If $f(x) = 2x^2+6x-1$, then the value of $\frac{f\left(\frac{3}{4}\right)+1}{f\left(\frac{3}{4}\right)-1}$ is

(a) 11/13 (b) 35/3
(c) 45/29 (d) None of these

Q.31. Let $\{x\}$ and $[x]$ denote the fractional and integral part of a real number x respectively. Solve $4\{x\} = x + [x]$.

(a) 2/3 (b) 4/7 (c) 8/3 (d) 5/3

Direction for questions 32 and 33: Read the passage below and solve the questions based on it.

Consider a function $f(n)$ defined for non-negative integer values of n.

$f(n) = (-1)^{S_{n+1}} + (-1)^{S_n} + f(n-1)$, where $f(0) = 1$ and $si = 1 + 2 + 3 + 4 + \ldots\ldots + i$, for all $i \in N$.

Q.32. Find $f(111)$.

(a) 1 (b) −1 (c) 2 (d) 0

Q.33. Evaluate $f(1) + f(2) + f(3) + \ldots f(25)$.

(a) −26 (b) −24
(c) −22 (d) None of these

Q.34. Let f be the greatest integer function and g be the modulus function then what is the value of $(gof)\left(-\frac{5}{3}\right)+(fog)\left(-\frac{5}{3}\right)$?

(a) 3/5 (b) 11/3 (c) 2/7 (d) 3

Q.35. If $f(x) = x^2 + x + 7$ then find $\frac{f(\alpha)+f(\beta)}{\alpha-\beta}$, $\alpha \neq \beta$.

(a) $\alpha^2 + \beta^2 - 2$ (b) $\alpha + \beta + 1$
(c) $\alpha + \beta + \beta^2$ (d) $\alpha + \beta - 1$

Q.36. Let $y = f(x) = \log_a x$ and $a > 1$. Then only one of the following statements is false. Which one is it?

(a) If $x = 1$, then $y = 0$
(b) If $x < 1$, then $y < 0$
(c) If $x = ½$, then $y = ½$
(d) If $x = 1$, then $y = 1$

Q.37. If $(x) = \log \frac{1+x}{1-x}$, then $f(x) + f(y)$ is

(a) $f(x+y)$ (b) $f\left(\frac{x+y}{1+xy}\right)$
(c) $(x+y)f\left(\frac{1}{1+xy}\right)$ (d) None of these

Q.38. If $f(x, y) = |x+y|$ and $g(x, y) = |x-y|$, how many ordered pairs of the form (x, y) would satisfy $f(x, y) = g(x, y)$?

(a) 1 (b) 2
(c) 4 (d) Infinitely many

Q.39. $f(x) = y + \frac{1}{y}$, where $y > 0$. If y increases in value, then $f(x)$

(a) Fluctuates up and down in value
(b) Decreases in values
(c) Increases in value but at a much higher rate than y
(d) Increases in value and tends towards y

Q.40. A function H is defined for all the positive integers that satisfy the following condition:

$H(1) + H(2) + H(3) + \ldots + H(x) = x^2 H(x)$

If $H(1) = 2006$, then find the value of H(2005).

(a) $\frac{2}{2005}$ (b) $\frac{2}{2006}$
(c) $\frac{2}{2006!}$ (d) $\frac{2}{2005!}$

ADVANCED

Q.1. If $a<b<c<d<e$ and $f(x) = (x-a)^2 (x-b) (x-c) (x-d) (x-e)$ which of the following is true?

(a) $f(x)>0$, for $(x<a)$ as well as for $(d<x<e)$
(b) $f(x)<0$, for $(a<x<b)$ as well as for $(d<x<e)$
(c) $f(x)<0$, for $(b<x<c)$ as well as for $(d<x<e)$
(d) None of these

Q.2. A function $f(x)$ is defined on the interval $0 \le x \le 2$ as $f(x)$ = integer closest to x, if $x \neq 0.5$ or 1.5. $f(0.5) = 1$ $f(1.5) = 2$

Find the area under the graph of this function above the x–axis and the lines $x = 0$ and $x = 2$.

(a) 2.5 sq. units (b) 1.5 sq. units
(c) 4 sq. units (d) None of these

Q.3. $[x]$ is defined as the greatest integer less than or equal to x and $\{x\}$ is defined as the least integer greater than or equal to x, for all real values of x. Consider the four statements.

1. $[x] + [-x] = -1$, if x is not an integer.
2. $\{x\} + [-x] = 0$
3. $\{[x]\} - [\{x\}] = -1$
4. $[2x] + \{3x\} \le 5x$

How many of the above statements are always true for all the real values of x?
(a) 1 (b) 2 (c) 3 (d) 4

Q.4. If $f(x) = \frac{x-1}{x+1}$ then $f(ax)$ in terms of $f(x)$ is equal to
(a) $\frac{f(x)+a}{1+af(x)}$ (b) $\frac{(a-1)f(x)+a+1}{(a+1)f(x)+a-1}$
(c) $\frac{(a+1)f(x)+a-1}{(a-1)f(x)+a+1}$ (d) None of these

Q.5. Let $f(1) = 1$ and $f(n) = \sum_{r=1}^{n-1} f(r)$. Then $\sum_{n=1}^{m} f(n)$ is equal to
(a) 3^{m-1} (b) 3 m
(c) 3^{m-1} (d) None of these

Direction for questions 6 and 7: Read the passage below and solve the question based on it.

A formula is given by $a(n)=a(a(n-1))+a(n-a(n-1))$, where n is any integer greater than 2.

Q.6. If it is known that $a(1) = 1$ and $a(2) = 1$, find the value of $a(4)$.
(a) 1 (b) 2 (c) 3 (d) 4

Q.7. If it is known that $a(2) = 1$ and $a(3) = 2$, find the value of $a(1)$.
(a) -1 (b) 0
(c) 1 (d) Cannot be determined

Q.8. Which of the following functions satisfies the condition $f\left(\frac{x+y}{x-y}\right) = \frac{f(x)+f(y)}{f(x)-f(y)}$?
(a) $f(x) = x$ (b) $f(x) = ax + b$
(c) $f(x) = 2^x$ (d) $f(x) = x^2$

Q.9. [N] denotes the largest integer less than or equal to N. V is a positive integer and $\left[\frac{V}{5}\right] - \left[\frac{V}{7}\right] = 1$. If the minimum value of V is r and the maximum value of V is s, then the value of r + s is equal to
(a) 40 (b) 33
(c) 35 (d) 34

Q.10. $f(x) = |x| + |y|$
$g(x) = \max(x + y, x - y)$
$h(x) = \min(x + y, x - y)$
i. $g(x) \ge f(x)$
ii. $g(x) + h(x) \ge f(x)$
iii. $g(x) > f(x)$
Which of the following are not necessarily true?
(a) i and ii (b) i and ii
(c) ii and iii (d) i, ii and iii

Direction for questions 11 and 12: Read the instructions given below and solve the questions based on it.

Following are the certain functions:
H(a, b, c) = Greatest common divisor of a, b and c
L(a, b, c) = Least common multiple of a, b and c
A(a, b, c) = Average of a, b, and c
Min(a, b, c) = Smallest value among a, b and c
Max(a, b, c) = Largest among a, b and c.

Q.11. Which of the following is true? (a, b and c are distinctly different positive numbers)
(a) H(a, b, c)xL(a, b, c) = abc
(b) H(a, b, c)>A(a, b, c)
(c) H(a, b, c)>Min(a, b, c)
(d) H(a, b, c)<A(a, b, c)<L(a, b, c)

Q.12. If Max (a, b, c) = Min(a, b, c), then
(a) A(a, b, c) = H(a, b, c)
(b) A(a, b, c) = L(a, b, c)
(c) A(a, b, c) = Min(a, b, c)
(d) All of these

Q.13. If the area of a circle increases at a uniform rate then the rate of increase of the perimeter is
(a) Directly proportional to the radius
(b) Inversely proportional to the radius
(c) Directly proportional to the square of the radius
(d) Inversely proportional to the square of the radius

Q.14. If $f(x) = 3^x + 4^x + 5^x - 6^x$, then $f(x) < f(3)$ for
(a) Only one value of x
(b) No value of x
(c) Only two values of x
(d) Infinitely many values of x

Direction for questions 15 and 16: Read the passage below and solve the questions based on it.

For a natural number x, if x has 2 or more than 2 digits, then $f(x)$ = the sum of all digits of x and $f(x) = x$, when x has only one digit. Also given that $f^n(x) = f\{f^{n-1}(x)\}$, where $n \ge 2$.

Q.15. If $x = 123456^{654321}$, then $f^{\infty}(x)$ is equal to
(a) 9 (b) 3 (c) 1 (d) 7

Q.16. If $x = 7^{183}$, then $f^{\infty}(x)$ is equal to
(a) 7 (b) 4 (c) 1 (d) 9

Direction for questions 17 and 18: Read the passage below and solve the questions based on it.

Following is a standard example of iterative function.

STEP 1	X = 0
STEP 2	Y = 2
STEP 3	If X < Y, X = X * Y + I
	Else if X > Y, Y = Y * Y
	Else stop
STEP 4	Z = X * Y + X
STEP 5	If Z = 15 stop
	Else Go To STEP 3

Q.17. How many times (iteration) will the process run?
(a) Once (b) Twice
(c) Thrice (d) None of these

Q.18. What will happen to the process if we make the following change in STEP 2?
Y = 1.
(a) Process will run ten times
(b) Process will run thrice
(c) Process will run infinite times
(d) Process will run once

Direction for questions 19 and 20: Read the passage below and solve the questions based on it.

A function f is defined for $x, y \in$ such that
$f(0, y) = y^2 - 1$
$f(x + 1, y) = f(x, y + 4)$

Q.19. Find the value of $f(7, 3)$.
(a) 728 (b) 840
(c) 960 (d) Cannot be determined

Q.20. Which of the following is true?
(a) $f(8, 4) = f(0, 32)$ (b) $f(0, 0) = f(-5, 20)$
(c) $f(-2, 9) = f(3, 29)$ (d) All of these

Direction for questions 21 and 22: Read the passage below and solve the questions based on it.

V(x) denotes the absolute value of x, $mn(x, y)$ denotes the minimum of x and y, $mx(x,y)$ denotes the maximum of x and y.

Q.21. Find mn [V (mx (x, y)), mx (z, V (mn(x, z)))]; for $x = -5$, $y = 3$, $z = -2$.
(a) 2 (b) 3 (c) 1 (d) 4

Q.22. When would $mn[x, mx(x, z)]$ be equal to $mx[z, mn(x, z)]$?
(a) $x = 3z$ (b) $x = \frac{1}{2}z$
(c) $x = -z$ (d) $x = z$

Q.23. The following computer programme is a very standard example of Iterative function:
A computer performs a certain algorithm to compute an output S for an input N, with both S and N being real values. The algorithm is as follows.
Step 1 : S = 0.
Step 2 : If (N ≤ −1 or 1 ≤ N), then S = S + 1 else go to step 5.
Step 3 : N = N ÷ 10
Step 4 : Go to step 2.
Step 5 : Print S.
Which of the following best describes the algorithm?
(a) It calculates the highest power of 10 that can perfectly divide N
(b) It calculates the number of digits in N
(c) It calculates the number of zeros on the extreme right of N
(d) It calculates the number of digits to the left of the decimal point in N

Q.24. If A = {$x \mid f(x) = 0$} and B = {$x \mid g(x) = 0$}, then A ∩ B will be the set of roots of the equation
(a) $[f(x)]^2 + [g(x)]^2 = 0$
(b) $[f(x)] + [g(x)] = 0$
(c) $[f(x)]^2 - [g(x)]^2 = 0$
(d) $[f(x)] - [g(x)] = 0$

Q.25. If $f(x, y) = 2x^2 + 4xy - y^2$, then what is the value of $f(x + 1, y - 1)$?
(a) $2y^2 + 4xy - x^2 - 6y + 3$
(b) $2x^2 - 4xy + y^2 + 6y - 3$
(c) $2y^2 + 4xy - x^2 + 6y - 3$
(d) $2x^2 - 2xy + 2y^2 - 6y + 3$

Direction for questions 26 to 29: Read the passage below and solve the questions based on it.

If P (x, y, z) = max (x, y) + min (y, z)
Q (x, y, z) = max (y, z) + min (x, y)
R (x, y, z) = max (x, y, z)
S (x, y, z) = min (x, y, z)
T (x, y, z) = max (x, z) − min (y, z)
U (x, y, z) = max (y, z) − min (x, y)

Q.26. For, $x = 1$, $y = 2$, $z = 3$ which of the following is equal to 1?
(a) $\frac{p(x,y,z)+s(x,y,z)}{2\times U(x,y,z)}$
(b) $\frac{Q(x,y,z)+S(x,y,z)}{2\times T(x,y,z)}$

(c) $\frac{Q(x,y,z)+U(x,y,z)}{2\times R(x,y,z)}$

(d) $\frac{R(x,y,z)+S(x,y,z)}{2\times P(x,y,z)}$

Q.27. For $x=1, y=2, z=3$, which of the following is greater than 1?

(a) $\frac{P(x,y,z)}{2\times S(x,y,z)}$ (b) $\frac{U(x,y,z)}{Q(x,y,z)}$

(c) $\frac{S(x,y,z)}{R(x,y,z)}$ (d) $\frac{T(x,y,z)}{R(x,y,z)}$

Q.28. For $x=1, y=2, z=3$, which of the following is equal to zero?

(a) $P(x, y, z) + Q(x, y, z)$

(b) $Q(x, y, z) - R(x, y, z) - S(x, y, z)$

(c) $R(x, y, z) - S(x, y, z) - T(x, y, z)$

(d) $T(x, y, z) - U(x, y, z) - S(x, y, z)$

Q.29. For $x=1, y=2, z=3$, which of the following is less than 1?

(a) $\frac{Q(x,y,z)}{R(x,y,z)+S(x,y,z)}$

(b) $\frac{Q(x,y,z)-U(x,y,z)}{2S(x,y,z)}$

(c) $\frac{Q(x,y,z)+U(x,y,z)}{R(x,y,z)+T(x,y,z)}$

(d) $\frac{U(x,y,z)}{R(x,y,z)+S(x,y,z)}$

Q.30. $f(x) = x^3 + ax^2 + bx + c$ and $g(x) = x^2 + 1$, where a, b and c belongs to $\{1, 2, 3, \ldots, 10\}$. What are the number of polynomials which are divisible by $g(x)$?

(a) 0 (b) 1 (c) 10 (d) 100

Direction for questions 31 to 34: Read the passage below and solve the questions based on it.

The pages of a book are numbered 0, 1, 2, ... P, P>0. There are four categories of instructions that direct a person in positioning the pages in the book. The instruction types with their meanings are given below:

i. Open Position of the book at page number 1.
ii. Close Position of the book at page number 0.
iii. Forward, n From the current page, move forward by n pages. If, in this process, page number P is reached, stop at P.
iv. Backward, n From the current page, move backward by n pages. If, in this process, page number 0 is reached, stop at page number 0.

In each of the following questions, you will find a sequence of instructions. Let p1 be the page number before the instructions are given and p2 be the page number after executing all the instructions given.

Q.31. Forward, 30; Backward, 12. Which of the following statements is true?

(a) p1 = p2, if P = 12 and p1 = 0

(b) P = 25, provided p1 > 0

(c) p1 > 35, provided P = 100

(d) p1 = 40, provided P = 30

Q.32. Backward, 7; Forward, 7. Which of the following statements is true?

(a) p1 = p2, if p1 ≥ 7

(b) p1 = p2, if p1 > 0

(c) p2 =7, provided P>0

(d) p1>p2, provided P>0

Q.33. Forward, 15; Forward, 15. Which of the following statements is true?

(a) p2 – p1 = 30, only if p1= 0

(b) p2 – p1 = 30, only if p1= 1 and P > 30

(c) p2 – p1 = 15, only if p1= 0 and P = 31

(d) p2 > p1, if P > 0

Q.34. Forward, 5; Backward,4. Which of the following statements is true?

(a) p2 = p1 + 4 provided 1 < p1<7

(b) p2 = p1, if P<6

(c) p2 = p1 + 1, if P – p1 ≥5

(d) p2 – p1 < 0, if P>0

Q.35. If $f(x) = a^{1+x}$, then $f(x).\ f(y).\ f(z) =$

(a) $f(x + y + z)$ (b) $f(x + y + z + 1)$

(c) $f(x + y + z + 2)$ (d) $f(x + y + z + 3)$

Q.36. A polynomial function $f(x)$ satisfies $f(x)\ f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$. If $f(10) = 1001$, then what is the value of $f(20)$?

(a) 2002 (b) 8004

(c) 8001 (d) None of these

Q.37. The function $f(x) = \max\{(1-x), (1+x), 2\}$, where x can take any real value, is equivalent to

(a) $f(x) = \begin{cases} 1-x & x \le -1 \\ 2 & -1 < x < 1 \\ 1+x & x \ge 1 \end{cases}$

(b) $f(x) = \begin{cases} 1+x & x \le -1 \\ 2 & -1 < x < 1 \\ 1-x & x \ge 1 \end{cases}$

(c) $f(x) = \begin{cases} 1-x & x \le -1 \\ 1 & -1 < x < 1 \\ 1+x & x \ge 1 \end{cases}$

(d) None of these

Q.38. The equivalent definition of $f(x) = ||x|-1|$

(a) $f(x) = \begin{matrix} x+1 & x \geq 0 \\ x+1 & x \leq 0 \end{matrix}$

(b) $f(x) = \begin{matrix} -x-1 & x \leq -1 \\ x+1 & -1 < x \leq 0 \\ x-1 & 0 \leq x \leq 1 \\ x+1 & x \geq 1 \end{matrix}$

(c) $f(x) = \begin{matrix} -x-1 & x \leq -1 \\ x+1 & -1 < x \leq 0 \\ 1-x & 0 \leq x \leq 1 \\ x-1 & x \geq 1 \end{matrix}$

(d) None of these

Q.39. If $f(x) = \big|\,|x|-1\big|$, f of (x) is

(a) $f(x) = \begin{matrix} |x|-2, & |x| \geq 2 \\ 2-|x| & 1 < |x| < 2 \\ |x| & |x| \leq 1 \end{matrix}$

(b) $f(x) = \begin{matrix} |x|+2, & |x| \geq 2 \\ |x|-2 & 1 \leq |x| \leq 2 \\ |x| & |x| \leq 2 \end{matrix}$

(c) $f(x) = \begin{matrix} |x|-2, & |x| \geq 2 \\ |x|+2 & 1 \leq |x| \leq 2 \\ |x| & |x| \leq 1 \end{matrix}$

(d) None of these

Q.40. If $f(x)$ is the greatest integer less than or equal to $x/2$, how many of the following statements are true?

1. $f(x)$ is half the least even integer greater than or equal to x
2. $f(x)$ is the least integer greater than or equal to $\frac{x}{2} - 1$.
3. $f(x)$ is one less than half the least even integer greater than x.

(a) 0 (b) 1 (c) 2 (d) 3

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(c)	(b)	(a)	(d)	(b)	(c)	(b)	(b)	(c)	(d)	(d)	(b)	(a)	(b)	(a)	(b)	(d)	(b)	(a)

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(a)	(d)	(b)	(d)	(d)	(b)	(a)	(c)	(b)	(d)	(d)	(d)	(d)	(c)	(b)	(c)	(c)	(b)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(d)	(f)	(c)	(d)	(c)	(c)	(a)	(b)	(b)	(c)	(c)	(a)	(c)	(b)	(a)	(b)	(a)	(b)	(c)

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(c)	(a)	(b)	(a)	(c)	(c)	(a)	(b)	(b)	(b)	(a)	(d)	(b)	(a)	(d)	(c)	(d)	(b)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(c)	(d)	(a)	(b)	(d)	(d)	(b)	(c)	(d)	(d)	(d)	(a)	(d)	(d)	(b)	(c)	(b)	(d)	(d)	(a)

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(d)	(d)	(c)	(b)	(b)	(c)	(a)	(d)	(d)	(d)	(d)	(a)	(d)	(a)	(c)	(c)	(d)	(c)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(b)	(d)	(a)	(a)	(c)	(c)	(a)	(b)	(d)	(c)	(a)	(a)	(b)	(c)	(c)	(c)	(a)	(c)	(a)	(d)

HINTS AND SOLUTIONS

Moderate

1. Go thorough the options.

[redacted]

$[x]$ = Greatest integer less than or equal to x
$\{x\}$ = Smallest integer more than or equal to x
Take x = P such that P = I + F, where I is the integral and F(F > 0) is the fractional part of the number P(P cannot be an integer).
$[x] = [I + F] = I$ and similarly $\{x\} = \{I + F\}$
$= I + 1$

3. Hence, $[x] - \{x\} = -1$

4. $[x] + \{x\} = I + I + 1 = 2I + 1$ = Odd integer
Alternatively, we can do both the problems by assuming any non-integral value of x (like 3.5 or 4.2 or −2.6 etc.)

5. For all the values of $n < 50$, $f(n) = 0$
And for all the $n \geq 50$, $f(n) = 1$. Hence, 51 such values are there.

6. $x/2 + x/3 + x/5 = 31x/30$
It means $[x/2] = x/2$, $[x/3] = x/3$ and $[x/5] = x/5$
Now [N] = N is possible only if N is an integer.
Hence, $x/2$, $x/3$ and $x/5$ are integers. So, x is divisible by 2, 3 and 5. Or, x is divisible by 30.
Total 33 values are possible.

7. We have to find out some powers of $f(1)$, which is in all the cases equal to 1

10. $x = 1$ and $x = -1$ are two roots of $f(x)$. Similarly, if we put x = 0, then $f(x) = -3$. Now solve the equation to get the value of a + c.

13. $f(x) = 1 + |x|$, $x < -1$ and $[x]$, $x \geq -1$,
So, $f(-2.3) = 1 + |-2.3| = 1 + 2.3 = 3.3$
$f\{f(-2.3)\} = f(3.3) = [3.3] = 3$

14. Expression under square root should be non-negative and the denominator cannot be equal to zero. Now find the common values.
Alternatively, we can do this problem by using options.

15. For any increase in the value of x, increase in the numerator will be more than the increase in the denominator. This can be verified through taking a few values of x. Alternatively, this can be verified through plotting the graphs. The slope of the graph of the numerator will be more than the slope of the graph in the denominator.
Hence, $f(x)$ will increase.

16. Assume some values of x, y and z and then go through options.

17. For ${}^{16-x}C_{2x-1}$ to be defined, $(16 - x) \geq (2x - 1)$ and similarly, for ${}^{20-3x}P_{4x-5}$ to be defined, $(20 - 3x) \geq (4x - 5)$.
Now solve.

20. In this case, $f(x) = g(x)$ is possible only if the domain of the respective functions are same. Now, either solve it or go through the options.

21. $g(x) = f(x + 1) = |x - 2 + 1| + |x - 3 + 1| + |x - 4 + 1| = |x - 1| + |x - 2| + |x - 3|$
Obviously, this is neither odd nor even.
Alternatively, we know the graph of this function will neither be symmetrical to axis or origin. [See the topic Graphs and Maxima Minima]

28. Let us find out (Mag (6, (Nag (7, −7)))
Nag (7, −7) = −7, so, (Mag (6, (Nag (7, −7))) = 6
Mag (8, (Nag (4, (Nag (7, −7))))) = 8, so the answer = b

30. Applying componendo and dividendo in $\frac{f\left(\frac{3}{4}\right)+1}{f\left(\frac{3}{4}\right)-1} =$

$[2.f(3/4)]/2 = f(3/4)$

$f(3/4) = 2(3/4)^2 + 6(3/4) - 1$

$f(x) = 2x^2 + 6x - 1$, then the value of $\frac{f\left(\frac{3}{4}\right)+1}{f\left(\frac{3}{4}\right)-1}$ is

So, the answer is option (d).

36. Go through the options.

38. $f(x, y) = g(x, y)$ is possible for all the sets of $(x, y) = (x, 0)$.

39. Draw the graph of y or see it by assuming different values of y.

40. Using $H(1) + H(2) + H(3) + \ldots + H(x) = n^2 H(x)$

For $n = 1$: $H(1) = 1 . H(1)$

For $n = 2$: $H(1) + H(2) = 4 \times H(2)$, or, $H(2) = [H(1)]/3$

For $n = 3$: $H(1) + H(2) + H(3) = 9 \times H(3)$, or, $H(3) = [H(1)]/2.3$

Now go through the same method to find H(2005)

Advanced

1. Use the following:

If $a < x$, then $(x - a) > 0$, $b < x$, then $(x - b) > 0$, $c < x$, then $(x - c) > 0$ and $d < x$, then $(x - d) > 0$, $e < x$, then $(x - e) > 0$. And $a > x$, then $(x - a) < 0$, similarly other also can be calculated. Now use the options. Hence, the answer is option (d) None of these.

3. Assume different values of x in an attempt to refute the statements given. After a bit of hit and trial, we can observe that none of the statements can be refuted. So, the answer is option (d).

8. Using the options, if $f(x) = x$, then $f[(x + y)/(x - y)] = (x + y)/(x - y)$ and similarly it can be seen that RHS will be same as LHS.

9. Minimum value of V = 5 and maximum value of V = 29. So, sum = 34

15. This question requires finding out the digital sum of x given.

Digital sum of $x = 123456^{654321}$ is same as digital sum of $x = (1 + 2 + 3 + 4 + 5 + 6)^{654321} = f^{\infty}(x) = 21^{654321} = 3^{654321}$

Now digital sum of any power of 3 greater than 1 is 9. So the digital sum of $3^{654321} = 9$

16. Digital sum of $7^1 = 7$
Digital sum of $7^2 = 49 = 4 + 9 = 1 + 3 = 4$
Digital sum of $7^3 = 343 = 1$
Digital sum of $7^4 = 2401 = 7$
Digital sum of $7^5 = 16807 = 4$
So, digital sum of $7^{183} = 1$

[redacted]

V(x) denotes the absolute value of x, $mn(x, y)$ denotes the minimum of x and y, $mx(x, y)$ denotes the maximum of x and y.

21. Find mn [V (mx (−5, 3)), mx (−2, V (mn (−5, −2)))];

Now solve it according to the conditions given.

25. Put $x + 1$ at the place of x and $y - 1$ at the place of y and check the options.

[redacted]

31. Checking the options:
Option (a). If $p1 = 0$, then after forward 30, page number 12 will be obtained. After this, if we apply Backward 12, we will again get back to $0 = p1 = p2$.

32. If $p1$ is greater than or equal to 7 pages, then $p1 = p2$.

33. Use options.

35. $f(x).f(y).f(z) = a^{1+x} \times a^{1+y} \times a^{1+z} = a^{3+x+y+z}$
This is equal to $f(x + y + z + 2)$.

36. Let $f(x) = x^n + 1$, so $f(1/x) = (1/x^n) + 1$

Checking it for $f(x)$ $f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ satisfies the value of $f(x)$ assumed.

$f(x) = x^n + 1 = 1001$, so $x^n = 1000$

Hence, $n = 3$.

$f(20) = 20^3 + 1 = 8001$

37. Go through the options.

38. Go through the options.

39. Go through the options.

40. Assume the values of x to check the options. All the three statements are true.

13

CHAPTER

Graphs and Maxima Minima



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Maxima and minima, and how they are calculated
- Some standard graphs and their shifting
- Definitions and properties
- Kinds of questions asked in the CAT
- Methods of solving questions

Introduction to the topic

In case of maxima, minima and graphs, the biggest mistake any student can make is to see this chapter as a purely mathematical concept. However, though there is some mathematics involved, as far as the questions asked in the CAT are concerned, with good logic and common sense, they can be handled without much effort. This chapter will help the students understand the basic concepts on which the problem-solving techniques are built.

MAXIMUM AND MINIMUM VALUE

MAXIMUM VALUE

Let $f(x)$ be a function with the domain D $\subset$ R. Then $f(x)$ is said to have attained the maximum value at a point 'a' if $f(a) \geq f(x)$ for all the values of x. In this case, 'a' is called the point of maximum and $f(a)$ is known as the maximum value or the greatest value or the absolute maximum value of $f(x)$.

For example, in case of $y = f(x) = |x|$, we will have the maximum value of y as $+\infty$.

MINIMUM VALUE

Let $f(x)$ be a function with domain D $\subset$ R. Then $f(x)$ is said to attain the minimum value at a point 'a' if $f(x) \geq f(a)$. In such a case, the point 'a' is called the point of minimum and $f(a)$ is known as the minimum value or the least value or the absolute minimum value of $f(x)$.

Let us first see the maximum and minimum values of some very basic functions

Example 1 Find the maximum and the minimum values of $y = f(x) = -|x - 1| + 4$.

	$y=f(x)$	$y=f(x)$	$y=f(x)$	$y=f(x)$	$y=f(x)$
	$=x$	$=x^2$	$=\frac{1}{x}$	$=\log_e x$	$=\lvert x-2\rvert$
Maximum	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$
Minimum	$-\infty$	0	$-\infty$	$-\infty$	0

Solution We have $f(x) = -|x - 1| + 4$ for all the real values of x.

$|x - 1| \geq 0$ for all real values of x

Hence, $-|x - 1| \leq 0$

$|x - 1| + 4 \leq 4$

So, the maximum value of $y = 4$

Now, $f(x) = 4 \Rightarrow -|x - 1| + 4 = 4 \Rightarrow |x - 1| = 0 \Rightarrow x = 1$.

Hence, $f(x)$ attains the maximum value 4 at $x = 1$.

To find out the minimum value, we can make $-|x-1|$ as small as possible by taking different values of x. So, the minimum value of y will be $-\infty$.

Example 2 What is the maximum and minimum value of $y = f(x) = |x + 3|$?

Solution We have $y = f(x) = |x + 3|$

$|x + 3| \geq 0$

$\Rightarrow f(x) \geq 0$ for all $x \in R$

So the minimum value of $f(x)$ is 0, which is attained at $x = -3$.

To find out the maximum value, it can be seen that $y = f(x) = |x + 3|$ can be made as big as possible. So, the maximum value of y is $+\infty$.

Finding Maxima and Minima in Some Typical Cases

1. Maxima and minima of quadratic equation

The standard methods of finding the maxima and the minima of all the quadratic equations, are as follows:

Let us discuss these methods with respect to a particular equation $y = x^2 - 5x + 6$.

(a) Graphical method

As we have seen in the concepts of quadratic equation, graph of $y = ax^2 + bx + c = 0$ will have its minimum value at $x = \frac{-b}{2a}$ and $y_{min} = \frac{-D}{4a}$.

So, the minimum value of $y = x^2 - 5x + 6$ will be $y_{min} = \frac{-D}{4a} = \frac{-1}{4}$

(b) Quadratic equation method

Suppose we have find the minimum value of $y = x^2 - 5x + 6$

We can find the minimum value by breaking this equation into the sum of a whole square and whatever is left i.e., $P^2 + Q$.

$x^2 - 5x + 6 = [x^2 - 2.x.5/2 + (5/2)^2] - ¼$

$= \left(x - \frac{5}{2}\right)^2 - \frac{1}{4}$

We know that the minimum value of $\left(x - \frac{5}{2}\right)^2$ is equal to zero, so the minimum value of $y = -\frac{1}{4}$

Example 3 Find the minimum value of $y = (x-a)(x-b)$.

(a) ab (b) $\frac{(a-b)^2}{4}$

(c) 0 (d) $-\frac{(a-b)^2}{4}$

Solution $x^2 - x(a + b) + ab$

$$= \left[x^2 - 2\frac{(a+b)}{2}x + \left(\frac{a+b}{2}\right)^2\right] - \frac{(a-b)^2}{4}$$

$$= \left[x - \left(\frac{a+b}{2}\right)\right]^2 - \frac{(a-b)^2}{4}$$

Now we know that the minimum value of any perfect square is 0.

So, the minimum value of $y = -\frac{(a-b)^2}{4}$

2. Maxima and minima of any modulus

We have seen above how to find the maxima and the minima of any modulus. Here we will discuss some more situations involving modulus. Let us see this with the help of an example:

Finding the maximum and the minimum value of $y = f(x) = |x + 3| + |x - 5| + |x - 7|$

Maximum value Obviously, the maximum value can be extended upto $+\infty$.

Minimum value In this case, all the three parts of $f(x)$ i.e., $|x + 3|$, $|x - 5|$ and $|x - 7|$ can not be simultaneously equal to zero. So, we are required to find the critical points here at first, and then these points in $f(x)$ to see that which one gives us the minimum value.

To obtain critical points, put all the three components $|x + 3|$, $|x - 5|$ and $x - 7|$ one by one equal to zero.

$|x + 3| = 0$, so, $x = -3$

$|x - 5| = 0$, so, $x = 5$

and $|x - 7| = 0$, so, $x = 7$

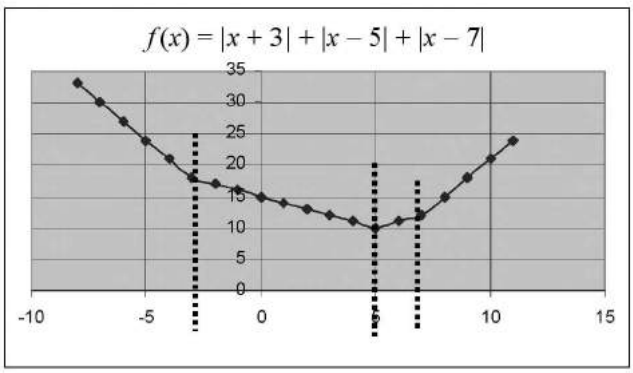
Now putting these values in $f(x)$ gives us the following result:

At $x = -3$, $f(x) = 18$

At $x = 5$, $f(x) = 10$

At $x = 7$, $f(x) = 12$

So, the minimum value of $f(x) = 10$



Alternatively, drawing the graph gives us the clear picture regarding the movement of the graph which gives us the minimum value of $f(x)$.

The dotted lines in the above graph show the movement of the graph. Obviously, at $x = 5$, graph is at its lowest point which is the minimum value of the graph.

The inspection of the graph gives us the maximum value of $f(x)$ which is $+\infty$.

3. Finding the maximum or the minimum value of the product/sum of two or more than two variables when the sum/product of these variables is given

1. If sum of two or more than two variables is given, then there product will be the maximum when the value of all the variables are equal.

Example 4 Find the maximum and the minimum value of xy subject to $x + y = 8$.

(a) 8 (b) 16 (c) 20 (d) 24

Solution The maximum value of xy will occur when $x = y = 4$

So, the maximum value of $xy = 4 \times 4 = 16$

To obtain the minimum value, we can take either x or y to be negative, and then the product of xy will be negative. And this process continues till $-\infty$. So, the minimum value of $xy = -\infty$

2. If the product of two or more than two positive variables is given, then there sum will be the minimum when the value of all the variables are equal.

Example 5 What is the minimum value of $f(x) = \dfrac{x^3 + x + 2}{x}$, $x > 0$?

Solution Since $\dfrac{x^3 + x + 2}{x} = x^2 + 1 + \dfrac{1}{x} + \dfrac{1}{x}$

Now, the product of all the terms x^2, 1, $1/x$ and $1/x$ is 1, the sum of these terms will have the minimum value for $x^2 = 1 = 1/x$ i.e., for $x = 1$. Hence, the minimum value of $f(x)$ is 4.

Example 6 If $ab^2c^3 = 27 \times 2^8$, then find the minimum value of $a + b + c$ (given that $a, b, c > 0$).

Solution To find the minimum value of $a + b + c$, we need to know the product of a, b and c.

Assuming $a = x$, $b = 2y$ and $c = 3z$, we have $x \times 4y^2 \times 27z^3 = 27 \times 2^8$. So now we have to calculate the minimum value of $x + 2y + 3z$.

Thus, $ab^2c^3 = 27 \times 2^8 \Rightarrow x \times y^2 \times z^3 = 2^6$.

Now, $x + 2y + 3z = x + y + y + z + z + z$ and we know that $x \times y \times y \times z \times z \times z = 2^6$. So, the least value will occur when all are equal to 2. Thus, the least sum will be 12.

Corollary For any positive value of x, the minimum value of the sum of x and its reciprocal will be 2.

$$\left(x + \frac{1}{x}\right)_{\min} = 2$$

Example 7 If $a > 1$, $b > 1$ then the minimum value of $\log_b a + \log_a b$ is

(a) 0 (b) 1

(c) 2 (d) None of these

Solution Assume $\log_a b = N$, so $\log_b a = 1/N$

So, the minimum value of $(\log_b a + \log_a b)$ = minimum value of $(N + 1/N) = 2$

4. Using arithmetic mean, geometric mean and harmonic mean to find the maxima and the minima

i. AM ≥ GM ≥ HM

ii. $(GM)^2 = AM \times HM$ (for two numbers only)

Example 8 If a, b, c and d are positive real numbers such that $a + b + c + d = 2$, then which of the following is true regarding the values of $N = (a + b)(c + d)$?

(a) $0 \le N \le 1$ (b) $1 \le N \le 2$

(c) $2 \le N \le 3$ (d) $0 \le N \le 1$

Solution Using AM ≥ GM

$\frac{1}{2}[(a+b)+(c+d)] \geq [(a+b)]^{1/2} = \frac{1}{2}(2) \geq (N)^{1/2}$

Also, $(a+b)(c+d) \geq 0$

So, $0 \leq N \leq 1$

GRAPHS

Sketching a Graph

To sketch any graph, we make a table of points on the X-axis and corresponding points on the Y-axis.

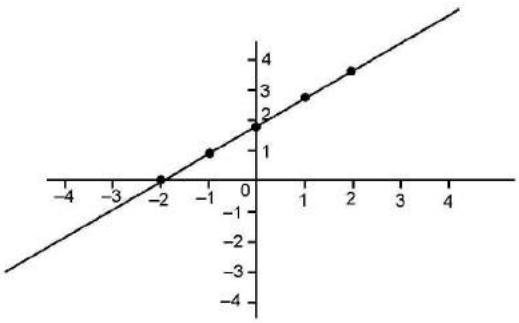
1. $y = x + 2$

Making a table of points for x and y

y	x
2	0
3	1
4	2
1	–1
0	–2

And hence the graph:

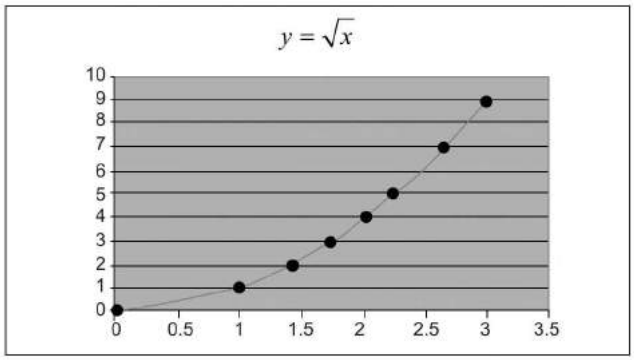
$y = x + 2$



2. $y = \sqrt{x}$

y	x
0	0
1	1
1.414214	2
1.732051	3
2	4

And hence the graph:



3. $y = |x-1| + |x-2|$

To draw the graph of any modulus function, we need to work with the help of the critical points, which help us in determining the intervals in which the modulus will be +ve or –ve.

Arranging the critical points on the number line:

$-\infty$ 1 2 $+\infty$

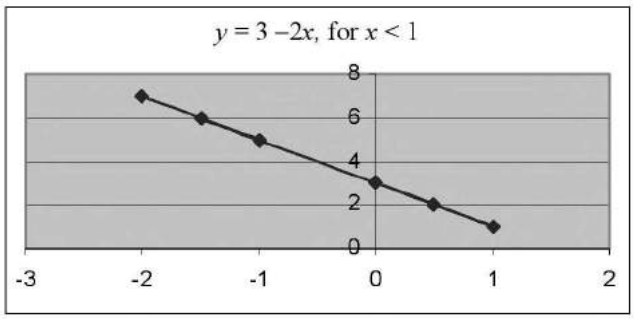
For all the values $-\infty < x < 1$

$|x-1| + |x-2| = (1-x) + (2-x) = 3 - 2x$

So, the table for values in between this interval is:

$y = 3 - 2x$	x
3	0
2	0.5
6	–1.5
5	–1
7	–2

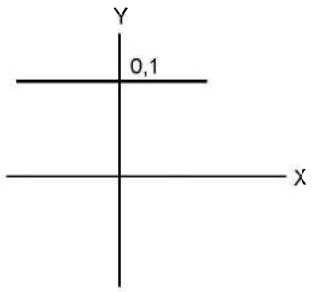
So, the graph for this interval will look like:



For all the values $1 < x < 2$

$|x-1| + |x-2| = (x-1) + (2-x) = 1$

So, the graph for this interval will look like:

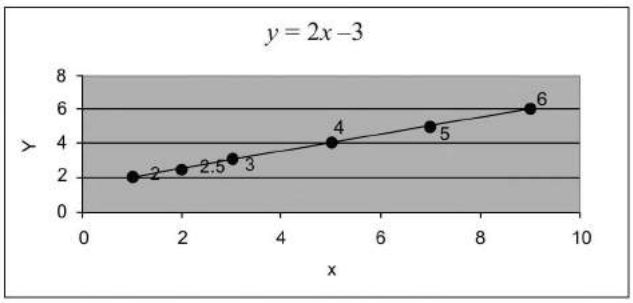


For all the values $x > 2$

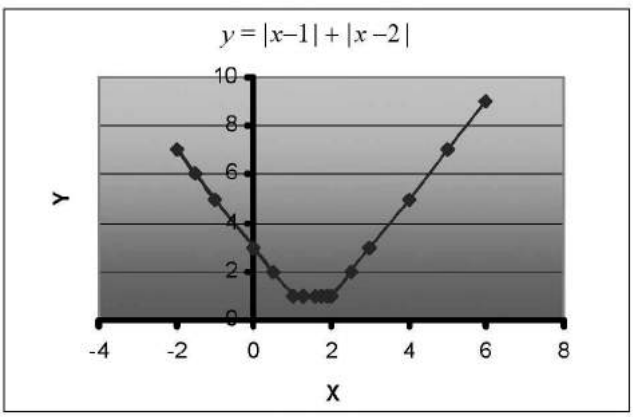
$|x-1| + |x-2| = (x-1) + (x-2) = 2x - 3$

$y = 2x - 3$	x
2	2.5
1	2
3	3
5	4
7	5
9	6

And its graph will look like:

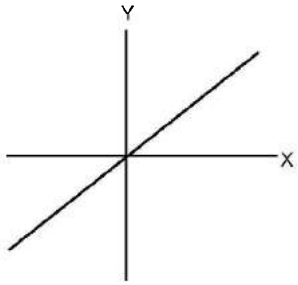


Combining all these graphs, we finally get,

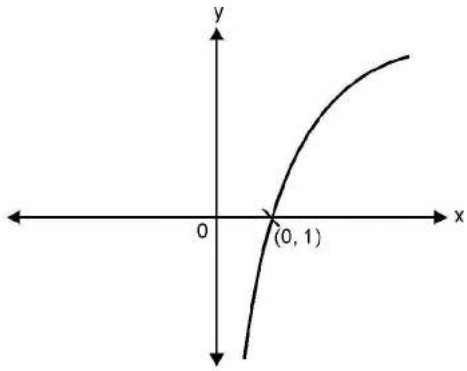


Some Standard Graphs

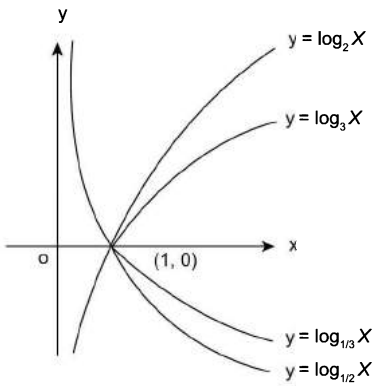
1. $y = x$



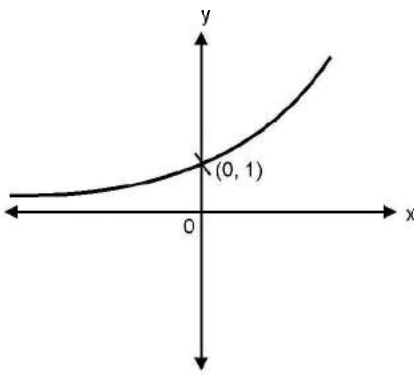
2. $y = \log_e x$



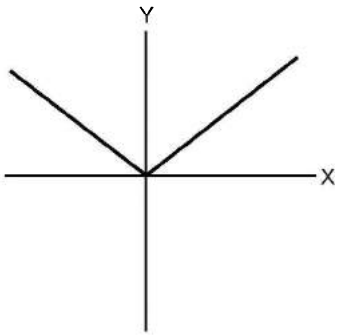
3. $y = \log_a x$



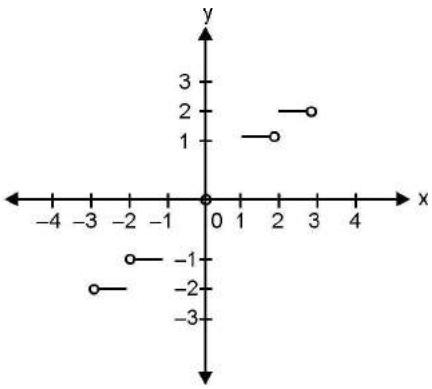
4. $y = e^x$



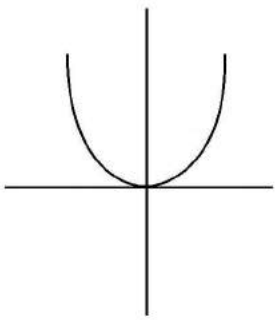
5. $y = |x|$



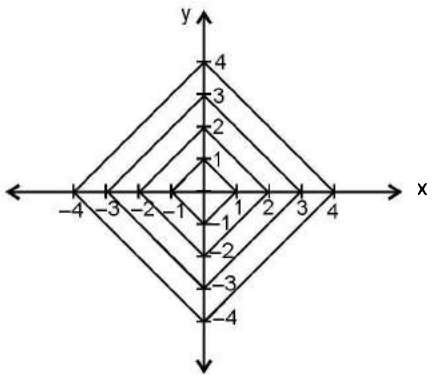
6. $y = [x]$, where $[x]$ represents the greatest integer less than or equal to x.



7. $y = x^2$



8. $|x| + |y| = K$



Shifting of a Graph

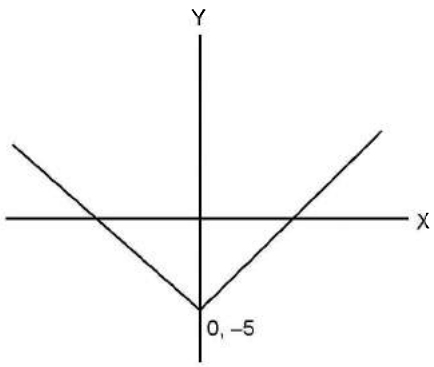
To know how a graph would shift, we need to know two things:

i. The shifting is either on the X-axis or on the Y-axis or on both of them,
ii. The shifting is in the +ve direction or in the –ve direction.

Let us understand this phenomena with the help of an example $y = f(x) = |x|$

Case 1

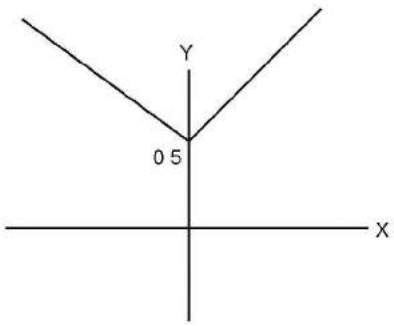
If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = f(x) - 5$	It will shift 5 units downwards, i.e., 5 units on the –ve Y axis



$y = |x| - 5$

Case 2

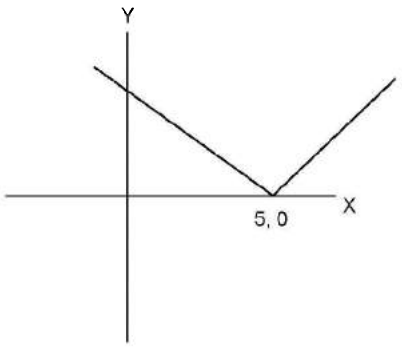
If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = f(x) + 5$	It will shift 5 units upwards, i.e., 5 units on the +ve Y axis



$y = |x| + 5$

Case 3

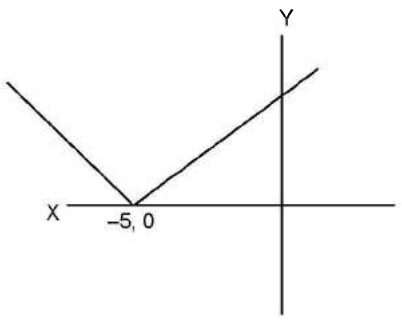
If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = f(x) - 5$	It will shift 5 units rightwards, i.e., 5 units on the +ve X axis



$y = |x - 5|$

Case 4

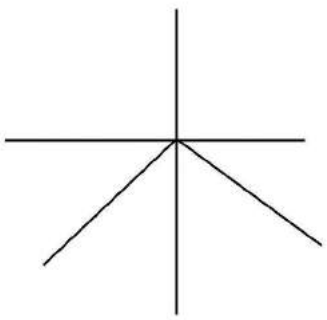
If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = f(x) - 5$	It will shift 5 units leftwards, i.e., 5 units on the +ve X axis



$y = |x + 5|$

Case 5

If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = -f(x)$	It will be reflected along the X-axis as if X-axis is a mirror



$y = -|x|$

Case 6

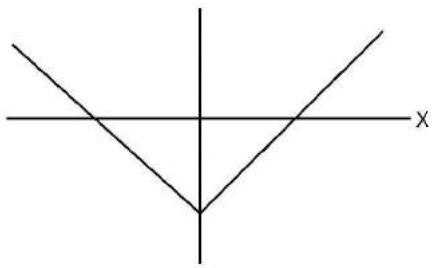
If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = \|f(x)\|$	Graph below the X-axis will be reflected along the X-axis on its above as if X-axis is a mirror

We can summarize the whole phenomena of the shifting of a graph as:

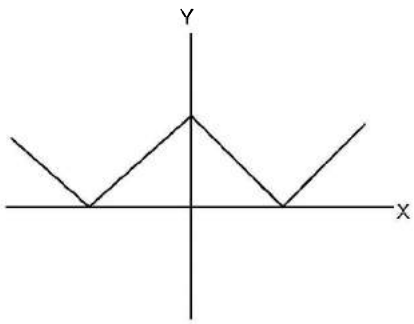
1. If there is a change in the value of X-axis, then it will move on the X-axis and similarly for the Y-axis. The graph of $y = f(x-5)$ has a change in the values of x where as graph of $y = f(x) - 5$, which can be written as $y + 5 = f(x)$ has a change in the values of y.
2. If there is an addition, then the graph will move on the −ve side, either on the X-axis or on the Y-axis and if there is a subtraction, then graph will move on the +ve side.

Example 9 The graph of $y = |x|$ is given. Draw the graph of $y = ||x|-2|$.

Solution Let us first sketch $|x|-2$.

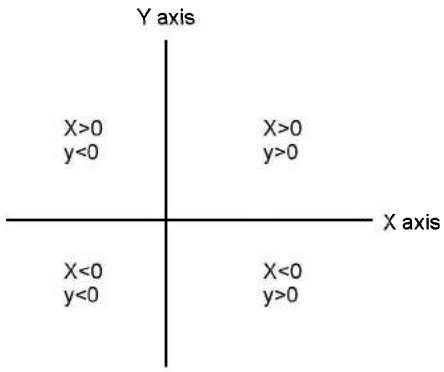


Now $||x|-2|$ will be having all the negative part of $|x| - 2$ on the positive side of Y axis, or above X-axis.



Area of a Graphs

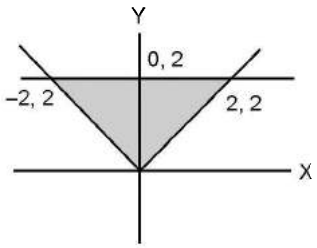
Before we proceed ahead with calculating the area of the combination of graphs, we should be clear with the quadrants and the signs of X and Y in the same.



To find the area of graphs, we first need to sketch the graphs of the equations and then by using geometry/ coordinate geometry we can find the area of the enclosed figure.

Example 10 Find out the area of the region enclosed by $y = |x|$ and $y = 2$.

Solution Following is the area enclosed by the equations given above:



The area of the enclosed figures will be,
Area = ½ × 4 × 2 = 4 sq. units

Example 11 In the X – Y plane, the area of the region bounded by the graph $|x + y| + |x - y| = 4$ is
(a) 8 (b) 12 (c) 16 (d) 20

Solution Let $x \geq 0, y \geq 0$ and $x \geq y$
Then $|x + y| + |x - y| = 4$
$\Rightarrow x + y + x - y = 4 \Rightarrow x = 2$
Similarly, $x \geq 0, y \geq 0, x \leq y$
$x + y + y - x = 4 \Rightarrow y = 2$
The area in the first quadrant is 4.
By using symmetry, the total area in all the four quadrants = 4 × 4 = 16 sq. units

Example 12 If p, q, and r are any real numbers, then
(a) max $(p, q) <$ max (p, q, r)
(b) min $(p, q) = \frac{1}{2}(p + q - |p - q|)$
(c) min $(p, q) <$ min (p, q, r)
(d) None of these

Solution If we take $r < p, q$, then (1) and (3) cannot hold. For (2), if $p \geq q$, then $|p - q| = p - q$.

$\frac{1}{2}(p + q - |p - q|) = \frac{1}{2}(p + q - p + q|) = q =$ min (p, q)

Similarly, If $p < q$ then $|p - q| = q - p$.

$(p + q - |p - q|) = \frac{1}{2}(p + q - q + p) = p =$ min (p, q)

So, option (b) is the answer.

Example 13 If $a\,b\,c\,d = 1, a > 0, b > 0, c > 0, d > 0$, then what is the minimum value of $(a + 1)(b + 1)(c + 1)(d + 1)$?
(a) 1 (b) 8
(c) 16 (d) None of these

Solution The minimum value will occur when $a = b = c = d = 1$

So, the minimum value of $(a + 1)(b + 1)(c + 1)(d + 1) = 16$

Example 14 Consider a triangle drawn on the X-Y plane with its three vertices (41, 0), (0, 41) and (0, 0), each vertex being represented by its (X,Y) coordinates. What is the number of points with integer coordinates inside the triangle (excluding all the points on the boundary)?
(a) 780 (b) 800 (c) 820 (d) 741

Solution The equation formed from the data is $x + y < 41$
The values which will satisfy this equation are
(1, 39), (1, 38) ...(1, 1)
(2, 38), (2, 37), ...(2,1)
(39, 1)

So the total number of cases are 39 + 38 + 37 ... + 1 $= \frac{39 \times 40}{2} = 780$

Example 15 A telecom service provider engages male and female operators for answering 1000 calls per day. A male operator can handle 40 calls per day whereas a female operator can handle 50 calls per day. The male and the female operators get a fixed wage of Rs 250 and Rs 300 per day respectively. In addition, a male operator gets Rs 15 per call he answers and female operator gets Rs 10 per call she answers. To minimize the total cost, how many male operators should the service provider employ assuming he has to employ more than 7 of the 12 female operators available for the job?
(a) 15 (b) 14 (c) 12 (d) 10

Solution First let us form both the equations:
40 m + 50 f = 1000
250 m + 300 f + 40 × 15 m + 50 × 10 × f = A
850 m + 8000 f = A
Where m and f are the number of males and females and A is the amount paid by the service provider.
Then the possible values for f are 8, 9, 10, 11, 12
If f = 8, then m = 15
If f = 9, 10 and 11 then m will not be an integer while f = 12 then m will be 10.
By putting f = 8 m = 15 and A = 18800. When f = 12 and m = 10 then A = 18100.
Hence, the number of males will be 10.

Example 16 If a, b and c are the sides of a triangle, then what is the maximum value of the expression $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}$?
(a) 1 (b) 3/2 (c) 2 (d) 5/2

Solution Assume 2s = $a + b + c$. We know that $b + c > a$, so we get $2(b + c) > a + b + c = 2s$
Hence, $b + c >$ s

Similarly, $c + a > s$, $a + b > s$

Hence, $\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b} < \frac{a+b+c}{s}=\frac{2s}{s}=2$

Example 17 At how many distinct points the graphs of $y = x^{-1}$ and $y = \log_e x$ intersect? (CAT 2003)

Solution We can see that the graphs of $y = x^{-1}$ and $y = \log_e x$ intersect just once.

Example 18 How many integral solution is/are possible for the equation $|y-18| + |y - 9| + |y + 9| + |y + 18| = 54$?

Solution Here, $|y-N|$ should be seen as nothing but the distance of the point y from the point N on the number line, a person standing at a point N. Hence $|y - 18| + |y - 9| + |y + 9| + |y + 18|$ is the sum of the distances of the point y from 18, 9, −9 and −18. Now, for any point y, where $p \le y \le q$, the sum of the distances from p and q is $q-p$. So, for point y, where $-9 \le y \le 9$, the sum of the distances of y from −18, −9, 9 and 18 is $[18-(-18)] + [9-(-9)] = 54$

For points outside this limitation, this expression will have different values. Hence, the required numbers are −9, −8, … , 8, 9. So, there will be a total of 19 values.

PRACTICE EXERCISES

WARM UP

Q.1. What is the minimum value of the expression $x^2 + 8x + 10$?
(a) 0 (b) $+\infty$
(c) −6 (d) +6

Q.2. What is the maximum value of the expression $x - 6x - x^2$?
(a) 0 (b) 14
(c) $-\infty$ (d) None of these

Q.3. What is the maximum value of the function $y=\frac{2x^2+3x+4}{x^2+x+3}$?
(a) 11/23 (b) 23/11
(c) 4/7 (d) 7/4

Q.4. In the above question, what is the minimum value of y?
(a) 0 (b) 1
(c) −1 (d) Cannot be determined

Q.5. If $f(x) = \max(4x + 3, 3x + 6)$ for $x \in ? [-6, 10]$, find the maximum value of $f(x)$?
(a) $+\infty$ (b) 36
(c) 43 (d) None of these

Q.6 What is the maximum value of $f(x) = \min(4 - 5x, x - 3)$ for every $x \in (0, 4)$?
(a) −1 (b) 2
(c) 4 (d) None of these

Direction for questions 7 to 11: Go through the following definitions and solve the questions based on it.

A $(x, y, z) = \min(x + y, y + z, z + x)$
B $(x, y, z) = \min(x - y, y - z, z - x)$
C $(x, y, z) = \max(A(x, y, z)\ B(x, y, z))$
D $(x, y, z) = \min(A(x, y, z), B(x, y, z))$

Q.7. What is the value of C (1, 2, 3)?
(a) 1 (b) 2
(c) 3 (d) 4

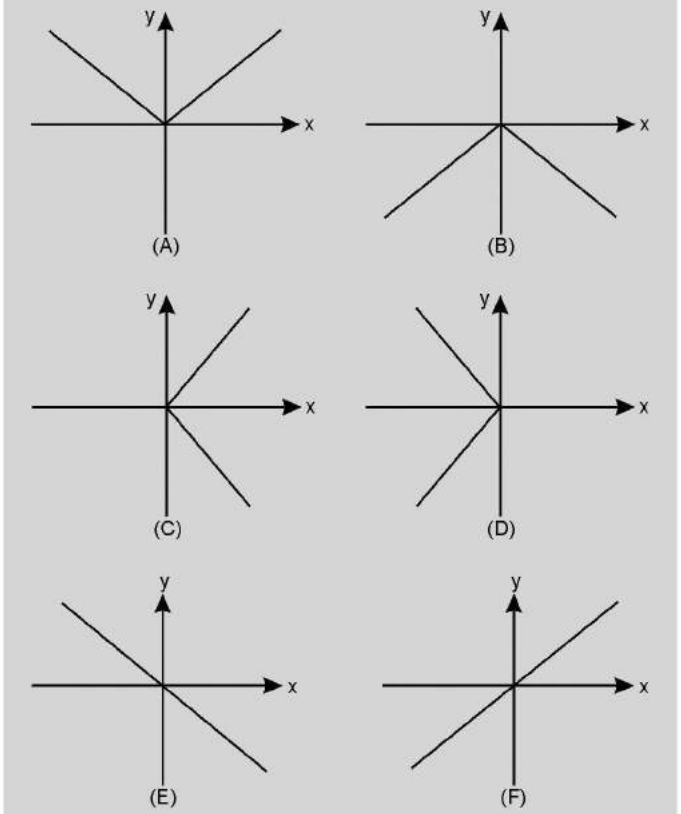
Q.8. Find the value of D (0, 1, 2).
(a) −1 (b) 1
(c) 0 (d) None of these

Q.9. Find the value of A (2, 3, 4) + B (2, 3, 4).
(a) 13 (b) 4
(c) 7 (d) None of these

Q.10. Find the value of C (0, 1, D (1, 2, 3)).
(a) 1 (b) 0
(c) −2 (d) None of these

Q.11. What is the value of D (1, 2, C (0, 1, 2))?
(a) 1 (b) 2
(c) 3 (d) 0

Direction for questions 12 to 16: Following graphs represent various functions. Match the figure with the appropriate function.



Q.12. $f(x) = -x$
(a) B (b) C (c) E (d) D

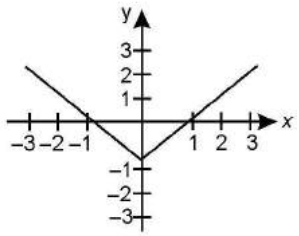
Q.13. $f(x) = |x|$
(a) A (b) B (c) C (d) D

Q.14. $f(x) = -|x|$
(a) E (b) B (c) A (d) F

Q.15. $f(x) = x$
(a) E (b) F
(c) A (d) B

Q.16. $f(x) = |x| - x$
(a) A (b) B
(c) C (d) None of these

Q.17. Which of the following represents the following graph?



(a) $|x - 1|$ (b) $|x + 1|$
(c) $|x| - 1$ (d) $|x| + 1$

Q.18. The graph of $y = (x+2)^2 - 3$ is the graph of $y = x^2 -$
(a) Shift 2 units up and 3 units rightwards towards +ve X axis.
(b) Shift 2 units rightwards towards +ve X axis and 3 units downwards towards –ve Y axis.
(c) Shift 2 units leftwards towards –ve X axis and 3 units downwards towards –ve Y axis.
(d) Shift 2 units up and 3 units rightwards towards +X axis.

Q.19. The graph of the function $y = f(x)$ is symmetrical about the line $x = 2$, then
(a) $f(x + 2) = f(x - 2)$ (b) $f(2 + x) = f(2 - x)$
(c) $f(x) = f(-x)$ (d) None of these

Q.20. If $x + y + z = 24$, then the maximum value of xyz is
(a) 215 (b) 512
(c) 125 (d) 576

FOUNDATION

Q.1. The minimum value of $4^x + 4^{1-x}$, $x \in R$, is
(a) 2 (b) 4
(c) 1 (d) None of these

Q.2. a, b and c are three positive numbers and abc^2 has the greatest value $\frac{1}{64}$. Then
(a) $1 = b = \frac{1}{2}, c = \frac{1}{4}$ (b) $a = b = \frac{1}{4}, c = \frac{1}{2}$
(c) $a = b = c = \frac{1}{3}$ (d) None of these

Q.3. The sum of the products of the ten numbers ± 1, ± 2, ± 3, ± 4, ± 5 taking two at a time is
(a) 165
(b) -55
(c) 55
(d) None of these

Q.4. If $ab = 2a + 3b$, $a > 0$, $b > 0$ then the minimum value of ab is
(a) 12 (b) 24
(c) $\frac{1}{4}$ (d) None of these

Q.5. If $a > 1$, $b > 1$ then the minimum value of $\log_b a + \log_a b$ is
(a) 0 (b) 1
(c) 2 (d) None of these

Q.6. What is the maximum value of $\frac{x}{y}$ for $(x-2)^2 = 4$ and $(y-3)^2 = 25$?
(a) $\frac{1}{2}$ (b) $\frac{5}{8}$ (c) $-\frac{1}{8}$ (d) $-\frac{5}{2}$

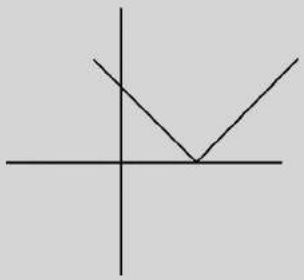
Q.7. If a and b are both positive real values, then the minimum value of $(a+b)\left(\frac{1}{a}+\frac{1}{b}\right)$ is
(a) 0 (b) 2
(c) -2 (d) 4

Q.8. If $|X + 2| > 3$, then the solution set for X is
(a) $X > -1$ (b) $X > 1$
(c) $X < -5$ (d) $X > 1$ or $X < -5$

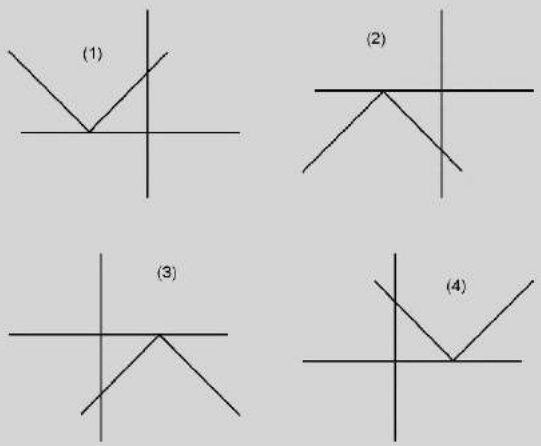
Q.9. If a and b are both positive real values, then the maximum value of $(a+b)\left(\frac{1}{a}+\frac{1}{b}\right)$ is
(a) 0 (b) 2 (c) 4 (d) ∞

Direction for questions 10 to 13: Read the passage below and solve the questions based on it.

The graph of $y = f(x)$ is given as



Following are the four graphs:



Q.10. Which of the following defines graph (1)?
(a) $f(x) - 1$ (b) $f(x - 1)$
(c) $f(x) + 1$ (d) $f(x + 1)$

Q.11. Which of the following defines graph (2)?
(a) $f(x) - 1$ (b) $f(x - 1)$
(c) $f(x) + 1$ (d) None of these

Q.12. Which of the following defines graph (3)?
(a) $-f(x)$ (b) $f(x-1)$
(c) $f(x) + 1$ (d) $f(x+1)$

Q.13. Which of the following defines graph (4)?
(a) $f(x) - 1$ (b) $f(x)$
(c) $f(x) + 1$ (d) $f(x+1)$

Q.14. What is the maximum value of $y = |x - 5| + |x - 7|$?
(a) -2 (b) 2 (c) 0 (d) ∞

Q.15. If a lies between 2 and 3, both included, and b lies between 4 and 6, both included, then what is the ratio of the minimum and the maximum limits of $a^2 - b^2$?
(a) -4 (b) 4 (c) $\frac{32}{7}$ (d) $-\frac{28}{6}$

Q.16. If $y = |x| - 5$, then the minimum possible value of y is
(a) 5 (b) -5
(c) 0 (d) Both a and b

Q.17. Given that $(x - 2)^2 = 9$ and $(y - 3)^2 = 25$. What is the minimum value of $\frac{x}{y}$?
(a) $\frac{1}{2}$ (b) $\frac{5}{8}$ (c) $-\frac{1}{8}$ (d) $-\frac{5}{2}$

Q.18. When $x^2 + 4xy + 4y^2$ takes a minimum value, then
(a) $x = -2y$ (b) $x = 2y$
(c) $2x = y$ (d) $-2x = y$

Q.19. A man finds that it costs Rs (300 + 4N) per day to make N articles. If the selling price of each article is Rs 7, then what is the minimum number of articles to be produced per day if he has to make a profit?
(a) 300 (b) 150
(c) 100 (d) None of these

Q.20. If a, b and c are all the real numbers, then the minimum value of $ab + bc + ca$ is
(a) 1 (b) $-\frac{1}{2}$
(c) $-\frac{3}{2}$ (d) None of these

Q.21. $|x + y| = 10$, where x and y are integers, then what is the minimum value of $x^2 + y^2$?
(a) 75 (b) 100
(c) 50 (d) None of these

Q.22. If $|x| + |y| = 7$, then what is the sum of the minimum and the maximum values of $x + y$?
(a) $\frac{3}{2}$ (b) -7 (c) 7 (d) 0

Q.23. If $|x| - |y| = 13$, then which of the following cannot be the value of $x-y$?
(a) -18 (b) -9
(c) -17 (d) None of these

Q.24. What is the minimum value of $\left(\frac{a+1}{a}\right)^2+\left(\frac{b+1}{b}\right)^2+\left(\frac{c+1}{c}\right)^2$?
(a) 1 (b) 4 (c) 8 (d) 12

Q.25. Let y = max $(x + 3)$, $(7 - 2x)$ what is the minimum value of y for $2 \geq \times \geq 1$?
(a) 4.50 (b) 4.00 (c) 4.67 (d) 4.33

Q.26. The minimum value of $3x + 3y + z$ subjected to the condition $xyz = 24$, where x, y and z are all positive real numbers, is
(a) $14\times3^{1/3}$ (b) 18 (c) 216 (d) 12

Q.27. Find the maximum and the minimum values of the function $\frac{x^2-x+1}{x^2+x+1}$ for real values of x.
(a) 3 and -3 (b) $\frac{1}{3}$ and $-\frac{1}{3}$
(c) 3 and $\frac{1}{3}$ (d) None of these

Q.28. A group of children walked to the Ramnarayan's shop and bought pastries for a total of Rs 168 and ice candies for a total of Rs 126. Each child has one pastry and one ice candy. How many children could be there in the group?
(a) 28 (b) 21 (c) 42 (d) 36

Q.29. Let x, y, z be three positive numbers such that $x + y + z = 1$. The minimum value of $\left(x+\frac{1}{x}\right)^2+\left(y+\frac{1}{y}\right)^2+\left(z+\frac{1}{z}\right)^2$ will be
(a) 16 (b) 24
(c) 33 (d) None of these

Q.30. If a, b and c are the positive real numbers, find the greatest value of $a^2\ b^3\ c^4$ subject to the condition $a + b + c = 18$.
(a) $+\infty$ (b) $4^2.6^3.8^4$
(c) 16686 (d) None of these

Q.31. What is the least value of $(x-2)\ (x-4)^2\ (x-6) + 6$, for real values of x?
(a) 6 (b) 4/3
(c) 4 (d) None of these

Q.32. Consider the following functions:
$f(x) = x^2 + 3x$
$g(x) = 3x + 4$
For what value of x will $f[g(x)]$ have its minimum value?
(a) 0 (b) $\frac{-11}{2}$
(c) $\frac{-11}{6}$ (d) None of these

Q.33. If a, b and c are all positive then what is the minimum value of $\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b}$?
(a) 1 (b) 3 (c) 9 (d) 27

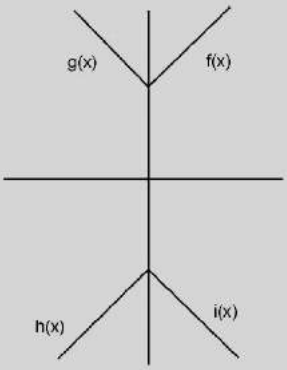
Q.34. If a lies between 2 and 3, both included, and b lies between 4 and 6, both included, then what is the ratio of the minimum and the maximum limits of $a^2 - b^2$?
(a) -4 (b) 4
(c) $\frac{-28}{5}$ (d) None of these

Q.35. The least value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ for positive x, y and z, satisfying the condition $x + y + z = 9$ is
(a) $\frac{15}{7}$ (b) $\frac{1}{9}$ (c) 3 (d) 1

Q.36. The minimum value of $3x + 4y$, subject to the conditions $x^2y^3 = 6$ and x and y are positive, is
(a) 10 (b) 14 (c) 13 (d) 13

Q.37. The minimum value of $f(x) = x^8 + x^6 - x^4\ 2x^3 - x^2 - 2x + 9$ is
(a) 5 (b) 1 (c) 0 (d) 9

Direction for questions 38 to 40: Go through the graph given below and solve the questions based on it (Neglect fine errors).



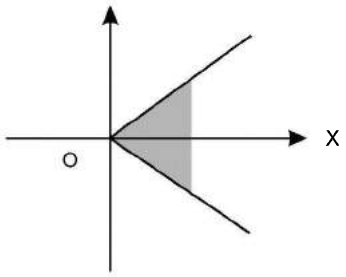
Q.38. Which of the following is true?
(a) $f(x) = g(-x)$ (b) $f(x) = -g(x)$
(c) $f(x) = -g(-x)$ (d) None of these

Q.39. Which of the following is true?
(a) $i(x) = g(-x)$ (b) $i(x) = -g(x)$
(c) $i(x) = -g(-x)$ (d) None of these

Q.40. How many of the following statements are true?
(a) $g(x) = h(-x)$ (b) $f(x) = -h(-x)$
(c) $h(x) = -i(x)$ (d) $f(x) = -i(x)$

(a) 1 (b) 2
(c) 3 (d) 4

Q.41. The shaded region in the diagram represents the relation

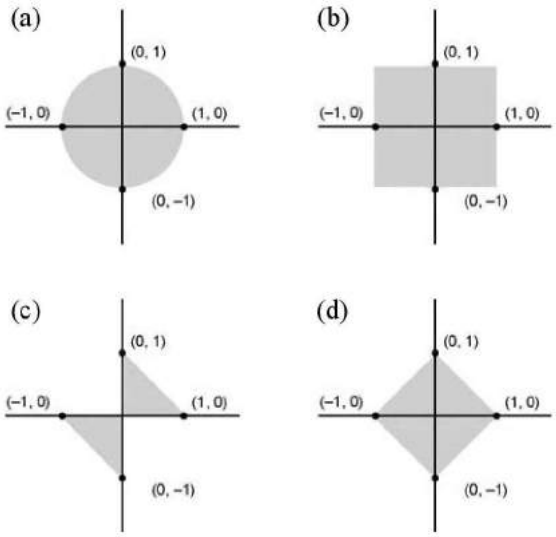


(a) $y \le x$ (b) $|y| < |x|$
(c) $y \le |x|$ (d) $|y| \le x$

Q.42. If P, Q and R are three vertices with coordinates (1,4), (4, 2) and (m, 2m−1) respectively, then the value of m for which PR + RQ is minimum is

(a) $\frac{17}{8}$ (b) $\frac{5}{2}$ (c) $\frac{7}{4}$ (d) $\frac{11}{8}$

Q.43. The set $\{(x, y): |x| + |y| \le 1\}$ is represented by the shaded region in one of the four figures. Which one is it?



Q.44. If x, y and z are three integers satisfying $x + y + z = 13$ and $xy + xz = 42$, what is the minimum possible value of $y^2 + z^2$?

(a) 18 (b) 22.5
(c) 16 (d) 24.5

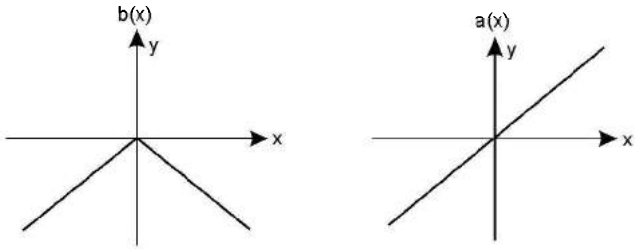
Q.45. What will be the maximum value of $\left(\frac{1}{x}\right)^x$?

(a) e (b) e^{-1}
(c) $e^{1/e}$ (d) $e^{-1/e}$

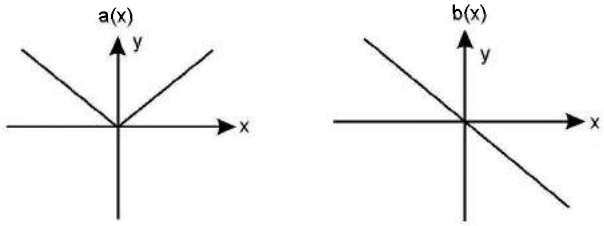
Direction for questions 46 to 48: Read the passage below and solve the questions based on it.

Mark option (a) as the answer if $a(x) = b(-x)$ for $x < 0$, and $a(x) = -b(-x)$ for $x > 0$
Mark option (b) as the answer if $a(x) = -b(x)$ for $x < 0$, and $a(x) = b(-x)$ for $x > 0$
Mark option (c) as the answer if $a(x) = -b(x)$ for all the values of x.
Mark option (d) as the answer if none of the above is true.

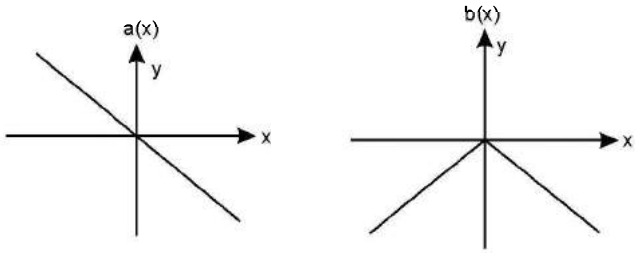
Q.46.



Q.47.



Q.48.



Q.49. Let a, b, c and d be the real numbers. If G = max{min (a, b), min(c, d)} and H = min {max (a, c), max(b, d)}. Then
(a) G ≤ H, for all values of a, b, c and d
(b) G = H, for all values of a, b, c and d
(c) G ≥ H, for all values of a, b, c and d
(d) Any one of the above three options can be true for some particular values of a, b, c and d.

Q.50. If $x = \frac{5}{a+b}$ and $y = 5(a + b)$, and if both x and y are greater than zero, then $(x + y)$ is:
(a) < 5 (b) ≥ 10
(c) Between 5 and 10 (d) None of these

MODERATE

Q.1. We have $|a + b| = |a| + |b|$, $|c + d| < |c| + |d|$, $|e + f| < |e| + |f|$. If $a, b, c, d, e, f \in$ R, then the product $abcdef$ is
(a) Equal to zero
(b) Necessarily non-negative
(c) Necessarily non-positive
(d) Either positive or negative

Q.2. For how many values of x, is the equality $|||x - 2| - 3| - 1| = 0$, true?
(a) 0 (b) 2
(c) 3 (d) None of these

Q.3. Let $f(x) = |x - a| + |x - 2a| + |x - 3a|$, where $a > 0$. Let the minimum value of $f(x)$ be denoted by m and the corresponding value of x be denoted by b. Which of the following is true?
(a) $m = a; b = a$ (b) $m = a; b = 2a$
(c) $m = 2a; b = a$ (d) $m = 2a; b = 2a$

Q.4. The minimum of $f(x) = |x - 1| + |x - 4| + |x - 5|$ is attained at $x =$
(a) 1 (b) 3 (c) 4 (d) 5

Q.5. Let $y = k^2(x^2+1) + 2k(3x + 4) + 27$. If k and x are real numbers, what is the minimum value of y?
(a) 1 (b) 3
(c) 4 (d) None of these

Q.6. If $x < -3$ and $y > 2$, then which of the following is true for all real values of x and y?
(a) $x + 2y < -1$ (b) $3y + 2x > 0$
(c) $2x - xy > 0$ (d) None of these

Q.7. If $a_1, a_2, \ldots a_n$ are n integers such that $a_1 a_2 \ldots a_n = 1$, then which of the following must be true?
(a) The minimum value of $(a_1 + a_2 + \ldots + a_n)$ is $-(n - 1)$, if n is odd.
(b) The minimum value of $(a_1 + a_2 + \ldots + a_n)$ is $(n - 2)$, if n is even.
(c) The minimum value of $(a_1 + a_2 + \ldots + a_n)$ is $(2 - n)$, if n is even.
(d) None of these

Q.8. If a and b are positive real values, and $ab + (a+b) = 21$, what is the minimum value of $(a + b)$?
(a) 6 (b) 8
(c) 12 (d) Cannot be determined

Q.9. The least integral value of k for which $(k - 2)x^2 + 8x + k + 4 > 0$ for all $x \in$ R, is:
(a) 5 (b) 4
(c) 3 (d) None of these

Q.10. If $f(x) = 2x^2 + 3x + 4$ and $g(x) = 5 - x^2$, the min $[f(x)] - \max[g(x)]$ is
(a) 0 (b) 5 (c) $\frac{17}{8}$ (d) $-\frac{17}{8}$

Q.11. If $a + b + c = 20$ and $\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) = 30$, then the value of $\frac{a}{b} + \frac{b}{c} + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b}$ is
(a) 597 (b) 8 (c) 350 (d) 441

Q.12. The minimum value of the expression $\frac{1+V^2}{1+V}$, for positive values of V, is
(a) $\sqrt{2}$ (b) $\frac{2}{\sqrt{2}+1}$
(c) 1 (d) None of these

Q.13. If p, q, r, s are real numbers such that $p^2 + q^2 = 1$, $r^2 + s^2 = 1$, then the maximum value of $pr + qs$ would be
(a) -1 (b) 2 (c) -2 (d) 1

Q.14. If p, q, r denote the sides of a triangle, then $\frac{p}{q+r} + \frac{q}{r+p} + \frac{r}{p+q}$ always lie between (both end values inclusive)
(a) 3 and 4 (b) 2 and 3
(c) 3.5 and 5 (d) 1.5 and 2

Q.15. p, q, r, s are any four positive real numbers, the minimum value of $\frac{p}{q} + \frac{q}{r} + \frac{r}{s} + \frac{s}{p}$ is
(a) 1 (b) 2 (c) $2\sqrt{2}$ (d) 4

Q.16. $p, q,$ and r are all positive numbers. If $p + q + r = 1$, then the least value of $\left(\frac{1}{p} - 1\right)\left(\frac{1}{q} - 1\right)\left(\frac{1}{r} - 1\right)$ is
(a) 10 (b) 25
(c) 24 (d) None of these

Q.17. Find the maximum value of $y = 5 - |x + 1| - |x - 3|$.
(a) 1 (b) 5 (c) 4 (d) ∞

Q.18. If three positive real numbers x, y and z are in an AP such that $xyz = 4$, then what will be the minimum value of y?
(a) $2^{1/3}$ (b) $2^{2/3}$
(c) $2^{1/4}$ (d) $2^{3/4}$

Q.19. Find the least value of the expression $2\log_{10} x - \log_x (.01)$ for $x>1$.
(a) 4 (b) 2
(c) 1 (d) None of these

Q.20. When the curve $y = x^2 + 4x - 5$ and $y = \frac{1}{(1+x^2)}$ are drawn in the $x - y$ plane, how many times do they intersect for values of $x > 0$?
(a) Never (b) Once
(c) Twice (d) More than twice

Q.21. If p, q, and r are three real number such that $p + q + r = 4$ and $p^2 + q^2 + r^2 = 6$, then what can be the maximum value of q?
(a) 1 (b) 2/3
(c) 4 (d) None of these

Q.22. In the above question, what will be the value of p when q is minimum?
(a) 2 (b) 1/3
(c) 2/3 (d) None of these

Q.23. Given $y = \min(x - 5, 7 - x)$. What is the minimum value of y?
(a) 0 (b) $-\infty$ (c) $+\infty$ (d) 1

Q.24. Given $y = \min(x - 5, 7 - x)$. What is the maximum value of y?
(a) 0 (b) $-\infty$ (c) $+\infty$ (d) 1

Q.25. Given $y = \max(x - 5, 7 - x)$. What is the minimum value of y?
(a) 0 (b) $-\infty$ (c) $+\infty$ (d) 1

Q.26. Given $y = \max(x - 5, 7 - x)$. What is the maximum value of y?
(a) 0 (b) $-\infty$ (c) $+\infty$ (d) 1

Q.27. Let y be the maximum of $\{(x + 5), (3 - x)$. What is the minimum value of y, if $-2 \le x \le 0$?
(a) 3 (b) 2
(c) 1 (d) None of these

Q.28. The maximum possible value of $x^2 + 3y^2 + 9z^2$, subject to $x + 2y + 3z = 12$, where x, y and z are real numbers, is
(a) 48 (b) 216
(c) 240 (d) None of these

Q.29. At how many different points, $y = x^{-1}$ and $y = \log_e x$ intersect?
(a) 0 (b) 1 (c) 2 (d) Infinite

Q.30. A rectangle of the largest area is inscribed in a semi-circle of radius r. What is the area of the rectangle?
(a) $1.5 \times r^2$ (b) $r^2 \times \sqrt{2}$
(c) r^2 (d) $r^2\sqrt{3}$

Q.31. The circumference of a circle, the perimeter of a pentagon and an octagon are the same. Their areas are denoted by C, P and O respectively. Which of the following is true for their areas?
(a) C > P > O (b) O>P>C
(c) O>C>P (d) C>O>P

Q.32. At what value of x, $\left(\frac{1}{x}\right)^x$ will attain the maximum value?
(a) e (b) e^{-1}
(c) e^{-2} (d) e^2

Q.33. The altitude of a right circular cone of the minimum volume circumscribed about a sphere of the radius r is
(a) 2r (b) 3r
(c) 4r (d) None of these

Q.34. The minimum value of $(x - a)(x - b)$ is
(a) ab (b) 0
(c) $\frac{(a-b)^2}{4}$ (d) $-\frac{(a-b)^2}{4}$

Q.35. If x, y and z are arbitrary positive real numbers satisfying the equation $4xy + 6xz + 8zx = 9$, find the maximum possible value of the product xyz.
(a) $\frac{1}{2\sqrt{2}}$ (b) $\frac{\sqrt{3}}{4}$
(c) $\frac{3}{8}$ (d) None of these

Q.36. The minimum value of the quantity
$$\frac{(a^2+3a+1)(b^2+3b+1)(c^2+3c+1)}{abc},$$
where a, b and c are positive real numbers, is
(a) $\frac{11^3}{2^3}$ (b) 125 (c) 25 (d) 27

Q.37. A right circular cylindrical container closed on both the sides contains a fixed volume of motor oil. Suppose its base has the diameter d and its height is h. The overall surface area of the container is minimum when
(a) $h = \frac{4}{3}\pi d$ (b) h = 2d
(c) h = d (d) None of these

Q.38. A rectangular box of volume 48 cu ft is to be constructed, so that its length is twice its width. The material to be used for the top and the four sides is three times costlier per ft^2 than that used for the bottom. Then, the height (in ft) of the box that minimizes the cost is equal to.
(a) $\frac{8}{27}$ (b) $\frac{8\sqrt[3]{4}}{3}$ (c) $\frac{4}{27}$ (d) $\frac{8}{3}$

Q.39. A truck is to be driven for 300 km on a highway at a constant speed of x km/h. The speed rules of

the highway requires $30 \le x \le 60$. The fuel costs Rs 10 per litre and is consumed at the rate of $2+\frac{x^2}{600}$ litres per hour. The wages of the driver are Rs 200 per hour. The most economical speed to drive the truck, in km/h, is

(a) 30 (b) 60

(c) $30\sqrt{3.3}$ (d) $20\sqrt{33}$

Q.40. A square tin sheet of side 12 inches is converted into a box with an open top in the following steps–The sheet is placed horizontally. Then, equal sized squares, each of side x inches, are cut from the four corners of the sheet. Finally, the four resulting sides are bent vertically upwards in the shape of a box. If x is an integer, then what is the value of x that maximizes the volume of the box?

(a) 3 (b) 4 (c) 1 (d) 2

Q.41. How many integral points are there within the graph of $|x| + |y| \le 4$?

(a) 32 (b) 41

(c) 9 (d) None of these

Q.42. The sides of a rectangle of the greatest area which can be inscribed into an ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1=1$ is:

(a) 5, 3 (b) $5\sqrt{2}$, 3

(c) 5, $3\sqrt{2}$ (d) $5\sqrt{2}$, $3\sqrt{2}$

(See geometry chapter for ellipse)

Q.43. The sets

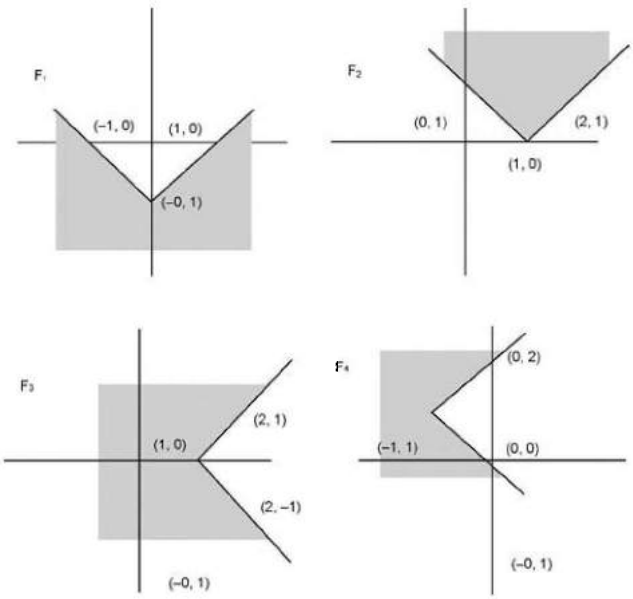
$\{[x, y] : |y-1|-x \ge 1\}$

$\{[x, y] : |x|-y \ge 1\}$

$\{[x, y] : |x-1|-y \le 1\}$

$\{[x, y] : |y-1|x-1| \ge 0\}$

are represented by the shaded regions in the figures given below in some order.



Then the correct order of the figures is

(a) F_4, F_1, F_2, F_3 (b) F_4, F_2, F_3, F_1

(c) F_1, F_4, F_3, F_2 (d) F_4, F_1, F_3, F_2

Direction for questions 44 to 46: Read the passage below and solve the questions based on it.

In each question, choose the function which best describes the graph.

$f(x) = (|x|-1)(|x|-2)$

$g(x) = ||x-1|-2|-2$

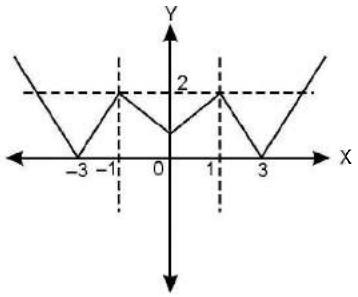
$h(x) = ||\,|x|-1|-2|$

$j(x) = (|x|+1)(|x|-2)$

$k(x) = |x-1|\,|x-3|$

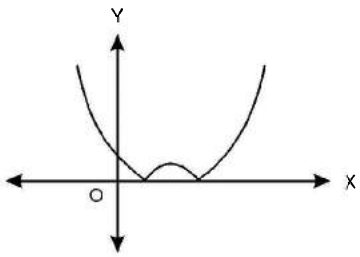
$l(x) = ||\,|x|-2|-1|$

Q.44.



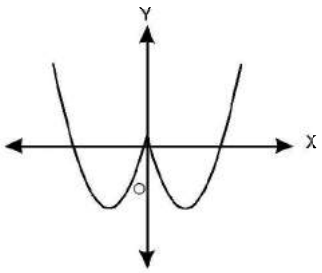
(a) $g(x)$ (b) $l(x)$ (c) $h(x)$ (d) $k(x)$

Q.45.



(a) $f(x)$ (b) $l(x)$ (c) $j(x)$ (d) $k(x)$

Q.46.



(a) $f(x)$ (b) $k(x)$ (c) $j(x)$ (d) $h(x)$

Q.47. From a square tin sheet of side 12 feet a box with its top open is made by cutting equal squares at the four corners and then bending the tin sheet so as to form the sides of the box. The side of the removed square

for which the box has the maximum possible volume in feet is.

(a) 3 (b) 1

(c) 2 (d) None of these

Q.48. From a circular sheet of paper of radius a, a sector with a central angle is cut out and folded into the shape of a conical funnel. The volume of this funnel is maximum when θ equals

(a) $\frac{2\pi}{\sqrt{2}}$ (b) $2\pi\sqrt{\frac{2}{3}}$

(c) $\frac{\pi}{2}$ (d) π

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(b)	(b)	(b)	(c)	(d)	(c)	(b)	(c)	(d)	(a)	(c)	(a)	(b)	(a)	(d)	(c)	(c)	(b)	(b)

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(b)	(b)	(c)	(a)	(d)	(d)	(d)	(d)	(d)	(a)	(b)	(d)	(c)	(c)	(b)	(a)	(d)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(c)	(d)	(b)	(d)	(d)	(b)	(c)	(c)	(d)	(b)	(d)	(c)	(c)	(d)	(d)	(a)	(a)	(a)	(c)	(b)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(d)	(a)	(d)	(a)	(c)	(d)	(d)	(b)	(a)	(b)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(d)	(d)	(c)	(d)	(b)	(d)	(a)	(a)	(d)	(a)	(b)	(d)	(d)	(d)	(d)	(a)	(b)	(a)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(d)	(b)	(d)	(d)	(c)	(d)	(d)	(b)	(c)	(d)	(b)	(d)	(d)	(c)	(b)	(c)	(d)	(b)	(d)
Q. No.	41	42	43	44	45	46	47	48												
Answer	(b)	(d)	(d)	(b)	(d)	(c)	(c)	(b)												

HINTS AND SOLUTIONS

Moderate

1. $|a + b| = |a| + |b|$ is possible only if both a and b are having the same signs or $a = b = 0$.
$|c + d| < |c| + |d|$ is possible only if both c and d are having opposite signs, and the same is true a bout $|e + f| < |e| + |f|$.
sign of ab = +ve or $ab = 0$
Sign of cd = –ve and sign of ef = –ve
Hence, the sign of $abcdef$ = Non negative.

2. $||x - 2| - 3|-1 = 0$, or, $||x - 2| - 3| = 1$, or,
$|x - 2| - 3 = 1$, or, $|x - 2| = 4$,
If $x - 2 > 0$, then $x = 6$, and if $x - 2 < 0$, then $-(x - 2) = 4$, so, $x = -2$.
Similarly,
If $x - 2 > 0$, then $|x - 2| = (x - 2)$
Or, $|x - 2| - 3 = -1$, or, $(x - 2) = 2$, or, $x = 4$.
If $x - 2 < 0$, $|x - 2| = -(x - 2)$
So, $|x - 2| - 3 = -1$, or, $-(x - 2) = 2$, or, $2 - x = 2$, so $x = 0$.
So, four values of x are possible.

3. Obviously, the minimum value will be attained at the median of the critical points which is x = 2a and the minimum value will be $f(x) = 2a$

4. Minimum value of $f(x)$ will be attained at the median of the critical points of $f(x)$ ie., $x = 4$.

6. Going through the options.
Checking option 2. $3y > 6$ and $2x < -6$, so $3y + 2x > 0$.

8. $ab + 2(a + b) = 21 \rightarrow ab + 2(a + b) + 4 = 21 + 4$.
So, $(a + 2)(b + 2) = 25$.
For minimum value of $(a + b)$, a and b should be equal.
Hence, $(a + 2) = (b + 2) = 5$, (we will not take – 5 because it will give negative values of a abd b).
So, $a = b = 3$
Alternatively, the quadratic equation can also be used to find out the minimum value of $(a + b)$.
Minimum value will occur if $a = b$
So, $ab + 2(a + b) = 21$ will be equal to $a^2 + 2(2a) - 21 = 0$
Or, $(a + 7)(a - 3) = 0$, hence, $a = 3 = b$

9. $D = 8^2 - 4(k - 2)(k + 4) > 0$
Or, $(k + 6)(k - 4) < 0$
Or, $4 < k < 6$. So, $k = 5$

10. min $[f(x)]$ = min $(2x^2 + 3x + 4) = 46/16$
max$[g(x)] = 5$

11. Use AM ≥ GM ≥ HM for all the four fractions.

15. Use AM ≥ GM ≥ HM for all the four fractions.

17. Find the maximum value of $y=5-[|x+1|+|x-3|]$
Assume $[|x + 1| + |x - 3|] = z$
So, $y = 5 - z$, y will attain the maximum value at minimum value of z.
Now, Min of z will occur at $x = -1$ [Using critical points]
So, minimum value of Z = 4
Maximum value of $y = 1$

18. $2y = x + z$
Now, AM of x, y and z will be more than their GM.
$Y \geq (xyz)^{1/3}$
Hence, $y_{min} = 4^{1/3} = 2^{2/3}$

19. $2\log_{10} x - \log_x (.01) = \log_{10} x^2 - \log_x 10^{-2} = 2[\log_{10} x + 1/\log_{10} x]$
Minimum value of $[\log_{10} x + 1/\log_{10} x] = 2$

23. Drawing the graphs of $x - 5$ and $7 - x$ tells us that both the graphs are heading towards $+\infty$ and $-\infty$, depending upon the values. So, minimum of y will be $-\infty$.

24. Now the direction of movement of both the graphs are opposite (one is going towards $+\infty$ and other one is going towards $-\infty$ for a particular set of values and vice-versa.)
For $x < 6$, $x - 5$ will provide us the minimum value and for $x > 6$, $7 - x$ will provide us with the minimum value and at $x = 1$, we will get the values equal. And this $x = 6$ is the point for which y_{max} = min $(x - 5, 7 - x)$ is obtained.

28. Any of x, y or z can be taken in such a way that $x^2 + 3y^2 + 9z^2$ can go to $+\infty$. Actually it was a problem of common sense, rather than maxima and minima (refer to introduction of this topic.)

29. Drawing the curves tells us that it will intersect at one and only one point.

31. We know that for a fixed perimeter of polygons, polygon with the more number of sides will have more area with respect to the polygon with the less number of sides. And similarly, if circumference of a circle and perimeter of polygon are same, then the area of circle will be more than area of any polygon. [Circle can be treated to be a polygon of n sides, n tending to infinity].

32. Go through the options.

34. Make a quadratic equation in the format of $z^2 + y$. Minimum value of $z^2 = 0$, so, minimum value = y.

36. $\dfrac{(a^2+3a+a)(b^2+3b+1)(c^2+3c+1)}{abc}$

$= [a + (1/a) + 3]\ [b + (1/b) + 3]\ [c + (1/c) + 3]$

Minimum value of $[a + (1/a)] = [b + (1/b)]$

$= [c + (1/c)] = 2$

39. Go through the options.

44. At $x = 0$, $y = 1$ is the key. Otherwise also, you don't have to draw the graph. Just check the options using the points given on the graph.

45. Since no part of the graph is negative, function has to involve an overall modulus. So, it cannot be $f(x)$ or $j(x)$. Now, this graph is having two points on the +ve side of X-axis.

Checking option (b):

$l(x) = 0 = |\,|x| - 2| - 1|$

Or, $|\,|x| - 2| = 1$

Or, $|x| - 2 = \pm 1$, or, $|x| - 2 = +1$, or, $|x| = 3$, or, $x = \pm 3$

Or, $|x| - 2 = -1$, or, $|x| = 1$, or, $x = \pm 1$

We get four points of x for $l(x) = 0$. So, not possible.

Checking option (d):

$K(x) = |x - 1|\,|x - 3| = 0$, Hence, $x = 1$ or $x = 3$

So, we get two points on +ve X-axis for $k(x) = 0$

Hence, option (d) is the answer.

Otherwise also, since we have eliminated the first three options, the fourth option has to be the answer.

46. Here, we get two values of x, one +ve and other one –ve, for $y = 0$.

Checking options:

Option 1 – $f(x) = (|x| - 1)\ (|x| - 2)$

We are getting four points on X-axis for $f(x) = 0$. So, not possible.

Option 2 – $k(x)$ is given in 45. please go through.

Option 3 – $j(x) = (|x| + 1)\ (|x| - 2) = 0$

So, either $(|x| + 1) = 0$ or $(|x| - 2) = 0$

If $(|x| + 1) = 0$, then $|x| = -1$, which is not possible.

If $(|x| - 2) = 0$, then $x = \pm 2$.

Hence, option (c) is the answer.

14

CHAPTER

Logarithm



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Meaning of logarithm and its bases
- Definitions and properties
- Value of the log pertaining to negative and positive values
- Methods of solving questions

Introduction to the topic

Considering logarithm in terms of the number of questions being asked in the CAT, this chapter is quite important. The CAT paper usually has 1–2 questions from this topic. The concept of logarithm is simple and a simple formula is sufficient to solve the questions without getting into lengthy derivations.

LOGARITHM

The logarithmic function is defined as the inverse of an exponential function.

If any number N is expressed in the form a^x then the index 'X' is called the logarithm of the number N to the base 'a'.

Thus, if $N = a^x$

Then $x = \log_a N$.

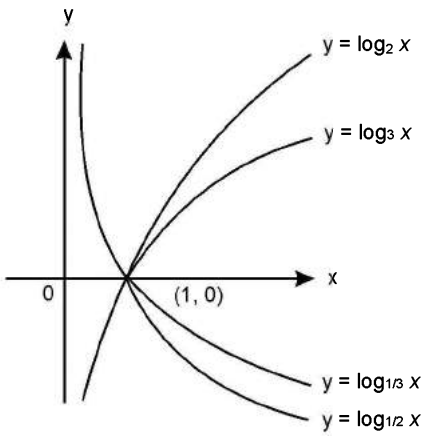
Generally, logarithm of any number is calculated to the base 10. When run base is not mentioned, it should be taken as 10.

Restrictions with logarithm of any number For logarithm of any number to be defined, the number should be greater than zero and base should be positive and not equal to 1.

For $\log_a x$ to be defined, $x > 0$ and $a > 0$ as well $a \neq 1$.

It can also be seen with the help of the graph of $\log x$ (given alongside).

Following observations can be made from this graph:



- Value of y can be negative for some value of x.
- Value of x cannot be negative in any case.
- For constant x, if base is lying in between 0 and 1, then $\log x$ becomes a decreasing function. Otherwise it is an increasing function.

Some Important Properties

In case of all the following properties the standard restrictions on logarithm will be used.

1. $\log_a(XY) = \log_a X + \log_a Y$
   If the restrictions given regarding log of any number are not used, then a good number of contradictions about the numbers can be seen. One of the examples of a similar nature is as follows:
   $\log(12) = \log(-4 \times -3) = \log(-4) + \log(-3)$
   Now on the left hand side, we have a defined value, but on the right hand side the value is not defined. It is due to the fact that $\log_a(XY) = \log_a X + \log_a Y$ is possible only if $X > 0$ and $Y > 0$.
2. $\log_a(X/Y) = \log_a X - \log_a Y$
3. $\log_a(X^k) = k \log_a X$, $\log_a \sqrt[k]{X} = 1/k \times \log_a X$
4. $\log_a 1 = 0$ (As $a^0 + 1$)
5. $\log_X X = 1$
6. $\log_a X = 1/\log_x a$
7. Base change rule
   $\log_a X = \log_b X/\log_b a$
   $\Rightarrow \log_a X \div \log_b a = \log_b X$
8. $a^{(\log_a x)} = x$
9. If $a > 1$ and $X > a$, $\log_a X > 0$
10. If $0 < x < y$ then $\log x < \log y$ (i.e., $\log x$ is an increasing function). In particular if $x > 1$ then $\log x > 0$ and if $0 < x < 1$ then $\log x < 0$.
11. If $a > 1$ and $0 < x < y$ then $\log_a x < \log_a y$ and $a^x < a^y$. If $0 < a < 1$ and $0 < x < y$ then $\log_a x > \log_a y$ and $a^x > a^y$.
12. $\log_a 1 = 0$
13. Log of 0 and negative numbers is not defined.

CHARACTERISTICS AND MANTISSA

The integral part of logarithm is called *Characteristic* and its decimal part is called *Mantissa*. Logarithms to the base 10 are called *Common logarithms*. The characteristic of common logarithm can be found out by a visual inspection. The characteristics of the logarithm (base 10) of a number greater than 1 is less by one than the number of digits in the integral part and is a positive. However, if a decimal fraction number is less than 1 but positive, its characteristic will be greater by unity than the number of consecutives zeros immediately after the decimal point and is a negative.

Example 1 If $5\log_{27}(y) + 2\log_9(81y) = 20$, then y is equal to

(a) 1/7 (b) 81 (c) 729 (d) 243

Solution $5\log_{27}(y) + 2\log_9(81y) = \frac{5}{3}\log_3(y) + \frac{2}{2}\log_3(81y)$

$= \log_3(81) + \log_3\left(y^{\frac{8}{3}}\right)$

$= 4 + \frac{8}{3}\log_3(y) = 20$

Hence, $\log_3 y = 6$

So, $y = 3^6 = 729$

Alternatively, this question can be solved by using options too.

Example 2 If $x \geq y$ and $y > 1$, then the value of the expression $\log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right)$ can never be

(a) -1 (b) -0.5
(c) 0 (d) 1

Solution $P = \log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right)$

$= \log_x X - \log_x y + \log_y y - \log_y y$

$= 2 - \log_x y - \log_y x$

Assume now that $\log_x y = t$

$p = 2 - \frac{1}{t} - t = -\left(\sqrt{t} + \frac{1}{\sqrt{t}}\right)^2$

This can never be equal to 1.

PRACTICE EXERCISES

FOUNDATION

Q.1. If $3^{x+1} = 6^{\log_2 3}$ then x is

(a) 2 (b) 3

(c) $\log_3 2$ (d) $\log_2 3$

Q.2. If $\log_4 5 = a$ and $\log_5 6 = b$ then what is the value of $\log_3 2$?

(a) $\frac{1}{2a+1}$ (b) $\frac{1}{2b+1}$

(c) $2ab + 1$ (d) $\frac{1}{2ab-1}$

Q.3. $\log_5 P \times \log_P x = 2$. What is the value of x?

(a) 125 (b) P^2 (c) 25 (d) $5^{1/2}$

Q.4. Find the value of $3\log\frac{81}{80} + 5\log\frac{25}{24} + 7\log\frac{16}{15}$.

(a) log 2 (b) log 3

(c) 1 (d) None of these

Q.5. If [N] = the greatest integer less than or equal to N, then $[\log_{10} 6730.4]$ is equal to

(a) 6 (b) 4

(c) 5 (d) None of these

Q.6. What is the value of x in the following expression?

$1 - \log_{10} 5 = \frac{1}{3}\left(\log_{10}\frac{1}{2} + \log_{10} x + \frac{1}{3}\log_{10} 5\right)$

(a) 1 (b) $16 \times 5^{-1/3}$

(c) $16 \times 5^{1/3}$ (d) None of these

Q.7. What is the number of zeros coming immediately after the decimal point in the value of $(0.2)^{25}$? ($\log_{10} 2 = 0.30103$)

(a) 15 (b) 16

(c) 17 (d) None of these

Q.8. What is the value of $\log_{32} 27 \times \log_{243} 8$?

(a) $\frac{\log 9}{\log 4}$ (b) $\frac{\log 3}{\log 2}$

(c) log 27 (d) None of these

Q.9. Find the value of $\log_{64}(1/16)$.

(a) –2/3 (b) 2/3

(c) 3/2 (d) –3/2

Q.10. Find the value of $\frac{1}{\log_2 X} + \frac{1}{\log_3 X} + \frac{1}{\log_4 X} + \frac{1}{\log_5 X} + \frac{1}{\log_7 X} + \frac{1}{\log_6 X}$.

(a) 1 (b) $\log_{27} X$

(c) $\log_{5040} X$ (d) $\log_x 5040$

Q.11. If $\log_y N = 5$ and $\log_{2y} 8N = 4$, then find the value of N.

(a) 16 (b) 32

(c) 5 (d) 25

Q.12 If $\log_y^3 N = \log_N^3 y$, then what is the value of $\log_N^3 y$?

(a) 1/3 (b) –1/3

(c) Either 1 or 2 (d) None of these

Q.13. If $\frac{\log X}{\log 4} = \frac{\log 343}{\log 49} = \frac{\log y}{\log 64}$, then what is the value of $x + y$?

(a) 520 (b) 740

(c) 880 (d) Cannot be determined

Q.14. Arrange the following in an ascending order

$A = \log_7 2401, B = \log_7 \sqrt{7} \times 343\ C = \log_{\sqrt{6}} 216, D = \log_2 32.$

(a) ABCD (b) BDCA

(c) BDAD (d) BADC

Q.15. If $\log_{10}\left[1 - \left\{1 - \left(1 - x^2\right)^{-1}\right\}^{-1}\right]^{-1/2} = 1$, then which of the following is the value of x.

(a) 1 (b) 2

(c) 10 (d) None of these

Q.16. What is the value of N in the following expression: $4^{\log 2^8} + 27^{\log 27^{81}} = 144 + \log_{10} N$?

(a) 1 (b) 10

(c) 12 (d) 100

Q.17. What is the value of x if $\log_3 x + \log_9 x + \log_{27} x + \log_{81} x = \frac{25}{4}$?

(a) 9 (b) 27
(c) 81 (d) None of these

Q.18. If $\log_y x = 8$ and $\log_{10y} 16x = 4$, then find the value of y.

(a) 1 (b) 2
(c) 3 (d) 5

Q.19. What is the value of x in the following expression: $\log_7 \log_5 \left[\sqrt{(x+5)} + \sqrt{x}\right] = 0$?

(a) 1 (b) 2
(c) 3 (d) 4

Q.20. What is the value of x in the following expression: $\log_2 (3-x) + \log_2 (1-x) = 3$?

(a) 1 (b) 0
(c) −1 (d) Not possible

Q.21. If $\log N^2 - \log 2N = 3 \log 3 - \log 6$, then what is the value of N?

(a) 1 (b) 2
(c) 9 (d) More than one value

Q.22. What is the value of $\log_{3\sqrt{3}} 27$?

(a) 2 (b) 3 (c) 4 (d) 5

Q.23. Evaluate $\log\left(36\sqrt{6}\right)$ to the base 6.

(a) 1/2 (b) 5/2
(c) 3/2 (d) 7/2

Q.24. What is the value of P if $\log_e 2 . \log_p 625 = \log_{10} 16 . \log_e 10$?

(a) 2 (b) 4
(c) 5 (d) 7

Q.25. If $3\log_{(3x^2)} 27 - 2\log_{(3x)} 9 = 0$, then what is the value of x?

(a) 1/243 (b) 1/7
(c) 1/49 (d) None of these

MODERATE

Q.1. N = n !, where $n>2$. Find the value of $(\log_2 N)^{-1} + (\log_3 N)^{-1} + (\log_4 N)^{-1} + \ldots (\log_n N)^{-1}$.

(a) 0 (b) 1
(c) 10 (d) N

Q.2. What is the value of x in the following expression? $x + \log_{10}(1 + 2^x) = x \log_{10} 5 + \log_{10} 6$.

(a) 1 (b) 0
(c) −1 (c) 3

Q.3. If $a_1, a_2, a_3, \ldots$ are positive numbers in GP then $\log a_n$, $\log a_{n+1}$, $\log a_{n+2}$ are in

(a) AP (b) GP
(c) HP (d) None of these

Q.4. If $x = \log_a (bc)$, $y = \log_b (ca)$ and $z = \log_c (ab)$ when which of the following is equal to 1?

(a) $x + y + z$
(b) $(1 + x)^{-1} + (1 + y)^{-1} + (1 + z)^{-1}$
(c) xyz
(d) None of these

Q.5. If $\log x : \log y : \log z = (y - z) : (z - x) : (x - y)$ then

(a) $X^y . Y^z . Z^x = 1$ (b) $X^x Y^y Z^z = 1$
(c) $x\sqrt{x}, y\sqrt{y}, z\sqrt{z} = 1$ (d) None of these

Q.6. If $\log_{0.04} (x - 1) \geq \log_{0.2} (x - 1)$ then x belongs to the interval

(a) (1, 2] (b) $(-\infty, 2]$
(c) $[2, +\infty)$ (d) None of these

Q.7. If $\log_k x . \log_5 k = 1 k \neq 1$, $k > 0$, then x is equal to

(a) k (b) $\frac{1}{5}$
(c) 5 (d) None of these

Q.8. If $\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b}$ then $a^a b^b c^c$ is

(a) 0 (b) 1 (c) 0.5 (d) 2

Q.9. If $x > 1$, $y > 1$, $z > 1$ are three numbers in GP then $\frac{1}{1+\log x}, \frac{1}{1+\log y}, \frac{1}{1+\log z}$ are in

(a) AP (b) HP
(c) GP (d) None of these

Q.10. If $\log_k N = 6$, and $\log_{25k}(8N) = 3$, then k is

(a) 12.5 (b) $(12.5)^2$
(c) $\frac{5^{\frac{2}{3}}}{2}$ (d) $(12.5)^{\frac{2}{3}}$

Q.11. If a, b and c are distinct positive numbers ($\neq 1$) such that

$(\log_b a \log_c a - \log_a a) + (\log_a b \log_c b - \log_b b) + (\log_a c \log_b c - \log_c c) = 0$. What is the value of abc?

(a) 1 (b) 0
(c) −1 (d) None of these

Q.12. If $4^{\log_9 3} + 9^{\log_2 4} = 10^{\log_x 83}, x \varepsilon R$, how many values can x take?
(a) 0 (b) 3
(c) 2 (d) None of these

Q.13. How many values of x ($x > 1$) satisfy the following equation:
$\log_2 \times .\log_4 \times .\log_6 x = \log_2 x.\log_4 x + \log_2 x.\ \log_6 x + \log_4 x.\log_6 x$?
(a) 0 (b) 1
(c) 2 (d) More than 2

Q.14. If $a^{\log b} = k\, b^{\log a}$, find the value of $\log_{ab} k$, where $a, b > 1$.
(a) 1 (b) 0
(c) 2 (d) Cannot be determined

Q.15. What is the value of x in the following expression $\log_{3/4} \log_2 (x^2 + 7) \log_{1/4} (x^2 + 7)^{-1} = -2$?
(a) +3 (b) −3
(c) ±3 (d) None of these

Q.16. If $2[\log (x + y) - \log 5] = \log x + \log y$, then what is the value of $x^2 + y^2$?
(a) $20 - xy$ (b) $23xy$
(c) $25 - xy$ (d) $28xy$

Q.17. The logarithm of a number to a certain base is 9. The logarithm of 64 times the number to a base which is 11 times the original base is 6. Find the actual base.
(a) 11/6
(b) 2.33/7
(c) 121/4
(d) Cannot be determined

Q.18. If $\log_{30} 3 = x$ and $\log_{30} 5 = y$, then $\log_8 30$ is equal to
(a) $3(1 - x - y)$ (b) $\dfrac{1}{3(1-x-y)}$
(c) $\dfrac{3}{(1-x-y)}$ (d) $\dfrac{1-x-y}{3}$

Q.19. What is the value of $\dfrac{1}{\log_2 n} + \dfrac{1}{\log_3 n} + \ldots + \dfrac{1}{\log_{40} n}$?
(a) $\dfrac{1}{\log(40)!n}$ (b) $\log(40!)n$
(c) 1 (d) None of these

Q.20. The number of solutions of $\log_2 (x + 5) = 6 - x$ is
(a) 2 (b) 0
(c) 3 (d) None of these

Q.21. x, y and z are three angles of a triangle. Which of the following set of values of x, y and z satisfies $\log (x \times y \times z) = 3 \log X + 4 \log 2$, given that x, y and z are integers?
(a) 30°, 70°, 80° (b) 30°, 60°, 90°
(c) 20°, 80°, 80° (d) 30°, 50°, 100°

Q.22. What is the value of $\log_3 2.\ \log_4 3.\ \log_5 4 \ldots \log_{16} 15$?
(a) 1/2 (b) 1/3 (c) 2/3 (d) 1/4

Q.23. If $\log_{12} 18 = a$, $\log_{24} 54 = b$, find the value of $ab + 5(a - b)$.
(a) −1 (b) 1 (c) 0 (d) 2

Q.24. What is the value of x in the following expression?
$x \log_{10} (10/3) + \log_{10} 3 = \log_{10} (2 + 3^x) + x$
(a) 1 (b) −2 (c) 0 (d) 2

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(d)	(c)	(a)	(d)	(b)	(c)	(d)	(a)	(d)	(b)	(c)	(a)	(d)	(c)	(b)	(d)	(b)	(d)	(c)
Q. No.	21	22	23	24	25															
Answer	(c)	(a)	(b)	(c)	(a)															

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(a)	(a)	(b)	(b)	(c)	(c)	(b)	(b)	(a)	(a)	(d)	(b)	(b)	(c)	(c)	(c)	(b)	(a)	(d)
Q. No.	21	22	23	24	25															
Answer	(c)	(d)	(b)	(c)	(a)															

HINTS AND SOLUTIONS

Foundation

1. $3^{x+1} = (3\times 2)^{\log_2 3}$

$\Rightarrow 3^{x+1} = 3^{\log_2 3} \times 2^{\log_2 3}$

$\Rightarrow 3^{x+1} = 3^{\log_2 3} \times 3$

$\Rightarrow 3^{x} = 3^{\log_2 3} \quad \Rightarrow x = \log_2 3$

2. $\log_4 5 = a$ and $\log_5 6 = b$

$\Rightarrow \log_4 5 \times \log_5 6 = ab$

$\Rightarrow \log_4 6 = ab \Rightarrow \frac{1}{2}\log_2 6 = ab$

$\Rightarrow (1 + \log_2 3) = 2ab$

$\therefore \log_2 3 = 2ab - 1$

$\therefore \log_3 2 = \frac{1}{2ab-1}$

3. $\log_5 P \times \log_P x = 2$

$\Rightarrow \log_5 x = 2$

$\therefore x = 5^2 = 25$

4. $\log\left(\frac{81}{80}\right)^3 + \log\left(\frac{25}{24}\right)^5 + \log\left(\frac{16}{15}\right)^7$

$= \log\left[\frac{3^{12}\times 5^{10}\times 2^{28}}{2^{12}\times 5^3\times 2^{15}\times 3^5\times 3^7\times 5^7}\right]$

$= \log 2$

5. $[\log_{10} 6730.4] = [\log_{10} 6.7304 \times 10^3]$

$= [3 + \log_{10} 6.7304]$

$= 3$ [Value of $\log_{10} n$ will lie in between 0 and 1 when $1<n<10$]

6. $1 - \log_{10} 5 = \frac{1}{3}\left[\log_{10}\left(\frac{x.5^{1/3}}{2}\right)\right]$

$= 3[\log_{10} 10 - \log_{10} 5] = \log_{10}\left(\frac{x.5^{1/3}}{2}\right)$

$\Rightarrow \log_{10} 2^3 = \log_{10}\left(\frac{x.5^{1/3}}{2}\right)$

$\Rightarrow 2^3 = \frac{x.5^{1/3}}{2}$

$\therefore x = 16 \times 5^{-1/3}$

7. $\log (0.2)^{25} = 25 \log\left(\frac{2}{10}\right)$

$= 25(\log 2 - \log 10)$

$= -25(1 - 0.30103)$

$= -25 \times 0.69897 = -17.47425$

$\therefore$ Number of zeros after the decimal point is 17.

8. $\log_{32} 27 \times \log_{243} 8 = \log_{2^5} 3^3 \times \log_{3^5} 2^3$

$= \frac{3}{5}\log_2 3 \times \frac{3}{5}\log_3 2$

$= \left(\frac{3}{5}\right)^2 \log_2 3 \times \log_3 2 = \frac{9}{25}$

9. $\log_{64}(1/16) = \log_{4^3} 4^{-2} = \frac{-2}{3}$

10. Expression is equal to $\log_x 2 \times 3 \times 4 \times 5 \times 6 \times 7 = \log_x 5040$

11. Go through options.

14. $A = \log_7 2401 = 4$

$B = \log_7(\sqrt{7}.343)$

$\Rightarrow B = \log_7(7^{1/2} \times 7^3) = 3\frac{1}{2}$

$C = \log_{\sqrt{6}} 216 = 2.\log_6 6^3 = 3$

$D = \log_2 32 = 5$

$\therefore$ Arranging in ascending order: $BADC$

15. $\log_{10}\left[1-\left\{1-(1-x^2)^{-1}\right\}^{-1}\right]^{-\frac{1}{2}}=1$

$\Rightarrow \log_{10}\left[1-\left\{1-\frac{1}{1-x^2}\right\}^{-1}\right]^{-\frac{1}{2}}=1$

$\Rightarrow \log_{10}\left[1-\left\{\frac{-x^2}{1-x^2}\right\}^{-1}\right]^{-\frac{1}{2}}=1$

$\Rightarrow \left[\frac{1}{x^2}\right]^{-\frac{1}{2}}=10 \Rightarrow x=10$

17. $\log_3 x+\log_9 x+\log_{27} x+\log_{81} x=\frac{25}{4}$

$\Rightarrow \log_3 x+\frac{1}{2}\log_3 x+\frac{1}{3}\log_3 x+\frac{1}{4}\log_3 x=\frac{25}{4}$

$\Rightarrow \log_3 x\,[1+1/2+1/3+1/4]=\frac{25}{4}$

$\Rightarrow \frac{25}{12}\times\log_3 x=\frac{25}{4}$

$\Rightarrow \log_3 x=3 \Rightarrow x=27$

18. $\log_y x=8 \Rightarrow y^8=x$ ..... (1)

$\log_{10y} 16x=4 \Rightarrow 10^4.y^4=16x$ ..... (2)

Dividing (2) by (1) $10^4.y^{-4}=16 \Rightarrow y=5$

19. Go through options. (Use log 1 = 0)

20. Going through options.

Option (a) is not possible.

Checking option (c): For $x=-1$, expression $=\log_2 4+\log_2 2=\log_2 8=3\log_2 2=3$

21. $\log N^2-\log 2N=3\log 3-\log 6$

OR, $\log\left(\frac{N^2}{2N}\right)=\log\frac{3^3}{6}$ or, $\frac{N}{2}=\frac{9}{2}$

$\therefore N=9$

22. $\log_{3\sqrt{3}} 27=\log_{3\sqrt{3}}(3\sqrt{3})^2=2$

23. $\log_6 36\sqrt{6}=\log_6 36+\log_6\sqrt{6}$

$=2+0.5=2.5$

24. $\log_e 2.\log_p 625=\log_{10} 16.\log_e 10$

$\Rightarrow \log_p 625=\log_{10} 2^4.\frac{\log_e 10}{\log_e 2}$

$\Rightarrow \log_p 5^4=4\log_{10} 2.\log_2 10$

or, $4\log_p 5=4$

$\therefore \log_p 5=1 \Rightarrow P=5$

25. $3\log_{3x^2} 27-2\log_{3x} 9=0 \Rightarrow 3\log_{3x^2} 3^3-2\log_{3x} 3^2=0$

$\Rightarrow 9\log_{3x^2} 3=4\log_{3x} 3$

$\Rightarrow \frac{9}{\log_{3x} 3}=\frac{4}{\log_{3x^2} 3}$

$\Rightarrow 9\log_3 3x=4.\log_3 3x^2 \Rightarrow \log_3(3x)9=\log_3(3x^2)^4$

$\Rightarrow 3^9\times x^9=3^4\times x^8 \Rightarrow x=3^{-5}$

$\therefore x=\frac{1}{243}$

Moderate

1. $N=n!$ (given)

$\frac{1}{\log_2 N}+\frac{1}{\log_3 N}+\frac{1}{\log_4 N}+\ldots\ldots\ldots\frac{1}{\log_n N}$

$=\log_N 2+\log_N 3+\log_N 4+\ldots\ldots+\log_N n$

$=\log_N n!=\log_N N=1$

2. $x+\log_{10}(1+2^x)=x\log_{10} 5+\log_{10} 6$

$\Rightarrow \log_{10} 10^x+\log_{10}(1+2^x)=\log_{10} 5^x.6$

$\Rightarrow 10^x(1+2^x)=5^x.6 \Rightarrow 2^x(1+2^x)=6$

$\therefore 2^x=2$, so $x=1$.

3. $a_1=a_1$; $a_2=a_1r$; $a_3=a_1r^2$; $a_4=a_1r^3$ are in G.P. with common difference 'r'

$\therefore \log a_1=\log a_1$

$\log a_2=\log a_1+\log r$

$\log a_3=\log a_1+2\log r$

$\log a_4=\log a_1+3\log r$

$\log a_{n+1}=\log a_1+n\log r$

$\log a_{n+2}=\log a_1+(n+1)\log r$

$\therefore \log a_1, \log a_2, \log a_3, \log a_4, \ldots\ldots\ldots \log a_{n+2}$ are in A.P. with common difference '$\log r$'.

Alternatively,

Assume $a_1=10$, $a_2=100$ and $a_3=1000$.

To find out about $\log a_n$, $\log a_{n+1}$, $\log a_{n+2}$, take $n=1$.

Hence we have to find that log 10, log 100, log 1000 are in which series.

log 10 = 1, log 100 = 2 and log 1000 = 3

So, log 10, log 100, log 1000 are in AP.

4. $x=\log_a bc \quad \therefore 1+x=\log_a abc$

$y=\log_b ca \quad \therefore 1+y=\log_b abc$

$z=\log_c ab \quad \therefore 1+z=\log_c abc$

$\therefore \frac{1}{1+x}+\frac{1}{1+y}+\frac{1}{1+z}=\log_{abc} a+\log_{abc} b+\log_{abc} c$

$=\log_{abc} abc=1$

5. $\log x : \log y : \log z = (y-z) : (z-x) : (x-y)$

$\therefore \frac{\log x}{y-z} = \frac{\log y}{z-x} = \frac{\log z}{x-y} = k$ (say)

$\therefore \log x = k(y-z)$

$\Rightarrow x \log x = kx(y-z)$

Similarly $y \log y = ky(z-x)$

$z \log z = kz(x-y)$

Adding all 3 equations

$x \log x + y \log y + z \log z = 0 \Rightarrow \log X^x . Y^y . Z^z = 0$

$\therefore X^x . Y^y . Z^z = 1$

6. $\log_{0.04}(x-1) \geq \log_{0.2}(x-1)$

$\Rightarrow \frac{\log_e(x-1)}{\log_e 0.04} \geq \frac{\log_e(x-1)}{\log_e 0.2}$

$\Rightarrow \frac{\log_e(x-1)}{\log_e (0.2)^2} \geq \frac{\log_e(x-1)}{\log_e 0.2}$

$\Rightarrow \frac{\log_e(x-1)}{2\log_e(0.2)} \geq \frac{\log_e(x-1)}{\log_e(0.2)}$

$\because \log_e 0.2 < 0$

$\therefore \log_e (x-1) \leq 2 \log_e (x-1)$

$\therefore x - 1 \geq 1$

$\therefore x \geq 2 \Rightarrow x \in [2, \infty]$

7. $\log_k x . \log_5 K = 1$

$\Rightarrow \log_5 x. = 1 \Rightarrow x = 5$

9. Solution 1:

x, y, z are in G.P.

$\therefore \log x, \log y, \log z$ are in A.P. as shown in solution of question No. 3 in moderate section

$\therefore 1 + \log x, 1 + \log y, 1 + \log z$ are in A.P.

$\therefore \frac{1}{1+\log x}, \frac{1}{1+\log y}, \frac{1}{1+\log z}$ are in H.P.

Solution 2:

Since the numbers are in GP, let us assume the numbers x, y and z to be equal to 10, 100 and 1000.

Then, $\frac{1}{1+\log x} = \frac{1}{2}$, $\frac{1}{1+\log y} = \frac{1}{3}$ and $\frac{1}{1+\log z} = \frac{1}{4}$

These are in HP.

10. $\log_K N = 6 \quad \Rightarrow K^6 = N$

$\log_{25K} 8N = 3 \quad \Rightarrow (25K)^3 = 8N$

$\therefore \frac{25^3 K^3}{K^6} = 8 \Rightarrow 25^3 . K^{-3} = 2^3$

$\therefore K = \frac{25}{2} = 12.5$

11. $\log_b a \log_c a + \log_a b \log_c b + \log_a c \log_b c = 3$

$\Rightarrow \frac{(\log a)^2}{\log b . \log c} + \frac{(\log b)^2}{\log a \log c} + \frac{(\log c)^2}{\log a \log b} = 3$

$\Rightarrow \frac{(\log a)^3 + (\log b)^3 + (\log c)^3}{\log a . \log b . \log c} = 3$

$\Rightarrow (\log a)^3 + (\log b)^3 + (\log c)^3 = 3 \log a . \log b . \log c$

We know, $x^3 + y^3 = z^3 = 3xyz$ when $x + y + z = 0$

$\log a + \log b + \log c = 0 \Rightarrow \log abc = 0 \Rightarrow abc = 1.$

13. $\log_2 x . \log_4 x . \log_6 x = \log_2 x . \log_4 x + \log_2 x . \log_6 x + \log_4 x . \log_6 x$

Dividing LHS and RHS by $\log_2 x . \log_4 x . \log_6 x$

$\Rightarrow 1 = \frac{1}{\log_6 x} + \frac{1}{\log_4 x} + \frac{1}{\log_2 x}$

$\Rightarrow 1 = \log_x 6 + \log_x 4 + \log_x 2$

$\Rightarrow \log_x 6 \times 4 \times 2 = 1$

$\Rightarrow x = 48. \therefore$ only 1 value of x.

14. $a^{\log b} = kb^{\log a}$

$\Rightarrow \log b . \log a = \log k + \log a . \log b$ {taking log on both sides}

$\Rightarrow \log k = O$

$\therefore K = 1$

$\therefore \log_{ab} k = O$

15. Go through the options.

16. $2[\log(x+y) - \log 5] = \log x + \log y$

$\Rightarrow \log \frac{(x+y)^2}{5^2} = \log xy$

$\Rightarrow \frac{(x+y)^2}{25} = xy \quad \Rightarrow x^2 + y^2 + 2xy = 25xy$

$\therefore x^2 + y^2 = 23xy$

17. Let the base be 'b'.

$\therefore \log_b N = 9$ and $\log_{11b} 64N = 6$

$\Rightarrow N = b^9$ and $64N = (11b)^6$

$\therefore 64 = \frac{11^6 b^6}{b^9} \qquad \therefore b^3 = \frac{11^6}{64}$

$\therefore b = \frac{11^2}{2^2} = \frac{121}{4}$

18. $\log_{30} 3 = x, \log_{30} 5 = y$

$\therefore x + y = \log_{30} 15$

$\Rightarrow x + y = \log_{30}(30/2)$

$\Rightarrow (x+y) = 1 - \log_{30} 2 \Rightarrow \log_{30} 2 = 1 - x - y$

$\Rightarrow 3 \log_{30} 2 = 3(1 - x - y) \Rightarrow \log_{30} 8 = 3(1 - x - y)$

$\therefore \log_8 30 = \frac{1}{3(1-x-y)}$

19. $\frac{1}{\log_2 n}+\frac{1}{\log_3 n}+\frac{1}{\log_4 n}+....\frac{1}{\log_{40} n}$

$= \log_n 2 + \log_n 3 + \log_n 4 + ... + \log_n 40$

$= \log_n (40!)$

$= \frac{1}{\log_{(40!)} n}$

20. $\log_2(x + 5) = 6 - x$

Draw the graph of $\log_2(x + 5)$ and $(6 - x)$:

$\therefore$ The two graphs will intersect at only one point.

21. It can be simply seen that in the expression $\log (x \times y \times z) = 3 \log X + 4 \log 2$, when we have replaced y and z on the LHS, answer is in constant with a term of x only. It means y and z has to be replaceable with each other. Only option 3 satisfies this relationship.

22. $\log_3 2.\log_4 3.\log_5 4.....\log_{16} 15.$

$= \frac{\log 2}{\log 3}.\frac{\log 3}{\log 4}.\frac{\log 4}{\log 5}.....\frac{\log 15}{\log 16} = \frac{\log 2}{\log 16} = \frac{\log 2}{\log 2^4}$

$= \frac{\log 2}{4\log 2} = \frac{1}{4}$

23. Put the values of a and b in $ab + 5(a - b)$ after converting them to same base.

24. Use options to solve this question.

25. Use options to solve this question.

15

CHAPTER

Permutation and Combination



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Difference between permutation and combination
- Different cases of permutation such as word formation
- Different cases of combination such as selection of a team
- Application of P and C to derive more formulae
- Kinds of questions asked in the CAT
- Methods of solving questions

Introduction to the topic

Permutations and combinations has lately emerged as an important topic for management entrance examinations. This is primarily because questions from this topic require analytical skills and a logical bent of mind. Even students who do not have math as a subject can handle them if they have a fairly good understanding of the concepts and their application. However, a point to note is that questions from permutations and combinations asked in the CAT are not based on mathematical concepts. Hence anyone who is well-versed in different methods of counting and basic calculations will be able to solve these problems easily.

MEANING OF PERMUTATION AND COMBINATION

If we go by the dictionary meaning of the words permutation and combination, then permutation is the number of ways in which a set or a number of things can be put in an order or arranged and; combination refers to the number of ways in which a group of things can be chosen from a larger group without regard to their arrangement.

Let us go through an example. Suppose there are four different batsmen A, B, C and D and we have to select a group of three batsmen out of these four. Now, we can select any combination of three batsmen so that no set of batsmen has all the same three batsmen. These set of batsmen will be — ABC, BCD, ABD, ACD. This is a case of combination as for every set of selection of three batsmen, order of selection does not play any role (i.e., we can select anybody—first or second or third—and it does not create any difference in the final selection as well as in the total number of selections).

Now, if we try to define their batting order also, i.e., who bats first and second and so on, then corresponding to every selection of a set of three batsmen, we will have six different arrangements of their batting order. It can be seen below that corresponding to the selection of ABC as a team, following is the list of different batting orders:

ABC, ACB, BAC, BCA, CAB, CBA.

This is a case of permutation since the order of occurrence has become important. Since there are four different ways of selecting a group of three batsmen and every selection can be arranged in 6 different ways, so the total number of ways of arranging 3 batsmen (or, distinct things) out of 4 batsmen (or, distinct things) = $4 \times 6 = 24$ ways.

Permutation and combination can be better understood through the examples of hand-shake and gifts exchange also. Assume that there are 20 persons in a party and everybody shakes hand with each other and also presents a gift. Now if we take a case of two persons A and B, then the event of shaking hand between them is a case of combination because when A shakes hand with B or B shakes hand with A, the number of hand shake is just one. So, there is no order as such and hence it is a case of combination.

Similarly, the event of presenting the gift is a case of permutation because the gift given to B by A and the gift given to A by B are two different gifts. So, the order of case plays a role here and hence it is a case of permutation.

$n!$ = Product of all the natural numbers from n to $1 = n(n-1)(n-2)(n-3)\ldots \times 3 \times 2 \times 1$.
$0! = 1$

Factorials are defined only for whole numbers, and not for negative numbers or fractions (≠ whole numbers).

FUNDAMENTAL PRINCIPLES OF COUNTING: TWO BASIC THEOREMS

1. Multiplication theorem

If there are two jobs in such a way that one of them can be done in m ways and when it is completed in any of the m ways, the second job can be completed in n ways, then the whole job can be done in $m \times n$.

2. Addition theorem

If there are two jobs in such a way that one of them can be done in m ways and the second one can be done in n ways independently, then either of the jobs can be done in $(m + n)$ ways.

Basically, there is one point where these two theorems differ—in multiplication the job does not get completed while in addition it gets completed. In a layman's language, we multiply the number of ways when the job has not been completed and we add the number of ways when the job has been completed.

Example 1 There are 10 girls and 15 boys in a class. In how many ways can

i. a class representative be selected?
ii. a team of two students be chosen with one girl and one boy?

Solution

i. A class representative can be a girl or a boy. Now, one girl can be selected from 10 girls in 10 ways (any of the girls can be selected) and one boy can be selected from 15 boys in 15 ways (any of the boys can be selected). So the ways of selecting a class representative includes either selecting a boy or a girl. So, the total number of ways of selecting a class representative $=10 + 15 = 25$
In this case, the moment a girl gets selected, the job is completed. There are some more ways of doing this by selecting a boy. So, it is a case of addition.

ii. One girl can be chosen from 10 girls in 10 ways. Now corresponding to every selected girl, any one of the 15 boys can be selected in 15 ways.
It can be seen in the following presentation:
Girl selected (Assume name of the girls are $G_1, G_2, G_3, \ldots G_9, G_{10}$) – G_1

Boy selected (Assume the names of the boys are $B_1, B_2, B_3, \ldots B_{14}, B_{15}$)– B_1, Or B_2, or B_3, or B_{15}.

So, corresponding to G_1, the total number of selection of a boy = 15

Corresponding to G_2, the total number of selection of a boy = 15

Corresponding to G_3, the total number of selection of a boy = 15

… … … … …

Corresponding to G_{15}, the total number of selection of a boy = 15

So, the total number of ways of selecting a team of one boy and a girl = the total number of ways of selecting a girl × the total number of ways of selecting a boy = 10 × 15 = 150

In this case, just by selecting a girl or a boy, work has not been completed. So, it is a case of multiplication.

Another example of multiplication theorem If there are three cities A, B and C located in such a way that there are 3 roads joining A and B, and 4 roads joining B and C, then the number of ways one can travel from A to C is 3 × 4, i.e., 12.

PERMUTATIONS

As we have seen, the arrangements made by taking some or all elements out of a number of things is called a permutation. Permutation implies "arrangement" where "order of the things" is important.

The permutations of three things, *a b* and *c*, taken two at a time are *ab, ba, ac, ca, cb* and *bc*. Since the order in which the things are taken is important, *ab* and *ba* are counted as two different arrangements.

The number of permutations of *n* things taking *r* at a time is denoted by ${}^{n}P_{r}$.

COMBINATIONS

As we have seen, the groups or selections made by taking some or all elements out of a number of things is called a combination. In combination, the order in which the things are taken is not important.

The combination of three things, *a*, *b* and *c*, taken two at a time are *ab*, *bc* and *ca*. Here, *ab* and *ba* are same because the order in which *a* and *b* are taken is not important. What is required is only a combination including *a* and *b*. The words "combination" and "selection" can be used without any differentiation.

The number of combinations of *n* things taking *r* at a time is denoted by ${}^{n}C_{r}$.

Approaching a Problem

Mostly the questions asked in the CAT are self-explanatory, i.e., they clearly mention what process is to be used—permutation or combination.

In case, the question does not specify this, you should try to find out whether it is a case of permutation or a case of combination. Sometimes the problem very clearly states whether it is the number of permutations (or arrangements) or the number of combinations (or selections) that has to be found out. The questions can be as follows:

For permutations:

"What is the number of permutations that can be done … " or "What is the number of arrangements that can be made…" or "Find the different numbers of ways in which something can be arranged… etc.

For combinations:

"What is the number of combinations that can be done … " or "What is the number of selections that can be made … " or "Find the different numbers of ways in which things can be selected etc.

Some other standard examples of permutation and combination are:

Permutation Word formation, number formation and circular permutation etc.

Combination Selection of a team, forming geometrical figures, distribution of things (except some particular cases).

Still, sometimes the questions may not explicitly state what you have to find—permutation or combination. In that case, the nature of what is to be found out will decide whether it is the number of permutations or the number of combinations.

See the example given below

I have to invite two of my eight friends to my anniversary party. In how many different ways can I do this?

Assume my eight friends are A, B, C, D, E, F, G and H. Whether the two friends that I call for the party are A and B or B and A, does not make any difference. As we have discussed before, that what matters most in case of permutation is the order of occurrence of things. As order does not play any role here, it is clearly the case of combination.

Meaning and Derivation of ${}^{n}P_{r}$ and ${}^{n}C_{r}$

Number of permutations of n different things taking r at a time = ${}^{n}P_{r}$

In this statement, we take the following two assumptions:

- All the *n* things are distinct (or no two things are of the same type)
- Each thing is used at most once (i.e., nothing is repeated in any arrangement)

Let us assume that there are *r* boxes and each of them can hold one thing. When all the *r* boxes are filled, what we have is an arrangement of *r* things taken from the given *n* things. So, each time we fill up the *r* boxes with things taken from the given *n* things, we have an arrangement of *r* things taken from the given *n* things without repetition. Hence, the number of ways in which we can fill up the *r* boxes by taking things from the given *n* things is equal to the number of permutations of *n* things taking *r* at a time.

The first box can be filled in *n* ways (because this box can be filled by any one of the *n* things given). After filling the first box, we now have only $(n-1)$ things to fill the second box; any one of these things can be used to fill the second box and hence the second box can be filled in $(n-1)$ ways; Similarly,

the third box can be filled in $(n-2)$ ways and so on the rth box can be filled in $(n-(r-1)$ ways, i.e., $[n-r+1]$ ways. Hence, from the fundamental rules of counting, all the r boxes together can be filled up in $\rightarrow n.\ (n-1).\ (n-2)\ldots\ (n-r+1)$ ways

Hence, ${}^nP_r = n.\ (n-1).\ (n-2)\ldots\ (n-r+1)$

This can be simplified by multiplying and dividing the right hand side be $(n-r)\ (n-\overline{r-1})\ldots 3.2.1.$

$${}^nP_r = n\ (n-1)(n-2)\ldots\ [n-\overline{r-1}]$$

$$= \frac{n!}{(n-r)!}$$

The number of arrangements of n distinct things taken r things at a time is

$${}^nP_r = \frac{n!}{(n-r)!}$$

If we take n things at a time, then we get ${}^nP_n$. From the discussion similar to that we had for filling the r boxes above we can find that ${}^nP_n$ is equal to $n!$ The first box can be filled in n ways, the second one in $(n-1)$ ways, the third one in $(n-2)$ ways and so on, then the *nth* box in 1 way; hence, all the n boxes can be filled in

$${}^nP_n = n!$$

But if we substitute $r = n$ in the formula for ${}^nP_n$ then we get ${}^nP_n = \frac{n!}{0!}$; since we already found that ${}^nP_n = n!$

We can conclude that $0! = 1$

The number of combinations of n distinct things taking r at a time $= {}^nC_r$

Let the number of combinations ${}^nC_r$ be S. Consider one of these S combinations. Since this is a combination, the order of the r things is not important. If we now impose the condition that order is required for these r things, we can get $r!$ arrangement from this one combination. So each combination can give rise to $r!$ permutations. S combinations will thus give rise to $S \times r!$ permutations. But since these are all permutations of n things taking r at a time, this must be equal to ${}^nP_r$. So,

$$S \times r! = {}^nP_r = \frac{n!}{(n-r)!}$$

$$\text{So, } S = {}^nC_r = \frac{n!}{(n-r)!} \times \frac{1}{r!}$$

It can also be deduced from here that the number of selections of n distinct things taken all at a time will be equal to 1 (since there is only one way in which all the articles can be selected).

$$\text{Alternatively } {}^nC_n = \frac{n!}{0! \times n!} = 1$$

Out of n things kept in a bag, if we select r things and remove them from the bag, we are left with $(n-1)$ things inside the bag i.e., whenever r things are selected out of n things, we automatically have another selection of $(n-1)$ things. Hence, the number of ways of making combinations taking r out of n things is the same as selecting $(n-r)$ things out of n given things, i.e.,

$${}^nC_r = {}^nC_{n-r}$$

Before we move ahead, let us once again make it clear that whenever we are using ${}^nC_r$ and ${}^nP_r$, our assumption is that all the things are distinct, i.e., no two of them are same.

Example 2 Munchun has 10 children. She takes 3 of them to the zoo at a time, as often as she can, but she does not take the same three children to the 0 more than once. How many times Munchun will be required to go to the zoo?

(a) 120 (b) 45 (c) 90 (d) 180

Solution Number of times (read ways) 3 children (read distinct things) can be selected from 10 children (read distinct things) $= {}^{10}C_3$.

So, she will be required to go to the zoo ${}^{10}C_3$ times.

So, option (a) is the answer.

Example 3 In the above question, how many times a particular child will go?

(a) 72 (b) 45 (c) 90 (d) 36

Solution Consider the case for any particular child C_1 Since C_1 has already been selected, so out of the rest 9 children Munchun will be required to select 2 more children. This can be done on ${}^9C_2$ ways.

So, option (d) is the answer.

Example 4 In a chess competition involving some boys and girls of a school, every student had to play exactly one game with every other student. It was found that in 45 games both the players were girls, and in 190 games both were boys. The number of games in which one player was a boy and the other was a girl is

(a) 200 (b) 216 (c) 235 (d) 256

Solution Let there be m boys and n girls

$${}^nC_2 = 45 = \frac{n(n-1)}{2} \Rightarrow n(n-1) = 90 \Rightarrow n = 10$$

$${}^mC_2 = 190 = \frac{m(m-1)}{2} = 190 \Rightarrow m\ (m-1) = 380 \Rightarrow m = 20$$

Number of games between one boy and one girl

$= {}^{10}C_1 \times {}^{20}C_1 = 10 \times 20 = 200$

Hence, option (a) is the answer.

Example 5 In how many ways can three persons be seated on five chairs?

Solution This question is a very fundamental problem of arrangements without repetition. The first person can sit in 5 ways (into any of the five chairs), the second person can take place in 4 ways (into any of the remaining 4 chairs) and the third person can sit in 3 ways.

So, the total number of ways in which these 3 persons can arrange themselves on 5 chairs is $5 \times 4 \times 3 = 60$.

Some Important Derivations

While deriving an expression for ${}^nP_r$, we imposed two constraints, viz. distinct things and repetition being not allowed over it and learned how to find the number of permutations. Let us now see what will happen if we do not impose these two restrictions on ${}^nP_r$.

Number of arrangements of *n* things of which *p* are of one type, *q* are of a second type and the rest are distinct When all the things are not distinct, then we cannot use the general formula for ${}^nP_r$ for any value of r. If we want to find out ${}^nP_r$ for a specific value of r in that given problem, we will be required to use it on the basis of the given situation.

The number of ways in which n things may be arranged taking all of them at a time, when p of the things are exactly alike of one kind, q of them exactly alike of another kind, r of them exactly alike of a third kind, and the rest all is distinct is

$$\frac{n!}{p!.q!.r!}$$

Number of permutations of *n* distinct things where each one of them can be used for any number of times (i.e., repetition allowed) Derivation for this is based upon common sense. If I have 5 friends and 3 servants and I have to send the invitation letters to all my friends through any of my servants, I obviously have 3 options for the invitation card to be sent to friend1, the same 3 options for the invitation card to be sent to friend2, and similarly 3 options for the invitation card to be sent to each of the friends. So the total number of ways of sending the invitation letters = 3^5 And it will not be 5^3, as friends are not going to the servants to get the letter.

In general, the number of perambulations of n things, taking r at a time when each of the thing may be repeated once, twice,... up to r times in any arrangement is n^r.

Total Number of Combinations

Out of n things, the number of ways of selecting one or more things:

where we can select 1 or 2 or 3 ... and so on n things at time; hence, the number of ways is ${}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots {}^nC_n = 2^n - 1$, where n is the number of things.

Above derivation can also be understood in the following manner:

Let there be n bags.

The first bag can be dealt in two ways—it is either included or not included. Similarly, the second bag can be dealt in two ways, the third one in two ways and so on, the *nth* bag in two ways. Using multiplication theorem of counting, the number of ways of dealing with all the bags together is $2 \times 2 \times 2 \times \ldots n$ times $= 2^n$ ways. But out of these, there is one combination where we do not include any of the bags. This is not allowed because we have to select at least one thing.

Hence, the number of ways of selecting one or more things from n given things is $2^n - 1$.

Distributing the given things ($m + n$) into two groups where one group is having m things and other one *n* things If we select m things (which can be done in ${}^{m+n}C_m$ ways), then we will be left with n things, i.e., we have two groups of m and n things respectively. So, the number of ways of dividing $(m + n)$ things into two groups of m and n things respectively is equal to ${}^{m+n}C_m$.

$$^{m+n}C_m = \frac{(m+n)!}{n!.m!}$$

If we take $m = n$, then the above expression will denote "Distributing 2m things equally between two distinct groups $= {}^{2m}C_m = \frac{(2m)!}{m!.m!}$

However, when the groups are identical, then we will be required to divide the above result by 2!.

Hence, in that case it becomes $\frac{(2m)!}{2!m!.m!}$

(Refer to word formation examples)

All the above derivations with their different applications can be seen below in a summarized form.

1. Fundamental Principle of counting:
   (a) *Multiplication rule* If a work is done only when all the number of works are done, then the number of ways of doing that work is equal to the product of the number of ways of doing separate works.
   (b) *Addition rule* If a work is done only when any one of the number of works is done, then the number of ways of doing that work is equal to the sum of the number of ways of doing separate works.
   Thus, if a work is done when exactly one of the number of works is done, then the number of ways of doing this work = sum of the number of ways of doing all the works.
2. If ${}^nC_x = {}^nC_y$ then either $x = y$ or $x + y = n$.
3. $\lfloor n = 1.2.3 \ldots n$; $\lfloor 0 = 1$
4. (a) The number of permutations of n different articles taking r at a time is denoted by ${}^nP_r$ and ${}^nP_r = \frac{\lfloor n}{\lfloor n-r}$
   (b) The number of permutations of n different articles taking all at a time = $\lfloor n$.
   (c) The number of permutations of n articles, out of which p are alike and are of one type, q are alike and are of second type and rest are all different = $\frac{\lfloor n}{\lfloor p = \lfloor q}$.

5. The number of permutations (arrangements) of n different articles taking r at a time when articles can be repeated any number of times = $n \times n \times \ldots$ r times = n^r.
6. Circular permutations:
   i. The number of circular permutations (arrangements) of n different articles = $\lfloor n-1$.
   ii. The number of circular arrangements of n different articles when clockwise and anticlockwise arrangements are not different i.e., when the observation can be made from both the sides $= \frac{\lfloor n-1}{2}$.
7. The number or combinations of n different articles taking r at a time is denoted by ${}^nC_r$ and ${}^nC_r = \frac{\lfloor n}{\lfloor r \lfloor n-r}$.
8. The number of selections of r articles ($r \leq n$) out of n identical articles is 1.
9. Total number of selections of zero or more articles from n distinct articles = ${}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots {}^nC_n = 2^n$.
10. Total number of selections of zero or more articles from n identical articles = 1 + 1 + 1 + ... to (n + 1) terms = n + 1.
11. The number of ways of distributing n identical articles among r persons when each person may get any number of articles = ${}^{n+r-1}C_{r-1}$.
12. The number of ways of dividing m + n different articles in two groups containing m and n articles respectively ($m \neq n$)

$$= {}^{m+n}C_n \times {}^mC_m = \frac{\lfloor m+n}{\lfloor m \lfloor n}.$$

13. The number of ways of dividing $2m$ different articles each containing m articles $= \frac{\lfloor 2m}{\lfloor m \lfloor m \lfloor 2}$.
14. The number of ways of dividing $3m$ different articles among three persons and each is getting m articles = $\frac{\lfloor 3m}{\lfloor m \lfloor m \lfloor m \lfloor 3} \lfloor 3$.
15. The number of ways of selecting n distinct articles taken r at a time when p particular articles are always included = ${}^{n-p}C_{r-p}$.
16. ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$
17. ${}^nP_r = r.{}^{n-1}P_{r-1} + {}^{n-1}P_r$

SOME STANDARD FORMATS OF QUESTIONS

Word Formation

As we know that order of occurrence of letters decide the formation of words, so word formation is one standard example of permutation.

Let us understand with the help of some examples:

Example 6 How many words can be formed with the word 'LUCKNOW' which have

i. no restriction
ii. L as the 1st letter of the word
iii. L and W as the terminal letters
iv. all the vowels together
v. L always occuring before U
vi. L always occuring before U and U always occuring before W

Solution

i. The total number of distinct letters = 7 (L, U, C, K, N, O, W)
So, the total number of words that can be formed = 7!
ii. Now we can arrange only 6 letters (as place of L is restricted),
So, the total number of words that can be formed = 6!
iii. Now we can arrange only 5 letters (as place of L and W are restricted),
So, the number of arrangements = 5!
But the place of L and W can be interchanged between themselves.
So, the total number of words that can be formed = 5! × 2!
iv. U and O should be together, so we will assume these two letters to be tied up with each other.
Now we have 6 distinct things to be arranged– (L, U, O, C, K, N, W)
So, the number of arrangements = 6!
But place of U and O can be interchanged between themselves.
So, the total number of words that can be formed = 6! × 2!
v. There is an equal likelihood occurrence of all the letters in the word, so in half of the cases L will occur before U and in the remaining half, U will occur before O.
So, the total number of words that can be formed = 7!/2
vi. There are six possible arrangements (3!) corresponding to L, U and W. However, only one
out of these six will be in the prescribed order: L always occurs before U and U always occurs before W.
So, corresponding to 7! arrangements, the number of ways in which the condition will be satisfied = 7!/3! ways.

Example 7 How many new words can be formed with the word 'PATNA'?

Solution Total number of letters— P, T, N occur once while A occurs twice.

So, the total number of words that can be formed = 5!/2! = 60

Total number of new words = 60 – 1 = 59

Example 8 How many words can be formed with the word 'ALLAHABAD'?

Solution Letters are— A – Four times
L – Twice
H, B and D occur once.

So, the total number of words = $\frac{9!}{4!2!}$

Example 9 How many 4-lettered distinct words can be formed from the letters of the word 'EXAMINATION'?

Solution Letters are: A – Twice
I – Twice
N – Twice
E, X, M, T, O – Once
Words will be of three types:
i. All distinct, ii. Two same, two distinct, iii. Two same and of one kind: two same and of other kind.

i. All distinct = ${}^8P_4$ (Distinct letters are – A, I, N, E, X, M, T and O)
ii. Two same, two distinct
Selection of one pair out of the three pairs (A, I, N) can be chosen in ${}^3C_1$ ways. Now rest of the two distinct letters can be chosen in ${}^7C_2$ ways.
Total number of words = ${}^3C_1 \times {}^7C_2 \times \frac{4!}{2!}$
iii. Two same and are of one kind, two same and are of other kind = Out of the three pairs of letters (A, I, N), we can select two pairs in ${}^3C_2$ ways.
Total number of words = ${}^3C_2 \times \frac{4!}{2! \times 2!}$

Number Formation

Number formation is another standard example of permutation. Here we will discuss the box diagram method of solving the questions.

If a three-digit number is to be constructed, then we will use

Hundred's place	Ten's place	Unit's place

If a four-digit number is to be constructed, then we will use

Thousand's place	Hundred's place	Ten's place	Unit's place

and so on.

While solving the questions related to number formation, we should know two things very clearly:

- While using the box diagram, we should start with the digit which has restriction, i.e., some condition is imposed on that digit.
- When nothing about the repetition of digits is mentioned in the question, we will assume that the repetition is allowed.

Example 10 How many different 3-digit numbers can be formed using the digits 1, 2, 3, 4 and 5?
i. When repetition is not allowed
ii. When repetition is allowed

Solution The box given below represents the respective positioning of digits in a three-digit number.

Hundred's place	Ten's place	Unit's place

i. Since repetition of the digits is not allowed, we can fill the unit's place in 5 ways, ten's place in 4 ways and hundred's place in 3 ways.

3	4	5

Using Multiplication Theorem, the total number of numbers which can be formed = $5 \times 4 \times 3 = 60$
Alternatively, 3 digits can be selected out of 5 digits in ${}^5P_3 = 60$
ii. Since repetition of the digits is allowed here, we can fill each of the hundred's, ten's and unit's place in 5 ways.

5	5	5

Using Multiplication Theorem, the total number of numbers which can be formed = $5 \times 5 \times 5 = 125$

Example 11 How many 4-lettered numbers divisible by 4 can be formed from the digits 0, 1, 2, 3, 4, 5?

Solution Any number divisible by 4 will have the number formed by its last two digits divisible by 4.

In this case, last two digits of the number can be 00, 04, 12, 20, 24, 32, 40, 44, 52.

Corresponding to any one of 00, 04, 12, 20, 24, 32, 40, 44, 52, we can have the following digits at its hundred's and thousand's place:

5	6	1	1

Thousand's place cannot be filled by 0, so it can be filled in 5 ways.

Hundred's place can be filled by any of the 0, 1, 2, 3, 4, 5; hence 6 ways.

So, corresponding to any one of 00, 04, 12, 20, 24, 32, 40, 44, 52, the total number of ways = $5 \times 6 = 30$

So, the total number of numbers which can be formed = $30 \times 9 = 270$

Example 12 In the above question, how many numbers can be formed if repetition of the digits is not allowed?

Solution Last two digits of this number can be—04, 12, 20, 24, 32, 40, 52.

At this point, we will have to divide the process of solving this question—one part will have those numbers which contain '0' as one of its last two digits viz., 04, 20, 40 and other part will have the remaining numbers viz., 12, 24, 32, 52.

1st part – Last two digits are 04, 20, 40.

4	3	1	1

$= 4 \times 3 = 12$ ways

Hence, the total number of numbers which can be formed $= 12 \times 3 = 36$

2nd part — Last two digits are 12, 24, 32, 52.

'0' cannot occur at thousand's place.

3	3	1	1

$= 3 \times 3 = 9$ ways

Hence, the total number of numbers which can be formed $= 9 \times 4 = 36$

Total numbers = 36 + 36 = 72

Example 13 How many odd integers from 1000 to 8000 have none of its digits repeated?

Solution There are two restrictions operating in this questions:

i. For a number to be odd, unit digit should be either 1 or 3 or 5 or 7 or 9.
ii. Thousand's place cannot be filled with 8 or 9.

For unit's digit — When it is filled with 9, thousand's place can be filled in 7 ways namely any digit from 1 to 7, and the remaining two places can be filled in $8 \times 7 = 56$ ways.

So, the total number of numbers formed in this way $= 56 \times 7 = 392$

Now, if the unit's place is filled with any of the four digits 1, 3, 5 or 7, the thousand's place can be filled in 6 ways (0 will be excluded), and the remaining two places can be filled in $8 \times 7 = 56$ ways.

So, the total number of numbers formed in this way $= 56 \times 6 \times 4 = 1344$

So, the total number of numbers = 392 + 1344 = 1736

Example 14 How many integers from 6000 to 6999 have atleast one of its digits repeated?

Solution Total number of numbers = None of its digits repeated numbers + atleast one of its digits repeated number (i.e., either the digits will be repeated or not repeated).

Total numbers with none of its digits repeated $= 1 \times 9 \times 8 \times 7 = 504$

So, the numbers having atleast one of its digits repeated $= 1000 - 504 = 496$

Example 15 How many natural numbers less than a million can be formed using the digits 0, 7 and 8?

Solution The numbers formed would be of a single digit, two digits, three digits, four digits, five digits and six digits.

Single-digit numbers = 7 and 8

For two-digit numbers,

2	3

$= 2 \times 3 = 6$ numbers

For three-digit numbers,

2	3	4

$= 2 \times 3 \times 3 = 18$ numbers

For four-digit numbers,an

2	3	3	3

$= 2 \times 3 \times 3 \times 3 = 54$ numbers

For five-digit numbers,

2	3	3	3	3

$= 2 \times 3 \times 3 \times 3 \times 3 = 162$ numbers

For six-digit numbers,

2	3	3	3	3	3

$= 2 \times 3 \times 3 \times 3 \times 3 \times 3 = 486$ numbers

So, the total number of numbers = 728

CIRCULAR PERMUTATION

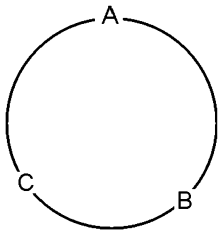
When n distinct things are to be arranged in a straight line, we can do this in $n!$ ways. However, if these n things are arranged in a circular manner, then the number of arrangements will not be $n!$.

Let us understand this:

The number of ways A, B and C can be arranged in a straight line $= 3! = 6$

The possible arrangements are — ABC, ACB, BAC, BCA, CAB, CBA

Now arrange these three people A, B and C in a circle



What we observe here is that the arrangements ABC, BCA and CAB are the same. And similarly the arrangements ACB, CBA and BAC are the same.

So, there are only two permutations in this case of circular permutation.

To derive the formula for circular permutation, we first fix the position of one thing then the remaining $(n - 1)$ things can be arranged in $(n - 1)!$ ways.

Hence, the number of ways in which n distinct things can be arranged in a circular arrangement is $(n - 1)!$

It can be seen in the following way also:

If n things are arranged along a circle, then corresponding to each circular arrangement the number of linear arrangement = n

So, the number of linear arrangements of n different things = $n \times$ (number of circular arrangements of n different things)

Hence, the number of circular arrangements of n different things = $(1/n) \times$ number of linear arrangements of n different things = $(1/n) \times n! = (n-1)!$

Clockwise and Anti-clockwise Circular Arrangements

If we take the case of four distinct things A, B, C and D sitting around a circular table, then the two arrangements ABCD (in clockwise direction) and ADCB (the same order but in anti-clockwise direction) will be different and distinct. Hence, we can conclude that the clockwise and anti-clockwise arrangements are different. However, if we consider the circular arrangement of a necklace made of four precious stones A, B, C and D, the two arrangements discussed as above will be the same because we take one arrangement and turn the necklace around (front to back), then we get the other arrangement. Hence, the two arrangements will be considered as one arrangement because the order of stones is not changing with the change in the side of observation. So, in this case there is no difference between the clockwise and the anti-clockwise arrangements.

Summarizing the above discussion, the number of circular arrangements of n distinct things is $(n-1)!$ if there is a difference between the clockwise and anti-clockwise arrangements and $(n-1)!/2$ if there is no difference between the clockwise and anti-clockwise arrangements.

Example 16 In how many ways 5 Indians and 4 Americans can be seated at a round table if

i. There is no restriction
ii. All the four Americans sit together
iii. No two Americans sit together
iv. All the four Americans do not sit together

Solution

i. Total number of persons = 9. These 9 persons can be arranged around a circular table in 8! ways.
ii. Assuming all the Americans to be one group, we have 6 things (5 Indians + 1 group) to be arranged around a circular table which can be arranged in 5! ways. However, these 4 Americans can be arranged in 4! ways among themselves.
So, the total number of arrangements = $5! \times 4!$
iii. Since there is no restriction on Indians. The 5 Indians can be seated around a table in 4! ways. The Americans will now be seated between two Indians, i.e., 5 places. 4 Indians can be seated on these 5 places in ${}^5P_4$ ways.
iv. The total number of arrangements when there is no restriction = 8! and the number of arrangements when all the four Americans sit together = $5! \times 4!$
So, the total number of arrangements when all the four Americans do not sit together = $8! - 5! \times 4!$.

PERMUTATION AND COMBINATION IN GEOMETRY

It is quite difficult to quantify the importance of P and C in geometry, a good number of P and C questions which use the concepts of geometry (and vice-versa) have been asked in the CAT and other premier B-school exams.

Example 17 How many diagonals will be there in an n-sided regular polygon?

Solution An n-sided regular polygon will have n vertices. And when we join any of these two vertices (${}^nC_2$) we get a straight line, which will be either a side or a diagonal.

So, ${}^nC_2$ = Number of sides + number of diagonals = n + number of diagonals

Hence, the number of diagonals = ${}^nC_2 - n = \dfrac{n(n-3)}{2}$

Above written result can be used as a formula also.

Example 18 Ten points are marked on a straight line and 11 points are marked on another parallel straight line. How many triangles can be constructed with vertices among the above points?

Solution Triangles will be constructed by taking one point from the 1st straight line and two more points from the 2nd straight line, and vice versa.

So, the total number of Δ formed = ${}^{10}C_2 \times {}^{11}C_1 + {}^{11}C_2 \times {}^{10}C_1 = 1045$

Example 19 There is an n-sided polygon ($n>5$). Triangles are formed by joining the vertices of the polygon. How many triangles can be constructed which will have no side common with the polygon?

Solution An n-sided polygon will have n vertices. Triangles constructed out of these n vertices will be of three types:

i. Having two sides common with the polygon,
ii. Having one side common with the polygon,
iii. Having no side common with the polygon. And total number of triangles formed will be ${}^nC_3$.

i. Having two sides common with the polygon — Out of total n vertices, any combination of three consecutive vertices will give us the triangle which is having two sides common with polygon = n
ii. Having one side common with the polygon — Number of selection of three vertices out of which two are consecutive (If we select A_5 and A_6 as the two vertices, then A_7 or A_4 should not be the third vertex because it will constitute the two sides of the common triangle). $= n \times {}^{(n-4)}C_1$

iii. So, the total number of triangles having no side common with polygon = ${}^{n}C_3 - n \times {}^{(n-4)}C_1 - n$

Some More Important Results:

1. Maximum No.of points of Intersection among n Straight Lines = ${}^{n}C_2$
2. Maximum No. of points of Intersection among n Circles = ${}^{n}P_2$

FINDING THE RANK OF A WORD

To find the rank of a word out of all the possibilities using all the letters given in the word is nothing but the extension of the concept of alphabetically arranging the words in a dictionary. However, unlike the case of the dictionary, we can have 'meaningless' words also in the case of finding the rank.

Example 20 All the letters of the word 'LUCKNOW' are arranged in all possible ways. What will be the rank of the word LUCKNOW?

Solution Alphabetical order of occurrence of letters— C, K, L, N, O, U, W.

Number of words starting with C = 6!
Number of words starting with K = 6!
All the words starting with LC – 5!
All the words starting with LK – 5!
All the words starting with LN – 5!
All the words starting with LO – 5!
Next word will start with LU – C – K – N – O – W.
So, rank of LUCKNOW – $2 \times 6! + 4 \times 5! + 1 = 1921$.

Example 21 In a chess competition involving some boys and girls of a school, every student had to play exactly one game with every other student. It was found that in 45 games both the players were girls, and in 190 games both were boys. The number of games in which one player was a boy and the other was a girl is

(a) 200 (b) 216 (c) 235 (d) 256

Solution Let there be m boys and n girls

$${}^{n}C_2 = 45 = \frac{n(n-1)}{2} \Rightarrow n(n-1) = 90 \Rightarrow n = 10$$

$${}^{m}C_2 = 190 = \frac{m(m-1)}{2} = 190 \Rightarrow m(m-1) = 380 \Rightarrow m = 20$$

Number of games between one boy and one girl
$= {}^{10}C_1 \times {}^{20}C_1 = 10 \times 20 = 200$

Hence, option (a).

Example 22 If each permutation of the digits 1, 2, 3, 4, 5, 6 is listed in the increasing order of the magnitude, the 289th term will be

(a) 326541 (b) 341256
(c) 356241 (d) 314256

Solution $289 = (2 \times 5!) + (2 \times 4!) + 1$

So, the number will be 341256.

Example 23 There are 12 intermediate stations between two places A and B. In how many ways can a train be made to stop at 4 of these 12 intermediate stations provided no two of them are consecutive?

Solution

1st Method Let $S_1, S_2, \ldots, S_8$ denote the stations where the train does not stop. The four stations where the train stops should be at any four of the nine places indicated by cross.

$$\therefore \text{Required number} = {}^{9}C_4 = \frac{9.8.7.6}{\underline{|4}} = 126$$

2nd Method Let S_1, S_2, S_3, S_4 be the four intermediate stations where the train stops.

$$\boxed{a}\, S_1\, \boxed{b}\, S_2\, \boxed{c}\, S_3\, \boxed{d}\, S_4\, \boxed{e}$$

Let a, b, c, d, e be the number of stations between A and S_1, S_1 and S_2, S_2 and S_3, S_3 and S_4 and B respectively.

Then, $a + b + c + d + e = 8$ ...(1)

Where $a \geq 0, b \geq 1, c \geq 1, d \geq 1, e \geq 0$

Let $x = a, y = b - 1, z = c - 1, t = d - 1, w = e$

Now $x + y + z + t + w = a + b + c + d + e - 3$
$= 8 - 3 = 5$

Or $x + y + z + t + w = 5$, where $x, y, z, t, w \geq 0$ ...(2)

Required number = number of non negative integral solutions

$= {}^{n+r-1}C_r = {}^{5+5-1}C_5 = {}^{9}C_5 = 126$

Example 24 Find the number of integral solutions of equation $x + y + z + t = 25$, $x > 0, y > 1, z > 2$ and $t \geq 0$.

Solution Given, $x + y + z + t = 25$, ...(1)

Where $x \geq 1, y \geq 2, z \geq 3, t \geq 0$

Let $p = x - 1, q = y - 2, r = z - 3, s = t$

Then $p + q + r + s = x + y + z + t - 6 = 25 - 6 = 19$, where, $p, q, r, s \geq 0$

$\therefore p + q + r + s = 19, p, q, r, s \geq 0$ ...(2)

$\therefore$ Required number = number of ways in which 19 identical things can be distributed among 4 persons when each person can get any number of things

$= {}^{n+r-1}C_{r-1} = {}^{22}C_3$

Example 25 There are 4 oranges, 5 apricots and 6 alphonso in a fruit basket. In how many ways can a person make a selection of fruits from among the fruits in the basket?

Solution Whenever we are talking about fruits, we assume them to be identical. However when we are talking about men, we treat them to be distinct.

Zero or more oranges can be selected out of 4 identical oranges in 4 + 1 = 5 ways.

Zero or more apricot can be selected out of 5 identical apricots in 5 +1 = 6 ways.

Zero or more can be selected out of 6 identical alphanso in 6 + 1 = 7 ways.

∴ The total number of selections when all the three types of fruits are selected (the number of any type of fruit may also be zero)

$= 5 \times 6 \times 7 = 210.$

But in one of these selections the number of each type of fruit is zero and hence this selection must be excluded.

∴ Required number = 210 – 1 = 209.

Example 26 Twelve different letters of alphabet are given. Words with six letters are formed from these given letters. Find the number of words which have at least one letter repeated.

Solution The total number of letters = 12. Words of six letters are to be framed.

The total number of words of 6 letters when any letter may be repeated any number of times (This also includes the number of words formed when no letter is repeated.)

$= 12 \times 12 \times 12 \times 12 \times 12 \times 12 = 12^6$

Number of words of 6 letters when no letter is repeated $= {}^{12}P_6$.

So, Number of words of 6 letters which have at least one letter repeated $= 12^6 - {}^{12}P_6$

PRACTICE EXERCISES

WARM UP

Q.1. How many numbers between 100 and 1000 can be formed using the digits 0, 2, 4, 6, 8, 5, if repetition of digits in a number is not allowed?
(a) 99 (b) 82 (c) 100 (d) 120

Q.2. How many numbers between 100 and 1000 can be formed using the digits 0, 2, 4, 6, 8, 5, if repetition of digits in a number is allowed?
(a) 164 (b) 180 (c) 192 (d) 100

Q.3. Two out of six paper sets for an examination are of mathematics. What is the number of ways in which the papers can be set so that the two mathematics papers are not together?
(a) 480 (b) 520 (c) 492 (d) 512

Q.4. In a letter lock, each of the three rings is marked with 15 letters. What is the maximum number of unsuccessful attempts that one has to make before the lock is opened?
(a) 3374 (b) 5284 (c) 8457 (d) 8101

Q.5. A certain code consists of 5 variables, with each variable having 4 different constant values possible. What is the total number of coded messages that can be sent with 5 constants one from each variable?
(a) 1024 × 5! (b) 1024 × 4!
(c) 1024 × 3! (d) None of these

Q.6. From 3 different soft drinks, 4 Chinese dishes and 2 ice-creams, how many different meals are possible if at least one of each of the three items is to be included, depending upon the number of people likely to turn up?
(a) 315 (b 282
(c) 864 (d) None of these

Q.7. The governing council of an institute has fifteen members and wants to hold its annual meeting. In how many ways can the council be seated around a round table if the chairman and the vice-chariman of the council are always seated together?
(a) 10 × 12! (b) 14 × 10!
(c) 13! (d) None of these

Q.8. Akshay is planning to give a birthday party at his place. In how many ways can he invite one or more of five friends and seat them at a circular table?
(a) 84 (b) 89 (c) 78 (d) 81

Q.9. In the above question, In how many ways can he invite one or more of his five friends and seat them at a circular table with him?
(a) 325 (b) 205 (c) 265 (d) 320

Q.10. A cricket team of eleven is to be chosen from 8 batsmen, 6 bowlers and 2 wicket-keepers. In how many ways can the team be chosen if there must be at least four batsmen, at least four bowlers and exactly one wicket-keeper?
(a) 1681 (b) 5304
(c) 1652 (d) None of these

Q.11. How many words can be formed using the letters of the word "CORRESPONDENCE"?
(a) $[(14!)/(2!)^4]$ (b) 14!
(c) $[(14!)/(2!)^4(3!)]$ (d) $[(14!)/(2!)^3(3!)]$

Q.12. How many words can be formed using the letters of the word "CORRESPONDENCE" if the consonants are always written together?
(a) 182 (b) 184
(c) $216 \times 9!$ (d) None of these

Q.13. If six persons are selected out of ten, in how many ways will a particular person be found among those six?
(a) 124 (b) 126 (c) 144 (d) 84

Q.14. A committee of five is to be chosen from among six men and four ladies. In how many ways can this be done in order to include at least one lady?
(a) 252 (b) 246 (c) 244 (d) 152

Q.15. How many ways are there to pick two different cards from a deck of 52 cards such that the first card is an ace and the second is not a queen
(a) 188 (b) 198
(c) 164 (d) None of these

Q.16. In the above question, what is the number of ways such that the first card is a spade and the second is not a queen?
(a) 612 (b) 612
(c) 164 (d) None of these

Q.17. How many different license-plates involving three letters and three digits are there if the three letters appear together, either at the beginning or at the end of the license?
(a) $2 \times 26^3 \times 10^3$ (b) 54102
(c) $4 \times 25^2 \times 10^4$ (d) None of these

Q.18. A joint student teacher committee of 5 members is to be formed with 4 teachers, 3 male students and 5 female students. How many different committees can be formed if the committee must consist of at least 2 teachers, 1 male student and 2 female students?
(a) 170 (b) 152 (c) 180 (d) 104

Q.19. There are 8 different locks, with exactly one key for each lock. All the keys have been mixed up. What is the maximum number of trials required in order to determine which key belongs to which lock?
(a) 44 (b) 28 (c) 24 (d) 32

Q.20. Hoppers' Stop stocks four styles of trousers. For each style, there are ten different possible waist sizes, six different trousers lengths and four colour choices. How many different types of trousers could the store have?
(a) 1024 (b) 960
(c) 921 (d) 924

FOUNDATION

Q.1. In a group of boys, the number of arrangements of 4 boys is 12 times the number of arrangements of 2 boys. The number of boys in the group is
(a) 10 (b) 8
(c) 6 (d) None of these

Q.2. From a group of persons the number of ways of selecting 5 persons is equal to that of 8 persons. The number of persons in the group is
(a) 13 (b) 40 (c) 18 (d) 21

Q.3. For Board of Cricket Control of India (BCCI), a selection committee is to be chosen consisting of 5 ex-cricketers. Now there are 10 representatives from four zones. It has further been decided that if Kapil Dev is selected, Sunil Gavaskar will not be selected and vice versa. In how many ways can this be done?
(a) 140 (b) 112 (c) 196 (d) 56

Q.4. At a get-together, it was found that a total of 66 handshakes took place at the conclusion of the party. If each guest shook hand only once with all the others, how many people were present.
(a) 33 (b) 22 (c) 12 (d) 13

Q.5. A committee is to be formed comprising of 7 members such that there is a majority of men and at least 1 woman in every committee. The shortlist consists of 9 men and 6 women. In how many ways can this be done?
(a) 3724 (b) 3630
(c) 3526 (d) 4914

Q.6. A number of 6-digit numbers that can be made with the digits 0, 1, 2, 3, 4 and 5, so that even digits occupy odd places, is
(a) 24 (b) 36
(c) 48 (d) None of these

Q.7. The Total number of ways in which six '+' and four '–' signs can be arranged in a line such that no two '–' signs occur together, is
(a) $\frac{7!}{3!}$ (b) $6! \times \frac{7!}{3!}$
(c) 35 (d) None of these

Q.8. Let A be the set of 4-digit numbers $a_1a_2a_3a_4$ where $a_1 > a_2 > a_3 > a_4$ then how many values of A are possible?
(a) 126 (b) 84
(c) 210 (d) None of these

Q.9. A number of 5-digit numbers in which no two consecutive digits are identical, is:
(a) $9^2 \times 8^3$ (b) 9×8^4
(c) 9^5 (d) None of these

Q.10. The total number of selections of at most n things from $(2n + 1)$ different things is 63. Then the value of n is
(a) 3 (b) 2
(c) 4 (d) None of these

Q.11. From 4 gentlemen and 6 ladies, a committee of five is to be formed. The number of ways in which the committee can be formed so that the gentlemen are in majority, is
(a) 66 (b) 156
(c) 60 (d) None of these

Q.12. The number of all four digit numbers, which are divisible by 4, that can be formed from the digits 1,2, 3, 4 and 5 is
(a) 125 (b) 30
(c) 95 (d) None of these

Q.13. The total number of integral solutions for (x, y, z) such that $xyz = 24$, is
(a) 36 (b) 90
(c) 120 (d) None of these

Q.14. The number of ways to give 16 different things to three persons A, B, C so that B gets 1 more than A and C gets 2 more than B, is
(a) $\frac{16!}{4!5!7!}$ (b) 4!5!7!
(c) $\frac{16!}{3!5!8!}$ (d) None of these

Q.15. The number of positive integral solutions of $x + y + z = n$, $n \in N$, $n \geq 3$, is
(a) ${}^{n-1}C_2$ (b) ${}^{n-1}P_2$
(c) $n(n-1)$ (d) None of these

Q.16. If ${}^nC_4$, ${}^nC_5$ and ${}^nC_6$ are in an AP then n is
(a) 8 (b) 9
(c) 14 (d) 10

Q.17. There are 20 questions in a question paper. If no two students solve the same combination of questions but solve equal number of questions, then the maximum number of students who appeared in the examination is
(a) ${}^{20}C_9$ (b) ${}^{20}C_{11}$
(c) ${}^{20}C_{10}$ (d) None of these

Q.18. The number of ways, to distribute 32 different things equally among 4 persons, is
(a) $\frac{32!}{(8!)^3}$ (b) $\frac{32!}{(8!)^4}$
(c) $\frac{1}{4}(32!)$ (d) None of these

Q.19. In a dinner party, there are 10 Indians, 5 Americans and 5 Englishmen. In how many ways can they be arranged in a row so that all persons of the same nationality sit together?
(a) 10!×5!×5! (b) 20!
(c) 3!×10!×5!×5! (d) 20! 3!

Q.20. In how many ways can 15 I.Sc. and 13 B.Sc. candidates be arranged in a line so that no two B.Sc. candidates may occupy consecutive positions?
(a) 15!×13! (b) $15! \times 16p_{12}$
(c) $13! \times {}^{16}P_{12}$ (d) 2!×15!×13!

Q.21. How many new words can be formed from the letters of the word 'circle' taken together?
(a) 720 (b) 719 (c) 360 (d) 359

Q.22. How many different words can be formed with the letters of the word 'Vice-Chancellor' so that the vowels are together?
(a) $10 \times 5!$ (b) $10! \times 5!$
(c) $5 \times 10!/2$ (d) $5 \times 10!$

Q.23. In how many ways can the letters of the word 'Lucknow' be arranged so that the two vowels do not come together?
(a) 720 (b) 1440
(c) 3600 (d) None of these

Q.24. How many words can be formed out of the letters of the word 'Article' so that the vowels occupy the even places?
(a) 72 (b) 144 (c) 288 (d) 36

Q.25. How many numbers greater than four millions (4000000) can be formed with the digits 2, 3, 0, 3, 4, 2, 5,?
(a) 280 (b) 380
(c) 360 (d) None of these

Q.26. How many signals can be made by hoisting 2 blue, 2 red and 5 black flags on a pole at the same time?
(a) 378 (b) 1512
(c) 756 (d) None of these

Q.27. Find the number of arrangements of the letters of the word 'Maruti' if u always comes before a.
(a) 60 (b) 180 (c) 420 (d) 360

Q.28. In a class of students there are 6 boys and 4 girls. In how many ways can they be seated, around a table so that all the 4 girls sit together?

(a) $6! \times 5!$ (b) $2 \times 6! \times 4!$
(c) $5! \times 4!$ (d) $6! \times 4!$

Q.29. A round table conference is to be held among 20 delegates of 20 countries. In how many ways can they and the host be seated if two particular delegates are always to sit on either side of the host?
(a) $19! \times 2!$ (b) $19!$
(c) $18! \times 2!$ (d) $20! \times 2!$

Q.30. Four gentlemen and four ladies are invited to a certain party. Find the number of ways of seating them around a table so that only ladies are seated on the two sides of each gentleman.
(a) 72 (b) 144 (c) 36 (d) 56

Q.31. Five men, 6 boys and 7 women be seated in a row so that the men, women and boys may not be separated in
(a) 5!×6!×7! ways (b) 6!6!7! ways
(c) 2!5!6!7! ways (d) 3!5!6!7! ways

Q.32. In how many ways 12 different books can be distributed equally among 4 persons?
(a) 195 (b) 154 (c) 210 (d) 256

Q.33. Eight chairs are numbered 1 to 8. Two women and three men wish to occupy one chair each. First two women choose the chairs from amongst the chairs marked 1 to 4; and then the men selected the chairs from amongst the remaining. Find the number of possible arrangement.
(a) 720 (b) 1440
(c) 2110 (d) None of these

Q.34. How many numbers between 100 and 1000 can be formed with the digits, 1, 2, 3, 4, 5, 6, 7; with no digit being repeated?
(a) 210 (b) 420 (c) 105 (d) 320

Q.35. How many different numbers greater that 5000 can be formed with the digits 0, 1, 5, 9, with no digit being repeated?
(a) 18 (b) 12 (c) 24 (d) 6

Q.36. How many numbers less than 1000 and divisible by 5 can be formed in which no digit occurs more than once in the same number?
(a) 154 (b) 162 (c) 176 (d) 132

Q.37. Find the number of numbers of 4 digits formed with the digits 1, 2, 3, 4, 5 in which 3 occurs in the thousands place and 5 occurs in the units place.
(a) 4 (b) 8
(c) 12 (d) 6

Q.38. Find the sum of all the four digit numbers that can be formed with the digits 3, 2, 3, 4.
(a) 40982 (b) 39996
(c) 41682 (d) None of these

Q.39. In how many ways can three prizes be given away to 5 students when each student is eligible for any of the prizes?
(a) 243 (b) ${}^5C_3$ (c) ${}^5P_3$ (d) 125

Q.40. A round table conference is to be held between 20 delegates. If two particular delegates are always to sit together, they can be seated in
(a) $2 \times 19!$ ways (b) 19! ways
(c) 20! ways (d) $2 \times 18!$ ways

Q.41. How many quadrilaterals can be formed joining the vertices of a polygon of n sides?
(a) ${}^nC_2 \times 2$ (b) ${}^nC_4$
(c) ${}^nC_4$ (d) None of these

Q.42. A delegation of 6 members is to be sent abroad out of the 12 members. In how many ways can the selection be made so that a particular member is included?
(a) 376 (b) 424 (c) 448 (d) 462

Q.43. The sides AB, BC, CA of a triangle ABC have 3, 4 and 5 interior points respectively on them. Find the number of triangles that can be constructed using given interior points as vertices.
(a) 210 (b) 195 (c) 205 (d) ${}^{12}C_3$

Q.44. Let S be the set of five-digit numbers formed by digits 1, 2, 3, 4 and 5, using each digit exactly once such that exactly two odd position are occupied by odd digits. What is the sum of the digits in the rightmost position of the numbers in S?
(a) 228 (b) 216 (c) 294 (d) 192

Q.45. Triplet x, y and z are chosen from the set {1, 2, 3, ... 24, 25} such that $x \leq y < z$. How many such triplets are possible?
(a) 25_{C_2} (b) 600
(c) $25_{C_2} + 25_{C_3}$ (d) 1200

Q.46. From 8 gentlemen and 4 ladies, a committee of 5 is to be formed. In how many ways can this be done so as to include at least one lady?
(a) 736 (b) 728 (c) 280 (d) 792

Q.47. Out of 7 men and 4 ladies a committee of 5 is to be formed. In how many ways can this be done so as to include at least 3 ladies?
(a) 84 (b) 91
(c) 90 (d) 102

Q.48. A person has 12 friends of whom 8 are relatives. In how many ways can he invite 7 friends such that at least 5 of them may be relatives?
(a) 284 (b) 384
(c) 456 (d) 512

Q.49. A person wishes to throw as many different parties as he can out of his 20 friends such that each party

consists of the same number of persons. The number of friends he should invite at a time is
(a) 5 (b) 10
(c) 8 (d) None of these

Q.50. If all permutations of the letters of the word *AGAIN* are arranged as in dictionary then fiftieth word is
(a) NAAGI (b) NAGAI
(c) NAAIG (d) NAIAG

MODERATE

Q.1. The number of distinct rational numbers x such that $0 < x < 1$ and $x = \frac{p}{q}$, where $p, q \in (1, 2, 3, 4, 5, 6)$, is
(a) 15 (b) 13 (c) 12 (d) 11

Q.2. The total number of 9-digit numbers of different digits is
(a) 10 (9!) (b) 8 (9!)
(c) 9 (9!) (d) None of these

Q.3. How many three-digit numbers with distinct digits can be formed such that the product of the digits is the cube of a positive integer?
(a) 21 (b) 24 (c) 36 (d) 30

Q.4. Seven different lecturers are to deliver lectures in seven periods of a class on a particular day. A, B and C are three of the lecturers. The number of ways in which a routine for the day can be made such that A delivers his lecture before B, and B before C, is
(a) 420 (b) 120
(c) 210 (d) None of these

Q.5. Let A = $\{x \mid x$ is a prime number and $x < 30\}$. The number of different rational numbers whose numerator and denominator belong to A is
(a) 90 (b) 180
(c) 91 (d) None of these

Q.6. Find the number of non-negative integral solutions to the system of equations $x + y + z + u + t = 20$ and $x + y + z = 5$.
(a) 228 (b) 336 (c) 448 (d) 528

Q.7. Find the number of integral solution to $|x|+|y|+|z| = 15$.
(a) 902 (b) 728 (c) 734 (d) 904

Direction for questions 8 and 9: Read the passage below and solve the questions based on it.

There are 6 boys and 4 girls sitting for a photo session. They were posing for the photograph standing in 2 rows one behind the other. There were 5 people sitting in the front row and 5 standing in the back row.

Q.8. If the boys were divided equally among the front and back rows, in how many ways can the photo session be arranged?
(a) $5! \times 5!$ (b) ${}^6C_3 \times {}^4C_2 \times 5! \times 5!$
(c) ${}^6C_3 \times {}^4C_2 \times 10!$ (d) ${}^6C_3 \times {}^4C_2 \times 5! \times 5! \times 2!$

Q.9. In how many ways can the photos be taken such that no two boys and no two girls are standing or sitting together
(a) ${}^6C_3 \times {}^4C_2 \times 10!$ (b) ${}^6C_3 \times {}^4C_2 \times (3!)2(2!)2$
(c) ${}^6C_3 \times {}^4C_2$ (d) None of these

Q.10. A teacher takes 3 children from her class to the zoo at a time as often as she can, but she does not take the same three children to the zoo more than once. She finds that she goes to the zoo 84 times more than a particular child goes to the zoo. The number of children in her class is
(a) 12 (b) 10
(c) 60 (d) None of these

Q.11. ABCD is a convex quadrilateral. 3, 4, 5 and 6 points are marked on the sides AB, BC, CD and DA respectively. The number of triangles with vertices on different sides is
(a) 270 (b) 220 (c) 282 (d) 342

Q.12. The number of triangles that can be formed with 10 points as vertices, n of them being collinear, is 110. Then n is
(a) 3 (b) 4 (c) 5 (d) 6

Q.13. If x, y, and z are integers and $x \geq 0$, $y \geq 1$, $z \geq 2$, $x + y + z = 15$ then the number of values of the ordered triplet (x, y, z) is
(a) 91 (b) 455
(c) ${}^{17}C_{15}$ (d) None of these

Q.14. If a, b, c are positive integers such that $a + b + c \leq 8$ then the number of possible values of the ordered triplet (a, b, c) is
(a) 84 (b) 56
(c) 83 (d) None of these

Q.15. The product of r consecutive integers is necessarily divisible by
(a) r (b) $\sum_{k=1}^{r-1} k$
(c) (r + 1)! (d) None of these

Q.16. A candidate is required to answer six out of ten questions which are divided into groups, each containing five questions and he is not permitted to attempt more than 4 from any group. In how many ways can he make up his choice?
(a) 184 (b) 192
(c) 200 (d) None of these

Q.17. In an examination, the question paper contains three different sections A, B and C containing 4, 5 and 6 questions respectively. In how many ways, a candidate can make a selection of 7 questions, selecting at least two questions from each section?
(a) 890 (b) 1420
(c) 2700 (d) 3140

Q.18. A tea party is arranged for 16 people along the two sides of a long table with 8 chairs on each side. Four men wish to sit on one particular side and two on the other side. In how many ways can they be seated?
(a) ${}^{10}C_4 \times 8!$ (b) ${}^{10}P_4 \times (8!)^2$
(c) $10C4 \times (8!)2$ (d) $4! \times 2! \times (8!)^2$

Q.19. A candidate is required to answer 7 questions out of 12 questions, which are divided into two groups, each containing 6 questions. He is not permitted to attempt more than 5 from either of the group. In how many different ways can he choose the seven questions?
(a) 640 (b) 780 (c) 920 (d) 1020

Q.20. The number of parallelograms that can be formed from a set of four parallel lines intersecting another set of three parallel lines, is
(a) 6 (b) 18 (c) 12 (d) 9

Q.21. We are required to form different words with the help of the word INTEGER. Let m_1 be the number of words in which I and N are never together and m_2 be the number of words which begin with I and end with R, then m_1/m_2 is equal to
(a) 42 (b) 30 (c) 6 (d) 50

Q.22. In an election there are five candidates and three members are to be elected, and an elector can vote for any number of candidates not greater than the number to be elected. Then the number of ways in which an elector may vote is
(a) 25 (b) 30
(c) 32 (d) None of these

Q.23. The greatest possible number of points of intersection of 8 straight lines and 4 circles is
(a) 32 (b) 64
(c) 76 (d) 104

Q.24. A lady gives a dinner party to 5 guests to be selected from nine friends. The number of ways of forming the party of 5, given that two particular friends A and B will not attend the party together is
(a) 56 (b) 126
(c) 91 (d) None of these

Q.25. In a plane there are 37 straight lines, of which 13 pass through the point A and 11 pass through the point B. Besides, no three lines pass through one point, no line passes through both points A and B, no two are parallel. Then the number of intersection points the lines have is equal to
(a) 535 (b) 601
(c) 728 (d) None of these

Q.26. If a denotes the number of permutations of $x+2$ things taking all at a time, b the number of permutations of x things taking 11 at a time and c the number of permutations of $x-11$ things taking all at a time such that $a = 182bc$, then the value of x is
(a) 15 (b) 12 (c) 10 (d) 18

Q.27. There are 4 letters and 4 directed envelopes. The number of ways in which all the letters can be put in the wrong envelope is
(a) 8 (b) 9
(c) 16 (d) None of these

Q.28. The total number of natural numbers of six digits that can be made with digits 1, 2, 3, 4, if all digits have to appear in the same number at least once, is
(a) 1560 (b) 840 (c) 1080 (d) 480

Q.29. In a club election the number of contestants is one more than the number of maximum candidates for which a voter can vote for. If the total number of ways in which a voter can vote is 62, then what is the number of candidates?
(a) 7 (b) 5
(c) 6 (d) None of these

Q.30. Seven women and seven men are to sit round a circular table such that there is a man on either side of every women then the number of seating arrangement is
(a) $(7!)^2$ (b) $(6!)^2$
(c) 6! x 7! (d) 7!

Q.31. How many 10 digits numbers can be written by using the digits 1 and 2
(a) ${}^{10}C_1 + {}^{9}C_2$ (b) 210
(c) ${}^{10}C_2$ (d) 10!

Q.32. All possible two factors products are formed from the numbers 1, 2, 3, 4 ..., 200. The number of factors obtained out of the total which are multiples of 5 is
(a) 5040 (b) 7180
(c) 8150 (d) None of these

Q.33 How many different flags can be made with 5 different stripes using them any number at a time?
(a) 3^5 (b) 5^3
(c) 385 (d) 625

Q.34. Three apples and two bananas have to be distributed among three boys—Anuj, Bairag and Chandan, and two girls—Mugdha and Vaidehi, such that each person gets one fruit. In how many different ways can this be done if at least one girl gets an apple?
(a) 6 (b) 5
(c) 9 (d) None of these

Q.35. Find the distinct number of 7 digit numbers the sum of whose digits is even.
(a) 90×10^6 (b) 45×105
(c) 640000 (d) None of these

Q.36. Find the number of non-congruent rectangles that can be found on a chessboard normal 8×8 chessboard.
(a) 24 (b) 36
(c) 48 (d) None of these

Q.37 A dinner menu is to be designed out of 5 different starters, 6 identical main courses and 4 distinct desserts. In how many ways menu be designed such that there is atleast one of each of the starters, main courses and desserts?
(a) $31 \times 6 \times 15$ (b) $32 \times 6 \times 16$
(c) $31 \times 7 \times 15$ (d) $5 \times 6 \times 4$

Q.38. A man has 7 relatives. 4 of them are ladies and 3 gentlemen; his wife also has 7 relatives. 3 of them are ladies and 4 gentlemen. In how many ways can they invite 3 ladies and 3 gentlemen for a dinner party so that there are 3 of the man's relatives and 3 of the wife's relatives?
(a) 395 (b) 465
(c) 485 (d) 515

Q.39. Find the number of integral solutions of equation $x + y + z + t = 29$, $x > 0$, $y > 1$, $z > 2$ and $t \geq 0$.
(a) $^{27}C_3$ (b) $^{28}C_3$
(c) 2600 (d) $^{29}C_4$

Q.40. Find the number of numbers between 2×10^4 and 6×10^4 having sum of the digits even.
(a) 20,000 (b) 19999
(c) 24000 (d) 25000

Q.41. Eight straight lines in a set are parallel to each other and the distance between any two adjacent lines in 1 cm. Another set of 6 straight lines are parallel to each other and the distance between any two adjacent lines is 1 cm. These 6 straight lines of second set intersect with the first set of 8 straight lines to form parallelograms. How many of such formed parallelograms will not be rhombuses?
(a) 285 (b) 365
(c) 335 (d) None of these

Q.42. Ten guests of which three are ladies are to be seated in a row. The ladies insist on sitting together while two of the gentlemen refuse to take consecutive seats. In how many ways can the guests be seated?
(a) 256 (b) 512
(c) 8!.3! −7!.2! (d) 8!.3!−7!.3!.2!

Q.43. Find the number of non-negative integral solutions of $2x + 2y + z = 10$
(a) $^{12}C_2$ (b) $^{14}C_4$
(c) $^{11}C_2$ (d) None of these

Q.44. How many committees of 11 persons can be made out of 50 persons if three particular persons A, B and C are not to be included together in the committee?
(a) $^{50}C_{11} - {}^{47}C_8$ (b) $^{50}C_{11} - {}^{47}C_9$
(c) $^{50}C_{11} \times {}^{47}C_8$ (d) None of these

Q.45. If each of the m points on the one straight line be joined to each of the n points on the other straight line, then excluding the points on the given two lines, number of points of intersection of these lines is
(a) $\frac{1}{4}mn(m-1)(n-1)$ (b) $^{m+n}C_2$
(c) $^{mn}C_2$ (d) None of these

ADVANCED

Q.1. There is a regular decagon. Triangles are formed by joining the vertices of the polygon. What is the number of triangles which have no side common with any of the sides of the polygon?
(a) 50 (b) 300 (c) 44 (d) 294

Q.2. There are 12 intermediate stations between two places A and B. In how many ways can a train be made to stop at 4 of these 12 intermediate stations that no two stations are consecutive?
(a) $^{15}C_3$ (b) $^{11}C_3$
(c) $^{9}C_4$ (d) $^{9}C_3$

Q.3. A conference attended by 200 delegates is held in a hall. The hall has 7 doors, marked A, B, ... G. At each door, an entry book is kept and the delegates entering through that door sign it in the order in which they enter. If each delegate is free to enter any time and through any door he likes, how many different sets

of seven lists would arise in all? (Assume that every person signs only at his first entry.)
(a) $^{206}C_6$ (b) $^{199}P_5$
(c) $^{199}C_5$ (d) $^{206}P_6$

Q.4. In an election for the post of president of students union, every student can vote in 56 ways. Every student votes for atleast one candidate and a student can vote for more than one candidate. If the total number of candidates exceeds the maximum number that can be elected by 2, then what is the number of candidates?
(a) 5 (b) 7
(c) 6 (d) None of these

Q.5. The number of positive integral solution of $abc = 30$ is:
(a) 27 (b) 81
(c) 243 (d) None of these

Q.6. A train going from Cambridge to London stops at nine intermediate stations. Six persons enter the train during the journey with six different tickets of the same class. How many different sets of tickets would they have?
(a) $^{11}C_5$ (b) $^{35}C_6$
(c) $^{45}C_6$ (d) None of these

Q.7. Five balls of different colours are to be placed in three boxes of different sizes. Each box can hold all five balls. In how many different ways can we place the balls so that no box remains empty?
(a) 119 (b) 150
(c) 180 (d) 210

Q.8. An eight-oared boat is to be manned by a crew chosen from 11 men of whom 3 can steer but cannot row and the rest cannot steer. In how many ways can the crew be arranged if two of the men can only row on bow side?
(a) $^3C_1.^4C_2.6!$ (b) $^3P_1.^4P_2$
(c) 3P1.4P2.6! (d) $^3C_1.^4C_2$

Q.9. Find the number of n-digit number formed with the digits 1, 2, 3 …, 9 in which no two consecutive digits are the same.
(a) 9^{n-1} (b) 9.8^{n-1}
(c) 9.8^n (d) 9^n

Q.10. If the number of ways in which n different things can be distributed among n persons so that at least one person does not get any thing is 232 then what is the value of n?
(a) 3 (b) 4
(c) 5 (d) None of these

Q.11. If m be the greatest value of k for which 100! is divisible by 3^k, then the number of ways of distributing k different things among 5 persons so that two of them get 10 things each, two get 12 things each and the fifth one gets four things is
(a) $\frac{48!}{(101)^2(121)^2 41}$ (b) $\frac{48!}{(10!)^2(12!)^2}$
(c) $\frac{48!}{(10!)^2(12!)^2 4}$ (d) $\frac{48!}{(10!)^2(12!)^2}.\frac{5}{4}$

Q.12. A bag contains n white and n red balls. Pairs of balls are drawn without replacement until the bag is empty. If the number of ways in which each pair consists of one red and one white ball is 14400, then what is the value of n?
(a) 5 (b) 10 (c) 15 (d) 20

Direction for questions 13 to 15: Read passage below and solve the questions based on it.

In a horticultural show, a participant arranged a total of 1000 fruits of four types in a long straight line. He first placed one apple, then two mangoes, three oranges, four custard apples, five apples, six mangoes, seven oranges, eight custard apples and so on (with apples followed by mangoes, oranges, custard apples in that order) upto the 1000th fruit.

Q.13. What was the 1000th fruit?
(a) Apple (b) Mango
(c) Orange (d) Custard apple

Q.14. What was the position of the 100th Mango?
(a) 387 (b) 437 (c) 417 (d) 363

Q.15. Among the last 700 fruits, which type of fruit occurred the greatest number of times?
(a) Apple (b) Mango
(c) Orange (d) Custard apple

Q.16. There is a framework of a cuboid of length 6, breadth 5 and height 7. The cuboid is only composed of a skeleton of 210 cubes of side 1. An insect is on one corner of the cube and it wants to travel to the opposite end of the longest diagonal. It can only travel along the sides of the small cubes and it always takes the shortest possible route.
Find the number of choices the insect has.
(a) $\frac{18!}{7!6!5!}$ (b) $\frac{15!}{6!5!4!}$
(c) $\frac{17!}{7!6!5!}$ (d) $\frac{16!}{6!5!4!}$

Q.17. In the above question, if the insect suddenly realizes that one of the faces on the opposite side of the cuboid having maximum area has been sprayed with pesticides due to which it can not reach the original destination and if the insect still wants to reach the opposite end of longest diagonal now possible, then in how many ways can it do so?

(a) $\frac{16!}{7!6!5!}$ (b) $\frac{18!}{6!5!4!}$

(c) $\frac{15!}{6!6!5!}$ (d) $\frac{16!}{6!5!5!}$

Direction for questions 18 and 19: Read the passage below and solve the questions based on it.

James Bond has to crack a password of 3 symbols. The symbols can be +, – or ×. Every time, he makes an attempt to crack the password, a feedback is given in the form of 'R' and 'W'. 'R' indicates that one of the symbols is in the right place. 'W' indicates that one of the digits is in the wrong position. All the codes are used.

This is his first two attempts with their respective feedbacks:

Attempt 1	Feedback
+ + ×	R R W
Attempt 1	
– + ×	R W W

Q.18. What is the minimum number of additional attempts required to crack the code?
(a) 0 (b) 1 (c) 2 (d) 3

Q.19. What is the right code?
(a) × – + (b) + – ×
(c) + + – (d) – + ×

Q.20. At the beginning of a party, each present person shook hands with all other people present, and in all there were 28 handshakes. In the midst of the party, 2 persons left due to an emergency. Now, the number of males and females present in the party was equal. At the end, each female shook hands only with every female present and each male shook hands only with every male present. What is the total number of handshakes that took place at the party?
(a) 52 (b) 40
(c) 34 (d) 36

Q.21. Consider S = (1, 2, 3, ... 10). In how many ways two numbers from S can be selected so that the sum of the numbers selected is a double digit number?
(a) 36 (b) 16
(c) 29 (d) ${}^{9}C_{2} - {}^{5}C_{2}$

Q.22. In a chess tournament every person played one game with every other person in the group. The total number of games that men played between themselves exceeded those played by men with women by 18. If there were 4 women in the tournament, in all how many games were played in the tournament?
(a) 114 (b) 66
(c) 78 (d) 120

Q.23. The number of ways of painting the faces of a cube with six different colour is
(a) 1 (b) 6
(c) 6! (d) None of these

Q.24. Find the number of non-negative integer solutions to the system of equations $a + b + c + d + e = 20$ and $a + b + c = 5$ is
(a) 240 (b) 336 (c) 672 (d) 1008

Q.25. If the number of ways of selecting k coupons out of an unlimited number of coupons bearing the letters A,T,C so that they can not be used to spell the word CAT is 93, then what is the value of k?
(a) 2 (b) 5
(c) 8 (d) None of these

Q.26. In a test of 10 multiple choice questions of one correct answer, each having 4 alternative answers, then the number of ways to put ticks at random for the answers to all the questions is
(a) 410 (b) 10^4
(c) $4^{10}-4$ (d) 10^4-10

Q.27. The streets of a city are arranged like the lines of a chessboard. There are m streets running north-south and n streets running in east-west direction. What is the number of ways in which a man can travel on these streets from the north-west to the south-east corner, going by the shortest possible distance.
(a) $(m + n - 2)!$
(b) $(m + n - 1)!/ [(m - 1)!]$
(c) $(m + n - 1)!/[(n - 1)!]$
(d) $(m + n - 2)!/[(m - 1!)(n - 1)!]$

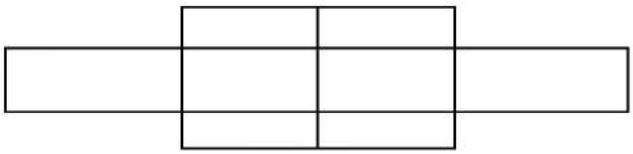
Q.28. Let $(1 + x + x^2)^9 = \alpha_0 + \alpha_1 x + ,,,,, \alpha_{18}x^{18}$, then which of the following is true?
(a) $\alpha_0 + \alpha_2 + \ldots + \alpha_{18} = \alpha_1 + \alpha_3 + \ldots + \alpha_{17}$;
(b) $\alpha_0 + \alpha_2 + \ldots + \alpha_{18}$ is even;
(c) $\alpha_0 + \alpha_2 + \ldots + \alpha_{18}$ is divisible by 9;
(d) $\alpha_0 + \alpha_2 + \ldots + \alpha_{18}$ is divisible by 3 but not by 9.

Q.29. If N be an element of the set A = {1, 2, 3, 5, 6, 10,15, 30} and P, Q and R are integers such that PQR = N, then the number of positive integral solutions of PQR = N, is
(a) 32 (b) 64
(c) 96 (d) 128

Q.30. In how many ways can one arrange letters in the word 'INSTITUTION' such that no two same letters comes together along with the following conditions:
(A) There is no 'T' which is immediately preceded as well as followed by 'N'.
(B) There is no 'T' which is immediately preceded as well as followed by 'T'.
(a) 32256 (b) 40320
(c) 60480 (d) None of these

Q.31. Consider S = (1, 2, 3, ... 10). In how many ways the two numbers from S can be selected so that the sum of the numbers selected is a double digit number?
(a) 36 (b) 16
(c) 29 (d) ${}^{9}C_{2} - {}^{5}C_{2}$

Q.32. Six X's have to be placed in the squares of the figure such that each row contains at least one X.



In how many different ways can this be done?
(a) 28 (b) 27
(c) 26 (d) None of these

Q.33. Number of ways in which n distinct things can be distributed among n persons so that atleast one person does not get anything is 232. Find n.
(a) 3 (b) 4 (c) 5 (d) 6

Q.34. Six persons A, B, C, D, E and F are to be seated at a circular table. In how many ways can this be done if A must always have either B or C on his immediate right and B must always have either C or D on his immediate right?
(a) 16 (b) 24 (c) 30 (d) 18

Q.35. The number of employees in a nationalised bank in a small town is 10, out of which 4 are female and the rest male. A committee of 5 is to be formed. If m be the number of ways to form such a committee in which there is atleast one female employee and n be the number of ways to form such a committee which includes at least two male employees, then find the ratio m:n.
(a) 3:2 (b) 5:1
(c) 1:1 (d) 0.8:0.9

Q.36. A question paper is split into two parts — Part A and Part B. Part A contains 5 questions and part B has 4 questions. Each question in part A has an alternative. A student has to attempt at least one question from each part. Find the number of ways in which the student can attempt the question paper?
(a) 1,830 (b) 2,440
(c) 3,630 (d) 4,260

Q.37. How many integers between 1 and 100000 have the sum of their digits equal to 18?
(a) 89,93 (b) 25,927
(c) 36,592 (d) 24,987

Q.38. A person invites his 10 friends on a party and places them 5 at one round table, 5 on the other round table. Find the number of ways in each case in which he can arrange the guests.
(a) ${}^{10}P_{5}/25$ (b) ${}^{10}P_{5} \times {}^{5}P_{5}/25$
(c) 10!/25 (d) None of these

Q.39. In the above question, find the number of ways in which he can arrange 4 at one round table and 6 at the other round table.
(a) ${}^{9}P_{5}/24$ (b) ${}^{9}P_{5} \times {}^{5}P_{5}/24$
(c) 10!/24 (d) None of these

Q.40. In how many ways 4 men and 4 women can be seated in a row so that men and women are alternate?
(a) 28 (b) 36
(c) 4!.4! (d) 2.4!.4!

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(b)	(a)	(a)	(a)	(a)	(d)	(b)	(a)	(c)	(c)	(d)	(b)	(b)	(a)	(a)	(a)	(c)	(b)	(b)

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(a)	(c)	(c)	(d)	(a)	(c)	(c)	(c)	(a)	(a)	(a)	(c)	(a)	(a)	(c)	(c)	(b)	(c)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(d)	(c)	(b)	(c)	(c)	(d)	(d)	(c)	(b)	(d)	(a)	(b)	(a)	(b)	(a)	(d)	(b)	(d)	(d)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(b)	(d)	(c)	(b)	(b)	(a)	(b)	(c)	(b)	(c)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(c)	(d)	(d)	(c)	(b)	(a)	(b)	(b)	(b)	(d)	(c)	(a)	(b)	(b)	(c)	(c)	(c)	(b)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(b)	(c)	(c)	(c)	(a)	(b)	(b)	(a)	(c)	(c)	(b)	(b)	(c)	(c)	(b)	(b)	(a)	(c)	(c)	(a)
Q. No.	41	42	43	44	45															
Answer	(c)	(d)	(a)	(a)	(a)															

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(c)	(d)	(c)	(c)	(c)	(b)	(c)	(b)	(b)	(d)	(a)	(a)	(b)	(d)	(a)	(d)	(a)	(b)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(c)	(d)	(a)	(b)	(b)	(a)	(d)	(b)	(b)	(c)	(c)	(c)	(b)	(d)	(c)	(c)	(b)	(c)	(c)	(d)

HINTS AND SOLUTIONS

Moderate

1. Total sets possible = ${}^6C_2 = 15$
   But ½ = 2/4 = 3/6
   1/3 = 2/6
   2/3 = 4/6
   So, distinct values of $x = 11$

2. It will be a number of 9 digits.
   $9 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 = 9 \times 9!$

3. Without any restriction, 7 can be arranged in 7! Ways.
   Now, keeping 4 of them fixed, A, B and C can be arranged in 3! Ways, i.e., out of every 3! Arrangement, we have one choice for which A-B-C order is followed.
   So, [7!]/[3!]
   It can be seen that – For AB = [7!]/[2!]
   – For A-B-C (or any three) = [7!]/[3!]
   – For A-B-C-D (or any four) = [7!]/[4!]

– For A-B-C-D-E (or any five) = [7!]/[5!]
– For A-B-C-D-E-F (or any six) = [7!]/[6!]
– For A-B-C-D-E-E-F = [7!]/[7!]

6. n identical things can be distributed among r persons in ${}^{n+r-1}C_{r-1}$

7. Same as 6.

10. ${}^{n}C_3 - {}^{n-1}C_2 = 84$, where n is the number of students.
Now use options.

13. Here, Since $x \geq 0, y \geq 1, z \geq 2$
Now, x can get any number of things, y should get a minimum of 1 thing and z should get a minimum of 2 things. So, let us give these to y and z.
Now, total things left = 12
Now use the formula n identical things can be distributed among r persons in ${}^{n+r-1}C_{r-1}$
Where n = 12, r = 3

17. Various ways of doing this is as follows:

A	B	C
2	2	3
2	3	2
3	2	2

Now do the selections.

18. Four persons have chosen to sit on one particular side (assume side A) and two persons on the other side (assume side B). So, we are supposed to select four persons for side A from the remaining 10 persons and remaining six persons will be sitting on side B.
Number of ways 4 persons can be selected from 10 persons = ${}^{10}C_4$
Number of ways 6 persons can be selected from the remaining 6 persons = ${}^{6}C_6$
Number of ways 8 persons can be arranged on side A = 8!
Number of ways 8 persons can be arranged on side B = 8!
Total number of ways = ${}^{10}C_4 \times {}^{6}C_6 \times 8! \times 8!$

20. This is nothing but ${}^{4}C_2 \times {}^{3}C_2$.

24. This can be done in three ways:
i. A is selected but B is not selected – A has already been selected, rest 4 is to be selected from 7 persons = ${}^{7}C_4 = 35$
ii. B is selected but A is not selected - ${}^{7}C_4 = 35$
iii. Neither A nor B is selected - ${}^{7}C_5 = 21$
Total = 91

26. $(x + 2)! = a$, ${}^{x}P_{11} = b$ and $(x - 11)! = c$
$(x + 2)! = 182 \times {}^{x}P_{11} \times (x - 11)!$
Now, go through the options.

31. $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$.

33. Case I. When a boy sits at the first place:
Possible arrangement will be of the form

B	G	B	G	B	G	B	G

Now there are four places namely 1st, 3rd, 5th and 7th for four boys, therefore four boys can be seated in $\lfloor 4$ ways. Again there are four places namely 2nd, 4th, 6th and 8th for four girls.
Four girls can be seated in $\lfloor 4$ ways.
Number of ways in this case = $\lfloor 4 \lfloor 4$.
And similarly G B G B G B G.B.
So, total = $2 \times 4! \times 4!$

36. Find 1×2 rectangles, 2×3 rectangles, 3×4 rectangles, 4×5 rectangles, 5×6 rectangles, 6×7 rectangles, 7×8 rectangles.

37. Given $x + y + z + u + t = 20$
$x + y + z = 5$
Given, system of equations is equivalent to $x + y + z = 5$ ... (1)
and $u + t = 15$ ... (2)
Number non-negative integral solutions of equation (1)
$= {}^{n+r-1}C_r = {}^{3+1-1}C_5 = {}^{7}C_5$,
Number of non-negative integral solution of equation (2)
$= {}^{n+r-1}C_r = {}^{2+15-1}C_{15} = {}^{16}C_{15}$
∴ Required number = ${}^{7}C_5 \, {}^{16}C_{15} = 336$

39. This situation is similar to distributing 29 identical things among 4 persons x, y, z and t in such a way that x is getting a minimum 1 thing, y is getting minimum 2 things, z is getting minimum 3 things and t is getting minimum 0 thing.
So, let us distribute 1 thing to x, 2 things to y, 3 things to z and 0 thing to t. After doing this distribution, we are free to distribute the remaining things among x, y, z and t in any ways.
Now the equation is: $x + y + z + t = 23$
The total number of ways of distributing 23 identical things among 4 persons in such a way that anybody can get any number of things = ${}^{26}C_3$

Advanced

1. Total number of triangles = Triangles having no sides common + Triangles having one side common + Triangles having two sides common + Triangles having three sides common. Now take help of Worked out Example 19.

2. Let $S_1, S_2, \ldots, S_8$ denote the stations where the train does not stop. The four stations where the train stops should be at any four of the nine places indicated by cross.

∴ Required number = ${}^{9}C_4 \dfrac{9.8.7.6}{\lfloor 4} = 126$

5. If $a = 1 = b$, then $c = 30$. Similarly, we can have $a = c = 1$ and $b = c = 1$. These are three sets.
Now, take $a = 1$, then $bc = 30 = 2^1 \times 3^1 \times 5^1$
Possible sets of b, c for the product $bc = 30$ is 13.
So, total number of sets for any one of these to be equal to zero = 39.
Now take $a = 2$, then $bc = 15 = 3^1 \times 5^1$. Now again find the sets of (b, c) and similarly check all the possibilities.

10. Go through the options using formula.

Questions 13 to 15

Fruits are ordered like

1. Apple 2. Mangoes
3. Oranges 4. Custard apple

Fruits are being put up first one, then two, then three, then four and so on.

13. $1 + 2 + 3 + \ldots + 44 = 990$
So, the 45th fruit will be the 100th fruit. Since we are having a set of four different types of fruits, 45th will be Apple.

14. Mangoes will come like 2, 6, 10, 14 etc.
100th mango will come $2 + 6 + 10 + 14 + 18 + 22 + 26 + 2 = 100$
100th mango will come in its 8th turn.
Before that apple must have got its 8th turn, orange and custard apple must have got their 7th turn.
Total apples displayed till now
$= 1 + 5 + 9 + \ldots + 29 = 120$
Total oranges displayed till now
$= 3 + 7 + 11 + \ldots + 27 = 105$
Total custard apples displayed till now
$= 4 + 8 + 12 + \ldots + 28 = 112$
So, total fruits displayed till now (other than mangoes) = 337
So, the position of the 100th mango = 437

Questions 18 and 19

Use the options of 19 to solve both the questions simultaneously.

21. If the sum is a two-digit number, then both the values simultaneously cannot be less than 5. Now start making the sets by taking different values greater than or equal to atleast one of the numbers.

22. ${}^{M}C_2 - {}^{M}C_1 \, {}^{W}C_1 = 18$, where M and W are the number of men and women respectively.

24. Given $a + b + c + d + e = 20$ ...(1)
$a + b + c = 5$ ...(2)
Given, system of equations is equivalent to $a + b + c = 5$ ...(3)
and $d + e = 15$
Number non-negative integral solutions of equation (3)
$= {}^{n+r-1}C_r = {}^{3+1-1}C_5 = 7\,C_5$,
Number of non-negative integral solution of equation (4)
$= {}^{n+r-1}C_r = {}^{2+15-1}C_{15} = {}^{16}C_{15}$
$\therefore$ Required number $= {}^{7}C_5 \, {}^{16}C_{15} = 336$

26. Each of the questions can be answered in 4 ways. So, 10 questions can be answered in 4^{10} ways.

28. Eliminate the options by assuming several values of $x = 1$ or 2 etc.

29. This is nothing but the application of the concept of number of factors (see Number System).

33. $n^n - n! = 232$. Use options now.

34. If B is sitting on the immediate right of A, then C or D can sit on the immediate right of B and the rest 3 places can be filled in 3! ways. So, the total number of ways when B is sitting to the immediate right of A $= 2 \times 3! = 12$.
Similarly, when C is sitting to the immediate right of A, then D has to sit on the immediate right of B, so B-D becomes a pair. Now for rest of 2 can be seated at three different positions in two ways each (This is dependent of the positions of B - C). Number of arrangements possible here = 6
So, total arrangements = 18

35.

Males	Females
6	4

Now take different cases to fulfill the conditions given.

40. Case I. When a boy sits at the first place:
Possible arrangement will be of the form

B	G	B	G	B	G	B	G

Now there are four places namely 1st, 3rd, 5th and 7th for four boys, therefore four boys can be seated in $\lfloor 4$ ways. Again there are four places namely 2nd, 4th, 6th and 8th for four girls.
$\therefore$ Four girls can be seated in $\lfloor 4$ ways.
$\therefore$ Number of ways in this case $= \lfloor 4 \, \lfloor 4$.
Similarly, another possible arrangement will be like:

B	G	B	G	B	G	B	G

Number of ways of doing this $= \lfloor 4 \, \lfloor 4$.
So, total ways $= 2 \, \lfloor 4 \, \lfloor 4$.

16

CHAPTER

Probability



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Probability
- Meaning and definitions of the different terms
- Different theorems
- Methods of solving questions

Introduction to the topic

Probability is an expectation of the happening of some event. In our day-to-day life, we often predict the outcome of several events, and often hear phrases such as probably it will be a hot day today or it will probably rain tomorrow. These statements involve an element of uncertainty and hence questions such as how to measure the uncertainty of the happening of something arises. Probability is just an attempt to quantify this method of predicting the future events with the help of some concepts. The theory of probability deals with those situations in which a particular result or outcome is not certain, but it can be any one of the several possible outcomes given to us for consideration. As a decision maker, we will be required to use this tool of probability to obtain called “satisfycing output”, keeping all the variables in mind.

PROBABLITY AND ITS APPLICATION

Probability is the measure of uncertainty. We turn our attention to one of the problems that is widely held responsible for the development of the theory of probability, namely, that of throwing a dice. As all of us know, a dice is a well-balanced cube with six faces marked with numbers from 1 to 6, one number on one face.

Let's throw a dice once. What are the possible outcomes? Clearly, a dice can fall with any of its faces uppermost. The number on each of the faces is, therefore a possible outcome and all the outcomes are equally probable. Hence, it is as likely to show up a number, say 2, as any other number 1, 3, 4, 5, or 6.

Since there are six equally likely outcomes in a single throw of a dice and there is only one way of getting a particular outcome (say 2). Therefore, the chance of the number 2 coming up on the dice is 1 by 6. In other words, the same phenomenon is known as probability of getting 2 in a single throw of dice is 1/6.

We write this as P(2) = 1/6

Similarly, when an ordinary coin is tossed, it may show up head (H) or tail (T). Hence, the probability of getting a head in a single toss of a coin is given by

$$P(H) = \frac{1}{2}$$

Before we define the process to finding out the probability, it is essential to understand the various terms associated with probability.

Trial and Elementary Events

When we repeat a random experiment under identical conditions, then the experiment is known as a trial and the possible outcomes of the experiment are known as elementary events.

For example,

- tossing of a coin is a trial and getting head or tail is an elementary event.
- throwing of a dice is a trial and getting 5 on its upper face is an elementary event.

Compound Events

Events obtained by combining two or more elementary events are known as the compound events. A compound event is said to occur if one of the elementary events associated with it occurs.

Exhaustive Number of Cases

The total number of possible outcomes of a random experiment in a trial is known as the exhaustive number of cases.

For example, in throwing of a dice the exhaustive number of cases is 6, since any one of the six face marked with 1, 2, 3, 4, 5, 6 may appear on its upper face.

Mutually Exclusive Events

Events are said to be mutually exclusive if the occurrence of any one of them prevents the occurrence of all the others, i.e., if no two or more than two events can occur simultaneously in the same trial.

Equally Likely Events

Events are equally likely if there is no reason for an event to occur in preference to any other event. For example, while throwing a dice, chances of occurring of Head or Tail are an equally likely event.

Favourable Number of Cases

The number of cases favourable to an event in a trial is the number of elementary events such that if any one of them occurs, we say that the event happens.

In other words, the number of cases favourable to an event in a trial is the total number of elementary events such that the occurrence of any one of them ensures the happening of the event.

For example, in throwing of a dice, the number of cases favourable to the appearance of a prime number is 3 viz., 2, 3 and 5.

Independent Events

Events are said to be independent if the happening (or non-happening) of one event is not affected by the happening (or non-happening) of others.

For example, if two cards are drawn from a well-shuffled pack of 52 cards one after the other with replacement, then getting an ace in the first draw is independent of getting a jack in the second draw. But, if the first card drawn in the first draw is not replaced, then the second draw is dependent on the first draw.

Sample Space

This is the most important factor in probability and is defined as the set of all possible outcomes of a random experiment associated with it.

These examples suggest the following definition of probability (assuming that outcomes are equally likely).

Now, Probability of an event E, written as P(E), is defined as

$$P(E) = \frac{\text{Number of outcomes favourable to E}}{\text{Total number of possible outcomes (sample space)}}$$

Points to Remember

1. $0 \le P(E) \le 1$
2. $P(E) + P'(E) = 1$

Example 1 In a single throw of two die, what is the probability that the sum on the top face of both the die will be more than 9?

Solution When two die are thrown, sum of the numbers appearing on the faces can be anything from 2 to 12. To find the number of favourable cases we will be required to find the cases in which the sum is more than 9.

Following are the cases—(5, 5), (6, 4), (4, 6), (6, 5), (5, 6), (6, 6)

So, the total number of favourable cases = 6

The total number of possible outcomes = $6 \times 6 = 36$

Hence probability = 6/36 = 1/6

Example 2 Six dice are thrown simultaneously. Find the probability that all of them show the same face.

Solution Sample space of throwing six dice = $6 \times 6 \times 6 \times 6 \times 6 \times 6 = 6^6$.

All dice are showing the same face implies that we are getting same number on the entire six dice. The number of ways for which is ${}^6C_1$.

Hence, the required probability = $\frac{{}^6C_1}{6^6} = \frac{1}{6^5}$

Example 3 In the above question, find the probability that all of them show a different face.

Solution The total number of ways in which all the dice show different numbers on their top-faces is the same as the number of arrangement of 6 numbers 1, 2, 3, 4, 5, 6 by taking all at a time.

So, the number of favourable cases = 6!

Hence, the required probability = $\frac{6!}{6^6}$

Example 4 Five persons enter a lift on the ground floor of an 8-floor apartment. Assuming that each one of them independently and with equal probability can leave the lift at any floor beginning with the first. What is the probability that all the five persons are leaving the lift at different floors?

Solution Apart from the ground floor, there are 7 floors.

A person can leave the lift at any of the seven floors. Hence the total number of ways in which each of the five persons can leave the lift at any of the 7 floors = 7^5.

So, the sample space = 7^5

Five persons can leave the lift at five different floor = ${}^7P_5$ ways.

So, the favourable number of ways = ${}^7P_5$.

Hence, the required probability = $\frac{{}^7P_5}{7^5}$

Example 5 If you have 3 tickets of a lottery for which 10 tickets were sold and 5 prizes are to be given, the probability that you will win at least one prize is: (JMET 2005)

(a) $\frac{7}{12}$ (b) $\frac{9}{12}$ (c) $\frac{1}{12}$ (d) $\frac{11}{12}$

Solution Probability that you will win at least one prize = 1 – probability that you will not win any prize.

$$= 1 - \frac{5C_3}{10C_3} = \frac{11}{12}$$

Odds in Favour and Odds Against

$$\text{Odds in Favour} = \frac{\text{Number of favourable cases}}{\text{Number of unfavourable cases}}$$

$$\text{Odds in Against} = \frac{\text{Number of unfavourable cases}}{\text{Number of favourable cases}}$$

Understanding And/Or To understand the role played by And/Or in our calculation, let us take the example of throwing an unbiased dice. Let A and B be the two events associated with it such that

A = getting an even number, B = getting a multiple of 3.

Then A = {2, 4, 6}, and B = {3, 6}.

We now define a new event "A or B" which occurs if A or B or both occur i.e., at least one of A or B occurs. Clearly the event "A or B" occurs if the outcome is any one of the outcomes {2,3,4,6}. Thus, the event "A or B" is represented by the subset A U B.

Similarly, "A and B" means occurrence of both A and B which is possible if the outcome is {6}.

Hence, it is represented by the subset $A \cap B$.

ADDITION THEOREM

If A and B are two events associated with a random experiment, then

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Corollary: If the events are mutually exclusive, then $P(A \cup B) = P(A) + P(B)$

Addition theorem can be extended for any number of events.

Example 6 A basket contains 20 apples and 10 oranges out of which 5 apples and 3 oranges are defective. If a person takes out 2 at random what is the probability that either both are apples or both are good?

Solution Out of 30 items, two can be selected in ${}^{30}C_2$ ways. So, exhausted number of cases = ${}^{30}C_2$.

Consider the following events:

A = getting two apples; B = getting two good items

Required probability = $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

There are 20 apples, out of which two can be drawn in ${}^{20}C_2$ ways.

$$\therefore P(A) = \frac{20C_2}{{}^{30}C_2}$$

There are 8 defective pieces and the remaining 22 are good. Out of 22 good pieces, two can be selected in $^{22}C_2$ ways.

$$\therefore \text{P(B)} = \frac{22C_2}{30C_2}$$

Since there are 15 pieces which are good apples of which 2 can be selected in $^{15}C_2$ ways, therefore

P (A∩B) = Probability of getting 2 pieces which are good apples = $\frac{15C_2}{30C_2}$

From (i)

Required probability = P(A) + P(B) – P(A∩B)

$$= \frac{20C_2}{30C_2} + \frac{22C_2}{30C_2} - \frac{15C_2}{30C_2} = \frac{316}{435}$$

CONDITIONAL PROBABILITY

Let A and B be the two events associated with a random experiment. Then, the probability of occurrence of A under the condition that B has already occurred and P (B) ≠ 0, is called the conditional probability and it is denoted by P(A/B).

Thus, P(A/B) = Probability of occurrence of A given that B has already happened.

Similarly, P (B/A) = Probability of occurrence of B given that A has already happened.

Sometimes, P (A/B) is also used to denote the probability occurrence of A when B occurs. Similarly, P(B/A) is used to denote the probability of occurrence of B when A occurs.

Following examples illustrate the various meanings of these notations:

Example 7 A bag contains 5 white and 4 red balls. Two balls are drawn form the bag one after the other without replacement. Consider the following events.

A = drawing a white ball in the first draw, B = drawing a red ball in the second draw.

Solution Now, P (B/A) = Probability of drawing a red ball in the second draw given that a white ball has already been drawn in the first draw.

Since 8 balls are left after drawing a white ball in the first draw and out of these 8 balls 4 balls are red, therefore

$$\text{P (B/A)} = \frac{4}{8} = \frac{1}{2}$$

Note that P (A/B) is not meaningful in this experiment because A cannot occur after the occurrence of B.

Example 8 One card is drawn from a pack of 52 cards, each of the 52 cards being equally likely to be drawn. Find the probability that the card drawn is

i. a king
ii. either a red or a king
iii. a red and a king

Solution Out of 52 cards, one card can be drawn in $^{52}C_1$ ways. Therefore, exhaustive number of cases = $^{52}C_1$ = 52

i. There are 4 kings in a pack of cards, out of which one can be drawn in $^4C_1$ ways. Therefore, the favourable number of cases = $^4C_1$ = 4, so the required probability = $\frac{4}{52} = \frac{1}{13}$

ii. There are 28 cards in a pack of cards which are either a red or a king. Therefore, one can be drawn in $^{28}C_1$ ways. Therefore, the favourable number of cases = $^{28}C_1$=28.

So the required probability = $\frac{28}{52} = \frac{7}{13}$

iii. There are 2 cards which are red and king, i.e., red kings. Therefore, the favourable number of cases = $^2C_1$ = 2. so the required probability = $\frac{2}{52} = \frac{1}{26}$

Example 9 An urn contains 9 blue, 7 white and 4 black balls. If 2 balls are drawn at random, find the probability that

(i) Both the balls are blue
(ii) One ball is white

Solution There are 20 balls in the bag out of which 2 balls can be drawn in $^{20}C_2$ ways. So the total number of cases(sample space) = $^{20}C_2$ = 190.

i. There are 9 blue balls out of which 2 balls can be drawn in $^9C_2$ ways. Therefore, the favourable number of cases = $^9C_2$ = 36. So the required probability = $\frac{36}{190} = \frac{18}{95}$

ii. There are 7 white balls out of which one white can be drawn in $^7C_1$ ways. One ball from the remaining 13 balls can be drawn in $^{13}C_1$ ways. Therefore, one white and one other colour ball can be drawn in $^7C_1 \times {}^{13}C_1$ ways. So the favourable number of cases = $^7C_1 \times {}^{13}C_1$ = 91.

So, the required probability = $\frac{91}{190}$

Example 10 Three persons A, B and C are to speak at a function along with five others. If they all speak in random order, the probability that A speaks before B and B speaks before C is

(a) 3/8 (b) 1/6
(c) 3/5 (d) None of these

Solution The total number of ways in which 8 persons can speak is $^8P_8$ = 8!. The number of ways in which A, B and C can be arranged in the specified speaking order is $^8C_3$. There are 5! ways in which the other five can speak. So, the favourable number of ways is $^8C_3$ = 5!.

Hence, the required probability = $\frac{^8C_3 \times 5!}{8!} = \frac{1}{6}$

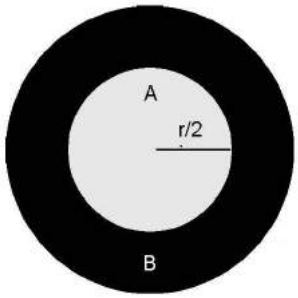
Example 11 There is a point inside a circle. What is the probability that this point is closer to the circumference than to the centre?

Solution

Assume that the radius of the bigger circle is r, and the radius of the inner circle is r/2. Point will be closer to the circumference than to the centre if the point is lying in the segment B.

Area of segment B = ¾ πr^2

So, the probability of point being closer to the circumference = ¾ $\pi r^2/\pi r^2$ = 3/4



PRACTICE EXERCISES

WARM UP

Q.1. What is the chance of drawing an ace from a deck of cards?
(a) 8/52 (b) 16/52
(c) 2/52 (d) 4/52

Q.2. A three digit number is to be formed by using the digits 1, 2, 3,... 9. What is the probability that the number formed is greater than 500, if repetition is not allowed?
(a) 280/504 (b) 54/280
(c) 58/204 (d) 24/504

Q.3. Mayank and Amit are throwing an unbiased dice. If Mayank throws 1, what is the Amit's chance of throwing a higher number?
(a) 1/5 (b) 5/6
(c) 2/12 (d) None of these

Q.4. A bag contains 4 red and 7 green balls. If three balls are drawn from the bag, replaced and once again three balls are drawn from the bag, what is the probability of obtaining 3 red balls on the first drawing and 3 green balls on the second drawing?
(a) 14/5445 (b) 14/27225
(c) 28/5445 (d) None of these

Q.5. One number is selected at random from the first 25 natural numbers. What is the probability that it is a multiple of either 5 or 7?
(a) 2/12 (b) 8/25
(c) 4/25 (d) None of these

Q.6. A bag contains 5 green apples and 7 red apples. If two apples are drawn from the bag, what is the probability that one is red and the other is green?
(a) 12/66 (b) 35/66
(c) 2/12 (d) 2/35

Q.7. The probability of a problem being solved by A is 1/4 the and B solving it is 1/5 the. If they try independently, then what is the probability that the problem is not solved?
(a) 2/5 (b) 4/9 (c) 3/5 (d) 2/9

Q.8. Ten horses are running in a race, the chances that A will win are 30%, that B will win are 20% and C will win are 10%. What is the probability that one of them will win?
(a) 0.689 (b) 0.598
(c) 0.498 (d) 0.398

Q.9. A's chance of winning a single game against B is 3/4 th. Find the chance that in a series of five games with B, A wins exactly 3 games?
(a) 135/1024 (b) 135/512
(c) 235/1024 (d) None of these

Q.10. It is known that at noon the sun is hidden by clouds on an average of two days out of every three. Find the probability that at noon on at least four out of five days the sun will be shining.
(a) 9/11 (b) 11/243
(c) 11/81 (d) None of these

Q.11. If on an average 1 vessel in every 10 is wrecked, find the chance that out of 5 vessels expected, at least 4 will arrive safely?
(a) 61246/50000
(b) 4592/5000
(c) 35927/50000
(d) None of these

Q.12. Two fair dice are thrown. What is the probability that the number of dots on the first dice exceeds 3 and that on the second exceeds 4?
(a) 2/6 (b) 3/6 (c) 1/6 (d) 5/6

Q.13. What is the probability that a card drawn at random from a pack of 52 cards is either a King or a Spade?
(a) 2/52 (b) 4/13
(c) 4/26 (d) None of these

Q.14. Nazneen was dialing a telephone number of six digits. She forgot the last three digits of the six digit telephone number but remembered that the number formed by the last three digits in the same order was a perfect square. What is the probability that the she dialed a right number?
(a) 1/32 (b) 2/32
(c) 2/16 (d) 1/23

Q.15. What is the probability that there are 53 Sundays and 53 Tuesdays in a leap year?
(a) 0 (b) 1
(c) 0.5 (d) None of these

Q.16. What is the probability that there are 52 Sundays in a leap year?
(a) 0 (b) 1
(c) 0.5 (d) None of these

Q.17. What is the probability that there are 53 Sundays and 53 Saturdays in a leap year?
(a) 1/7 (b) 2/7
(c) 0 (d) None of these

Q.18. What is the probability that there are 53 Sundays in a leap year?
(a) 2/7 (b) 1/7
(c) 0 (d) None of these

FOUNDATION

Q.1. Five boys and three girls are seated at random in a row. The probability that no boy sits between two girls is
(a) $\frac{1}{56}$ (b) $\frac{1}{8}$
(c) 3/28 (d) None of these

Q.2. In a convex hexagon two diagonals are drawn at random. The probability that the diagonals intersect at an interior point of the hexagon is
(a) $\frac{5}{12}$ (b) $\frac{7}{12}$
(c) $\frac{2}{5}$ (d) None of these

Q.3. Two cards are drawn at random from a pack of 52 cards. The probability of getting at least a spade and an ace is
(a) $\frac{1}{34}$ (b) $\frac{8}{221}$ (c) $\frac{1}{26}$ (d) $\frac{2}{51}$

Q.4. If the letters of the word ATTEMPT are written down at random, the chance that all the Ts are consecutive is
(a) $\frac{1}{42}$ (b) $\frac{6}{71}$
(c) $\frac{1}{7}$ (d) None of these

Q.5. Seven white balls and 3 black balls are placed in a row at random. The probability that no two black balls are adjacent is
(a) $\frac{1}{3}$ (b) $\frac{7}{15}$ (c) $\frac{2}{15}$ (d) $\frac{1}{3}$

Q.6. Four gentlemen and 4 ladies take seats at random round on a table. The probability that they are sitting alternately is
(a) $\frac{4}{35}$ (b) $\frac{1}{70}$ (c) $\frac{2}{35}$ (d) $\frac{1}{35}$

Q.7. Let $x = 33^n$. The index n is given a positive integral value at random. The probability that the value of x will have 3 in the units place is
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{1}{3}$ (d) None of these

Q.8. Three dice are thrown simultaneously. The probability of getting a sum of 15 is
(a) $\frac{1}{72}$ (b) $\frac{5}{36}$
(c) $\frac{5}{72}$ (d) None of these

Q.9. Three dice are thrown. The probability of getting a sum which is a perfect square is
(a) $\frac{2}{5}$ (b) $\frac{9}{20}$
(c) $\frac{1}{4}$ (d) None of these

Q.10. A box contains 6 red balls, 7 green balls and 5 blue balls. Each ball is of a different size. The probability that the red ball being selected is the smallest red ball is

(a) $\frac{1}{18}$ (b) $\frac{1}{3}$ (c) $\frac{1}{6}$ (d) $\frac{2}{3}$

Q.11. Two distinct numbers are selected at random from the first twelve natural numbers. The probability that the sum will be divisible by 3 is

(a) $\frac{1}{3}$ (b) $\frac{23}{66}$

(c) $\frac{1}{2}$ (d) None of these

Q.12. Three different numbers are selected at random from the set A={1, 2, 3, ... 10}. The probability that the product of two of the numbers is equal to the third is

(a) $\frac{3}{4}$ (b) $\frac{1}{40}$

(c) $\frac{1}{8}$ (d) None of these

Q.13. There are 7 seats in a row. Three persons take seats at random. The probability that the middle seat is always occupied and no two persons are consecutive is

(a) $\frac{9}{70}$ (b) $\frac{9}{35}$

(c) $\frac{4}{35}$ (d) None of these

Q.14. From a group of 10 persons consisting of 5 lawyers, 3 doctors and 2 engineers, four persons are selected at random. The probability that the selection contains at least one of each category is

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) $\frac{2}{3}$ (d) None of these

Direction for questions 15 and 16: Read the passage below and solve the questions based on it.

A and B are two possible events of an experiment such that P(A∪B) = 0.7 and P(A) = 0.4, then find P(B) given that

Q.15. A and B are mutually exclusive events

(a) 0.6 (b) 0.3 (c) 0.2 (d) 0.5

Q.16. A and B are independent events.

(a) 0.5 (b) 0.3 (c) 0.6 (d) 0.2

Q.17. The odds against an event are 3 to 4 and the odds in favour of another independent event are 2 to 5. The probability that exactly one of them occurs is

(a) $\frac{23}{49}$ (b) $\frac{26}{49}$ (c) $\frac{6}{49}$ (d) $\frac{8}{49}$

Q.18. In the previous question, what is the probability that atleast one of them occurs is

(a) $\frac{34}{49}$ (b) $\frac{15}{49}$ (c) $\frac{23}{49}$ (d) $\frac{26}{49}$

Q.19. Chunmun picks a letter from the set of English alphabet and finds it to be a vowel. What is the probability that the letter is E?

(a) 1/26 (b) 5/6
(c) 1/5 (d) None of these

Q.20. There are 3 red, 3 blue and 3 green balls. What is the probability that out of 3 selected balls 2 are red?

(a) 1/4 (b) 2/5
(c) 3/7 (d) None of these

Q.21. In the horse race, the horse Exaclibur's chances of winning are $\frac{1}{6}$, the horse Noorjehan's chances are $\frac{1}{10}$ and those of the horse Bombay are $\frac{1}{8}$. If the race starts with twelve horses and only one horse wins, then what are the chances that one of these three horses will win?

(a) $\frac{57}{120}$ (b) $\frac{47}{120}$ (c) $\frac{37}{120}$ (d) $\frac{27}{120}$

Q.22. If a year has 360 days and all the months with 30 days. What is the probability that your birthday falls on a Monday and that is an even day of an even month, given that January 1 is a Monday?

(a) $\frac{1}{10}$ (b) $\frac{13}{360}$

(c) $\frac{1}{28}$ (d) $\frac{11}{360}$

Q.23. If 4 whole numbers are taken at random, and multiplied together, then the probabaility that the last digit in the product is 1, 3, 7 or 9 will be

(a) 13/976 (b) 17/529
(c) 16/625 (d) 13/625

Q.24. There are 500 students in an examination. 150 students passed the first paper, 350 students passed the second paper and 50 students passed both the papers. Find the probability that a student selected at random has failed in both the papers.

(a) 1/5 (b) 1/10
(c) 3/10 (d) 3/5

Q.25. A person is asked to randomly pick two balls from a bag which has 15 yellow & 5 red balls. The probability that the person picks two balls of the same colour is

(a) 33/38 (b) 23/38
(c) 38/43 (d) 15/38

MODERATE

Q.1. If the integers m and n are chosen at random between 1 and 100, then the probability that a number of the form $7^m + 7^n$ is divisible by 5 is

(a) $\frac{1}{5}$ (b) $\frac{1}{7}$ (c) $\frac{1}{4}$ (d) $\frac{1}{49}$

Q.2. If the probability of A failing in an examination is $\frac{1}{5}$ and that of B failing the same examination is $\frac{3}{10}$, then the probability that either A or B fails is

(a) $\frac{1}{2}$ (b) $\frac{11}{25}$

(c) $\frac{19}{50}$ (d) None of these

Q.3. India plays two matches each with West Indies and New Zealand. In any match the probabilities of India getting 0,1 and 2 points are 0.45, 0.05 and 0.50 respectively. Assuming that the outcomes are independent, the probability of India getting at least 7 points is

(a) 0.0875 (b) $\frac{1}{16}$

(c) 0.1125 (d) None of these

Q.4. Let A and B be two independent events such that P(A) = $\frac{1}{5}$, $P(A \cup B) = \frac{7}{10}$ Then $P(\bar{B})$ is equal to

(a) $\frac{3}{8}$ (b) $\frac{2}{7}$

(c) $\frac{7}{9}$ (d) None of these

Q.5. Let A and B be two independent events in such a way that their probabilities are $\frac{3}{10}$ and $\frac{2}{5}$ respectively. The probability of exactly one of the events happening is

(a) $\frac{25}{30}$ (b) $\frac{1}{2}$

(c) $\frac{31}{50}$ (d) None of these

Q.6. A, B, C are three events for which P(A) = 0.6, P(B) = 0.4, P(C) = 0.5, P(A∪B) = 0.8, P(A∩B) = 0.3 and P(A∩B∩C) = 0.2. If Π(A∪B∪C) ≥ 0.85, then the interval of values of P(B∩C) is

(a) [0.2,0.35] (b) [0.55, 0.7]

(c) [0.2,0.55] (d) None of these

Q.7. A dice is thrown $2n+1$ times, $n \in N$. The probability that faces with even numbers shown odd number of times is

(a) $\frac{2n+1}{4n+3}$ (b) Less than $\frac{1}{2}$

(c) Greater than $\frac{1}{2}$ (d) None of these

Q.8. A shooter firing at a target has 10% chance of hitting the target in one shot. The number of times he must fire at the target to have above 50% chance of hitting the target is

(a) 11 (b) 9 (c) 7 (d) 5

Q.9. Let A = {2, 3, 4, ... , 20, 21}. A number is chosen at random from the set A and it is found to be a prime number. The probability that it is more than 10 is

(a) $\frac{9}{10}$ (b) $\frac{1}{10}$

(c) $\frac{1}{5}$ (d) None of these

Q.10. The probabilities that a student passes in mathematics, physics and chemistry are m, p and c respectively. Of these subjects, a student has a 75% chance of passing in at least one, a 50% chance of passing in at least two, and a 40% chance of passing in exactly two subjects. Which of the following relations are true?

(a) $p+m+c = \frac{19}{20}$ (b) $p+m+c = \frac{27}{20}$

(c) $pmc = \frac{1}{20}$ (d) $pmc = \frac{1}{4}$

Q.11. If A and B are two events such that P(A∪B) ≥ $\frac{3}{4}$ and $\frac{1}{8} \le P(A \cap B) \le \frac{3}{8}$ then

(a) $P(A) + P(B) \le \frac{11}{8}$

(b) $P(A).P(B) \le \frac{3}{8}$

(c) $P(A) + P(B) \ge \frac{7}{8}$

(d) None of these

Q.12. Three natural numbers are taken at random from the set A={$x|x, 1 \le x \le 100, x \in N$}. The probability that the AM of the numbers taken is 25 is

(a) $\frac{^{77}C_2}{^{100}C_3}$ (b) $\frac{^{25}C_2}{^{100}C_3}$

(c) $\frac{^{74}C_{72}}{^{100}C_3}$ (d) None of these

Q.13. The probability of a number n showing in a throw of a dice marked 1 to 6 is proportional to *n*. then the probability of the number 3 showing in throw is

(a) $\frac{1}{2}$ (b) $\frac{1}{6}$ (c) $\frac{1}{7}$ (d) $\frac{1}{21}$

Direction for questions 14 to 16: Read the passage below and solve the questions based on it.

Three mountaineers Amit, Vinit and Nishit are climbing up a mountain with their respective probabilities of reaching the summit being $\frac{1}{3}, \frac{1}{5}$ and $\frac{1}{4}$ respectively. What is the probability that

Q.14. All of them reach the summit?

(a) $\frac{3}{60}$ (b) $\frac{1}{60}$ (c) 1 (d) 0

Q.15. At least one of them reaches the summit?

(a) $\frac{59}{60}$ (b) $\frac{36}{60}$ (c) $\frac{17}{60}$ (d) $\frac{1}{60}$

Q.16. Exactly one of them reaches the Summit?

(a) $\frac{13}{30}$ (b) $\frac{17}{30}$ (c) $\frac{19}{30}$ (d) $\frac{11}{30}$

Q.17. The coach of the football team says that the probability that player A plays and team wins is $\frac{1}{4}$, player B plays and team wins is $\frac{1}{2}$ and both play and team wins is $\frac{1}{4}$. He also added that the team has $\frac{3}{4}$ chances to win. What is the probability that when either player A or player B played, the team won?

(a) $\frac{2}{3}$

(b) $\frac{1}{2}$

(c) Cannot be determined

(d) None of these

Direction for questions18 and 19: Read the passage below and solve the questions based on it.

Two teams Patel Nagar and Boring Road are playing a cricket tournament. If team Patel Nagar wins the toss, probability of it winning the match is 0.55. If Boring Road wins the toss, probability of its winning the match is 0.65. Probability of winning the toss is same for both the teams Patel Nagar and Boring Road.

Q.18. In a five-matche tournament, what is the probability of Patel Nagar winning all the matches, if it wins at most two tosses?

(a) 0.45 (b) 0.41 (c) 0.39 (d) 0.51

Q.19. What is the probability of Boring Road winning all the matches, if Patel Nagar wins 3 tosses in a five match tournament?

(a) 0.16

(b) 0.14

(c) 0.21

(d) None of these

Q.20. Rakesh went to take an exam. After the exam, while explaining the structure of the question paper, the teacher said that 40% of the total questions were difficult and 20% of those were on algebra and 30% of those were on geometry. Also, 10% and 20% of easy questions were on algebra and geometry respectively. All the questions that Rakesh attempted were either on algebra or on geometry. What is the probability that Rakesh found all the questions difficult (Questions are either easy or difficult)?

(a) $\frac{14}{100}$ (b) $\frac{14}{27}$

(c) $\frac{20}{27}$ (d) None of these

Direction for questions 21 and 22: Read the passage below and solve the questions based on it.

In a game, one can score as many points as he gets in every throw till he gets a score (in one throw) which is not a multiple of points scored in the previous throw.

Q.21. If a person scores 5 points in total, what is the probability that the dice was thrown 2 times?

(a) $\frac{3}{4}$ (b) $\frac{3}{5}$

(c) $\frac{1}{2}$ (d) None of these

Q.22. If a person throws the dice 3 times, what is the probability that the total score is 11?

(a) 5/43

(b) 5/41

(c) 5/37

(d) None of these

Q.23. Triangles are formed by joining vertices of an octagon. Any one of those triangle is selected at random. What is the probability that the selected triangle has no side common with the octagon?

(a) 3/7 (b) 2/7 (c) 5/7 (d) 1/7

Q.24. What is the probability that a 4 digit number of the type abcd (a, b, c, d are all different) is a multiple of 11?
(a) 0.142 (b) 0.048
(c) 0.072 (d) 0.091

Q.25. A month is randomly selected from the months in a non leap year and it is found that it has five Sundays. What is the probability that it has five Mondays?
(a) 1/7 (b) 2/7 (c) 2/3 (d) 20/33

ADVANCED

Q.1. Manoj throws a fair dice. He is promised an amount thrice the value of the number showing up if the number showing up is odd and an amount twice the value of the number showing up if it is even. What is the maximum amount Manoj is willing to pay each time to throw the dice, if in the long run he wants to make an average profit of Rs 5 per throw?
(a) Rs 3.50 (b) Rs 8.5
(c) Rs 5 (d) None of these

Direction for questions 2 to 4: Read the passage below and solve the questions based on it.

A shopkeeper inspected some items in a box containing a, b and c number of items of type P, Q and R respectively. He picked up an item randomly, and put it back into the box if he found that the item selected was regularly sold, he also added the equal number of items of the same type as those previously in the box otherwise he removed all the items of that type.

Q.2. If exactly one of the three items is regularly sold then what is the probability that the item selected in the second draw is similar to that in the first draw? (Given that the item selected in the second draw is a regularly sold item.)

(a) $2\left(\frac{a}{a+b+c}+\frac{b}{a+b+c}+\frac{c}{a+b+c}\right)$

(b) $\left(\frac{2a^2}{2a+b+c}+\frac{2b^2}{a+2b+c}+\frac{2c^2}{a+b+2c}\right)\frac{1}{a+b+c}$

(c) $\left(\frac{a}{2a+b+c}+\frac{b}{1+2b+c}+\frac{c}{a+c+2c}\right)\frac{1}{a+b+c}$

(d) None of these

Q.3. In the above problem, what is the probability that the item selected in the second draw is not a regularly sold item?

(a) $\frac{a}{a+b+c}+\frac{b}{a+b+c}+\frac{c}{a+b+c}$

(b) $\frac{b+c}{2a+b+c}+\frac{a+c}{a+2b+c}+\frac{a+b}{a+b+2c}$

(c) $\frac{a}{b+c}+\frac{b}{a+c}+\frac{c}{a+b}$

(d) None of these

Q.4. What is the probability that the items selected in the second draw is different from that in the first draw, given that the item selected in the first draw is a regularly sold item? (Assume exactly two of the three items are not regularly sold items.)

(a) $\left(\frac{a(a+c)}{2a+b+c}+\frac{b(a+c)}{a+2b+c}+\frac{c(a+b)}{a+b+2c}\right)\frac{1}{a+b+c}$

(b) $\left(\frac{a(b+c)}{2a+b+c}+\frac{b(a+c)}{a+2b+c}+\frac{c(a+b)}{a+b+2c}\right)\frac{1}{a+b+c}$

(c) $\left(\frac{a}{2a+b+c}+\frac{b}{a+2b+c}+\frac{c}{a+b+2c}\right)\frac{1}{a+b+c}$

(d) 0

Q.5. There is a frame of a cuboid of length 6, breadth 5 and height 7. The cuboid is only composed of skeleton of 210 cubes of side 1. An insect is on one corner of the cube and it wants to travel to the opposite end of the longest diagonal. It can only travel along the sides of the a small cubes and it always takes the shortest possible route. Find the probability that it passes through atleast one of the corners.

(a) $\frac{1}{6}$ (b) $\frac{7}{12}$
(c) $\frac{5}{18}$ (d) None of these

Q.6. A coin is so biased that the heads occurs four times as frequently as tails. Another coin is biased such that the heads occurs 65% of the times. When the two coins are tossed simultaneously, what is the probability of at least one tail turning up?
(a) 35% (b) 87% (c) 48% (d) 73%

Q.7. Only one bomb is needed to destroy a bridge and there are three bombers engaged in an attempt to destroy the bridge. The probabilities of the bombers hitting

the bridge are respectively 0.3, 0.4 and 0.7. What is the probability of the bridge being destroyed?
(a) 0.874 (b) 0.126 (c) 0.916 (d) 0.084

Q.8. A pack of playing cards was found to contain only 51 cards. If the first 13 cards, which are examined, are all red, what is the probability that the missing card is black?
(a) 3/7 (b) 5/7 (c) 2/3 (d) 3/8

Q.9. A speaks truth in 75% of the cases and B in 80% of the cases. In what percentage of cases are they likely to contradict each other in stating the same fact?
(a) 15% (b) 20% (c) 5% (d) 35%

Direction for questions 10 to 12: Read the passage below and solve the questions based on it.

In a manufacturing unit machines A, B, C manufacture 15%, 25%, 60% of the total production of bolts respectively. Of all the bolts manufactured by the machine 4% by A, 2% by B and 3% by C are defective respectively. A bolt is drawn at random and is found to be defective.

Q.10. What is the probability that it was produced by machine A?
(a) 6/29 (b) 5/29 (c) 18/29 (d) 4/29

Q.11. What is the probability that it was produced by machine B?
(a) 6/29 (b) 5/29 (c) 18/29 (d) 4/29

Q.12. What is the probability that it was produced by machine C?
(a) 6/29 (b) 5/29 (c) 18/29 (d) 4/29

Q.13. There is a 30% chance that it rains on any particular day. What is the probability that there is at least one rainy day within a period of 3 days?
(a) $(0.7)^3$ (b) $1-(0.7)^3$
(c) 0.3^3 (d) None of these

Q.14. The numbers are selected at random, one at a time, from the two-digit numbers 00, 01, 02, ... 99 with replacement. An event E occurs only if the product of the two digits of a selected numbers is 18. If four numbers are selected, find the probability that the event E occurs at least 3 times.
(a) 0.0025 (b) 0.00025
(c) 0.025 (d) 0.25

Q.15. A and B are two independent events. The probability that both A and B occur is 1/6 the and the probability that neither of them occurs in 1/3rd. Find the probability of the occurrence of A.
(a) 1/6
(b) 1/4
(c) 1/3
(d) Cannot be determined

Q.16. A biased dice is tossed and the respective probabilities for various faces to turn up are

Face	1	2	3	4	5	6
Probability	0.1	0.24	0.19	0.18	0.15	0.14

If an even face has turned up, then the probability that it is face 2 or face 4 is
(a) 0.25 (b) 1.42 (c) 0.75 (d) 0.9

Q.17. A natural number x is chosen at random from the first one hundred natural numbers. What is the probability that $x+\frac{100}{x}>50$?
(a) 13/20 (b) 3/5 (c) 9/20 (d) 11/20

Q.18. A dice is loaded in such a way that the probability of getting i on the dice is proportional to i. This dice is rolled three times. What is the probability of getting a product of 15?
(a) 10/1029 (b) 14/1029
(c) 16/1029 (d) 18/1029

Q.19. A target may be either at point I with probability $\frac{8}{9}$ or at point II with probability $\frac{1}{9}$. We have 21 shells, each of which can be fired either at point I or at point II. Each shell may hit the target, independently of the other shells, with probability $\frac{1}{2}$ How many shells must be fired at point I to hit the target with maximum probability?
(a) 10 (b) 11
(c) 9 (d) 12

Q.20. Let N cells be numbered 1, 2, 3, ... N. We randomly throw balls into them. The process is continued until any one of the cells receives two balls. What is the probability that more than n throws will be necessary?
(a) ${}^{N}P_{N}/N^{n}$
(b) ${}^{N}/N^{n}$
(c) ${}^{N}C_{n}/N^{n}$
(d) None of these

Q.21. The probability that a man will live 10 more years is $\frac{1}{4}$ and the probability that his wife will live 10 more years is $\frac{1}{3}$. Then the probability that neither will be alive in 10 years is
(a) $\frac{5}{12}$ (b) $\frac{7}{12}$
(c) $\frac{1}{2}$ (d) $\frac{11}{12}$

Q.22. A locker can be opened by dialing a fixed three-digit code between 000 and 999. A stranger, who does not know the code, tries to open the locker by dialing three digits at random. Find the probability that the stranger succeeds at the sth trial. (Assume that the stranger does not repeat unsuccessful combinations.)
(a) 2s/1000 (b) 2s/999
(c) s/1000 (d) s/999

Q.23. The probability of a bomb hitting a bridge is $\frac{1}{2}$ and two direct hits are needed to destroy it. The least number of bombs required so that the probability of the bridge being destroyed is greater than 0.9 is
(a) 8 (b) 9
(c) 10 (d) 11

Q.24. From a bag which is known to contain 4 balls each of which is just as likely to be black as white, a ball is drawn and is found to be white. What is the probability that the bag contained 3 white and 1 black balls?
(a) 3/7 (b) 5/8
(c) 3/8 (d) 4/7

Q.25. On a loaded dice, any even number occurs thrice as frequently as an odd number. If the dice is thrown twice, what is the probability that the sum is 10 or more?
(a) 55/72 (b) 17/72
(c) 39/72 (d) 23/72

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Answer	(d)	(a)	(b)	(c)	(b)	(b)	(c)	(d)	(b)	(b)	(d)	(c)	(b)	(a)	(a)	(b)	(a)	(a)

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(a)	(c)	(c)	(b)	(d)	(a)	(d)	(d)	(c)	(a)	(b)	(c)	(a)	(b)	(a)	(b)	(a)	(c)	(d)
Q. No.	21	22	23	24	25															
Answer	(b)	(b)	(c)	(b)	(b)															

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(b)	(a)	(a)	(a)	(d)	(d)	(c)	(d)	(c)	(d)	(c)	(c)	(b)	(b)	(a)	(a)	(b)	(d)	(b)
Q. No.	21	22	23	24	25															
Answer	(b)	(a)	(b)	(a)	(d)															

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(b)	(d)	(a)	(d)	(c)	(c)	(c)	(d)	(a)	(b)	(c)	(b)	(b)	(d)	(d)	(d)	(a)	(d)	(c)
Q. No.	21	22	23	24	25															
Answer	(c)	(c)	(b)	(c)	(b)															

HINTS AND SOLUTIONS

Moderate

1. For the sum to be divisible by 5, unit digit of $7^m + 7^n$ should be either 0 or 5.
   Different unit digit possible for 7 = 7, 9, 3 and 1.
   So, if 7^m is giving unit digit 7, 7^n should give 3, or vice-versa.
   Similarly, if 7^m is giving unit digit 9, 7^n should give 1, or vice-versa.

2. Use P(AUB) = 1/5 + 3/10 – (1/5) × (3/10)

5. Do as given in 2.

9. Total number of primes = 8 and number of numbers more than 10 = 11

14. All of them reach the summit = 1/3 × 1/5 × 1/4 = 1/60

15. To find out "At least one of them reaches the summit", we will find the probability of nobody reaching the summit = 2/3 × 4/5 × 3/4 = 2/5. Hence, the probability of at least one of them reaching the summit = 1 – 2/5 = 3/5

16. This is same as: Amit reaches the summit, but Vinit and Nishit do not reach the summit. Or, Vinit reaches the summit, but Amit and Nishit do not reach the summit. Or, Nishit reaches the summit, but Vinit and Amit do not reach the summit.
   Amit reaches the summit, but Vinit and Nishit do not reach the summit = 1/3 × 4/5 × 3/4 = 1/5 and similarly we will find all the other possibilities.

23. Total number of triangles formed = $^8C_3 = 56$
   Triangles having two sides common = 8
   Triangles having one side common = $^8C_1 \times {}^4C_1 = 32$
   Triangles having three sides common = 0
   Triangles having no side common = 56 – 40 = 16
   So, probability = 16/56 = 2/7.

Advanced

7. It is nothing but [1 – probability of bridge not getting destroyed] = 1 – 0.084

9. Contradiction can occur if A speaks truth and B lies, B speaks truth and A lies.
   So, required probability = 0.75 × 0.2 + 0.8 × 0.25 = 0.35

13. Probability that it rains = 0.3, so probability that it does not rain = 0.7.

15. 1 – Neither of them occurs = At least one of them occurs
   So, 2/3 = P(A) + P(B) – P(A ∩ B), So, it cannot be determined.

17. $x^2 - 50x + 100 > 0$. Find the range of the values of x and solve the question.

21. The probability that a man will not live 10 more years = 3/4 and the probability that his wife will not live 10 more years =2/3. Then the probability that neither will be alive in 10 years = 3/4 × 2/3.

BENCHMARKING TEST 3

This test paper contains 30 questions of 4 marks each. One-third of the mark allotted to a particular question will be deducted in case of wrong answer.

Time Given: 70 Minutes **Total Marks: 120** **Qualifying Marks: 55 Marks**

Break-up of the Expected Percentile
(assuming this to be a CAT paper)

Marks	*Expected Percentile*
88+	99+
83	98.5+
78	98
55	92
40	85
30	80

Q.1. The first term, a term of an AP = the common ratio of a GP and the first term of the GP = common difference of the AP. If the sum of the first two terms of the GP is equal to the sum of the first 2 terms of the AP, then the ratio of first term of GP to the first term of AP is
(a) > 1
(b) Independent of first term of AP
(c) Independent of first term of GP
(d) < 1

Q.2. In a college of 300 students, every student reads 5 newspapers and every newspaper is read by 60 students. The number of newspapers is
(a) At least 30 (b) At most 20
(c) Exactly 25 (d) None of these

Q.3. The condition that both the roots of a quadratic equation $ax^2 + bx + c = 0$ are positive, is
(a) a and c have opposite sign that of b
(b) b and c have opposite sign that of a
(c) a and b have opposite sign that of c
(d) None of these

Q.4. The 4th degree polynomial equation $p(x) = x^4 - 2x^3 + 3x^2 - 4x + 5 = 0$ has
(a) At least one real root
(b) No real root
(c) Two real roots
(d) All real roots

Q.5. Let $g(x)$ be a function such that $g(x+1) + g(x-1) = g(x)$ for every real x. Then for what value of p is the relation $g(x+p) = -g(x)$ necessarily true for every real x?
(a) 5 (b) 3 (c) 2 (d) 6

Q.6. If f is real valued function and $f(x+y) = f(y)\ \forall\ x \in R$. Given $f(1) = 3$ and $\sum_{r=1}^{r=n} f(r) = 1092$. Find the value of n.
(a) $n = 5$ (b) $n = 6$
(c) $n = 7$ (d) None of these

Q.7. Let $f(x) = 0$ be the polynomial equation of the least possible degree with rational coefficients. One of the roots of this equation is $\sqrt{7} + \sqrt[3]{49}$. What is the product of all the roots of $f(x) = 0$?
(a) 21 (b) 35 (c) 42 (d) 56

Q.8. If $f(x)$ is a real valued function and $f\{x + f(y)\} = x + f(y)$. Then $f(100)$ is
(a) 100 (b) 1000
(c) 10000 (d) 100000

Q.9. The quadratic equation $px^2 + qx + r = 0$ satisfies the following conditions:
1. p, q and r are distinct
2. p, q, r $\{1, 2, 3, 4, 5, 6\}$
3. $(x + 1)$ is a factor of quadratic equation $px^2 + qx + r = 0$

How many such quadratic equations are possible?
(a) 8 (b) 12
(c) 16 (d) None of these

Q.10. Given are $f(x) = 4^x + 4^{-x}$ and $g(x) = 5/4$. At how many points graphs of both the functions will intersect?
(a) 0 (b) 1 (c) 2 (d) Infinite

Q.11. How many values of p ($p > 1$) satisfy the following equation:
$\log_2 p \cdot \log_4 p \cdot \log_6 p = \log_2 p \cdot \log_4 p + \log_2 p \cdot \log_6 p + \log_4 p \cdot \log_6 p$?
(a) 0 (b) 1
(c) 2 (d) More than 2

Q.12. If $\sum_{r=0}^{\infty} \frac{1}{(2r-1)^2} = N$, then which of the following is equal to $\sum_{r=1}^{\infty} \frac{1}{r^2}$?
(a) $(3/4)N - 1$ (b) $(4/3)N + 1$
(c) $3/4(N + 1)$ (d) $4/3 (N - 1)$

Q.13. Let p, q, r and s be real numbers. If $N = \max[\min(p, q), \min(r, s)]$ and $S = \min[\max(p, r), \max(q, s)]$. Then which of the following is true?
(a) $N \le S$, for all values of p, q, r and s
(b) $N \ge S$, for all values of p, q, r and s
(c) $N = S$, for all values of p, q, r and s
(d) Cannot be uniquely determined

Q.14. The maximum possible value of $x^2 + 4y^2 + 9z^2$ is N. What is the value of N if $x + 2y + 3z = 12$ (x, y and z are real numbers)?
(a) 224 (b) 16
(c) 4096 (d) Infinite

Q.15. x, y, z and t are positive numbers which satisfy the following conditions:
1. If $x > y$, then $z > t$ and
2. If $x > z$, then $y < t$

Then which of the following is true?
(a) If $x < y$, then $z < t$
(b) If $x < z$, then $y > t$
(c) If $x > y + z$, then $z > y$
(d) If $x > y + z$, then $z < y$

Q.16. If $h(x)$ is the greatest integer less than or equal to $h/2$, then how many of the following statements is/are true?
1. $h(x)$ is half the least even integer greater than or equal to h.
2. $h(x)$ is the least integer greater than or equal to $\frac{h}{2} - 1$.
3. $h(x)$ is one less than half the least even integer greater than h.
(a) 0 (b) 1 (c) 2 (d) 3

Q.17. The number of ways in which three distinct numbers in AP and GP can be selected from 1, 2, 3, ... 24 is A and G respectively. Which of the following is the value of A : G?
(a) 21 (b) 12
(c) 9 (d) None of these

Q.18. $N_1 + N_2 + N_3 + \ldots + N_N = 100$, where N_1, N_2, N_3 ... and N_N are $n(n>1)$ consecutive natural numbers such that $N_1 < N_2 < N_3 < N_4 \ldots < N_N$. If $N_1 > 8$, then how many values of n is/are possible?
(a) 4 (b) 8
(c) 2 (d) Infinite

Q.19. $P = \frac{10^{999} + 9.10^{998} + 9^2.10^{997} + \ldots + 9^{999}}{10^{10^3} - 9^{10^3}}$, then find the value of $\frac{(P+1)(P+100)}{P(P-2)}$?
(a) 1
(b) 101
(c) 202
(d) None of these

Q.20. Let $f(x)$ be a function such that $f(x - 1) + (x + 1) = \sqrt{2} f(x)$. Then for what value of y is the relation $f(x + y) = f(x)$ necessarily true for every real x?
(a) 4 (b) 6 (c) 8 (d) 12

Q.21. The sum of all the real roots of the equations $|x - 2|^2 + |x - 2| - 2 = 0$ is
(a) 2 (b) −4 (c) −2 (d) 4

Q.22. Let p and q be roots of the equation $x^2 - 2x + A = 0$ and let r and s be the roots of the equations $x^2 - 18x + B = 0$. If $p < q < r < s$ are in an arithmetic progression then what is the value of A + B?
(a) −77 (b) 74 (c) −74 (d) 77

Q.23. Let 't' be any root of the equation $x^n + x^{n-1} + x^{n-2} + \ldots + 1 = 0$. Then $(t^{2n+2} + 3)(t^{3n+3} - 4)$ equals
(a) 12 (b) −12 (c) 6 (d) −6

Q.24. An n – digit number is a positive number with exactly n digits. Nine hundred distinct n – digit numbers are to

be formed using only the three digits 2, 5 and 7. The smallest value of n for which this is possible is
(a) 6 (b) 7 (c) 8 (d) 9

Q.25. Let $a_1, a_2, \ldots a_{10}$ be in A.P. and $h_1, h_2 \ldots, h_{10}$ be in H.P. If $a_1 = h_1 = 2$ and $a_{10} = h_{10} = 3$, then $a_4 h_7$ is
(a) 2 (b) 3 (c) 5 (d) 6

Q.26. If the integers m and n are chosen at random between 1 and 100, then the probability that a number of the form $7^m + 7^n$ is divisible by 5 equals
(a) $\frac{1}{4}$ (b) $\frac{1}{7}$
(c) $\frac{1}{8}$ (d) $\frac{1}{49}$

Q.27. Let $S_1, S_2 \ldots$ be squares such that for each $n \geq 1$, the length of a side of S_n equals the length of a diagonal of S_{n+1}. If the length of a side of S_1 is 10 cm, then for how many values of n is the area of S_n less than 1 cm^2?
i. 7 ii. 8 iii. 9 iv.10
(a) 1 (b) 2
(c) 3 (d) 4

Q.28. If a, b, c, d are positive real numbers such that $a + b + c + d = 2$, then $M = (a + b)(c + d)$ satisfies the relation
(a) $0 \leq M \leq 1$ (b) $1 \leq M \leq 2$
(c) $2 \leq M \leq 3$ (d) $3 \leq M \leq 4$

Q.29. How many different nine digit numbers can be formed from the number 223355888 by rearranging its digits so that the odd digits occupy even positions?
(a) 16 (b) 36
(c) 60 (d) 180

Q.30. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is
(a) 1 (b) 2
(c) $2\sqrt{2}$ (d) 4

ANSWERS

■■■ Benchmarking Test 3

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(c)	(a)	(b)	(b)	(b)	(d)	(a)	(b)	(a)	(b)	(d)	(a)	(d)	(c)	(d)	(b)	(c)	(d)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30										
Answer	(d)	(b)	(b)	(b)	(d)	(a)	(c)	(a)	(c)	(b)										

Module 4

Measurement

- Geometry
- Mensuration
- Co-ordinate Geometry

17

CHAPTER

Geometry



LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Triangles, polygons, circles, and quadrilaterals
- Different theorems and their applications
- Similarity and conguency
- Kinds of questions asked in the CAT
- Methods of solving questions

Introduction to the topic

Over the years, Geometry has become as important as the number system. In other words, it is one of those areas which can help students in getting good marks in QA on its own. In CAT, questions for close to 20% marks are asked from geometry alone. However, this is the only the secondary reason of this topic being so significant; the primary reasons being (i) creation of level-playing field for everyone as all the questions asked from this chapter pertain to close10 level maths only and (ii) most of the questions can be solved by the method of measurement of simple construction, without using any theorem of geometry.

Geometry, deals with measures and properties of points, lines, surfaces and solids. Here we will be discussing:

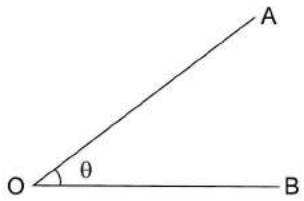
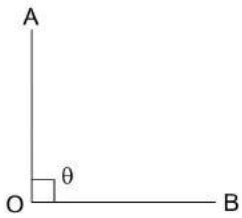
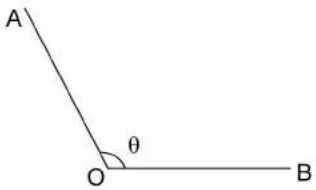
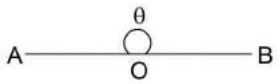
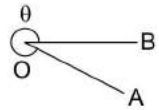
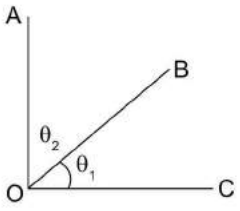
1. Lines and their properties
2. Polygons and their properties
3. Triangles and their properties
4. Quadrilaterals and their properties
5. Circles and their properties

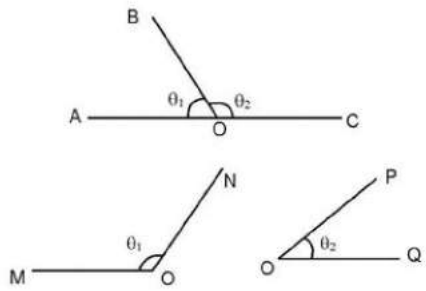
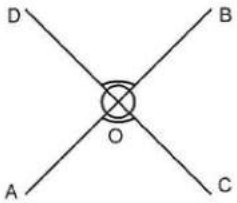
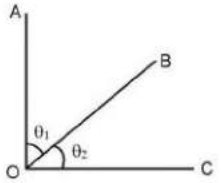
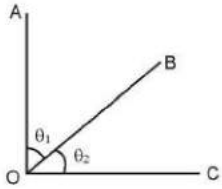
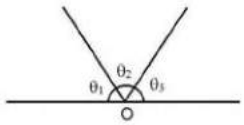
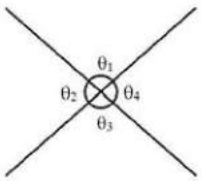
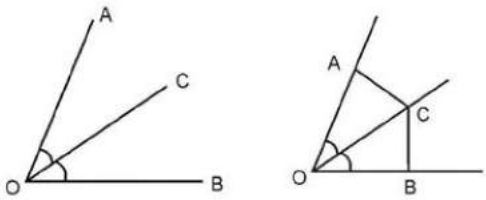
LINES AND THEIR PROPERTIES

A line is a set of points placed together that extends into infinity is both directions.

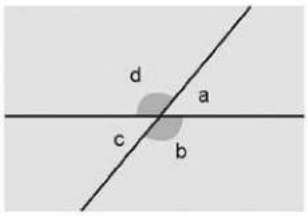
a) Different angles and pairs of angles

Measurement and nomenclature

Types of Angles	*Property*	*Diagram*
Acute Angle	$0° < \theta < 90°$ ($\angle AOB$ is an acute angle)	
Right Angle	$\theta = 90°$	
Obtuse Angle	$90° < \theta < 180°$	
Straight Line	$\theta = 180°$	
Reflex Angle	$180° < \theta < 360°$	
Complementary Angle	$\theta_1 + \theta_2 = 90°$ Two angles whose sum is 90° are complementary to each other	

Supplementary Angle	$\theta 1 + \theta 2 = 180°$ Two angles, whose sum is 180°, are supplementary to each other	
Vertically Opposite Angle	$\angle DOA = \angle BOC$ and $\angle DOB = \angle AOC$	
Adjacent Angles	$\angle AOB$ and $\angle BOC$ are adjacent angles Adjacent angles must have a common side (e.g., OB)	
Linear Pair	$\angle AOB$ and $\angle BOC$ are linear pair angles. One side must be common (e.g., OB) and these two angles must be supplementary.	
Angles on One Side of a Line	$\theta_1 + \theta_2 + \theta_3 = 180°$	
Angles Round the Point	$\theta_1 + \theta_2 + \theta_3 + \theta_4 = 360°$	
Angle Bisector	OC is the angle bisector of $\angle AOB$. i.e., $\angle AOC = \angle BOC = \frac{1}{2}(\angle AOB)$ When a line segment divides an angle equally into two parts, then it is said to be the angle bisector (OC)	 (Angle bisector is equdistant from the two sides of the angle) i.e.,

b) Angles associated with two or more straight lines



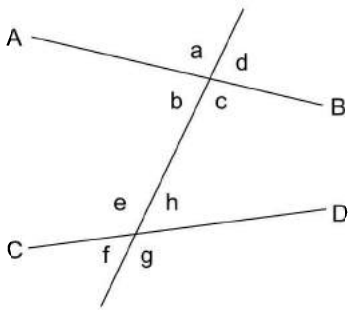
When two straight lines cross each other, ∠d and ∠b are the pair of vertical angles.

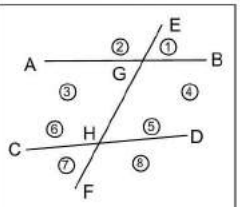
∠a and ∠c are the pair of vertical angles.

Vertical angles are equal in value.

Alternate angles and corresponding angles

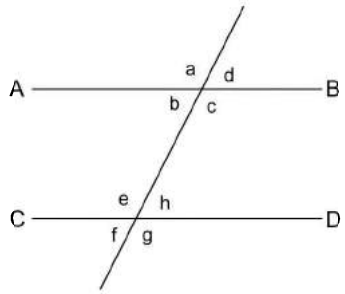
In the figure given below, corresponding angles are ∠a and ∠e, ∠b and ∠f, ∠d and ∠h, ∠c and ∠g. The alternate angles are ∠b and ∠h, ∠c and ∠e.



Corresponding angles	When two lines are intersected by a transversal, then they form four pairs of corresponding angles (a) ∠AGE, ∠CHG = (∠2, ∠6) (b) ∠AGH, ∠CHF = (∠3, ∠7) (c) ∠EGB, ∠GHD = (∠1, ∠5) (d) ∠BGH, ∠DHF = (∠4, ∠8)	

c) Angles associated with parallel lines

A line passing through two or more lines in a plane is called a transversal. When a transversal cuts two parallel lines, then the set of all the corresponding angels will be equal and similarly, the set of all the alternate angles will be equal.



In the figure given above, corresponding ∠a = ∠e and corresponding ∠b = ∠f

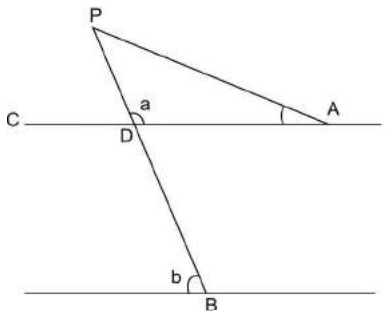
Similarly, alternate ∠b = ∠h and alternate ∠c = ∠e.

Now, ∠b + ∠c = 180°, so ∠b + ∠e = ∠h + ∠c = 180°

So we can conclude that the sum of the angles on one side of the transversal and between the parallel lines will be equal to 180°.

Converse of the above theorem is also true. When a transversal cuts two lines, and if the corresponding angles are equal in size, or if alternate angles are equal in size, then the two lines are parallel.

Example 1 In the figure given below, find the value of ∠b in terms of ∠a.



Solution In the given figure, ∠b = Alternate ∠PDC = 180° − ∠PDA = 180° − a

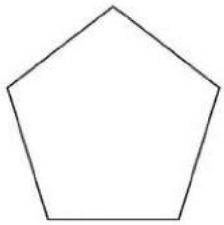
POLYGONS AND THEIR PROPERTIES

Any closed plane figure with *n* sides is known as a polygon. If all the sides and the angles of this polygon are equivalent, the polygon is called a regular polygon. Polygons can be convex or concave. The word "polygon" derives from the Greek *poly*, meaning "many," and *gonia*, meaning "angle." The most familiar type of polygon is the regular polygon, which is a convex polygon with equal side lengths and angles.

The generalization of a polygon into three dimensions is called a *polyhedron*, and into four dimensions is called a *polychoron*.

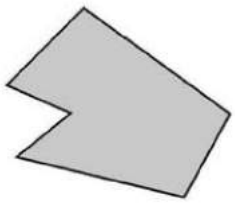
A **convex polygon** is a simple polygon that has the following features:

- Every internal angle is at most 180 degrees.
- Every line segment between the two vertices of the polygon does not go outside the polygon (i.e., it remains inside or on the boundary of the polygon). In other words, all the diagonals of the polygon remain inside its boundary.
- Every triangle is strictly a convex polygon.



Convex Polygon

If a simple polygon is not convex, it is called **concave**. At least one internal angle of a concave polygon is larger than 180°.



Concave Polygon

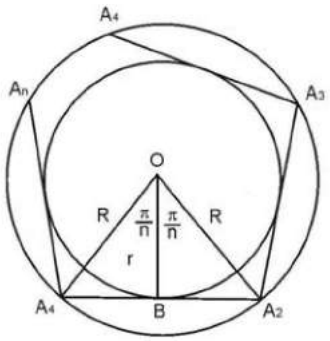
As we can see in the above figure, one of the internal angles is more than 180°.

Here onwards, all the discussion about polygon refer to regular polygons only.

Polygons are named on the basis of the number of sides they have. A list of some of the polygons are given below:

Number of sides	*Name of the polygon*
3	Triangle
4	Quadrilateral
5	Pentagon
6	Hexagon
7	Heptagon
8	Octagon
9	Nonagon
10	Decagon

Area and Perimeter of a Regular Polygon



Given, $A_1 A_2 A_3 A_4 \ldots A_n$ is a with regular polygon, 'n' sides.

$A_1A_2 = A_2A_3 = A_3A_4 = \ldots = A_{(n-1)}A_n$ = a units

OB (in radius) = r and $OA_1 = OA_2$ (Circumradius) = R

(i) Perimeter (P) = na

(ii) Area $= \frac{na}{2} \times r = \frac{p}{2} \times r$

(iii) Area $= \frac{na}{2}\sqrt{R^2 - \left(\frac{a}{2}\right)^2} = \frac{p}{2}\sqrt{R^2 - \left(\frac{a}{2}\right)^2}$

(iv) Area $= \frac{na^2}{4} \times \cot\left(\frac{\pi}{n}\right)^\circ$

(v) Area $= nr^2 \times \tan\left(\frac{\pi}{n}\right)^\circ$

(vi) r (in radius) $= \frac{a}{2}\cot\left(\frac{\pi}{n}\right)^\circ$

Properties of a Polygon

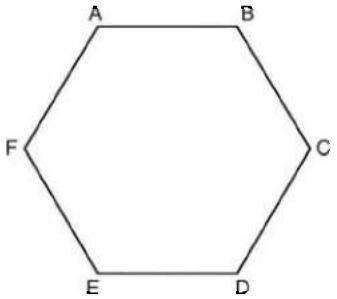
1. Interior Angle + Exterior Angle = 180°
2. The number of diagonals in an n-sided polygon = $n(n-3)/2$
3. The sum total of all the exteriors angles of any polygon = 360°
4. The measure of each exterior angle of a regular polygon $= \frac{360^\circ}{n}$
5. The ratio of the sides of a polygon to the diagonals of a polygon is $2 : (n-3)$
6. The ratio of the interior angle of a regular polygon to its exterior angle is $(n-2) : 2$
7. The sum total of all the interior angles of any polygon $= (2n-4) \times 90^\circ$

   So, each interior angle $= \frac{(2n-4)90^\circ}{n}$

Some frequently used polygons

Apart from triangles and quadrilaterals, regular hexagon and regular octagon are also worth mentioning.

Regular hexagon In the figure given below, ABCDEF is a regular hexagon with each side measuring a unit. Point O inside the hexagon is the centre of the hexagon.



Sum of the interior angles = 720°

Each interior angle = 120°

Each exterior angle = 60°

$$\text{Area} = \frac{3\sqrt{3}}{2}a^2$$

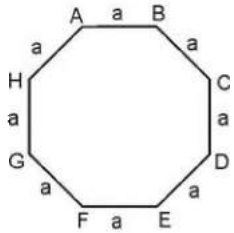
Regular octagon In the figure given below, ABCD-EFGH is a regular octagon with each side measuring 'a' unit.

Sum of the interior angles = 1080°

Each interior angle = 135°

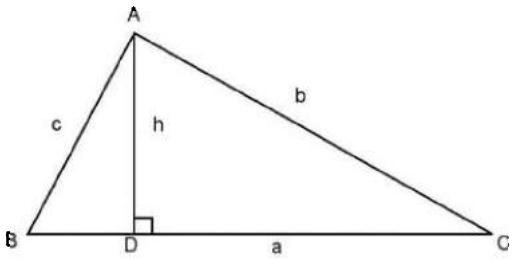
Each exterior angle = 45°

$$\text{Area} = 2a^2(1+\sqrt{2})$$



TRIANGLES AND THEIR PROPERTIES

A triangle is a figure enclosed by three sides. In the figure given below ABC is a triangle with sides AB, BC and CA measuring c, a and b units respectively. Line AD represents the height of the triangle corresponding to the side BC and is denoted by h.



In any triangle ABC

$$\text{Area} = \frac{1}{2} \times BC \times AD = \frac{1}{2}\, a \times h$$

Properties of a Triangle

- The sum of all the angles of a triangle = 180°
- The sum of lengths of the two sides > Length of the third side
- The difference of any two sides of any triangle < length of the third side
- The area of any triangle can be found by several methods:

  (a) Area of any triangle $= \frac{1}{2} \times$ base × perpendicular to base from the opposite vertex.

  (b) Area of any triangle $= \sqrt{S(S-a)(S-b)(S-c)}$, where S is the semi-perimeter of the triangle and a, b and c are the sides of a triangle.

  (c) Area of any triangle $= \left(\frac{1}{2}\right) \times$ bc Sin A

  Besides, there are some formulae which we use exclusively in some particular cases.

Example 2 What is the number of distinct triangles with integral valued sides and perimeter as 14?

(a) 6 (b) 5 (c) 4 (d) 3

Solution The sum of the lengths of the two sides > the length of the third side

So, the maximum length of any particular side can be 6 units.

Now if a = 6, then b + c = 8, so the possible sets are (6, 6, 2), (6, 5, 3) and (6, 4, 4)

If a = 5, then b + c = 9, so the possible set is (5, 5, 4)

So, the number of distinct triangles = 4

CLASSIFICATION OF TRIANGLES

Based Upon Sides

1. Scalene triangle

A triangle whose all sides are of a different length is a scalene triangle.

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}, \text{ where}$$

$$\text{S (semi perimeter)} = \frac{a+b+c}{2}$$

Example 3 What is the area of the triangle with side lengths 4 units, 5 units and 10 units?

Solution This triangle is not possible. (Since the sum of lengths of the two sides > Length of the third side)

2. Isosceles triangle

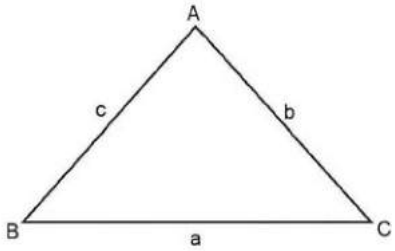
A triangle whose two sides are of an equal length is an isosceles triangle.

$$\text{Height} = \sqrt{\frac{4a^2 - b^2}{2}}$$

$$\text{Area} = \frac{b}{4}\sqrt{4a^2 - b^2}$$

3. Equilateral triangle

A triangle whose all sides are of an equal length is called an equilateral triangle.



In any equilateral triangle, all the three sides are of an equal length, so $a = b = c$

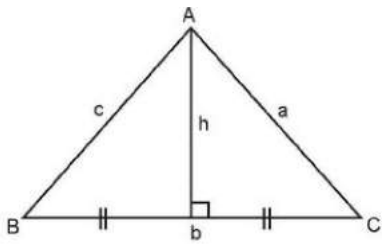
$$\text{Height} = (\text{side})\ \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}a$$

$$\text{Area} = \frac{\sqrt{3}}{4}(\text{side})^2 = \frac{\sqrt{3}}{4}a^2$$

Based Upon Angles

1. Right-angled triangle

A triangle whose one angle is of 90° is called a right-angled triangle. The side opposite to the right angle is called the hypotenuse.



$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{perpendicular}$$

Pythagoras Theorem

Pythagoras theorem is applicable in case of right-angled triangle. It says that, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$

$a^2 + b^2 = c^2$



For more proofs of Pythagoras theorem, go to www.cut-the-knot.org/pythagoras/index.shtml

The smallest example is $a = 3$, $b = 4$ and $c = 5$. You can check that

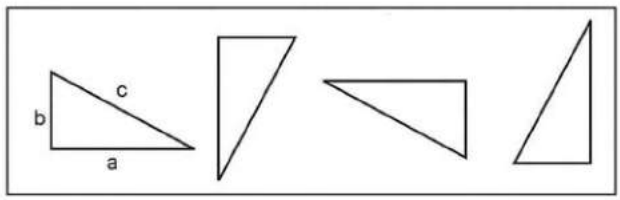
$3^2 + 4^2 = 9 + 16 = 25 = 5^2$.

Sometimes we use the notation (a, b, c) to denote such a triple.

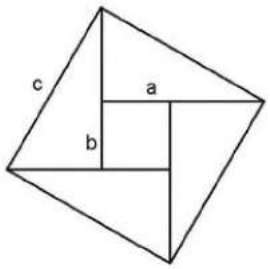
Notice that the greatest common divisor of the three numbers 3, 4 and 5 is 1. Pythagorean triples with this property are called *primitive*.

Proofs of Pythagoras theorem

Proof 1



Now we start with four copies of the same triangle. Three of these have been rotated at 90°, 180°, and 270°, respectively. Each has the area ab/2. Let's put them together without additional rotations so that they form a square with side c.

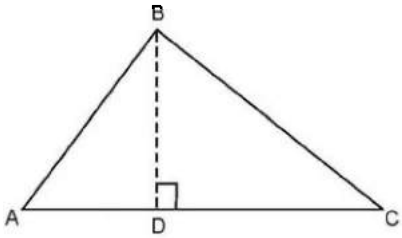


The square has a square hole with the side $(a - b)$. By summing up its area $(a - b)^2$ and 2ab, the area of the four triangles (4·ab/2), we get

$C^2 = (a-b)^2 + 2ab = a^2 + b^2$. QED

Proof 2 ABC is a right-angled triangle at B

To Prove: $AC^2 = AB^2 + BC^2$



Construction: Draw BD ⊥ AC

Proof: ΔADB ~ΔABC (Property 8.5)

$\therefore \frac{AD}{AB} = \frac{AB}{AC}$ (Sides are proportional)

Or $AB^2 = AD \times AC$ (1)

Also, $\Delta CDB \sim \Delta CBA$

$\therefore \frac{CD}{BC} = \frac{BC}{CA}$

$\therefore \frac{BC^2}{AC^2} = \frac{BD \times BA}{AB \times AD}$ (Sides are proportional)

or $BC^2 = CD \times CA$

Adding (1) and (2)

$AB^2 + BC^2 = AD \times AC + CD \times AC$

$= AC\,[AD + CD]$

$= AC \times AC = AC^2$ QED.

Pythagorean triplets

A *Pythagorean triplet* is a set of three positive whole numbers a, b and c that are the lengths of the sides of a right triangle.

$a^2 + b^2 = c^2$

It is noteworthy to see here that all of a, b and c cannot be odd simultaneously. Either of a or b has to be even and c can be odd or even.

The various possibilities for a, b and c are tabled below:

a	b	c
Odd	Odd	Even
Even	Odd	Odd
Odd	Even	Odd
Even	Even	Even

Some Pythagoras triplets are:

3	4	5	$(3^2 + 4^2 = 5^2)$
5	12	13	$(5^2 + 12^2 = 13^2)$
7	24	25	$(7^2 + 24^2 = 25^2)$
8	15	17	$(8^2 + 15^2 = 17^2)$
9	40	41	$(9^2 + 40^2 = 41^2)$
11	60	61	$(11^2 + 60^2 = 61^2)$
20	21	29	$(20^2 + 21^2 = 29^2)$

Note If each term of any pythagorean triplet is multiplied or divided by a constant (say P, P > 0) then the triplet so obtained will also be a pythagorean triplet. This is because if $a^2 + b^2 = c^2$, then $(Pa)^2 + (Pb)^2 = (Pc)^2$, where P > 0.

For example, 3×2 4×2 5×2 gives

6 8 10 $(6^2 + 8^2 = 10^2)$

Using Pythagoras theorem to determine the nature of triangle

If $c^2 = a^2 + b^2$, then the triangle is right-angled triangle.

If $c^2 > a^2 + b^2$, then the triangle is an obtuse-angled triangle.

If $c^2 < a^2 + b^2$, then the triangle is an acute-angled triangle.

Mechanism to derive a Pythagorean triplet If the length of the smallest side is odd, assume the length of the smallest side = 5

Step 1 Take the square of 5 (length of the smallest side) = 25

Step 2 Break 25 into two parts P and Q, where P – Q = 1. In this case, P = 13 and Q = 12. Now these two parts P and Q along with the smallest side constitute pythagorean triplet.

However, there is another general formula for finding out all the primitive pythagorean triplets:

$a = r^2 - s^2$,
$b = 2rs$,
$c = r^2 + s^2$,
$r > s > 0$ are whole numbers,
$r - s$ is odd, and
The greatest common divisor of r and s is 1.

Table of small primitive Pythagorean triplets Here is a table of the first few primitive Pythagorean triplets:

r	s	a	b	c
2	1	3	4	5
3	2	5	12	13
4	1	15	8	17
4	3	7	24	25
5	2	21	20	29
5	4	9	40	41
6	1	35	12	37
6	5	11	60	61
7	2	45	28	53

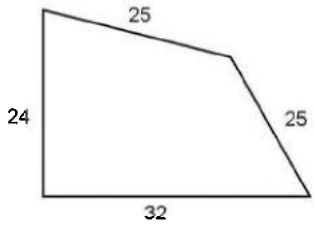
Perimeter, area, inradius and shortest side

The perimeter P and area K of a Pythagorean triple triangle are given by

$P = a + b + c = 2r(r + s)\,d$.

$K = ab/2 = rs\,(r^2 - s^2)\,d^2$.

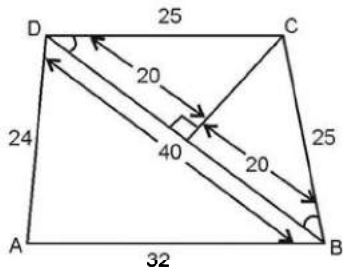
Example 4 Two sides of a plot measure 32 m and 24 m and the angle between them is a right angle. The other two sides measure 25 m each and the other three angles are not right angles.



What is the area of the plot (in m^2)?

(a) 768 (b) 534 (c) 696 (d) 684

Solution The figure given above can be seen as



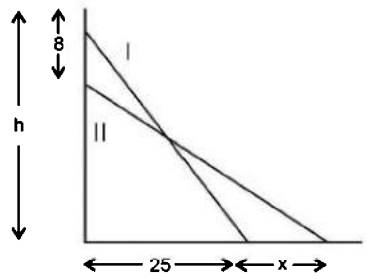
Since ABD is a right-angled triangle, so it will satisfy the Pythagoras theorem. And the triplet used here is – 3(×8), 4(×8) and 5(×8). Similarly the other part of the figure can also be bifurcated by drawing a perpendicular from C on BD.

So, the area of the plot is:

Area (ΔABD) + Area (ΔCBD) $= \frac{1}{2} \times 24 \times 32 + 2 \times (1/2 \times 20 \times 15) = 684$ m^2

Example 5 A ladder of length 65 m is resting against a wall. If it slips 8 m down the wall, then its bottom will move away from the wall by N m. If it was initially 25 m away from it, what is the value of x?

(a) 60 m (b) 39 m (c) 14 m (d) 52 m



Solution Using Pythagorean triplets, (5, 12, 13),

$\Rightarrow$ h = 60

After it has slipped by 8 m, the new height = 52 m, and the length of the ladder = 65 m.

So $25 + x = 39$ (3, 4, 5 triplet)

$\Rightarrow x = 14$ m

2. Obtuse-angled triangle

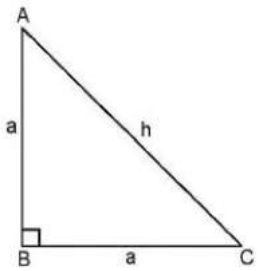
If one of the angles of the triangle is more than 90°, then the triangle is known as an Obtuse angled triangle. Obviously in this case, rest of the two angles will be less than 90°.

3. Acute-angled triangle

If all the angles of the triangle are less than 90°, then the triangle is known as acute angled triangle.

4. Isosceles right-angled triangle

A right angled triangle whose two sides containing the right angle are equal in length, is an isosceles right triangle.



In this case, Hypotenuse (h) $= a\sqrt{2}$

Perimeter $= 2a + h = 2a + a\sqrt{2}$

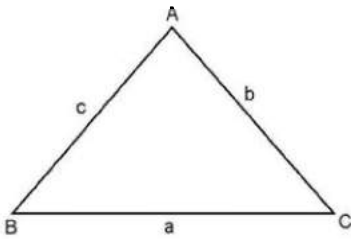
$= a\sqrt{2}(\sqrt{2} + a)$

$= h(1 + \sqrt{2})$

= Hypotenuse $(1 + \sqrt{2})$

Trigonometric formulae

In any Δ ABC,



Area of a triangle

Area of $\Delta = \frac{1}{2}$ bc sin A, where $\angle A = BAC$

Area of $\Delta = \frac{1}{2}$ ac Sin B, where $\angle B = ABC$

Area of $\Delta = \frac{1}{2}$ ab Sin C, where $\angle C = ACB$

Cosine Rule and Sine Rule

In any Δ, we have six quantities namely the three angles and the three sides. Using the following rules, we can find any

of the three quantities if we are provided with the remaining three quantities.

$$\text{Cos } A = \frac{b^2 + c^2 + a^2}{2bc}$$

$$\text{Cos } B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\text{Cos } C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\text{Sine Rule } \frac{a}{\text{Sin } A} = \frac{b}{\text{Sin } B} = \frac{c}{\text{Sin } C}$$

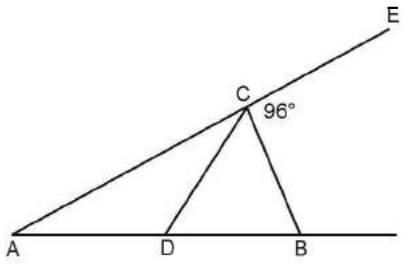
Sine and Cosine formulae are particularly more important in cases where we have one side and two angles out of the three angles of the triangle and we have to find out the value of all the sides and angles.

$$\frac{AB}{\sin \angle ACB} = \frac{BC}{\sin \angle BAC} = \frac{AC}{\sin \angle ABC} = 2R$$

Two important points

- The side opposite to the largest angle will be the largest.
- The side opposite to the smallest angle will be the smallest.

Example 6 In the figure given below, AD = CD = BC. What is the value of ∠CDB?

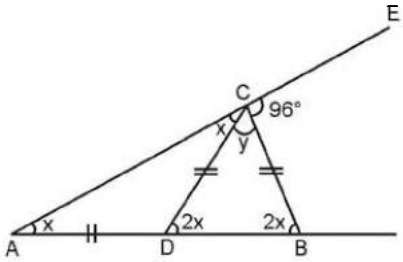


Solution $x + y = 180 - 96 = 84°$ (i)

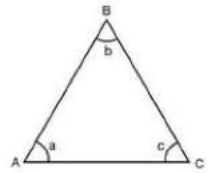
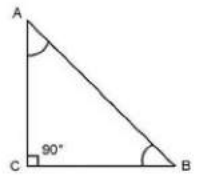
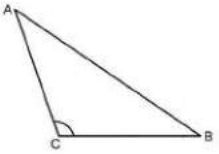
In ΔCDB, $4x + y = 180$ (ii)

Solving (i) and (ii), $x = 32°$

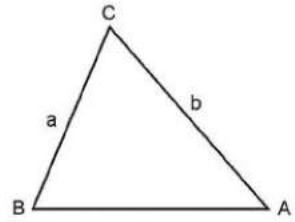
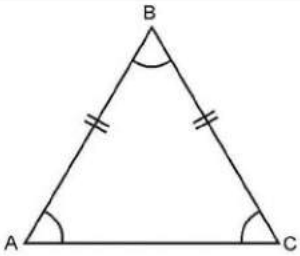
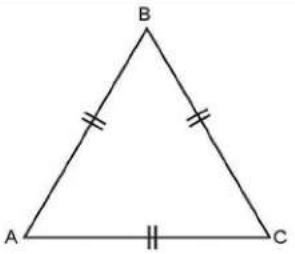
So, $2x = 64°$



Summarizing the Above Classification

Types of Triangles	*Property/Definition*	*Diagram*
Acute-angle Triangle	Each angle of a triangle is less than 90°, i.e., a < 90°, b < 90°, c < 90°	 {∠a, ∠b, ∠c} < 90°
Right-angled Triangle	If one of the angles is equal to 90°, then it is called a right-angled triangle. The rest two angles are complementary to each other.	 ∠C = 90°
Obtuse-angle Triangle	If one of the angles is obtuse (i.e., greater than 90°), then it is called an obtuse-angle triangle	 ∠C > 90°

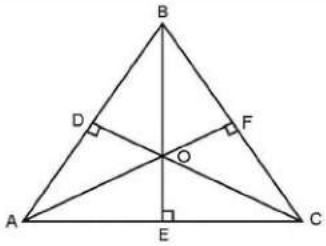
(b) According to the length of sides

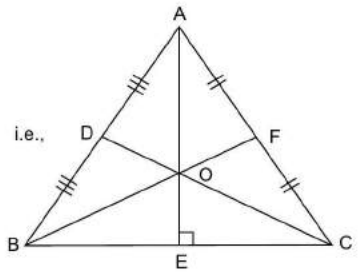
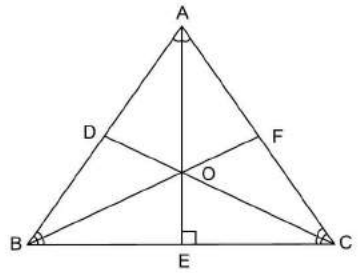
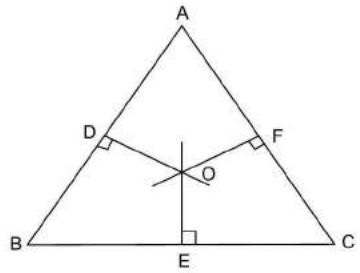
Types of Triangles	*Property/Definition*	*Diagram*
Scalene Triangle	A triangle in which none of the three sides are equal is called a scalene triangle (all the three angles are also different).	 $a \neq b \neq c$
Isosceles Triangle	A triangle in which two sides are equal is called an isosceles triangle. In this triangle, the angles opposite to the congruent sides are also equal.	 $AB = AC$ $\angle B = \angle C$
Equilateral Triangle	A triangle in which all the three sides are equal is called an equilateral triangle. In this triangle each angle is congruent and equal to 60°.	 $AB = BC = AC \ \angle A = \angle B = \angle C = 60°$

Points of a Triangle

Before we move ahead to discuss different points inside a triangle, we need to be very clear about some of the basic definitions.

Basic Definitions

Nomenclature	*Property/Definition*	*Diagram*
Altitude (or height)	The perpendicular drawn from the opposite vertex of a side in a triangle is called an altitude of the triangle. There are three altitudes in a triangle.	 AF, CD and BE are the altitudes

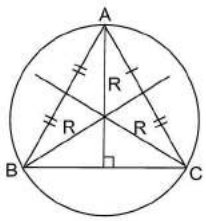
Nomenclature	*Property/Definition*	*Diagram*
Median	The line segment joining the mid-point of a side to the vertex opposite to the side is called a median. There are three medians in a triangle. A median bisects the area of the triangle. Area (ABE) = Area (AEC) $=\frac{1}{2}$ Area (ΔABC) etc.	 i.e., AE, CD and BF are the medians (BE = CE = AD = BD, AF = CF)
Angle bisector	A line segment which originates from a vertex and bisects the same angle is called an angle bisector. ($\angle BAE = \angle CAE = \frac{1}{2}\angle BAC$) etc.	 AE, CD and BF are the angle bisectors
Perpendicular bisector	A line segment which bisects a side perpendicularly (i.e., at right angle) is called a perpendicular bisector of a side of triangle. All points on the perpendicular bisector of a line are equidistant from the ends of the line.	 DO, EO and FO are the perpendicular bisectors

Circumcentre

Circumcentre is the point of intersection of the three perpendicular bisectors of a triangle. The circumcentre of a triangle is equidistant from its vertices and the distance of the circumcentre from each of the three vertices is called circumradius (R) of the triangle. These perpendicular bisectors are different from altitudes, which are perpendiculars but not necessarily bisectors of the side.

The circle drawn with the circumcentre as the centre and circumradius as the radius is called the circumcircle of the triangle and it passes through all the three vertices of the triangle.

The circumcentre of a right-angled triangle is the mid-point of the hypotenuse of a right- angled triangle.



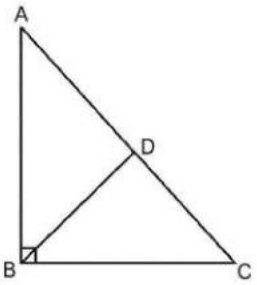
AB = c, BC = a, AC = b

The process to find the circumradius (R) For any triangle $R=\frac{abc}{4A}$, where a, b and c are the three sides and A = area of a triangle.

For equilateral triangle $R=\frac{\text{Side}}{\sqrt{3}}$

Positioning of the circumcentre

- If the triangle is acute-angled triangle, then the circumcentre will lie inside the triangle.
- If the triangle is obtuse-angled triangle, then the circumcentre will lie outside the triangle.
- If the triangle is a right-angled triangle, then the circumcentre will lie on the mid-point of the hypotenuse. This can be seen through the following diagram:

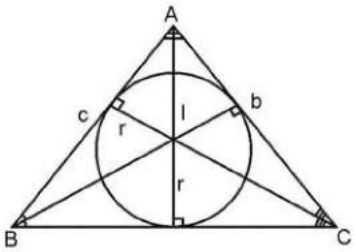


Here D is the circumcentre. So, AD = CD = BD

Incentre

Incentre is the point of intersection of the internal bisectors of the three angles of a triangle. The incentre is equidistant from the three sides of the triangle i.e., the perpendiculars drawn from the incentre to the three sides are equal in length and are called the inradius of the triangle.

The circle drawn with incentre as the centre and inradius as the radius is called the incircle of the triangle and it touches all the three sides from the inside.



AB = c, BC = a, CA = b

To find inradius (r)

For any triangle $r = \frac{A}{S}$, where

A = Area of triangle and

S = Semi-perimeter of the triangle $\frac{(a+b+c)}{2}$

For equilateral triangle $r = \frac{\text{side}}{2\sqrt{3}}$

$\angle BIC = 90° + \angle A/2$

Important derivation – In a right-angled triangle, Inradius = Semi-perimeter – Length of Hypotenuse.

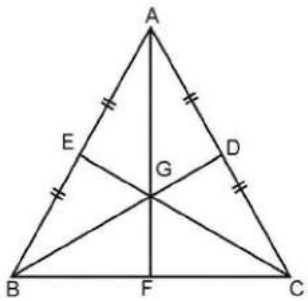
Euler's formula for inradius and circumradius of a triangle Let O and I be the Circumcentre and Incentre of a triangle with circumradius R and inradius r. Let d be the distance between O and I. Then

$d^2 = R(R - 2r)$

From this theorem, we obtain the inequality $r \geq 2r$. This is known as Euler's Inequality.

Centroid

Centroid is the point of intersection of the three medians of a triangle. The centroid divides each of the medians in the ratio 2 : 1, the part of the median towards the vertex being twice in length to the part towards the side.



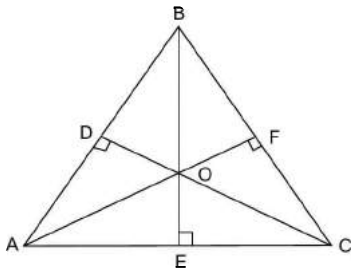
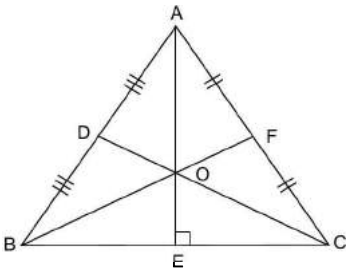
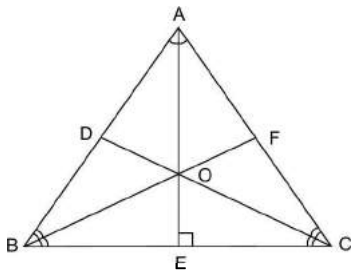
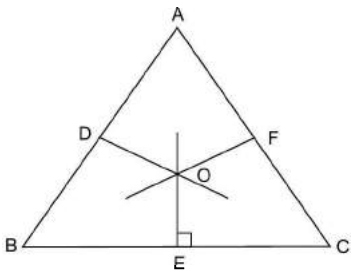
$$\frac{AG}{GF} = \frac{BG}{GD} = \frac{CD}{GE} = \frac{2}{1}$$

Median divides the triangle into two equal parts of the same area.

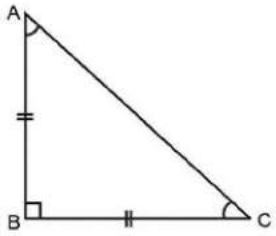
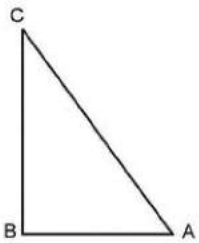
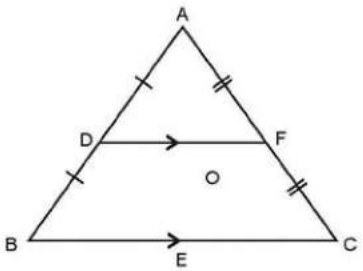
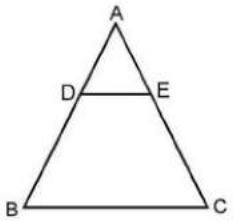
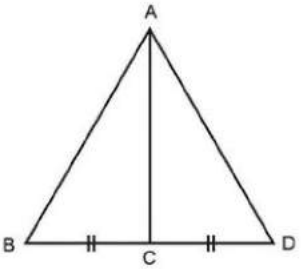
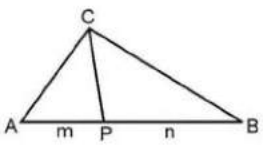
Orthocentre

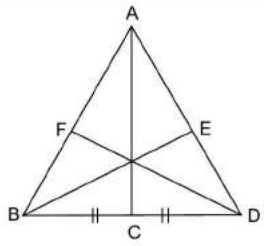
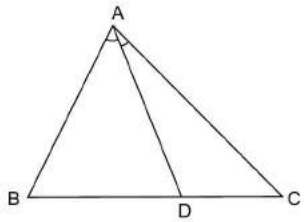
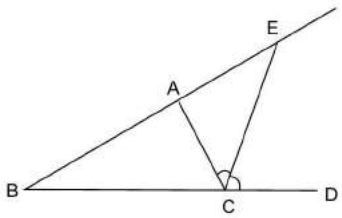
The point of concurrency of the altitudes is known as the orthocentre.

Summarizing the above discussion regarding the points of the triangle:

Orthocentre	The point of intersection of the three altitudes of the triangle is known as the orthocenter. ∠BOC = 190 − ∠A ∠COA = 190 − ∠ B ∠AOB = 180 − ∠ B	 'O' is the orthocenter
Centroid	The point of intersection of the three medians of a triangle is called the centroid. A centroid divides each median in the ratio 2 : 1 (vertex: base) $\frac{AO}{OE}=\frac{CO}{OD}=\frac{BO}{OF}=\frac{2}{1}$	 'O' is the centroid
Incentre	The point of intersection of the angle bisectors of a triangle is known as the incentre. Incentre O is the always equidistant from all three sides i.e., the perpendicular distance between the sides.	 'O' is the incentre
Circumcentre	The point of intersection of the perpendicular bisectors of the sides of a triangle is called the circumcentre. OA = OB = OC = (circum radius) Circumcentre O is always equidistant from all the three vertices A, B and C Perpendicular bisectors need not be originating from the vertices.	 'O' is the incentre

Important Theorems Related to Triangle

Theorem	*Statement/Explanation*	*Diagram*
45° – 45° – 90°	If the angles of a triangle are 45°, 45° and 90°, then the hypotenuse (i.e., longest side) is $\sqrt{2}$ times of any smaller side. Excluding hypotenuse rest two sides are equal. i.e., AB = BC and AC $= \sqrt{2}$ AB $= \sqrt{2}$ BC AB : BC : AC = 1 : 1 : $\sqrt{2}$	 ∠A = 45° ∠B = 90° ∠C = 45°
30° – 60° – 90°	If the angles of a triangle are 30°, 60° and 90°, then the sides opposite to 30° angle is half of the hypotenuse and the side opposite to 60° is $\frac{\sqrt{3}}{2}$ times the hypotenuse, e.g., AB $= \frac{AC}{2}$ and $= \frac{\sqrt{3}}{2}$ AC AB : BC: AC = 1: $\sqrt{3}$: 2	 ∠C = 30°, ∠B = 90°, ∠A = 60°
Basic Proportionality Theorem (BPT) or	Any line parallel to one side of a triangle divides the other two sides proportionally. So if DE is drawn parallel to BC, it would divide sides AB and AC proportionally i.e $\frac{AD}{DB} = \frac{AF}{FC}$ or $\frac{AD}{AB} = \frac{AF}{AC}$ $\frac{AD}{DF} = \frac{AB}{BC}$.	
Mid-point Theorem	Any line joining the mid-points of two adjacent sides of a triangle are joined by a line segment, then this segment is parallel to the third side, i.e., if AD = BD and AE = CE then DE||BC.	
Apollonius' Theorem	In a triangle, the sum of the squares of any two sides of a triangle is equal to twice the sum of the square of the median to the third side and square of half the third side. i.e., $AB^2 + AD^2 = 2(AC^2 + BC^2)$	
Stewarts Theorem/ Generalization of Apollonius Theorem	If length of AP = m and PB = n, then m × CB^2 + n × AC^2 = (m + n) PC^2 + mn (m + n) Also understand that *m* and *n* here are length of segments, and not their ratio.	

Theorem	*Statement/Explanation*	*Diagram*
Extension of Apollonius' Theorem	In the given Δ ABC, AC, BE and DF are medians. 3 (Sum of squares of sides) = 4 (Sum of squares of medians) $3(AB^2 + AD^2 + DB^2) = 4(AC^2 + EB^2 + FD^2)$	
Interior Angle Bisector Theorem	In a triangle the angle bisector of an angle divides the opposite side to the angle in the ratio of the remaining two sides. i.e., $\frac{BD}{CD} = \frac{AB}{AC}$ and BD × AC − CD × AB = AD^2	
Exterior Angle Bisector Theorem	In a triangle the angle bisector of any exterior angle of a triangle divides the side opposite to the external angle in the ratio of the remaining two sides i.e., $\frac{BE}{AE} = \frac{BC}{AC}$	

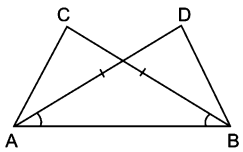
Congruency of Triangles

Two figures are said to be congruent if, when placed one over the other, they completely overlap each other. They would have the same shape, the same area and will be identical in all respects.

So, we can say that all congruent triangles are similar triangles, but vice-versa is not always true.

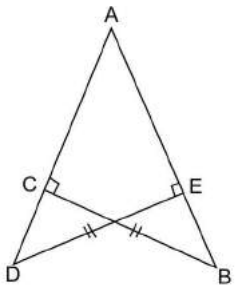
Rules for two triangles to be congruent

1. S – S – S
   If in any two triangles, each side of one triangle is equal to a side of the other triangle, the two triangles are congruent. This rule is S – S – S rule.
2. S – A – S
   In Δ ABC and Δ ABD,
   AB = AB (common side)
   ∠ABC = ∠ BAD (given)
   BC = AD (given)

   Thus by rule S – A – S the two triangles are congruent. This rule holds true, when the angles that are equal have to be included between the two equal sides (i.e., the angle should be formed between the two sides that are equal).
3. A – S – A
   In Δ ABC and Δ ADE,
   ∠ACB = ∠ AED (given)
   ∠ BAC = ∠ DAE (common angle)
   BC = DE (given)

   Thus by rule A – S – A the two triangles are congruent. For this rule, the side need not be the included side.

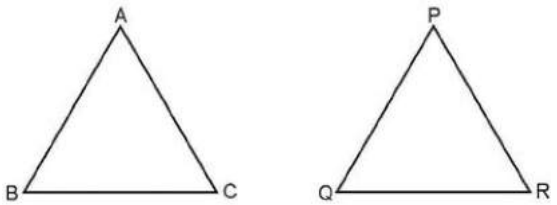
   A – S – A can be written as A – A – S or S – A – A also.
4. R – H – S
   This rule is applicable only for right-angled triangles.
   If two right-angled triangles have their hypotenuse and one of the sides as same, then the triangles will be congruent.

Similarity of the Triangles

If we take two maps of India of different sizes (breadths and lengths), then the map of all the 28 states of India will cover proportionally the same percentage area in both the maps.

Lets see this in geometry:

Criteria for similarity of two triangles



Two triangles are similar if (i) their corresponding angles are equal and/or (ii) their corresponding sides are in the same ratio. That is, if in two triangles, ABC and PQR, (i) $\angle A = \angle P$, $\angle B = \angle Q$, $\angle C = \angle R$, and/or (ii) $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}$, the two triangles are similar.

All regular polygons of the same number of sides such as equilateral triangles or squares, are similar. In particular, all circles are also similar.

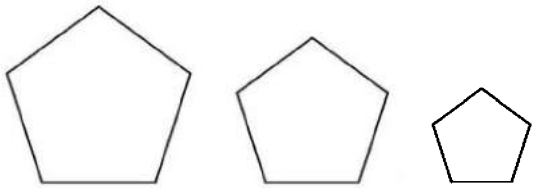
Theorems for Similarity

1. If in two triangles, the corresponding angles are equal, then their corresponding sides will also be proportional (i.e., in the same ratio). Thus the two triangles are similar.
   This property is referred to as the AAA similarity criterion for two triangles.
   Corollary If two angles of a triangle are respectively equal to two angles of another triangle, then the two triangles are similar. This is referred to as the AA similarity criterion for the two triangles. It is true due to the fact that if two angles of one triangle are equal to the two angles of another triangle, then the third angle of both the triangles will automatically be the same.
2. If the corresponding sides of two triangles are proportional (i.e., in the same ratio), their corresponding angles will also be equal and so the triangles are similar. This property is referred to as the SSS similarity criterion for the two triangles.
3. If one angle of a triangle is equal to one angle of the other and the sides including these angles are proportional, then the triangles are similar. This property is referred to as the SAS similarity criterion of the two triangles.
4. The ratio of the areas of the two similar triangles is equal to the ratio of the squares of their corresponding sides.
5. If a perpendicular is drawn from the vertex of the right angle of a right angled triangle to the hypotenuse, the triangles on each side of the perpendicular are similar to the whole triangle and to each other.

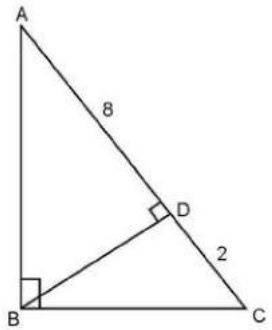
Similar Polygons

Two polygons of the same number of sides are similar, if

(i) Their corresponding angles are equal (i.e., they are equiangular) and
(ii) Their corresponding sides are in the same ratio (or proportional).
This can be seen in the figures given below:



Example 7 ΔABC is a right angled triangle $BD \perp AC$. If AD = 8 cm and DC = 2 cm, then BD = ?



(a) 4 cm (b) 4.5 cm
(c) 5 cm (d) Cannot be determined

Solution $\Delta ADB \sim \Delta BDC$

$\therefore \frac{AD}{BD} = \frac{BD}{DC}$

$\therefore BD^2 = AD \times DC = 8 \times 2$

$\therefore BD^2 = 16$

$\therefore BD = 4$ cm

Important Result of this question $BD^2 = AD \times DC$ can be used as a standard result also.

Example 8 Circles with radii 3, 4 and 5 units touch each other externally. If P is the point of intersection of the tangents to these circles at their point of contact, find the distance of P from the point of contacts of the circles.

Solution Let A, B and C be the centres of the three circles. So, the point P will be the incentre of DABC and distance of P from the point of contacts of the circles will be the inradius (r).

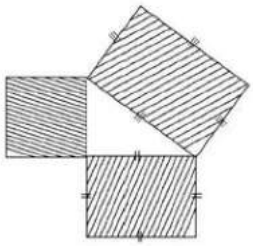
So, $r = \frac{A}{S}$

Sides of DABC will be 7 units, 8 units and 9 units.

So, $r = \sqrt{5}$

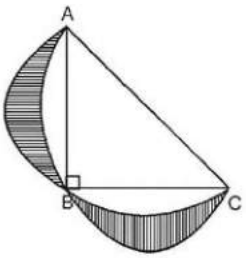
Some interesting facts

1. If we draw regular polygons on all the sides of a right angled triangle, taking the sides of the triangle as one of the sides of the figures, then the area of the shaded portion is equal to twice the area of the figure on the hypotenuse. This will also hold true for semi-circles whose diameters will be the side of the right-angled triangle.

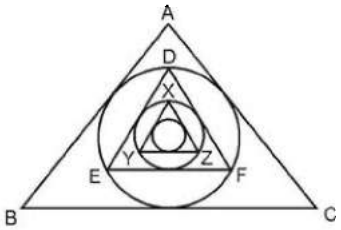


Area of the shaded region = 2 (area of figure on hypotenuse)

2. In the figure below, two semi-circles are drawn with diameters equal to the sides of the right angled triangle. The area of the shaded region (the crescents) is equal to the area of the right-angled triangle. Area of the shaded region (is equal to) the area of the right angled triangle.

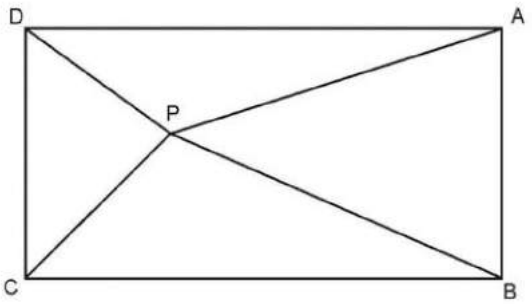


3. In the figure given below all triangles are equilateral triangles and circles are inscribed in these triangles. If the side of triangle ABC = a, then the side of triangle DEF $= \frac{a}{2}$ and the side of triangle $xyz = \frac{a}{4}$.



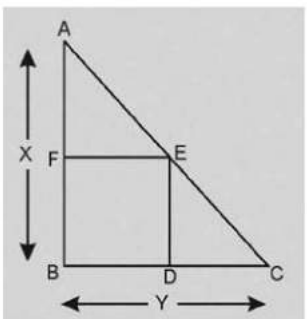
In other words, we can say in order to obtain the side of the next inner triangle divide the side of the immediate outer triangle by 2. The same algorithm holds true for the inscribed circles.

4. In the figure given below, if P is any point inside rectangle ABCD, then $PA^2 + PC^2 = PB^2 + PD^2$

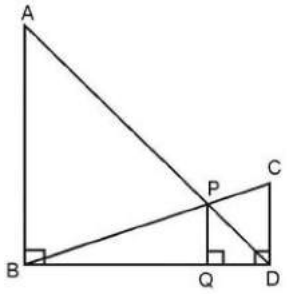


Some Important Points

1. (a) In an acute angled triangle, the circumcentre lies inside the triangle.
   (b) In a right-angled triangle, the circumcentre lies on the middle point of the hypotenuse.
   (c) In an obtuse angled triangle, the circum centre lies outside the triangle.
2. (a) In an acute angled triangle, the orthocenter lies inside the triangle.
   (b) In a right-angled triangle, the orthocenter lies on the vertex is where the right angle is formed i.e., the vertex opposite to the hy potenuse.
   (c) In an obtuse angled triangle, the orthocenter lies outside the triangle.
3. In a right angled triangle the length of the median drawn to the hypotenuse is equal to half the hypotenuse. This median is equal to the circumradius (R) of the right angled triangle.
4. In the figure given below triangle ABC is a right angle triangle, right angled at B. Side AB measures x units and BC measures y units. If a square (BDEF) the maximum area is inscribed in the triangle as shown, below then the side of the square is equal to $\frac{xy}{x+y}$.



Example 9 In the figure given below, ∠ABD = ∠CDB = ∠PQD = 90°. If AB:CD = 3:1, then what is the ratio of CD: PQ?

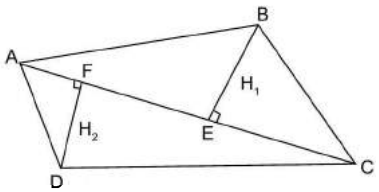


Solution Using the proportionality theorem, AB/PQ = BD/QD and PQ/CD = BQ/BD. Multiplying both these equations, we get AB/CD = BQ/QD = 3:1

Hence, CD/PQ = BD/BQ = 4 : 3.

QUADRILATERALS AND THEIR PROPERTIES

A quadrilateral is a figure bounded by four sides. In the figure given below ABCD is a quadrilateral. Line AC is the diagonal of the quadrilateral (denoted by d) and BE and DF are the heights of the triangles ABC and ADC respectively (denoted by h_1 and h_2).



AC = d, BE = h, and DE = h_2

(i) Area $= \frac{1}{2} \times$ one diagonal × (sum of perpendiculars to the diagonal from the opposite vertexes) $= \frac{1}{2}$ d $(h_1 + h_2)$

(ii) Area $= \frac{1}{2} \times$ product of diagonals × sine of the angle between them

(iii) Area of the cyclic quadrilateral $= \sqrt{(s-a)(s-b)(s-c)(s-d)}$ where *a*, *b*, *c* and *d* are the sides of quadrilateral and *s* = semi-perimeter $= \frac{a+b+c+d}{2}$

(iv) *Brahmagupta's formula* For any quadrilateral with sides of length *a*, *b*, *c* and *d*, the area *A* is given by

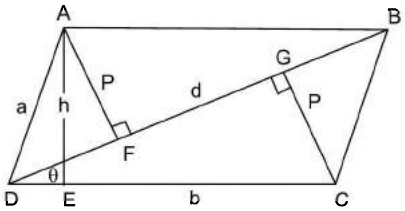
$$A = \sqrt{(s-a)(s-b)(s-c)(s-d) - abcd\cos^2\frac{1}{2}(A+B)}$$

Where $s = \frac{a+b+c+d}{2}$ is known as the semi-perimeter, A is the angle between sides *a* and *d*, and B is the angle between the sides *b* and *c*.

Different Types of Quadrilaterals

Parallelogram

A parallelogram is a quadrilateral when its opposite sides are equal and parallel. The diagonals of a parallelogram bisect each other.



Given: AD = BC = a and AB = DC = b

BD = d

AF (height of ΔABD) = CG (height of ΔCBD) and AE = height of the parallelogram = h

∠ADC = θ

(i) Area = base × height

(ii) Area = (any diagonal) × (perpendicular distance to the diagonal from the opposite vertex)

(iii) Area = (product of adjacent sides) × (sine of the angle between them) Area = AB sin θ

(iv) Area $= 2\sqrt{s(s-a)(s-b)(s-d)}$, where a and b are the adjacent sides and d is the diagonal.

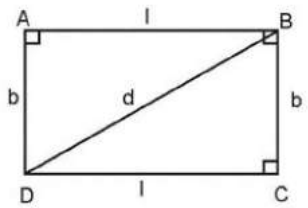
(v) $AC^2 + BD^2 = 2(AB^2 + BC^2)$

(vi) The parallelogram that is inscribed in a circle is a rectangle.

(vii) The parallelogram that is circumscribed about a circle is a rhombus.

(viii) A parallelogram is a rectangle if is diagonals are equal.

Rectangle



A rectangle is a quadrilateral when its opposite sides are equal and each internal angle equals 90°. The diagonals of a rectangle are equal and bisect each other.

Given: AD = BC = b and AB = DC = l, BD = d

(i) Area = length × breadth Area = lb

(ii) Perimeter = 2 (length + breadth) Perimeter = 2 (l + b)

(iii) Diagonal^2 = length^2 + breadth^2 (Pythagoras Theorem)

$d^2 = l^2 + b^2$ $d = \sqrt{l^2 + b^2}$

(iv) Finding area using Brahmagupta' Formula In this case, we know that $a = c$ and $b = d$, and A + B = π So, area of Rectangle

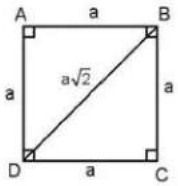
$= \sqrt{(a+b-a)(a+b-b)(a+b-a)(a+b-b) - a.b.a.b\cos^2 90°} = ab$

$= ab$

(v) The quadrilateral formed by joining the mid points of intersection of the angle bisectors of a parallelogram is a rectangle.

Square

A square is a quadrilateral when all its sides are equal and each internal angle is of 90°. The diagonals of a square bisect each other at right angles (90°)



Given: AB = BC = CD = DA = a

BD (diagonal) = $a\sqrt{2}$

(i) Area = (side)^2 $= \frac{(\text{diagonal})^2}{2} = \frac{(\text{perimeter})^2}{16}$

Area $= a^2 = \frac{d^2}{2} = \frac{P^2}{16}$

(ii) Using Brahmagupta's formula to find out the area of a square:

We know that $a = b = c = d$ and A + B = π

So, area of square

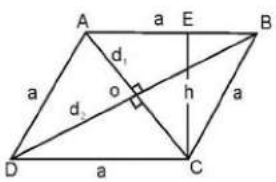
$= \sqrt{(s-a)(s-b)(s-c)(s-d) - abcd\cos^2\frac{1}{2}(A+B)} =$

$\sqrt{(2a-a)(2a-a)(2a-a)(2a-a) - a.a.a.a.\cos^2 90°} = a^2$

(ii) Perimeter = 4 (side) ⇒ Perimeter = 4a

Rhombus

A rhombus is a quadrilateral when all sides are equal. The diagonals of a rhombus bisects each other at right angles (90°)



Given = AB = BC = CD = DA = a

∠AOB = ∠BOC = ∠COD − ∠DOA = 90°

AC = d, (AO = OC) and BD = d_2 (BO = OD) CE (height) = h

(i) Area $= \frac{1}{2} \times$ (product of the diagonals)

Area $= \frac{1}{2}\ d_1\ d_2$

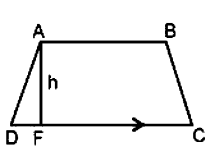
(ii) Area = base × height

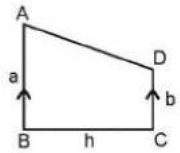
Area = a × h

(iii) A parallelogram is a rhombus if its diagonals are perpendicular to each other. Remember, the sum of the square of the diagonals is equal to four time the square of the side i.e., $d_1^2 + d_2^2 = 4a^2$

Trapezium

A trapezium is a quadrilateral in which only one pair of the opposite sides is parallel





Given: AB = a and CD = b

In Fig. 1, AF (height) = h and in Fig. 2, BC (height) = h

(i) Area $= \frac{1}{2} \times$ (sum of the parallel sides) × (distance between the parallel sides)

Area $= \frac{1}{2}\ (a + b)\ h$

(ii) The line joining the mid-points of the non-parallel sides is half the sum of the parallel sides and is known as Median.

(iii) If we make non-parallel sides equal, then the diagonals will also be equal to each other.

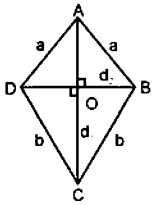
(iv) Diagonals intersect each other proportionally in the ratio of the lengths of the parallel sides.

(v) If a trapezium is inscribed inside a circle, then it is an isosceles trapezium with oblique sides being equal.

Kite

Kite is a quadrilateral when two pairs of adjacent sides are equal and the diagonals bisect each other at right angles (90°).

Given: AB = AD = a and BC = DC = b



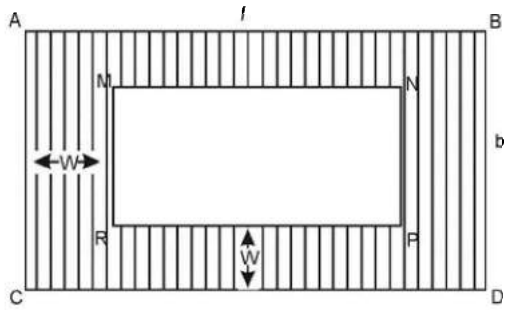
$AC = d_1$ (AO = OC) and $BD = d_2$ (BO = OD)
$\angle AOB = \angle BOC = \angle COD = \angle DOA = 90°$

(i) Area $= \frac{1}{2} \times$ (Product of the diagonals)

Area $= \frac{1}{2} d_1 d_2$

Area of Shaded Paths

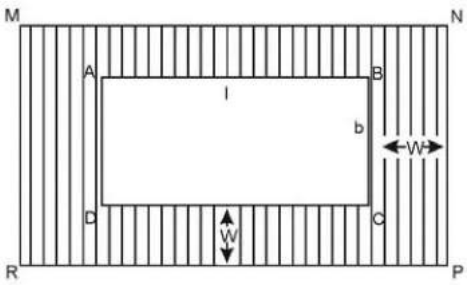
Case 1 When a pathway is made outside a rectangle having length = l and breadth = b



ABCD is a rectangle with length = l and breadth = b, the shaded region represents a pathway of uniform width = W

Area of the shaded region/pathway = 2w (l + b − 2w)

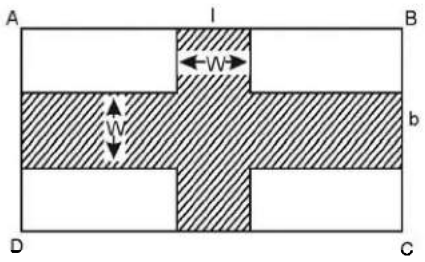
Case 2 When a pathway is made inside a rectangle having length = l and breadth = b



ABCD is a rectangle with length = l and breadth = b, the shaded region represents a pathway of uniform width = w

Area of the shaded region/pathway = 2w (l + b + 2w)

Case 3 When two pathways are drawn parallel to the length and breadth of a rectangle having length = l and breadth = b



ABCD is a rectangle with length = l and breadth = b, the shaded region represents two pathways of a uniform width = w

Area of the shaded region/pathway = W (l + b − w)

From the above figure we can observe that the area of the paths does not change on shifting their positions as long as they are perpendicular to each other.

We can conclude from here that:

1. Every rhombus is a parallelogram but the converse is not true.
2. Every rectangle is a parallelogram but the converse is not true.
3. Every square is a parallelogram but the converse is not true.
4. Every square is a rhombus but the converse is not true.
5. Every square is a rectangle but the converse is not true.

Construction of new figures by joining the mid-points

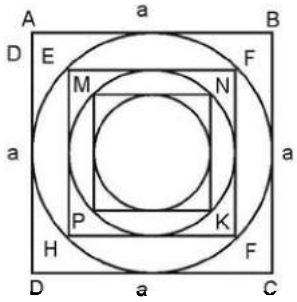
	Original Figure		Resulting Figure
Lines joining the mid-points of adjacent sides of	Quadrilateral	form	Parallelogram
	Parallelogram		Parallelogram
	Rectangle		Rhombus
	Rhombus		Rectangle
	Trapezium		Four similar Δ

Properties of diagonals

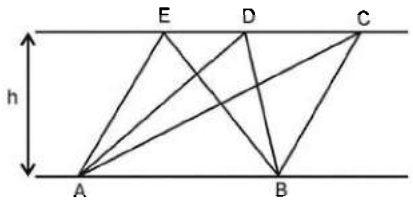
Properties		*Types of Quadrilaterals*				
Sl. No.		*Square*	*Rectangle*	*Parallelogram*	*Rhombus*	*Trapezium*
1.	Diagonals are equal	Y	N	N	Y	N
2.	Diagonals bisect each other	Y	Y	Y	Y	N
3.	Diagonals bisect vertex angles	Y	N	N	Y	N
4.	Diagonals are at right angles	Y	N	N	Y	N
5.	Diagonals make congruent triangles	Y	N	N	Y	N

Some important points

1. In the figure given below, all the side quadrilaterals are squares and circles are inscribed in these squares. If the side of the square ABCD = a, then the side of square EFGH $= \frac{a}{\sqrt{2}}$ and the side of square MNKP $= \frac{a}{2}$. In other words we can say that in order to obtain the side of the next inner square, divide the side of the immediate outer square by $\sqrt{2}$. The same procedure will be applied for the inscribed circles i.e., we divide the radius of the immediate outer circle by $\sqrt{2}$ to obtain the radius of the next inner circle.



2. Triangles on the same base and between the same parallel lines are equal in area.

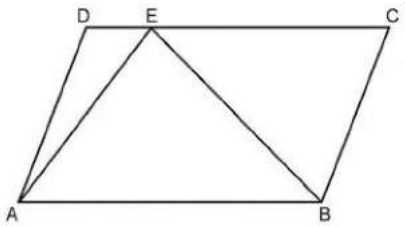


Area ($\angle$ABC) = Area (ΔABD) = Area (ΔABE)

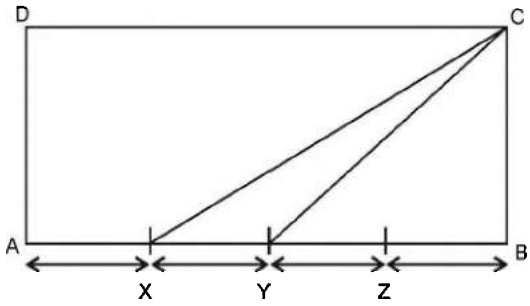
$= \frac{1}{2}$ (AB) × h

(Base AB is the same and the height 'h' is also the same.)

3. If a parallelogram and a triangle are drawn on the same base and between the same parallel lines, then the area of the parallelogram is twice the area of the triangle.
Area of ABCD = 2 × (Area of Δ ABE)



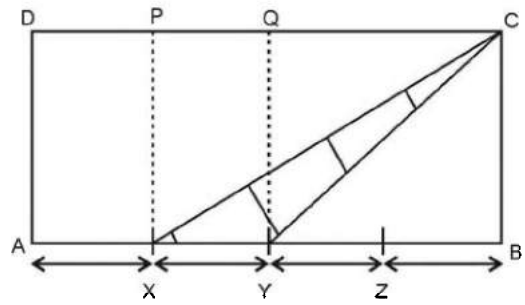
Example 10 Side AB of a rectangle ABCD is divided into four (4) equal parts as shown in the figure. Find the ratio of the area (ΔXYC) and area (€ABCD)?



Solution Let the area of the rectangle ABCD = A

Area of the rectangle XYQP = A/4

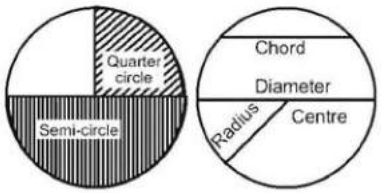
Rectangle XYQP and ΔXYC are on the same base and between the same parallel lines.



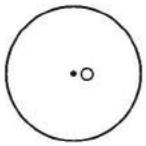
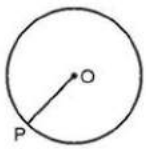
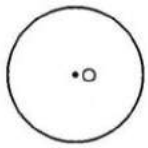
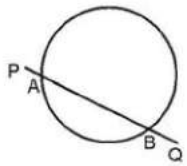
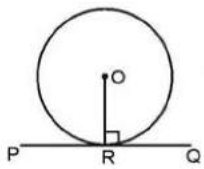
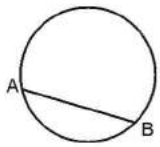
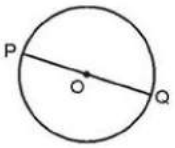
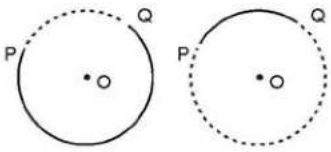
Thus area (XYC) = A/8

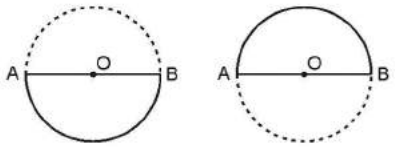
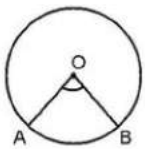
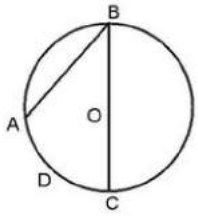
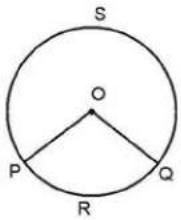
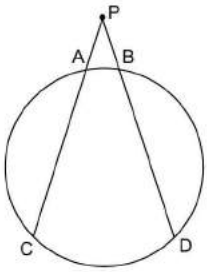
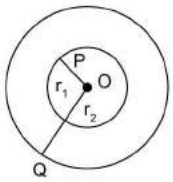
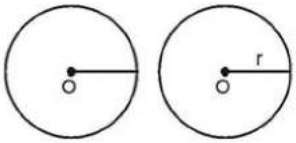
CIRCLES AND THEIR PROPERTIES

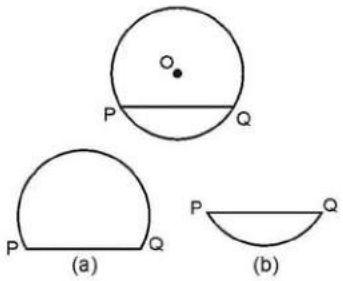
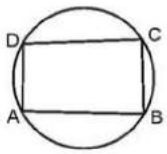
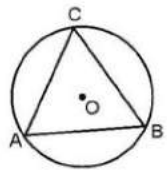
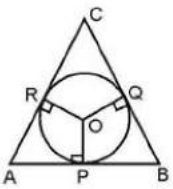
A circle is the path travelled by a point which moves in such a way that its distance from a fixed point remains constant. The fixed point is known as the centre and the fixed distance is called the radius.



Before we move ahead, let us understand the basics definitions of circle.

Nomenclature	*Definition*	*Diagram*
Centre	The fixed point is called the centre. In the given diagram 'O' is the centre of the circle.	
Radius	The fixed distance is called a radius. In the given diagram OP is the radius of the circle. (point P lies on the circumference)	
Circumference	The circumference of a circle is the distance around a circle, which is equal to $2\pi r$. (r →radius of the circle)	
Secant	A line segment which intersects the circle in two distinct points, is called as secant. In the given diagram secant PQ intersects circle at two points at A and B.	
Tangent	A line segment which has one common point with the circumference of a circle, i.e., it touches only at only one point is called as tangent of circle. The common point is called as point of contact. In the given diagram, PQ is a tangent which touches the circle at a point R.	 (R is the point of contact) Note: Radius is always perpendicular to tangent.
Chord	A line segment whose end points lie on the circle. In the given diagram AB is a chord.	
Diameter	A chord which passes through the centre of the circle is called the diameter of the circle. The length of the diameter is twice the length of the radius. In the given diagram PQ is the diameter of the circle. (O→is the centre of the circle)	
Arc	Any two points on the circle divides the circle into two parts the smaller part is called as minor arc and the larger part is called as major arc. It is denoted as ''. In the given diagram PQ is arc.	

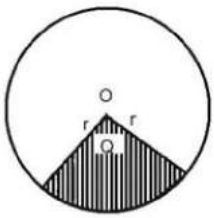
Nomenclature	*Definition*	*Diagram*
Semicircle	A diameter of the circle divides the circle into two equal parts. Each part is called a semicircle.	
Central Angle	An angle formed at the centre of the circle, is called the central angle. In the given diagram ∠AOB in the central angle.	
Inscribed Angle	When two chords have one common end point, then the angle included between these two chords at the common point is called the inscribed angle. ∠ABC is the inscribed angle by the arc ADC	
Measure of an Arc	Basically, it is the central angle formed by an arc. e.g., (a) measure of a circle = 360° (b) measure of a semicircle = 180° (c) measure of a minor arc = ∠POQ (d) measure of a major arc = 360 − ∠POQ	 m(arc PRQ = m ∠POQ m(arc PSQ) = 360° − m (arc PRQ)
Intercepted Arc	In the given diagram, AB and CD are the two intercepted arcs, intercepted by ∠CPD. The end points of the arc must touch the arms of ∠CPD, i.e., CP and DP.	
Concentric Circles	Circles having the same centre at a plane are called the concentric circles. In the given diagram, there are two circles with radii r_1 and r_2 having the common (or same) centre. These are called as concentric circl Ces.	
Congruent Circles	Circles with equal radii are called as congruent circles.	

Nomenclature	*Definition*	*Diagram*
Segment of a Circle	A chord divides a circle into two regions. These two regions are called the segments of a circle: (a) major segment (b) minor segment.	
Cyclic Quadrilateral	A quadrilateral whose all the four vertices lie on the circle.	
Circumcircle	A circle which passes through all the three vertices of a triangle. Thus the circumcentre is always equidistant from the vertices of the triangle. OA = OB = OC (circumradius)	
In Circle	A circle which touches all the three sides of a triangle i.e., all the three sides of a triangle are tangents to the circle is called an incircle. Incircle is always equidistant from the sides of a triangle.	

Now come to different formula and theorems attached to circle:

Circumference of a circle = $2\pi r$

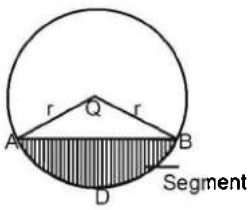
Area of a circle = πr^2, where r is the radius.



Area of a sector $= \pi r^2 \frac{\theta}{360^\circ}$

Circumference of a sector $= 2\pi r \frac{\theta}{360} +$

Perimeter of a sector $= 2r\left(\frac{\pi\theta}{360} + 1\right)$



Area of a segment = Area of a sector OADB – Area of triangle OAB

Area of a segment $= \pi r^2 \frac{\theta}{360} - \frac{1}{2} r^2 \text{Sin}\theta$

Common Tangents and Secants of Circles

Depending upon the positioning of the circles, two or more than two circles can have a common tangent. Following is a list indicating the number of common tangents in case of two circles: